

Organic Technology



Univ.-Prof. Dipl.-Ing. Dr. Christian Paulik

Topics to be covered

- Industrial chemistry
- Raw materials and energy
- Synthesis gas
 - C1 building blocks
- Petrochemistry
 - Important products and intermediates
 - C2, C3 compounds
 - Aromatics
- Polymers & recycling of polymers

Literature

- Hans-Jürgen Arpe, Industrielle Organische Chemie
 - Mark Benvenuto, Industrial Chemistry
 - Harold Wittcoff, Industrial Organic Chemicals
 - Robert Meyers, Handbook of Petrochemicals Production Processes
 - Winnacker-Küchler
 - Ullmann
-
- And of course the 4w
www & Wikipedia

Exams

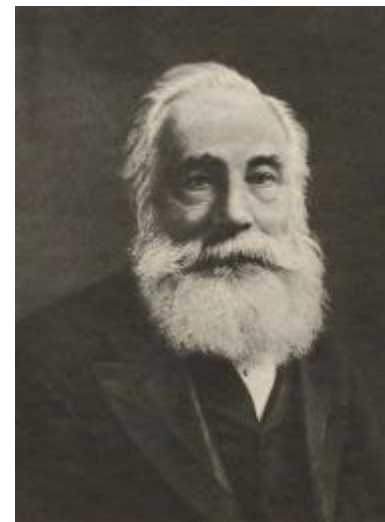
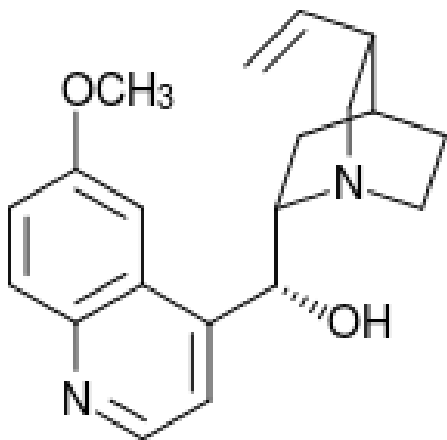
- Oral exams
- Basically every 2nd Monday
- Dates to be agreed with cto-office@jku.at
- Cancelling the appointment within the last week before the exam without a medical attest (cto-office@jku.at), will be blocked in 2 followed exam dates or at least 4 weeks!

What is industrial chemistry?

The development, optimization and monitoring of fundamental chemical processes used in industry for transforming raw materials and precursors into useful commercial products for society.

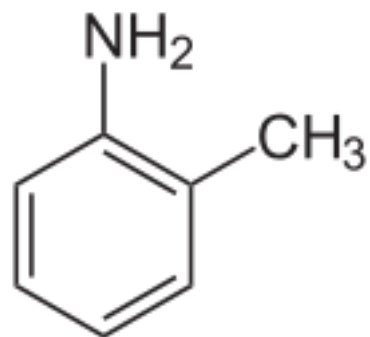
William Henry Perkin

- 1856 discovered Mauveine the first synthetic dye
- Age of 18, during Easter vacation from the Royal College of Chemistry (assistant to A.W. Hofmann)
- On the quest for quinine
(malaria – only from the bark of the cinchona tree grown in the East Indies)

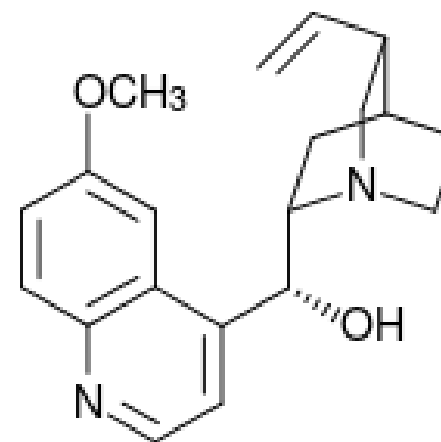
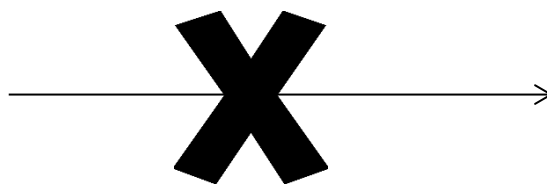


1838 - 1907

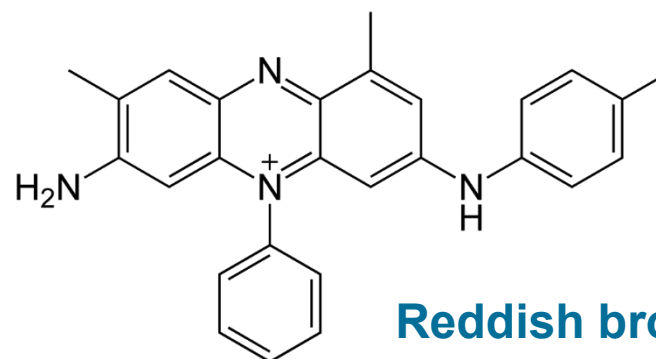
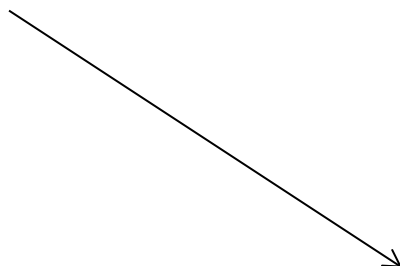
William Henry Perkin



Tolidine



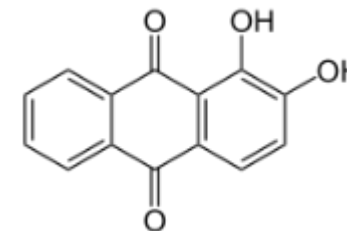
Quinine



Reddish brown sludge

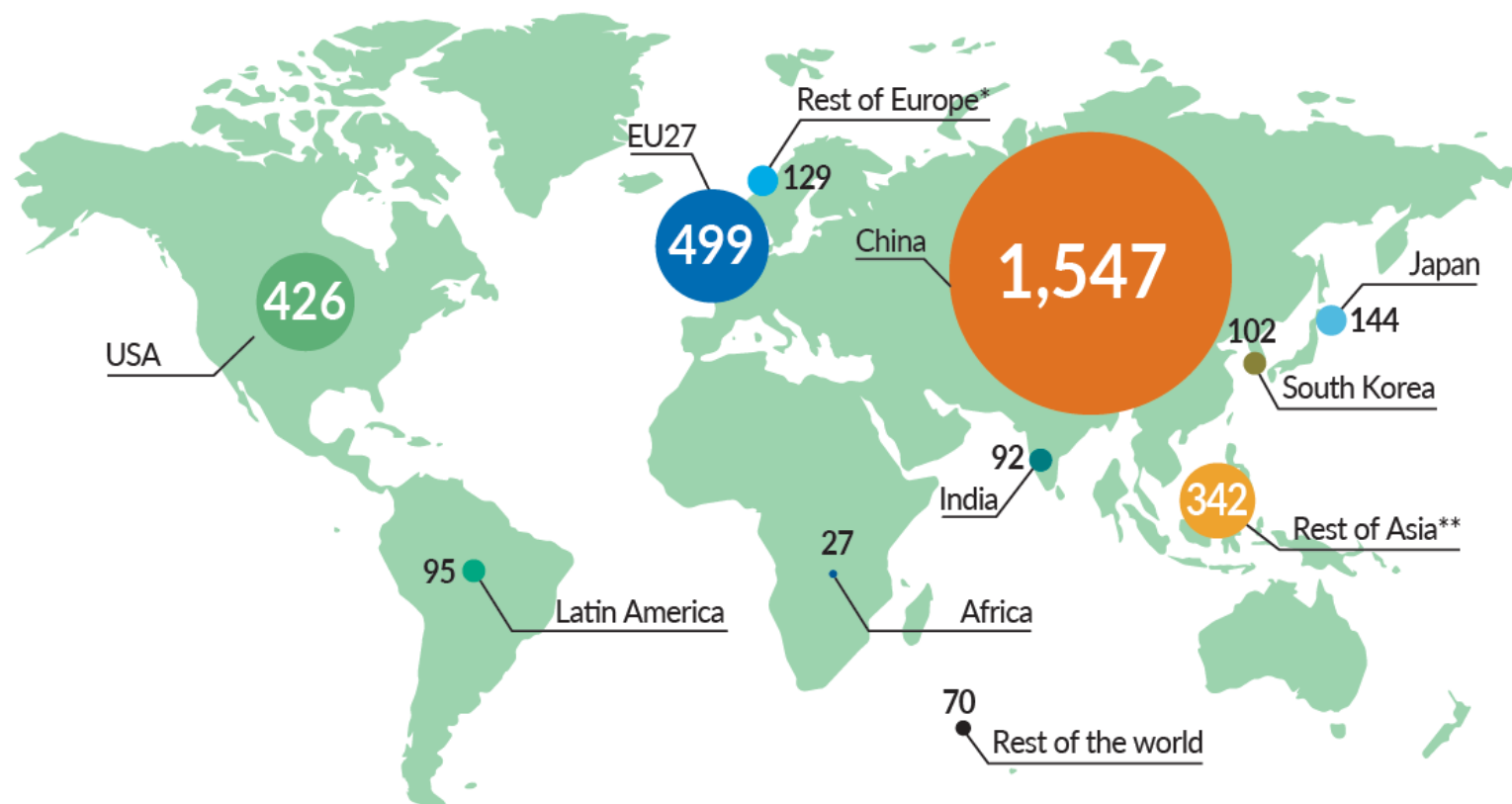
William Henry Perkin

- Perkin tested the purple solutions and found they dye cloth
 - Worked first for silk and later also for cotton
 - Patented the dye
 - Built a factory – solved the problems of up-scaling
 - Great success
 - Color was called: aniline purple, Tyrian purple, mauve
-
- Also Alizarin (red dye) was developed by Perkin
 - With 36 he sold his business and focused only on research



Europe is the second-largest chemicals producer in the world

World chemical sales (2020, €3,471 billion)



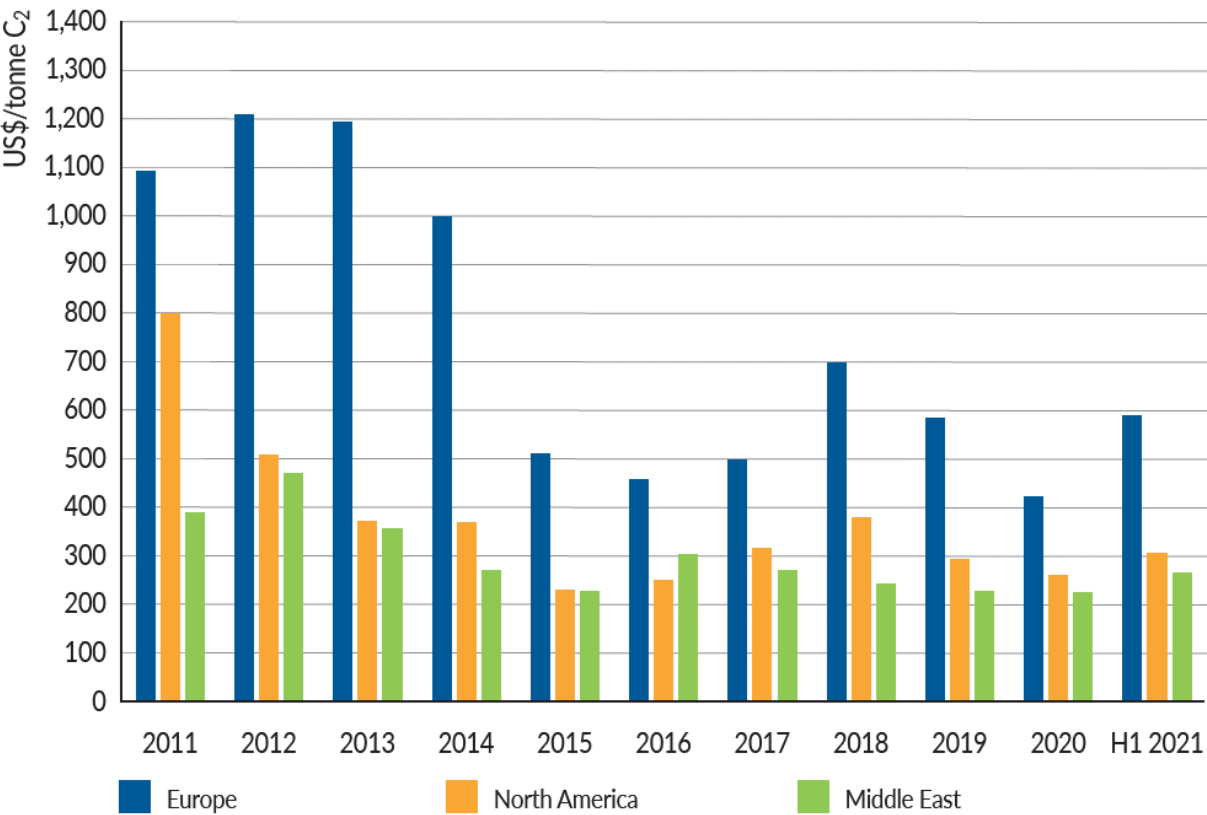
Source: Cefic Chemdata International

* Rest of Europe covers UK, Switzerland, Norway, Turkey, Russia and Ukraine

** Asia excluding China, India, Japan and South Korea

Europe at a competitive disadvantage compared to the USA and the Middle East

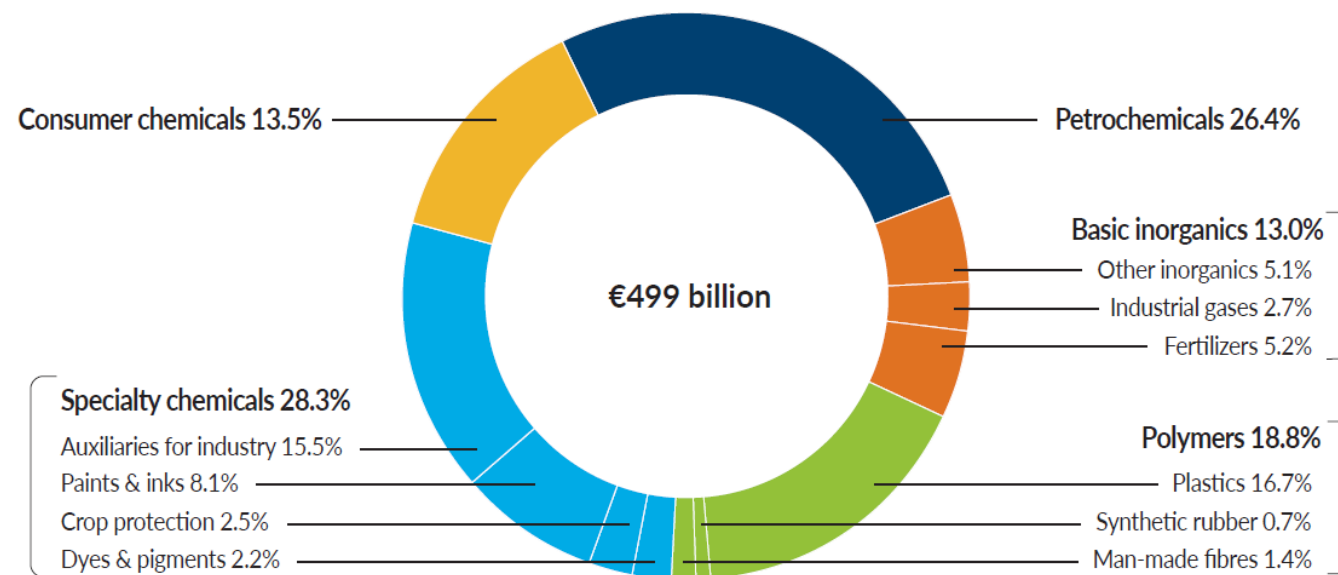
Ethylene cash cost of regional steam crackers



Source: ICIS

Base chemicals account for about 58% of EU27 chemicals sales

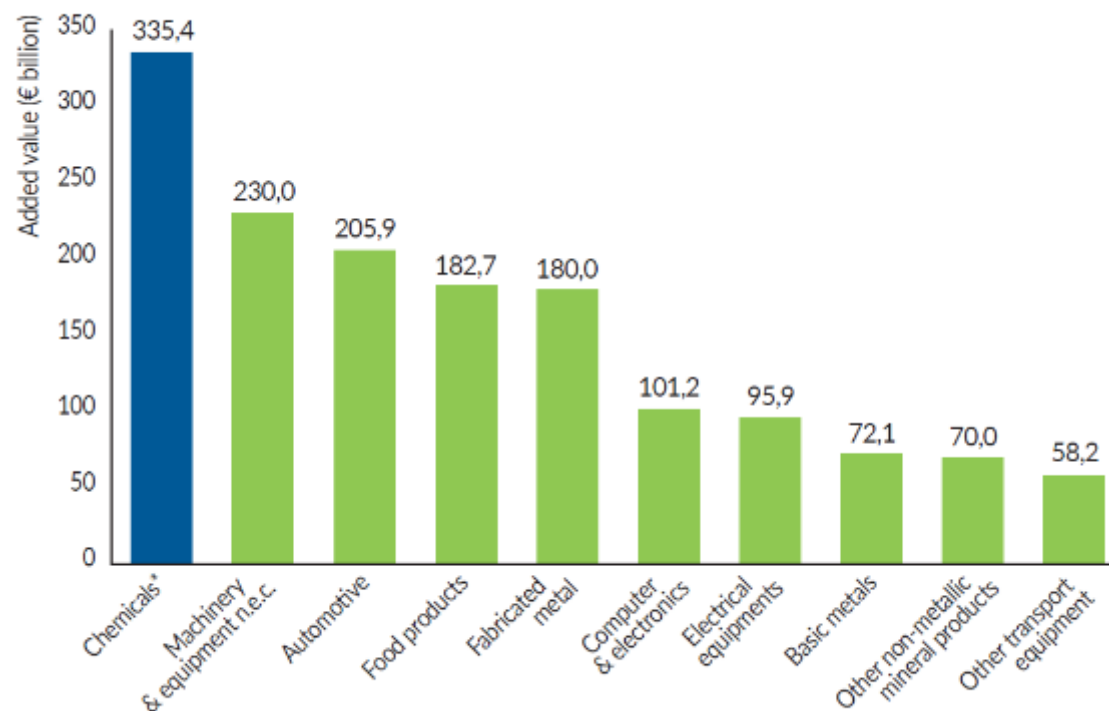
EU27 chemical sales 2020



Source: Cefic Chemdata International

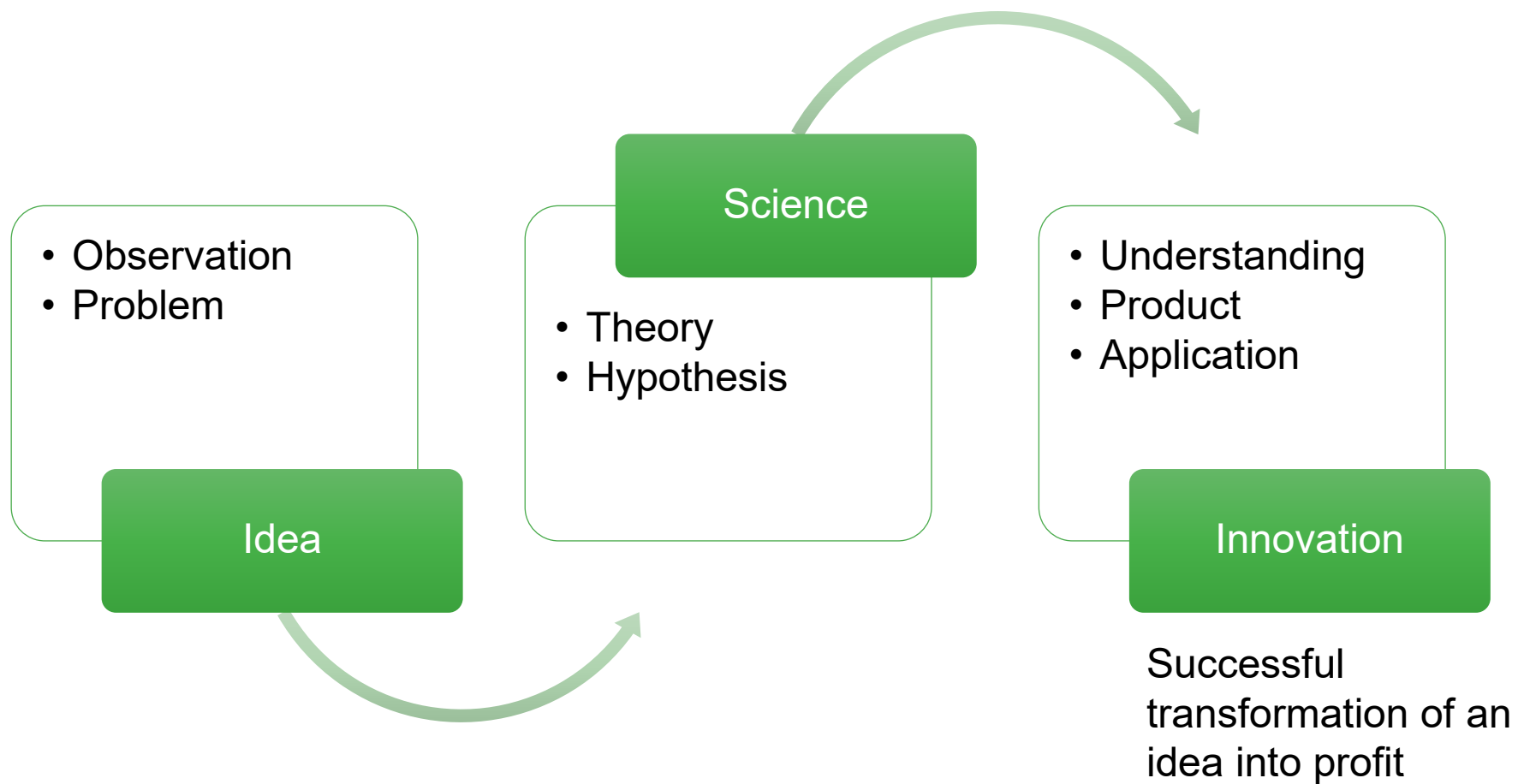
Chemicals* is the leading sector accounting for 17.7% of EU27 manufacturing added value

Top 10 Sectors: added value (€ billion, 2018)



Source: Eurostat
*Pharmaceuticals and rubber & plastics included

The role of science & technology

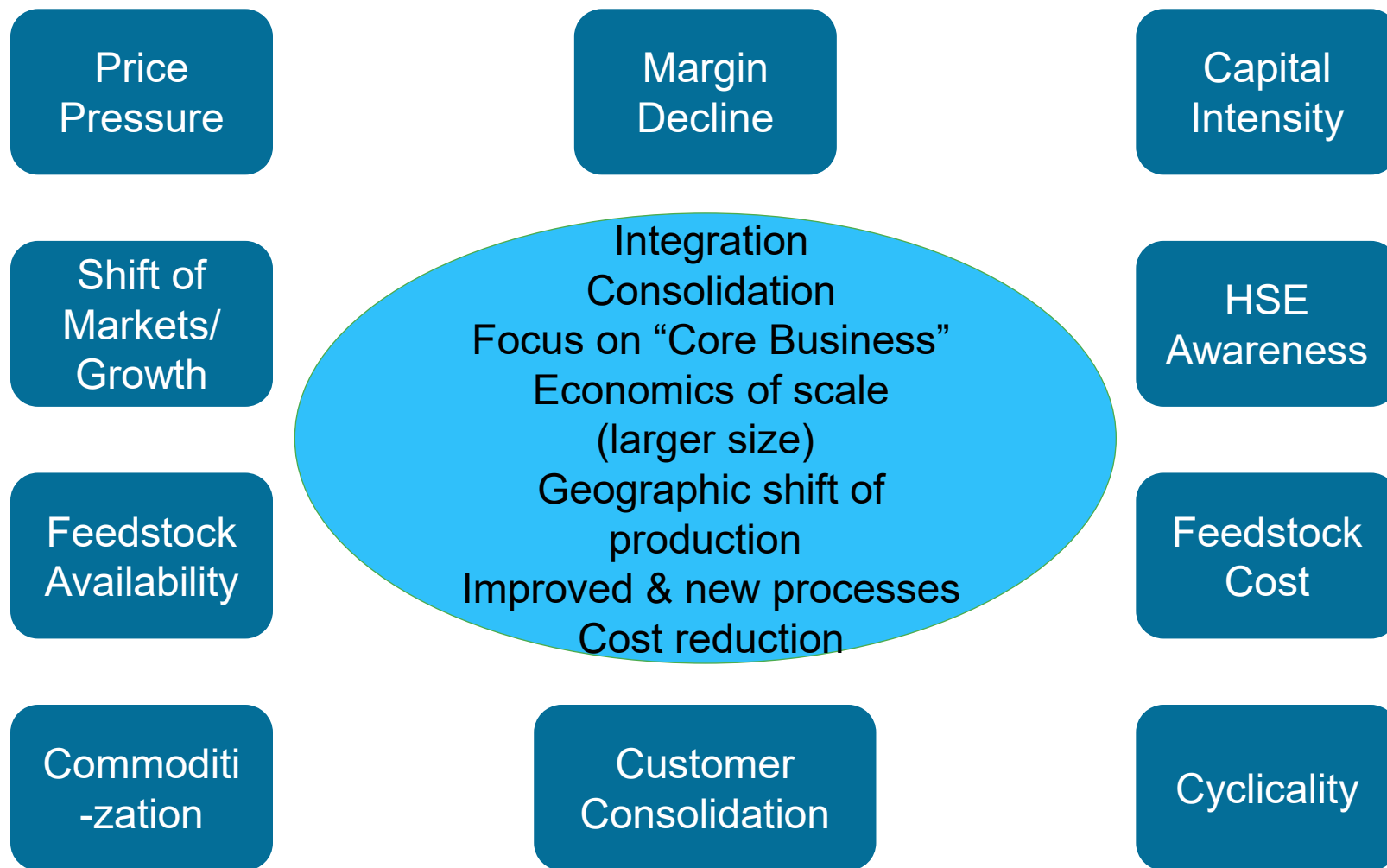


Citius, Altius, Fortius *more, better, cheaper*

- Strong application orientation vs. other sciences
- Strong exposition to needs, environment, markets and economics causes dependency
- Addresses scarcity and performance: “More, Better and Cheaper”



Factors impacting our industry



Circular Economy

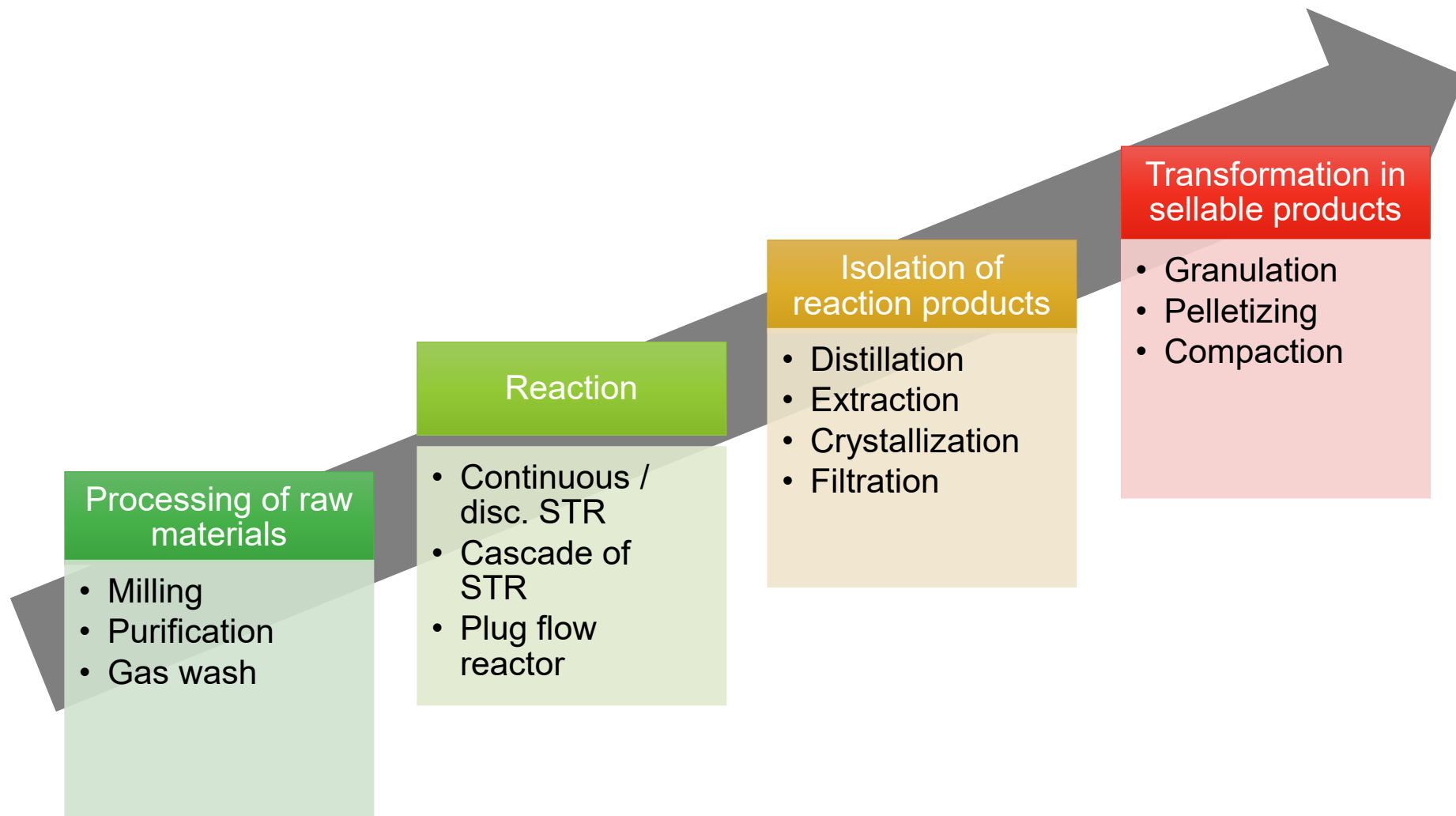


EU Circular Economy Package



Source: United Nations

Basic operations



Location issues

Depending on type of industry:

- Petrochemicals
- Pharmaceuticals
- Size: SME to large industry

- Transportation (easy access)
 - Sea (river), train, road
- Energy (cheap!)
- Water
 - Water supply for cooling, boiler water
 - Waste water (river, sea, ...)
- Workforce (qualified)



Oil pipelines in Austria

Label	Name	From	To	Diameter Inches (cm)	Length (km)	Capacity k b/d (hl/d)
E1	Adria	Wurmlach	Schwechat	18 (45,72)	419	150 (238.480,9)
E2		Zistersdorf	Schwechat	2*8 (20,32)	59	
E21		Schwechat	St. Valentin	16 (40,64)	170	60 (95.392,4)

1 barrel = 158,987295 liter

E1, E2: oil

E21: products

Large industry versus SME

Large industry

- Separate energy supply, short transportation ways
- Vicinity to river
- For electrochemical processes (Chlorine, NaOH):
cheap electricity (nuclear power plants)
- Transportation (railroad, port)

SME

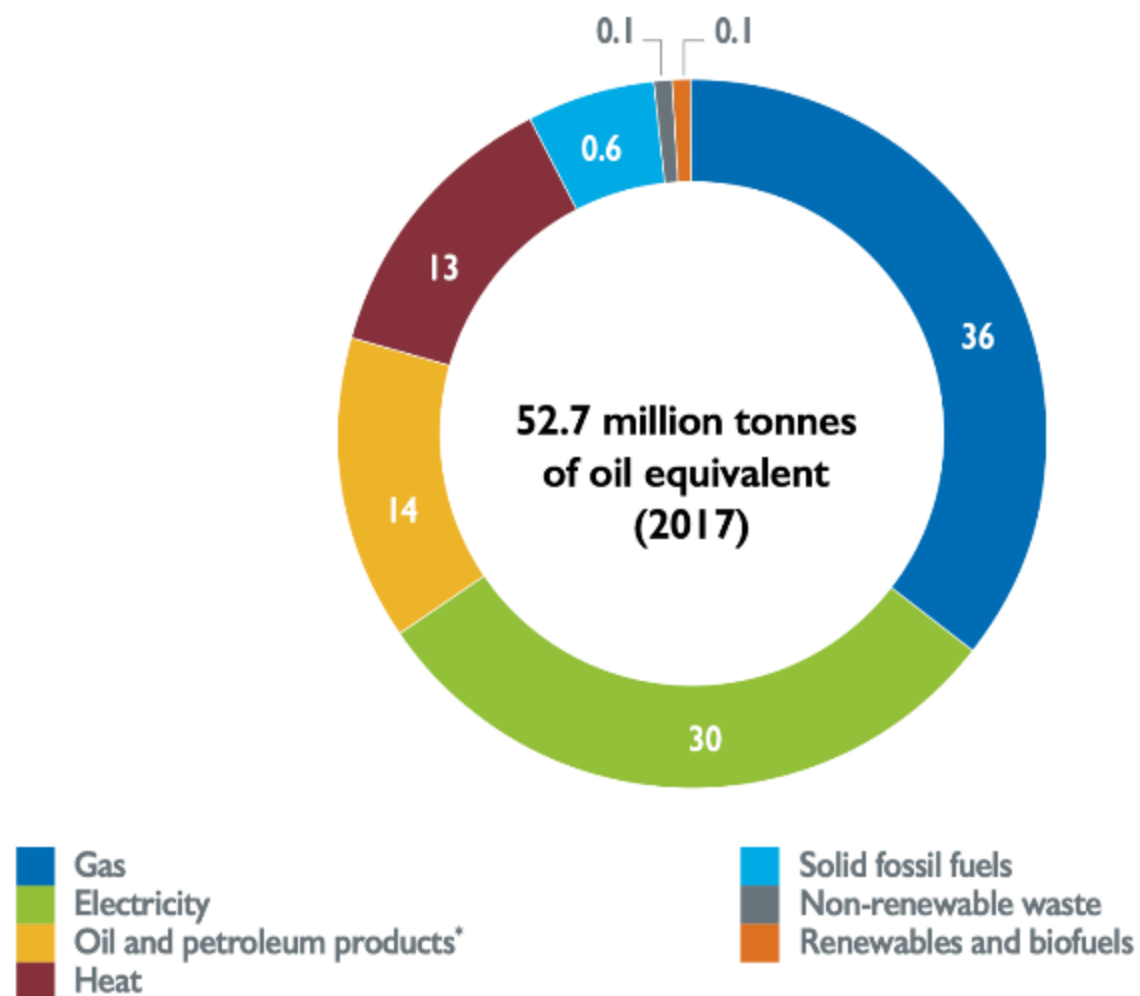
- Smaller volumes
→ railroad & road sufficient
- No separate energy supply necessary (maybe steam generator)

Location economies

- Costs (raw materials, energy, waste, taxes, wages and salaries, ...)
- Infrastructure / transportation
- Price for m² of developed site
- Subsidies
- Duration for authority approvals
- Environmental issues
- Political stability and social climate
- Acceptance of chemical industry / research



Total energy consumption* in the EU chemical industry by source



What is the response of industry?

- Use of a limited number of easily available monomers and/or low molecular weight reactive building blocks
- Production units are linked via pipelines (on site or off site) to allow for a maximum usage of products, by-products and co-products – integration
- Key raw material for building blocks/monomers are:
Ethane (Gas) for C2 and LPG/LNG/Naphtha for C2 and > C2
- C1 Chemistry is based on natural gas via synthesis gas
- Almost all key chemicals are produced from this feedstock via a series of key processes
- On site utility integration (power, steam, air, nitrogen/oxygen etc.) is optimized
- Optimized usage and efficiency

Production processes are big

- >> 200 kt/a
- Capital Cost >> 200 Mio € - steadily increasing
- 50–70% of capital is related to separations/purification
- Construction time > 24 months without engineering
- 8.400 hrs/year operating time
- Highly automated – minimum shift personnel
- Inherent process safety
- Minimum maintenance – high reliability
- Ideally > 3 yrs shut-down/inspection intervals

The “ideal” production process

- High yield (selectivity, conversion)
- Minimum recycle
- Simple Integration (raw materials, by-products)
- Low complexity
- Easy scale-up & good “turn-down” ratio
- Incrementally expandable
- Low cost, low capital
- No HSE issues
- “Transportable”
- “Forgiving”

Process development stages

Process economics / check vs. targets

Lab / bench scale

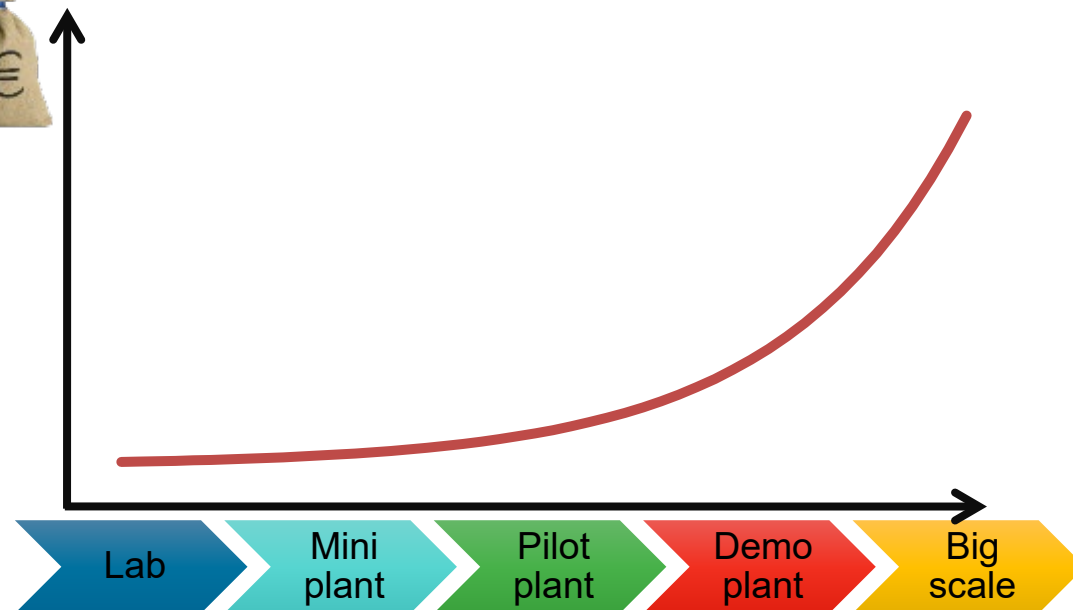
Process concept

Process model

Mini plant / pilot plant

Process package

Investment project



Critical issues in process development

- Product yield, purity
- Process analytics / on-line analysis
- Catalysts
(lifetime, separation)
- By-product separation – product purification
- Recycle streams
- Trace impurities and removal
- Physical / chemical data
- Models
(reactor, separations, process)
- Non-ideality of mixtures
- Liquid/liquid and
solid/liquid separations
- Solvents
- Emissions, waste, regulator

Phases in the realization

- Preparation of the site
- Civil works (basic concrete structures, roads, buildings, etc.)
- Frames, piping, reactors, pumps
- Controls
- Isolation, painting, etc.



Start-up

- Project controlling
- Cleaning (purging and drying)
- Testing of all equipments
- Filling with media
- Test run
- Site acceptance test (72h)
- Standard operation

Energy

- Chemical industry extremely dependant on cheap energy and raw materials
 - High energy demand
 - Dependent on fossil products as raw materials
- Needs to be decoupled
- CO₂ emissions
- Sustainability



Aufschlüsselung nach Ländern (Mtoe) i

Trend anzeigen

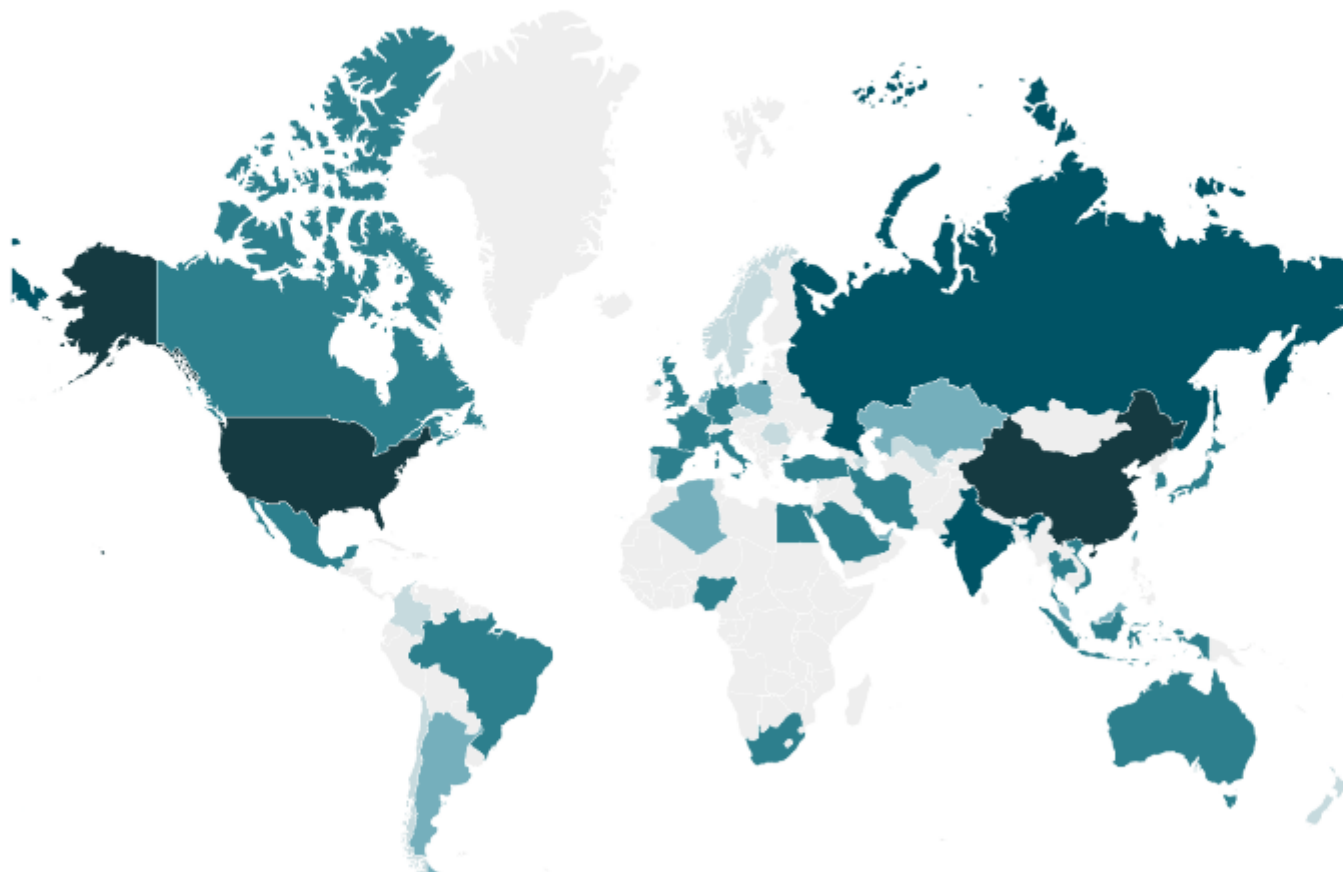
Jahr : 2023



Bild

Einheit: Mtoe

Am höchsten ▼



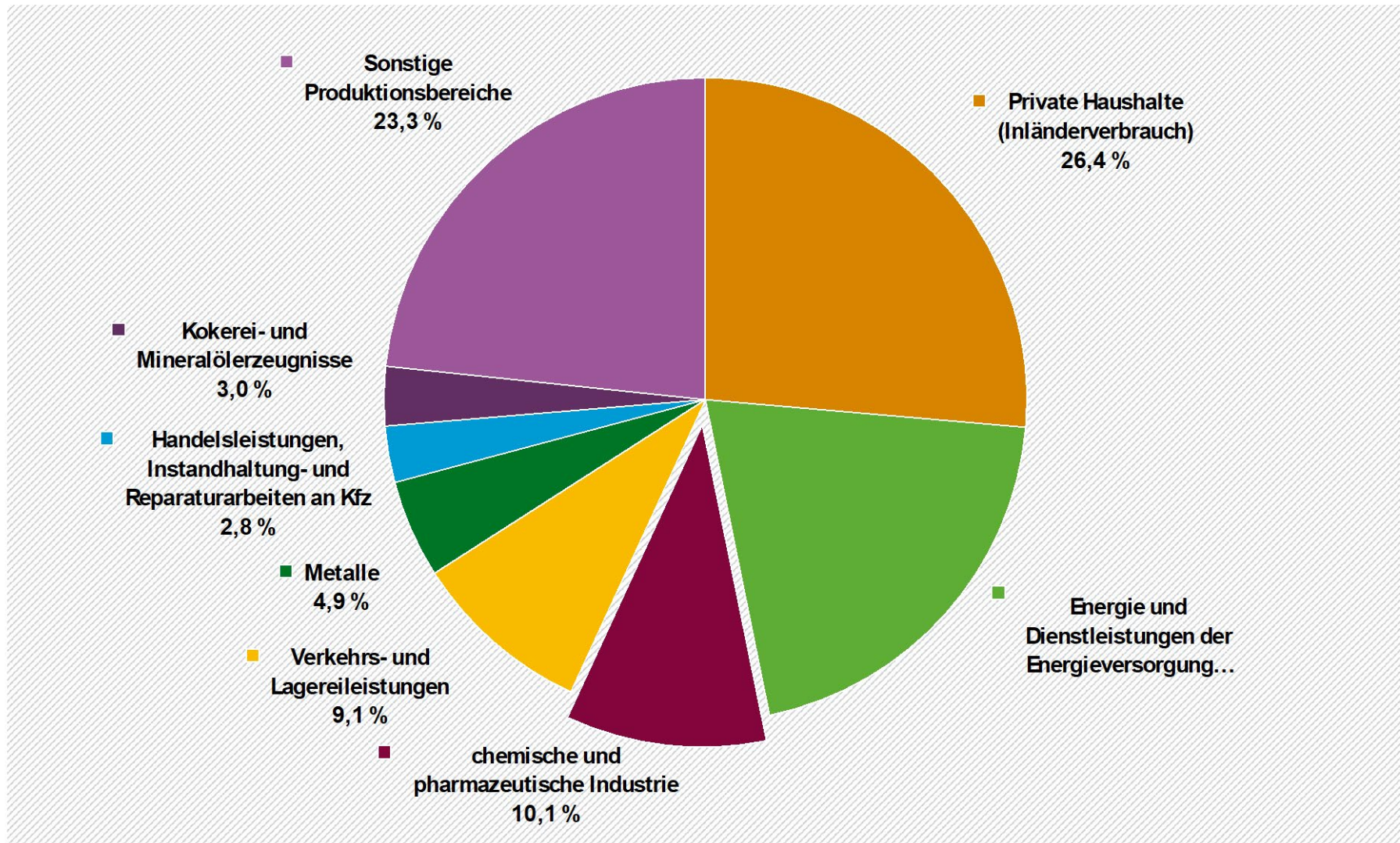
Unten 50 50 Zu 100 100 Zu 500 500 Zu 2000 Über 2000

Um die Statistiken zu einem bestimmten Land anzuzeigen, klicken Sie darauf.

China	4,060
Vereinigte Staaten	2,172
Indien	1,135
Russland	838
Japan	391
Brasilien	336
Iran	317
Indonesien	298
Kanada	297
Südkorea	291
Saudiarabien	279
Deutschland	246

Total energy includes coal, gas, oil, electricity, heat and biomass

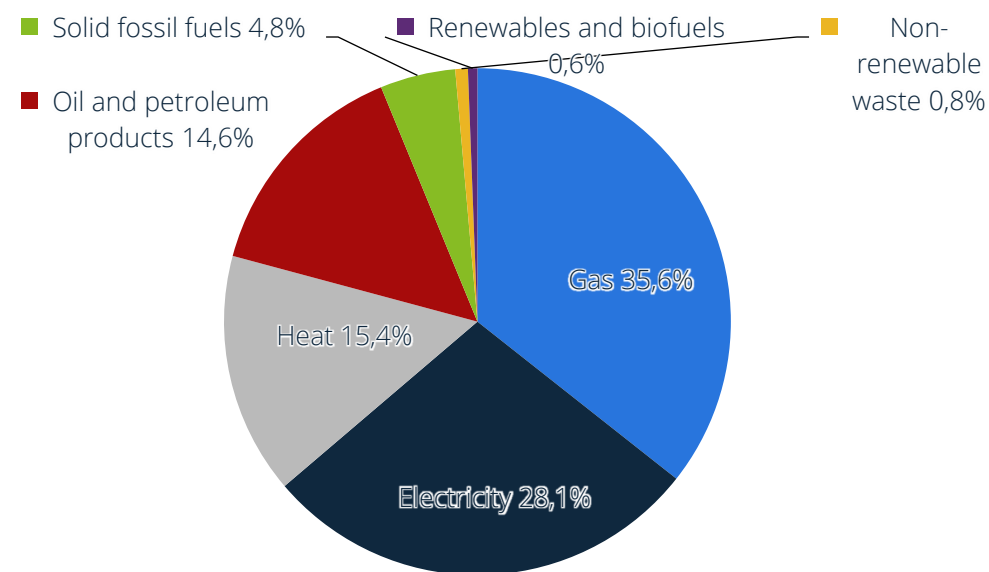
Primärenergieverbrauch nach Sektoren in Deutschland 2016



Quelle: Statistisches Bundesamt 2018, Umwelnutzung und Wirtschaft, Tabellen zu den Umweltökonomischen Gesamtrechnungen, Teil 2 Energie, Tabelle 3.3.4 "Primärennergieverbrauch im Inland - Kraftwerksverluste und Eigenverbrauch beim Energieerzeuger" <https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Umwelt/Publicationen/Umwelnutzung-Wirtschaft/umwelnutzung-und-wirtschaft-energie-xlsx-5850014.html>

Distribution of total energy consumption of the European Union's chemical industry in 2019, by source

EU-27 chemical industry energy consumption share 2019, by source



statista

1st law of thermodynamics

The law of conservation of energy states that the total energy of an isolated system is constant. Energy can be transformed from one form to another, but can be neither created nor destroyed.

The basic measure of energy (E) = Joule (J)

Chemical industry depends on energy

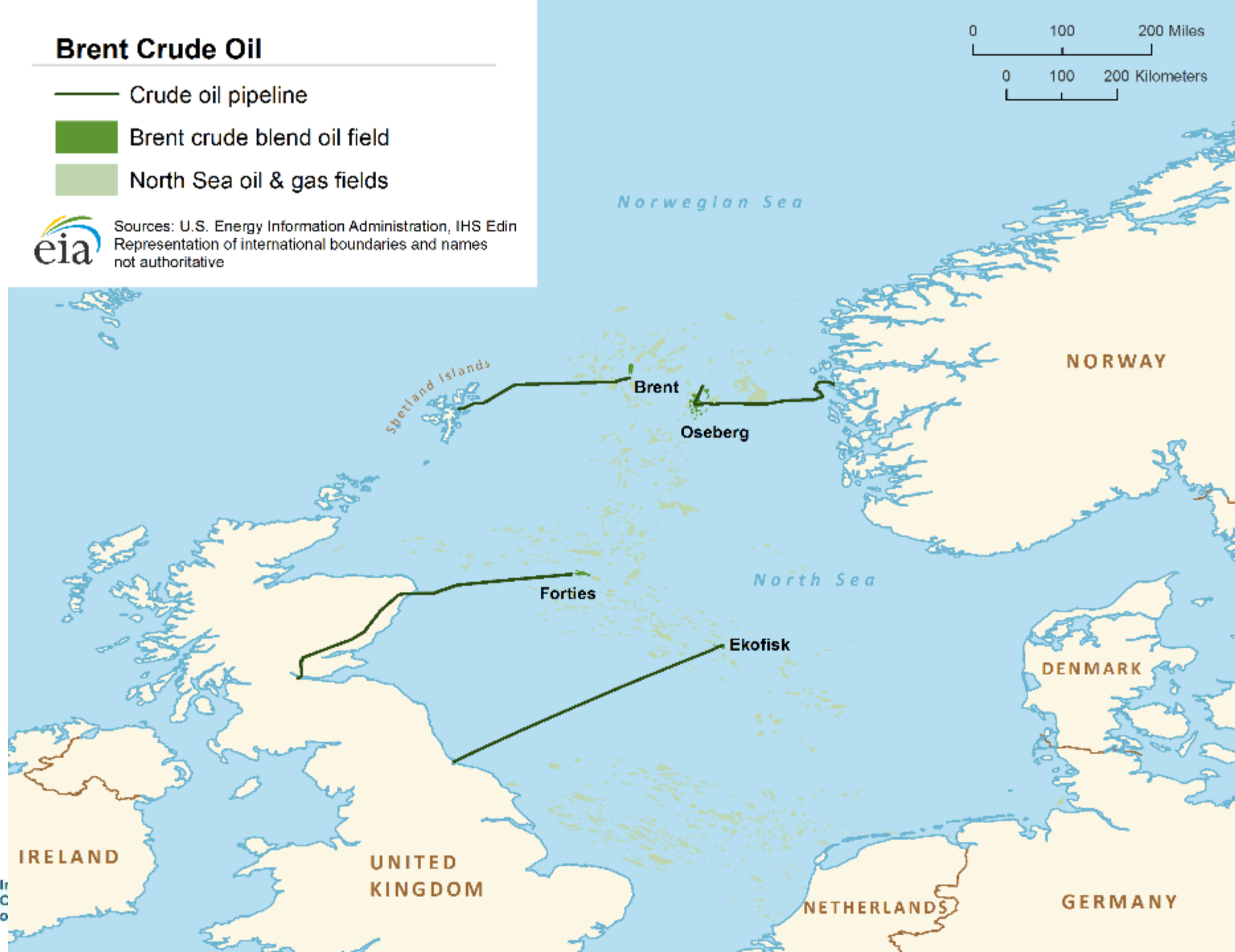
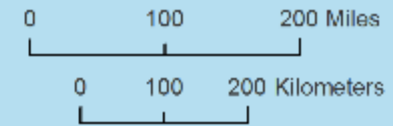
- Chemical industry extremely dependent on cheap energy and raw materials
 - High energy demand
 - Dependent on fossil products as raw materials
- Needs to be decoupled

Brent Crude Oil

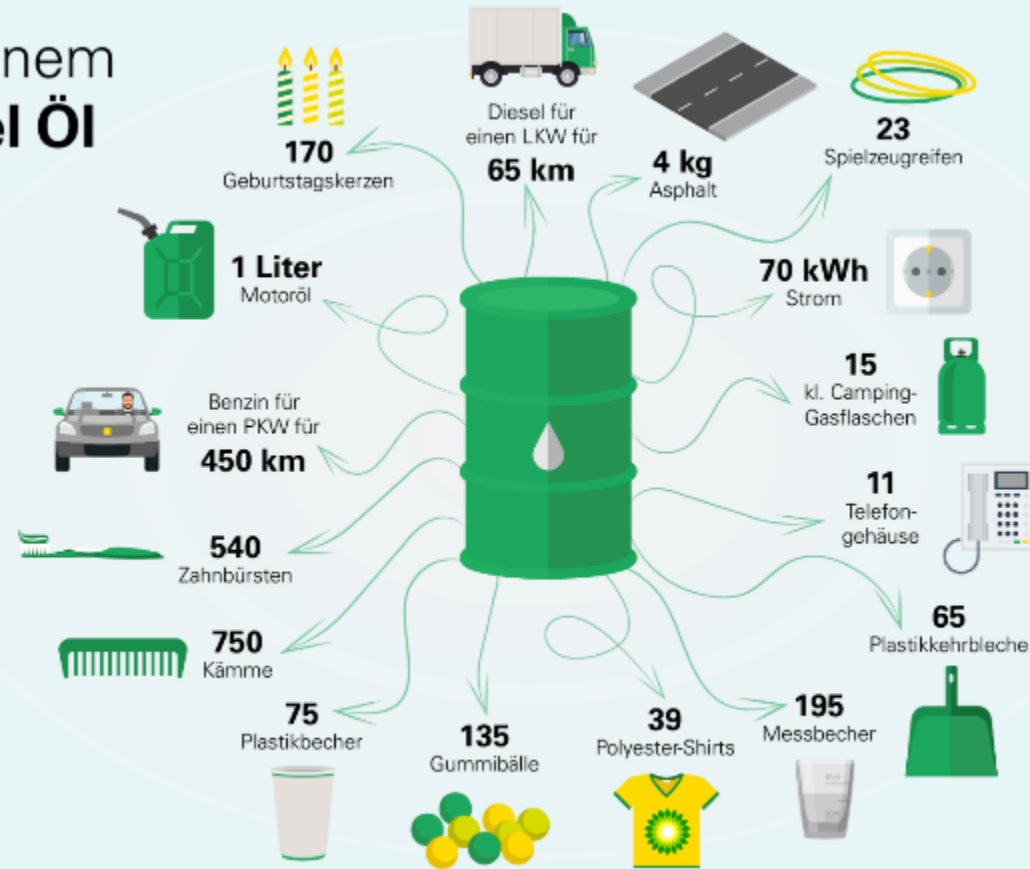
- Crude oil pipeline
- Brent crude blend oil field
- North Sea oil & gas fields



Sources: U.S. Energy Information Administration, IHS Edin
Representation of international boundaries and names
not authoritative



Aus einem Barrel Öl

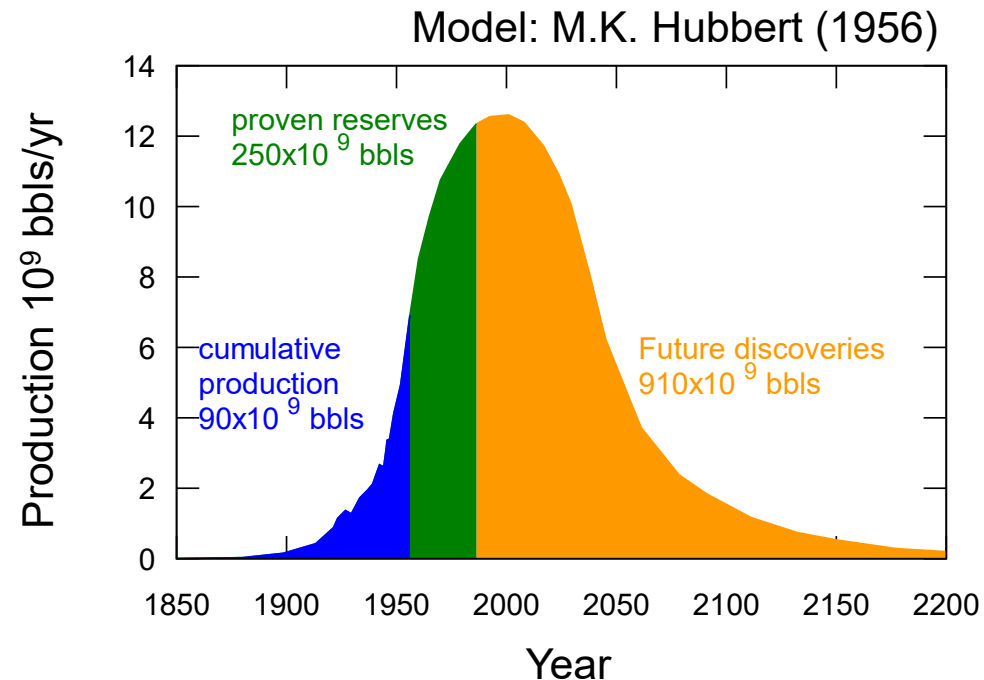


Quelle: Exxon Mobil

Peak oil

Christoph Rühl, chief economist BP

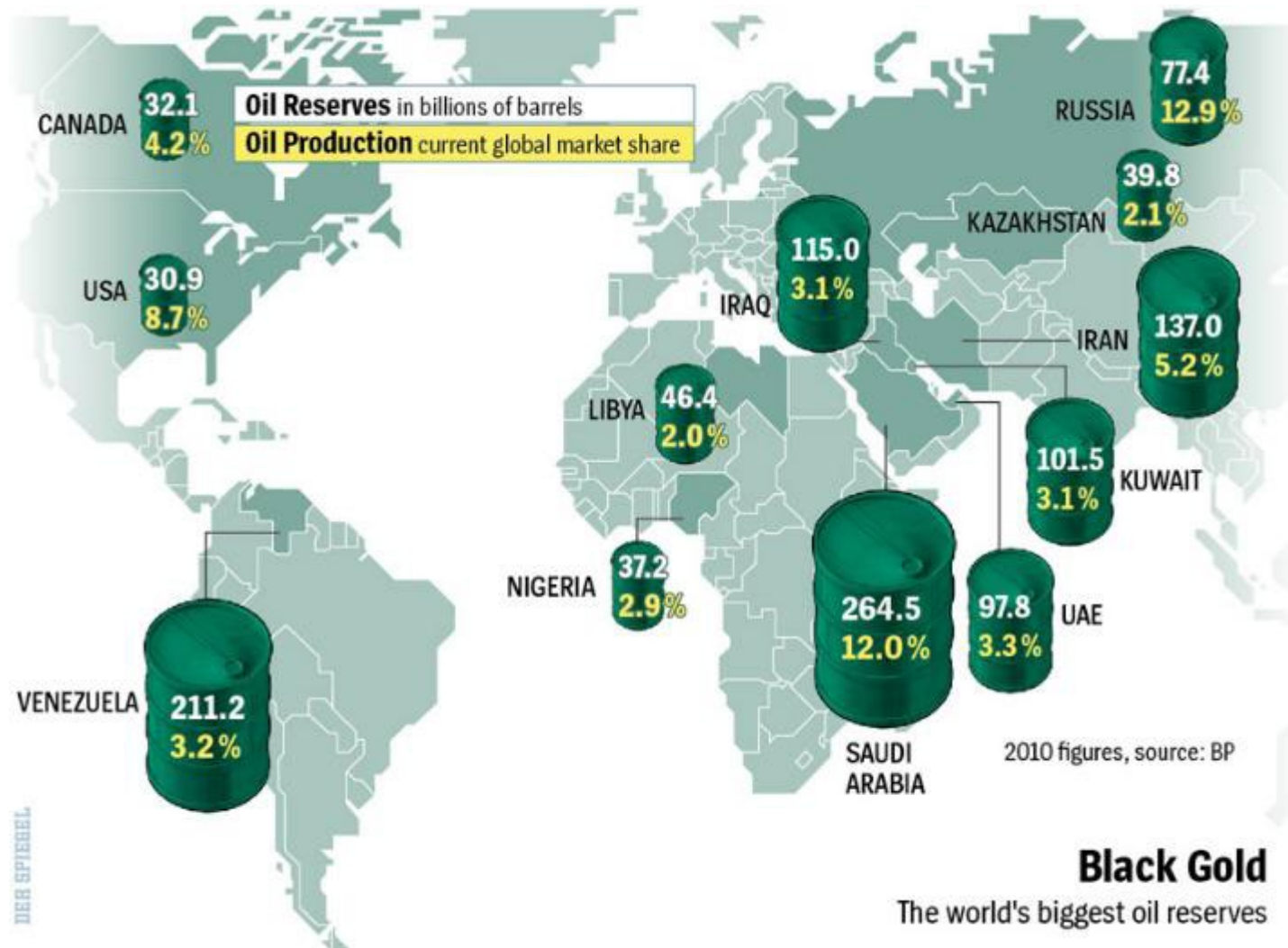
*„Therefore **there will never be a moment when the world runs out of oil** because there will always be a price at which the last drop of oil can clear the market. And **you can turn anything into oil** if you are willing to pay the financial and environmental price.“*



Oil reserves

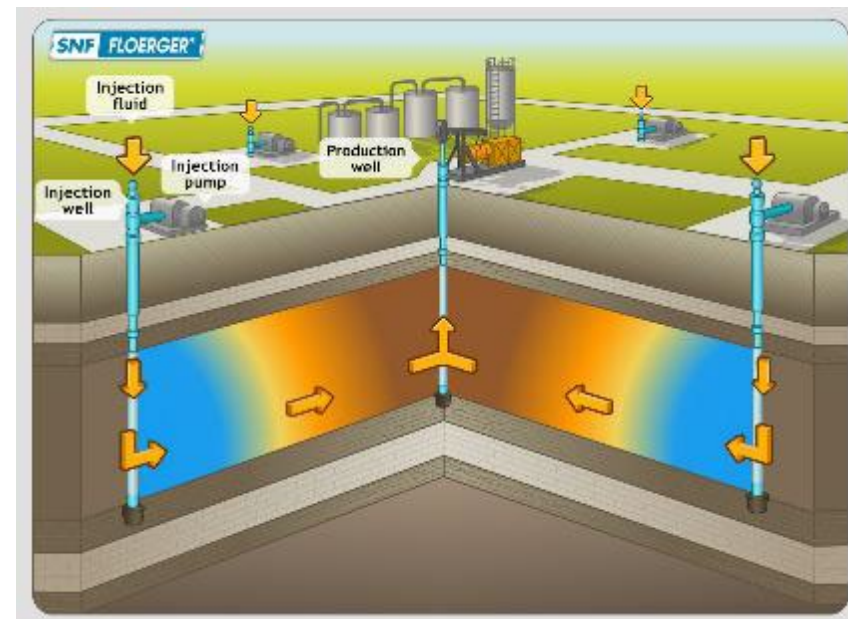
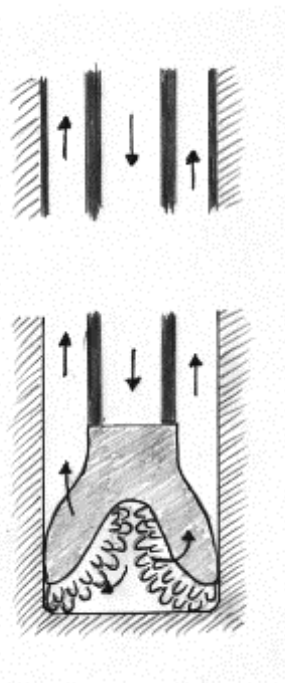
- Oil Reserves (barrels): 1.650.585.140.000
- Oil Consumption:
barrels per year 35.442.913.090
barrels per day 97.103.871
- Reserves / Consumption 47 years left

Where are the proven reserves?



Oil drilling

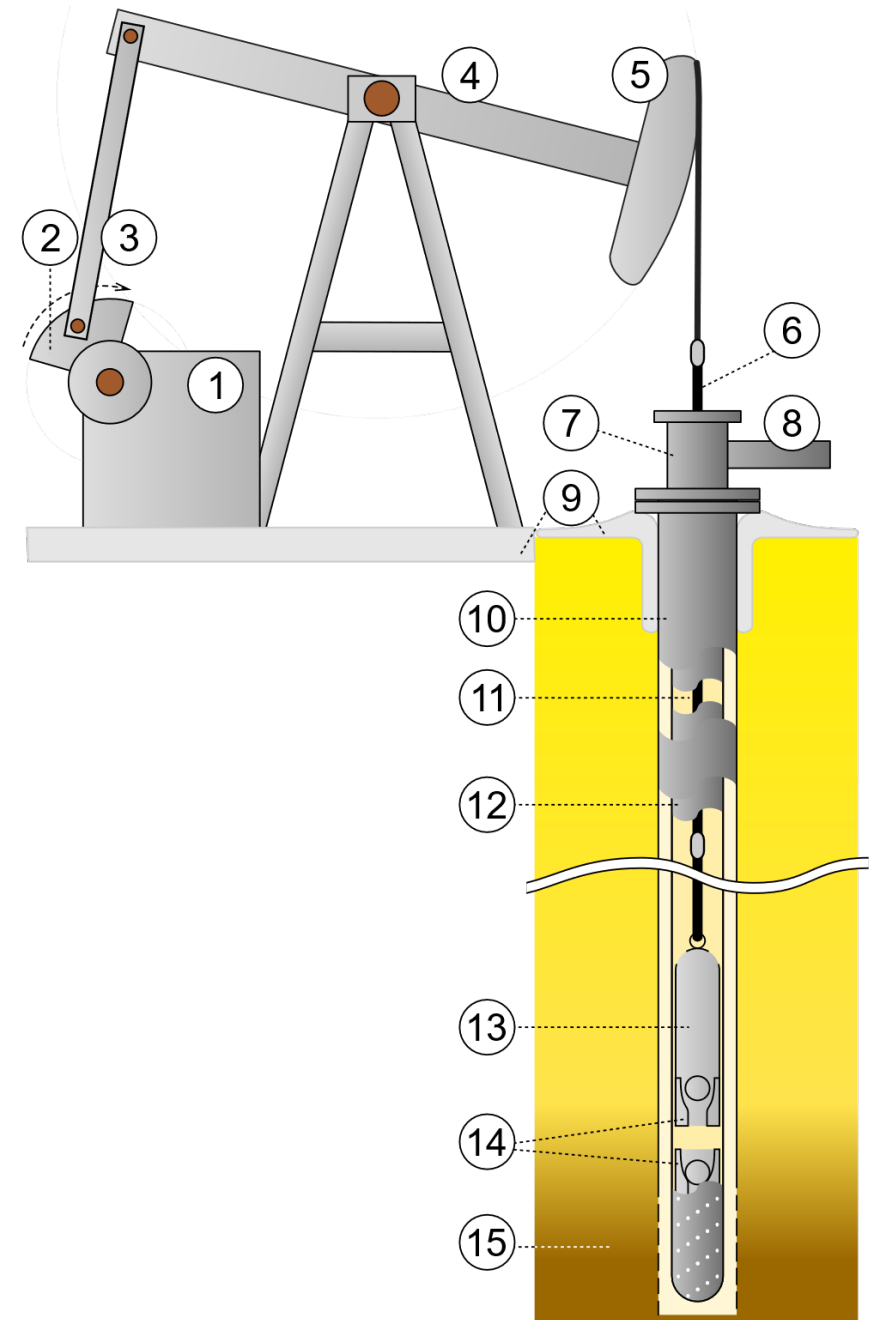
Phase	Enhanced oil recovery	Exploitation [%]
Primary	Inherent pressure	10-20
Secondary	Water/gas	>30
Tertiary	Polymers, surfactants	>50



Oil drilling

Prinzip einer Gestängetiefpumpe zur Erdölförderung:

1. Antriebsmotor
2. Gegengewicht
3. Pleuelstange
4. Balancier
5. Pferdekopf
6. Kolbenstange
7. Dichtung als Bestandteil des Eruptionskreuzes
8. Rohrleitung
9. Betonhinterfüllung des Bohrloches
10. äußere Bohrlochverrohrung
11. Pumpengestänge
12. innere Bohrlochverrohrung
13. Tiefpumpe
14. Pumpenventile
15. ölführende Gesteinsschicht



Alternatives

- Kerogen (oil sands)
- Gas / shale gas
- Coal
- Renewables

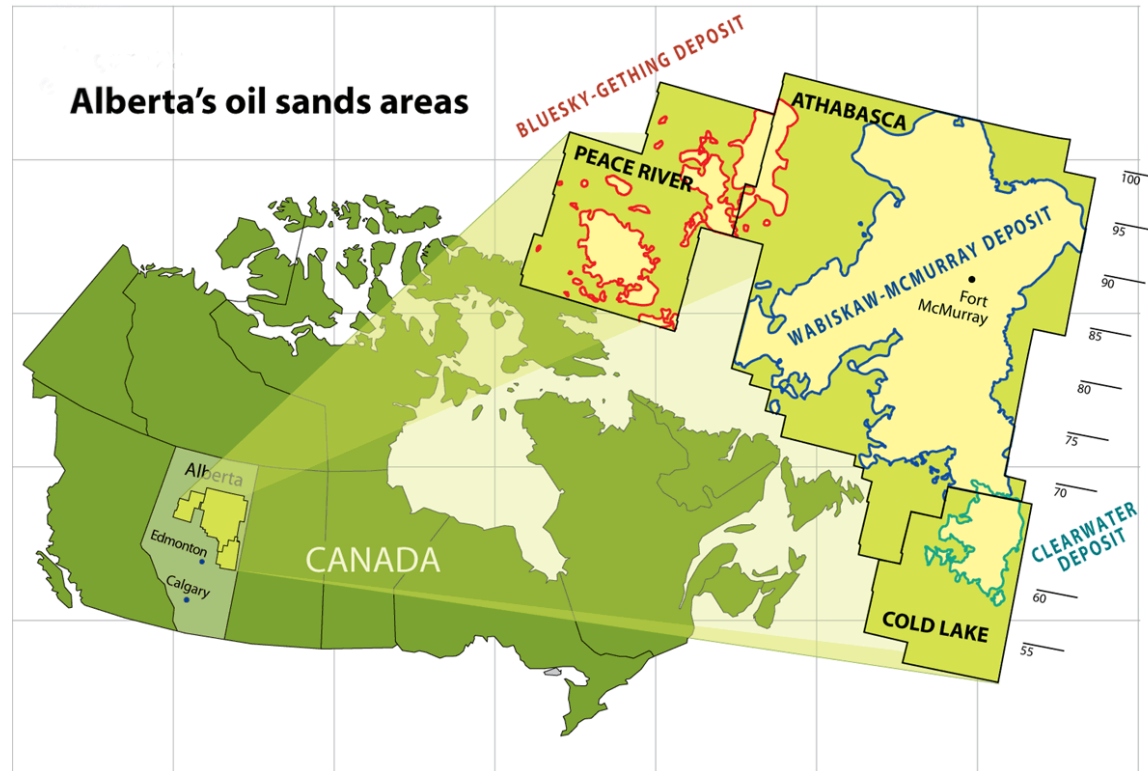
Oil sands

- Polymeric wax in minerals
- Transformation into synthetic oil
 - Thermal $>500^{\circ}\text{C}$
 - Hydration



The largest of these deposits is in Canada

- The Athabasca Oil Sands (representing about 80% of Canada's bitumen reserves)
- 164 billion barrels reserves
- 3,2 million barrels per day
- Shallow enough for surface mining (primary method)
- Oil sands have been surface mined from this location since 1967



Types of Kerogen

Type I: Sapropelic

- Provide the highest yield of oil and are the most promising sources
- $H/C > 1,25$
- $O/C < 0,15$
- It derives principally from algae, has few cyclic or aromatic structures, formed mainly from proteins and lipids

Type II: Planktonic

- Based on marine organic materials
- $H/C < 1,25$
- O/C 0,03 – 0,18

Types of Kerogen

Type III: Humic

- Land plants (coastal)
- $H/C < 1$
- O/C 0,03 – 0,3
- Material is thick, resembling wood or coal, it forms from cellulose

Type IV: Residue

- $H/C < 0,5$
- Mostly decomposed organic matter in the form of polycyclic aromatic hydrocarbons - no potential to produce hydrocarbons

Natural gas

Gas reserves somewhat bigger than oil reserves

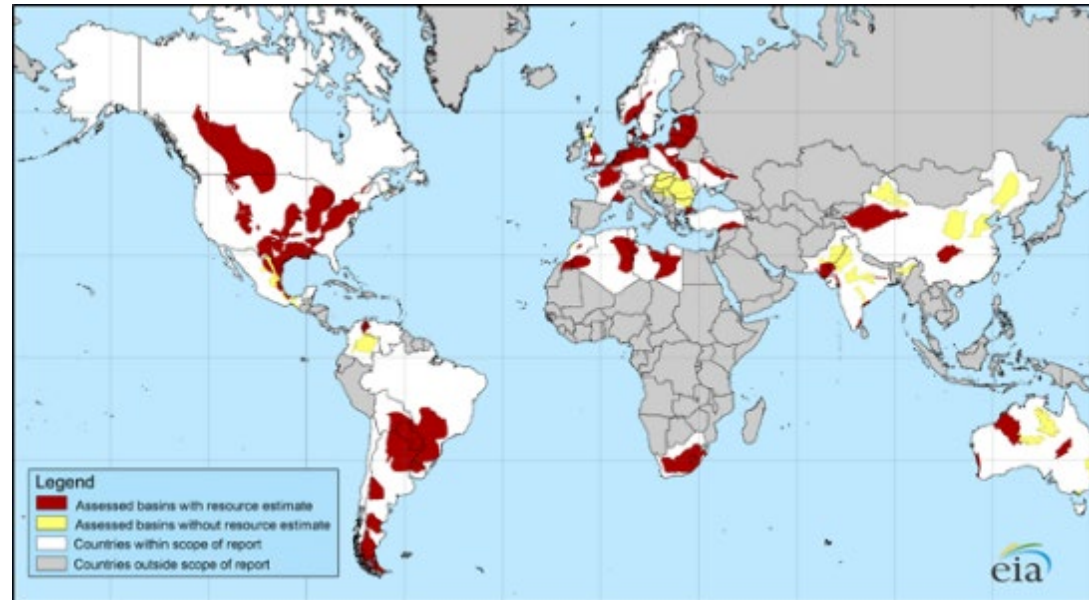
	Million cubic Ft	Million BOE
Gas reserves	6,922,922,000	1,153,820
Gas consumption	132,290,211	22,048

BOE: barrels of oil equivalent

- There are 6,923 trillion cubic feet of proven gas reserves in the world as of 2017
- The world has proven reserves equivalent to 52.3 times its annual consumption
- This means it has about 52 years of gas left (at current consumption levels and excluding unproven reserves)
- Russia holds about 24% of the worlds gas reserves, followed by Iran (17%)

Shale gas

- Shale gas is natural gas formed from being trapped within shale formations
- Shale gas has become an increasingly important in the US
- In 2023, about 77% (29.35 trillion cubic feet) of total U.S. dry natural gas production was from shale formations
- Environmental issues



Fracking fluids

- Acids: HCl, HAc (5 - 28%)
- NaCl
- Polyacrylamide (friction reducers)
- Ethylene glycol (scale prevention)
- Borate salts (fluid viscosity)
- Na- and K carbonates (maintaining effectiveness of crosslinkers)
- Glutaraldehyde (disinfectant of the water)
- Guar gum and others (increases viscosity of the fracturing fluid)
- Citric acid (corrosion prevention)
- Isopropanol (increases the viscosity of the fracture fluid)

The most common chemical used for hydraulic fracturing in the United States in 2005–2009 was: Methanol, followed by isopropyl alcohol, 2-butoxyethanol, and ethylene glycol

Coal is the most fossil source for energy

- Numbers for reserves do not consider the geological accessibility
- Energy demand of the world could be covered for some more thousand years with coal



Carnot heat engine

The maximum extractable work from the ideal engine is given by:

$$\frac{W \text{ (the most that can be produced by the heat engine)}}{Q_A \text{ (heat leaving A and going to the engine)}} = \frac{W_{\text{max, produced}}}{Q_{\text{from hot source}}} = \frac{T_A - T_B}{T_A}$$

positive

positive

where the temperature T is measured in absolute or Kelvin (K) units

T_A high temperature

T_B low temperature

In getting to this equation Carnot invented the concept of entropy, led the way to the Kelvin temperature scale, and came up with what we call the 2nd law of thermodynamics.

When run backwards the engine is called **Carnot Heat Pump**.

Carnot Heat Engine / Heat Pump

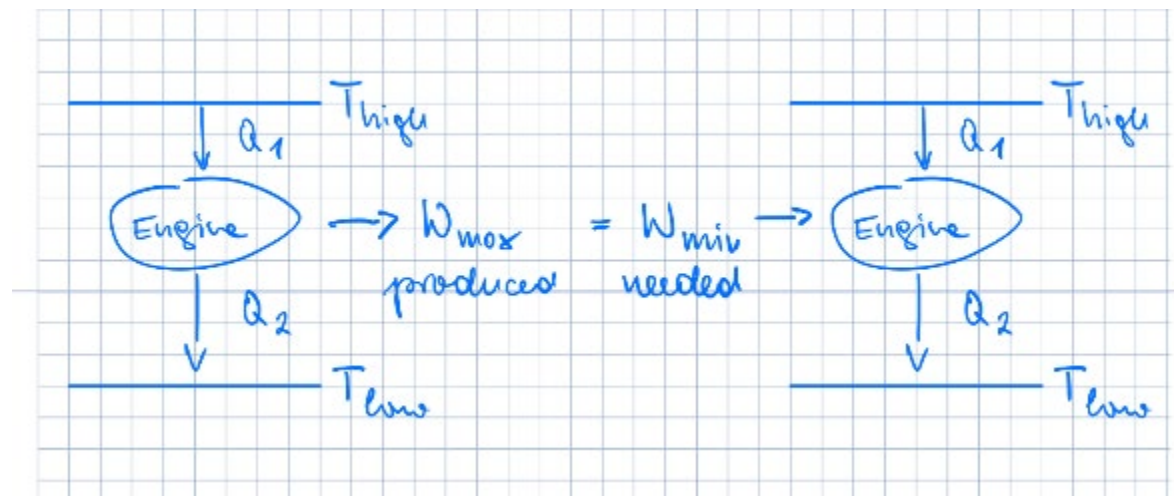
The minimum amount of work needed to remove heat from the cold region is given by:

$$\text{COP} = \frac{Q \text{ (T pumped out from the cold region)}}{W \text{ (minimum needed)}} = \frac{\overset{\text{positive}}{Q_2}}{\underset{\text{positive}}{W_{\min, \text{ideal}}}} = \frac{T_2}{T_1 - T_2}$$

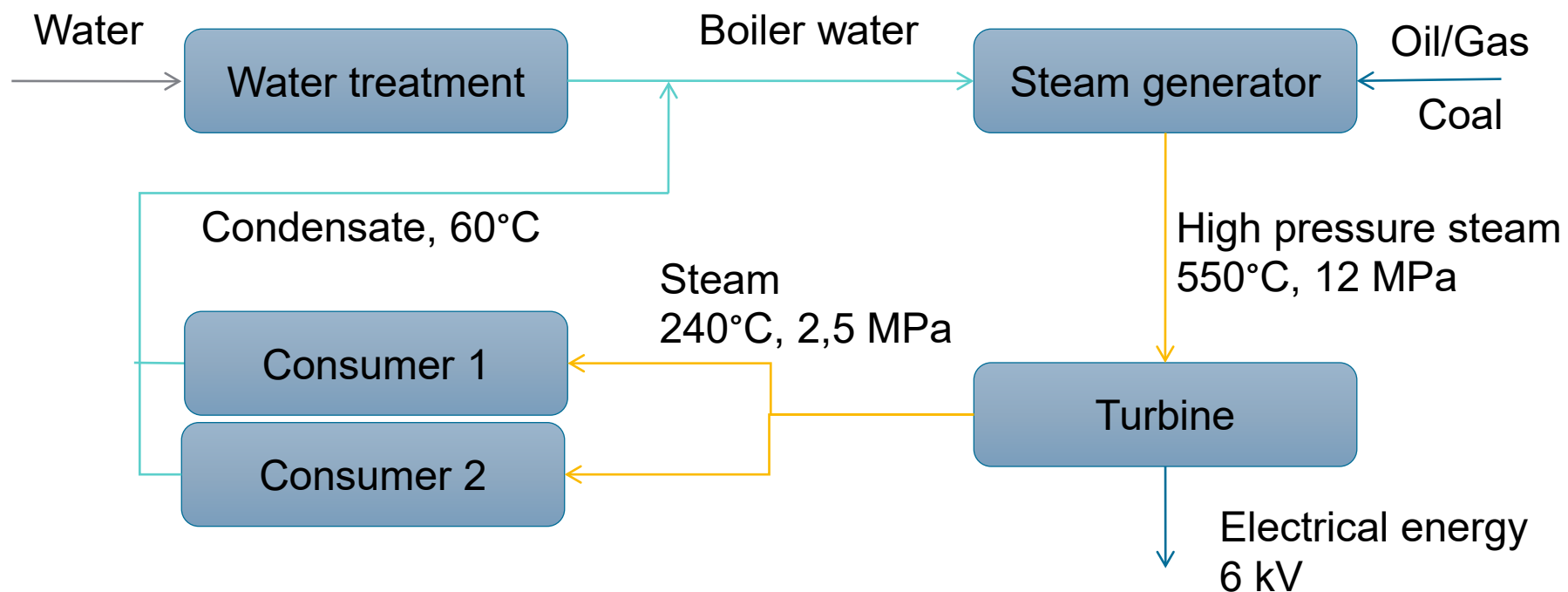
T_1 hot temperature

T_2 cold temperature

We want to maximize COP,
the coefficient of performance



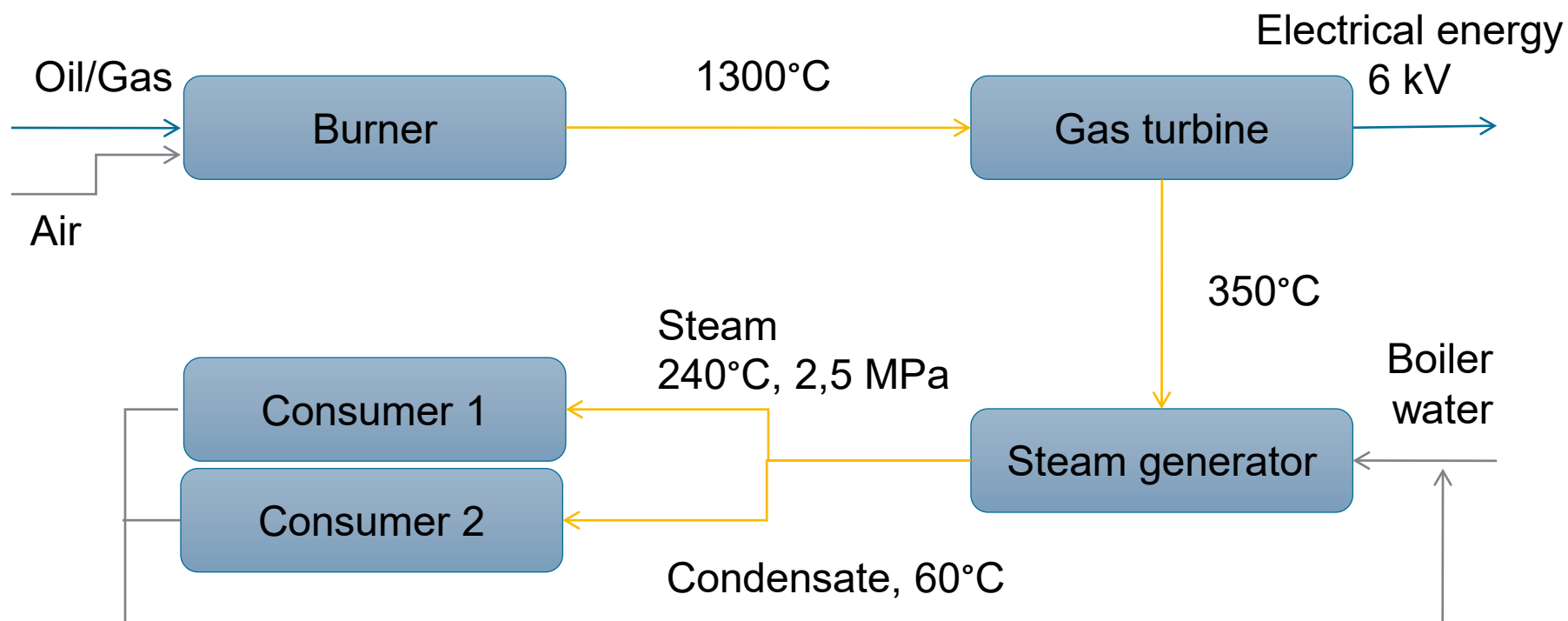
Heat & power generation



CARNOT equation:

$$\text{Efficiency: } \eta = (T_E - T_A)/T_E = (823 - 513)/823 = 0,38$$

Heat & power generation II



CARNOT equation:

$$\text{Efficiency: } \eta = (T_E - T_A)/T_E = (1573 - 623)/1573 = 0,60$$

Coal

- Black coal, anthracite coal, blue coal
- Brown coal
- In 2023, the EU production of hard coal was 50 million tonnes, 82% less than the 277 million tonnes of 1990
- From 2018 to 2023, the EU reduced its consumption of hard coal by 42% and of brown coal by 40%
- In 2022, solar overtook hard coal as a source of electricity for the first time in the EU

Unit: Mt

Highest ▼

China	4,705
India	1,045
Indonesia	752
United States	528
Australia	439
Russia	429
South Africa	222
Kazakhstan	109
Germany	102
Poland	89
Turkiye	66
Colombia	57

Benefits of bio mass

Plus

- Saving green house gases
- Use of side / waste streams from wood and agricultural industry
- Saving fossil resources / less dependant
- Using nature

Minus

- Less energy per unit of mass
- Solids
- High ash content
- Costs for transportation
- Supply security
- Competition to food

Raw materials

- Till the 50ies coal
- Since then dominated by oil
- At the moment about 96% of primary petrochemical products based on oil
- Chemistry depends on oil
- Mainly Naphtha (Rohbenzin)
 - Olefins
 - BTX
- Olefin market size
 - 2024: 265.56 billion USD
 - 2033: 418.01 billion USD
 - CAGR: 5.17 %

Oil is raw material and energy source

- Needs to be decoupled
 - Use or alternative energy sources
 - Better efficiency
- Oil drilling countries moving down the value chain
 - Saudi Arabia
 - Emirates
 - C_2 , C_2H_5OH , Ethylene glycol, Dichloromethane, Vinyl chloride, Styrene
 - Plastics

Coal

- At the moment used to produce Benzene, Naphthalene (and bigger aromatics)
- Basically oil could be replaced by coal - question of price
- Existing processes:
 - Gasification of coal to Syngas
 - Hydration or hydrating extraction of black coal
 - Low temp carbonization (coking)
 - Acetylene via Ca-carbide from coal and chalk

Basic products of industrial organic synthesis

- Syngas

C1 chemistry:

- Methanol
- Formaldehyde
- Formic acid
- Hydrocyanic acid
- Methane
- Halogen derivatives of methane
- Reactions with CO and CO₂

Synthesis gas - Syngas

CO/H₂ mixture with different compositions

Different names

1. Origin:

- **Water gas** (CO + H₂): water vapor and coal (Wassergas)
- **Reformed gas** (CO + 2H₂): CH₄ scission with H₂O (Spaltgas)

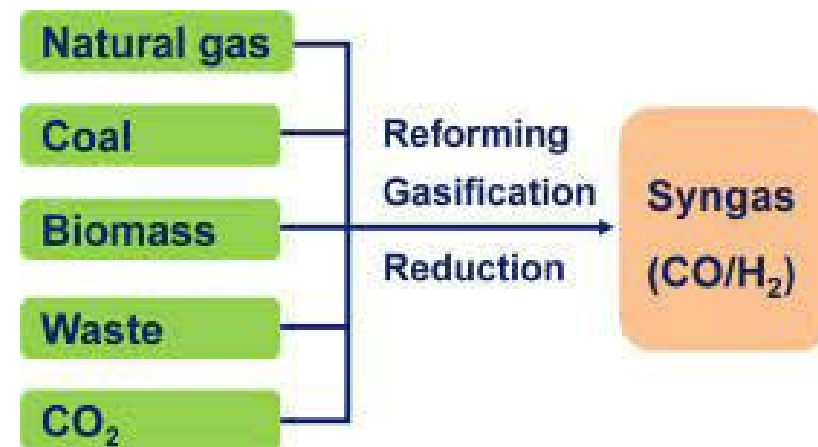
2. Use:

- **Methanol synthesis gas** (CO + 2H₂): for CH₃OH
- **Oxo gas** (CO + H₂): for hydro formulation

Synthesis gas - Syngas

Raw materials for syngas:

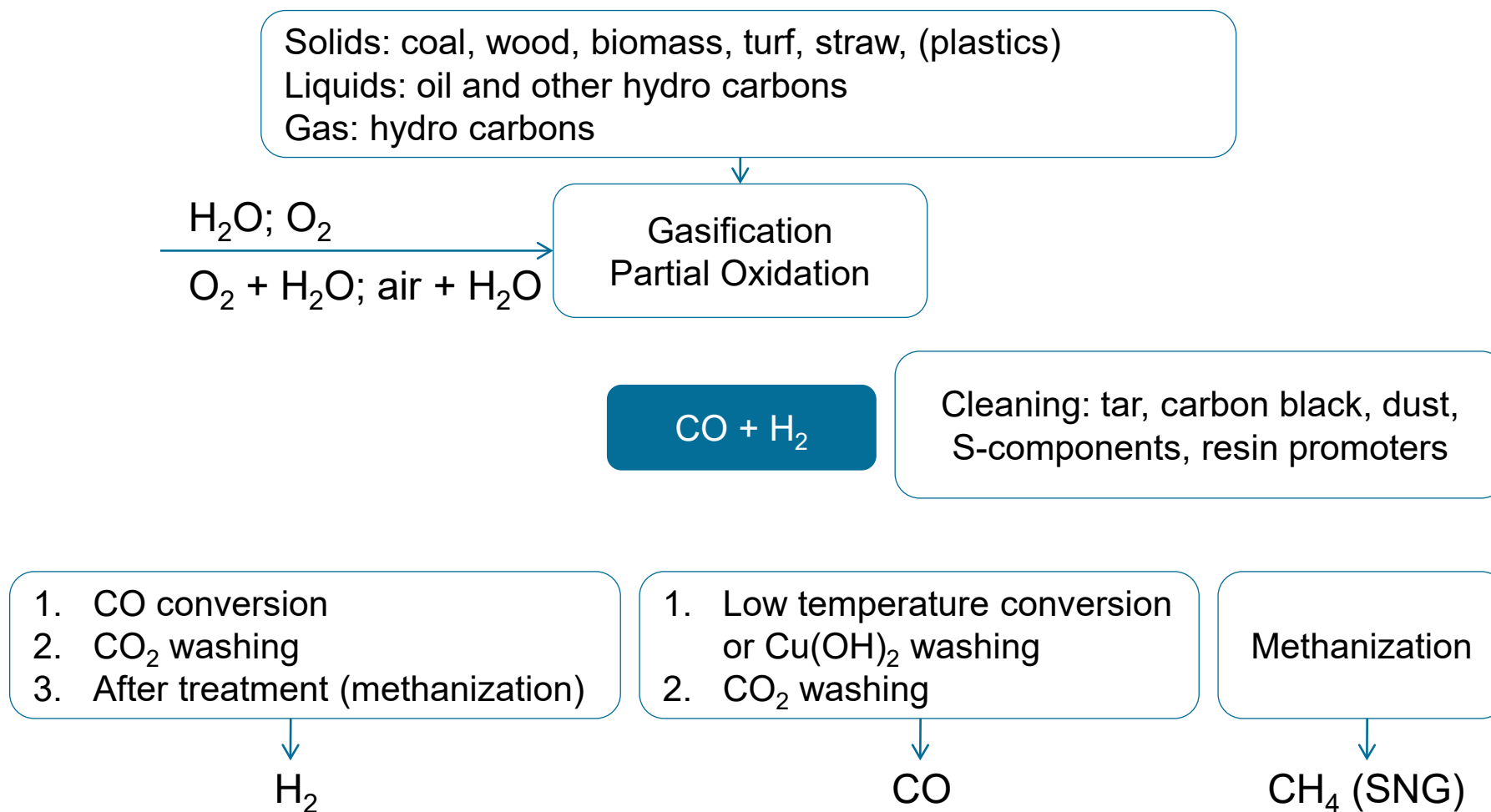
- Coal (black & brown)
- Gas, oil, oil fractions
- Bio sources (e.g. Güssing)



Gas and light oil fractions best → higher H₂ content [H/C]

Coal 1:1 / Oil 2:1 / Gasoline 2,4:1 / Gas 4:1

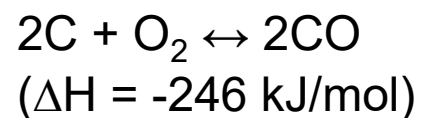
General flow diagram



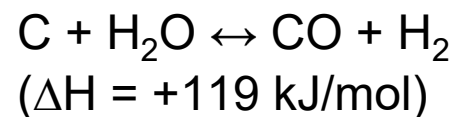
Coal gasification: gas/solid reaction

- Physics: gas/solid reaction
- Chemistry: partial oxidation of C or a reduction of H₂O with C
- Complex process with many parallel and dependent reactions

- **Partial combustion:**

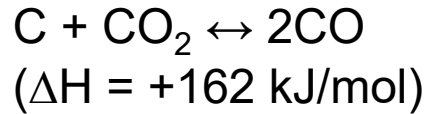


- **Heterogeneous water-gas reaction:**

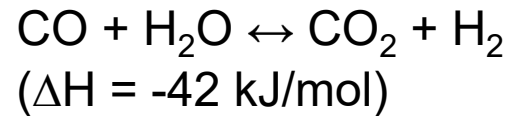


important equilibria

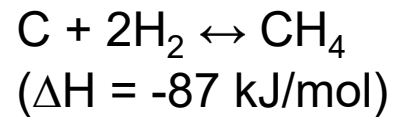
- **Boudouard reaction:**



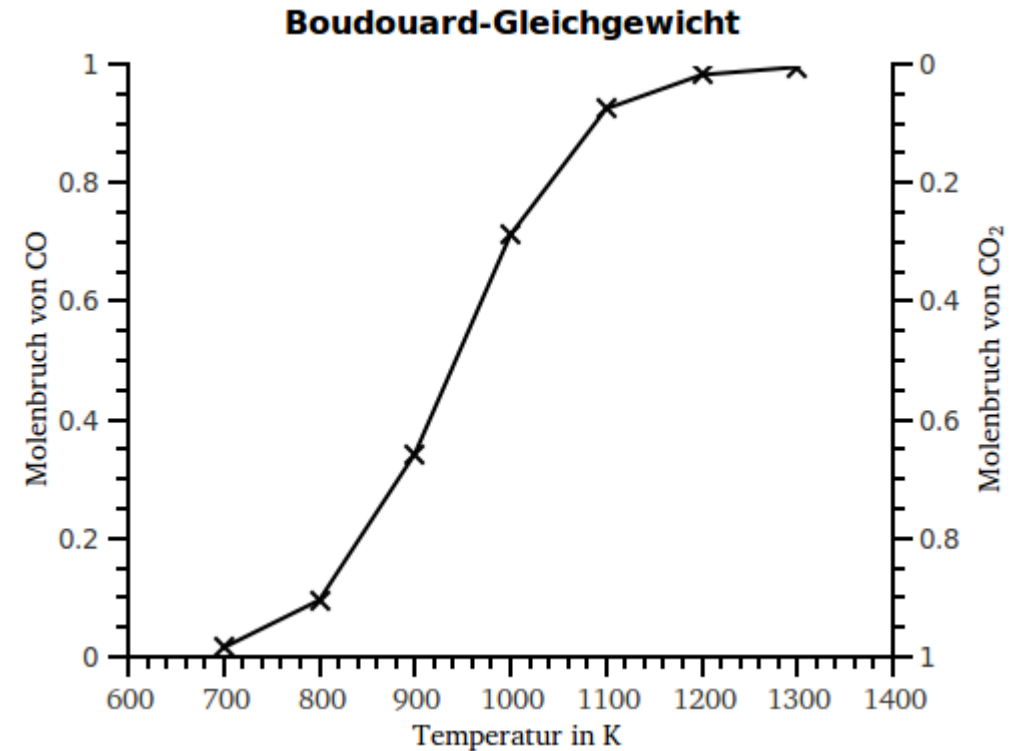
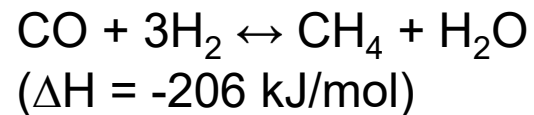
- **Homogenous water-gas reaction (conversion):**



- **Hydrating gasification:**



- **Methanization:**



High temperatures needed

- For C gasification high energy (heat) input at high temperatures (900 – 1000 °C) needed because:
 - Heterogeneous water-gas reaction is highly endothermic with high activation energy
$$\text{C} + \text{H}_2\text{O} \leftrightarrow \text{CO} + \text{H}_2$$

($\Delta H = +119 \text{ kJ/mol}$)
 - Reaction speed in commercial processes needs to be high enough

Coal gasification depends on

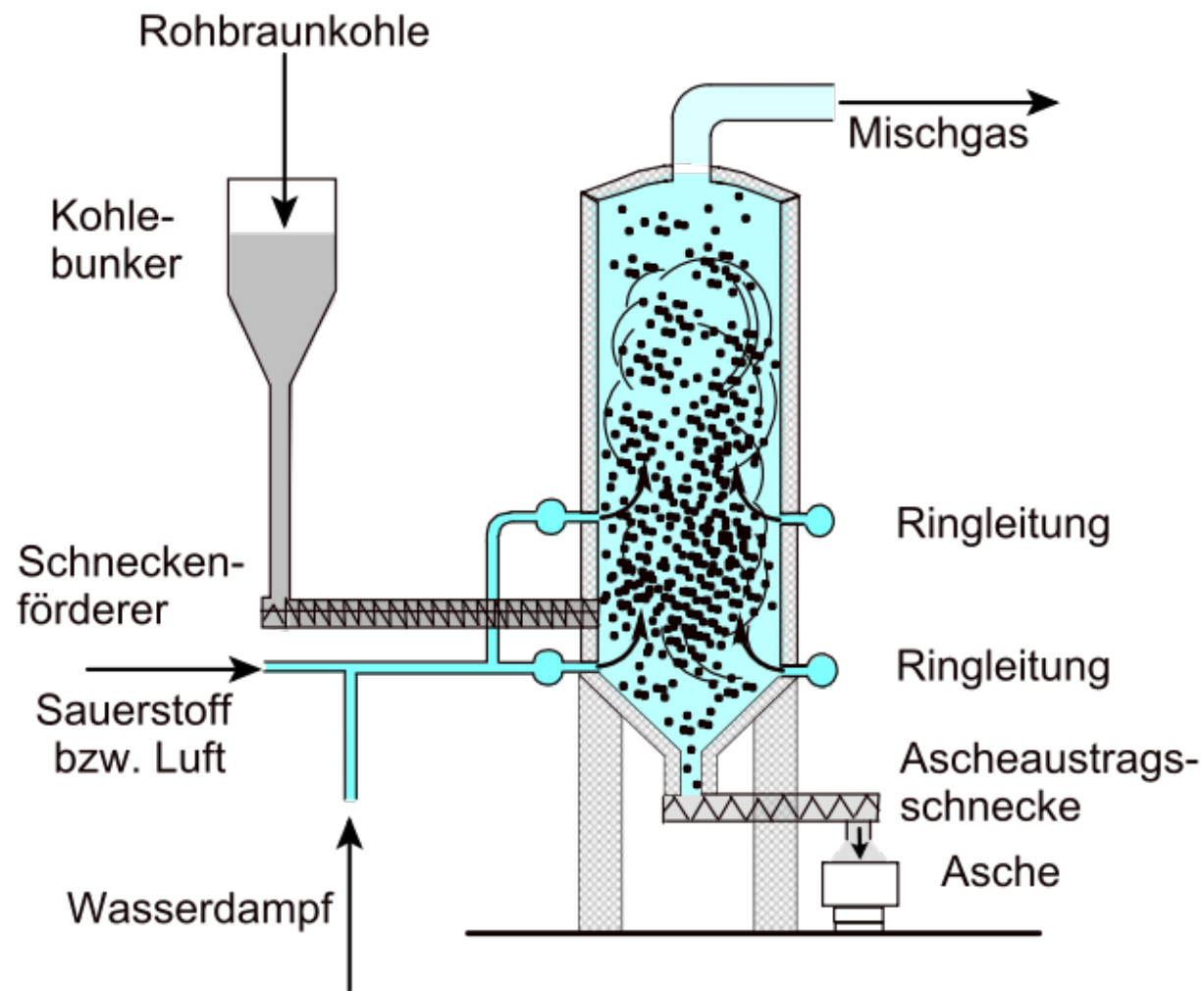
- Physical and chemical properties of the used coal
- Heat input (allothermal $\neq$ autothermal)
- Type of reactor
- Gasification agent
- Process parameters

Processes for coal gasification

- Industrial processes:
 - Winkler, Koppers-Totzek, Lurgi Druckvergasung
 - Hydrating gasification (Rheinbraun)
 - coal gasification in a Na_2CO_3 melt (Kellog)
 - coal gasification in an iron melt (Sumitomo)
- Multi stage processes for the production of SNG (substitute natural gas)
 - Synthane, Bi-Gas, Hy-Gas, U-Gas
- For all processes: part of the coal (30-40%) is used to generate the needed heat

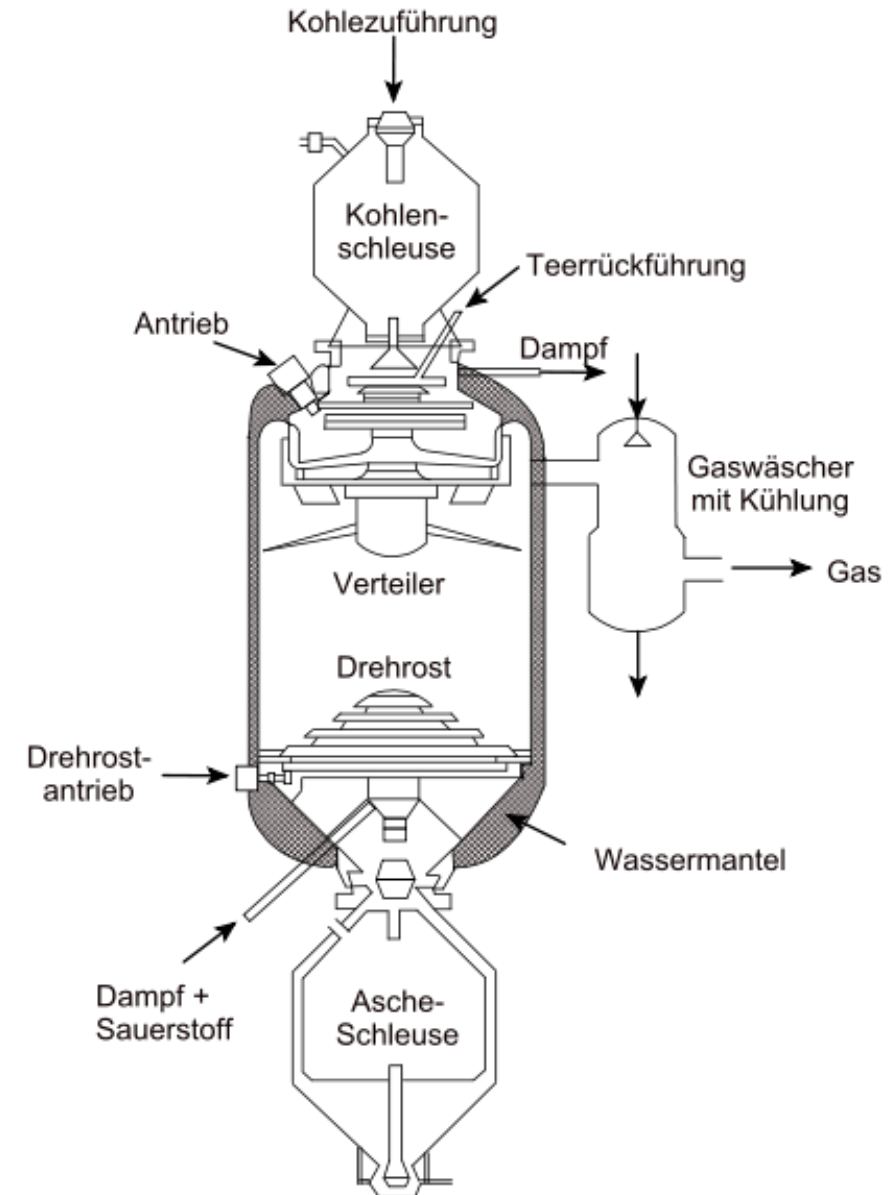
Winkler process

- Fluidized bed
- $O_2 + H_2O$
- 800 - 1000°C
atmospheric pressure
- Fine brown coal
- H_2/CO in product: 1,4:1
- Developed in D in the 30ies
- New developments:
higher p (25bar)
950°C → better economics
- Ca. 40 t/h coal → 60.000 m³ gas



Lurgi Druckvergasung

- Moved solid bed reactor
- 2 characteristic Temp:
 - 600 – 750 °C pre degassing
 - 1000 – 1200 °C gasification (depends on O_2/H_2O)
- Advantage:
 - Gas is pressurized
→ further use (SNG, etc.)
 - Already high CH_4 content

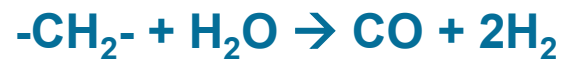


Syngas from oil or gas cleavage

- Similar to coal gasification



$$\Delta H = -92 \text{ kJ/mol}$$



$$\Delta H = +151 \text{ kJ/mol}$$

- Coupled exo- and endothermic gasification reaction

Allotherm or autotherm

Basically 2 methods:

1. **Allothermal** steam reforming:
catalytic cleavage in the presence of H_2O
(heat demand covered by external sources)
2. **Autothermal** cleavage:
heat generated by partial burning of educts used for cleavage
(in the presence of H_2O or recycled CO_2 to adjust CO/H_2)

Principle 1: steam reforming

- Allothermal process
- Most known process developed by ICI
- Based on Naphtha
- 3 steps
 1. Hydrating desulphurization
 H_2 , 350-450°C, $\text{CoO-MoO}_3/\text{Al}_2\text{O}_3$
 H_2S adsorbed on ZnO , parallel hydration of olefins
 2. Catalytic cleavage on a primary reformer
700-830°C, 15-40 bar, $\text{Ni-K}_2\text{O}/\text{Al}_2\text{O}_3$
 3. Post-cleavage of CH_4 in a second reformer

Principle 2: autothermal cleavage

- Partial oxidation of oil fractions
- BASF/Lurgi, Texaco, Shell
- Raw materials: all types of hydrocarbons (methane to heavy oil)
- Shell:
 - Pre-heated educts react with stoichiometric amounts of O_2 and H_2O
 - 30-80 bar, 1200 - 1500 °C, no catalyst
 - Small amount of carbon black → sold as fuel oil pellets

Work-up of syngas is necessary

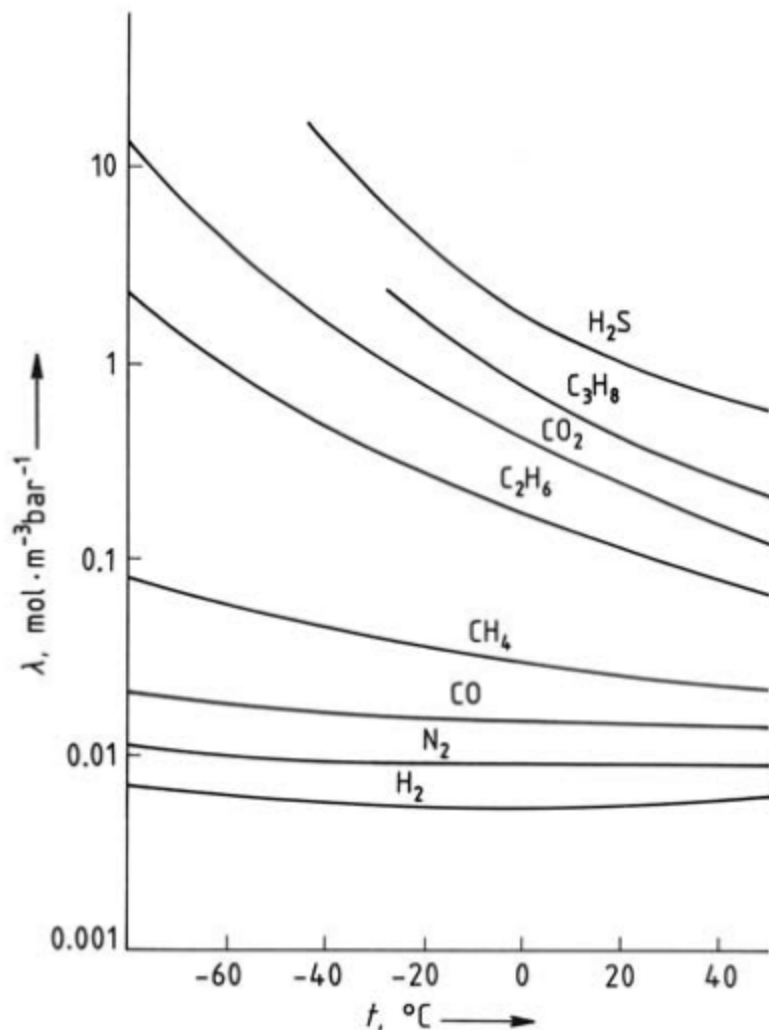
- Removal of impurities
 - H₂S, COS: can poison catalysts
 - CO₂: can influence reactions or build an inert

Raw material:	Coal		Vacuum residue		Natural gas	
Example: Production process:	1 Koppers – Totzek ex gasifier	2 Lurgi Pressure Gasification	3 Shell Gasification Process downstream of scrubber	4 Texaco Gasification Process downstream of CO shift	5 upstream of CO shift	6 Tubular reformer downstream of CO shift
Pressure, MPa	0.105	2.9	5.4	8.5	2.2	2.1
Temperature, °C	45	45	35	35	350	230
Water dew point, °C	45	45	35	35	170	162
Analysis, vol %						
H ₂	28.7	43.2	45.9	63.0	76.1	78.3
CO	57.0	11.6	48.6	1.8	10.6	0.4
CH ₄	0.1	10.7	0.5	0.3	3.3	3.0
C ₂₊		1.0				
N ₂ + Ar	1.4	0.3	0.2	0.1		
CO ₂	12.6	32.9	4.0	34.2	10.0	18.3
H ₂ S	0.2	0.3	0.8	0.6		
COS, ppm	300	150	500	10		
Other components	O ₂ , SO ₂ , HCN, NH ₃	O ₂ , C _m H _n , HCN, NH ₃	O ₂ , HCN, NH ₃	NH ₃		

Processes for syngas work-up

- Rectisol (Lurgi, Linde): washing with MeOH
- Selexol (UCC): p dependant solubility of acid gases in polyglycols/dimethylethers
- Sulfinol (Shell): sulfolane/diisopropanolamine/water mixtures
- Washing with K_2CO_3 (Benfield, Catacarb)
- Adsorption on mol sieves (UCC)
- Claus (OxyClaus, SuperClaus): $H_2S \rightarrow S$

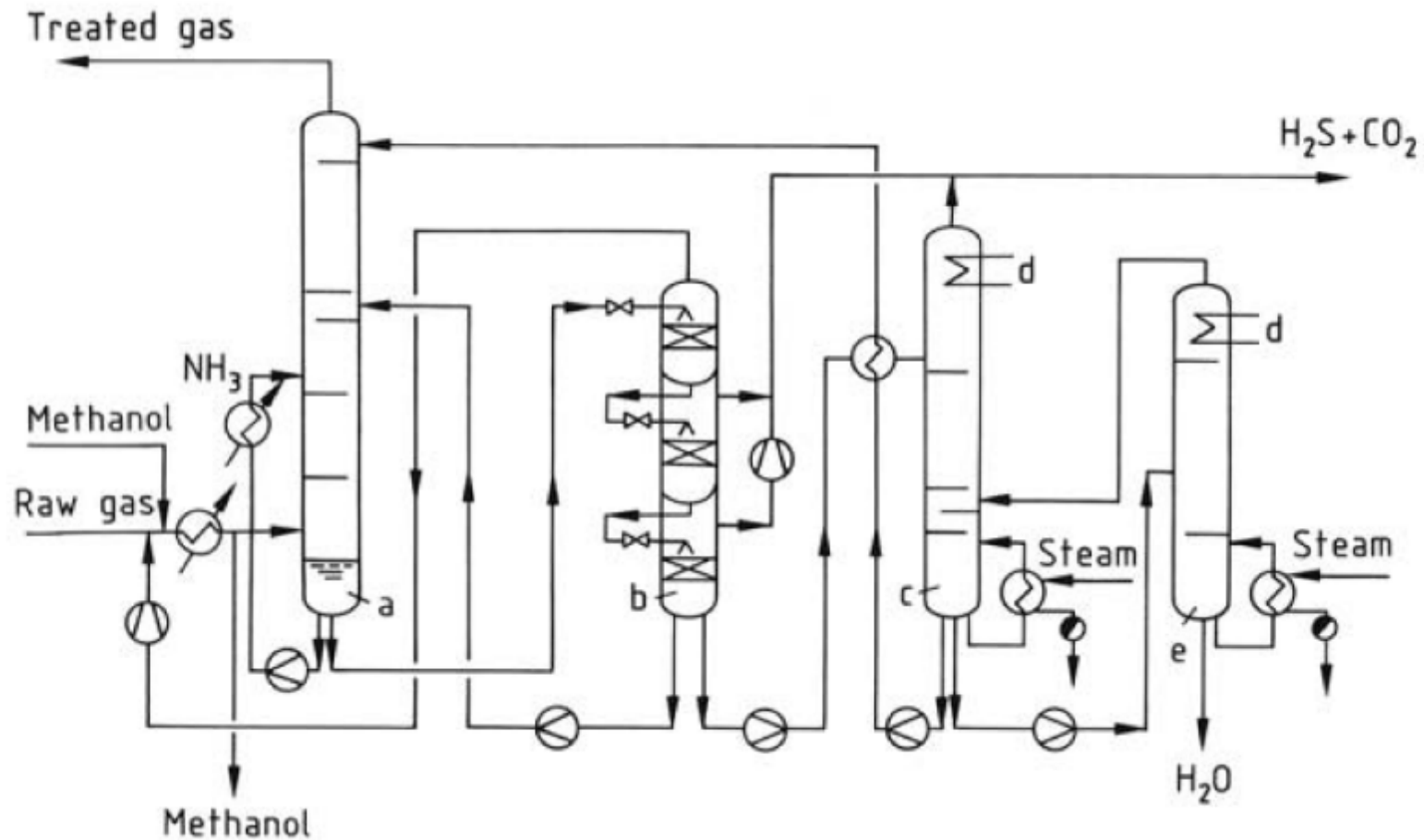
Rectisol - basic principle



- The solubility of acid gas in methanol increases strongly with decreasing temperature
- The solubility of the valuable light components is almost temperature independent
- Temperatures as low as -75°C are applied in order to minimize the required solvent circulation rate and the loss of valuable gas components

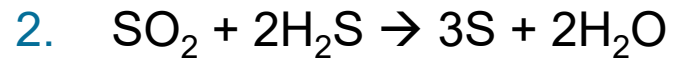
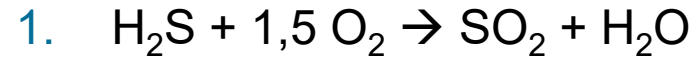
Absorption coefficient λ of various gases in methanol at low pressure (1 bar)

Rectisol process

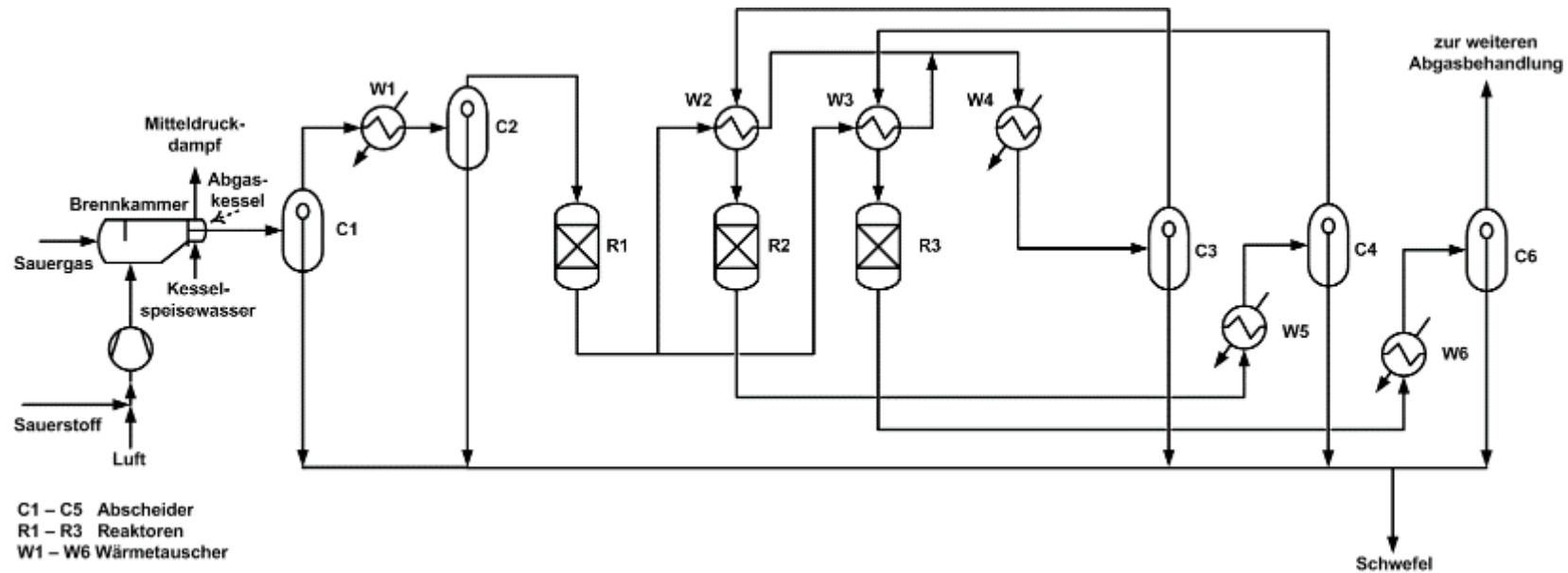


Standard Rectisol process for simultaneous removal of H_2S and CO_2
a) Absorber; b) Flash tower; c) Hot regeneration; d) Condenser; e) Methanol – water column

Claus process



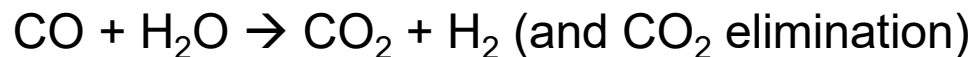
(with the use of a Al_2O_3 based catalyst)



Adjustment of CO/H₂

- The pure syngas needs to be adjusted with respect to the use
- Can be done during the gasification (H₂O and O₂)

- CO conversion



- Conversion catalysts:

- High temperature conversion:

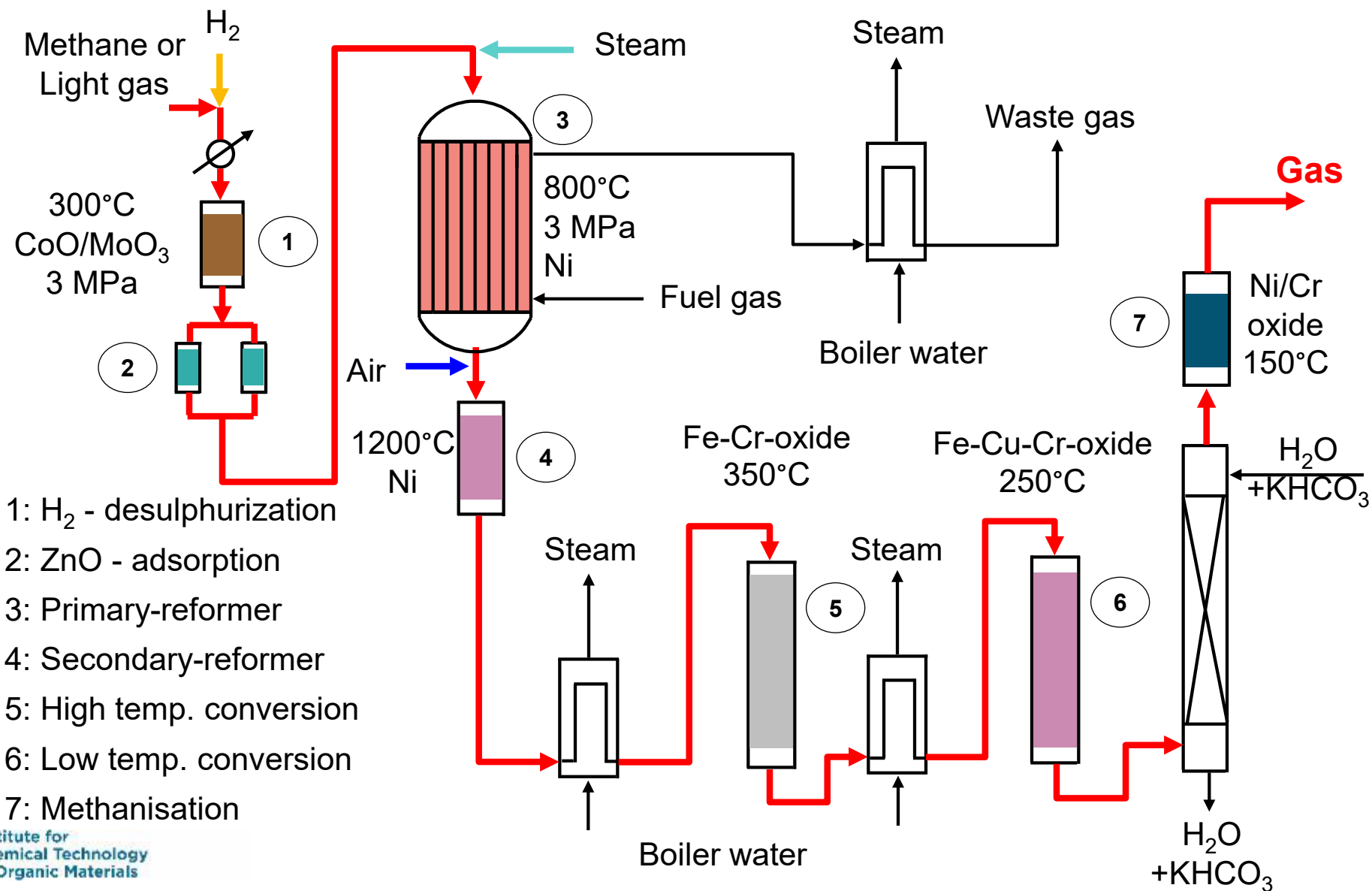
Fe-Cr oxides: in several steps at 350 – 400 °C (CO up to 3%vol)

- Low temperature conversion:

CuO-ZnO: single step at 190 – 260 °C (CO up to 0,1%vol)

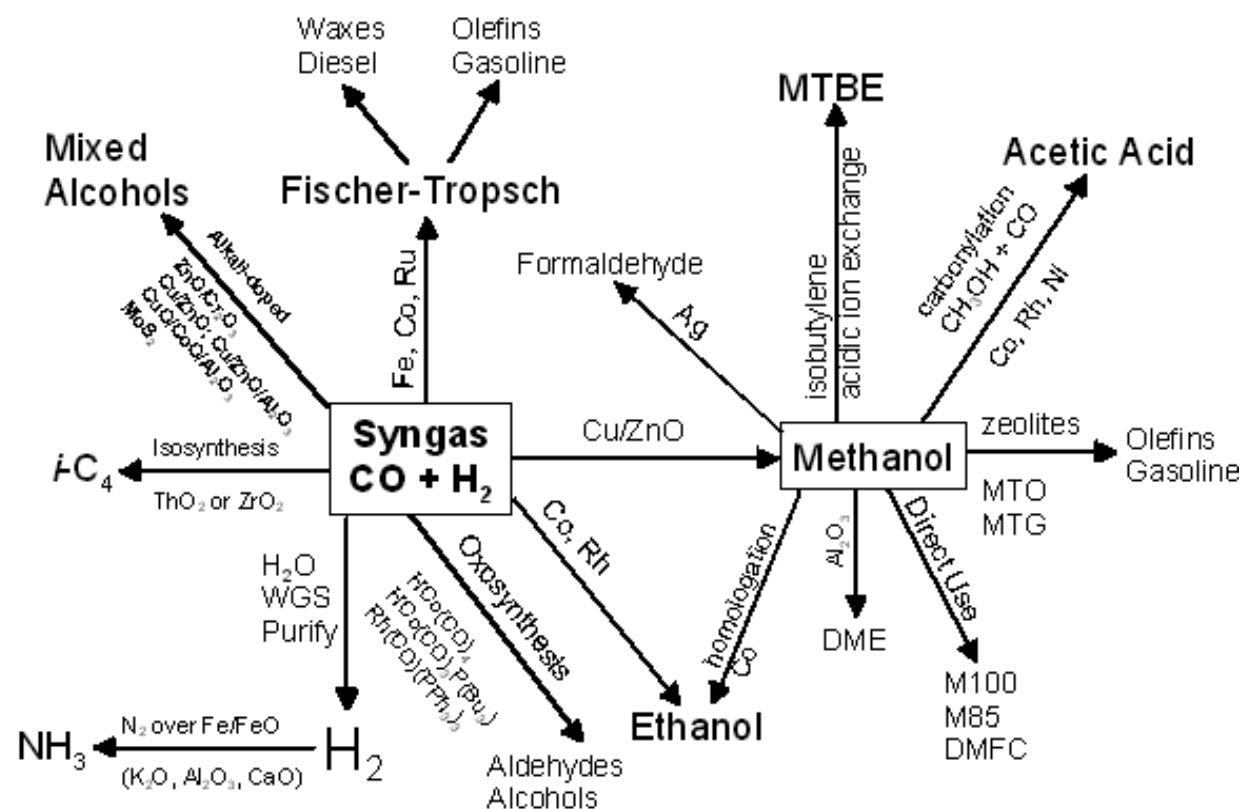
- Latter used for the production of H₂ (only NH₃ production)

Syngas from CH₄ or light gas



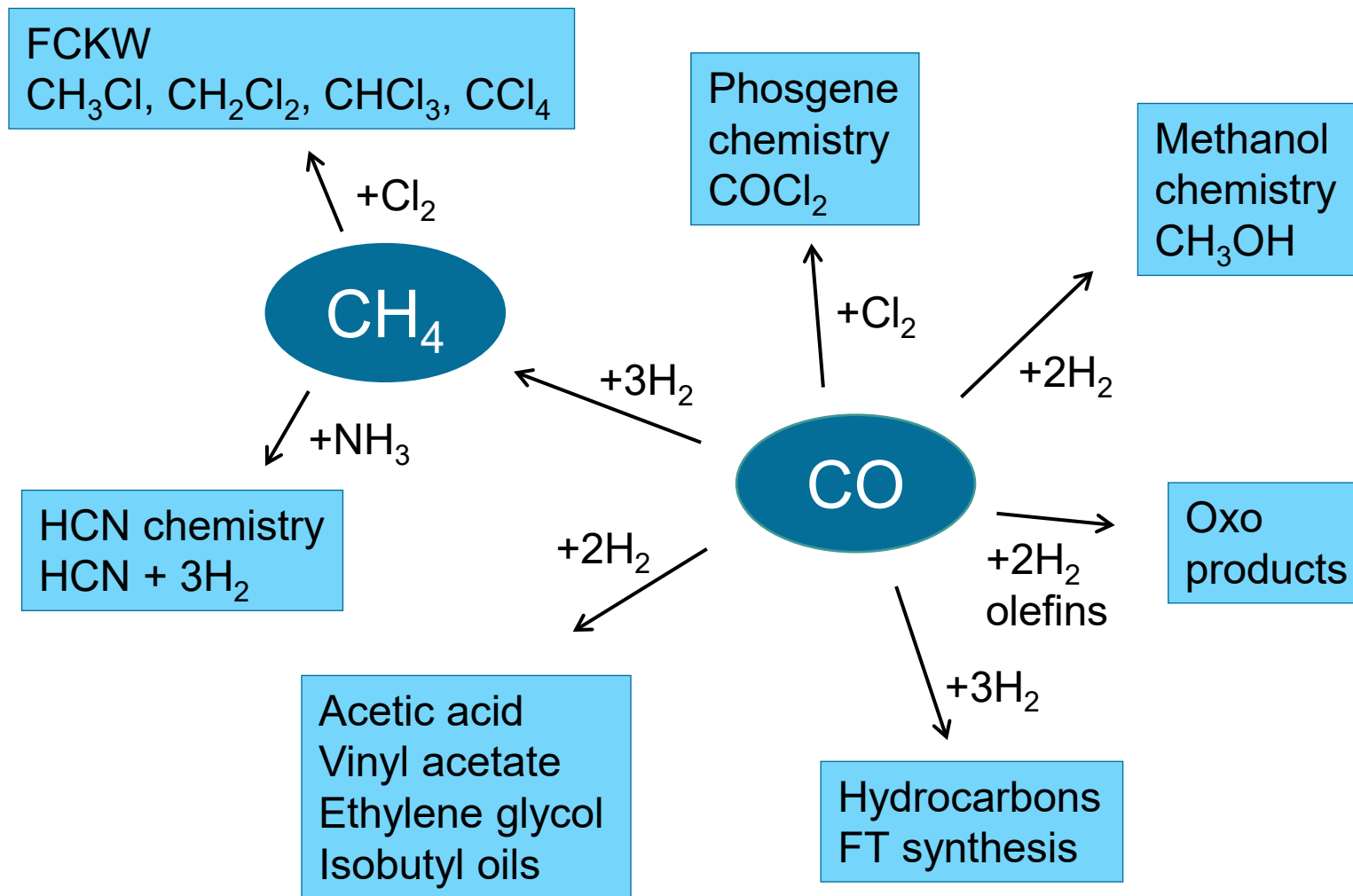
Products from syngas

Figure 1—Products from syngas



Source: P.L. Spath and D.C. Dayton, Preliminary screening—technical and economic assessment of synthesis gas to fuels and chemicals with emphasis on the potential for biomass-derived syngas, National Renewable Energy Laboratory, NREL/TP-510-34929, December, 2003.

C1 chemistry / building blocks



Basic products of industrial organic synthesis

- Syngas

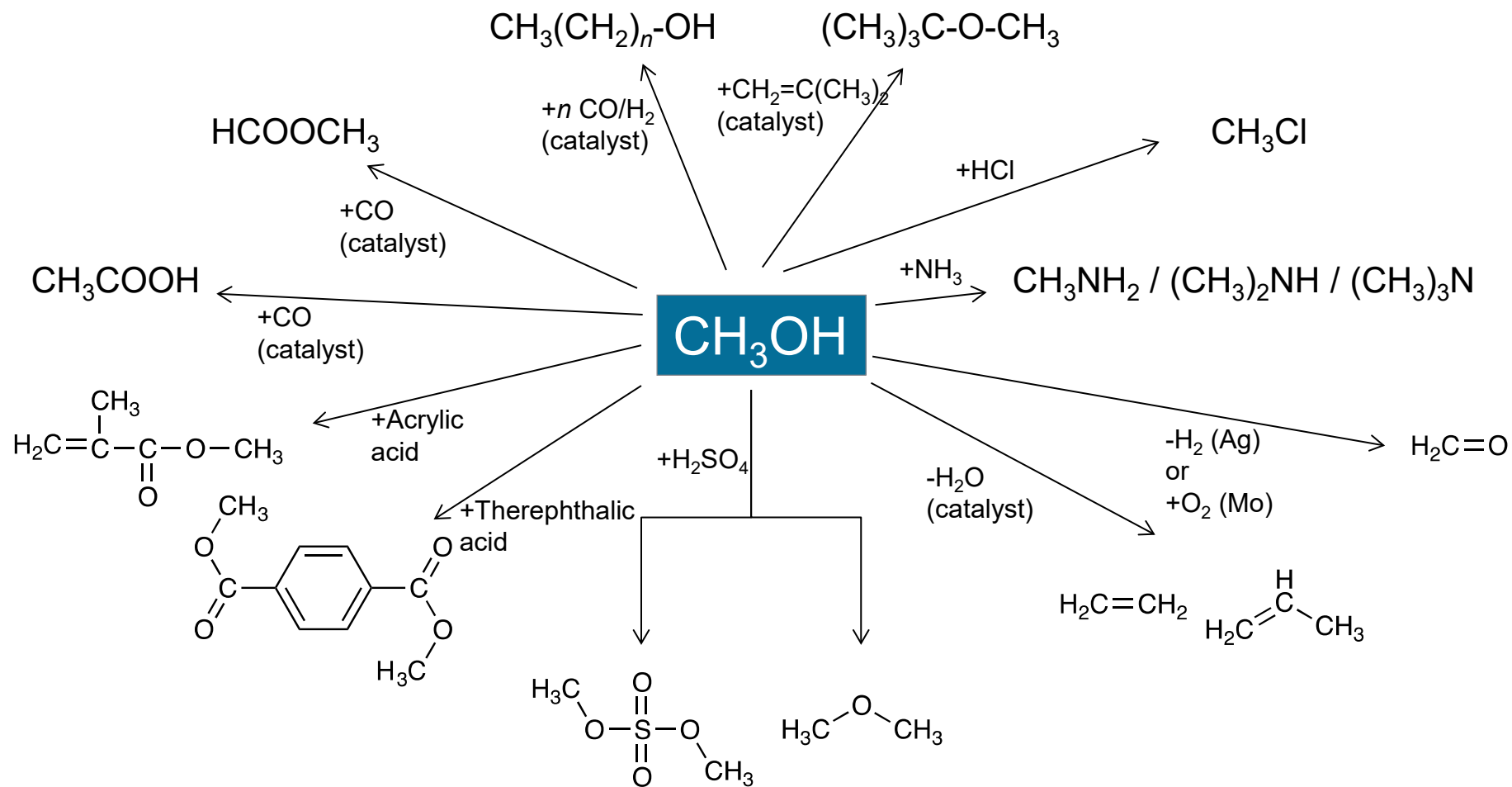
C1 chemistry:

- **Methanol**
- Formaldehyde
- Formic acid
- Hydrocyanic acid
- Methane
- Halogen derivatives of methane
- Reactions with CO and CO₂

Methanol

- Molar mass: 32,04 g/mol
- Density: 0,7918 g/cm³
- T_m: -97°C
- T_b: 64,7°C
- Acidity (pK_a): ~ 15,5
- Highly toxic
- ca. 250 €/t
- One of the largest raw material for synthesis
- Roughly 110 Mio t produced worldwide
- 90% in the chemical industry, rest as energy source
- Large capacity increase in the middle east
- Saudi Arabia up to 4 Mio t in 2003
- Largest producer: Methanex (2005: 4,3 Mio t/a capacity)
 - Trinidad: 2,6 Mio t/a

Industrially important reactions of CH₃OH

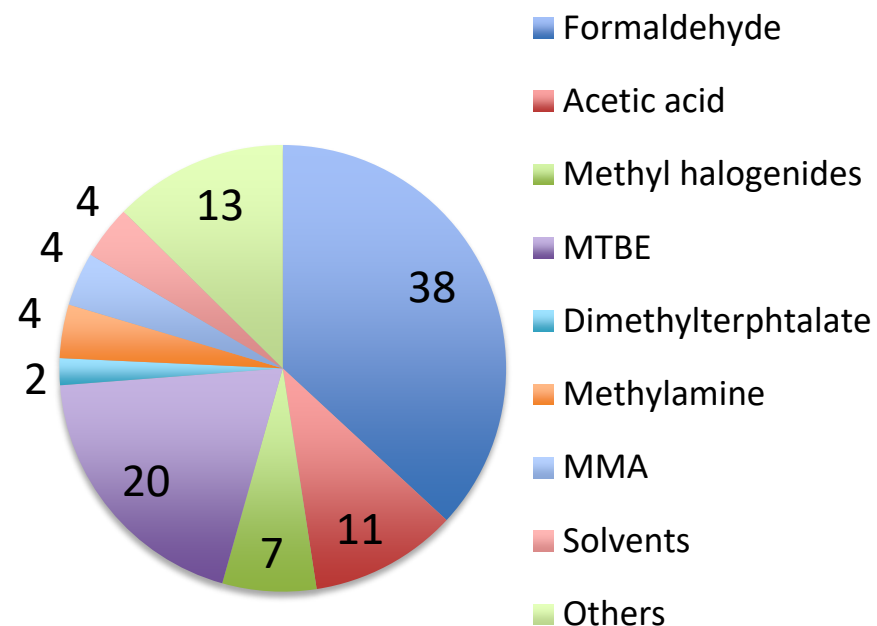


Use of CH₃OH

- 90% in the chemical industry, rest as energy source
- Fuel
- Component for fuels
- Source for proteins

Growing demand – Mega plants
(up to 10 Mio t/a capacity in planning)

World 2005 [%]



Methanol

Reactions:

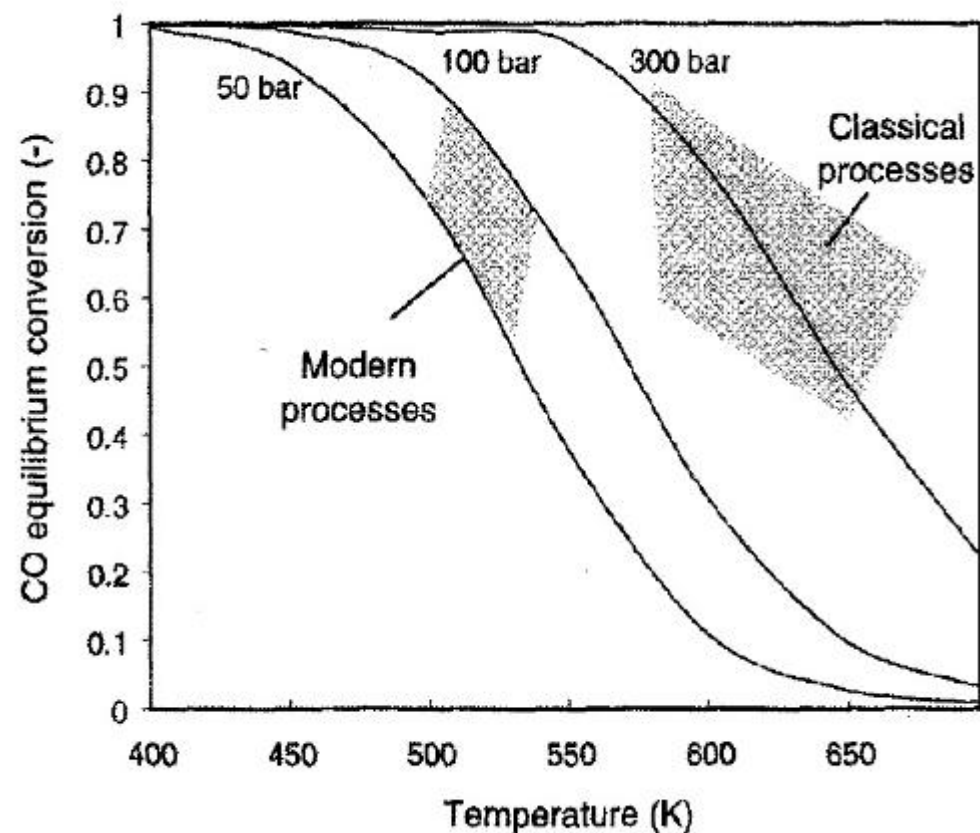
- $\text{CO} + 2\text{H}_2 \leftrightarrow \text{CH}_3\text{OH}$
($\Delta H = -92 \text{ kJ/mol}$)

When Methane rich fuels are used for syngas production, CO_2 is added:

- $\text{CO}_2 + 3\text{H}_2 \leftrightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$
($\Delta H = -50 \text{ kJ/mol}$)
- $\text{CO} + \text{H}_2\text{O} \leftrightarrow \text{CO}_2 + \text{H}_2$

Production processes for methanol

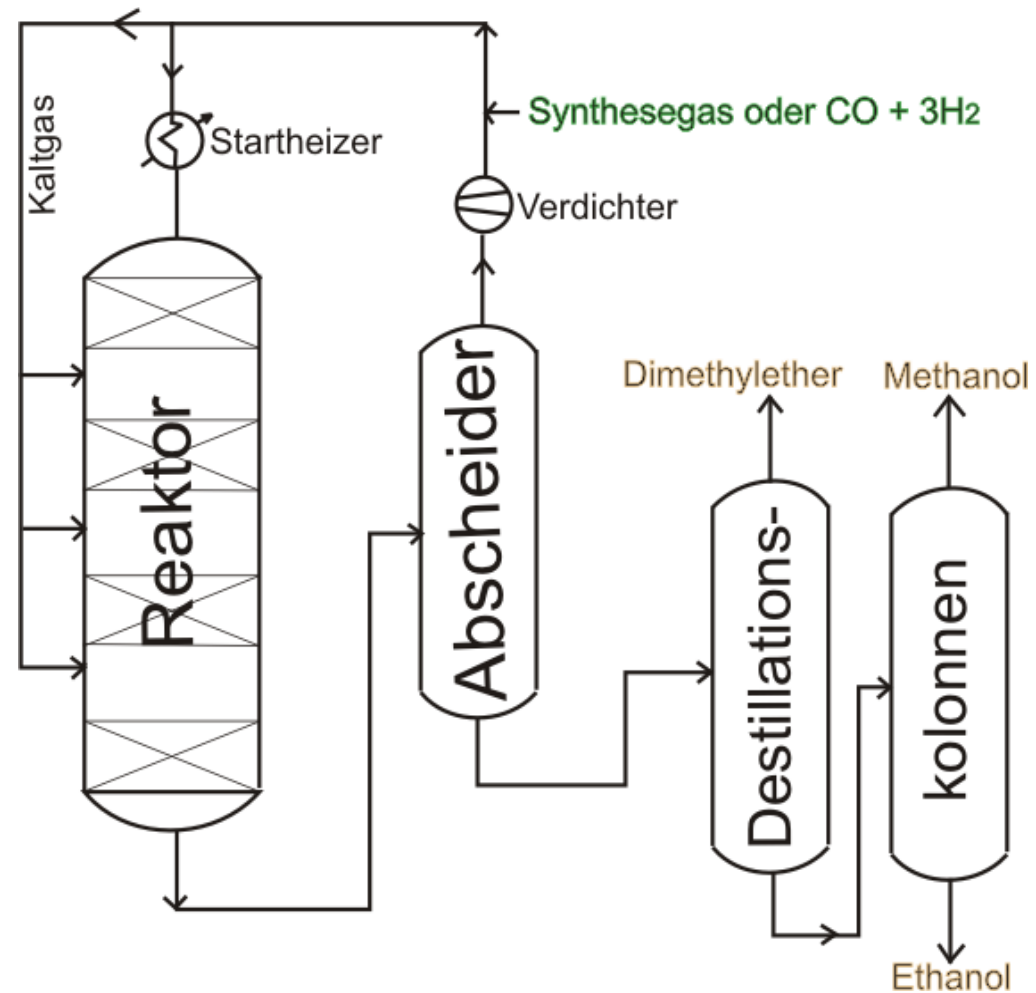
1. BASF high pressure process
2. ICI low pressure process
3. Lurgi low pressure process



1 - BASF high pressure process

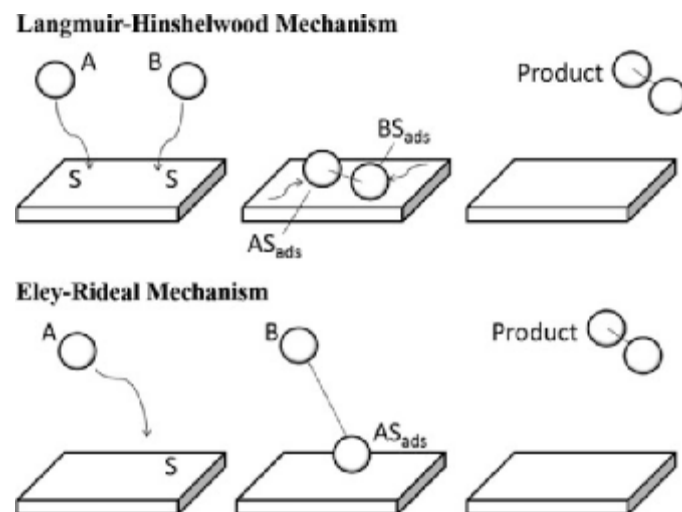
- Mittasch, BASF (Leuna), 1913:
found some oxygen containing compounds in NH_3 synthesis investigations
- 1923: first industrial production of CH_3OH based on syngas
- $\text{ZnO-Cr}_2\text{O}_3$ catalysts ($\text{Zn/Cr} = 70:30$)
- 320 - 380 °C
- 200 - 350 bar
- Multi layer reactor with cold gas feed

BASF CH₃OH process

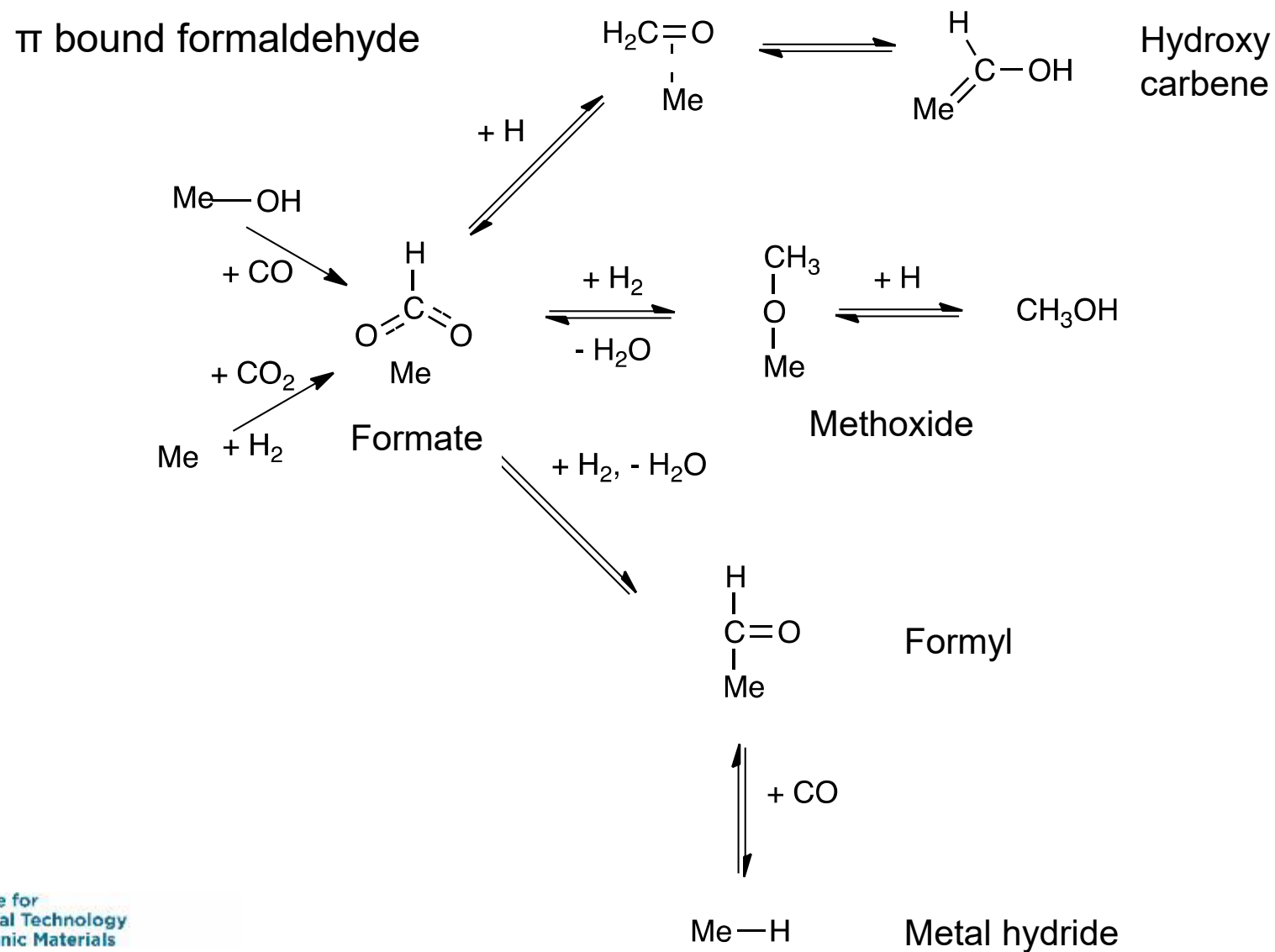


1 - BASF high pressure process

- Cu/ZnO catalysts on Al_2O_3
- Reaction kinetics follow Langmuir-Hinshelwood-mechanism
- CO and H_2 adsorb on the catalyst surface and react fast to a formyl species, which is then reacting with H_2 to a methoxy species
- Dominating step: reaction of the methoxy with adsorbed H_2 to methanol



Mechanism of CH₃OH synthesis



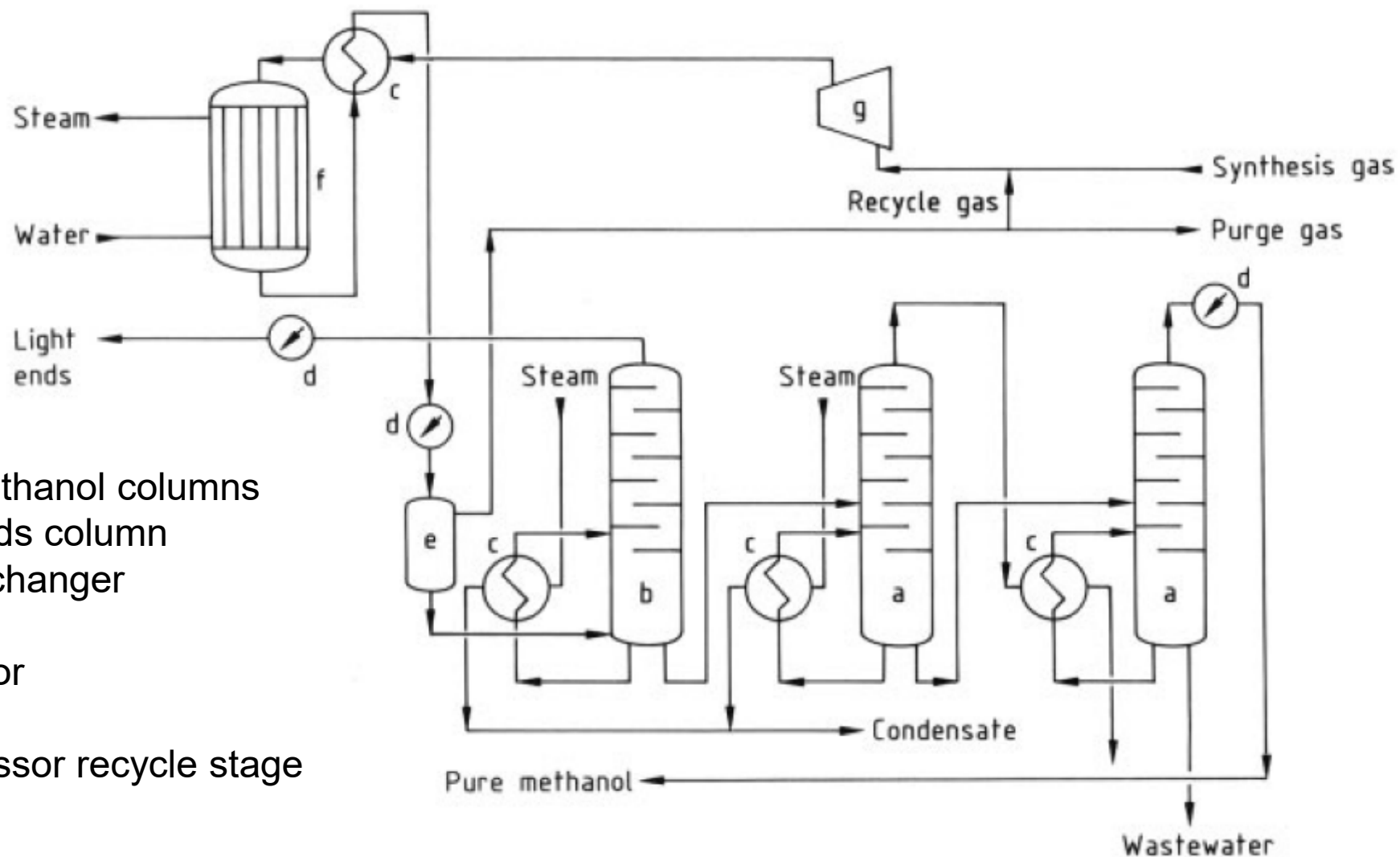
2 - ICI low pressure process

- 1966 first technical realization
- 1999 already 54 plants operating using the ICI process
- Capacities up to 1 Mio t/a
- Cu-Zn-Al-oxide catalyst (needs S and Cl free syngas!)
- 240 – 260 °C
- 50 – 100 bar
- Simple reactor design, only one catalyst bed, multiple cold gas feed
- Very pure (99,99%w) CH₃OH

3 - Lurgi low pressure process

- 1973: 200 kt/a plant taken in operation
(in combination with a NH_3 plant)
- 1999 already 29 plants operating using the Lurgi process
- Trinidad plant with 1,8 Mio t/a capacity
- Modified CuO-ZnO catalyst (needs S and Cl free syngas!)
- 250 – 260 °C
- 50 – 80 bar
- Pipe bundle reactor
- Temperature control via boiling water

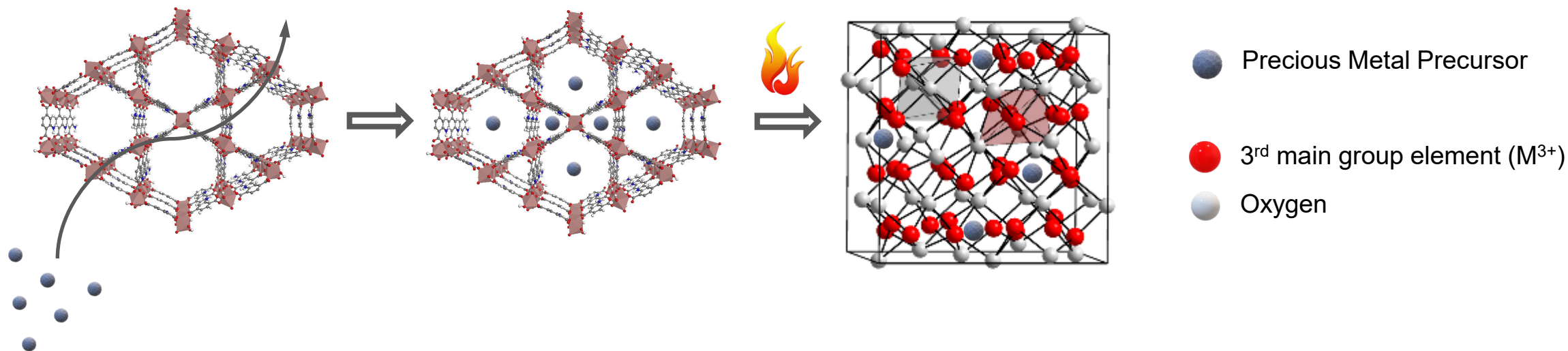
3 - Lurgi CH_3OH process



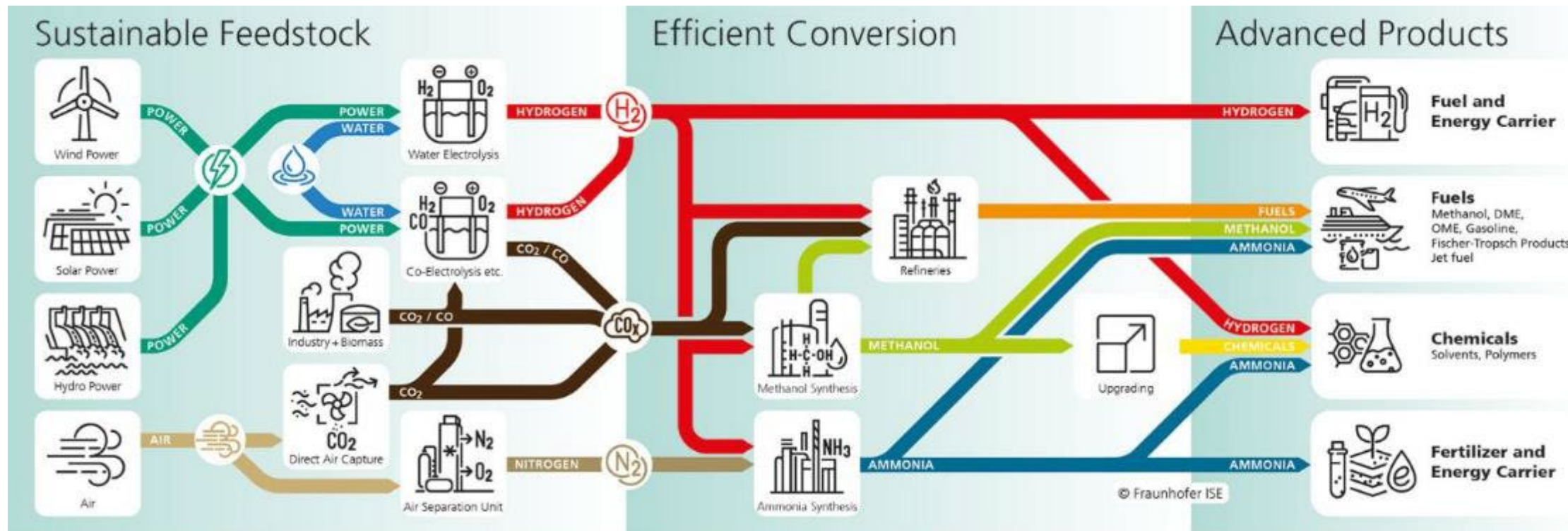
- a) Pure methanol columns
- b) Light ends column
- c) Heat exchanger
- d) Cooler
- e) Separator
- f) Reactor
- g) Compressor recycle stage

New approaches for CO₂ to MeOH

- E.g. project at our institute together with CHASE & OMV
- Metal-organic frameworks (MOF) pores serve as means to achieve atomic dispersion of the noble metal promoter (Pd) after calcination
- Maximizes metal-support-interaction and abundance of catalytic facets/centers

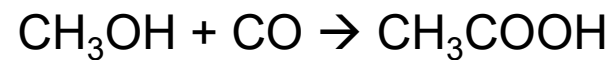


CH₃OH important intermediate



Methanol chemistry

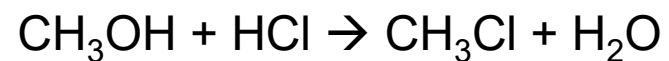
- Synthesis of **acetic acid**:



100°C

RhJ₃ catalysts

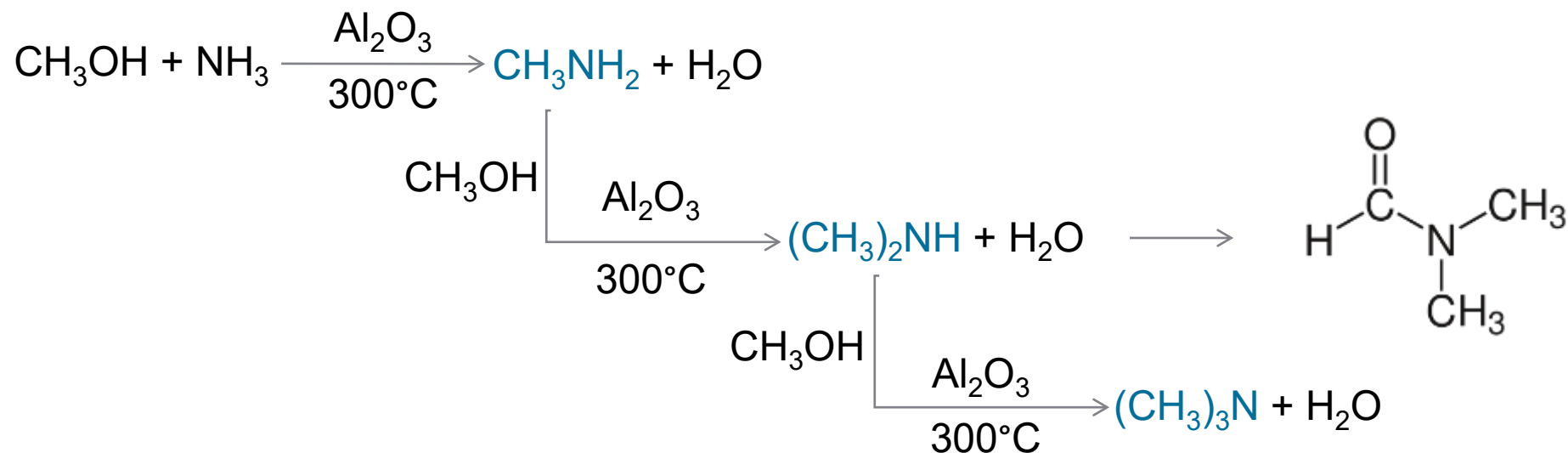
- Synthesis of **methyl chloride**:



250°C

ZnCl₂ catalysts

Synthesis of methyl amines



Use:

- Solvents
- Pesticides to Pharmaceuticals and Detergents
- Dimethylamine → Dimethylformamide & Dimethylacetamide (solvent)

CH₃OH as fuel component

- CH₃OH can be used directly or as in combination
- M15 feasible
- M100 with some minor changes in the carburetor and a bigger tank
 - Bigger heat of evaporation and lower energy content compared to standard fuel
 - + High Octane number (>110)
 - + Less pollutants in off gas
 - Caution: avoid second water phase in fuel tank
(e.g. via the use of sec. or tert. butanole)

CH₃OH to higher alcohols

Homologation:



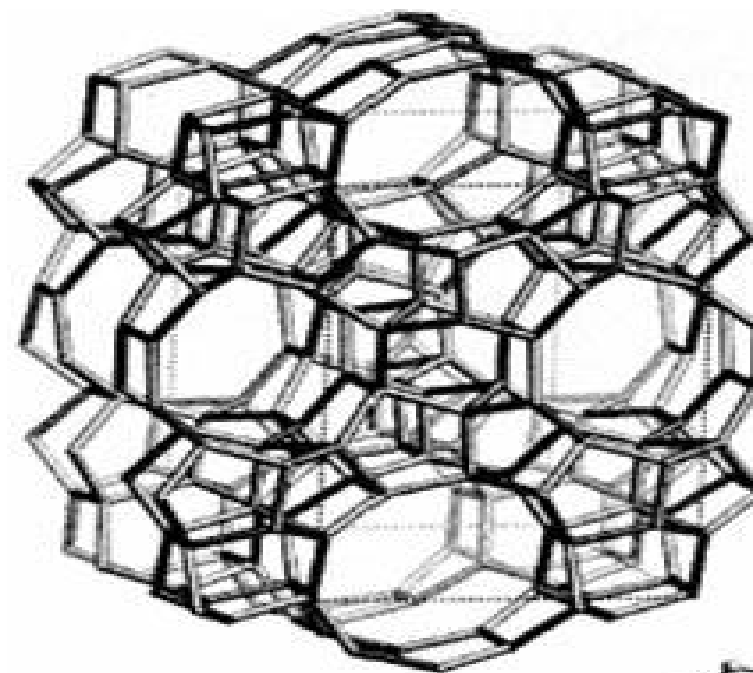
Co/Rh catalysts

Octanmix (Lurgi) process:

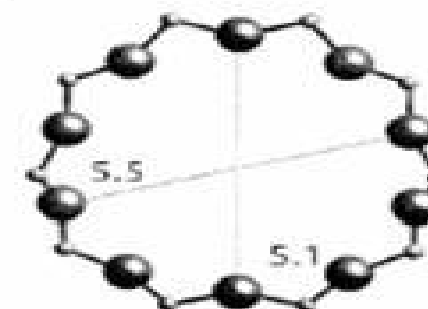
- Modified Cu catalyst, 300°C, 50 – 100 bar
- H₂/CO ratio → 1 (much smaller than for normal CH₃OH synthesis)
- 50%w CH₃OH, 45%w higher alcohols

MTG process

- **Methanol to Gasoline**
- Developed by Mobil via a special zeolite catalyst ZSM-5
(zeolite synthesized by Mobil)
- Pore diameter fits the size and shape up to C10 carbohydrates (shape selectivity)
- $n\text{CH}_3\text{OH} \rightarrow \text{DME} \rightarrow \text{gasoline}$



ZSM-5 (MFI)



Basic products of industrial organic synthesis

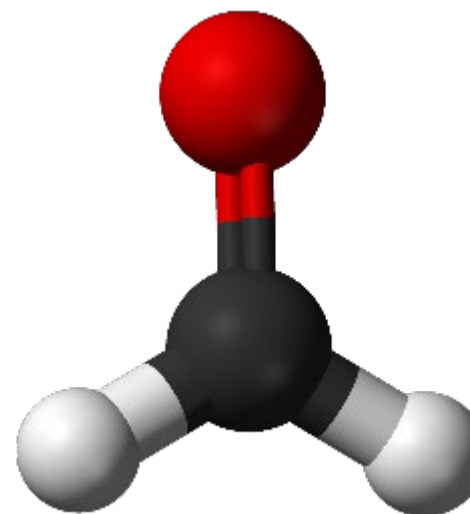
- Syngas

C1 chemistry:

- Methanol
- **Formaldehyde**
- Formic acid
- Hydrocyanic acid
- Methane
- Halogen derivatives of methane
- Reactions with CO and CO₂

Formaldehyde

- Molar mass: 30,026 g/mol
- Tm: -92°C / Tb: -21°C
- Toxic
- Producers: Borden and BASF
- Plant capacities up to 200 kt
- 400 – 500 \$/t
- Annual production roughly 21 Mill. t
- Colorless liquid, polymerizes easily
 1. 35% solution in water (hydrate)
 2. Trioxane
 3. Paraformaldehyde



Use of CH₂O

- Condensation products:
 - Phenol resins (Novolak, Resole, Bakelite)
 - Aminoplast, ...
 - Trioxane for polyoxymethylene
 - Aldol reaction:
 - Pentaerythrite, trimethylolpropane, neopentylglycol
 - Butindiol
 - Others (e.g. glycol acid)
-
- World: ca. 21 Mill t

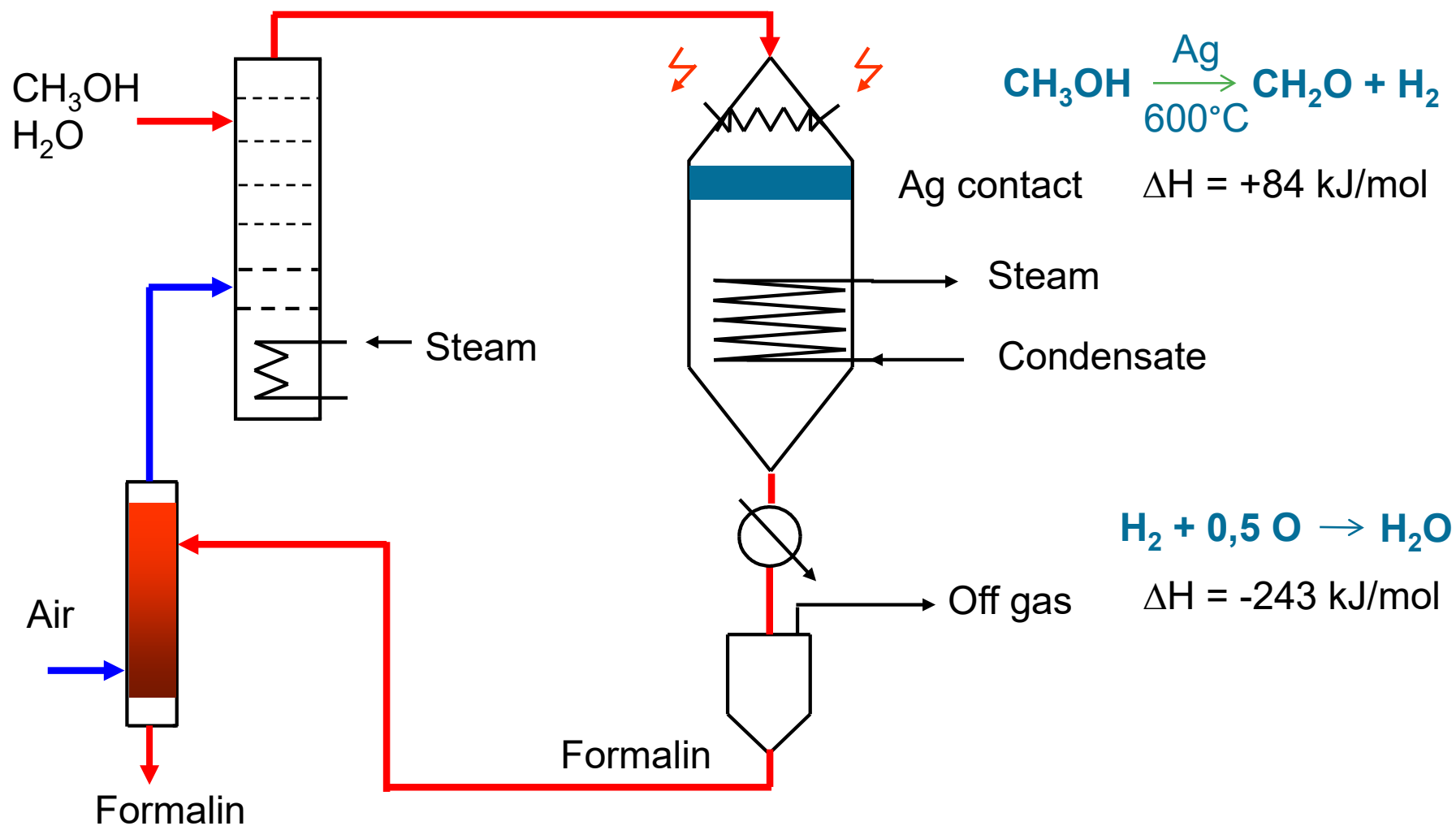
World [%w]	1984	2003
Urea resins	32	46
Phenol resins	11	12
Melamine resins	4	6
Pentaerythrite	6	5
1,4 Butanediol	2	4
Methylendiisocyanate	2	4
Others	43	23

Production of CH_2O

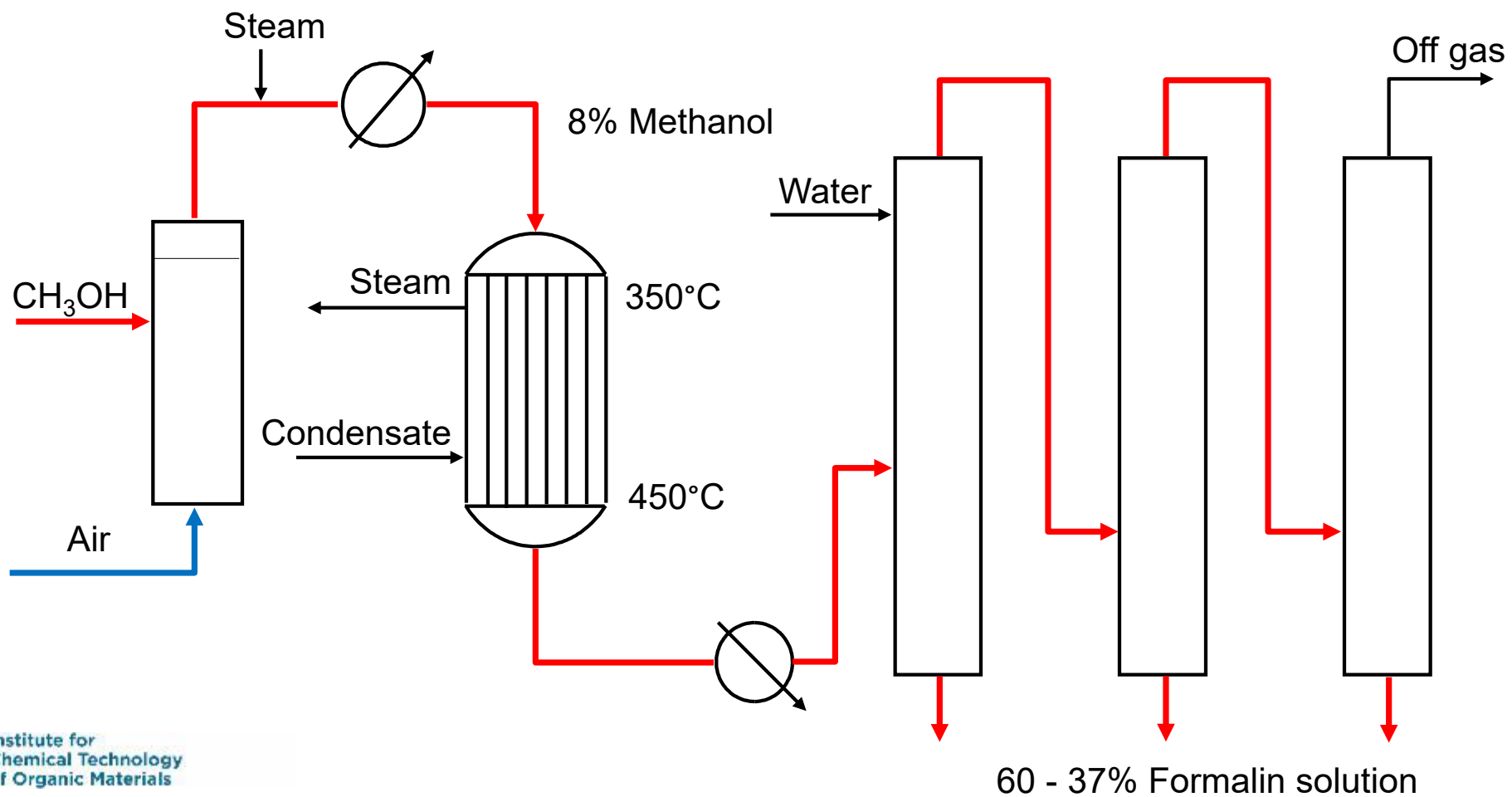
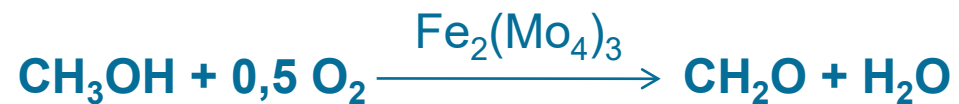
Based on CH_3OH

1. Dehydration on Ag or Cu catalysts
2. Oxidation on MoO_3 and Fe_2O_3

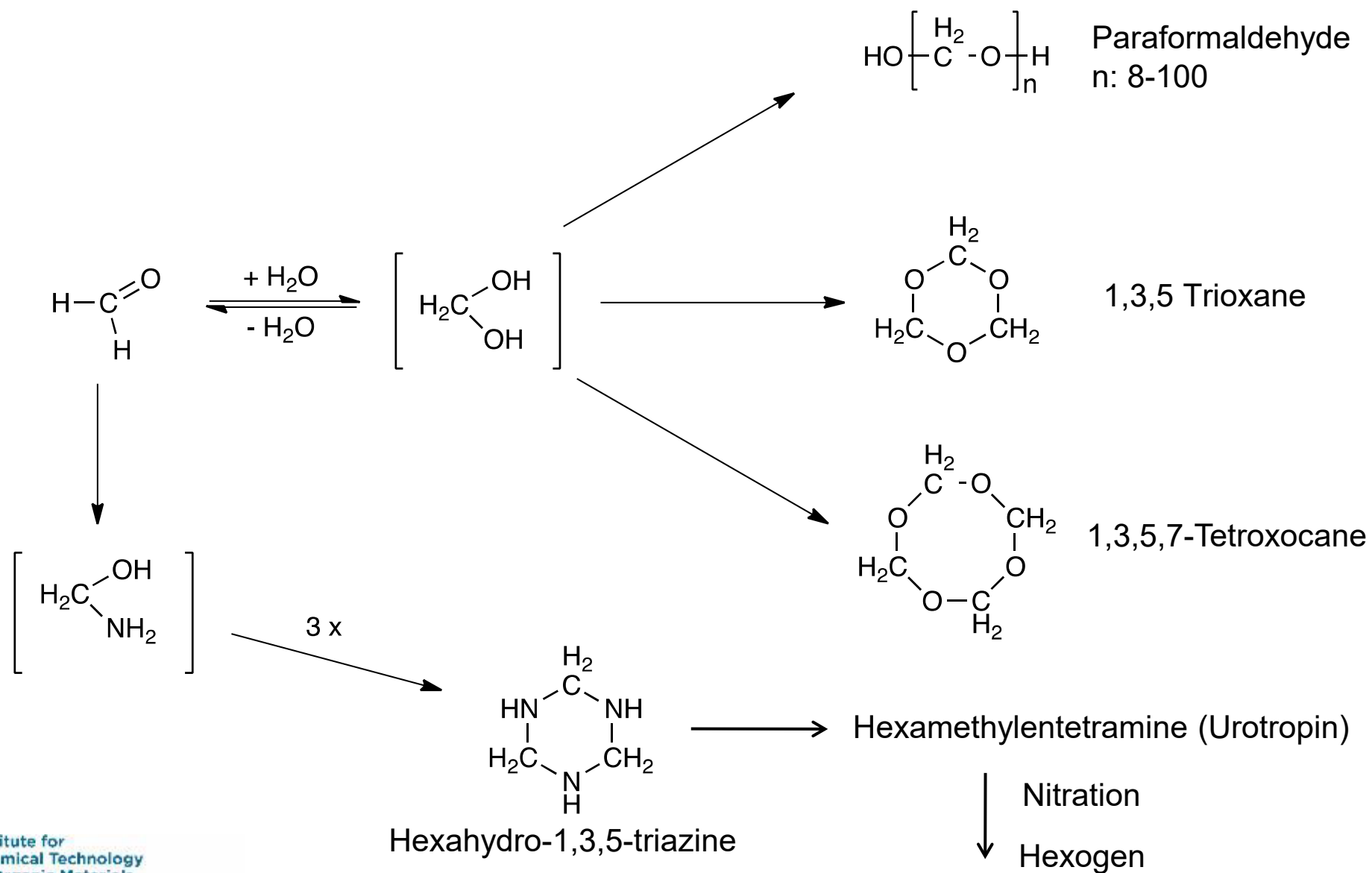
1 – Dehydration



2 – Oxidative process (Formox)

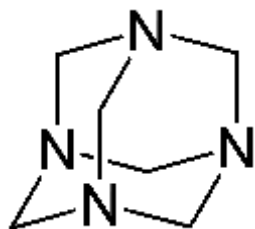


Chemistry of CH₂O



Urotropine

Hexamethylenetetramine



Base for:

- Amino- and phenol resins
- Food conserving (E239)
- Esbit
- Explosives
- Duff & Mannich reaction

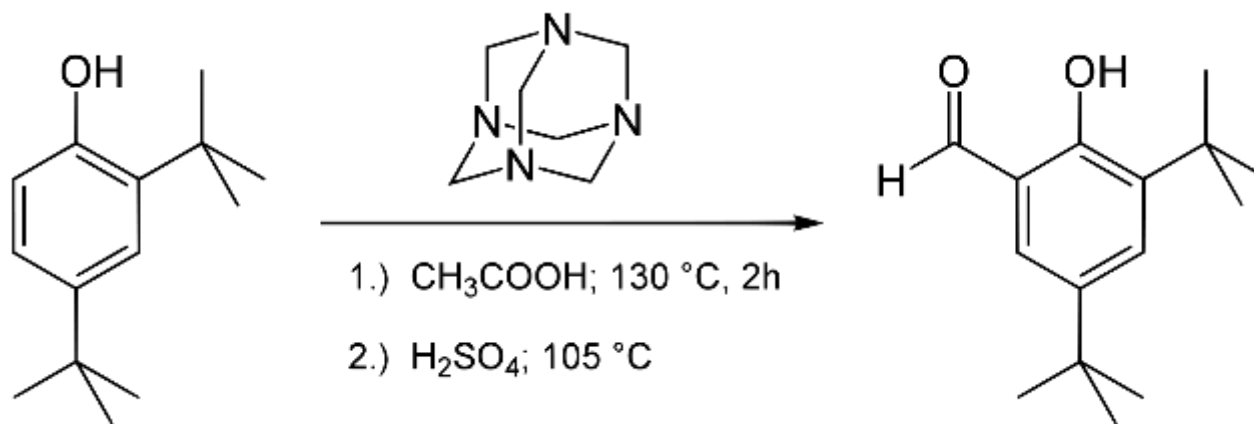


Provolone cheese



Duff reaction

- The Duff reaction or hexamine aromatic formylation is a formylation reaction used in organic chemistry for the synthesis of benzaldehydes with hexamine as the formyl carbon source
- It is named after James Cooper Duff, who was a chemist at the College of Technology, Birmingham



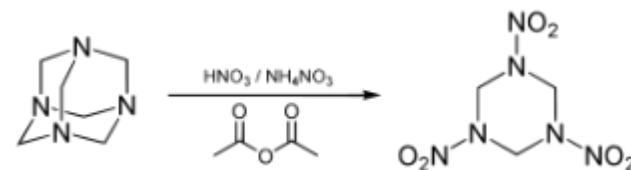
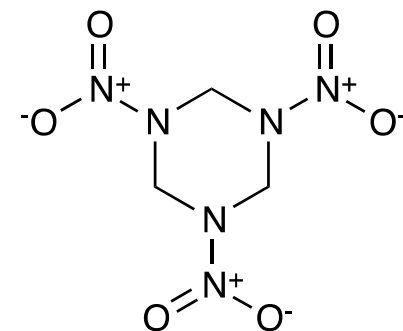
Explosives - Hexogen

Hexogen

Cyclotrimethylenetrinitramin, Cyclonit, T4 and RDX

(Research Department Explosive / Royal Demolition Explosive)

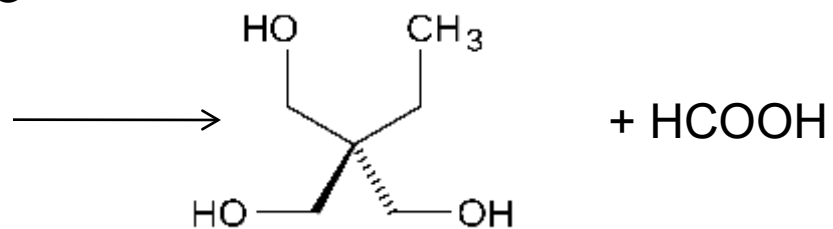
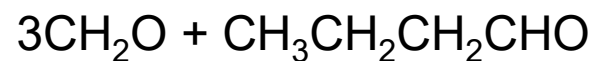
Hexahydro-1,3,5-trinitro-1,3,5-triazin



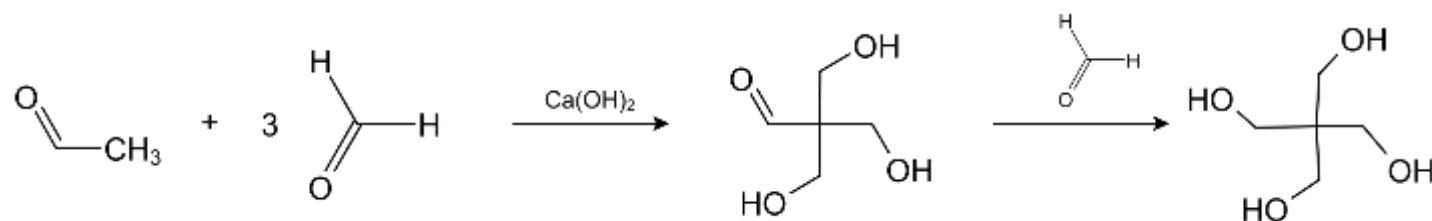
Poly-alcohols from CH₂O

Synthesis of poly-alcohols

- Trimethylolpropane:



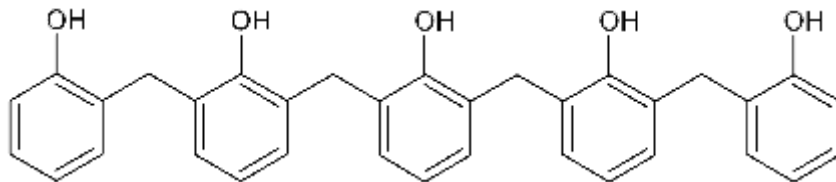
- Pentaerythrite:



CH₂O as resin base

Resins: formaldehyde plus

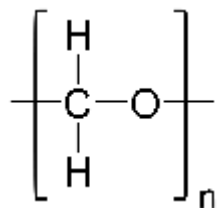
1. Phenol: phenol resins (Bakelit), Novolak (structure below)



2. Urea: urea resins
3. Melamine: melamine resins

Polymerization of CH₂O

Polyoxymethylene POM:



- Good mechanical strength
- Good electrical properties
- Excellent dimensional stability

Polyoxymethylene POM

- Stabilization through esterification of end groups with acetic anhydride or copolymerization with ethylene oxide
- Used in the automotive industry and for precision parts
- Developed by DuPont in 1956
- Acetal

CH₂O and the environment

In nature:

- Released into the atmosphere as a result of the combustion, degradation, and photochemical decomposition of organic materials
- CH₂O washed out of the air by rain is degraded by bacteria to CO₂ and H₂O

Emission source	Formaldehyde level
Natural gas combustion	
Home appliances and industrial equipment	2400 – 58 800 µg/m ³
Power plants	15 000 µg/m ³
Industrial plants	30 000 µg/m ³
Fuel-oil combustion	0.0 – 1.2 kg/barrel oil
Coal combustion	
Bituminous	< 0.005 – 1.0 g/kg coal
Anthracite	0.5 g/kg coal
Power plant, industrial, and commercial combustion	2.5 mg/kg coal
Refuse incinerators	
Municipal	0.3 – 0.4 g/kg refuse
Small domestic	0.03 – 6.4 g/kg refuse
Backyard (garden refuse)	up to 11.6 g/kg refuse
Oil refineries	
Catalytic cracking units	4.27 kg/barrel oil
Thermoform units	2.7 kg/barrel oil
Automotive sources	
Automobiles	0.2 – 1.6 g/L fuel
Diesel engines	0.6 – 1.3 g/L fuel
Aircraft	0.3 – 0.5 g/L fuel

CH₂O and the environment

In confined areas:

1. Smoking of cigarettes and tobacco products
2. Urea-, melamine-, and phenol– formaldehyde resins in particle board and plywood furniture
3. Urea – formaldehyde foam insulation
4. Open fireplaces, especially gas fires and stoves
5. Disinfectants and sterilization of large surfaces (e.g., hospital floors)

A maximum limit of 0.1 ppm formaldehyde in indoor living and recreation areas has been recommended by the BGA (German Federal Health Office)

CH₂O and the environment

- Cigarette smoke contains 57–115 ppm of formaldehyde and up to 1,7 mg of formaldehyde can be generated while one cigarette is smoked
- If five cigarettes are smoked in a 30m³ room, with a low air-change rate of 0,1 (i.e., 10 %) per hour, the formaldehyde concentration reaches 0,23 ppm



Basic products of industrial organic synthesis

- Syngas

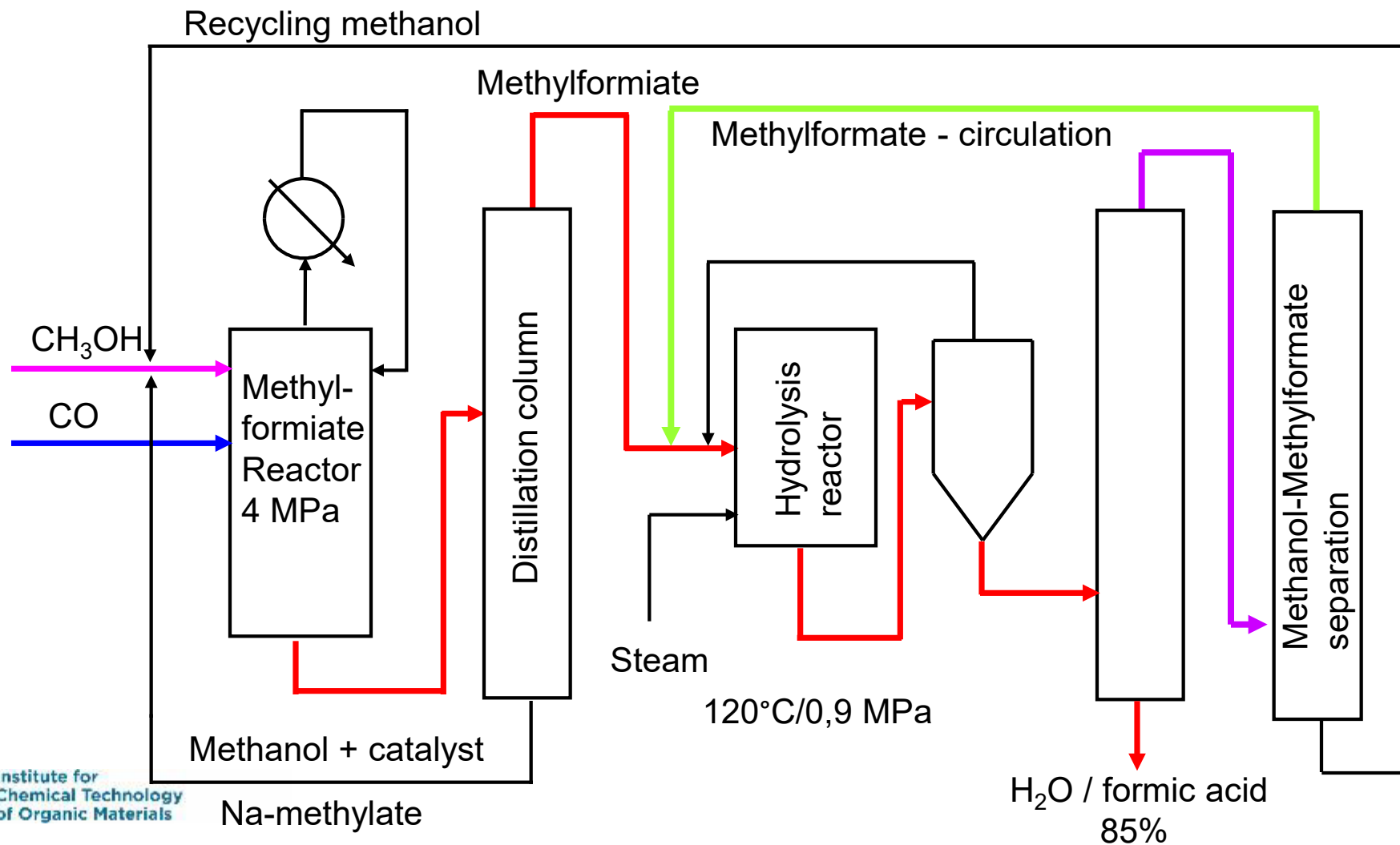
C1 chemistry:

- Methanol
- Formaldehyde
- **Formic acid**
- Hydrocyanic acid
- Methane
- Halogen derivatives of methane
- Reactions with CO and CO₂

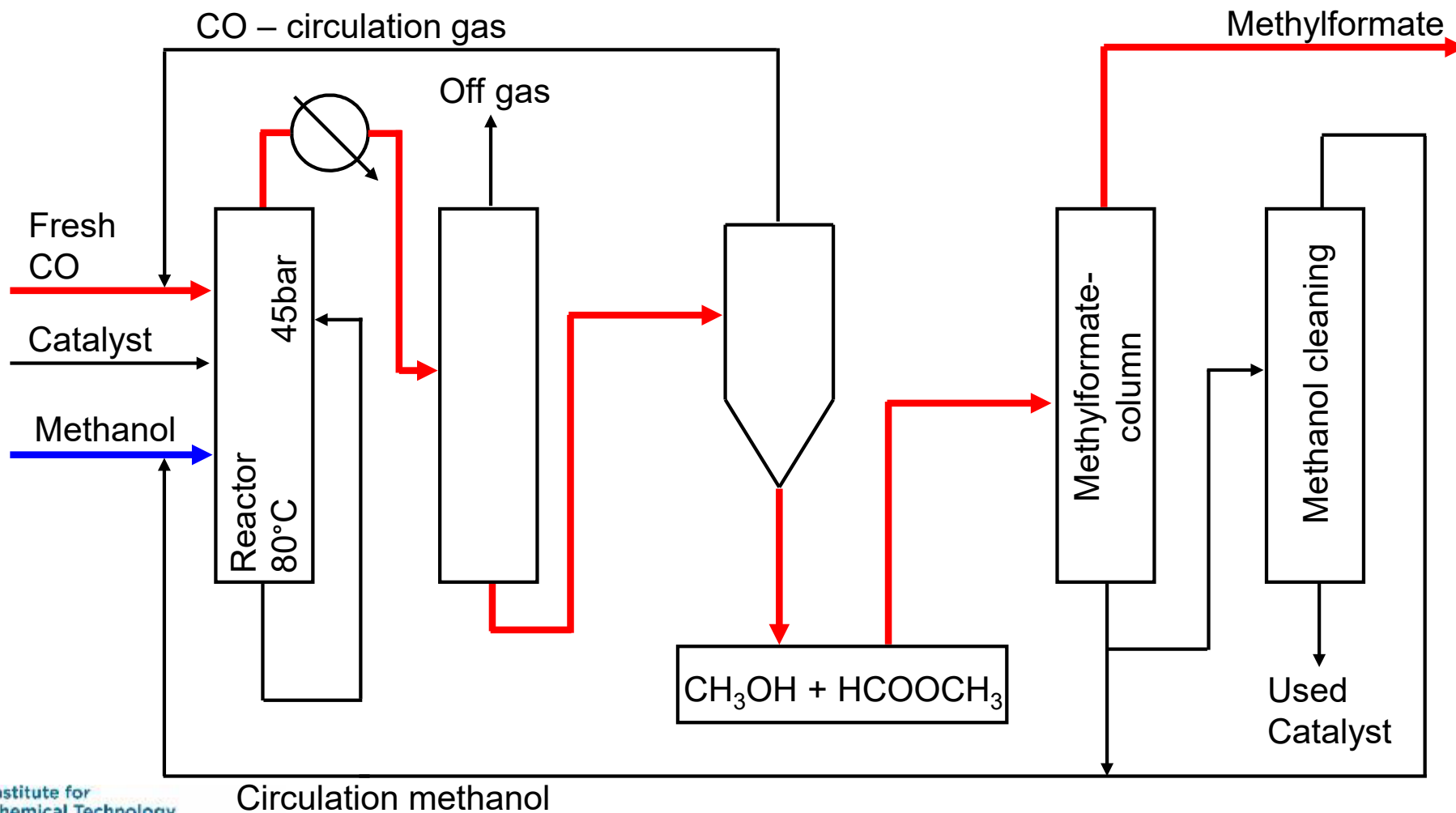
Formic acid

- Molar mass: 46,03 g/mol
- Density: 1,22 g/cm³
- T_m: 8°C
- T_b: 101°C
- Acidity (pK_s): 3,77
- Simplest carbon acid
- Use builds on its properties as carbon acid but also on its reducing effect
- Roughly 1 Mill t produced worldwide
- BASF, BP, Kemira

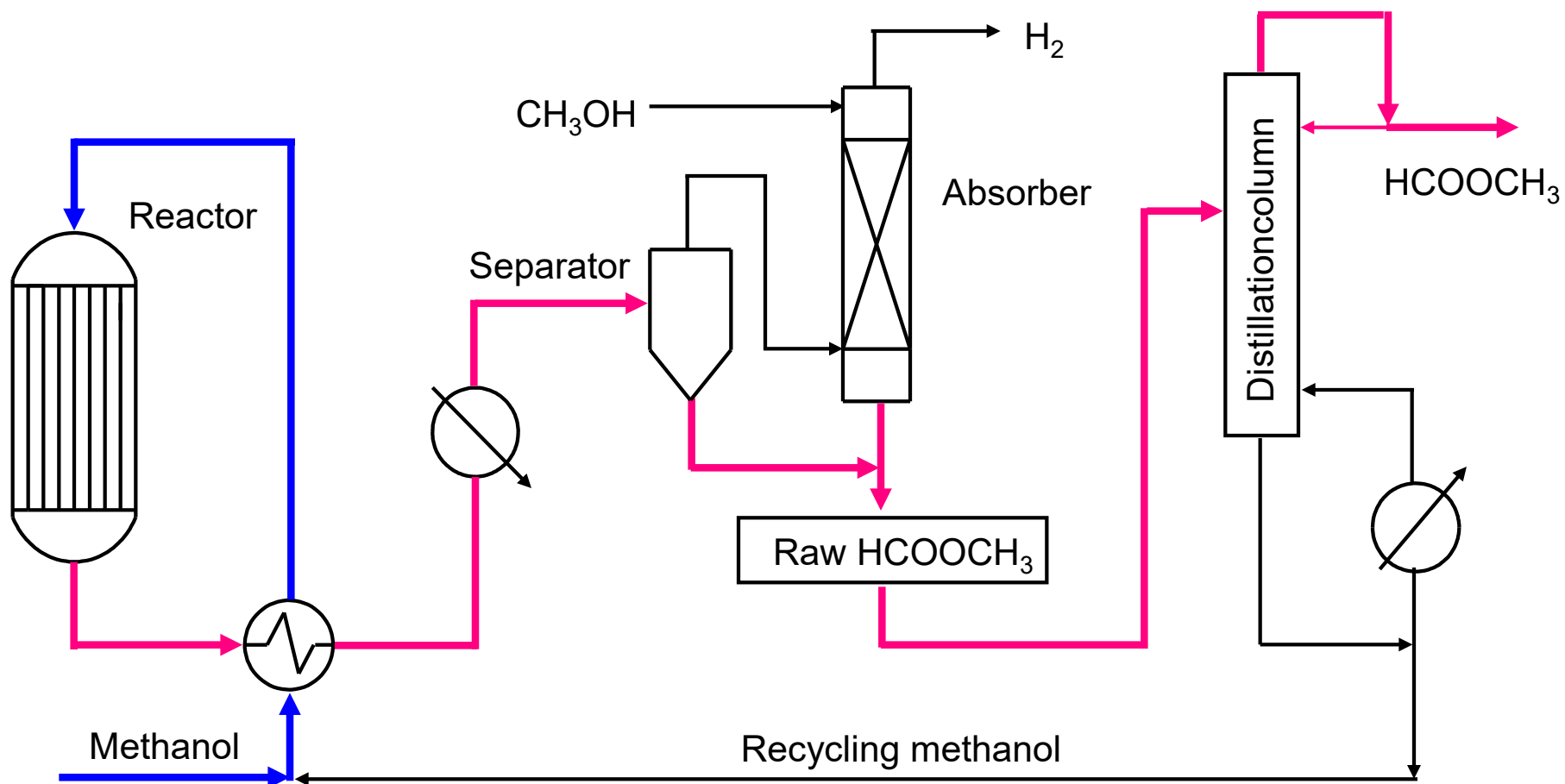
Leonard process



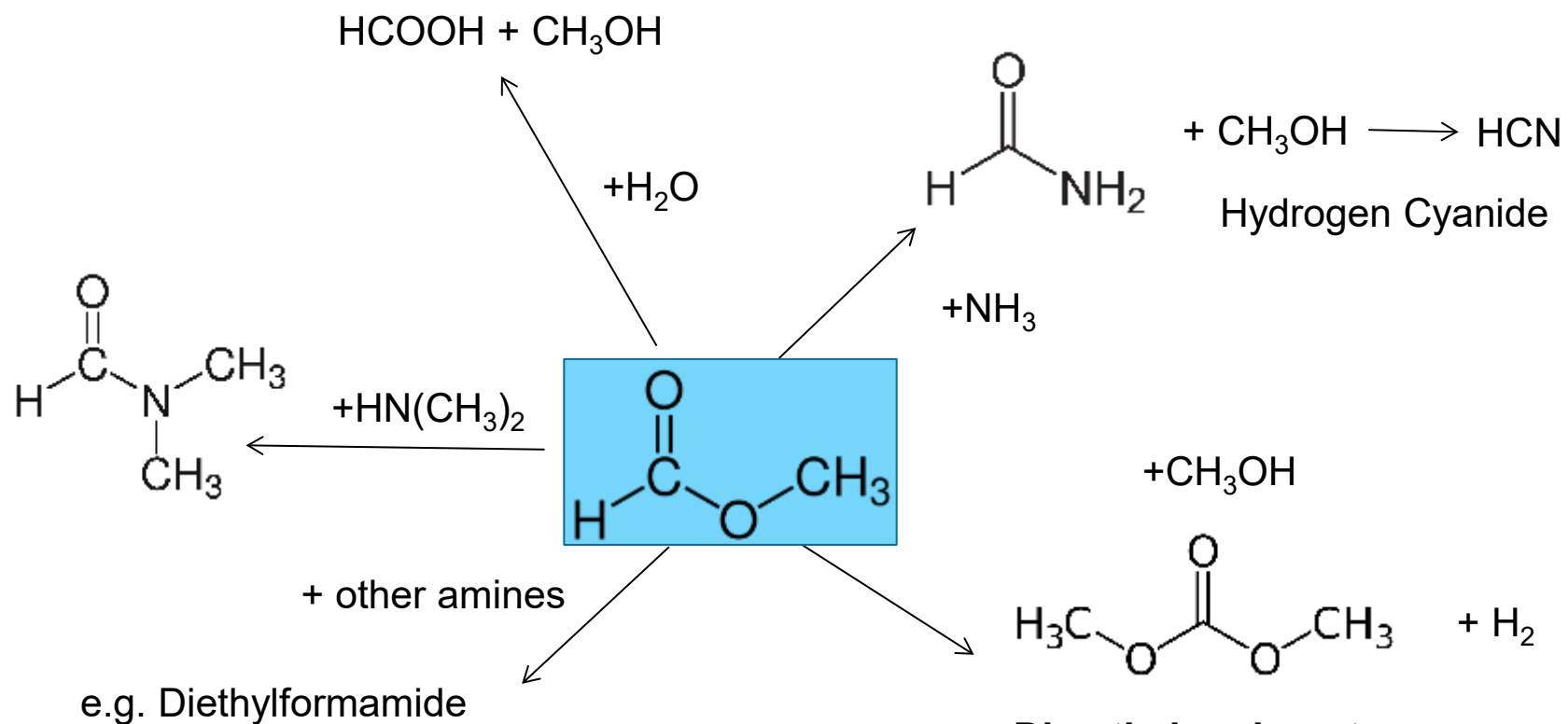
Carbonization of CH₃OH (BASF)



Dehydration of Methanol



Synthesis with Methyl formate



Dimethyl carbonate

- Methylating agent
- Solvent
- Polycarbonate synthesis
- Alternative fuel additive

Basic products of industrial organic synthesis

- Syngas

C1 chemistry:

- Methanol
- Formaldehyde
- Formic acid
- **Hydrocyanic acid**
- Methane
- Halogen derivatives of methane
- Reactions with CO and CO₂

Hydrocyanic acid

- Basic product of industrial organic synthesis
- Molar mass: 27,03 g/mol
- Density: 0,69 g/cm³
- Tb: 26°C

Production processes:

1. Formamide dehydration
2. Ammon oxidation
3. Condensation of CH₄ and NH₃
4. By-product in the ammon oxidation of propylene



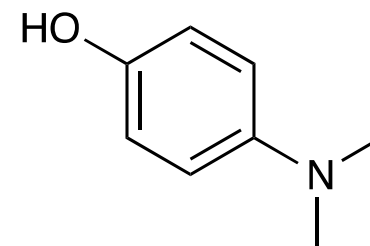
HCN poison

- Already 1-2 mg/kg body weight are lethal
- Swallow, respiration, skin
- Blocking of the oxygen-binding site by an irreversible binding of cyanide to the central Fe(III) ion of the hemoglobin of the respiratory chain in the mitochondria of the cell
- By the inactivation of this enzyme, cell respiration is stopped, the cell can no longer use the oxygen to produce energy and so it comes to so-called 'internal suffocation'

Antidote

4-Dimethylaminophenol

- Converts $\text{Fe(II)} \rightarrow \text{Fe(III)}$
- Leads to a met-hemoglobin formation
- Met-hemoglobin binds the cyanide ions



Production processes for HCN

1. HCONH_2 dehydration
2. Ammon oxidation (Andrussow)
3. Condensation von CH_4 and NH_3
4. Ammon oxidation of propene

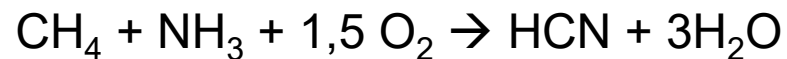
1 - HCN via HCONH₂ dehydration



$$\Delta H = +75 \text{ kJ/mol}$$

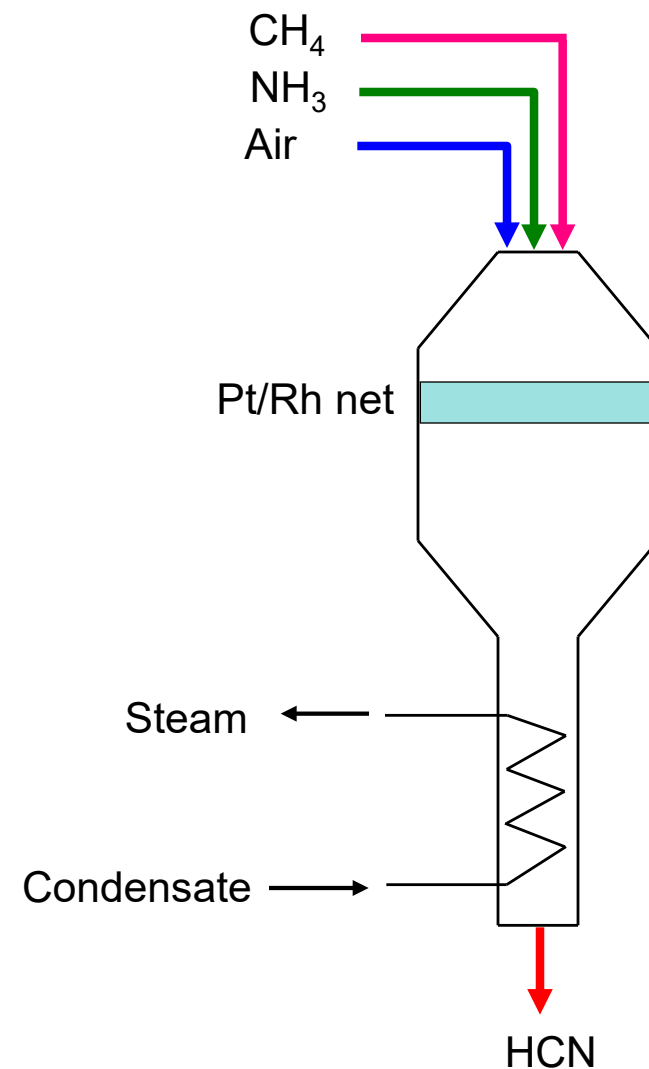
- 380 – 430 °C
- Reduced pressure (vacuum)
- Pipe reactor
- Catalyst: Fe or Al phosphate
- Reaction gas: 60 – 70 %v HCN (can be liquefied directly)
- High selectivity: 92-95%
- BASF

2 – Ammon oxidation (Andrussow)

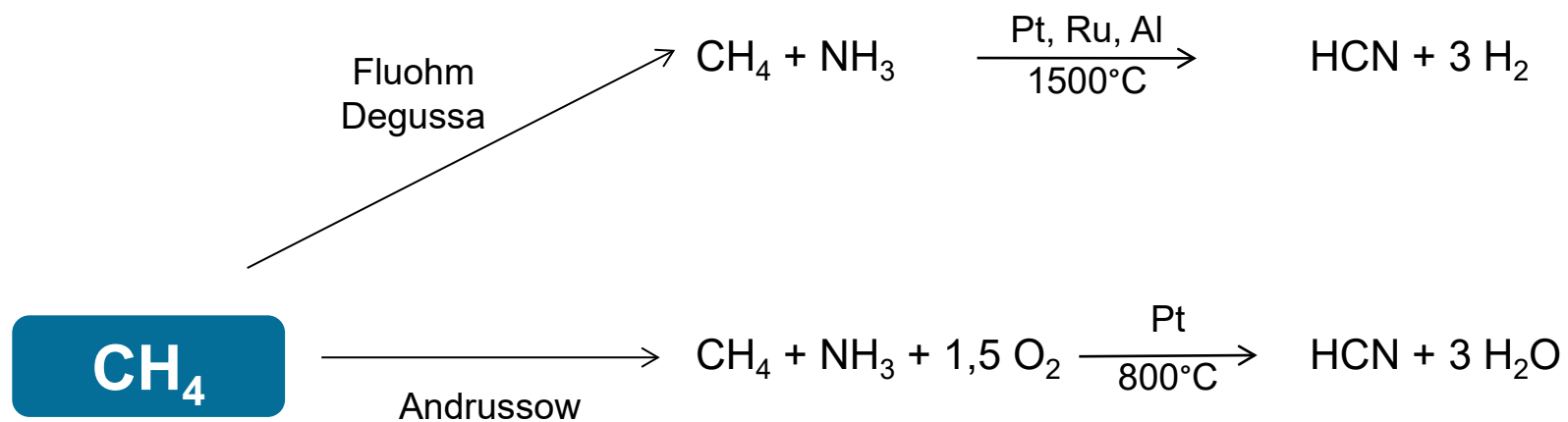


$$\Delta H = -473 \text{ kJ/mol}$$

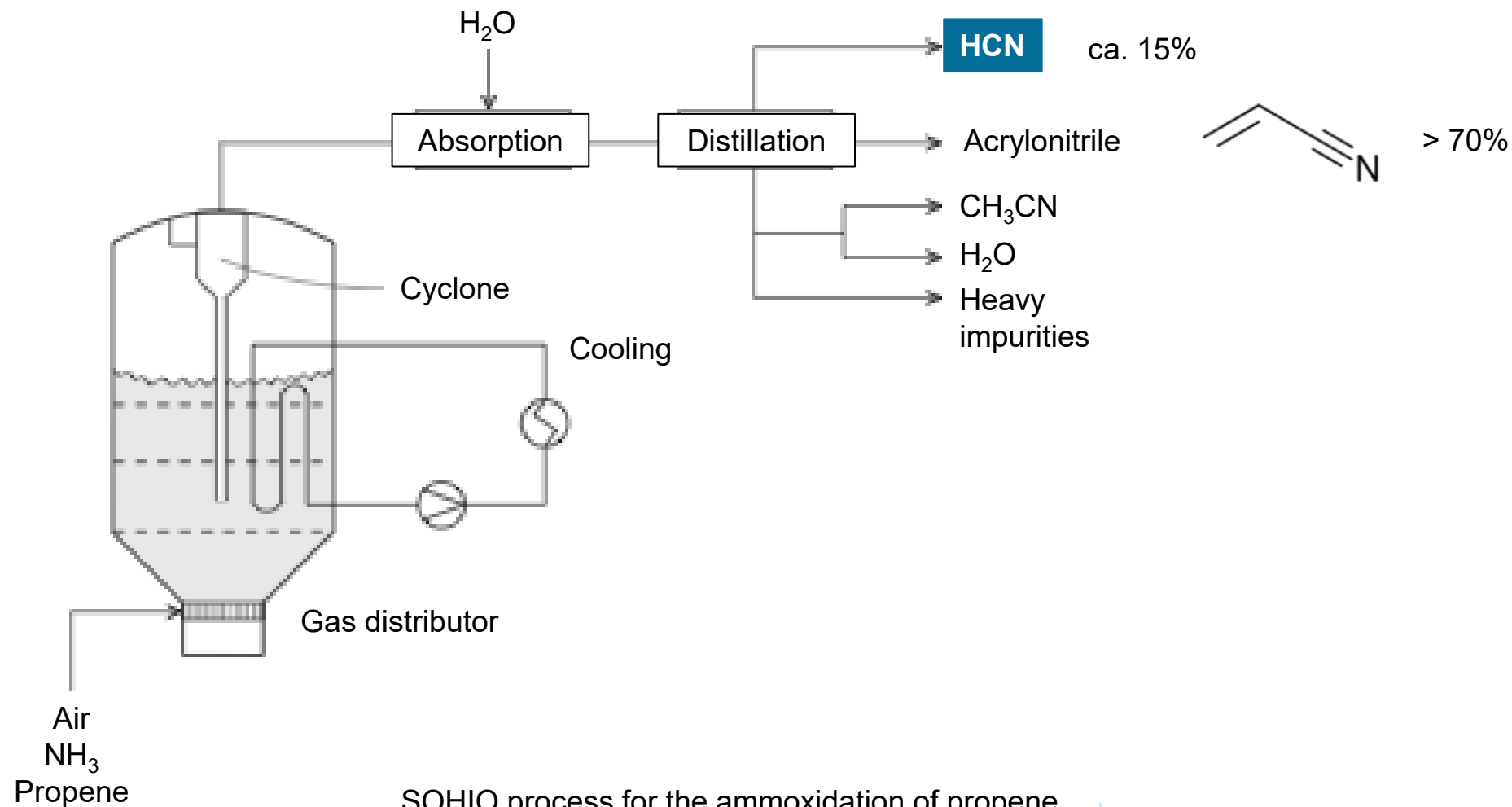
- Developed by BASF
- Catalyst: Pt/Rh
- At ambient pressure
- 1000 – 1200 °C
- Short residence times
- Fast quench of reaction gases to avoid HCN decomposition



3 – based on CH₄

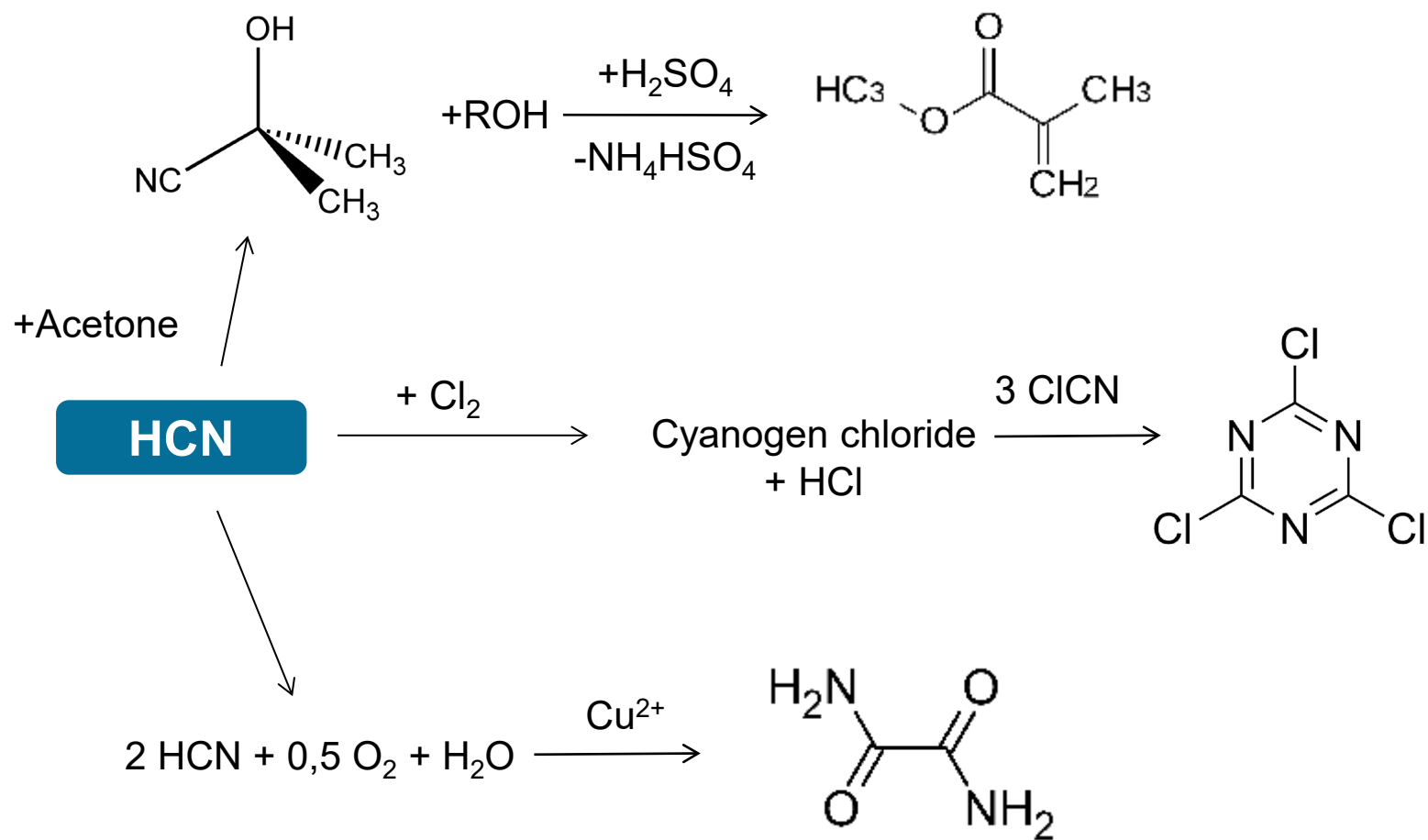


4 – Ammonoxidation of Propene

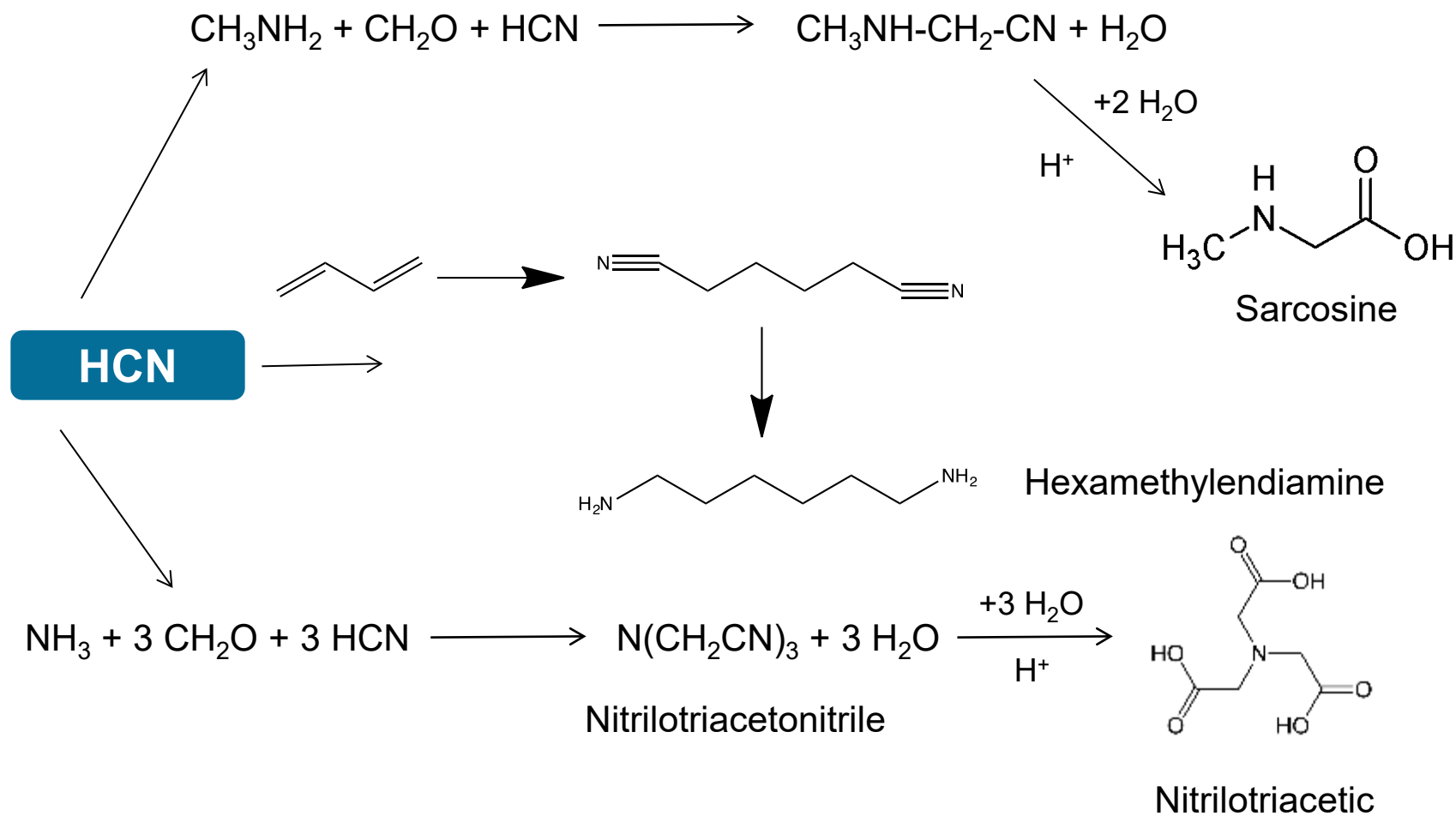


SOHIO process for the ammoxidation of propene

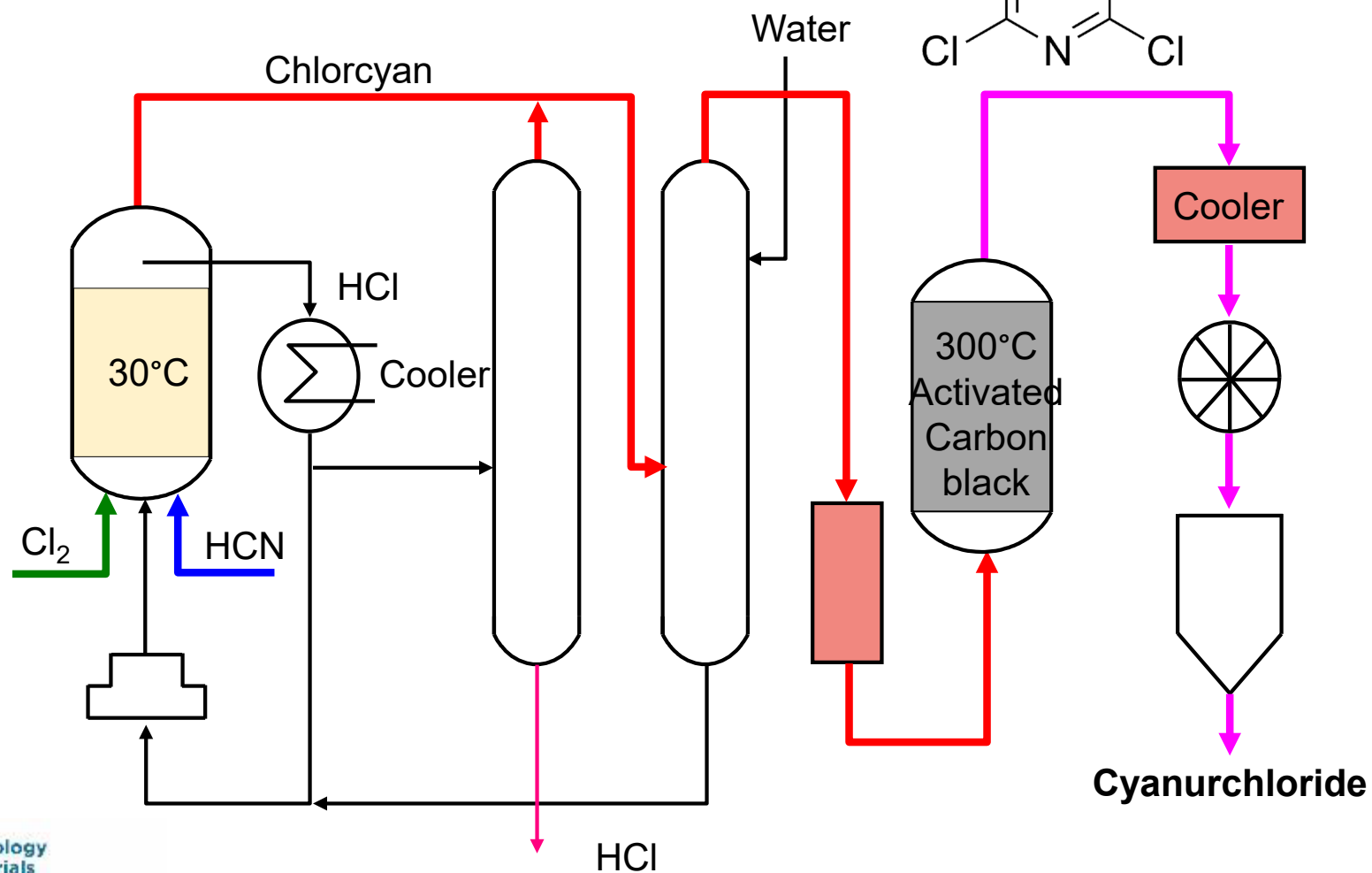
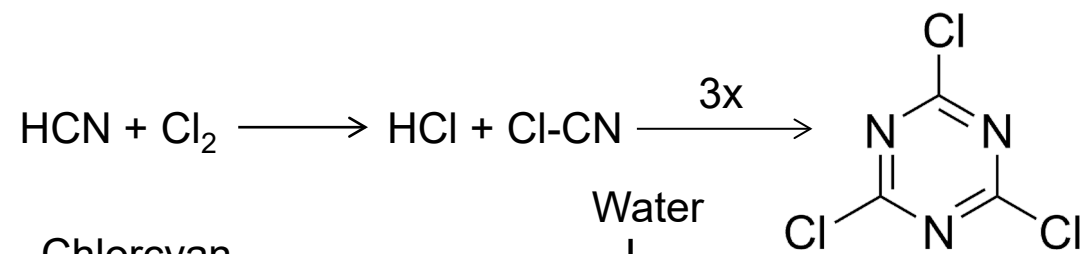
Synthesis with HCN



Synthesis with HCN (ctd)



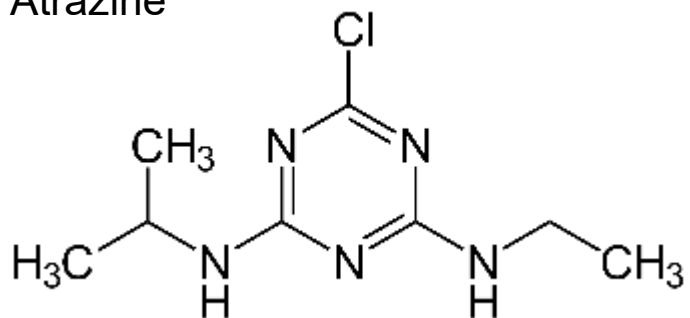
Synthesis of Cyanurichloride



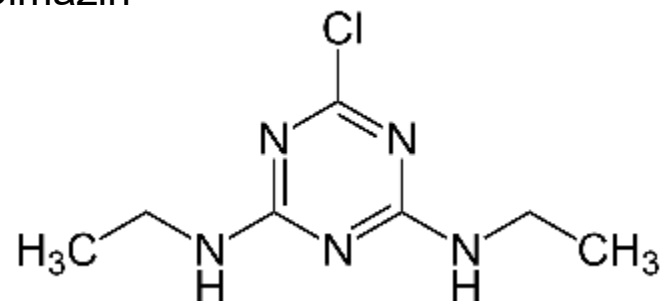
Cyanurichloride

- About 80% of the production goes into 1,3,5 Triazine herbicides

Atrazine



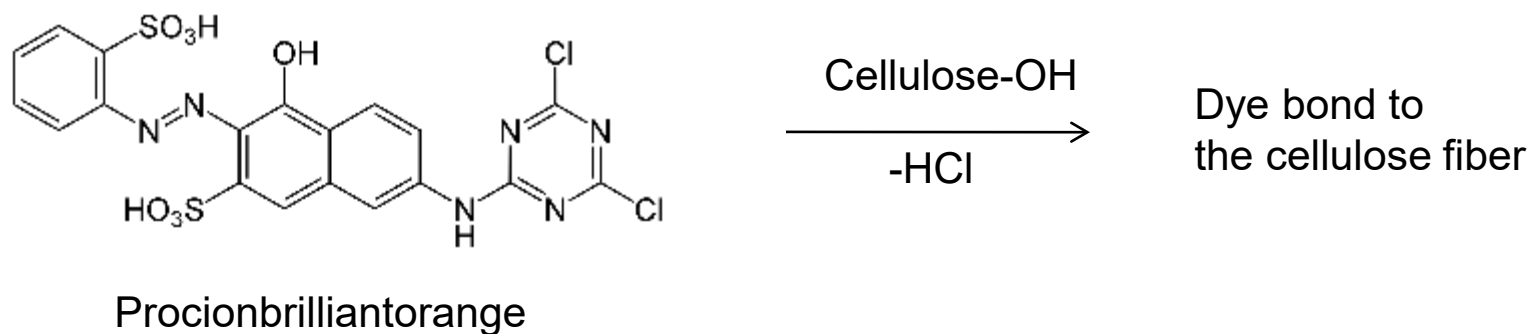
Simazin



- Roughly 10% are used for brighteners
- Pesticides, UV stabilizers, cross-linking agent, reactive dyes

Cyanurchloride

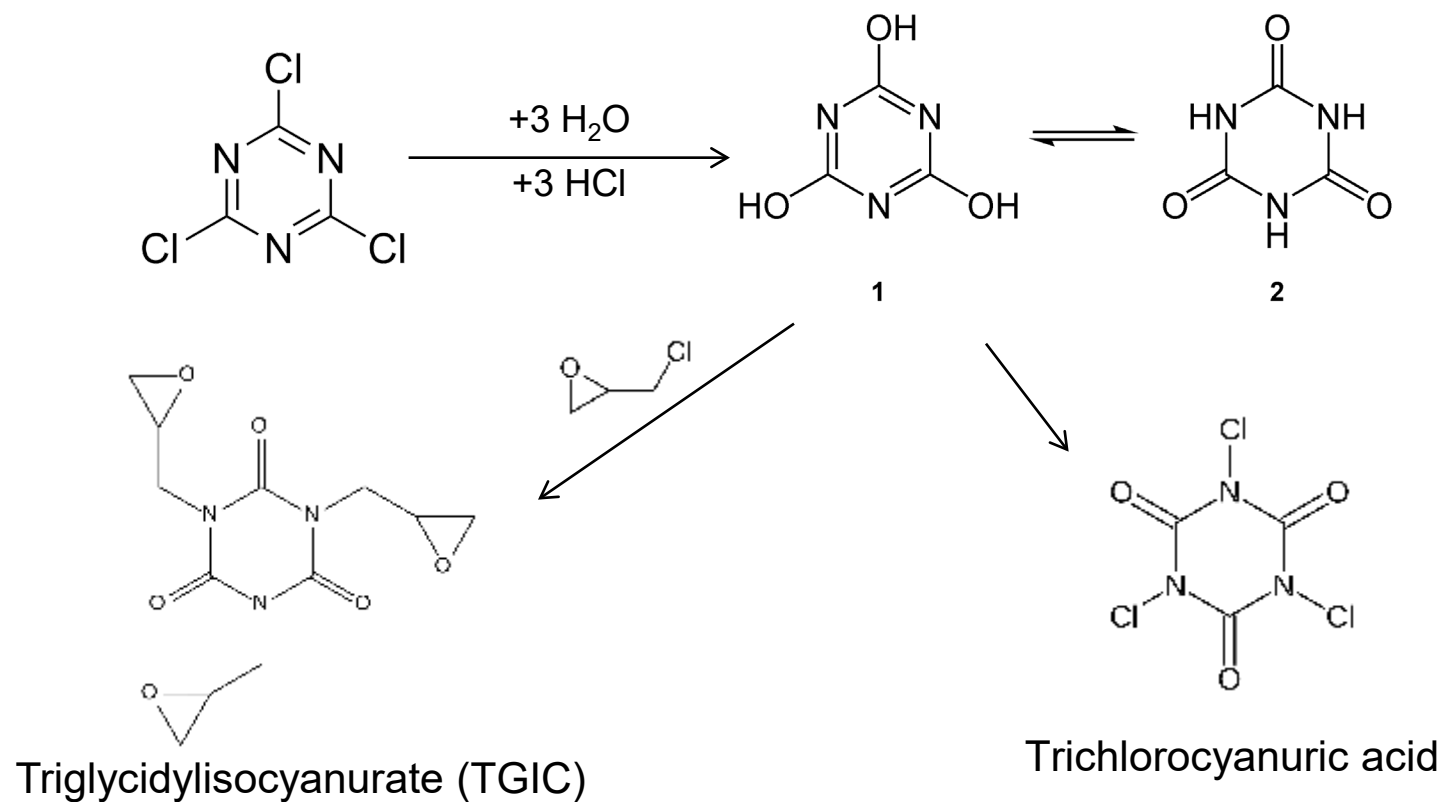
- Reactive dyes: dye is chemically bond to the textile fibers



- Of course reaction with NH₃ to **Melamine** (but this will be a separate lecture)

Cyanuric acid

- Hydrolysis of melamine
- Hydrolysis of cyanurichloride



Basic products of industrial organic synthesis

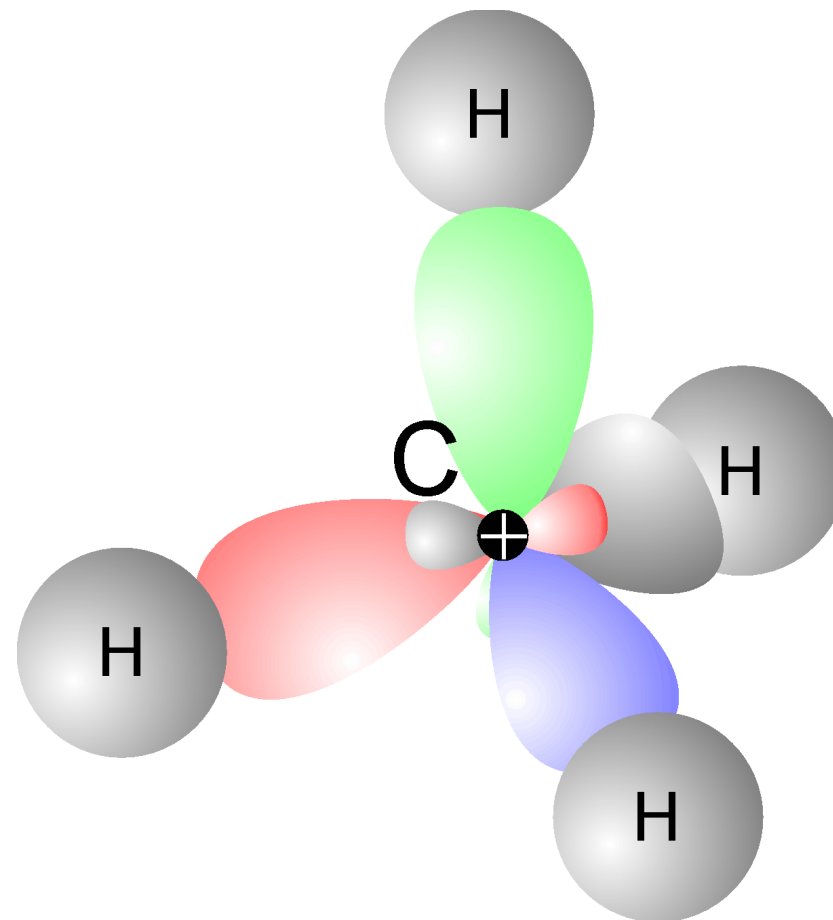
- Syngas

C1 chemistry:

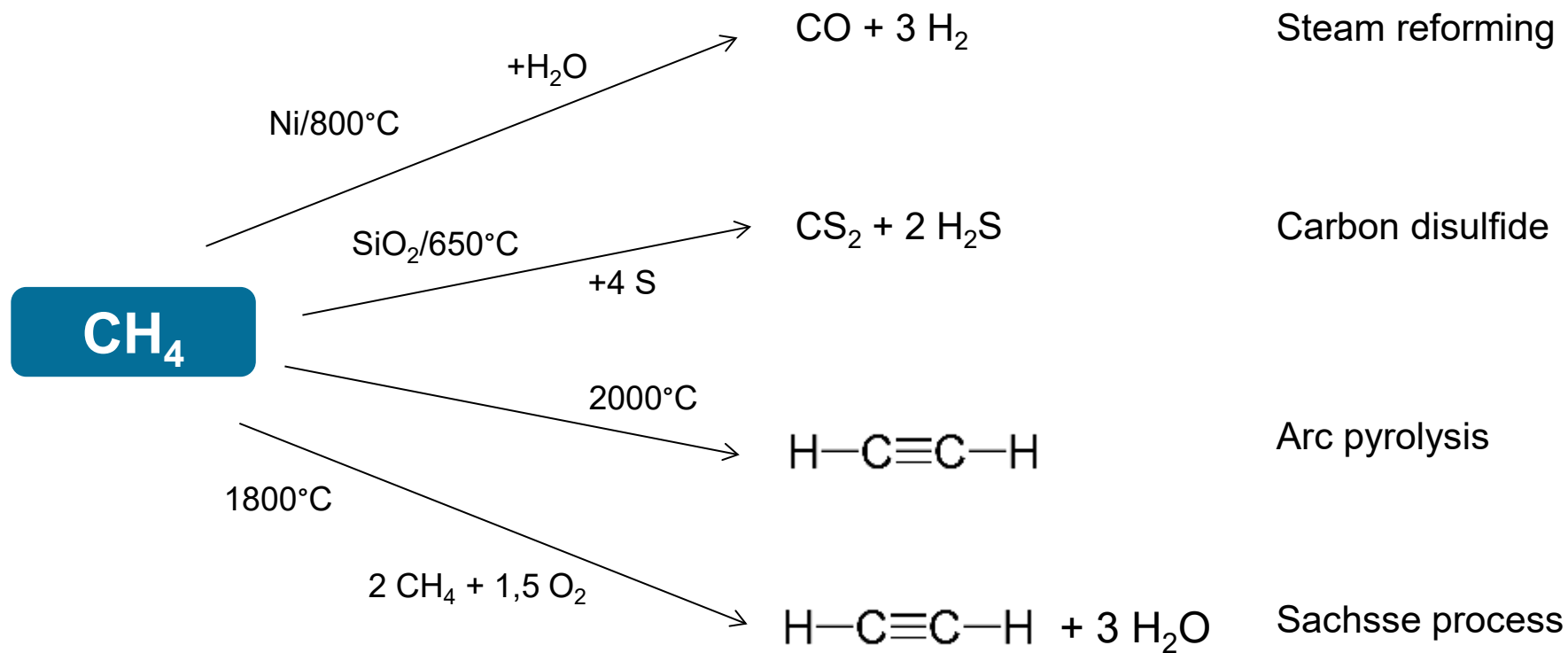
- Methanol
- Formaldehyde
- Formic acid
- Hydrocyanic acid
- **Methane**
- Halogen derivatives of methane
- Reactions with CO and CO₂

Methane

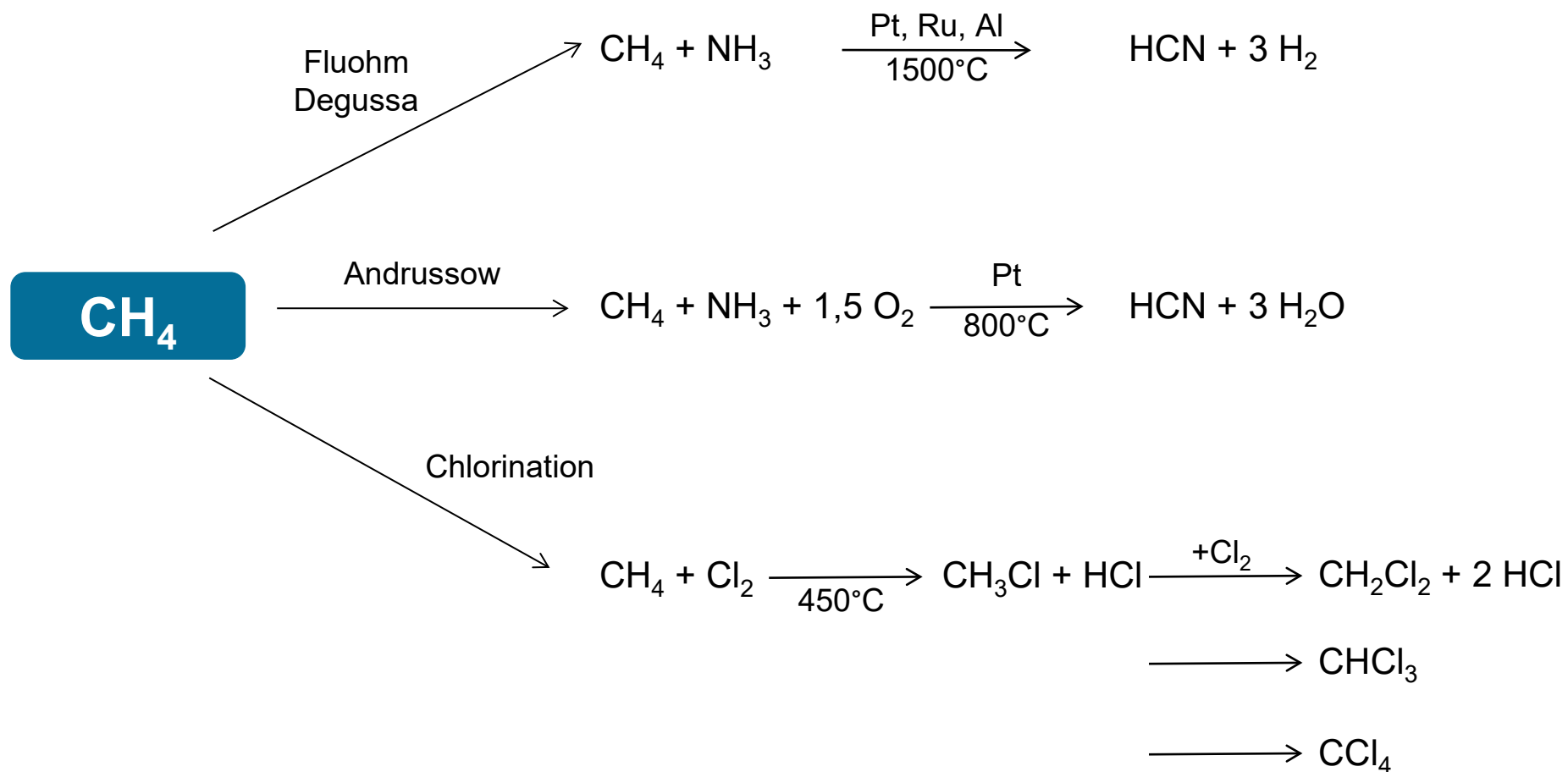
- Molar mass: 16,04 g/mol
- ρ_g : 0,72 g/dm³
- ρ_l : 0,42 g/cm³
- T_m: -182°C
- T_b: -162°C
- Green house gas



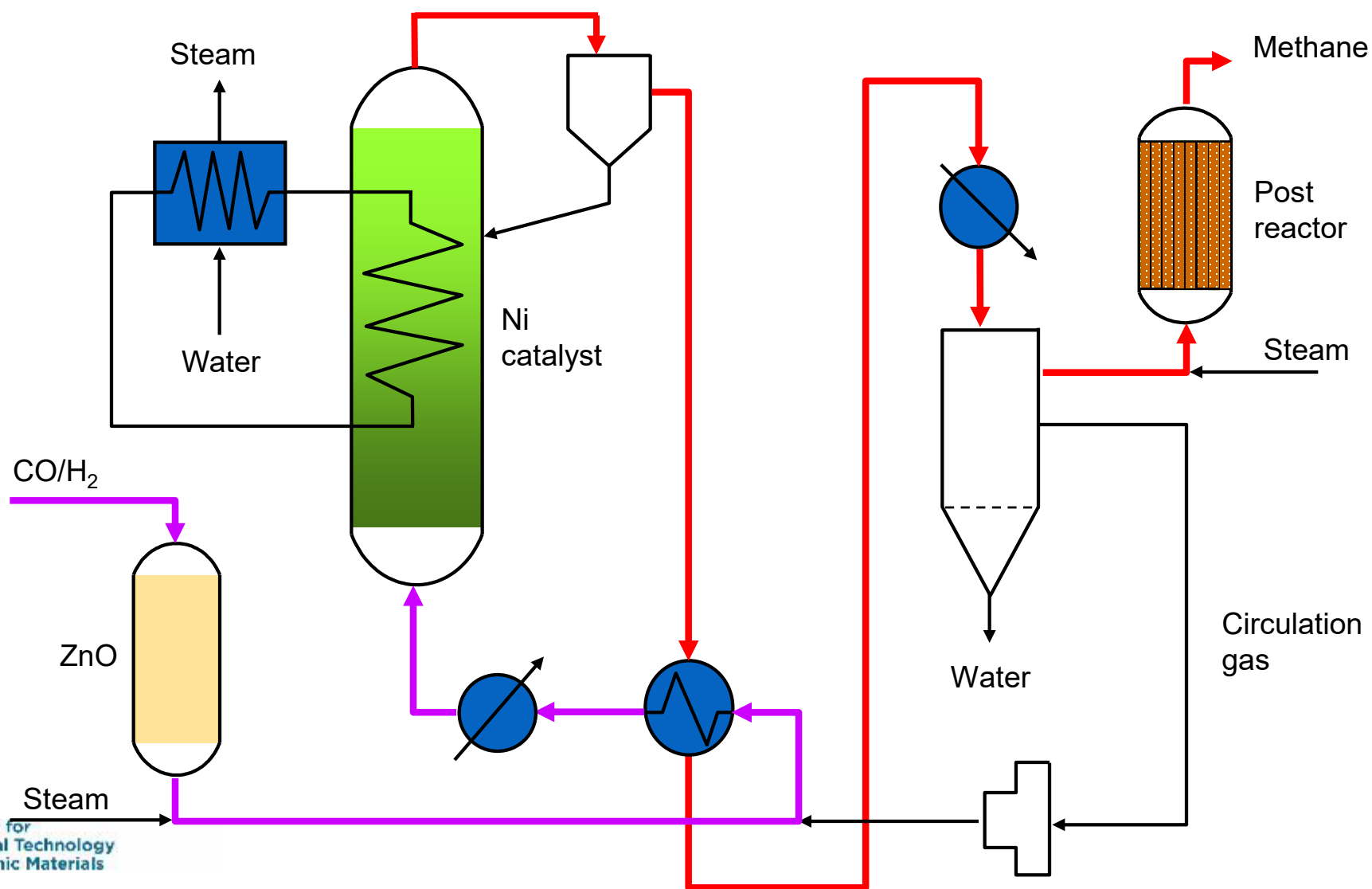
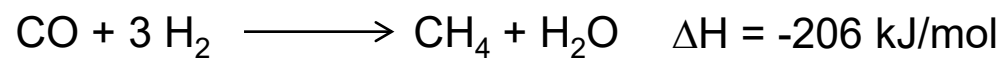
Chemistry with CH₄



Chemistry with CH₄ (ctd)



CH₄ from syngas (SNG)



Basic products of industrial organic synthesis

- Syngas

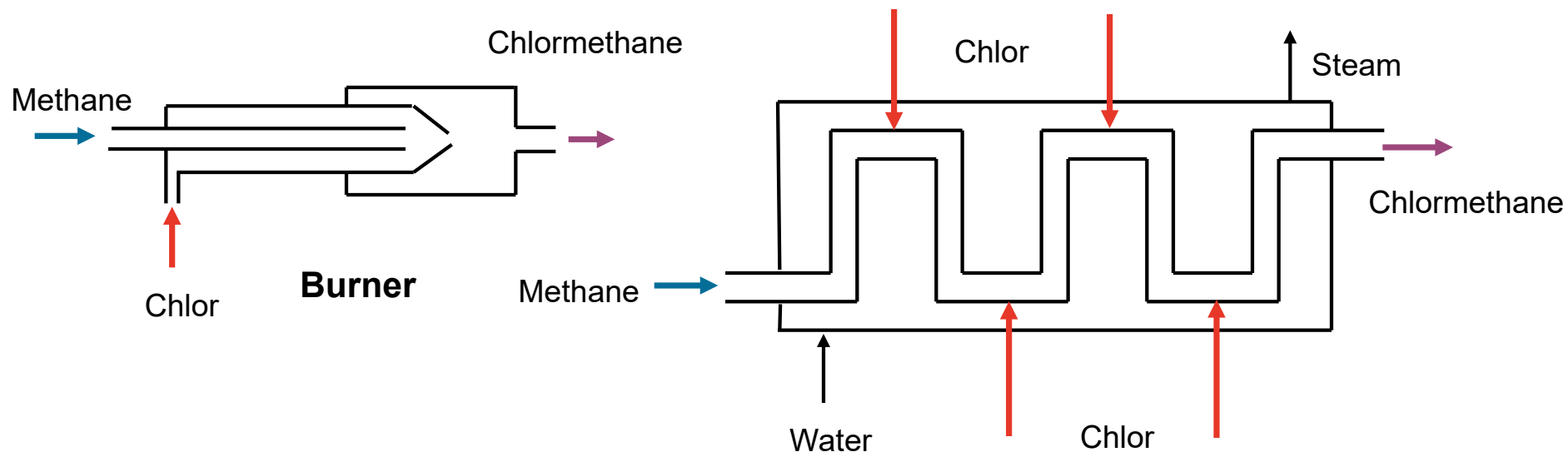
C1 chemistry:

- Methanol
- Formaldehyde
- Formic acid
- Hydrocyanic acid
- Methane
- **Halogen derivatives of methane**
- Reactions with CO and CO₂

Halogen derivatives

- Cl and F have been the technically most important halogens used in the H substitution in CH₄
- CFC chlorofluorocarbons (CFKW)
- Main applications: solvents, washing, cooling agents, extinguishing agent, foams
- Halon ban act from 1991
- Since 1995 no production of CFC in D
- Reducing green house gas emissions, saving ozone layer

Chlorination of CH₄

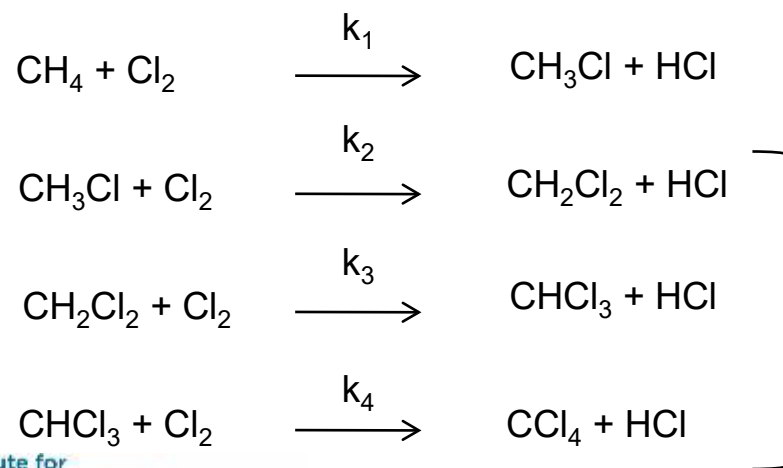


Pipe reactor with spitted dosing

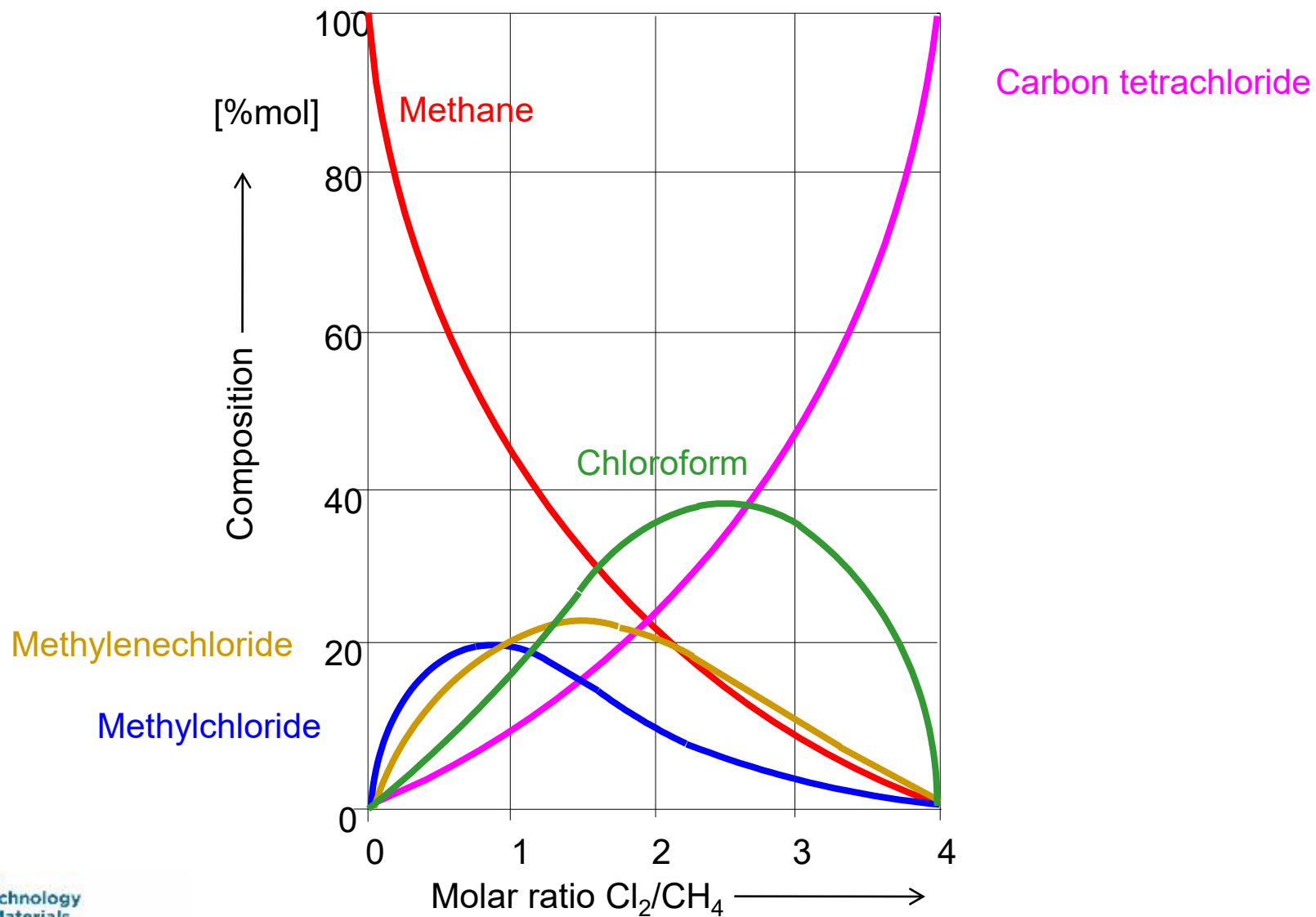
$$\Delta H = -105 \text{ kJ/mol}$$

$$k_1 : k_2 : k_3 : k_4 = 1 : 2,22 : 1,08 : 0,25$$

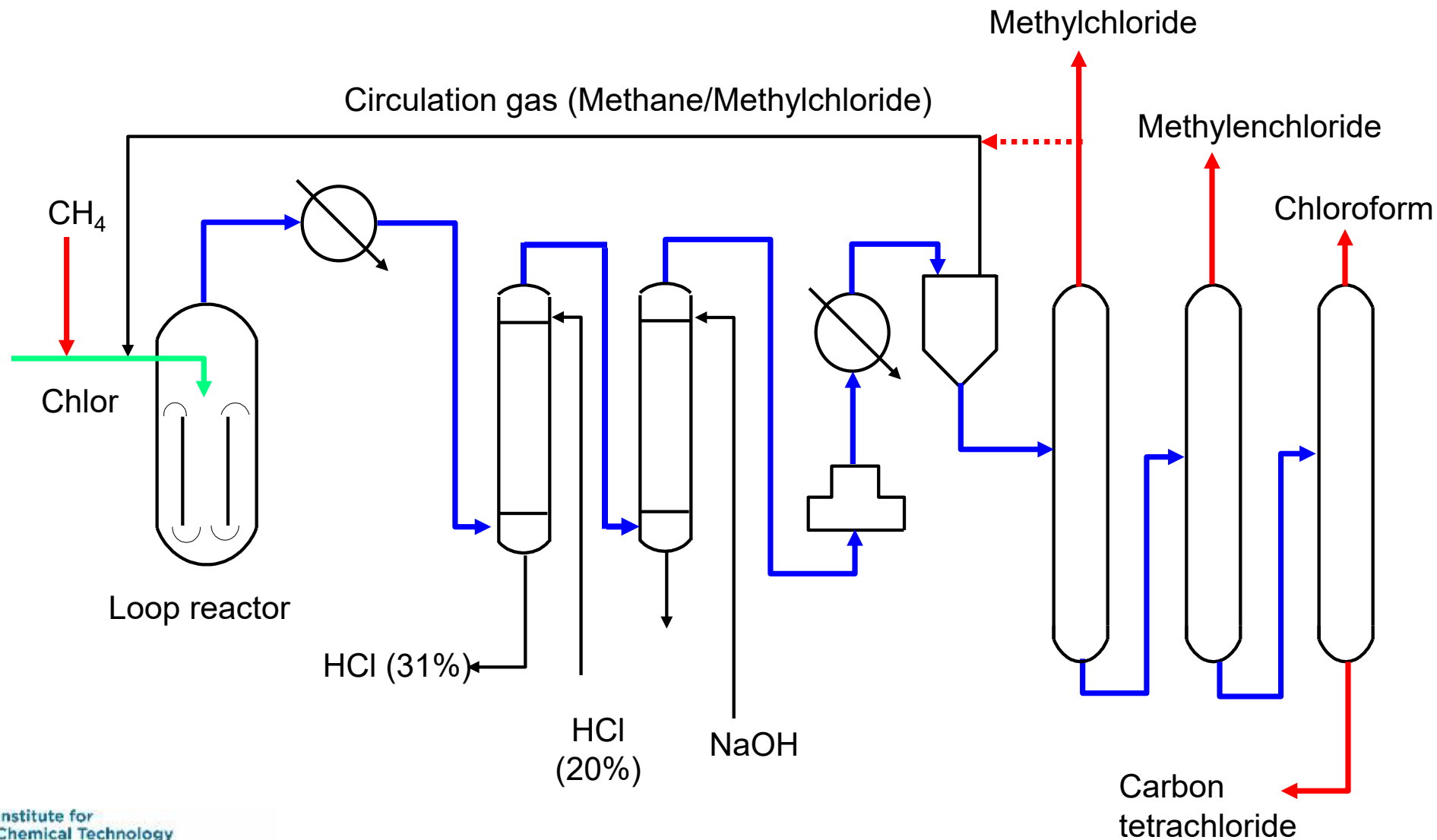
$$\Delta H = -96 \text{ kJ/mol}$$



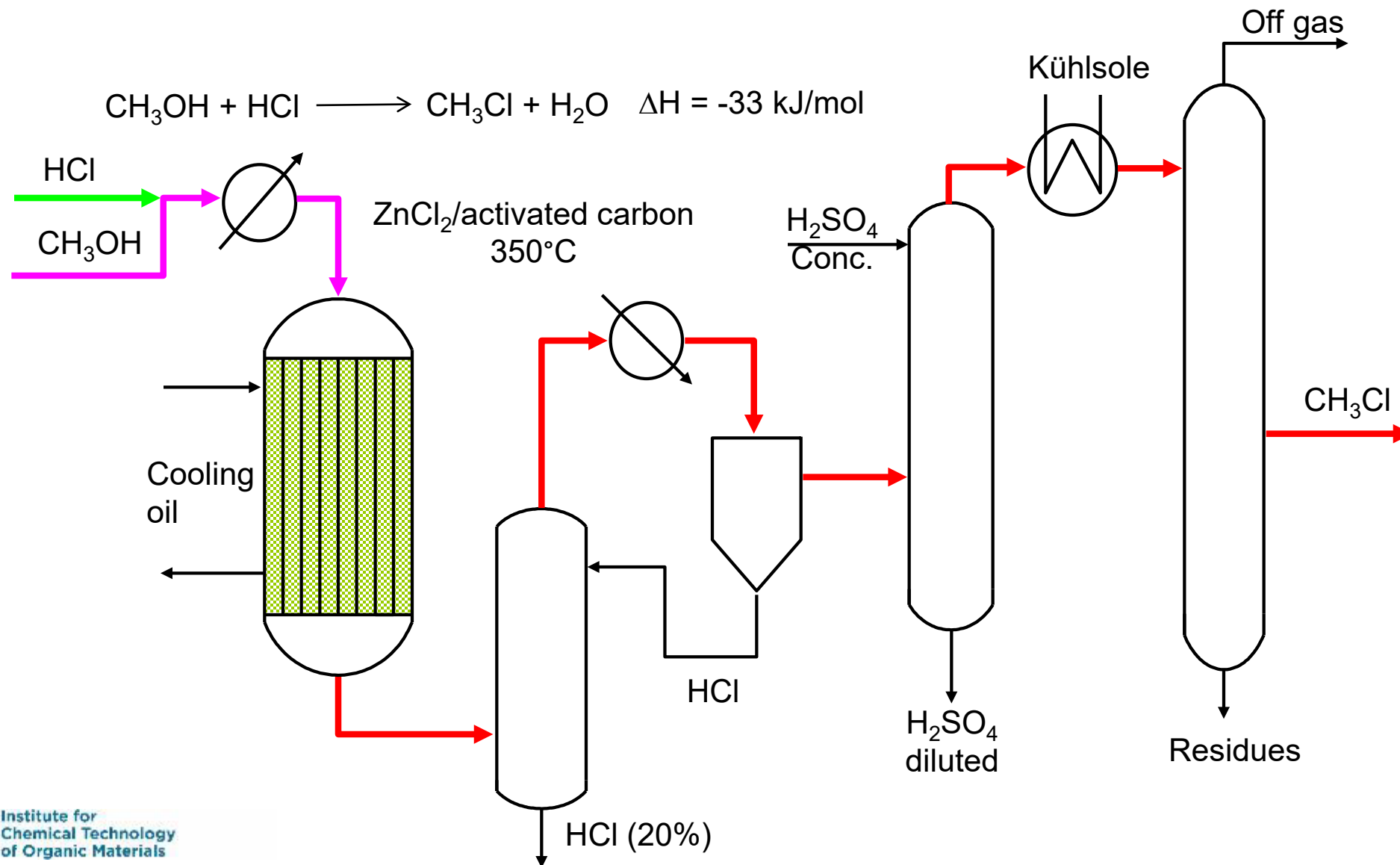
Product split in CH₄ chlorination



Chlorination of CH₄ (Hoechst)



Hydrochlorination of methanol

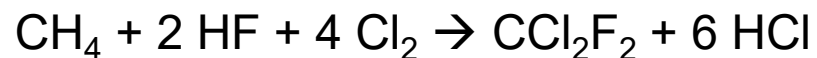
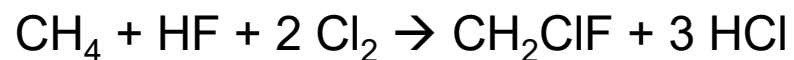


Rochow synthesis

- $2 \text{CH}_3\text{Cl} + \text{Si} \rightarrow (\text{CH}_3)_2\text{SiCl}_2$
- Co catalyst, fluidized bed reactor, 300°C
- Dimethyldichlorosilane hydrolyzes to linear / cyclic silicones

Fluor compounds of methane

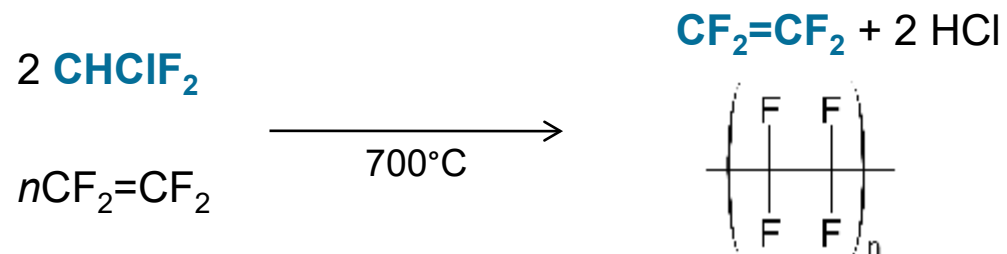
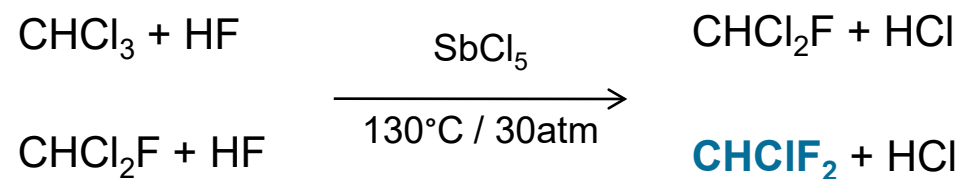
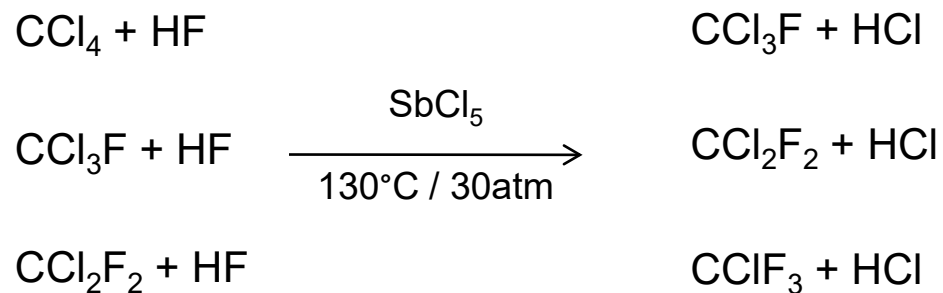
- No direct fluorination of methane because of strong exothermic reaction
- Chloro-fluorination at about 300°C possible



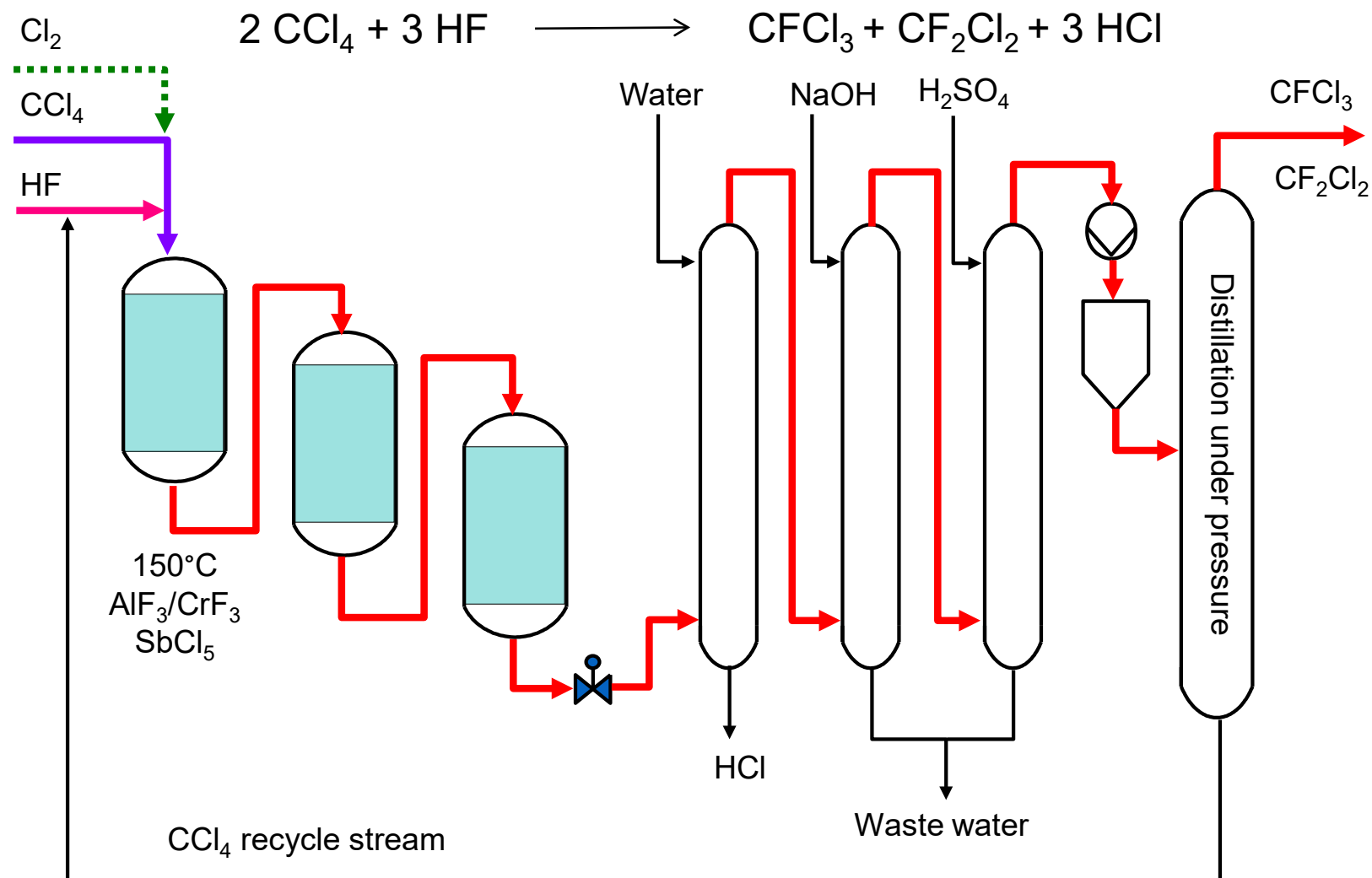
- Fluorination of Cl compounds with SbF_5 as catalyst



Chemistry of CH₄ - chlorination products



Fluorodichloro- and difluorodichloro methane



Nomenclature

R_{x,y,z}

X: number of C atoms -1

Y: number of H atoms +1

Z: number of F atoms

Rest: Cl

e.g. R22 = CHF₂Cl (R022)

R12 = CF₂Cl₂ (or R012)

R114 = CF₂Cl-CF₂Cl

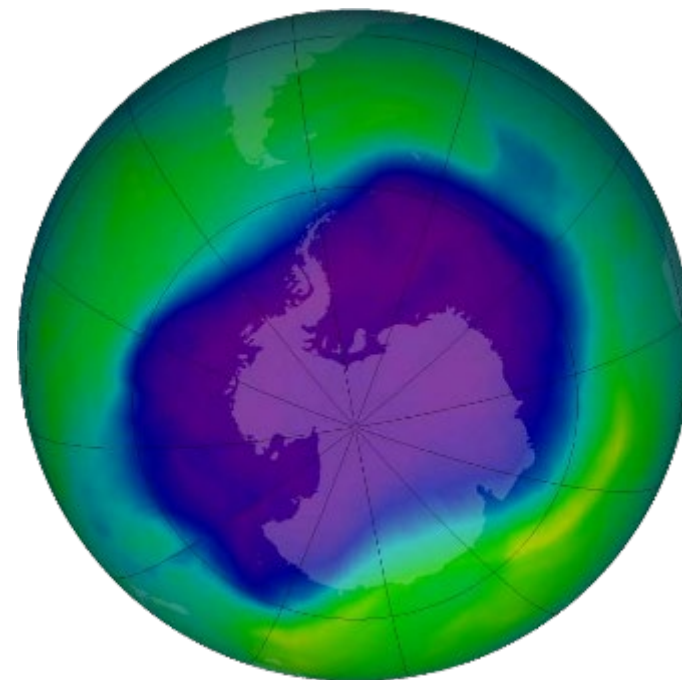
F114B2 = 1,2-Dibromo-tetrafluoroethane

Compounds with halogen functionality

Compound	Classification	T _B [°C]	Use
Dichlorofluoromethane	R21	+8,9	Blowing agent
Chlorodifluoromethane	R22	-40,8	Cooling agent
Trifluoromethane	R23	-82,1	Cooling agent
Trichlorofluoromethane	R11	+23,7	Blowing/cleaning agent
Dichlorodifluoromethane	R12	-29,8	Cooling agent
Chlorotrifluoromethane	R13	-81,1	
Tetrafluoromethane	R14	-128,0	
Bromotrifluoromethane	R13B1	-57,8	Fire extinguisher (Halon 1302)
1,2-Dichloro-tetrafluoroethane	R114	+4,0	Cooling/blowing agent
Tetrafluoroethylene	-	-75,6	Teflon

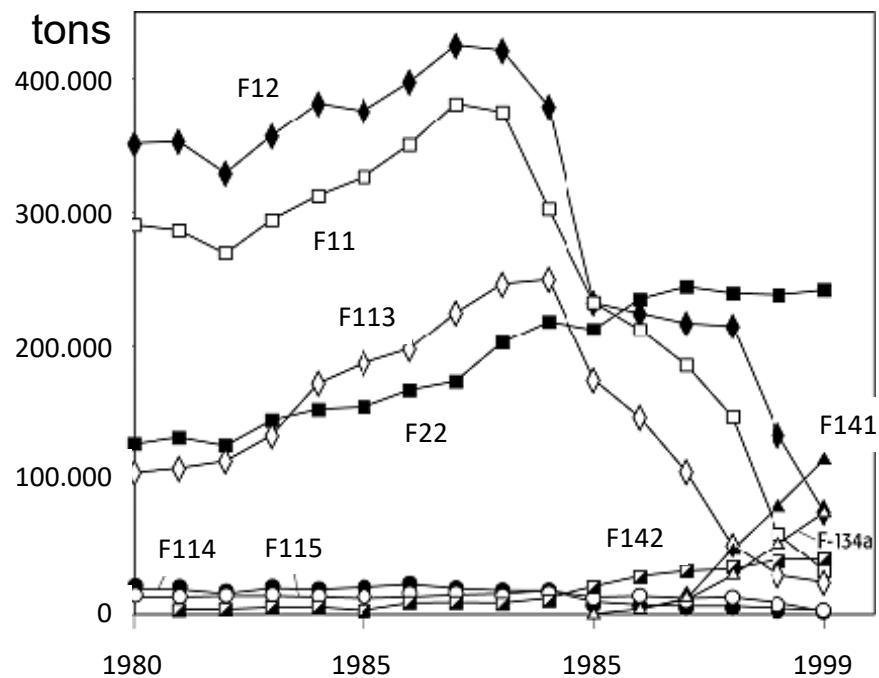
Ozone depletion

- The term ozone hole refers to a severe thinning of the ozone layer
- First observed in 1985 at the South Pole over Antarctica, and in early 2020 for the first time over the Arctic (North Pole)
- The causes of ozone destruction are mainly radical chlorine atoms from chlorinated organic compounds, (chlorofluorocarbons or CFCs)
- The thinned ozone layer allows more of the UV-B portion of solar radiation to pass through to the ground: Ultraviolet radiation can be carcinogenic to living organisms
- Climate change is causing an accelerated depletion of the ozone layer over the Arctic



Largest extent of Antarctic ozone hole 24.09.2006

FCKW (CFC) production rates



Classification	Compound
F11	CCl_3F
F12	CCl_2F_2
F22	CHClF_2
F113	CCl_3CF_3
F114	CCl_2FCF_3
F115	CClF_2CF_3
F141	$\text{CH}_3\text{CCl}_2\text{F}$
F142	CH_3CClF_2
F134	CH_2FCF_3

01.01.1989: Montreal Protocol

HF resistance

- Passivation of standard steel and stainless steel through formation of a pore free fluorine layer
- Standard steels are resistant against dry F up to 200 °C
Si < 0,01% → gaseous SiF₄
- Steel / stainless steel is not resistant against reducing HF
- Coatings with rubber, PP, PE or Teflon

Basic products of industrial organic synthesis

- Syngas

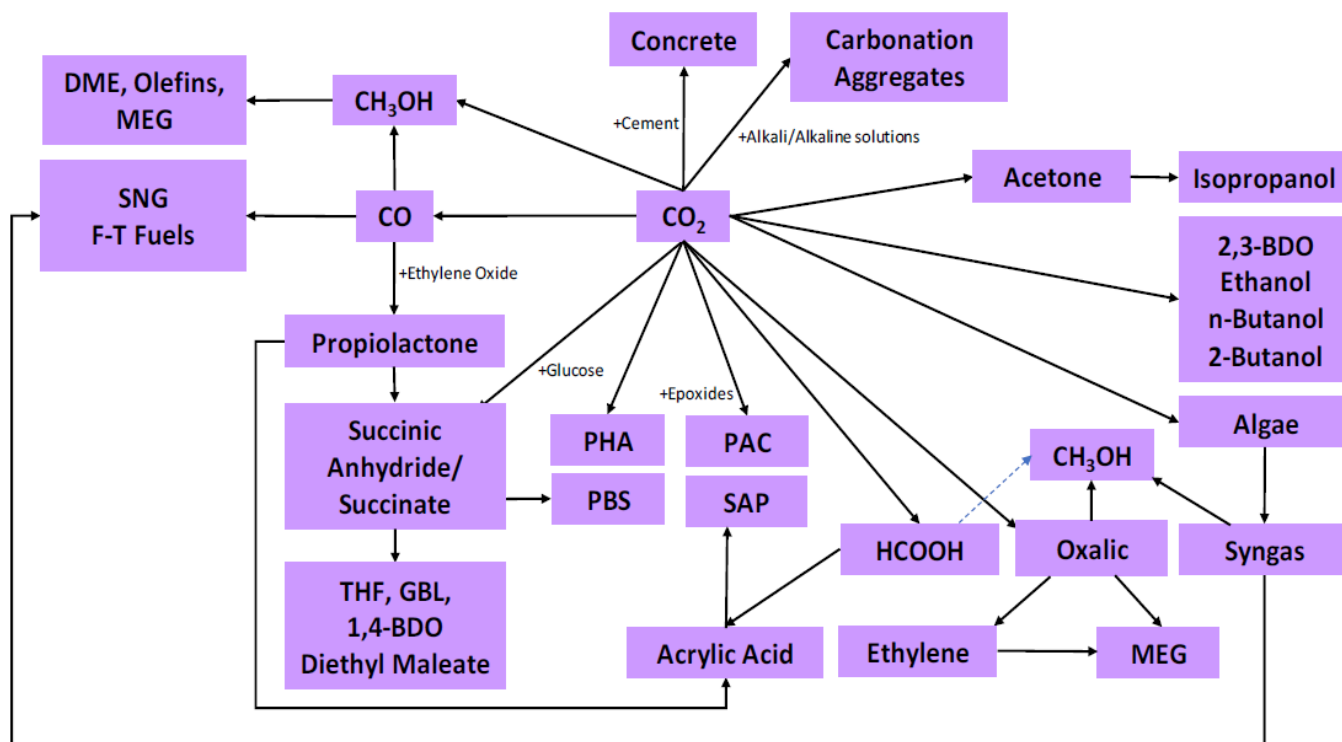
C1 chemistry:

- Methanol
- Formaldehyde
- Formic acid
- Hydrocyanic acid
- Methane
- Halogen derivatives of methane
- **Reactions with CO and CO₂**

CO₂ as chemical feedstock

- CO₂ is a chemical feedstock which due to its abundant availability allows to partially replace fossil feedstocks in chemical production
- Thermodynamically it is the end product of hydrocarbon combustion
- Very stable molecule with a standard Gibbs free energy of formation of -394,4 kJ/mol
- Conversion reactions of CO₂ are endothermic, require substantial amounts of energy of co-reactants with high Gibbs free energy (CH₄ or H₂)
- In addition many reactions have a high kinetic barrier → need efficient catalysts

CO₂ applications



Ammonium carbamate:

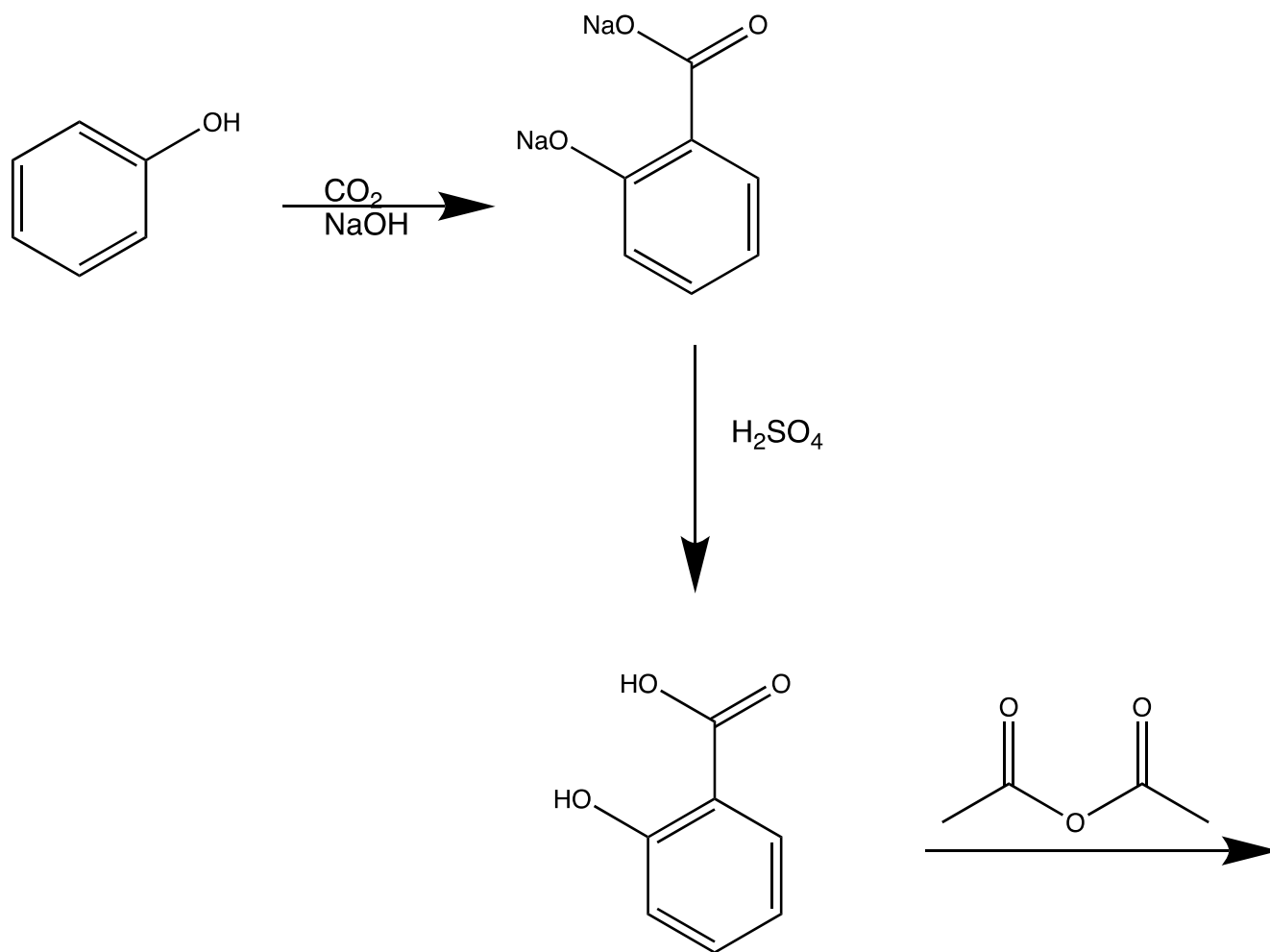


Urea:



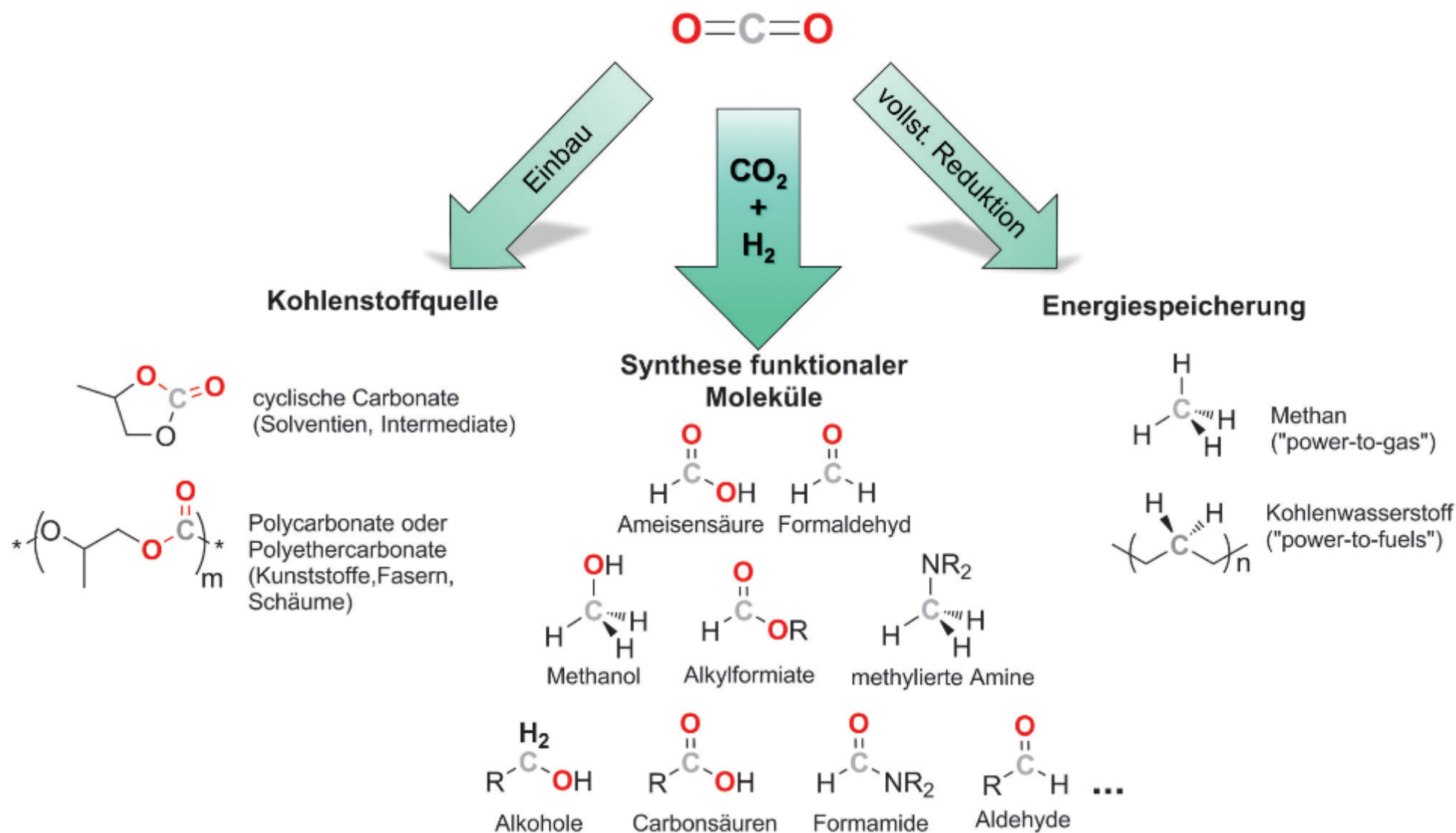
Source: The Catalyst Group Resources (TCGR), CO₂ Start-ups for Venture Capital; 2017; CO₂CC Program, Author: Dr M.K. Lynch, Enabled Future Limited
ed Future Limited

CO₂ application – salicylic acid



Roughly 41.000 t/a (2023)

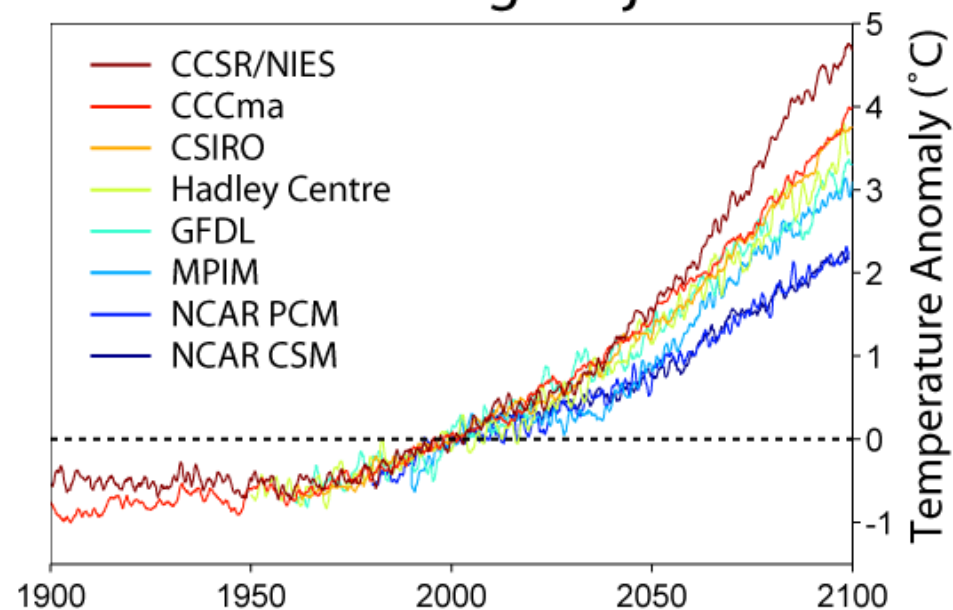
CO₂ as a building block



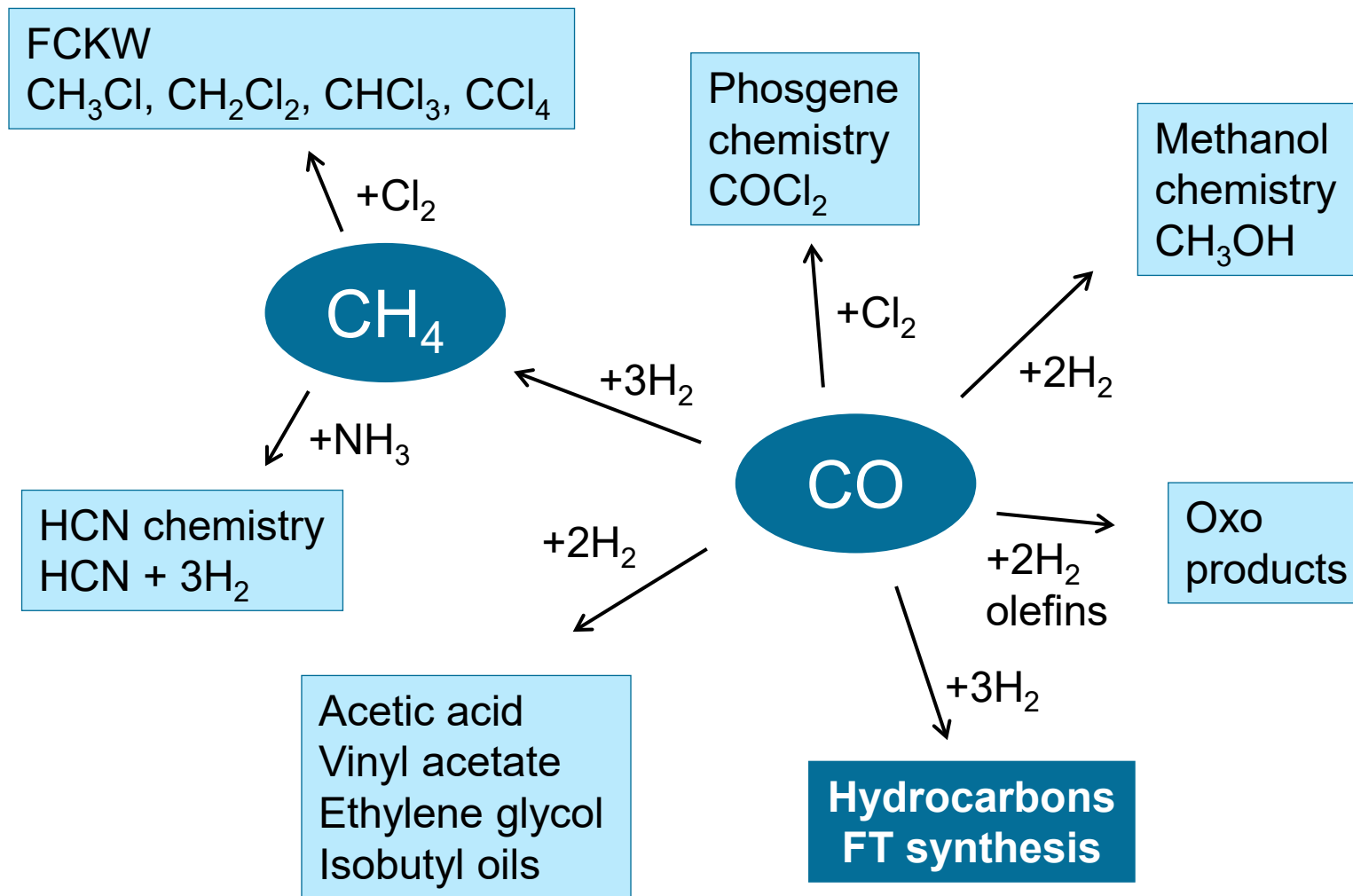
Green house gases

- CO_2 , CH_4 , N_2O , H_2O , O_3
- Absorb IR radiation from earth and by that lead to a warming
- Close to the surface
 - Average of $+15^\circ\text{C}$ (288 K)
 - Compared to -15°C (255 K) w/o natural green house gas effect

Global Warming Projections



C1 chemistry / building blocks



Fischer Tropsch synthesis

Two main characteristics of Fischer-Tropsch synthesis (FTS)

- **Unavoidable production of a wide range of hydrocarbon products** (olefins, paraffin's, and oxygenated products)
- **Liberation of a large amount of heat** from the highly exothermic synthesis reactions

FTS history

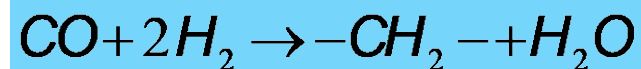
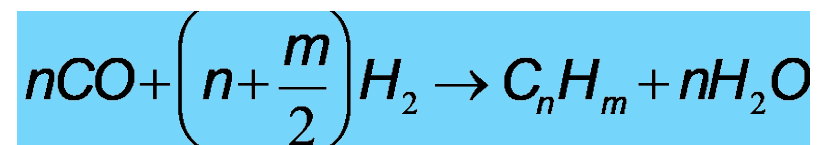
- The synthesis of hydrocarbons from CO hydrogenation over transition metal catalysts was discovered in 1902
Sabatier and Sanderens produced CH_4 from H_2 and CO mixtures passed over Ni, Fe, and Co catalysts
- In 1923, Fischer and Tropsch reported the use of alkalized Fe catalysts to produce liquid hydrocarbons rich in oxygenated compounds (Synthol process)
- The development started about 1925 in Germany
- The experiments took place in Franz Fischer's laboratory at the Kaiser Wilhelm Institute for Coal Research, and developed to an industry with 600.000 tons per year in 1945
- At those times strategic reasons for liquid fuel production from coal exceeded economic aspects

FTS history

- In the last decades the interest in FT synthesis has changed as a result of environmental demands, technological developments and change in fossil energy reserves
- 1955 – 1970: only a marginal interest in FT synthesis
- High oil prices increase the focus at alternative fuels
- Likewise as carbon dioxide concentration concern arises, being related to global warming
- Today the driving forces are environmental concern, but also higher oil price, limited oil reserves and increased focus at stranded gas
- DG Fuels will build the largest FT sustainable aviation fuel (SAF) production facility in the world, with a planned capacity of 13,000 b/d (600,000 t/y)
 - Waste biomass as feedstock
 - Start-up 2028

Fischer Tropsch synthesis

- The FT synthesis is in principle a carbon chain building process, where CH_2 groups are attached to the carbon chain
- Which reactions exactly taking place and how, is a matter of controversy, as it has been the last century since 1930's
- The resulting overall reaction can be presented as follows:



$$\Delta H_{FT}^0 = -165 \text{ kJ/mol}$$

Fischer Tropsch synthesis

- There are also other reactions taking place in the reactor, but the detailed behavior of the reactions is not known and is a theme of controversy
- The reactions reported are:

Reaction	Enthalpy ΔH_{300K} [kJ/mol]
$\text{CO} + 2\text{H}_2 \rightarrow -\text{CH}_2- + \text{H}_2\text{O}$	-165,0
$2\text{CO} + \text{H}_2 \rightarrow -\text{CH}_2- + \text{CO}_2$	-204,7
$\text{CO} + \text{H}_2\text{O} \rightarrow \text{H}_2 + \text{CO}_2$	-39,8
$3\text{CO} + \text{H}_2 \rightarrow -\text{CH}_2- + 2\text{CO}_2$	-244,5
$\text{CO}_2 + 3\text{H}_2 \rightarrow -\text{CH}_2- + 2\text{H}_2\text{O}$	-125,2

- These reactions are highly exothermic, and to avoid an increase in temperature, which results in lighter hydrocarbons, it is important to have sufficient cooling, to secure stable reaction conditions
- The total heat of reaction amounts to ca. 25 % of the heat of combustion of the synthesis gas, and lays thereby a theoretical limit on the maximal efficiency of the FT process

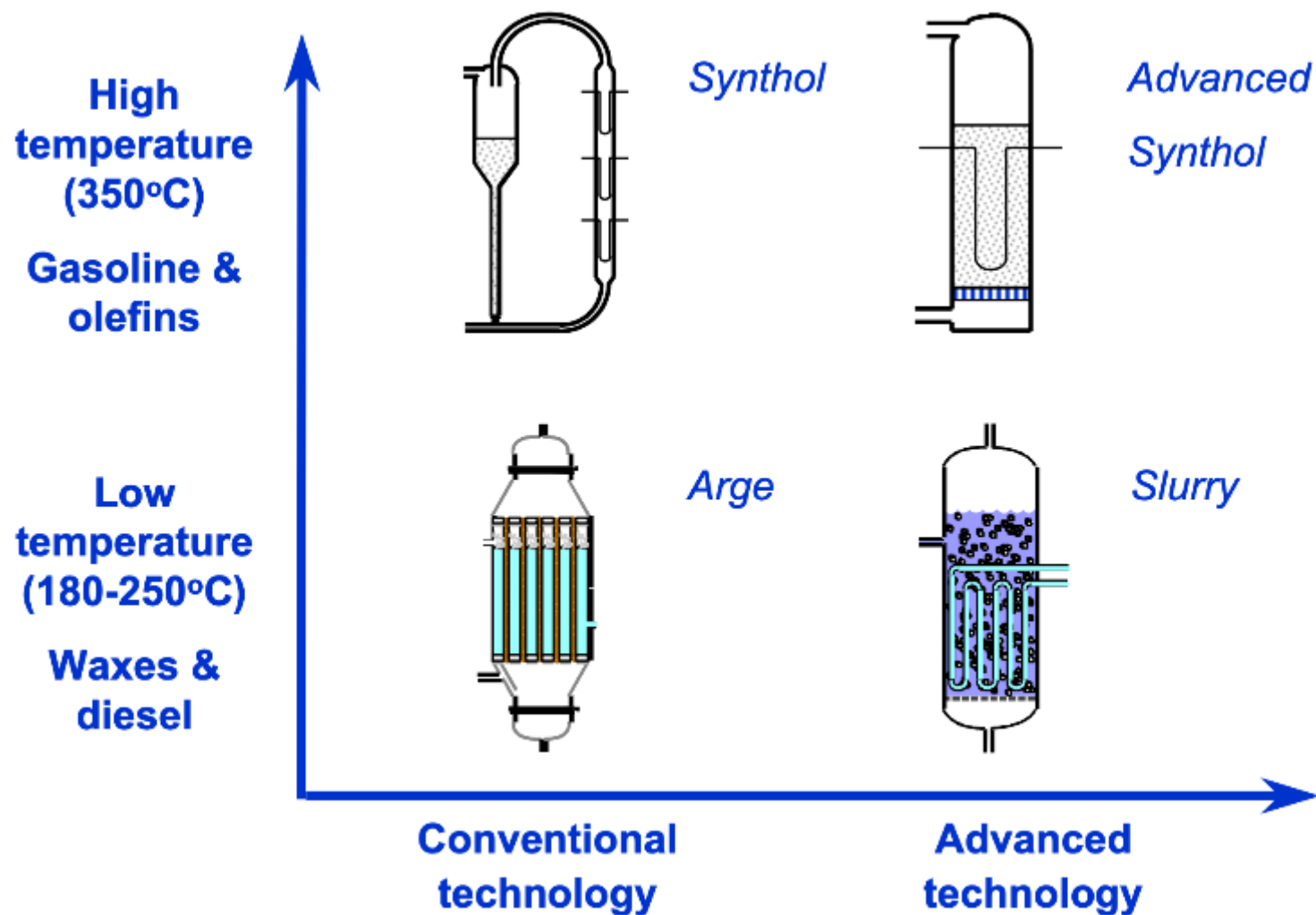
LT/HT FT

- Catalyst: mostly Fe or Co
- There is either a **low or high temperature process (LTFT, HTFT)**
- LTFT: temperatures ranging between 200-240°C, Co catalyst
- HTFT: 300-350°C, Fe catalyst for HTFT
- Iron is cheap, but cobalt has the advantage of higher activity and longer life, though it is on a metal basis 1000 times more expensive than iron catalyst

Commercial FT reactors

- For large-scale commercial FT reactors heat removal and temperature control are the most important design features to obtain optimum product selectivity and long catalyst lifetimes
- Over the years, basically four FT reactor designs have been used commercially
- These are the “multi-tubular fixed bed”, the “slurry” or the “fluidized bed” (with either fixed or circulating bed) reactor

FT reactor types



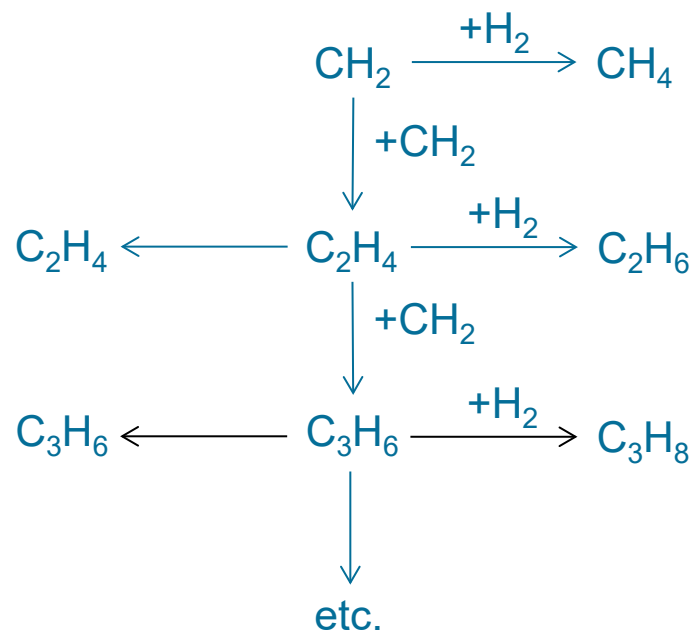
FT products

- For optimal chain growth, there is a need for 2 hydrogen molecules for each CO molecule
- The H_2/CO ratio differs in the range between 1,7:1 and 3:1
- The FT process produces different olefins and paraffins of different length
- The process is basically a chain-building process, where the chain either gains length by adsorbing another CO group, or terminates and leaves the catalyst as either paraffin or olefin

FT products

There are two possibilities:

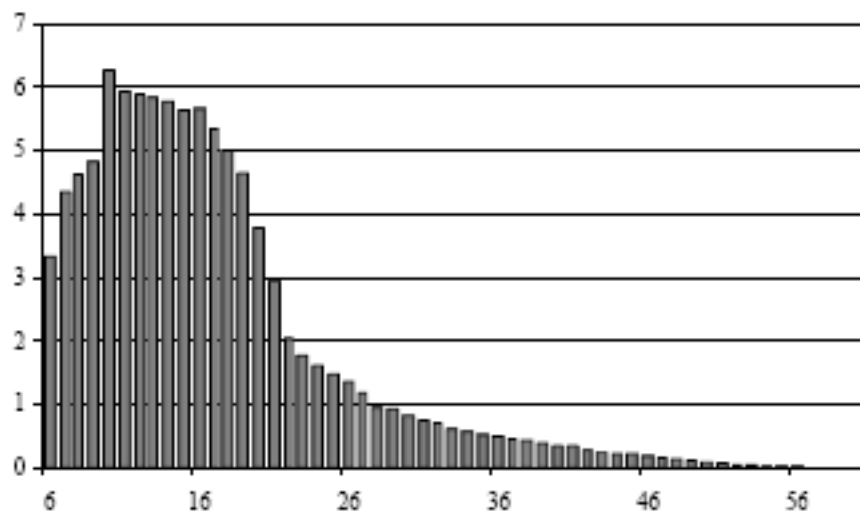
- Termination to a paraffin (right side) or olefin (arrow to the left)
- Or further growth with the absorption of CO and H₂ as CH₂



FT product distribution

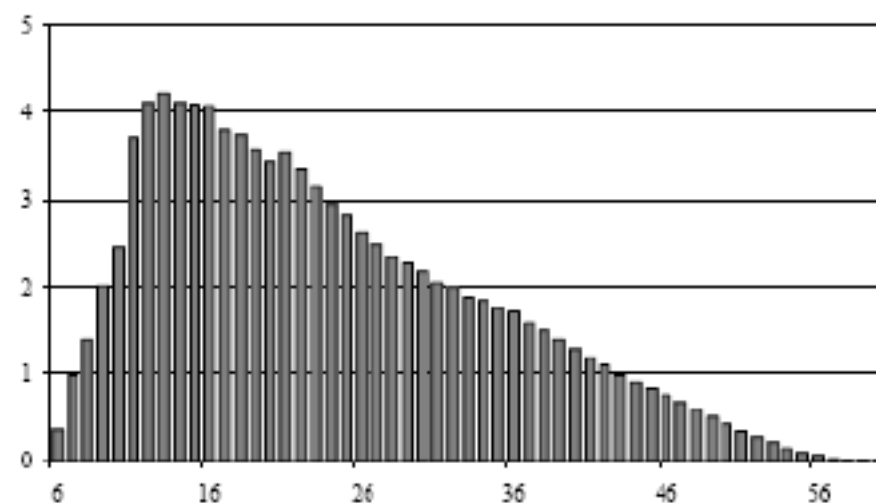
x-axis: chain length

y-axis: percentage on weight basis



Fe catalyst (30 bars and 280°C)

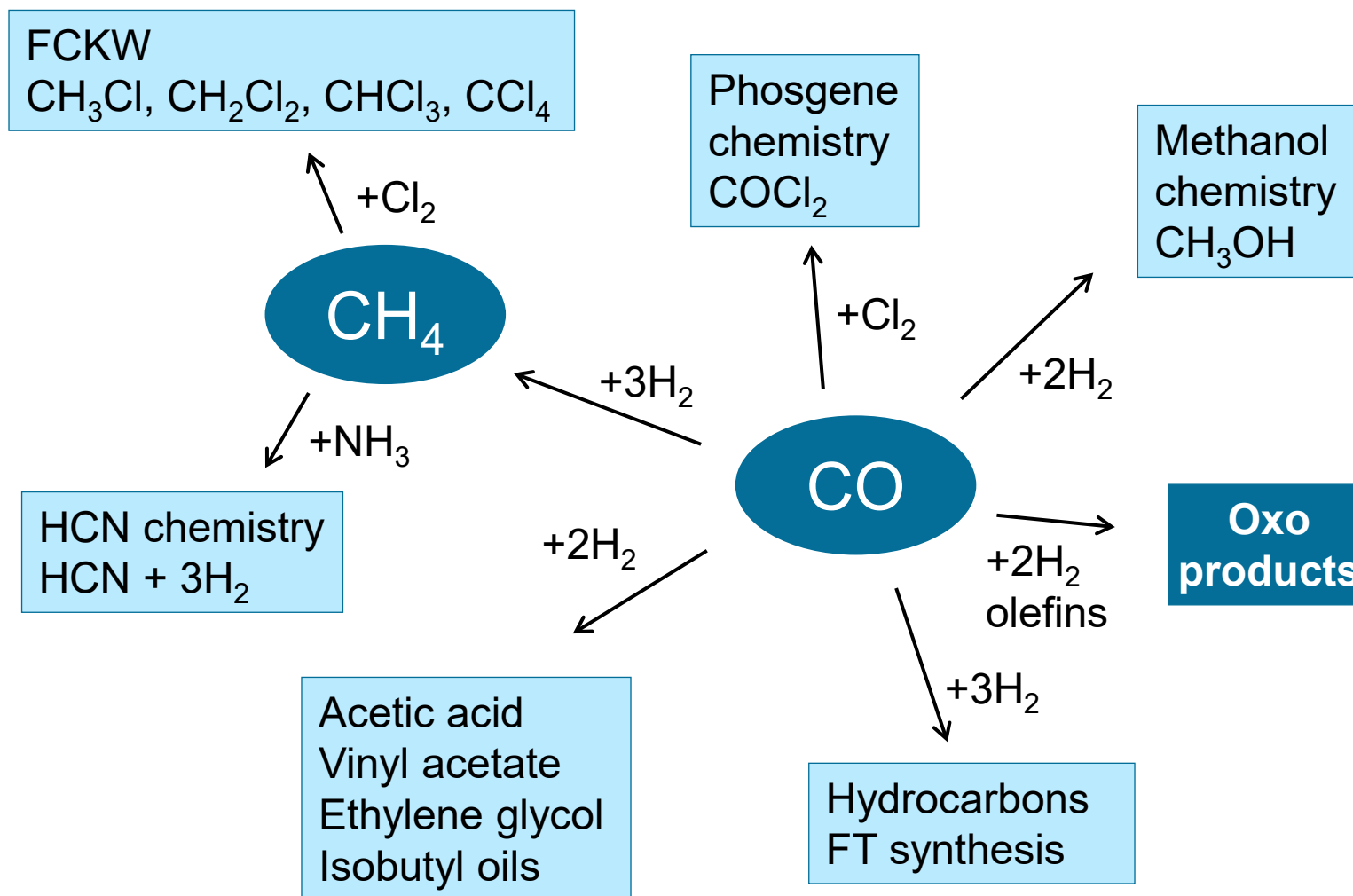
The iron catalyst provides high selectivity in the important interval between C10–C18, which means a high yield of diesel



Co catalyst (30 bars and 240°C)

Provides a higher growth probability, as heavier products are produced. The heavier and longer hydrocarbons can be easily cracked into desired products.

Reactions with CO



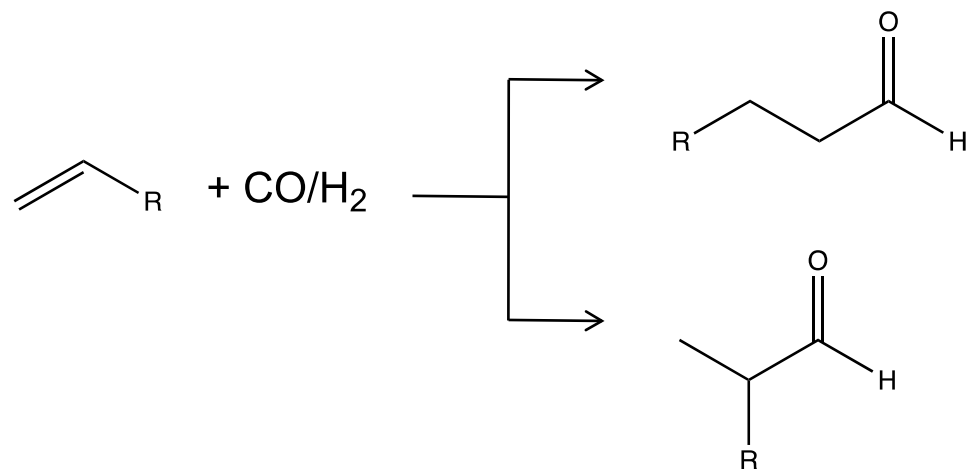
Hydroformylation

- Hydroformylation or Oxo synthesis is a large scale process for the production of aldehydes from olefins, CO and H₂
- Basic principle:
Catalytic addition of H₂ and CO to an olefin with simultaneous increase of chain length by one carbon
- Developed by O. Roelen (Ruhrchemie) in 1938:
Reaction of C₂H₄ with CO and H₂ to propionaldehyde at elevated temp and pressure and Co/Th as catalysts
- Fast up-scaled to a technical process for the production of higher alcohols for washing powders (C₁₂-C₁₄ alcohols)

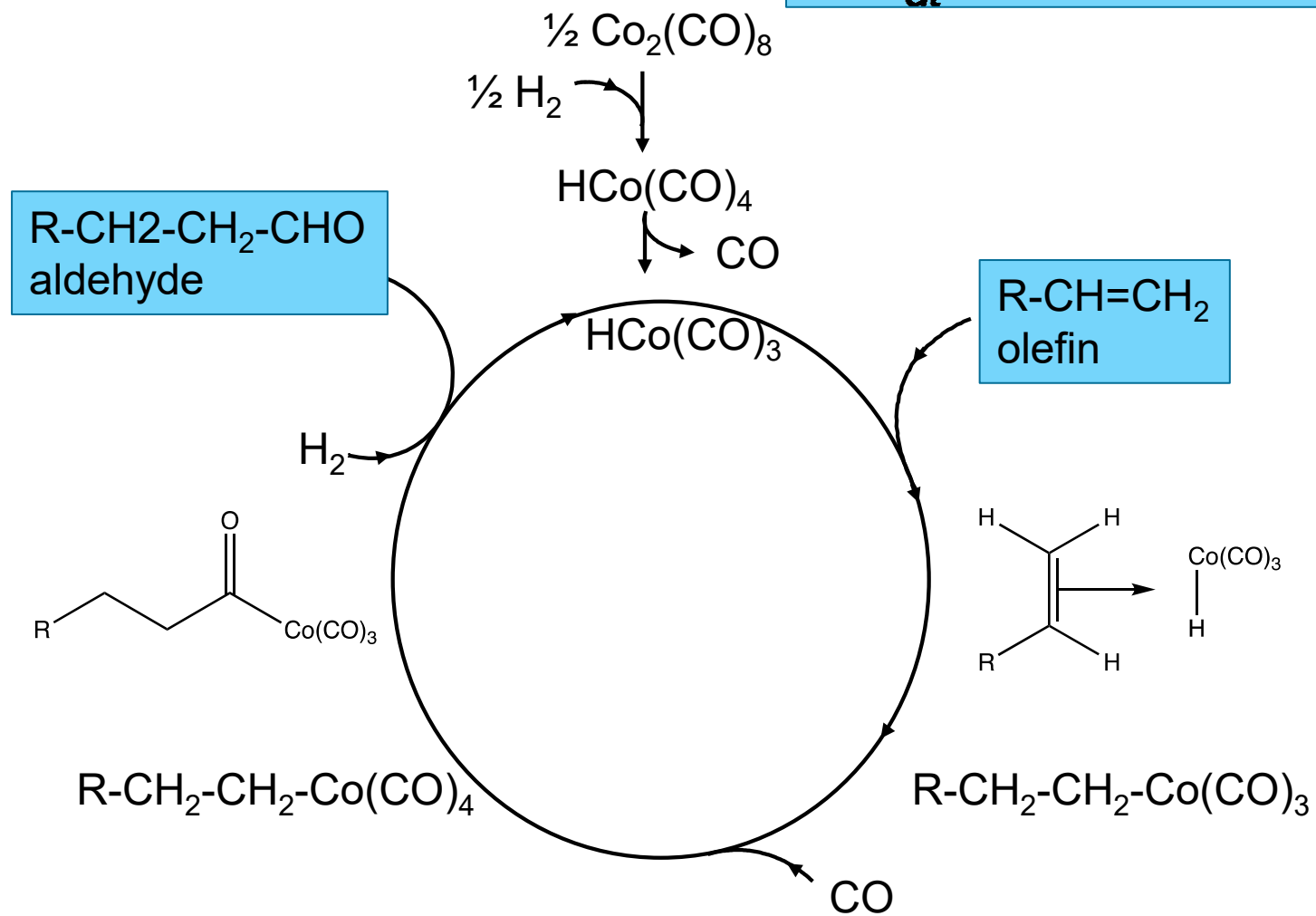


Hydroformylation of olefins

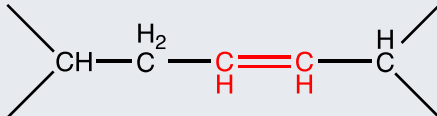
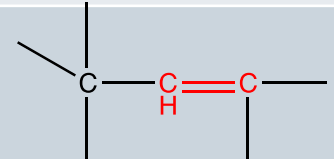
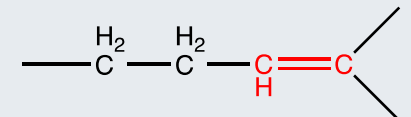
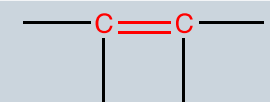
- World capacity of all hydroformylation products has reached 10,8 Mill t in 2002
- Most important educt: propylene
- Products: butanol, iso-butanol, 2-ethyl hexanol



$$\frac{d[\text{aldehyde}]}{dt} = k[\text{olefin}][\text{Co}]p_{\text{H}_2}p_{\text{CO}}^{-1}$$



Olefin structure - reaction speed

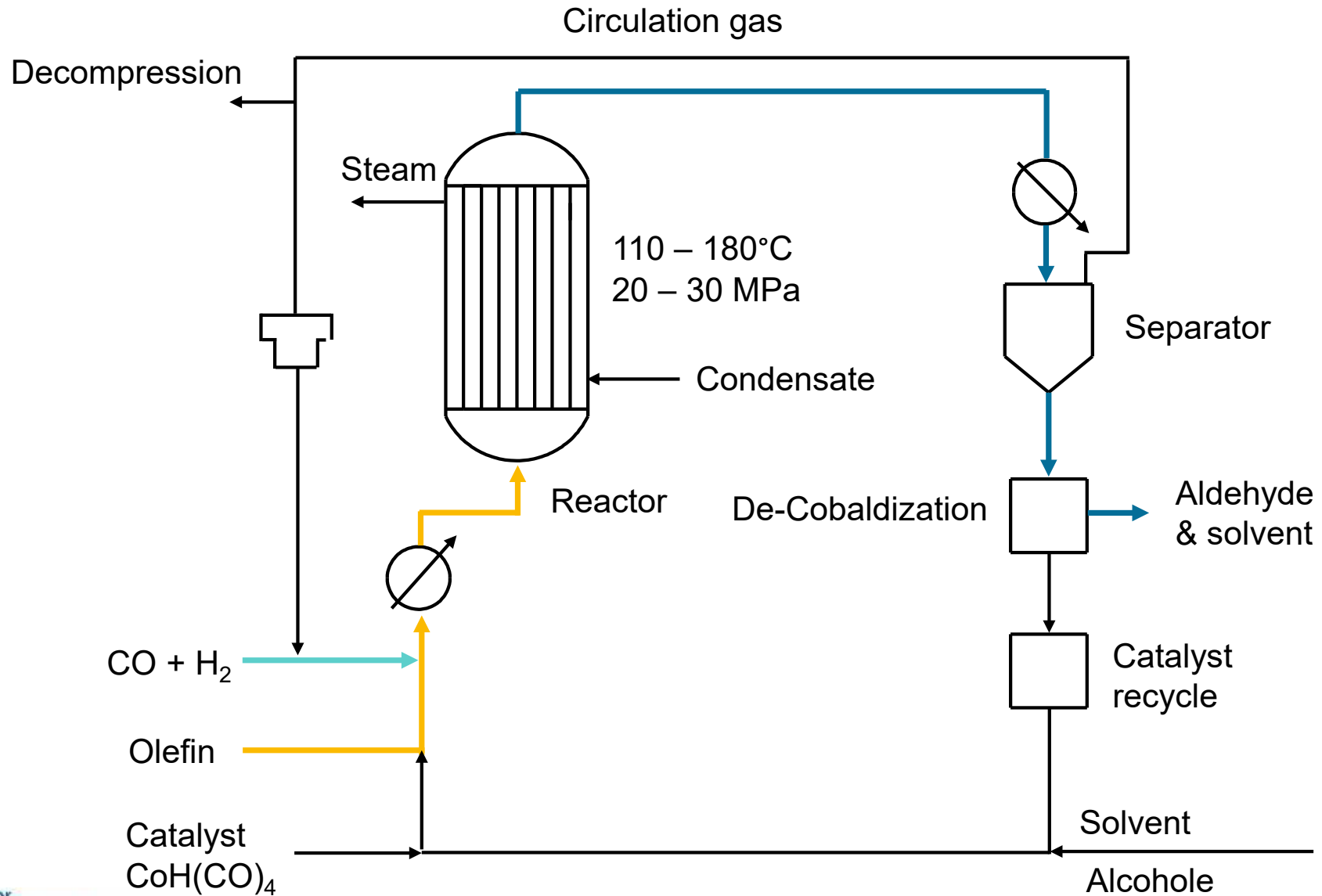
Olefin structure	Reaction speed $k \cdot 10^3 \text{ [min}^{-1}\text{]}$	Relative reaction speed
<chem>C-C-C-C-C=C</chem>	66	1
<chem>C-C-C-C=C-C</chem>	19	0,29
<chem>C-C-C=C-C-C</chem>	19	0,29
	6,2	0,09
	4,9	0,07
	2,3	0,03
	1,4	0,02

Influencing the n/i ratio

The ratio of n/i Aldehydes in the product can be influenced through

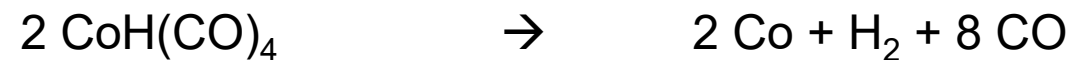
- Variation of ligands at the metal
- Tertiary amines, phosphites or phosphines
(tributyl- or triphenylphosphin) increase n-Aldehydes
- Rh basically reduces the n-Aldehydes but increases the reaction speed and the selectivity
- Temperature and CO partial pressure

Oxo process

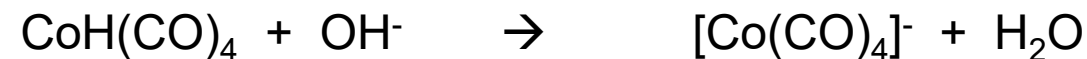


Catalyst recycle

1. Reduction of CO partial pressure



2. Dissociation with soda



3. Dissociation with formic acid



Catalyst recycle

Recirculation of separated Co

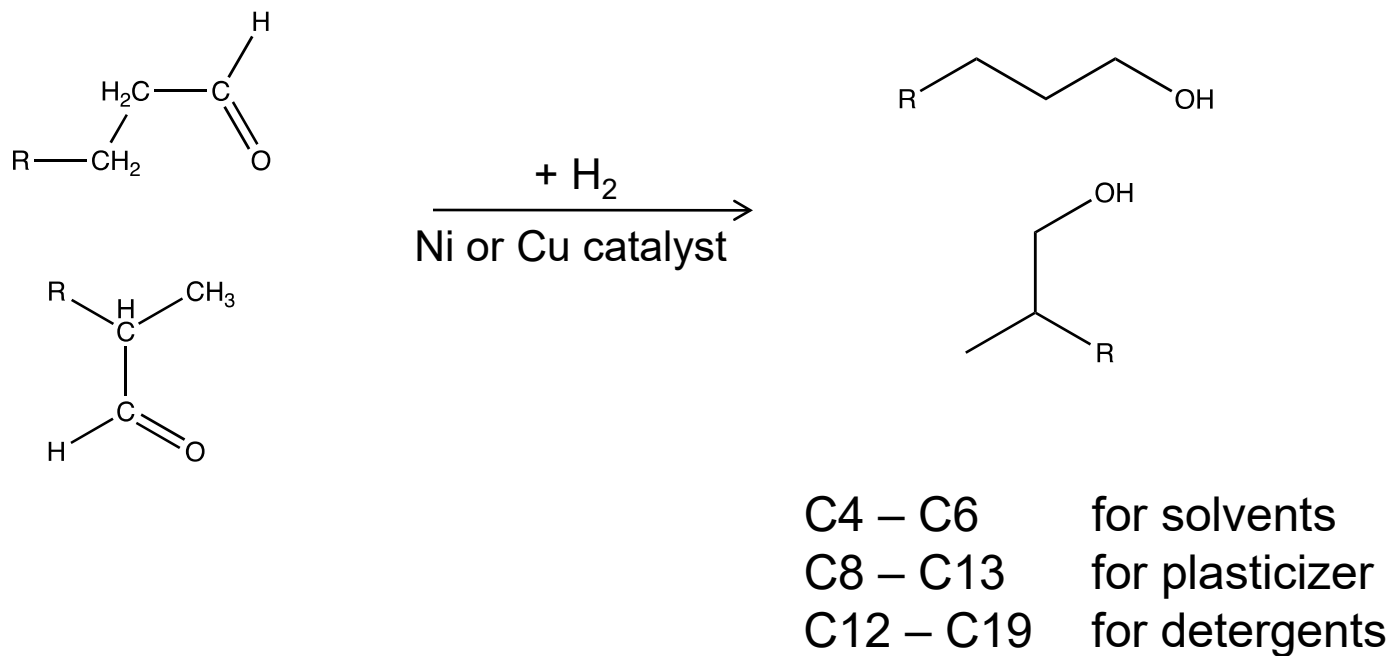


Or



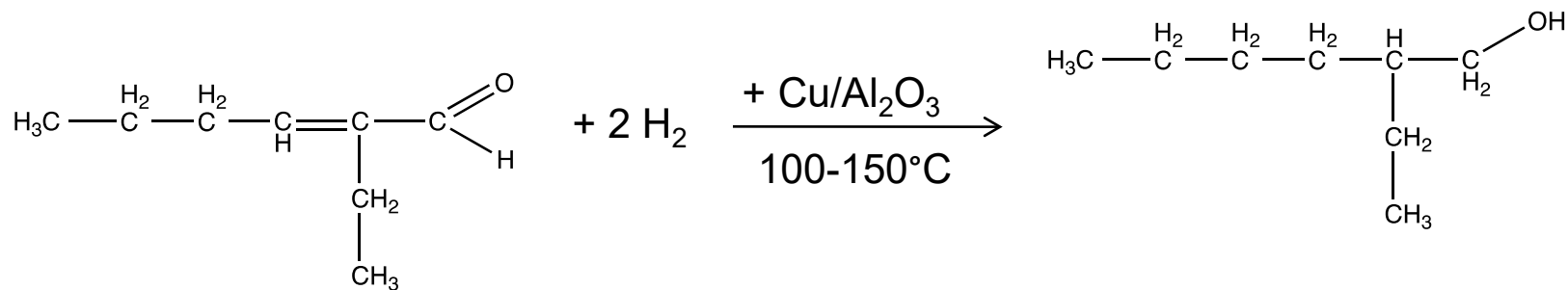
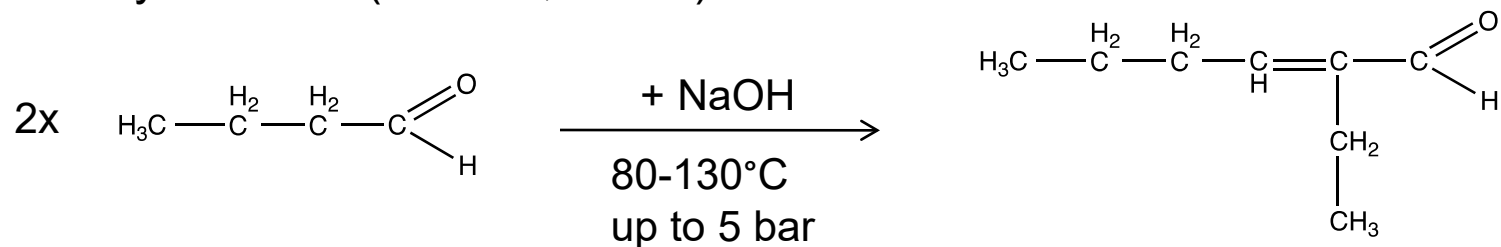
Oxo alcohols

Oxo alcohols via hydrogenation of aldehydes



Aldol condensation of oxo aldehydes

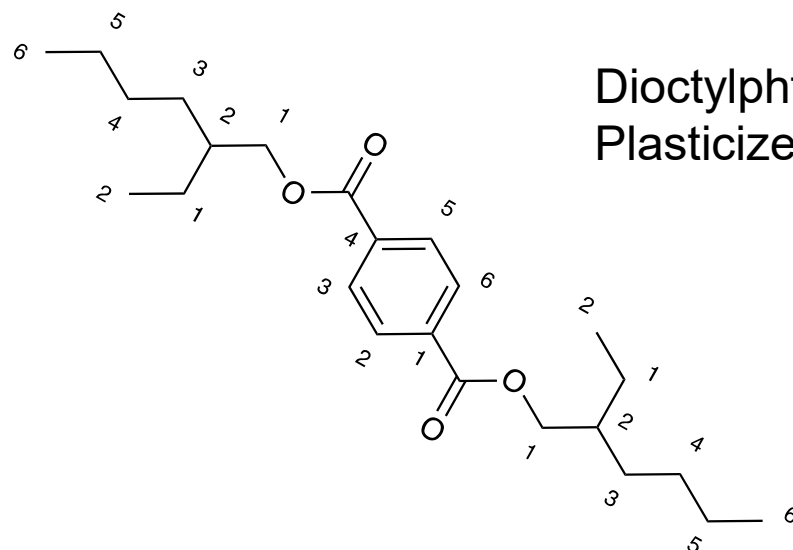
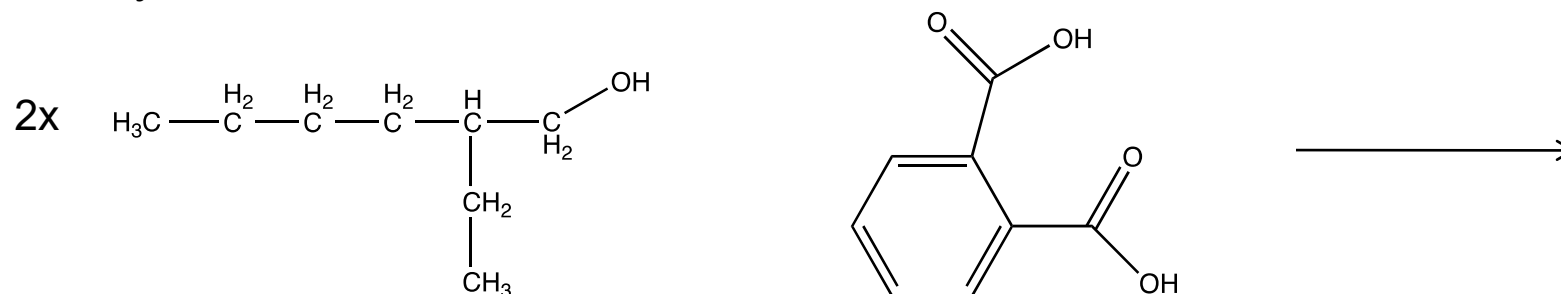
2-Ethyl-hexanol (2022: 4,5 Mill t)



Plasticizer for plastics:
Esterification with phthalic acid,
adipic acid and sebacic acid

Aldol condensation of oxo aldehydes

2-Ethyl-hexanol to DOP



Diethylphthalate, actually Bis-2-ethylhexylterephthalate
Plasticizer for plastics (PVC)

Butanol - oxo alcohols

- Most important oxo alcohol
- About 50% used for esterification with carboxylic acids
 - Acetic acid, butyric acid, valeric acid, glycolic acid, lactic acid
 - Solvents, fats, oil, waxes, plastics, natural resins
 - E.g. Di-butyl-phthalate was the plasticizer for PVC

Olefins

- Light olefins: C2, C3 and C4
almost half of the global chemical market
- Feedstock for polymers and synthetic rubbers
- Ethylene by far the dominating olefin
- Produced in steam crackers
- If at all only in small amounts in crude oil

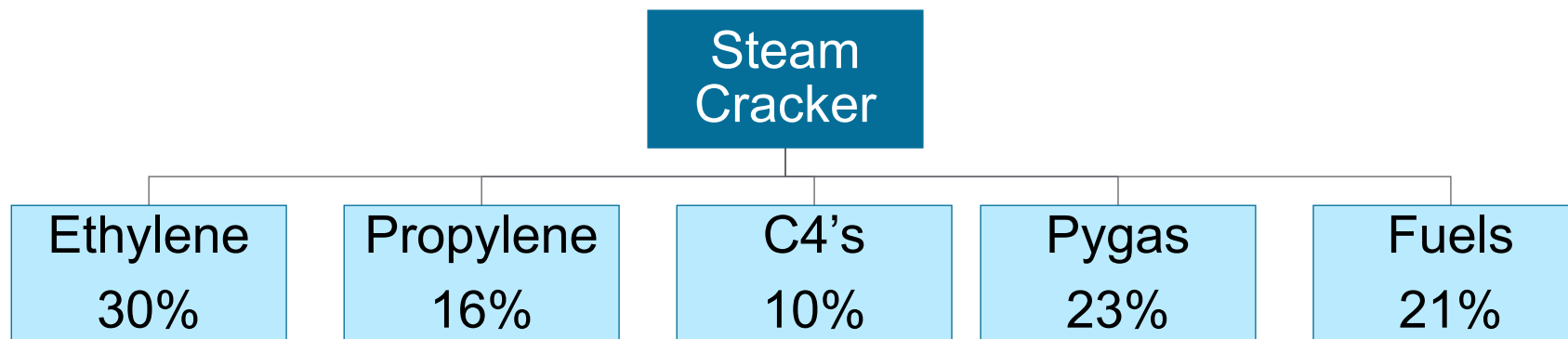
Olefins

- Cracking
- Production processes
- Ethylene
- Propylene
- Butylene
- Higher olefins
- Olefin Metathesis

Olefins

Basic cracking technologies:

- Catalytic cracking
- Hydro cracking
- Steam cracking



Catalytic cracking

- Today mostly FCC (fluid catalytic cracking)
- Fluidized bed reactor or reactor with rising catalyst (riser cracking)
- Transformation of higher boiling fractions into saturated, branched paraffin's, naphtha or aromatics
- Only small amounts of olefins → FCC used to generate fuels
- To be discussed together with the refinery in the summer term

Catalytic cracking

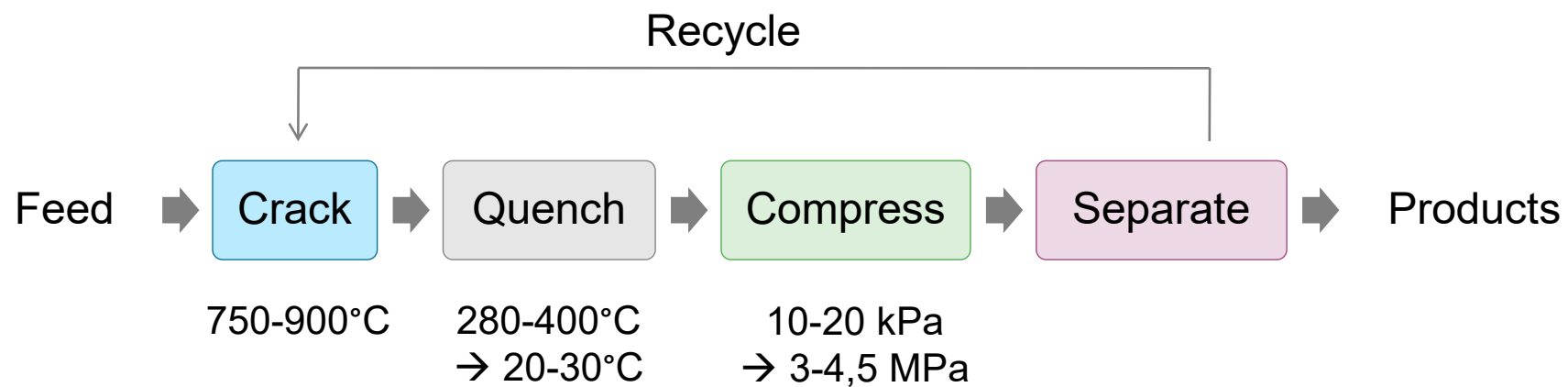
- Catalysts: Zeolites
 - Protonated form or Kat ions exchanged with rare-earth metals
 - Shape selectivity: pore shape leads to better selectivity
 - Catalysts get inactive due to coking → regeneration via off-burning coke (450-500°C in air)
- Catalytic cracking: biggest use of Zeolites
- ABB Lummus, Exxon, Gulf, Kellogg, Stone & Webster, Texaco

Hydro cracking

- Catalytic cracking in presence of H_2
- Product composition controlled through feed, used catalyst and process conditions
 - LPG, isobutene, fuel, naphtha, heating oil
- No olefins in the reaction product
- Catalysts: bi-functional systems
 - Metallic hydrogenation/de-hydrogenation (e.g. Co-Mo or Pd, Pt) and acid crack component (e.g. $Al_2O_3 \cdot SiO_2$ Zeolite)
- 270-450°C and 80-200 bar
- Per t feed oil 300 to 500 m³ H_2 are needed

Steam cracking

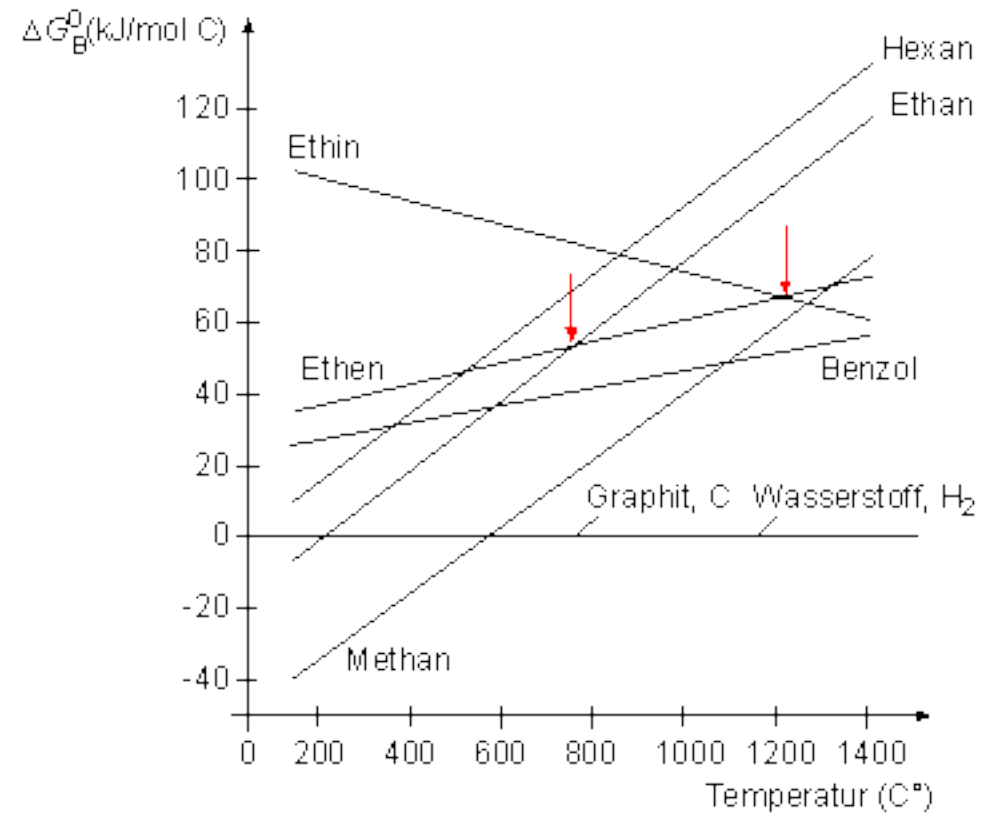
Simple block flow diagram:



Dissociation energies

Reaction		Energy [kJ/mol]
C – C scission		
$\text{H}_3\text{C} - \text{CH}_3$	2CH_3^*	350
$\text{H}_3\text{CCH}_2 - \text{CH}_2\text{CH}_3$	$2 \text{CH}_2\text{CH}_3^*$	327
$(\text{H}_3\text{C})_3\text{C} - \text{C}(\text{CH}_3)_3$	$2 (\text{H}_3\text{C})_3\text{C}^*$	283
$\text{H}_2\text{C} = \text{CH} - \text{CH}_3$	$\text{H}_2\text{C} = \text{CH}^* + \text{CH}_3^*$	386
$\text{H}_2\text{C} = \text{CHCH}_2 - \text{CH}_2\text{CH} = \text{CH}_2$	$2 \text{H}_2\text{C} = \text{CHCH}_2^*$	159
$\text{H}_3\text{C} - \text{CH} = \text{CH}^*$	Acetylene + CH_3^*	134
$\text{H}_3\text{C} - \text{CH}_2\text{CH}_2^*$	$\text{H}_2\text{C} = \text{CH}_2 + \text{CH}_3^*$	107
C – H scission		
$\text{H}_3\text{C} - \text{H}$	$\text{CH}_3^* + \text{H}^*$	430
$(\text{H}_3\text{C})_2\text{CH} - \text{H}$	$(\text{H}_3\text{C})_2\text{CH}^* + \text{H}^*$	396
$(\text{H}_3\text{C})_3\text{C} - \text{H}$	$(\text{H}_3\text{C})_3\text{C}^* + \text{H}^*$	381
$\text{H}_2\text{C} = \text{CHCH}_2 - \text{H}$	$\text{H}_2\text{C} = \text{CHCH}_2^* + \text{H}^*$	356

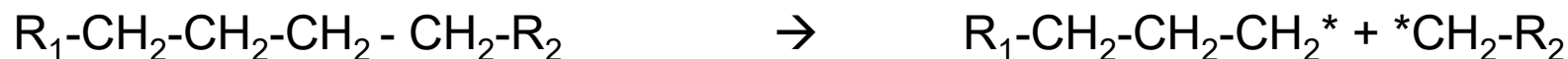
Formation enthalpy



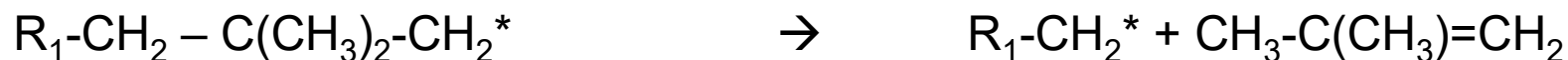
Pyrolysis

Starting reaction:

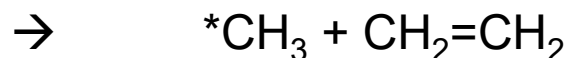
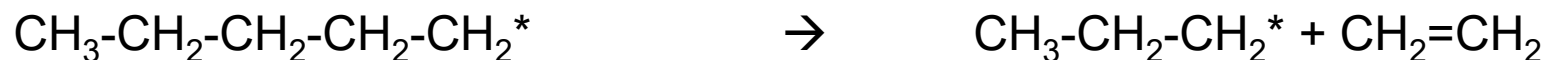
C – C bond energy ≈ 335 kJ/mol



Radical fragmentation (unimolecular β scission):



Unimolecular β scission can occur several times:

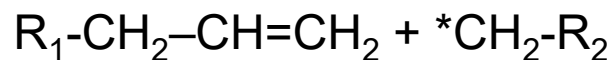


Pyrolysis (ctd.)

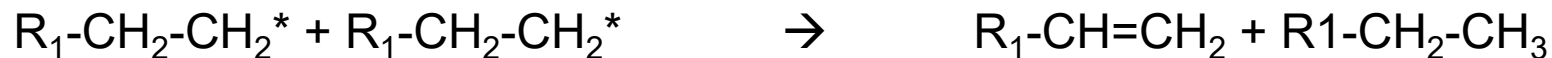
Bimolecular C-H scission:



Unimolecular β scission:

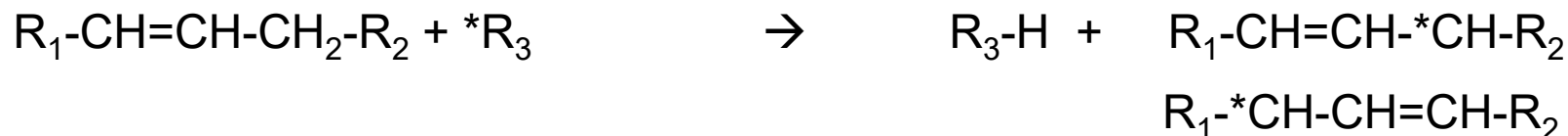


Chain termination:

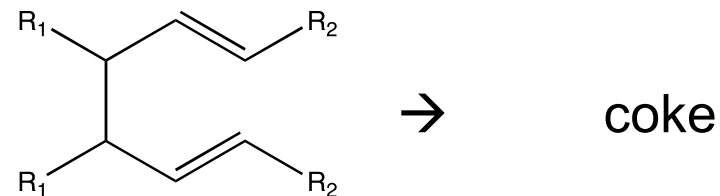
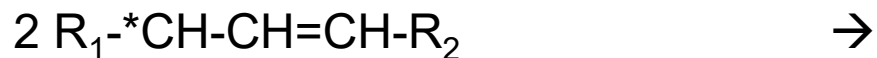
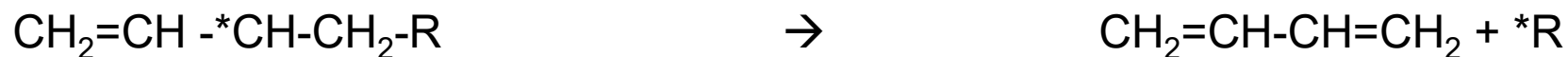


Pyrolysis (ctd.)

Formation of allyl radicals:



Reactions of allyl radicals:

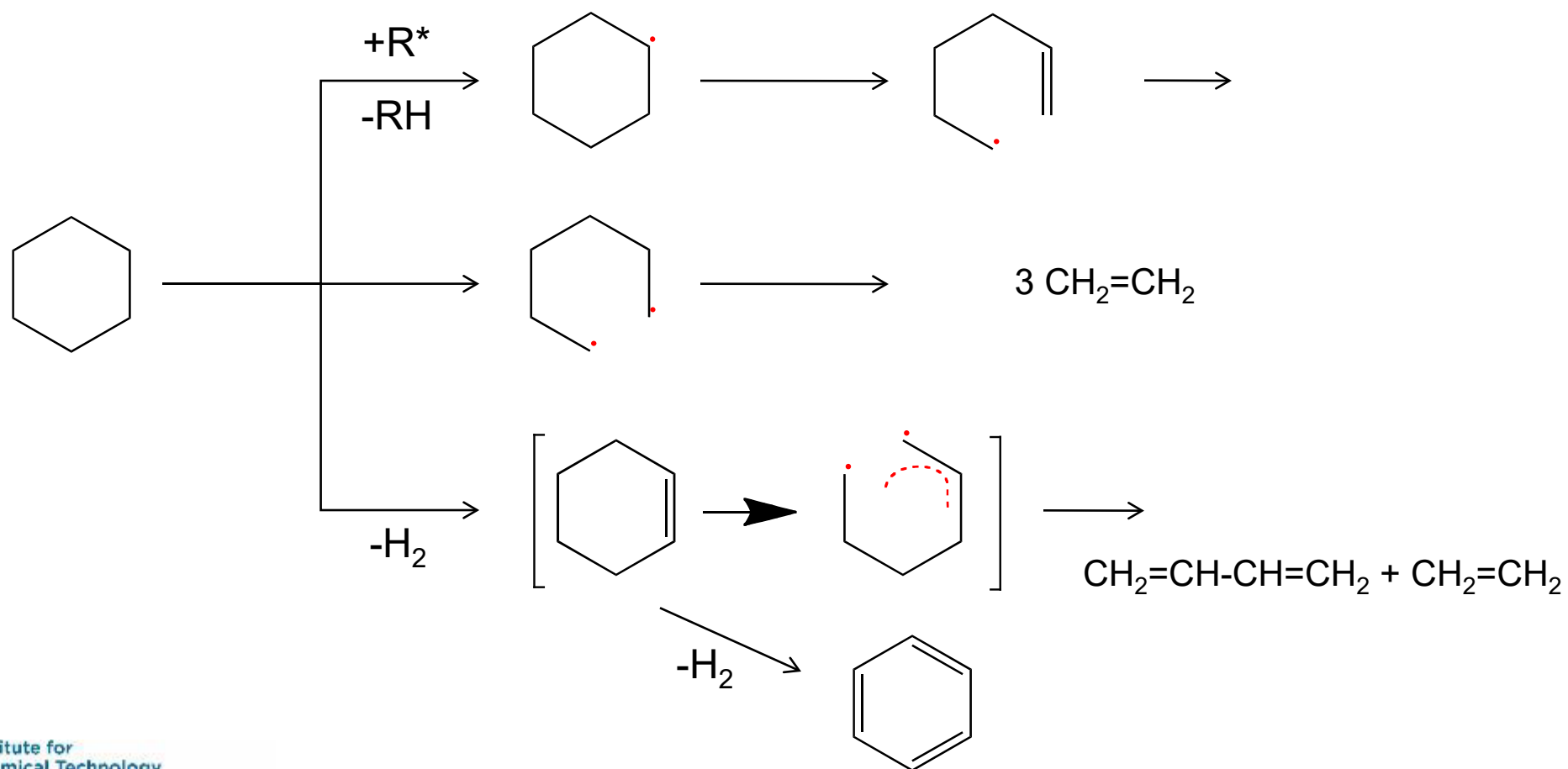


Dehydration:



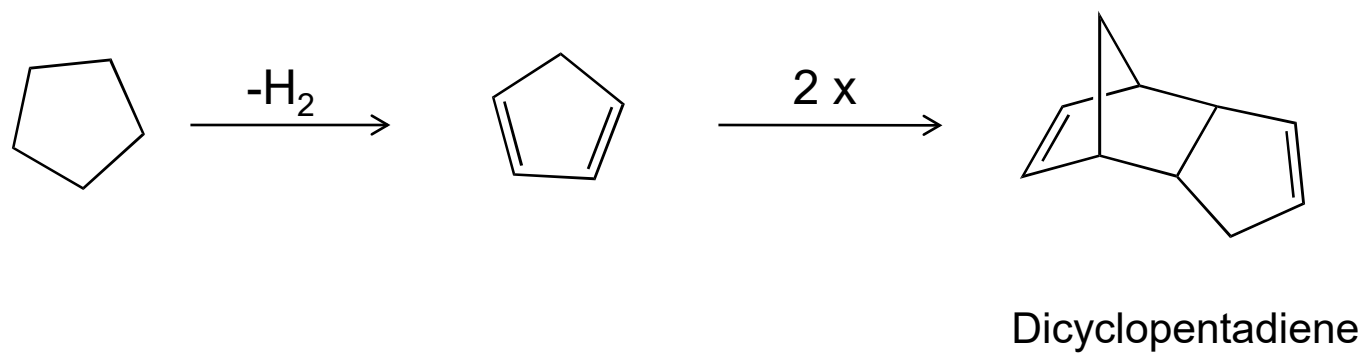
Pyrolysis (ctd.)

Reactions of naphthenic 5- and 6-rings:



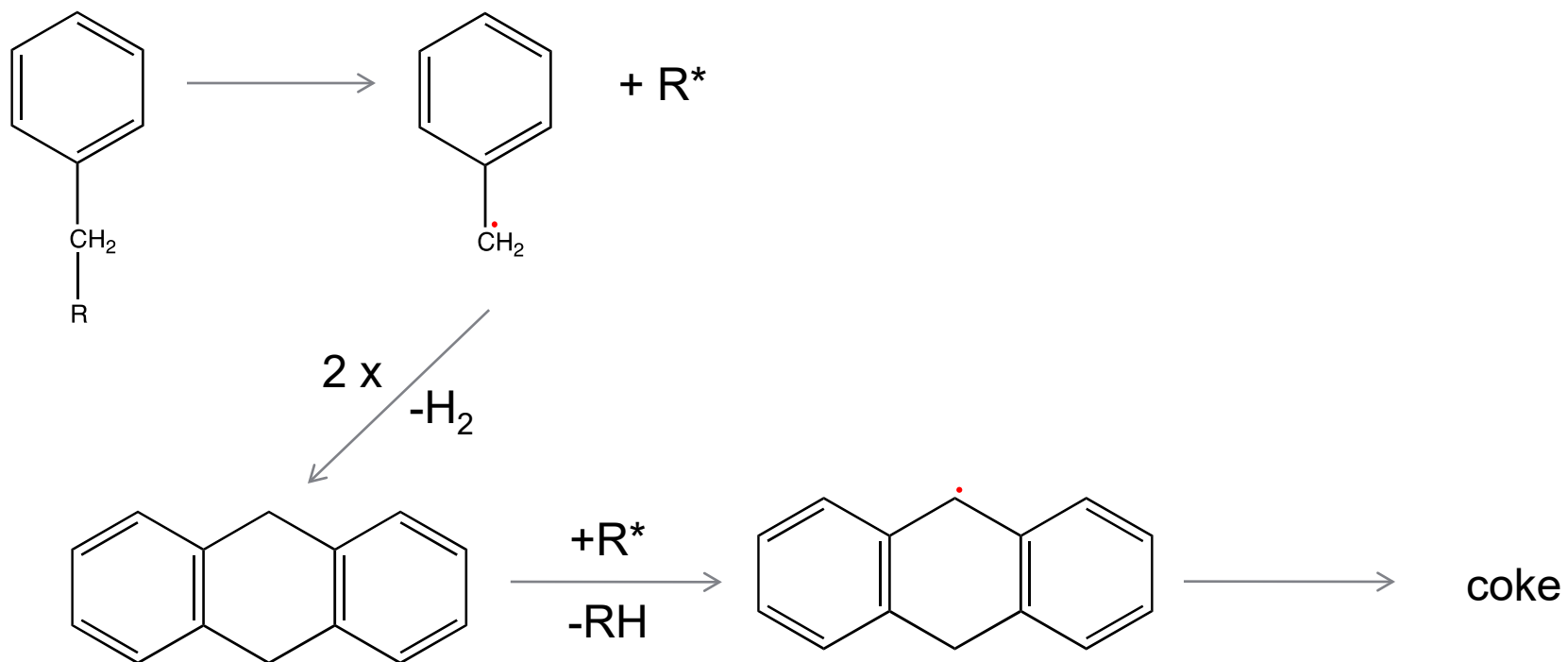
Pyrolysis (ctd.)

Reactions of naphthenic 5- and 6-rings:

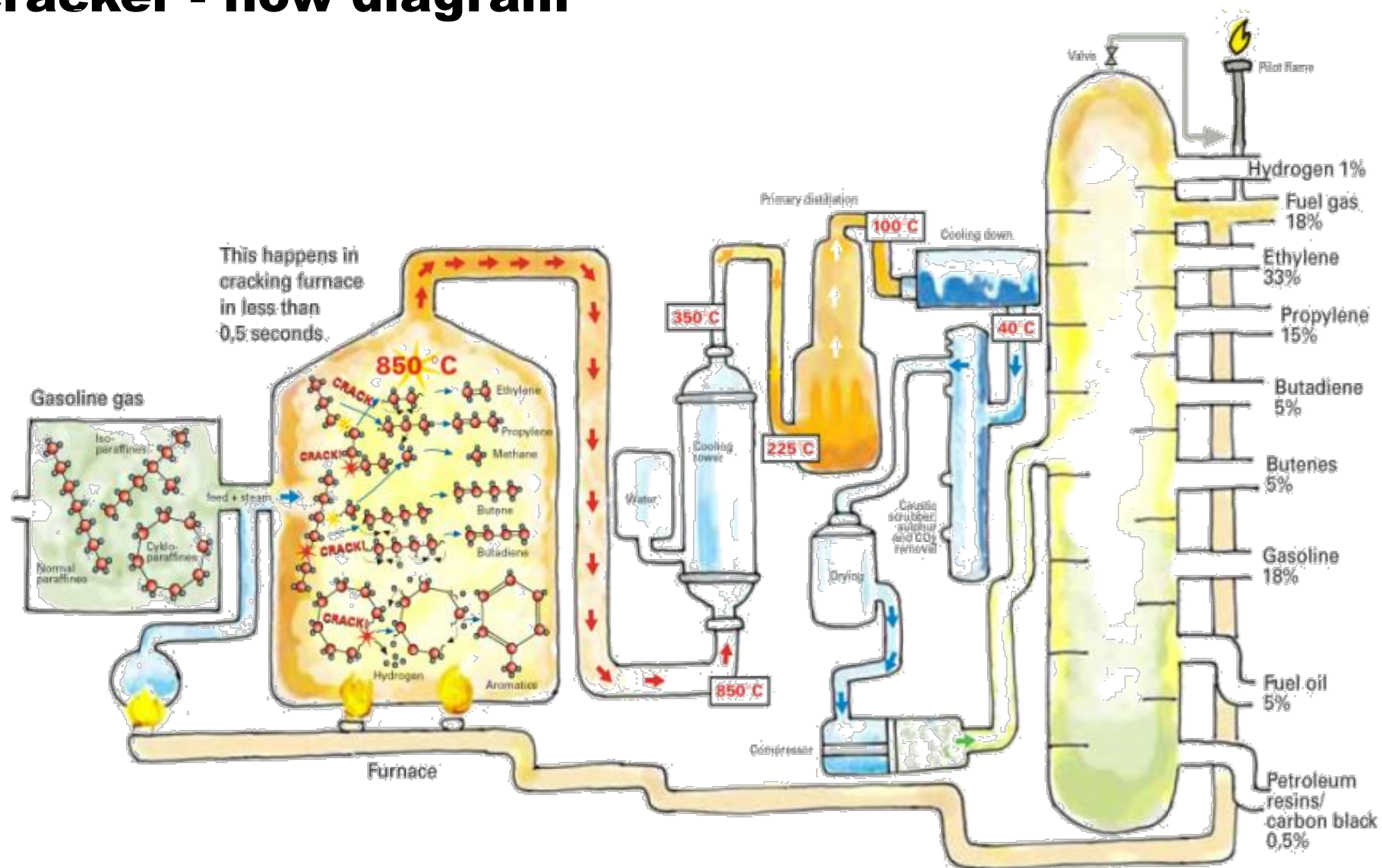


Pyrolysis (ctd.)

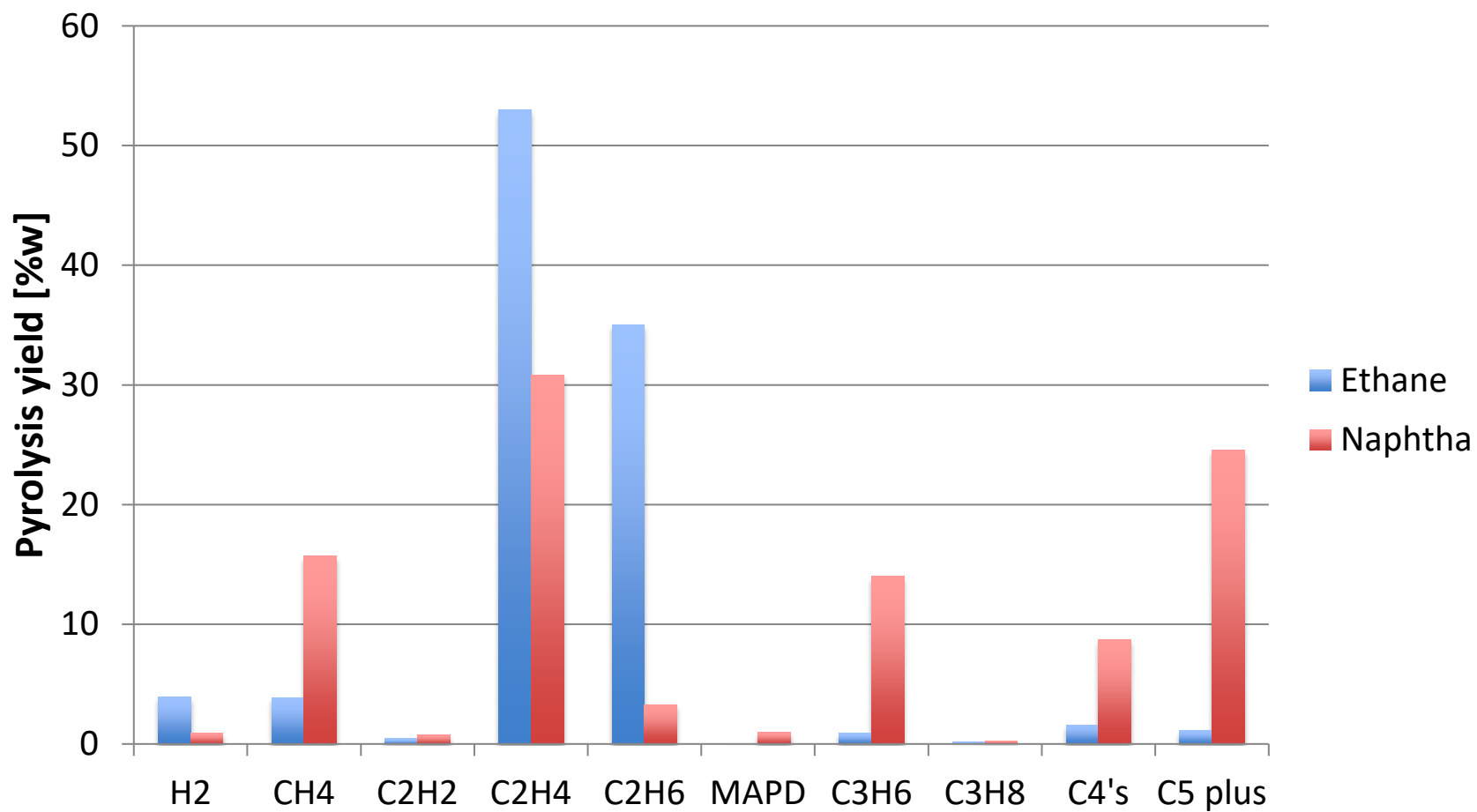
Formation of benzyl radicals:

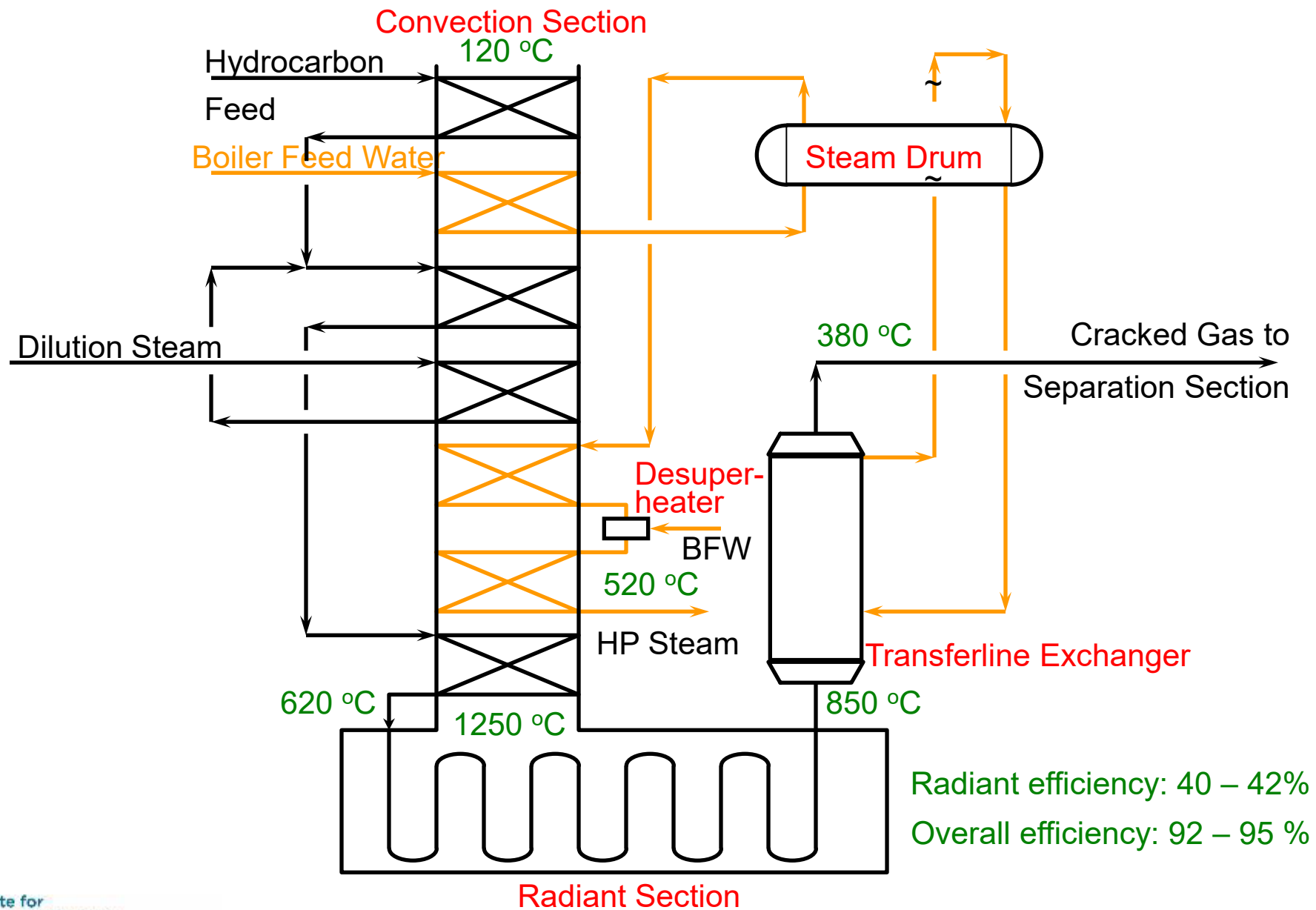


Steam cracker - flow diagram

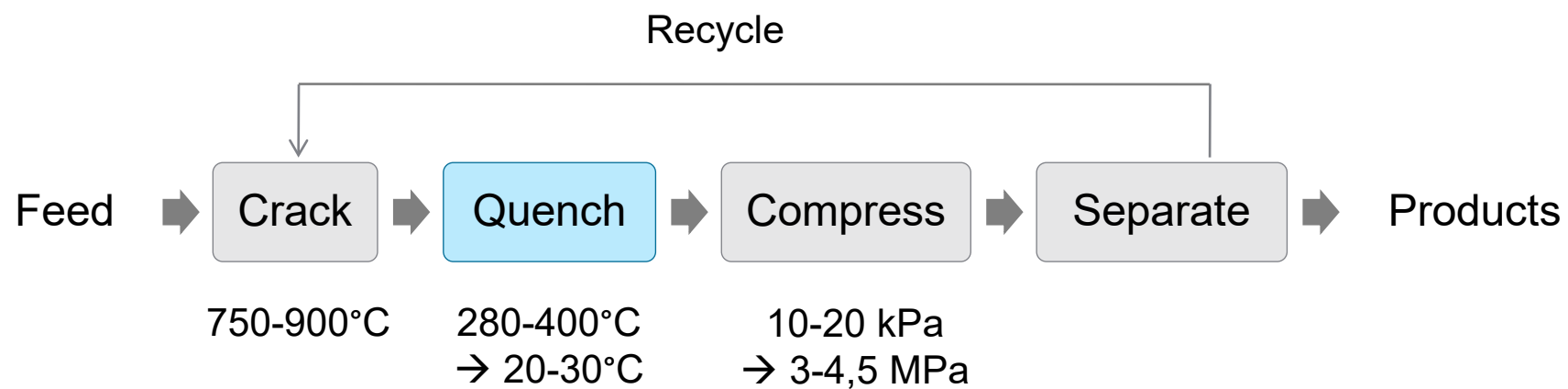


Steam cracking - products



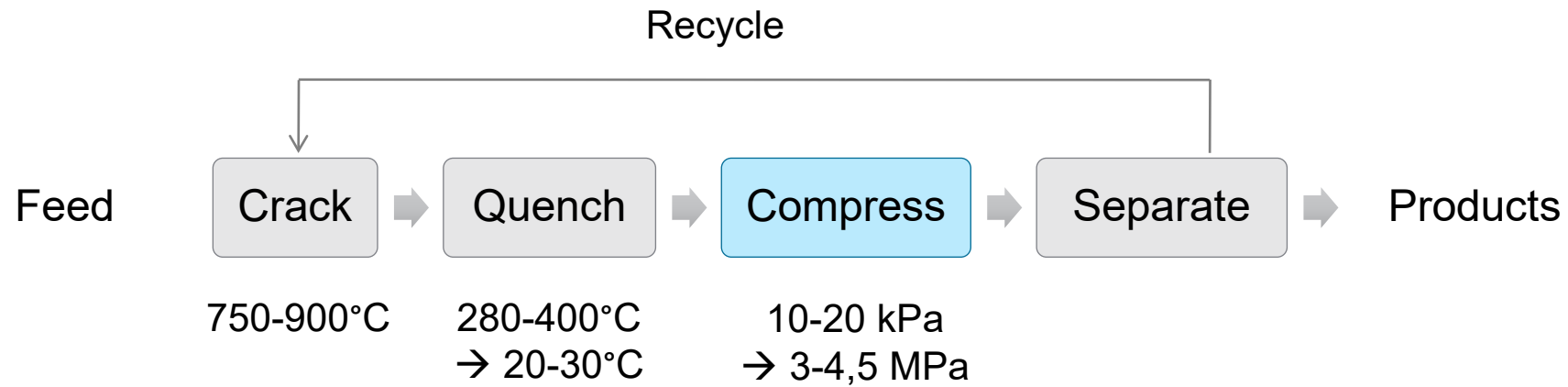


Quench



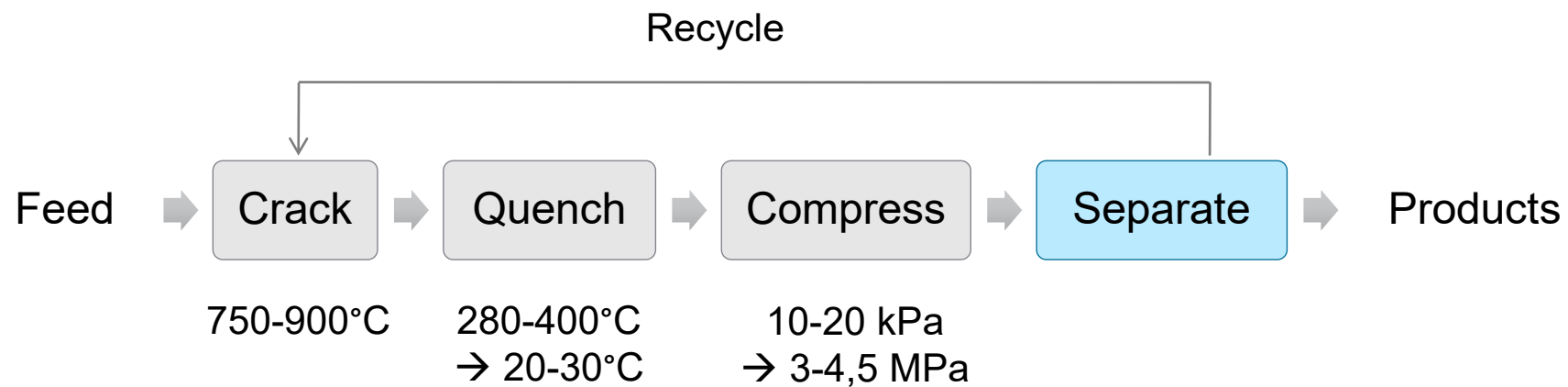
Cool down the furnace effluent as most effectively as possible

Cracked gas compression



- Preparation for downstream separation
- Can be 3, 4 or 5 stage compression processes
- Biggest steam utility user, a consideration when designing steam system
- Come with electric drives as standby

Separation



Separation – acid gas removal

- Acid gas removal is a scrubbing process
- Most of these technologies are dependent on solvent formulations, and recovery configurations
- Process followed by drying of gas
- Main chemical and process technology suppliers are: DOW, Coastal AGR, Lurgi, UOP and Fluor

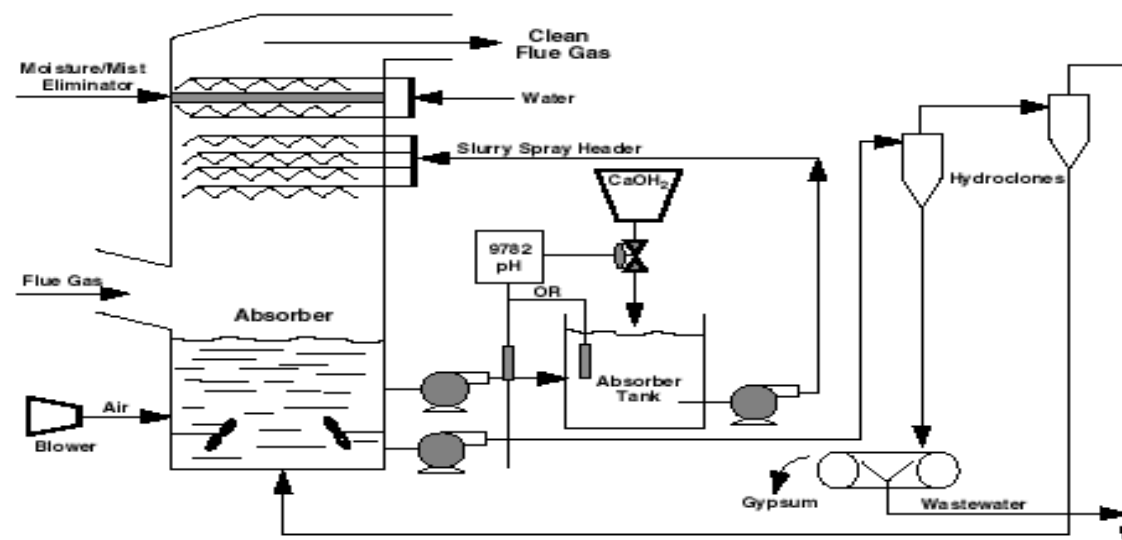


figure 1 - typical wet scrubbing process: limestone non-regenerable

Hydrogenation

- The other reactor in the olefins process line, after cracking
- One of the most important sources of different olefin plant configurations
- Reactors designed in as modular units
- Conversion of acetylene and MAPD

Hydration of C3 fraction

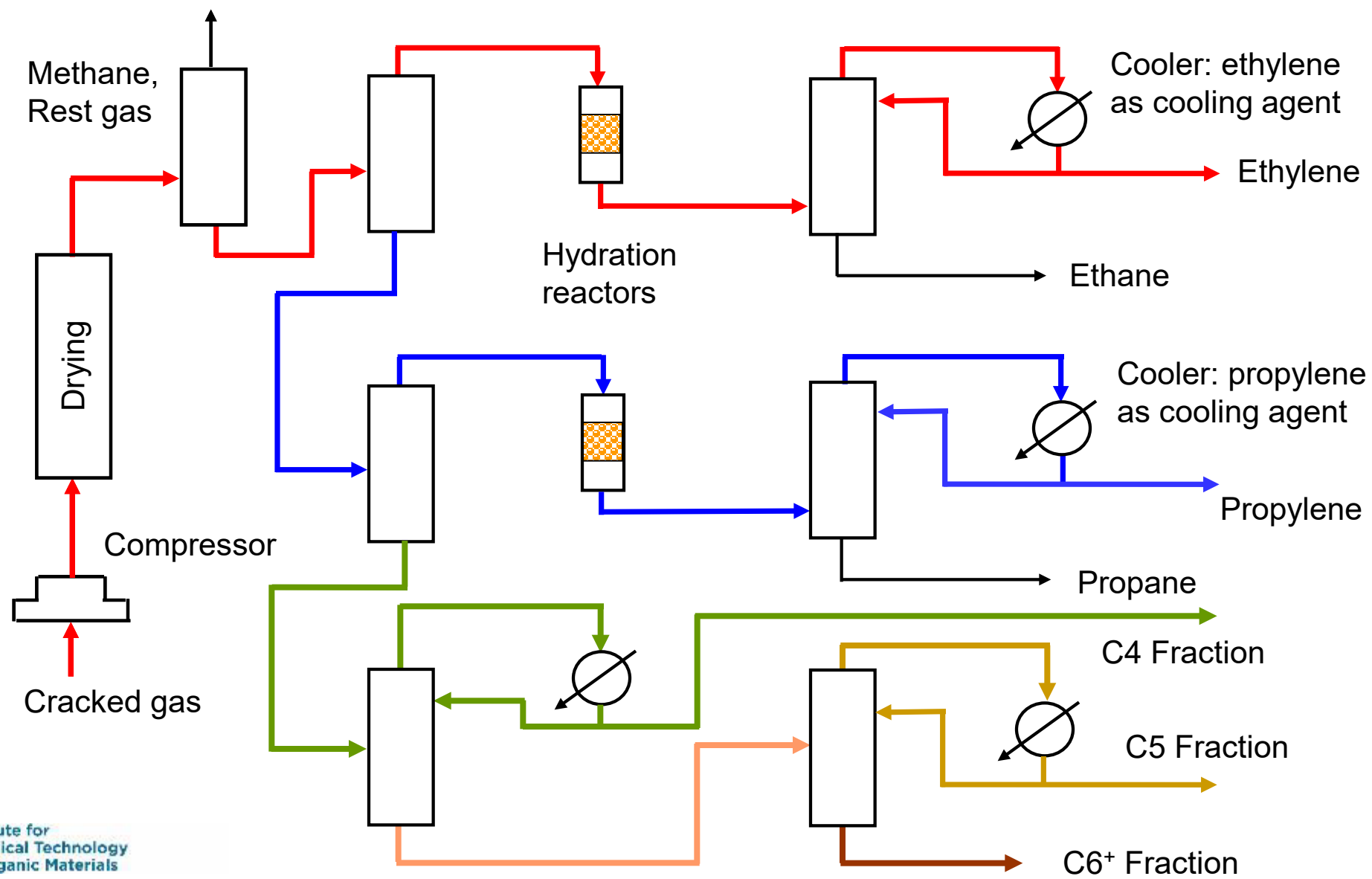
Compound	Before hydration [%v]	After hydration [%v]
C2	0,14	0,14
Allylene	2,58	< 0,001
Propadiene	1,73	< 0,001
Propylene	91,19	93,29
Propane	4,17	6,27
C4	0,19	0,30

Distillation

- A source of process configuration variation
- Part of it distils C1 and lighter cracked gas products
- Other stages separate C2, C3 and C4s i.e. deC2, deC3, deC4
- Splitters are expensive!

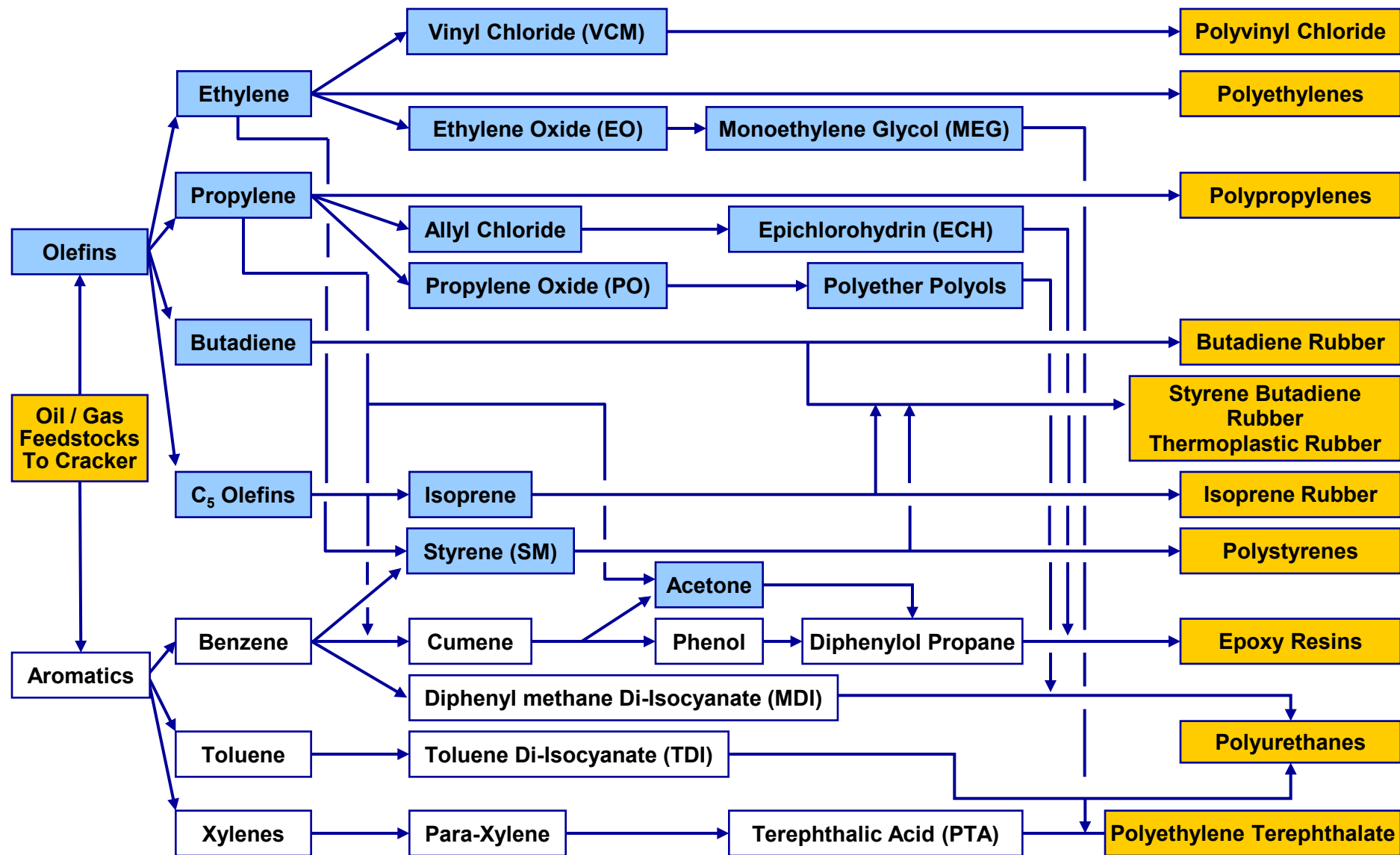


Low T & p distillation



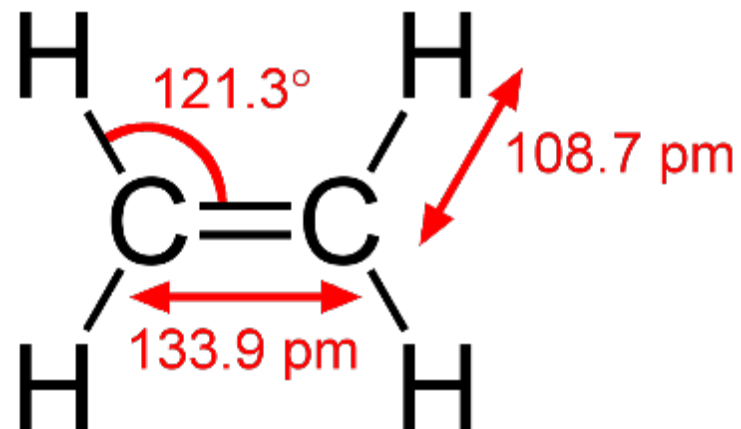
Feedstock Challenges

- Oil price
- Choice of feedstock will be influenced by – technology, location, cost
- Or other feedstock e.g. BASF is researching use of syngas from natural gas



Ethylene

- Molar mass: 28,05 g/mol
- Density: 1,178 kg/m³ (15°C)
- T_m: -169,18°C
- T_b: -103,72°C
- Acidity pK_a: 44
- Flash point: -136°C



Flash point

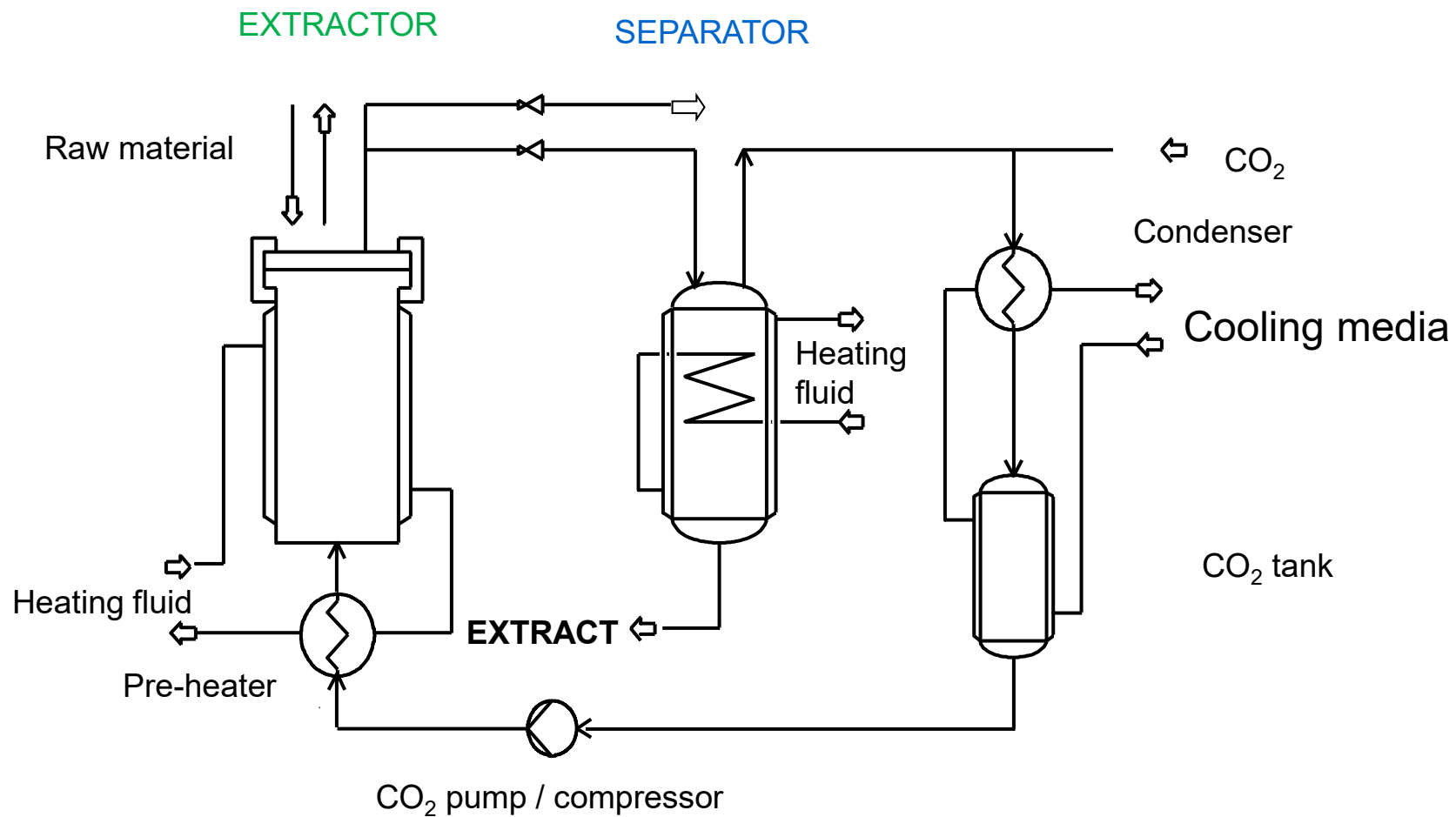
Liquids with a flash point $< 60,5^{\circ}\text{C}$ (or $37,8^{\circ}\text{C}$) are flammable, and liquids with a flash point above those temperatures are combustible

Fuel	Flash point	Auto ignition temperature
Ethanol (70%)	$16,6^{\circ}\text{C}$	363°C
Gasoline (petrol)	$- 43^{\circ}\text{C}$	246°C
Diesel	$> 62^{\circ}\text{C}$	210°C
Jet fuel	$> 60^{\circ}\text{C}$	210°C
Vegetable oil (canola)	327°C	
Biodiesel	$> 130^{\circ}\text{C}$	

Supercritical fluid

- Temp. / and p above a critical point
- Distinct liquid and gas phases do not exist
- Can effuse through solids like a gas, and dissolve materials like a liquid
- In addition, close to the critical point, small changes in pressure or temperature result in large changes in density, allowing many properties of a supercritical fluid to be "fine-tuned"
- Supercritical fluids are suitable as a substitute for organic solvents in a range of industrial and laboratory processes
- Carbon dioxide and water are the most commonly used supercritical fluids, being used for decaffeination and power generation, respectively

Basic principle of SFE



Lab extraction



Solubility in CO₂

Soluble	Reduced solubility	Not soluble
Ester, alcohols	Cooking oil from oil seeds	Sugars
Aldehydes, ketones	Wax	Fruit acids
Essential oils	Polyphenols	Starch
Flavors		Proteins
Mono- and sesquiterpenes		Mineral salts
		Glycosides

SCF applications

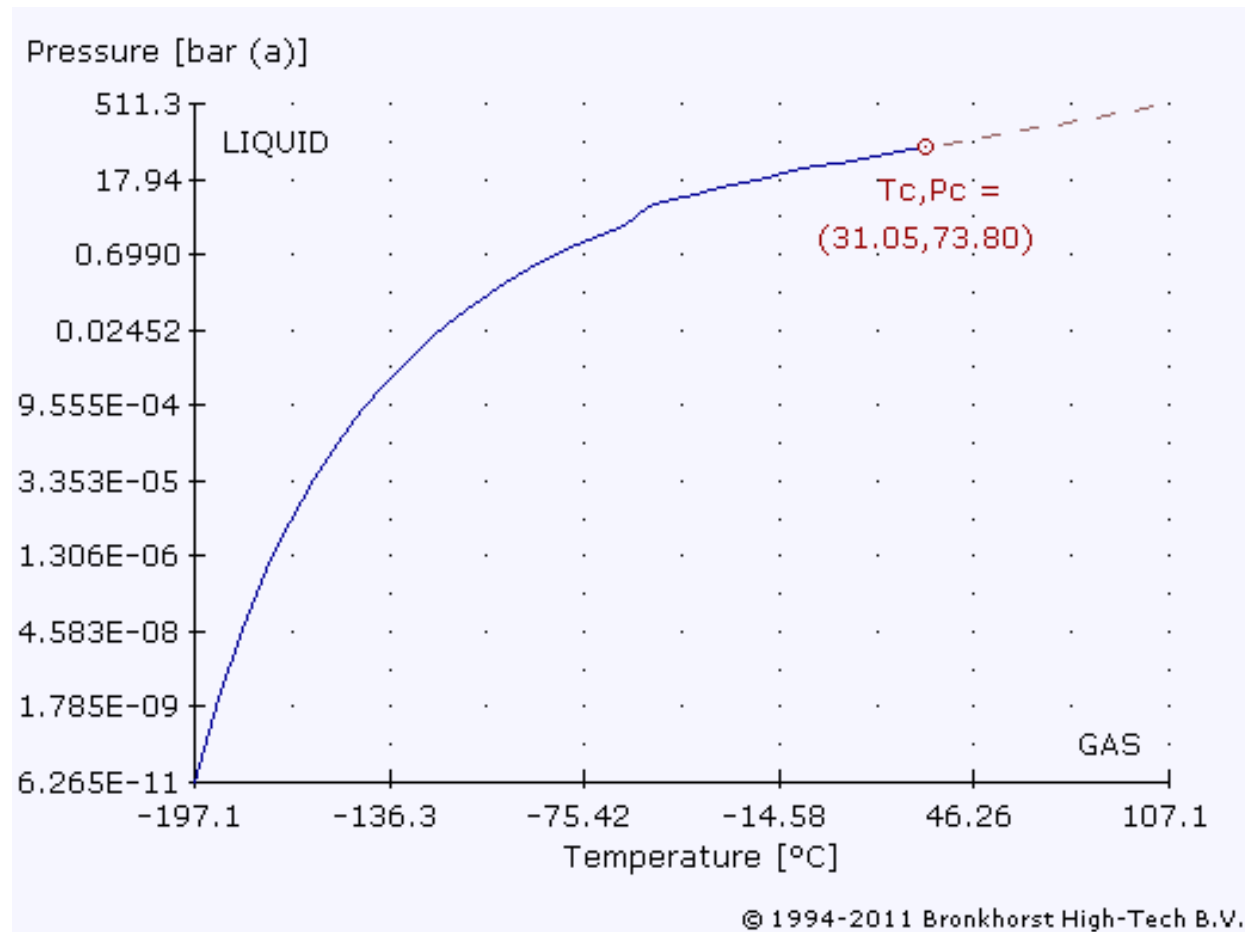
- Food industry
 - Coffee, tea, hops, herbs & spices, flavors, antioxidants
 - De-oiling of press cakes
- Pharmaceutical industry
 - Extraction of vegetable based pharmaceuticals
- Cosmetics industry
 - Active substances for skin lotions, aromas
- New applications:
 - Coloring of textiles
 - Impregnation of wood and plastics
 - Cleaning of parts
 - Removal of pollutants
 - Thin film extraction

Hop extraction

- Constant product quality
- Equaling out seasonal and geographical differences in hop quality
- Extract
- In the past: benzene, ethanol or dichloromethane
(residues of benzene or dichloromethane in product, ethanol also extracts pesticides etc.)

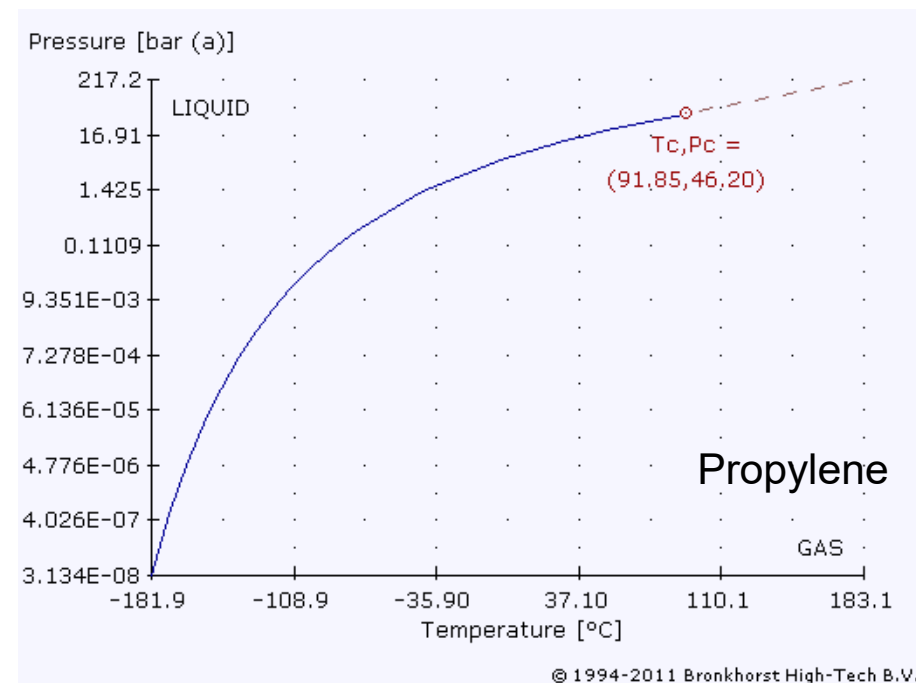
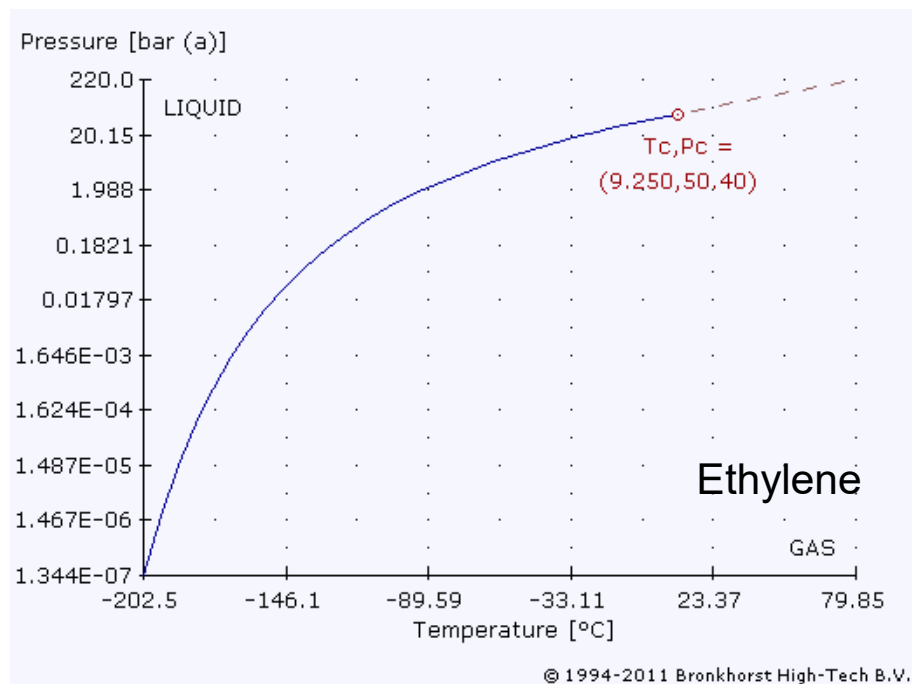


CO₂



<http://www.fluidat.com>

Ethylene / Propylene

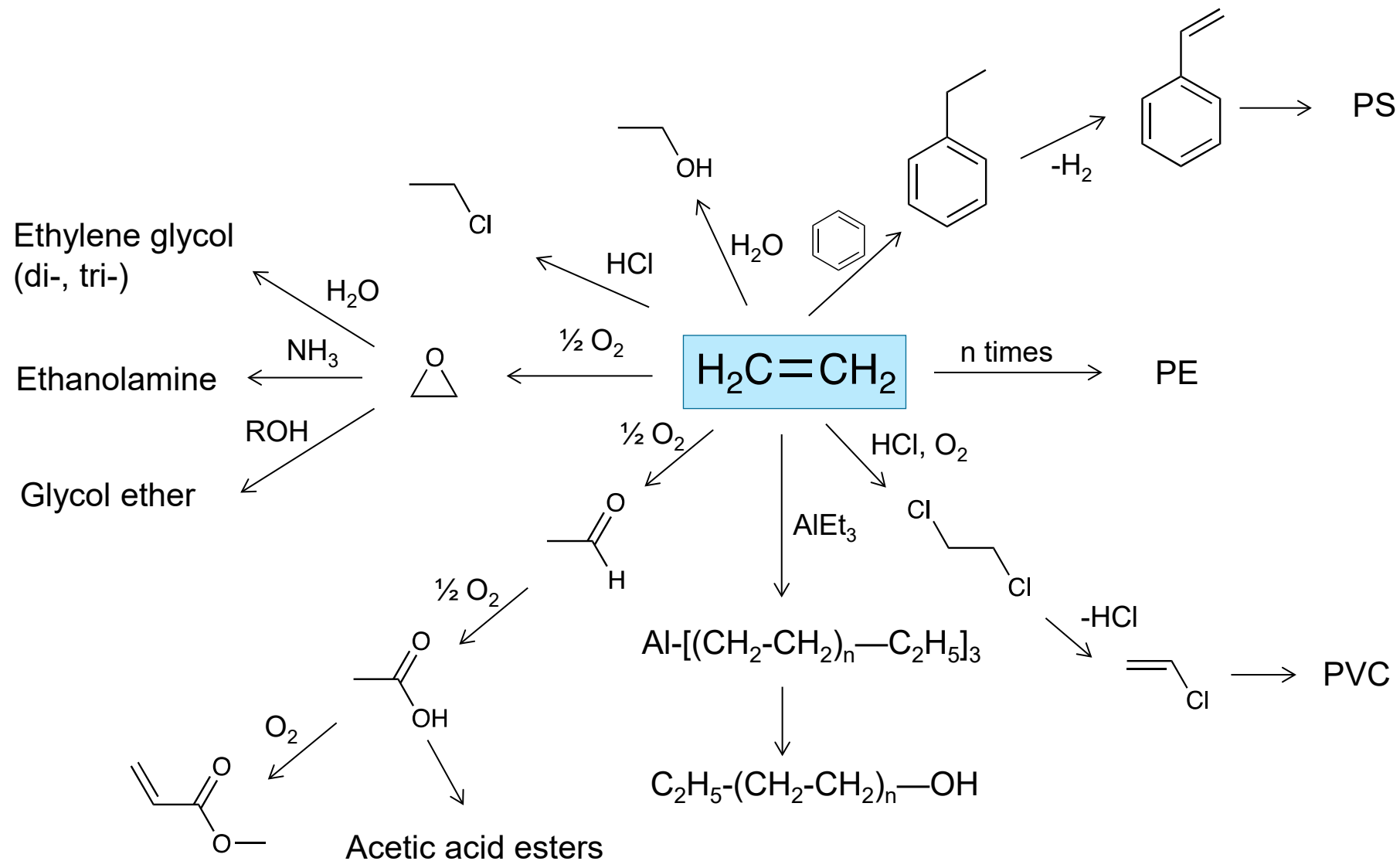


Ethylene – alternative ways

Dehydrogenation of ethanol

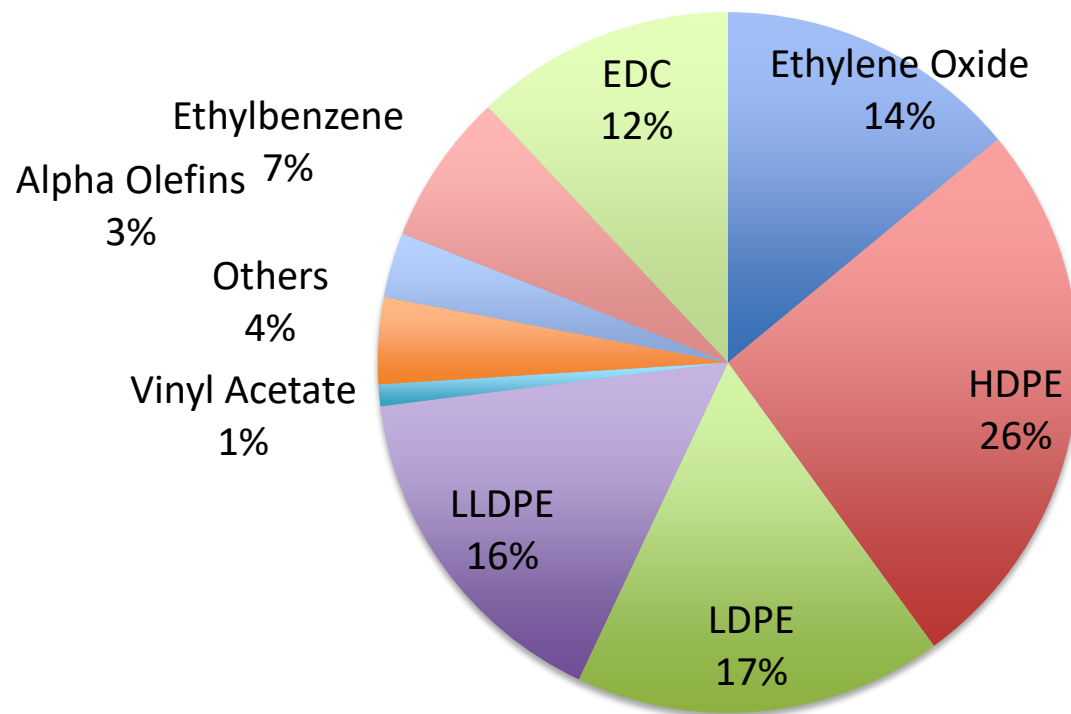
- Proven already 40 years ago
- In the 1980ies (Halcon) with Al/Mg oxide catalysts
300°C
Selectivity > 97%
Problem: purity for polymerization grade
- Catalysts:
Zeolites, Metal oxides, Al oxides, W phosphoric acids

Ethylene family tree



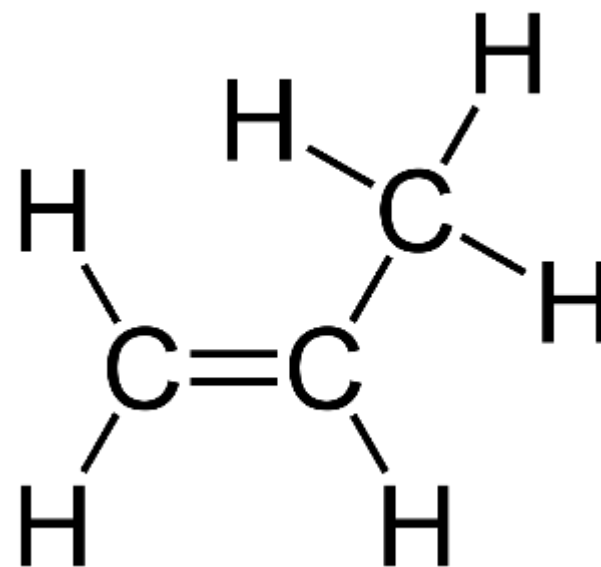
Ethylene – world demand by sector

Total demand roughly 225 Mill tons

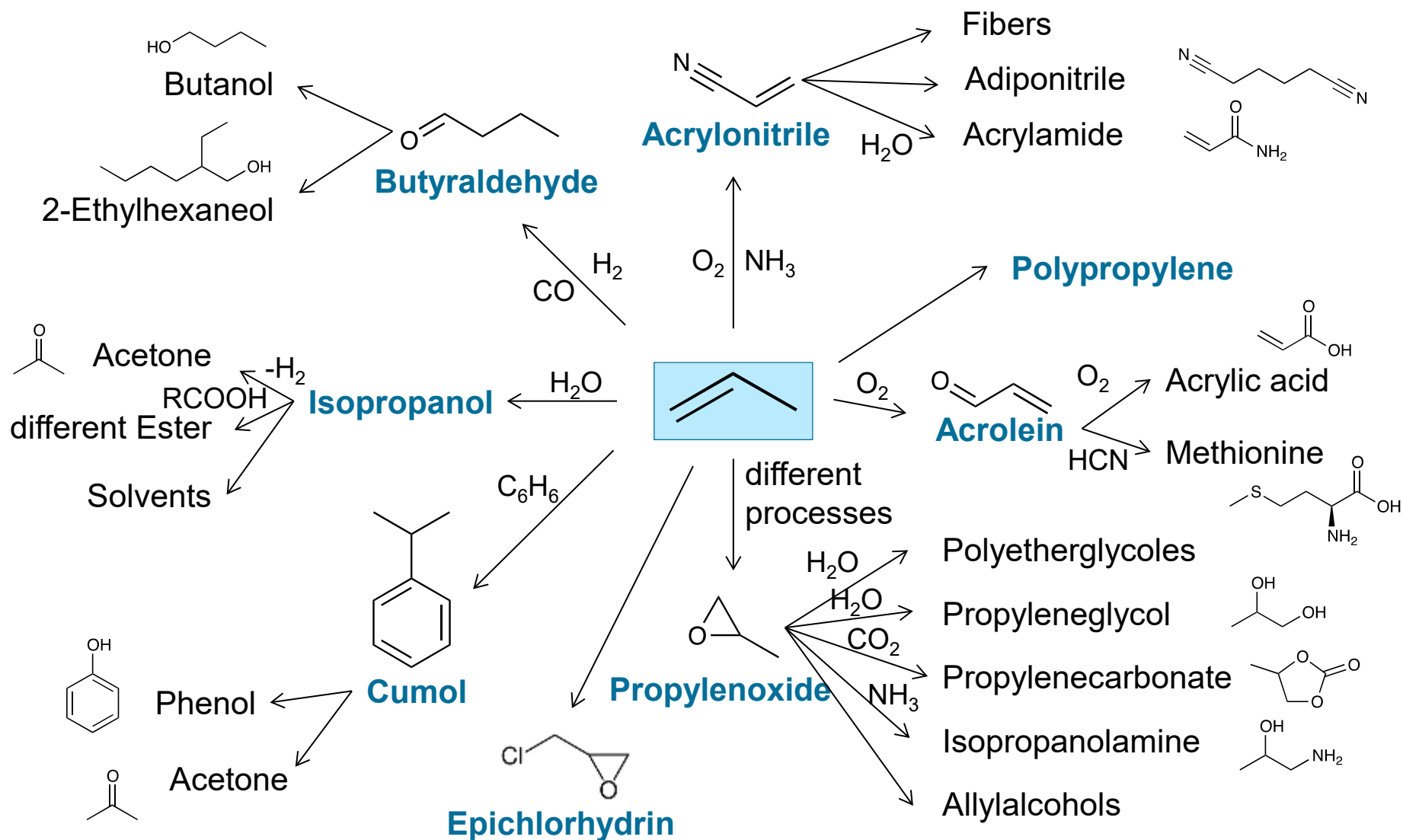


Propylene

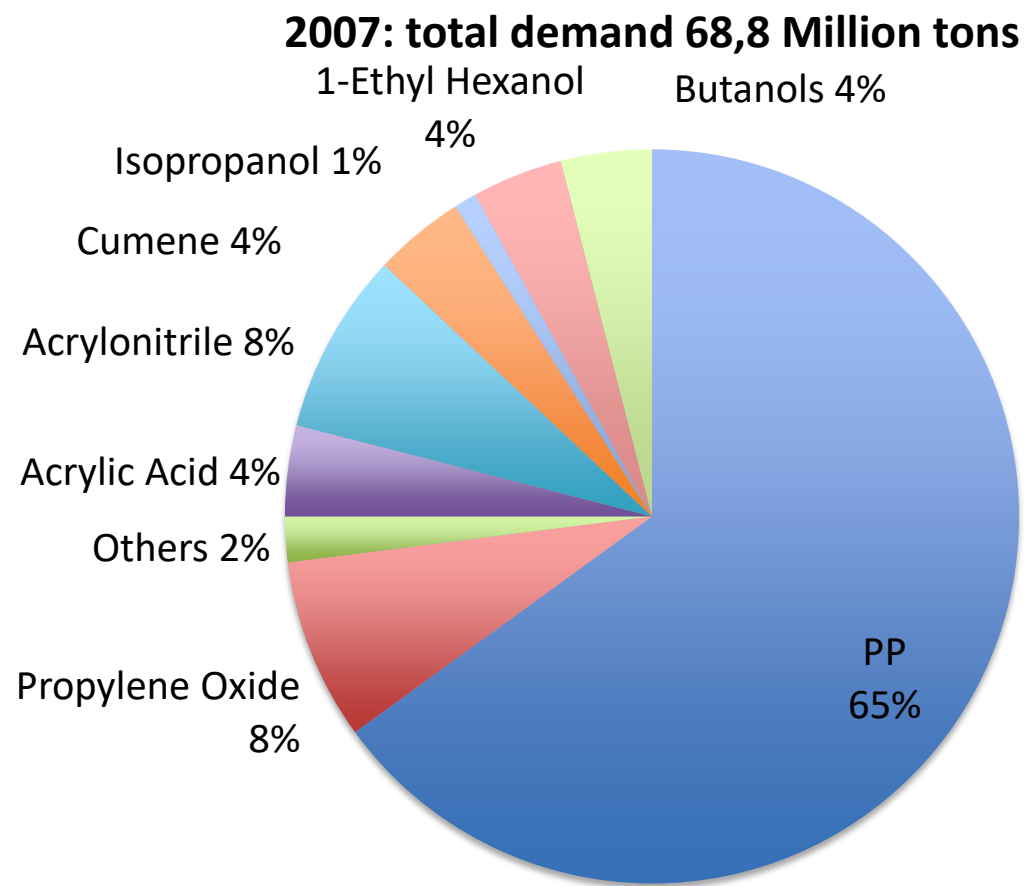
- Molar mass: 42,08 g/mol
- Density: 1,91 kg/m³ (0°C)
- T_m: -185,3°C
- T_b: -47,7°C
- Flash point: -108°C



Propylene family tree

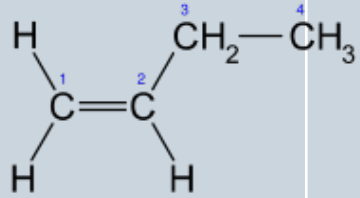
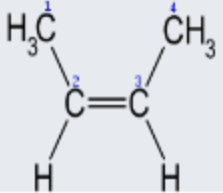
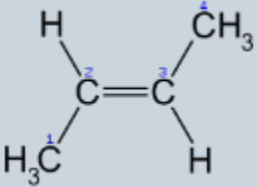
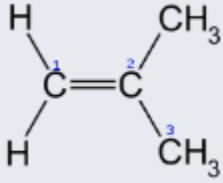


Propylene: world demand by sector



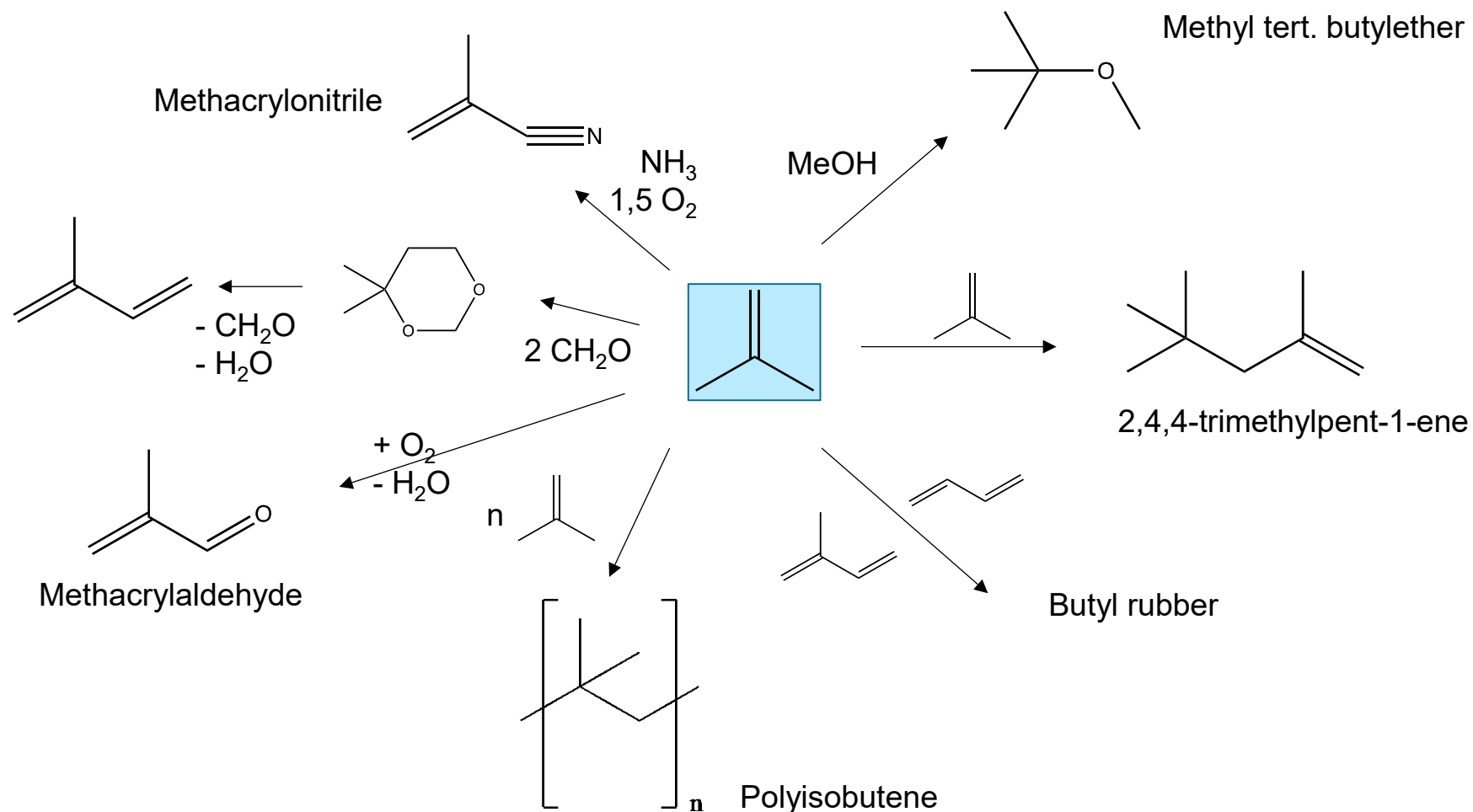
Polymerization and chemical grade

Butylene

IUPAC name	Common name	Structure	Tb [°C]
But-1-ene	α -Butylene		-6.26
Z-But-2-ene	Z- β -Butylene		3.78
E-but-2-ene	E- β -Butylene		0.88
2-methylpropene	Isobutylene		-6.90

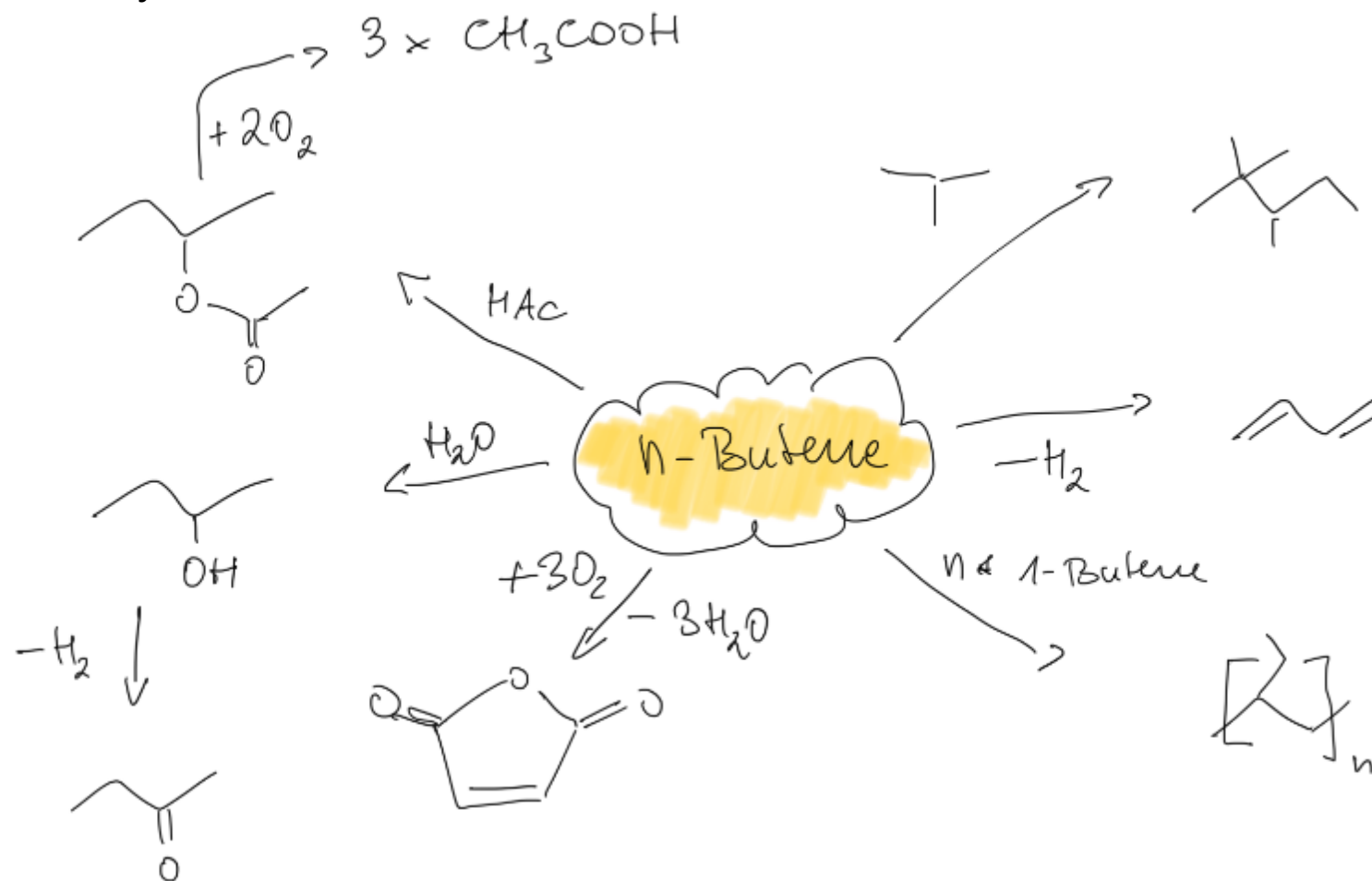
Isobutene

Industrially relevant derivatives



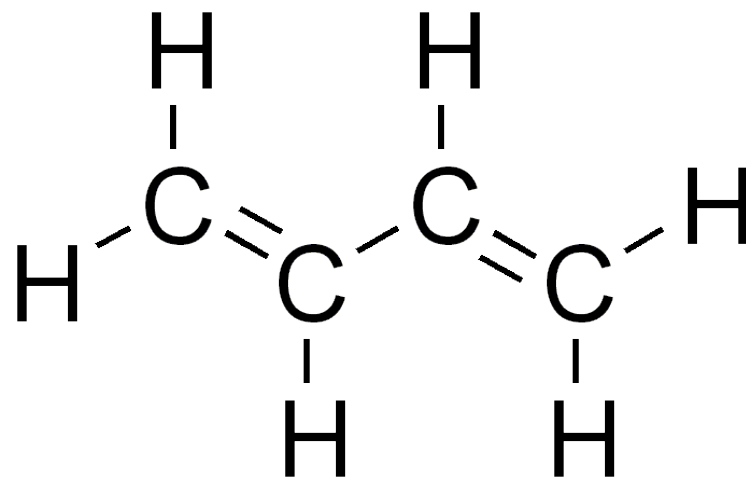
n-Butene

Industrially relevant derivatives



Butadiene

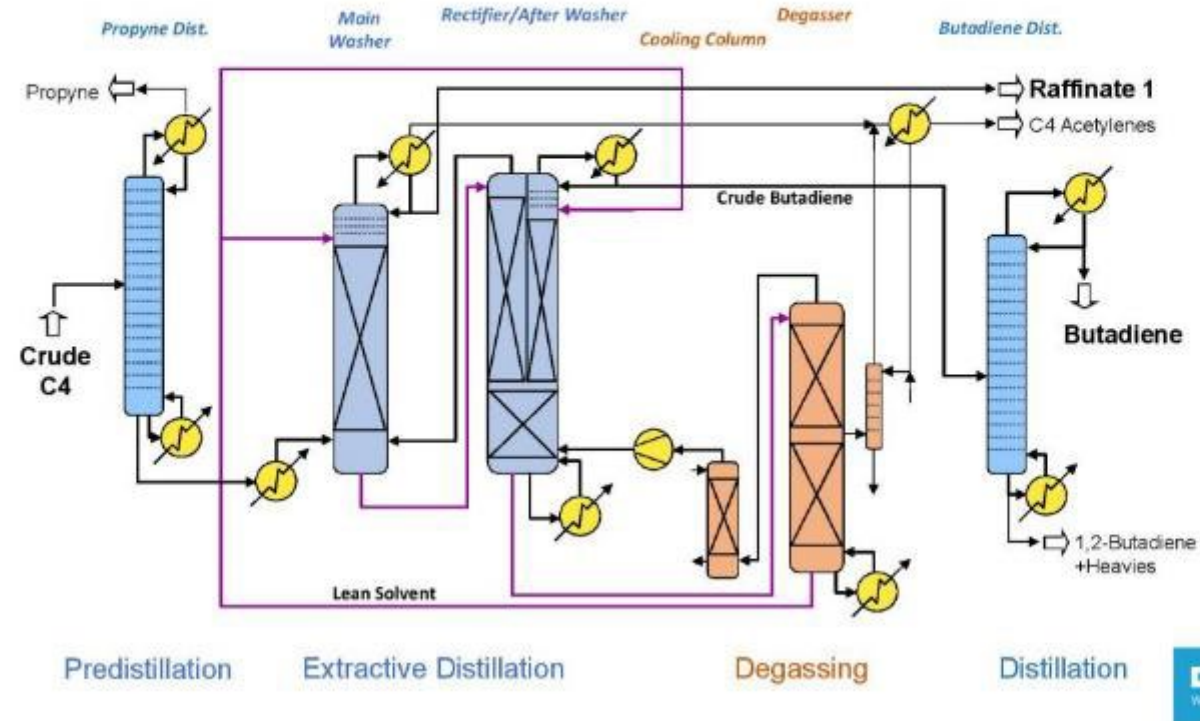
- Molar mass: 54,09 g/mol
- Density: 0,64 g/cm³ (-6°C)
- T_m: -108,9°C
- T_b: -4,4°C
- Produced by extraction from the C₄ fraction of the naptha steam cracker (crude C₄)



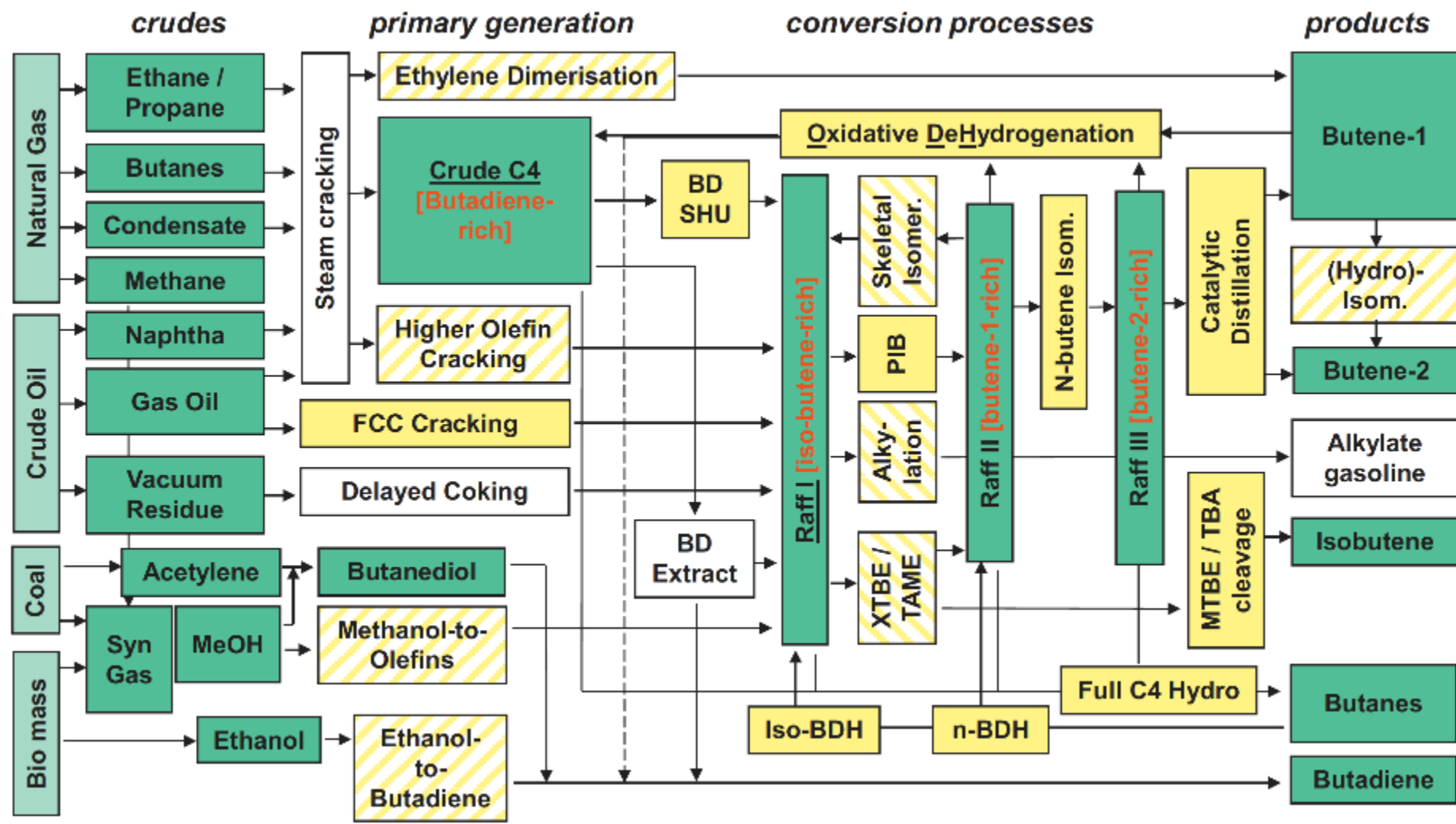
Extractive distillation

BASF Butadiene Extraction Technology

BASF BDE-Process: Standard Configuration (Option 1)

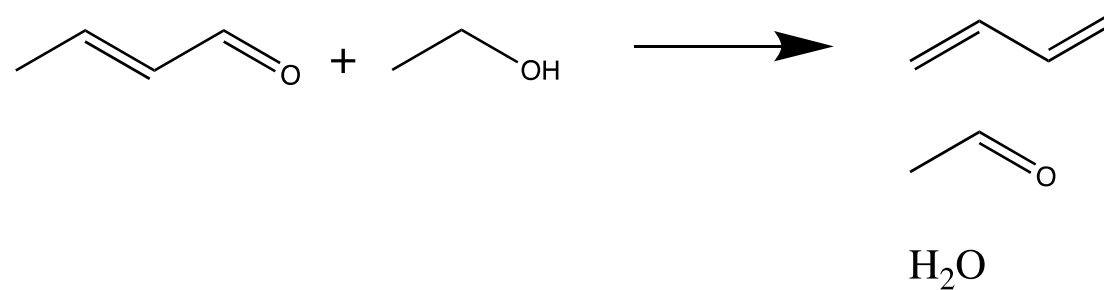
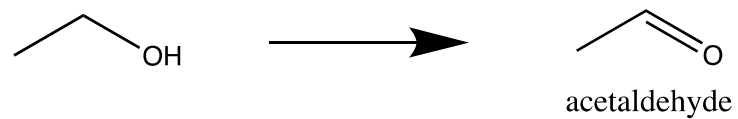


The two-stage extractive distillation process operates at pressures of 4-5 atm and utilizes aqueous n-methyl-2-pyrrolidone (NMP) or dimethylformamide (Leuna, BUTEX process) as a solvent

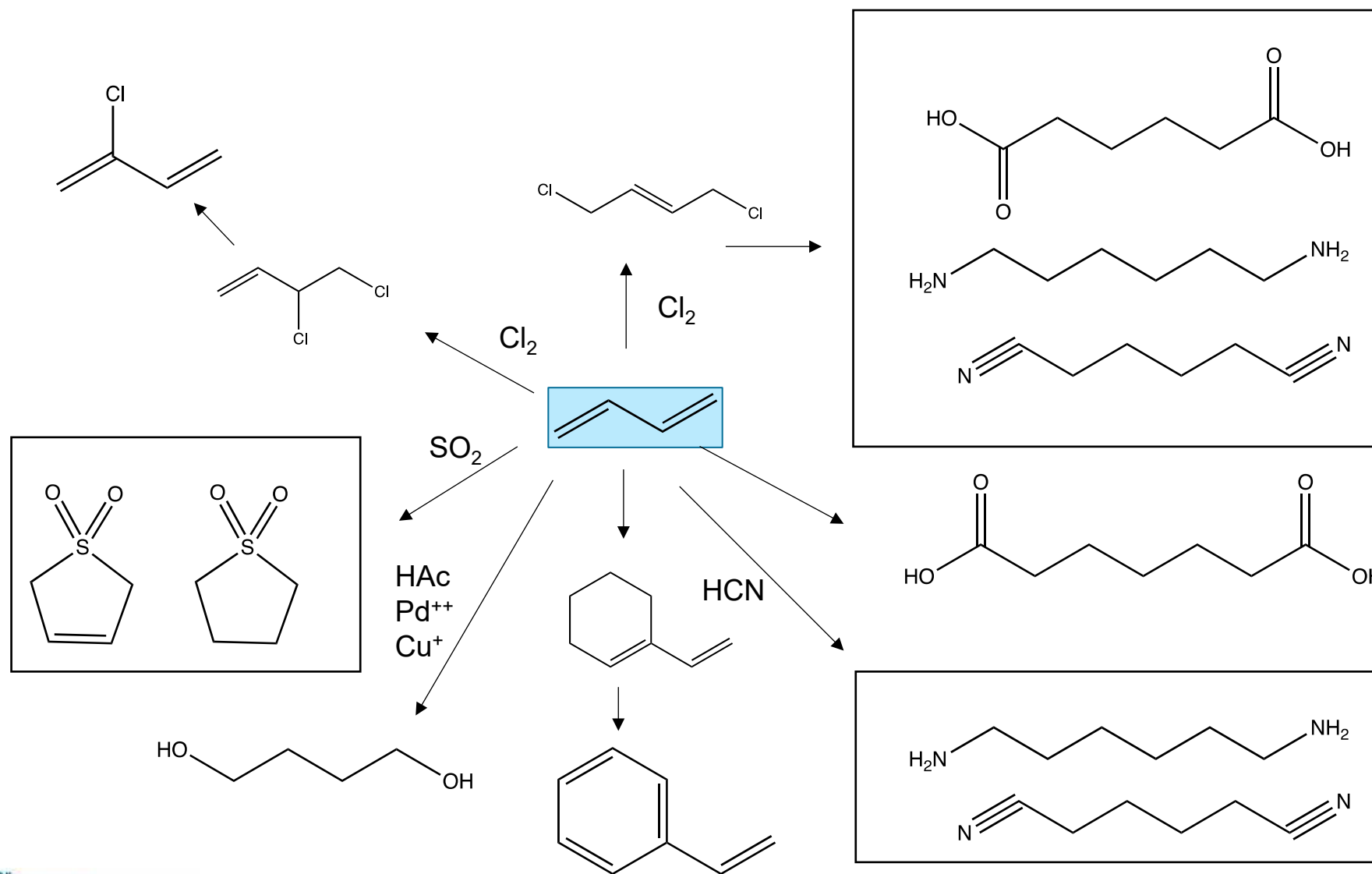


Lebedew Processes

- Catalytic dehydration of ethanol
- Developed in Russia in the 1920ies
- No commercial importance



Butadiene: relevant derivatives

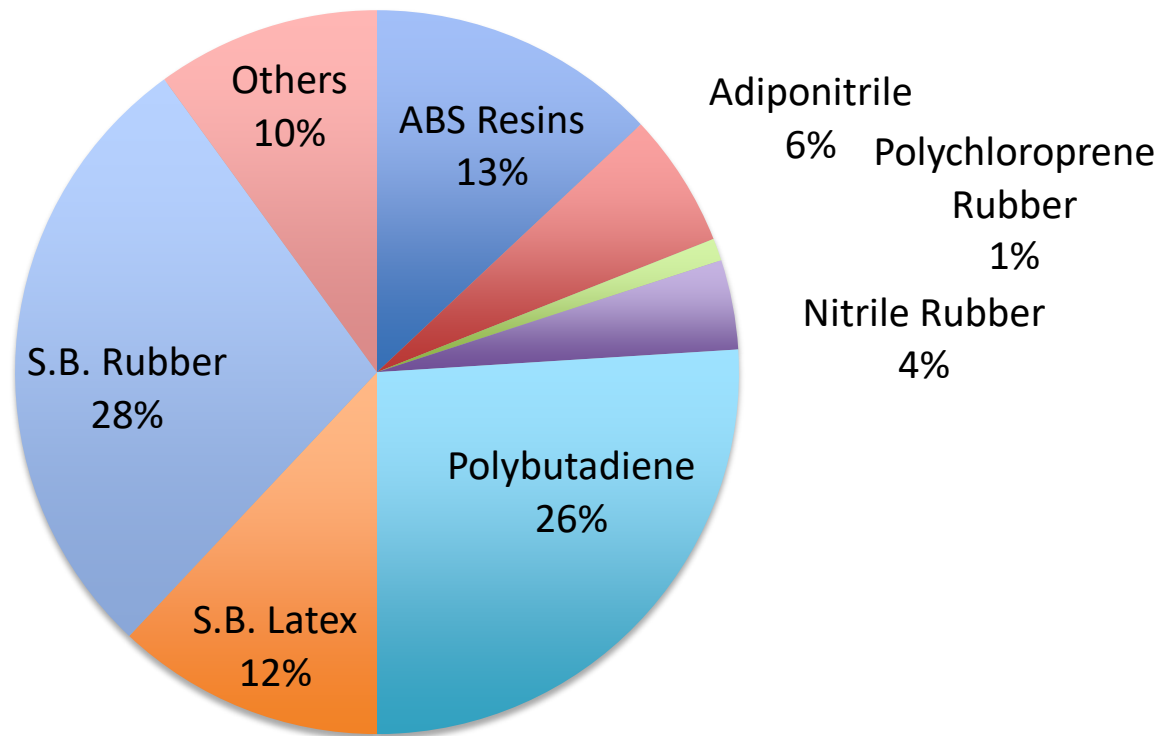


Butadiene: relevant derivatives

- Stereospecific polymerization: Polybutadiene Rubber (BR)
- Copolymerization with
 - Styrene → HIPS
 - Acrylonitrile → SAN
 - Styrene & Acrylonitrile → ABS

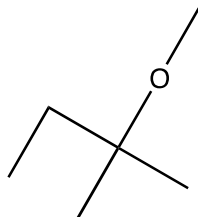
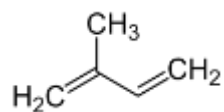
Butadiene: world demand by sector

2007: total demand 9,7 Million tons



Higher olefins

- Olefins with more than 4 carbons have an increasing number of isomers
- Separation of individual substances does not make sense
- C5 fraction: pentane, isopentane, pentene, isopentene, **isoprene**, cyclopentene, **cyclopentadiene**, pentadiene
- Isoprene: rubber, TAME

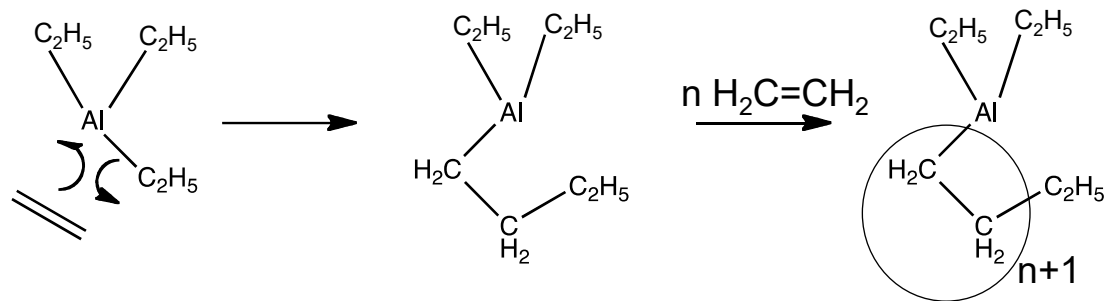


Linear higher olefins

- Technically of interest to C18
- Only mixtures (e.g. C₆-C₉, C₁₀-C₁₃, C₁₄-C₁₈)
- Characterized by location of double bond
 - Terminal (α)
 - Internal
- 2022: 6,5 Mio t world capacity (growth rate 6%)
- Oligomerization of ethylene
- Dehydration of paraffin's

Alfen process

- Synthesis of higher α -Olefins based on Ziegler (Alfen) process
 - 2 step process
1. Build-up reaction: 90-120°C, 100 bar insertion of ethylene in TEAL and growth of alkyl chain



2. Displacement reaction: 200-300°C, 50 bar release of α -olefins (even number of carbons)

Karl Ziegler

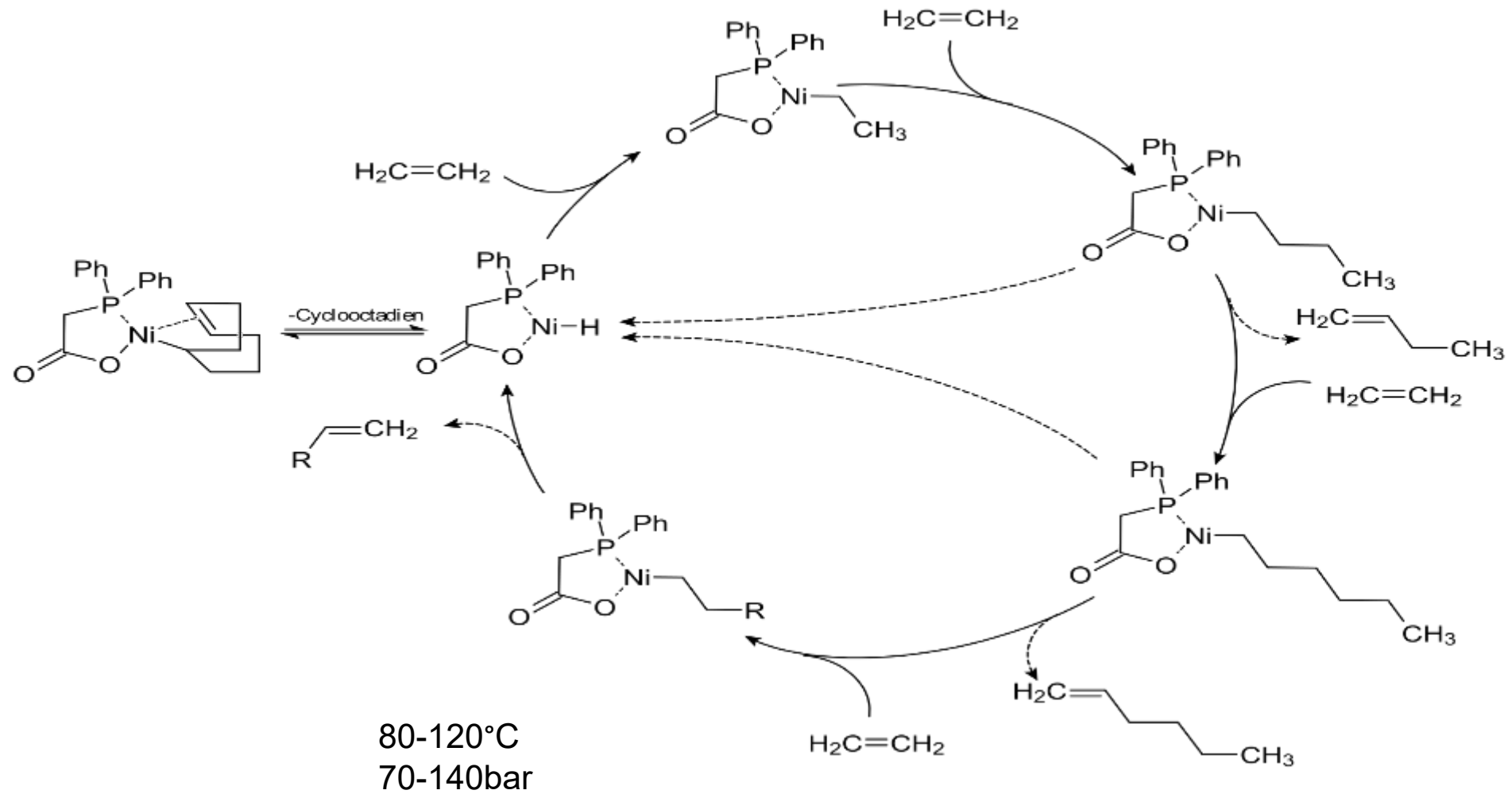
- 1898 – 1973
- Nobel prize 1963 (together with G. Natta for their works on the polymerization of olefins – difficult relationship)
- 1943 – 1969: head of “Kohleforschungsinstitut” Mühlheim (since 1949 Max Planck Institute)
- Research target: Polyethylene
- Until 1953 only small molecular weights (Al catalysts)
- Development of co-catalyst system (by chance)



Shell higher olefin process (SHOP)

- Ethylene is converted to α -olefins with a statistical distribution
- Lower oligomers are favored (Schulz–Flory distribution)
- 80–120°C and 70–140 bar in the presence of a nickel catalyst with phosphine ligands such as $\text{Ph}_2\text{PCH}_2\text{COONi}$
- The product mixture is separated into C_{4-10} , C_{12-18} , and C_{20+} fractions by distillation

SHOP process

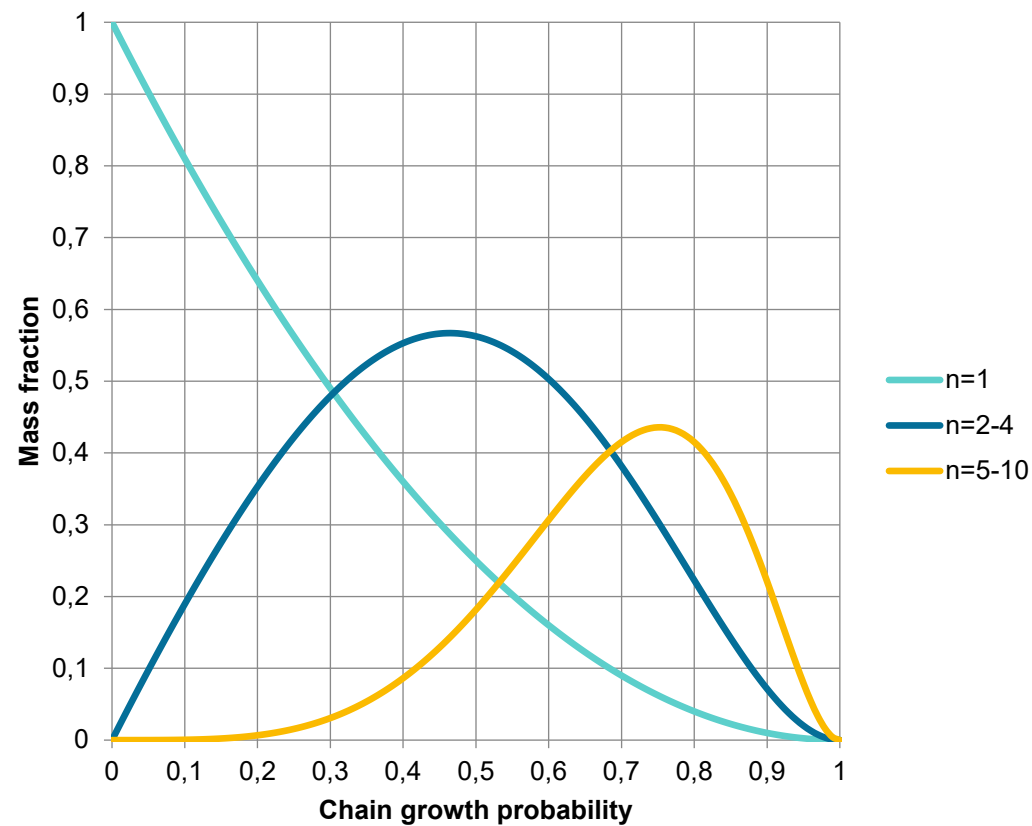


SHOP – product distribution

Anderson Schulz Flory distribution

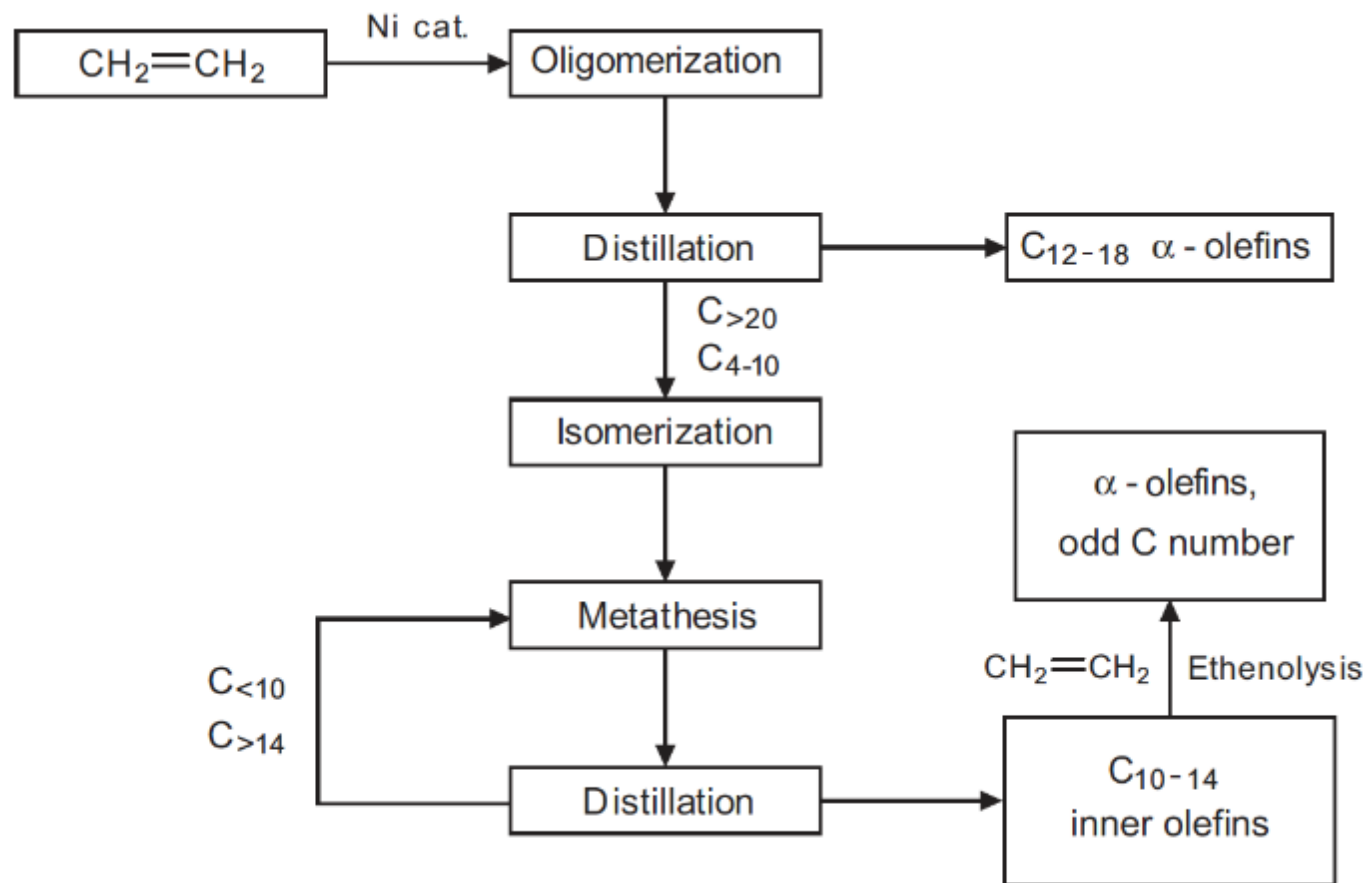
$$W_n = n(1-\alpha)^2\alpha^{n-1}$$

Plot of calculated selectivity
(% carbon atom basis) of
carbon number as function
of probability of chain growth



SHOP

- The C_{12-18} fraction contains α -olefins with the desired chain length for the detergent industry
- The top and bottom olefins are subjected to a combination of double bond isomerization and metathesis
- Isomerization gives a mixture of inner olefins with a statistical distribution of the double bond, metathesis of which gives a new mixture of inner olefins from which the C_{10-14} olefins can be separated by distillation



Branched higher olefins

Mainly via oligomerization of lower olefins such as propylene, isobutene and butene

Educts	Process	Products
Propylene	Dimerization with AlR_3	2-Methyl-1-pentene
Propylene	Dimerization with alkali metals	4-Methyl-1-pentene
Propylene	Tri- and tetramerization at H_3PO_4	Isononene, Isododecene
Propylene and Butene	Co-dimerization	Isoheptene
Isobutene	Dimerization on H_3PO_4 /carrier or with H_2SO_4	Diisobutene

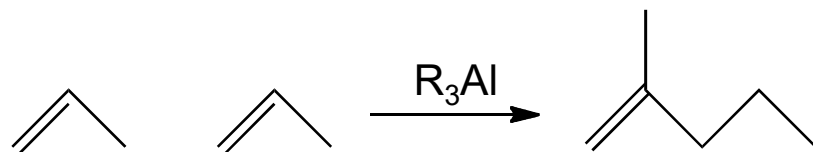
Branched higher olefins

Used catalyst systems:

- a) Acid catalysts with mineral acids
(H_3PO_4 , H_2SO_4 or acid ion exchanger)
Diisobutene: hydration, fuel additive
Diisobutene: hydroformilation and hydration to isononanol, plasticizer
- b) Metal organic catalysts with Al-alkyls (co-catalysts)
Dimerisolprocess
- c) Metal organic catalysts with alkali metals
Production of 4MP1 – polymerized to Poly-4-methyl-1-pentene (TPX), highly transparent, T_m 220°C

Dimerisol process

- Dimerisation of olefins
- Homo- or co-dimerization of propylene and butylene



- Continuous process: 60°C, 15 bar, 50 % selectivity
- Relatively low branched isoheptenes
- Low selectivity also because of homodimerization of propylene and butene
- If only Homodimerization: selectivity 98-92 %
- With C6, C7, C8 isoolefins: fuels and further use in hydroformylation and hydration to isoalcohols (dialkylphthalates – plasticizer)

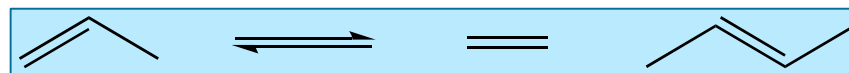
Olefin metathesis

- Derived from the Greek words **meta (change)** and **thesis (position)**
metathesis is the exchange of parts of two substances.
- In the reaction:
 $AB + CD \rightarrow AC + BD$
B has changed position with C
- 2005 Nobel prize for Yves Chauvin, Richard R. Schrock and Robert Grubbs
- Through carbene (alkylidene) exchange between the two starting olefins two new olefins have been formed:



Olefin metathesis

- Metathesis was discovered in the industry following observations in the 1950s of the polymerization of ethylene by Ziegler
(Nobel Prize in Chemistry 1963)
- In the beginning the mechanisms were not understood
- 1957 Du Pont: formation of **unsaturated polymers**
from highly-strained norbornene when it was added to molybdenum oxide on alumina combined with lithium aluminium hydride
- In the same year novel transformation **disproportionation of olefins**
conversion of propylene into ethylene and butene upon treatment with a mixture of triisobutylaluminum and molybdenum oxide on alumina



Olefin metathesis

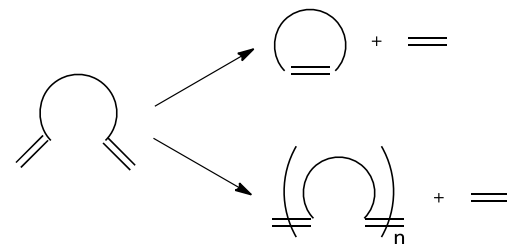
- In 1966 G. Natta (Nobel Prize in Chemistry 1963) and co-workers showed that combinations of tungsten hexachloride with either triethylaluminum or diethylaluminum chloride can polymerize cycloheptene, cyclooctene and cyclododecene
- Calderon suggested that the polymerization of cyclic alkenes to polyalkenemers and the disproportionation of acyclic alkenes are the same type of reaction and **named the reaction olefin metathesis**

Olefin metathesis

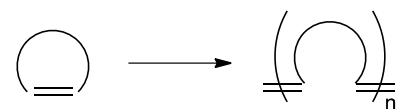
Cross metathesis (CM)



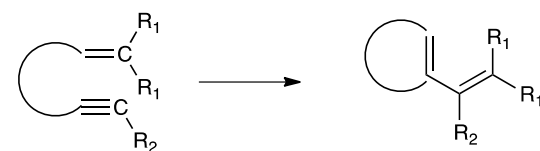
Ring-closing metathesis (RCM)



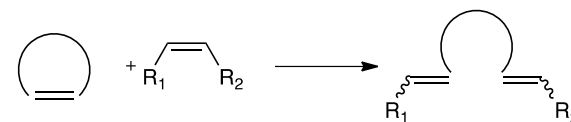
Acyclic diene metathesis polymerization (ADMP)



Ring-opening metathesis polymerization (ROMP)



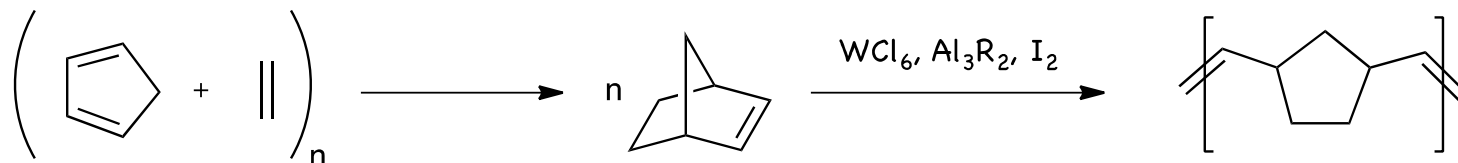
Enyne metathesis (EYM)



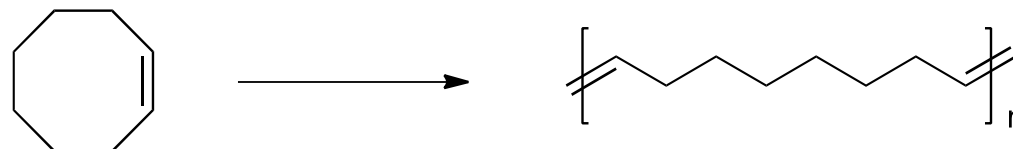
Ring-opening cross metathesis (ROCM)

Olefin metathesis: examples

- Polynorbornene – Norsorex process (CdF Chemie/Nippon Zeon):
Rubber like material: noise and vibration damping, oil spill sponge

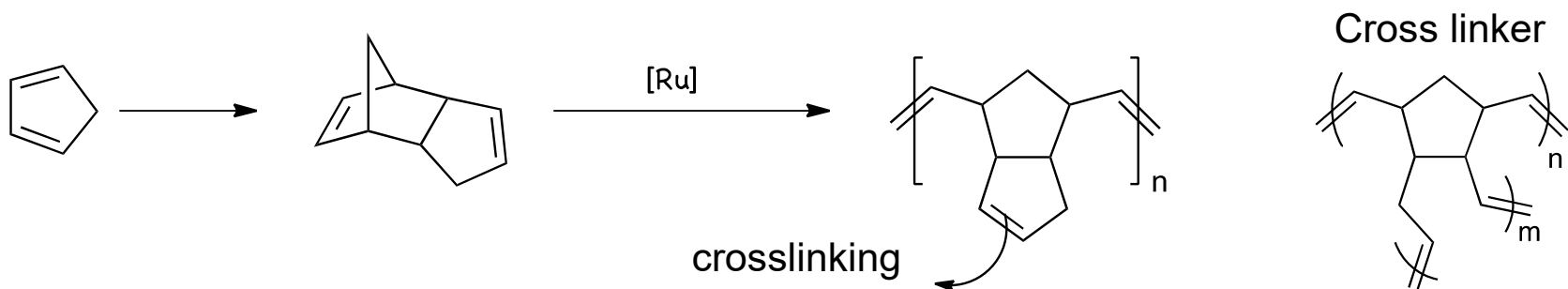


- Polyoctenamer – Hüls, Vestenamer:
Polymerization of cyclo octene to polyoctenamer
Cross linkable elastomer (rubber like material)



Polymerization of dicyclopentadiene

- ROMP of monomers with unsaturated and stressed rings



- Polydicyclopentadiene: sports (baseball bat), technical parts for automotive etc., bullet proof vest

Acetylene

- Molar mass: 26,04 g/mol
- Density: 1,18 g/dm³ (0°C)
- Ts: -83,8°C
- Roughly 12 Mill t produces worldwide
- Applications
 - Chemical synthesis
 - Welding and cutting
 - Metal fabrication
 - Others

Acetylene: world demand by sector

Product [%w]	USA 1982	USA 1997	WE 1982	WE 1997
Vinylchloride	43	32	33	-
1,4 Butanediol	28	59	12	42
Vinylacetate	17	-	10	41
Carbon black	8	5	45	4
Others	4	4		13
Total [1000 t]	115	151	ca. 320	150

Acetylene is replaced by olefins because they are cheaper and easier (safer) to handle.

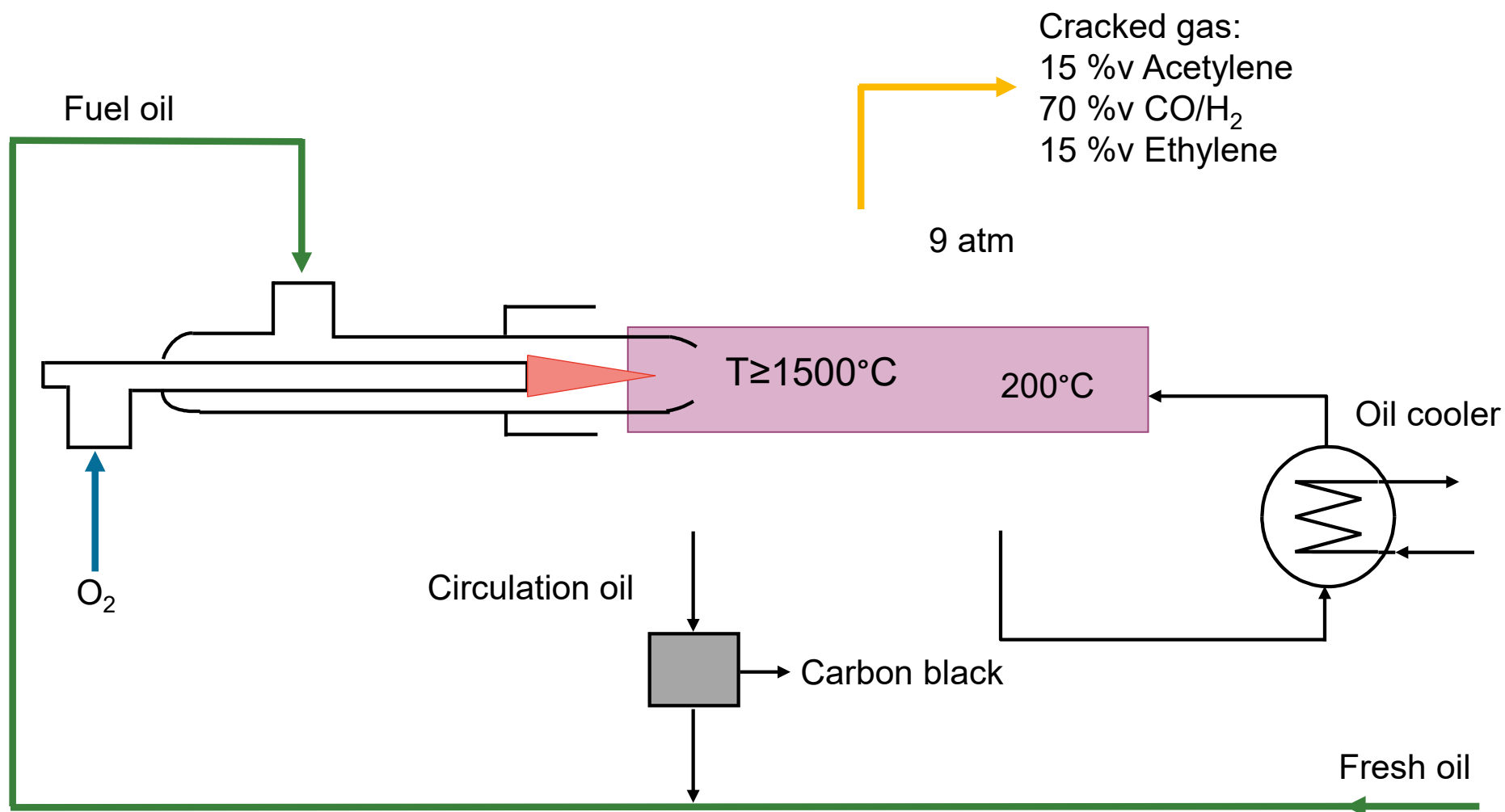
However, they are not as reactive!

Acetylene

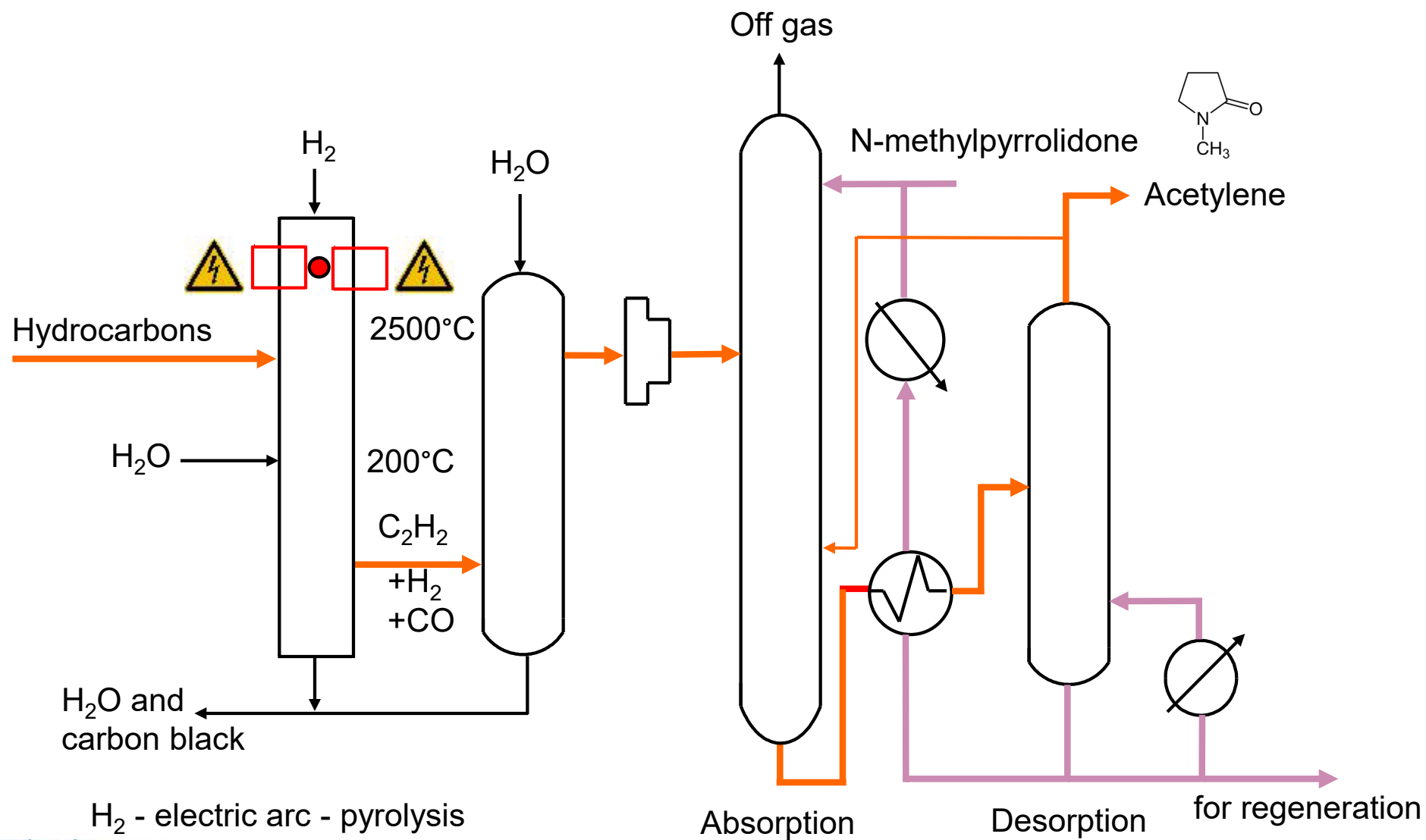
Reasons why still in use:

1. Existing plants, new route would mean high investments
2. Small product volumes and C2 share is small as well, e.g. vinyl ester and ether of higher alcohols
3. No economically feasible alternatives
e.g. 1,4 Butanediol or acetylene carbon black
4. Coal as source for acetylene reduces dependence on oil

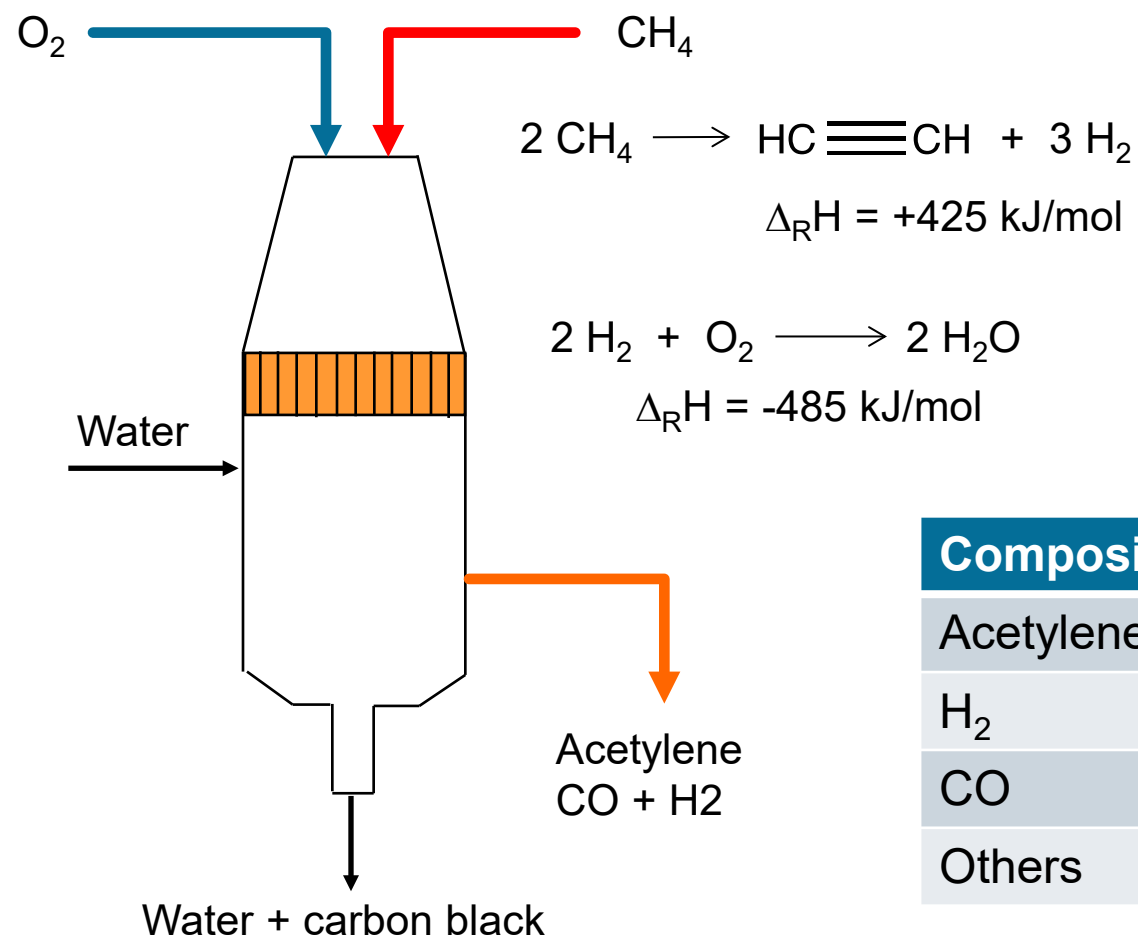
Acetylene production – diving flame



Acetylene production – electric arc



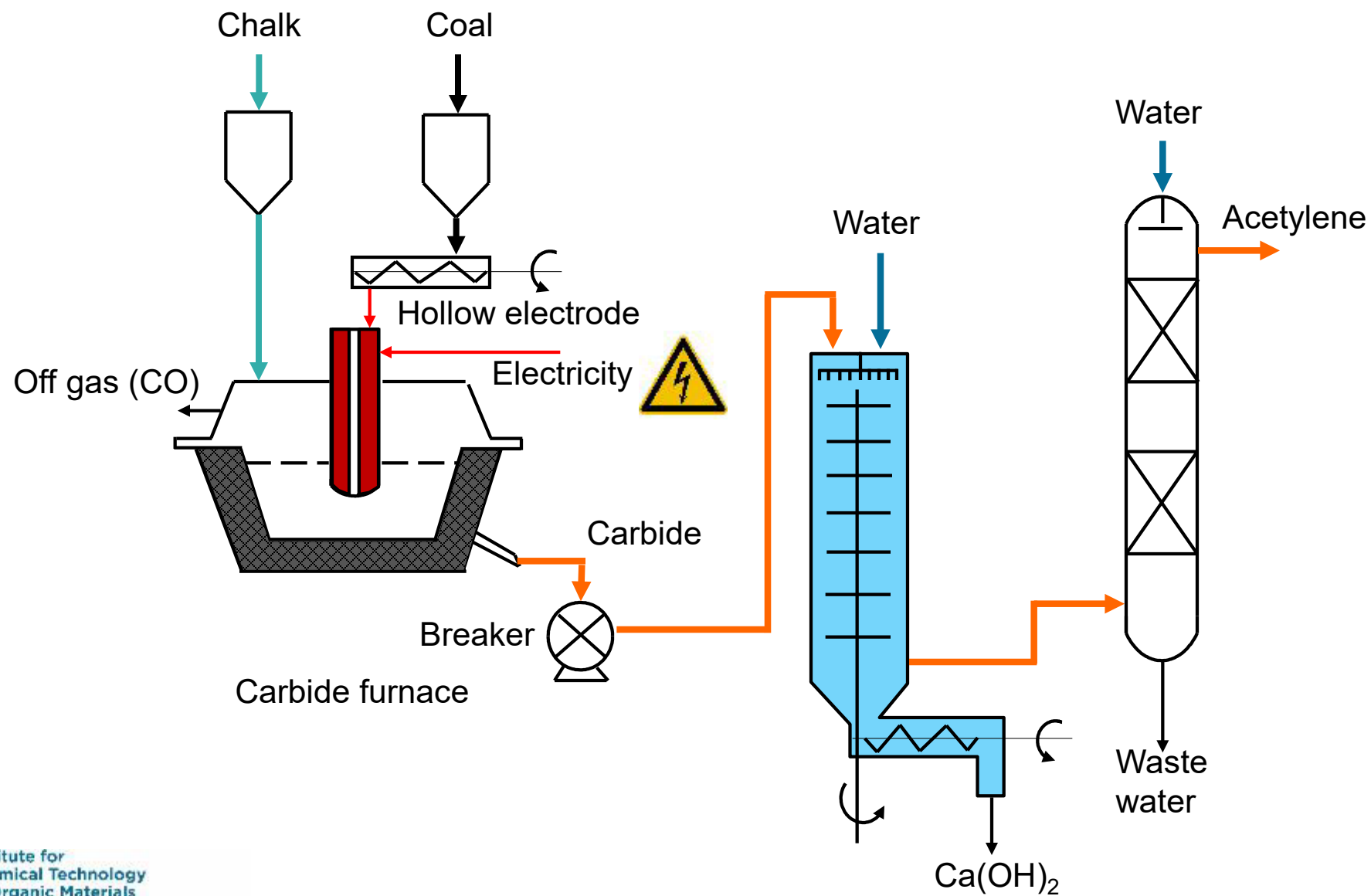
Acetylene production - Sachsse



Composition of reaction gas [%v]

Acetylene	10
H ₂	55
CO	25
Others	10

Acetylene production – CaC_2



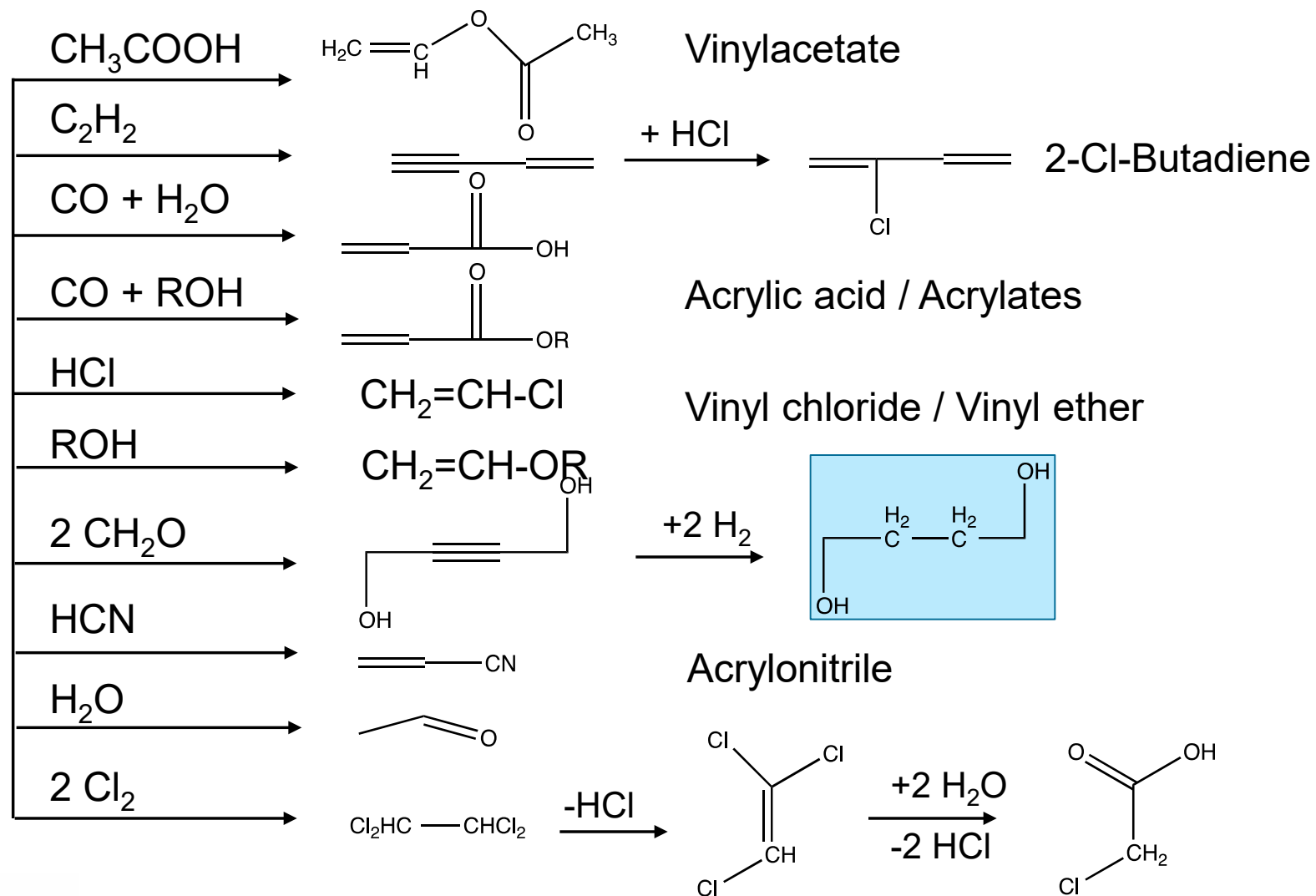
Reactions with acetylene - Reppe chemistry

- Reppe began his interest in acetylene in 1928
- BASF: forbidden to compress acetylene over 1.5 bar at BASF
- BASF was warned of Reppe in a letter to blow up the factory
- In fact in 1950 he did it by throwing a cigar into a sink in a laboratory
- His interest was the synthesis of small molecules, monomers for new polymers



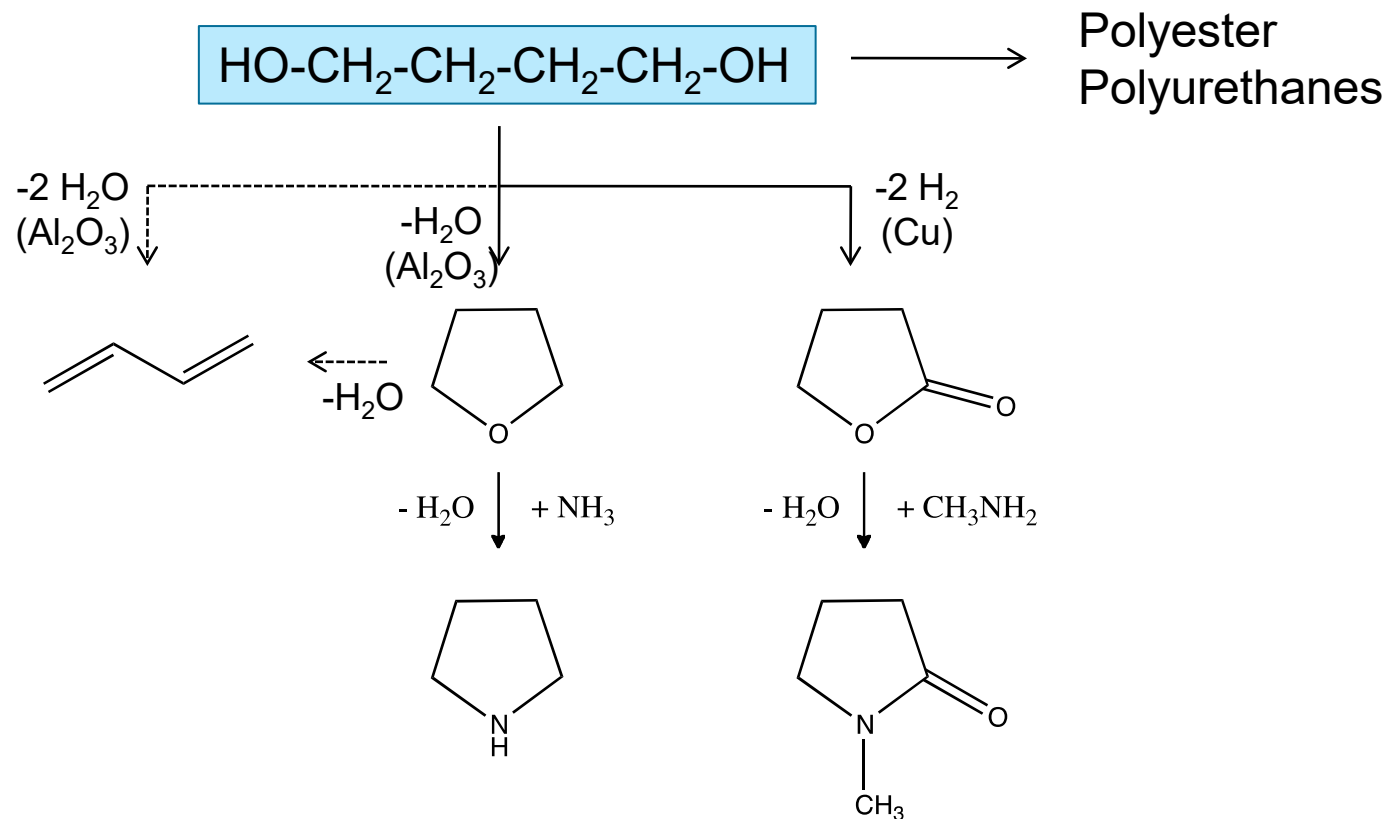
Reactions with acetylene

Acetylene



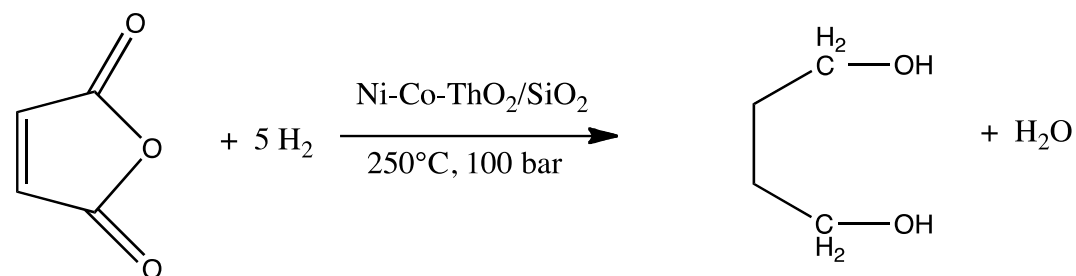
Butanediol

Technically important reactions

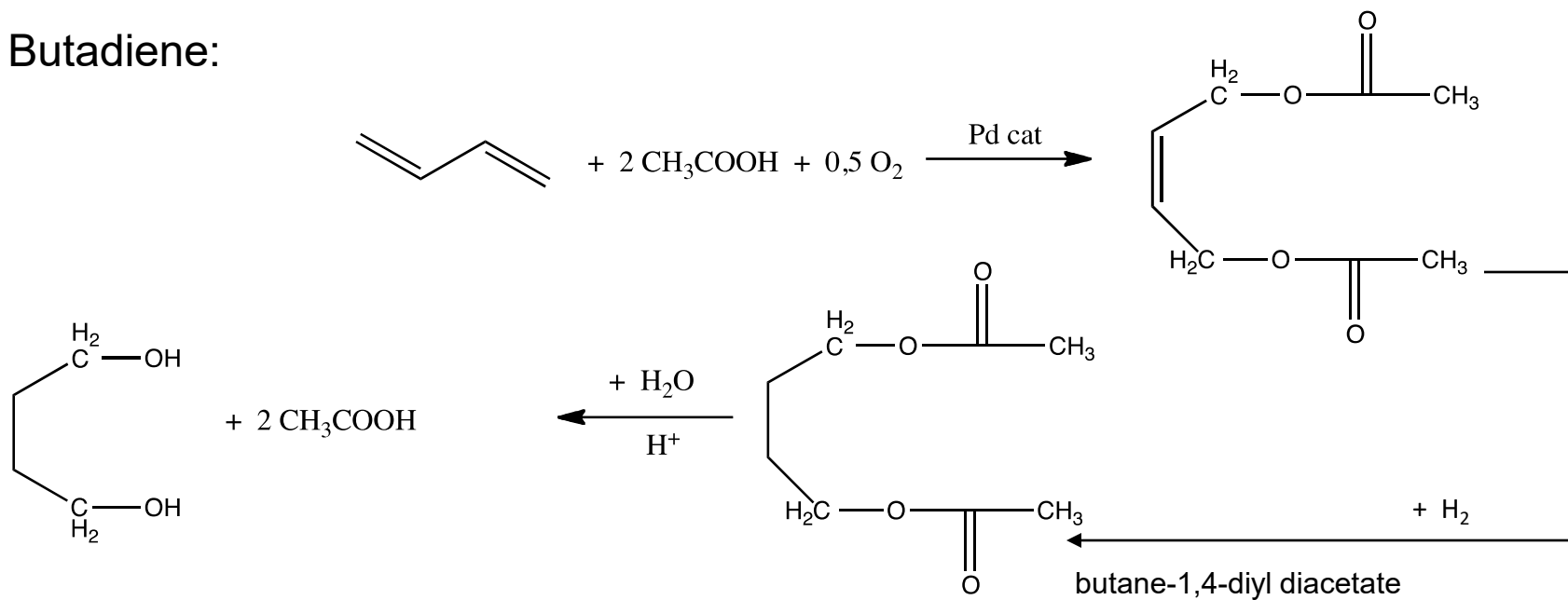


Butanediol

- MSA (ICI) process:



- Butadiene:



Caution

When working with acetylene:

- Avoid pressures > 2 bar when working with pure acetylene
- Acetylene concentration to be $< 70\%$ for p between 2 and 20 bar (dilution with N_2)
- Do not pressurize acetylene containing mixtures above 20 bar
- Design vessels to withstand 12 to 13 times the working pressure (rupture discs)
- Fill big reactors with packing material (“wall effects”)
- Do not use copper or copper containing materials (Cu-acetylide)
- Explosion limits: 2,5 – 80%



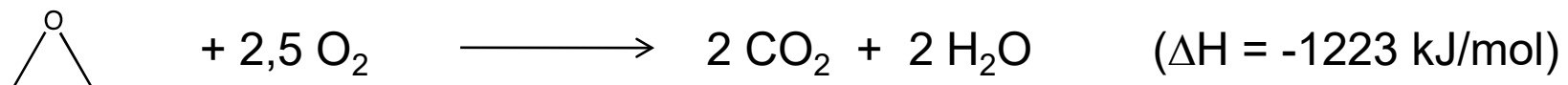
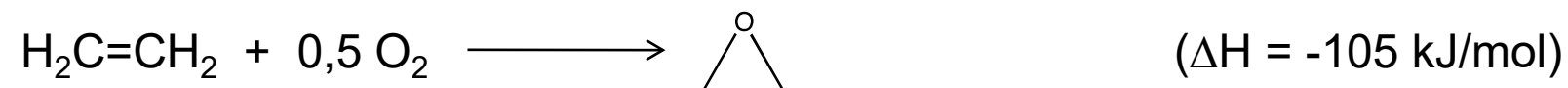
Oxidation products of ethylene

Ethylene oxide

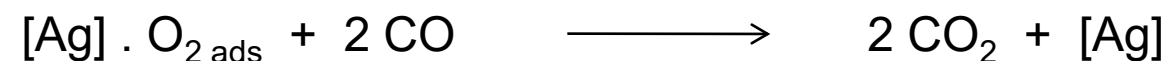
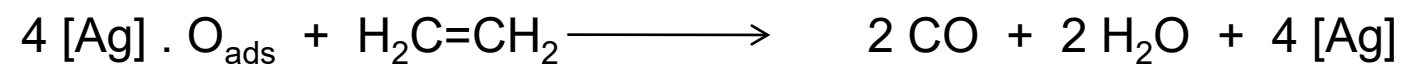
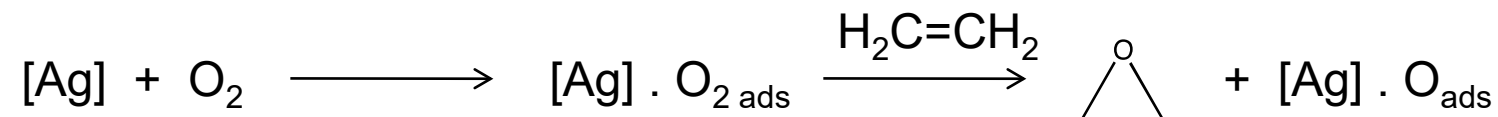
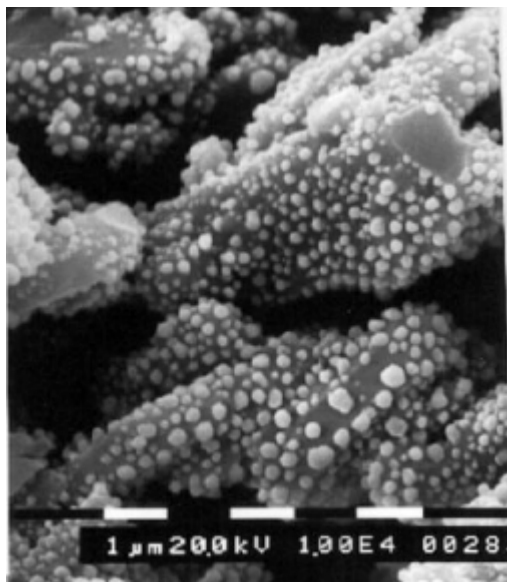
- Molar mass: 44,05 g/mol
- Density: 1,93 kg/m³ (15°C)
- Ts: -112,55°C
- Tb: 10,45°C
- Roughly 32 Mill t produced worldwide



Ethylene oxidation - basics



Ethylene oxidation - mechanism



EO production

T: 270-290°C

p: 30 bar

Ag/Al₂O₃

Selectivity: 75%

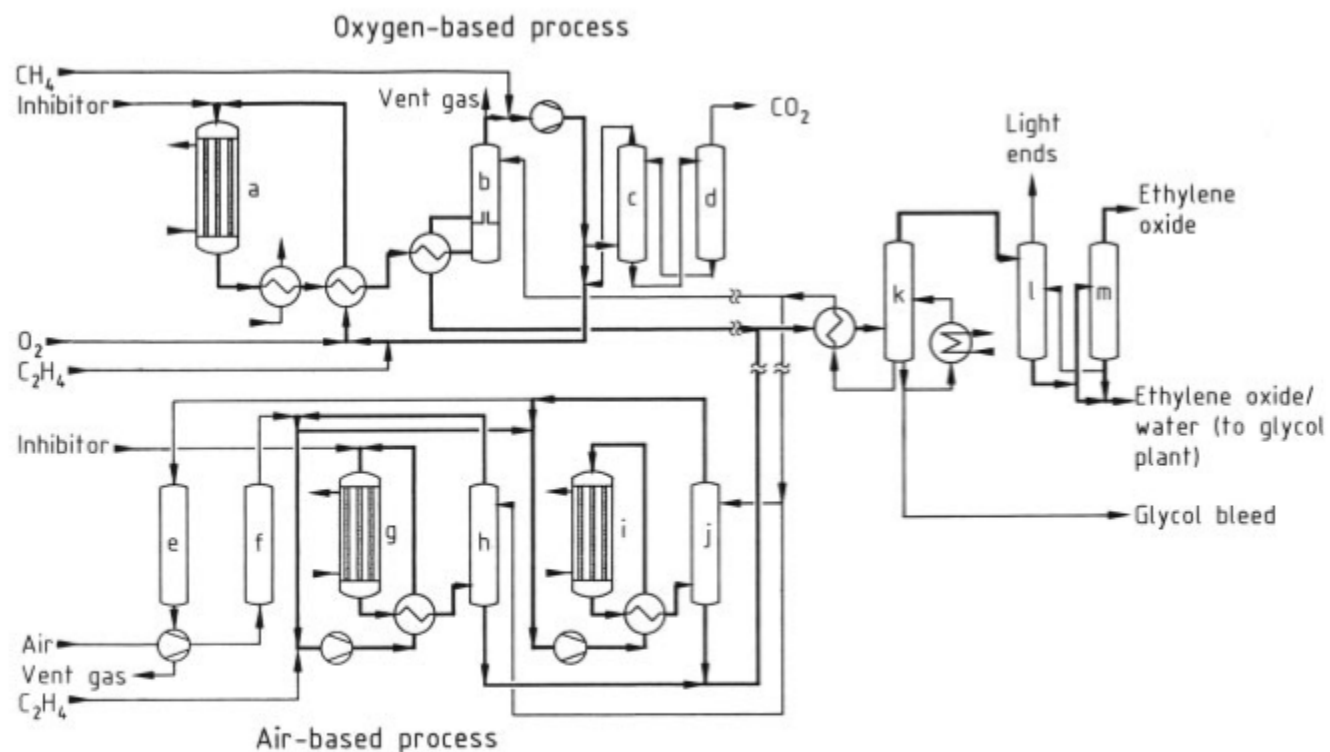


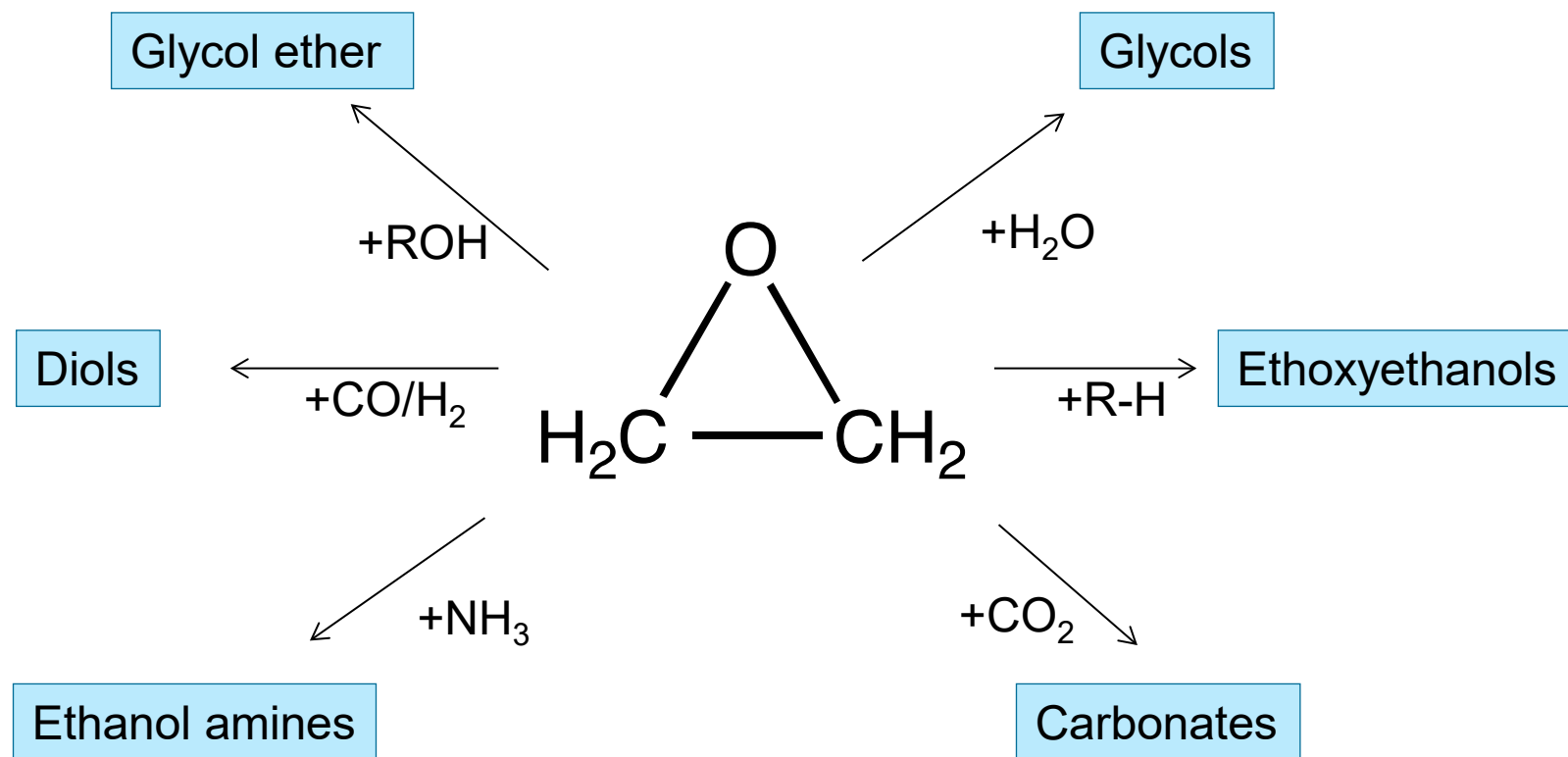
Figure 5. Flow scheme for ethylene oxide production by the oxygen-based or air-based oxidation of ethylene
a) Reactor; b) Ethylene oxide scrubber; c) CO₂ scrubber; d) CO₂ desorber; e) Off-gas purification; f) Air purification; g) Primary reactor; h) Primary ethylene oxide scrubber; i) Secondary reactor; j) Secondary ethylene oxide scrubber; k) Ethylene oxide desorber; l) Stripping column; m) Ethylene oxide distillation

Space-time-yield: 600-700 g EO / (l_{catalyst} h)

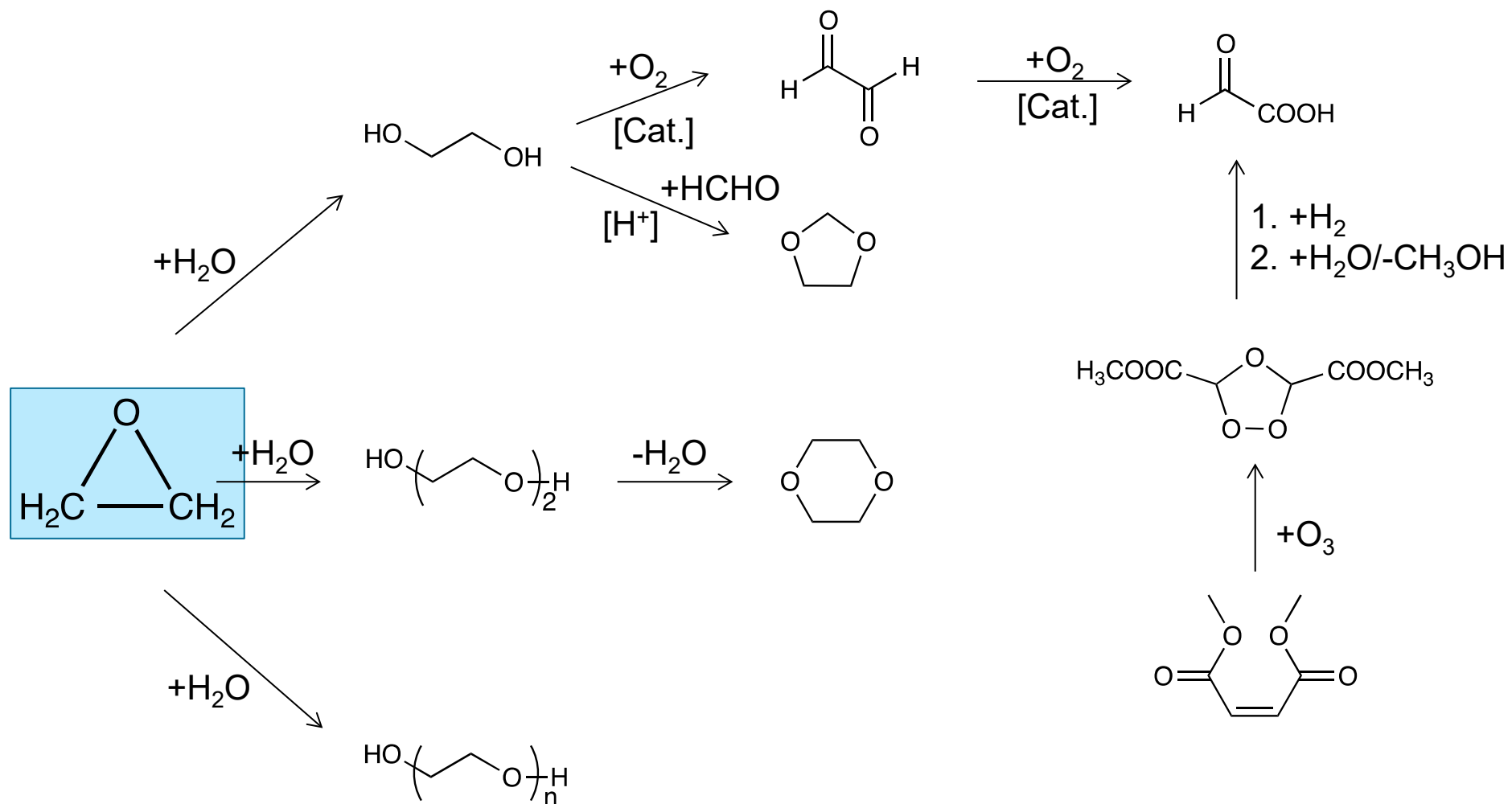
Use of EO

- EO only in small amounts directly used (e.g. for pest control, sterilizing agent, fermentation inhibition)
- Importance due to reactivity of oxirane ring – enables many intermediates and end products
- Basic principle: exothermal ring opening with nucleophile partners
 - H_2O , ROH , NH_3 , RNH_2 , RCOOH , phenols, mercaptans
 - Normally water solubility increases because of the addition of a hydroxyethyl group
 - Reaction speed can be increased with acid as well as alkaline catalysts (mineral acids or ion exchanger are preferred)
- Primary products of the addition have a reactive hydroxyl
- Can add additional EO -> di-, tri- and polyethoxy products
- For mono ethoxylation products EO to be used under stoichiometric ratios

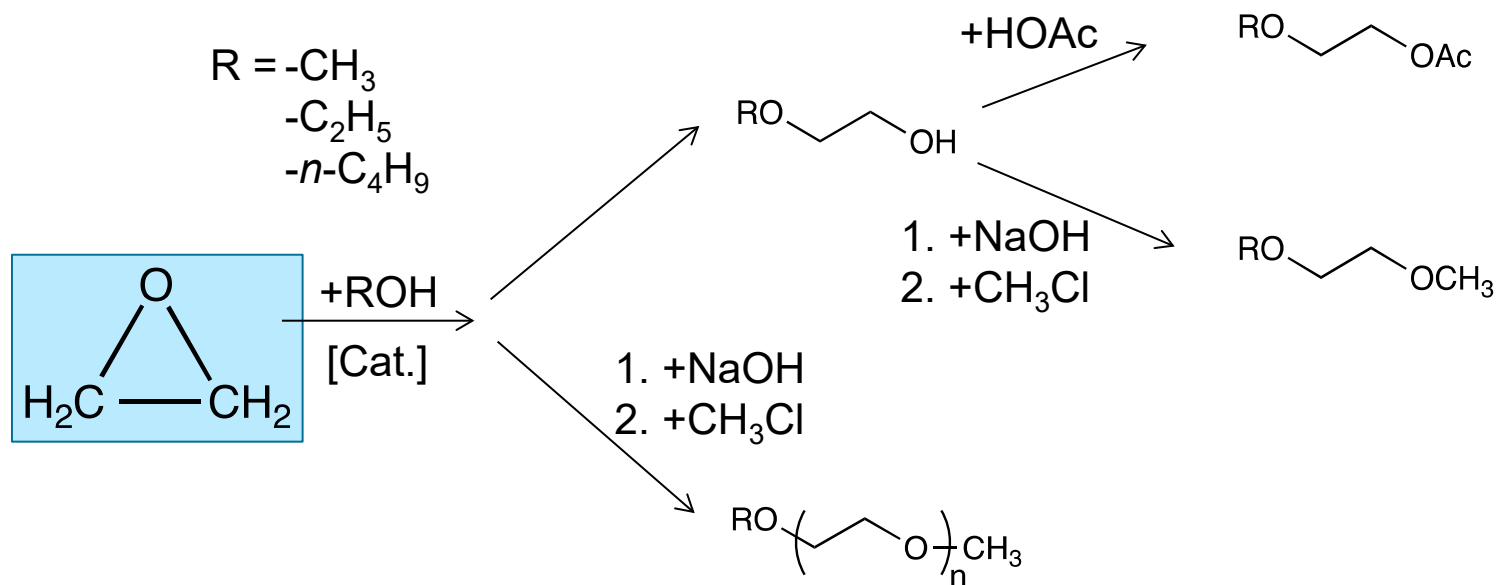
EO family tree



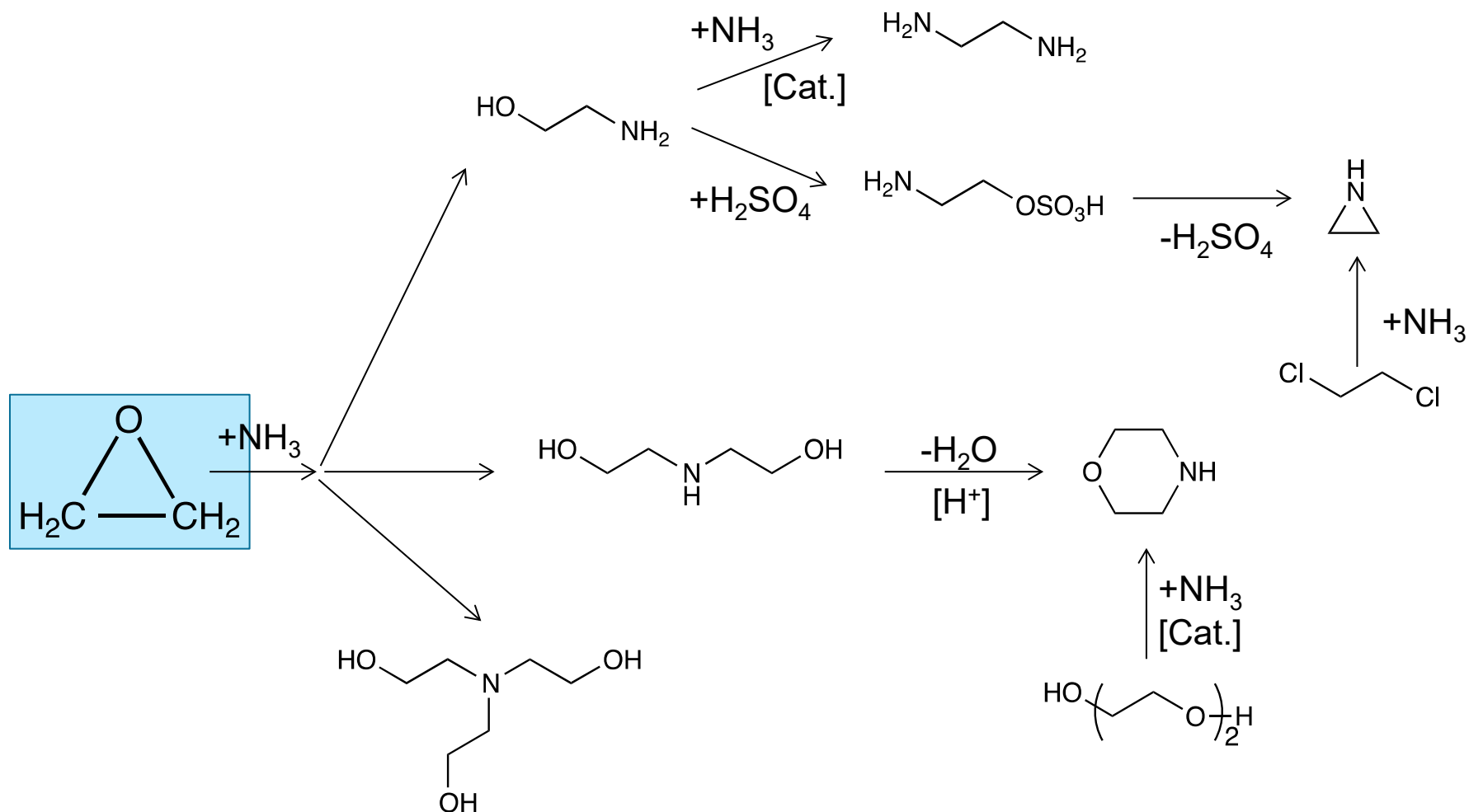
E0 family tree - glycols



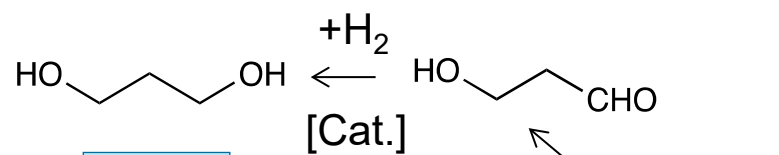
EO family tree – glycol ether



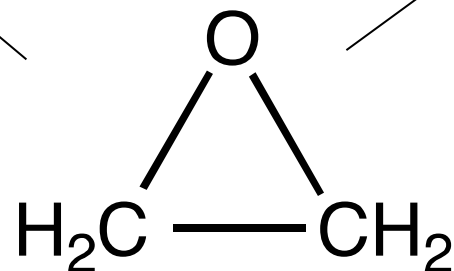
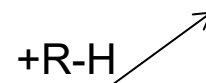
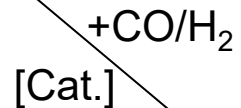
EO family tree – ethanol amines



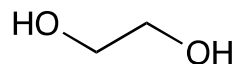
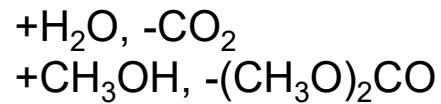
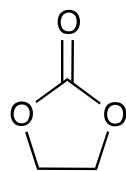
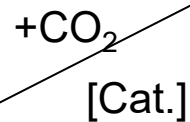
EO family tree



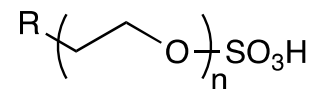
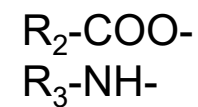
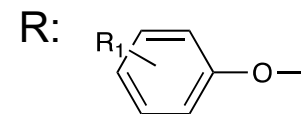
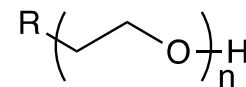
Diols



Carbonates



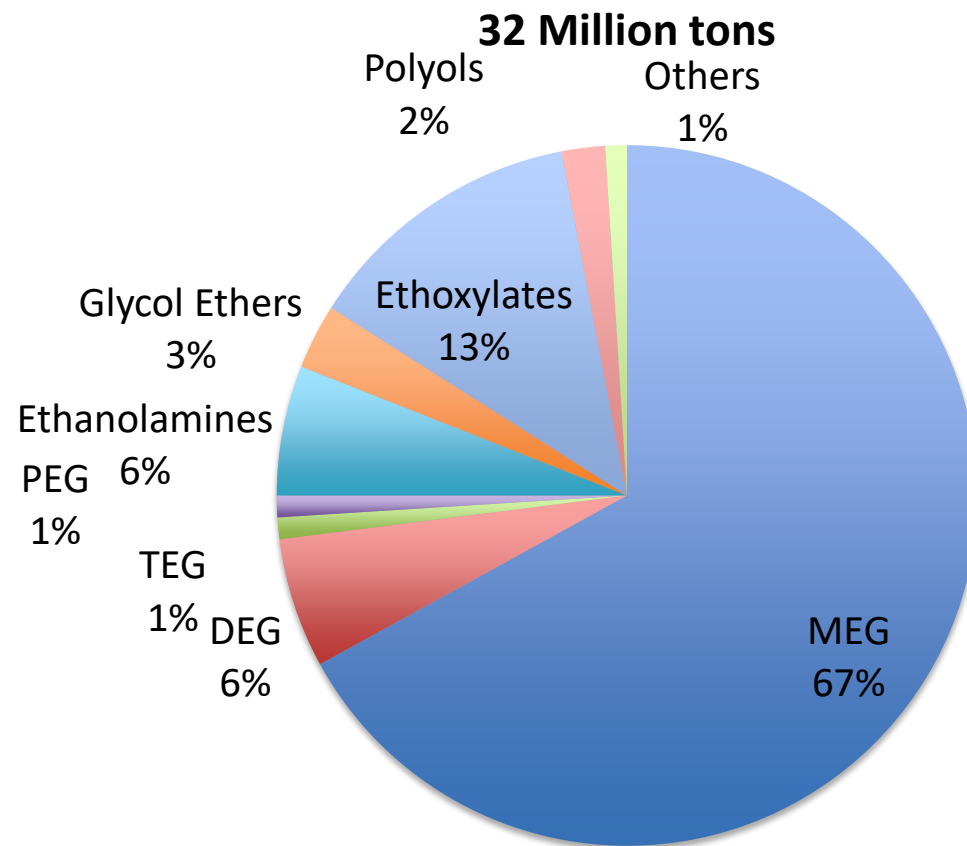
Ethoxyethanols



Use of EO

Partner	Product	Secondary product
H ₂ O	Ethylene glycol Di ethylene glycol Poly ethylene glycol	Glyoxal, Dioxane
Alkyl phenols Fatty alcohols, acids, amines	Poly ethoxylates	
NH ₃	Mono-, di- and tri ethanolamines	Ethylene imine Morpholin
ROH R: H, CH ₃ , n-C ₃ H ₇	Glycol mono alkyl ether Diglycol mono alkyl ether	Glycol dialkyl ether Ester of glycol mono alkyl ether
CO ₂	Ethylene carbonate	Carbamate
Syn gas	1,3 propane diol	Polyester

Ethylene oxide: world demand by sector



Ethylene glycol

- Molar mass: 62,07 g/mol
 - Density: 1,11 g/cm³
 - T_m: -16°C
 - T_b: 197°C
 - 2022: 57 Mill t
- Production in Mill. t

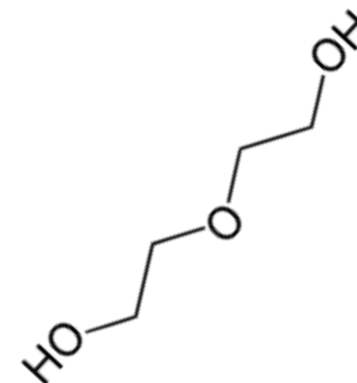


Use of ethylene glycol

2 main application areas

1. Diol for polyester (PET)
2. Anti freeze in engine cooling systems
3. In Austria: wine scandal

1997 [%-w]	World	USA	WE
Anti freeze	16	26	24
Poly ester fiber	57	24	30
Poly ester film	5	4	31
Others	22	46	15
Total [mill. t]	9,1	2,52	1,50

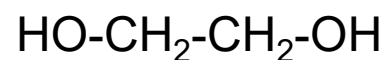


Ethylene glycol

- EO plus water (about 10 times)
- Liquid phase
- Normal pressure (20-40 bar)
- 50-70°C (140-230°C)
- Presence of acid catalyst (H_2SO_4)
- Raw glycol solution through re-boiler concentrated to 70%
- Series of vacuum distillation columns
- Selectivity for mono ethylene glycol ca. 90% (9% di-, 1% tri-)
- Total conversion 95-96%
- Purity: especially important for polymerization grades (99,9%)

Ethylene glycol: syngas

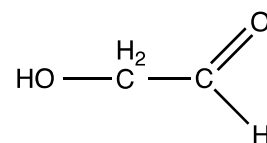
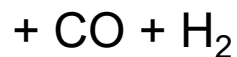
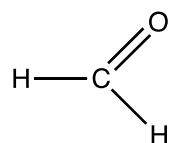
Direct:



Rh – carbonyl
cluster

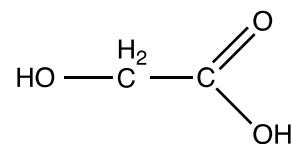
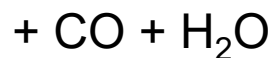
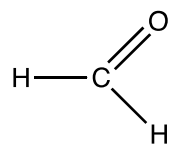
UCC

Indirect:



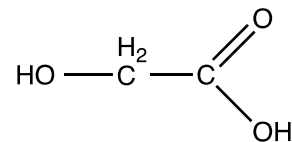
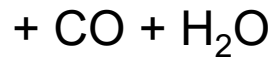
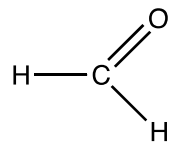
Co, Rh – carbonyl
complexes

Ajinomoto
Chevron



Rh – carbonyl
complexes

Chevron

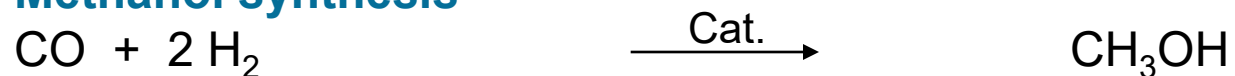


Strong acids
e.g. HF, BF₃/HF

Chevron

Ethylene glycol: formaldehyde

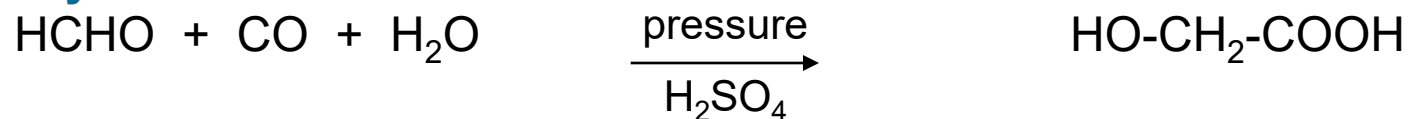
Methanol synthesis



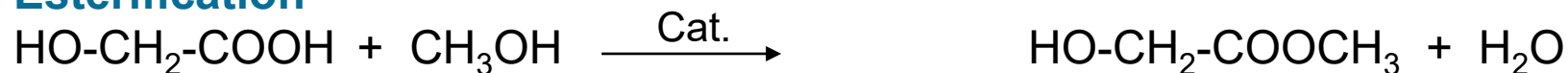
Formaldehyde via dehydration



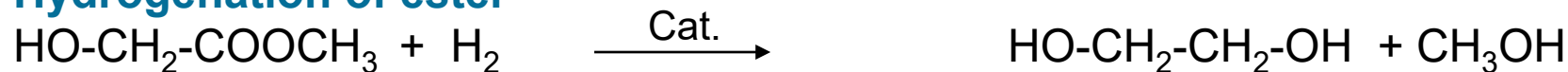
Glycolic acid



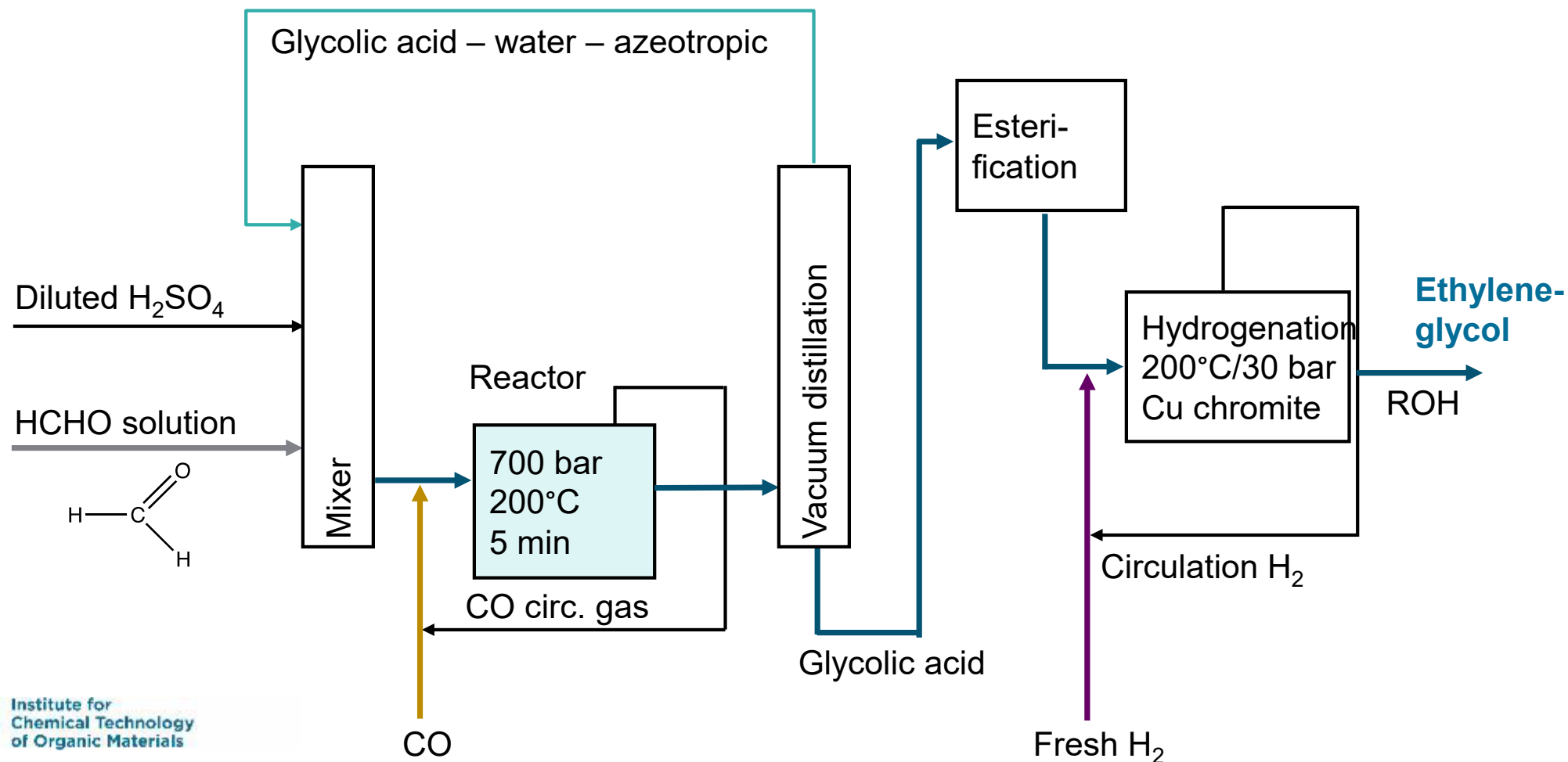
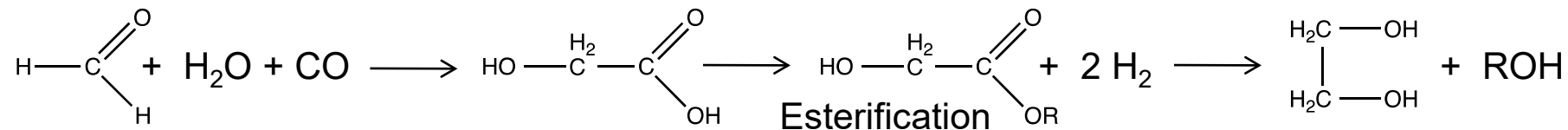
Esterification



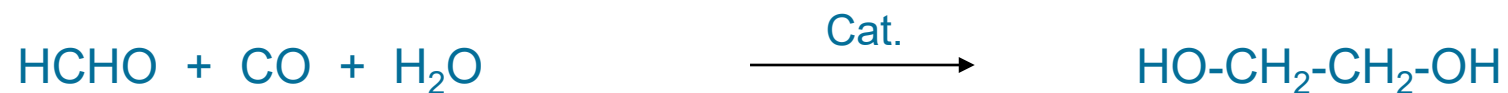
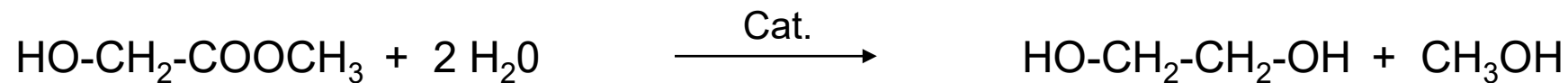
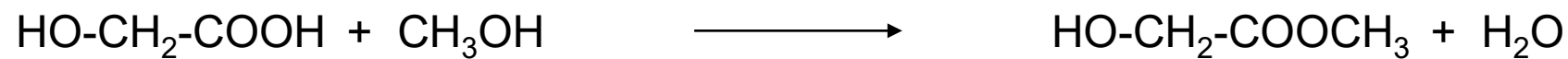
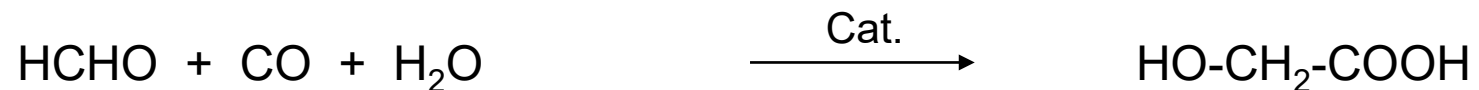
Hydrogenation of ester



Ethylene glycol: formaldehyde

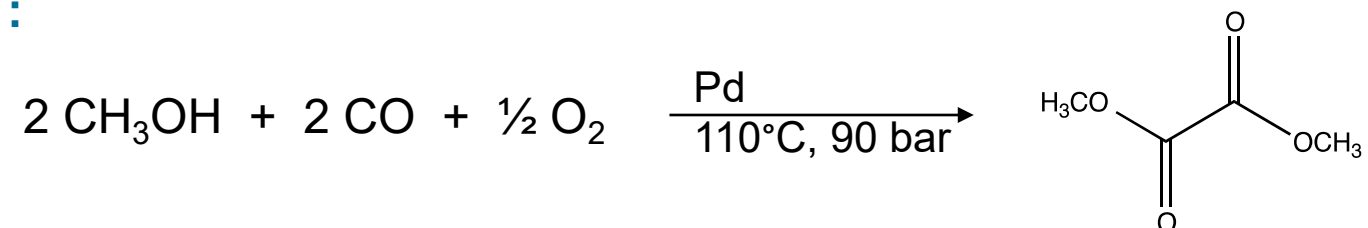


Ethylene glycol: syngas

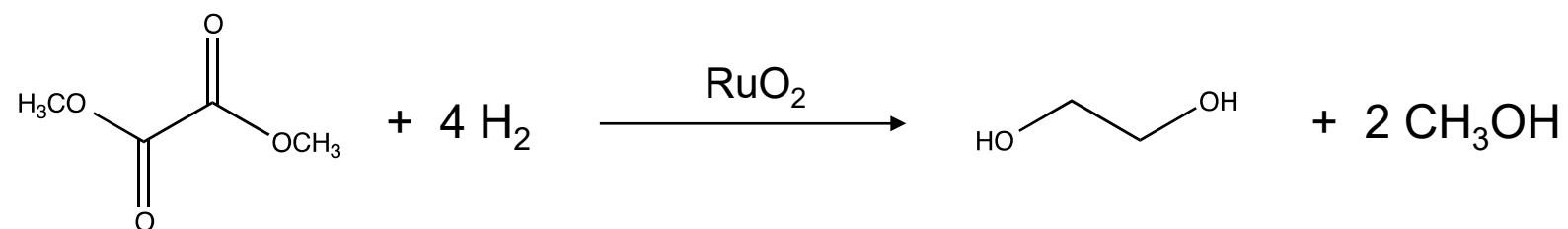


Ethylene glycol: carbonylation

Step 1:



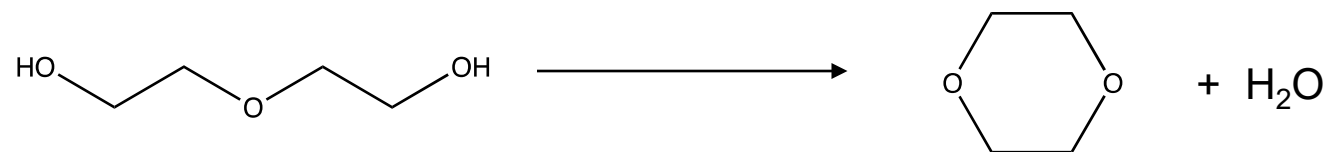
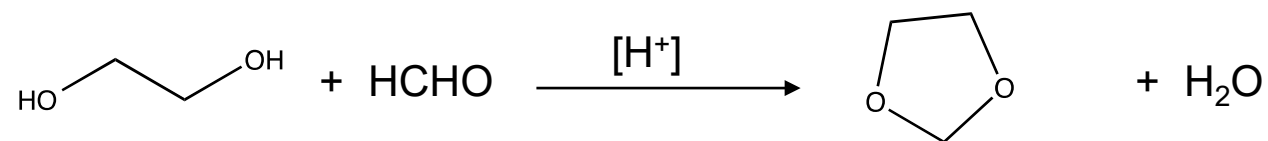
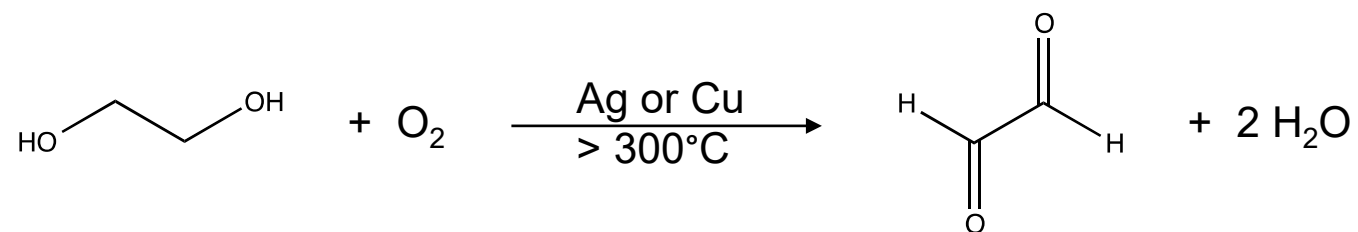
Step 2:



Instead of CH_3OH also higher alcohols (butanol) can be used

Derivatives

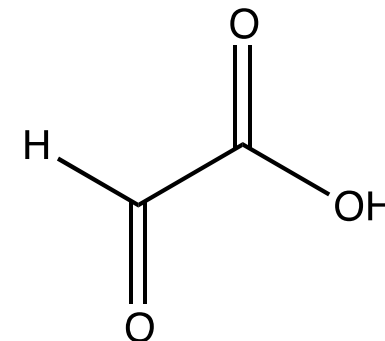
- Glyoxal
- Dioxolan
- 1,4-Dioxan



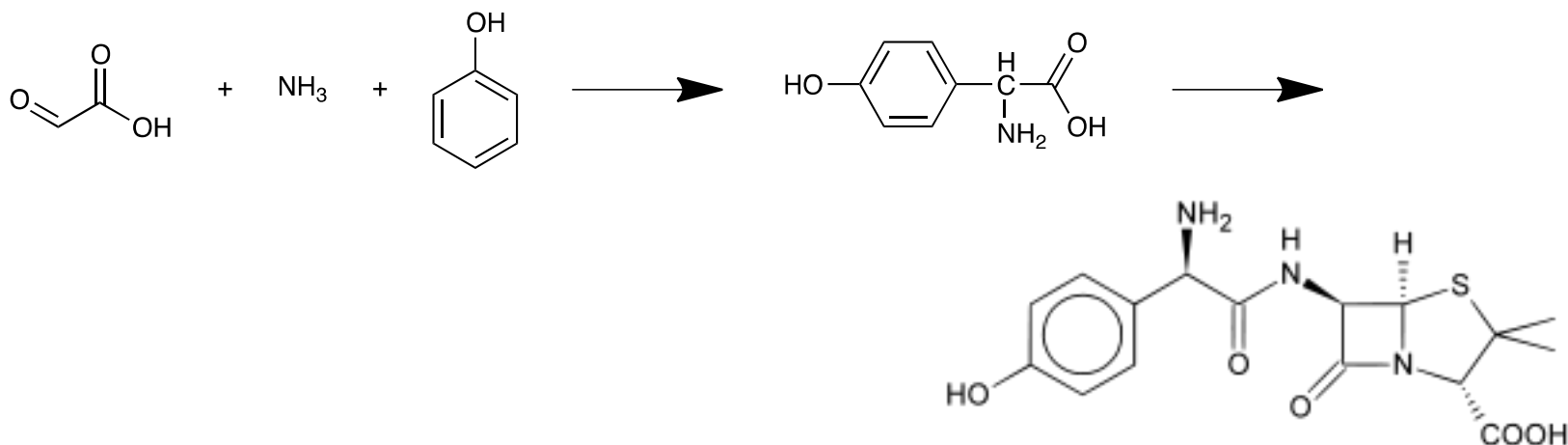
Glyoxylic acid



- Simplest α -oxocarboxylic acid
- Strongly hydrophilic
- 2 functional groups



- Glyoxylic acid, phenol and ammonia react to give 4-hydroxyphenylglycine $\rightarrow$ intermediate for penicillin (Amoxicillin)

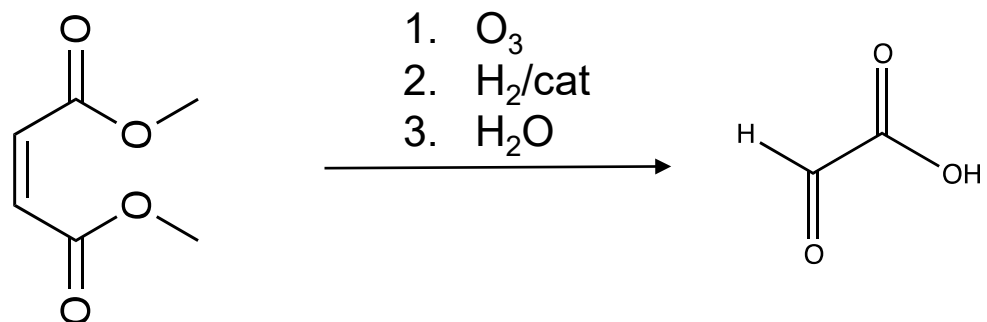


Glyoxylic acid

- Industrial production via oxidation of glyoxal with nitric acid

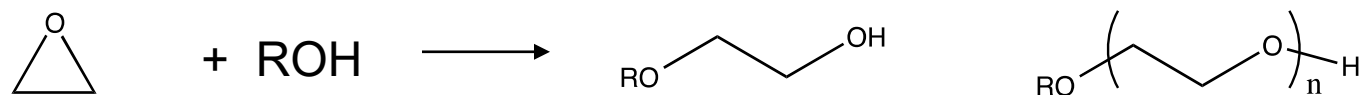
Or the “Chemie Linz” (DSM) process

- Oxidative cleavage of maleic acid or its esters by ozone



13.08.2003
09.08.2004

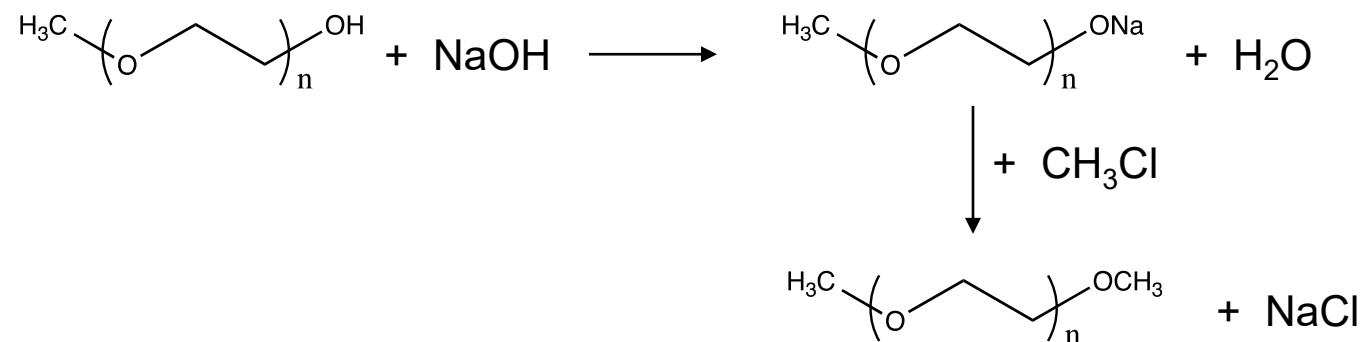
Ethylene glycol ether



- CH₃OH, C₂H₅OH and C₃H₇OH mostly used alcohols
- Alkali catalysts (alkali hydroxide, alkali alcoholate, Al₂O₃)
- 170-190°C, 10 to 15 bar
- Excess on ROH -> but still one gets a mixture of di-, tri- and higher ethylene glycol mono alkyl ether
- Mono alkyl ether of ethylene glycol:
 - Cellosolve, Carbitole, Dowanole
 - Solvent for lacquer, viscosity adjustment for brake fluids, emulsifying agent

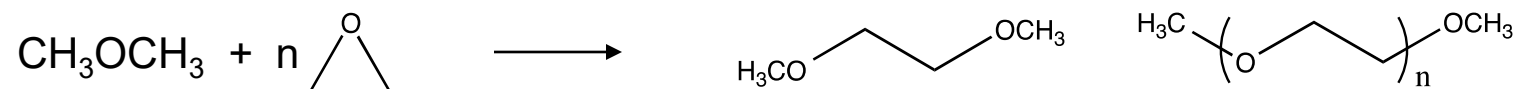
Ethylene glycol ether

- The global glycol ether market stood at approximately 4700 thousand t in 2022 and is anticipated to grow at a CAGR of 5.4%
- Ethylene glycol dialkyl ether (mostly di methyl) can be made in 2 stage process:



Ethylene glycol ether

Hoechst process:

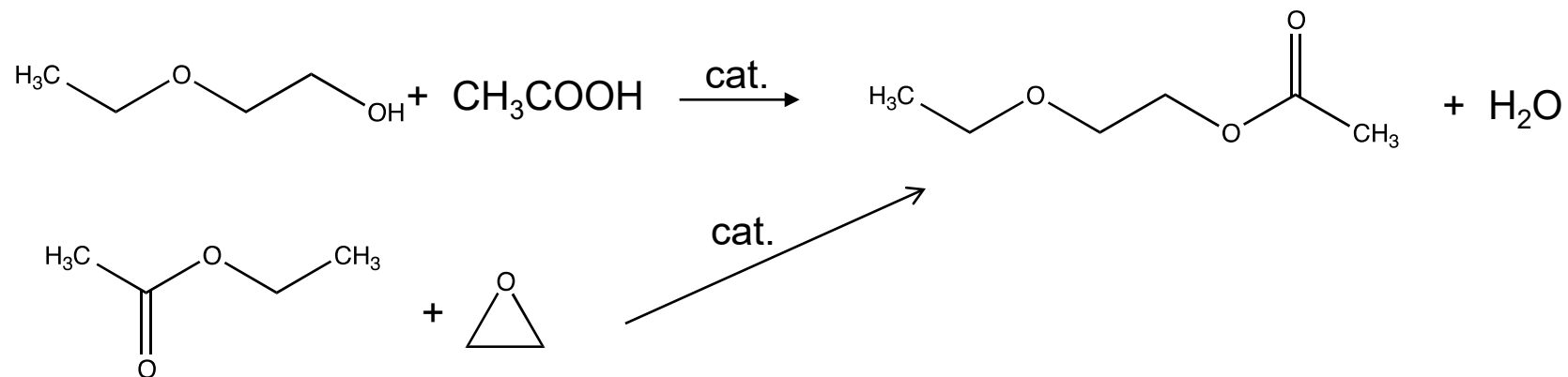


Di methyl ether plus EO with Boron tri fluoride as catalyst

Ethylene glycol dialkyl ether:

- A-protic solvent (e.g. for Grignard)
- Special solvent for resins, plastics, lacquer, paints (e.g. PU lacquer)
- Special fluids for brakes and hydraulic systems
- Cooling agent
- Extraction solvent (Selexol process)

Ethylene glycol ether



Esterification of free OH group

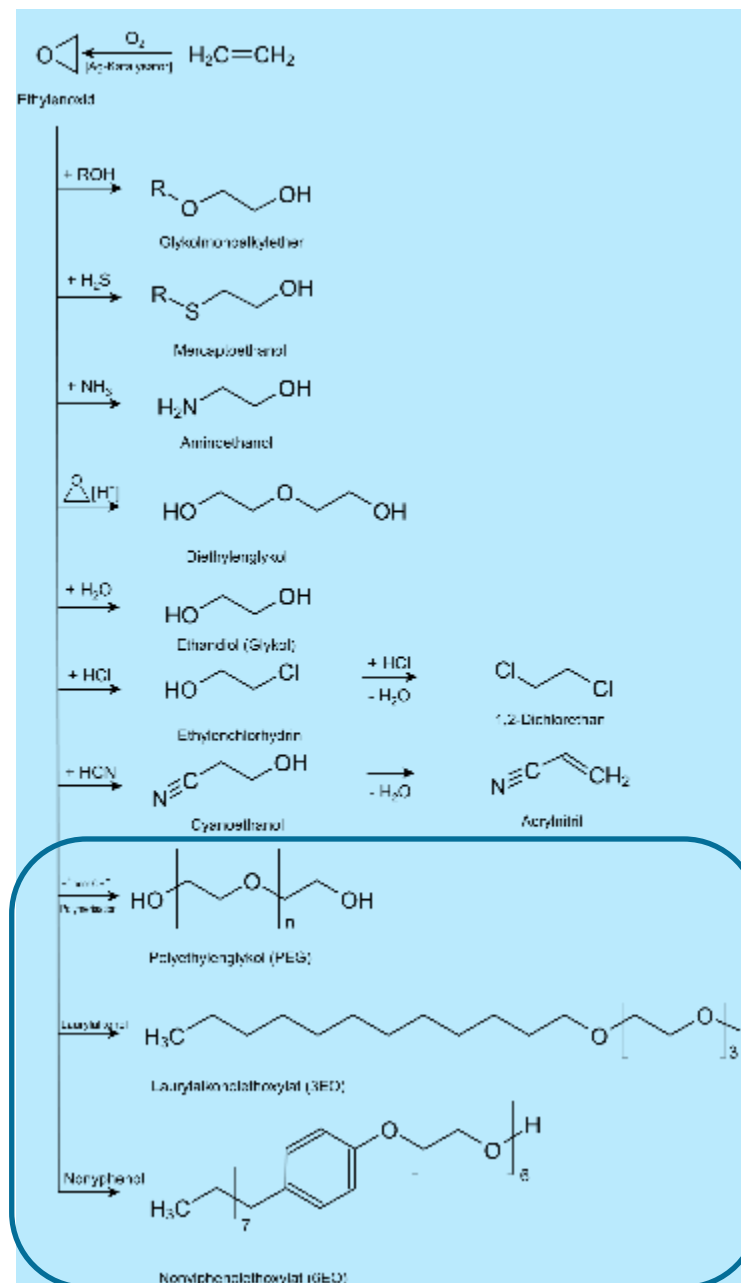
- Carboxylic acids (acetic acid)
- Catalyst: Friedel-Crafts-systems e.g. $\text{AlCl}_3 \cdot \text{N}(\text{C}_2\text{H}_5)_3$ (N or P ligands)
- Ethyl glycol acetate used as solvent for nitrocellulose and cellulose ether for preparation of lacquers

Polyethoxylate

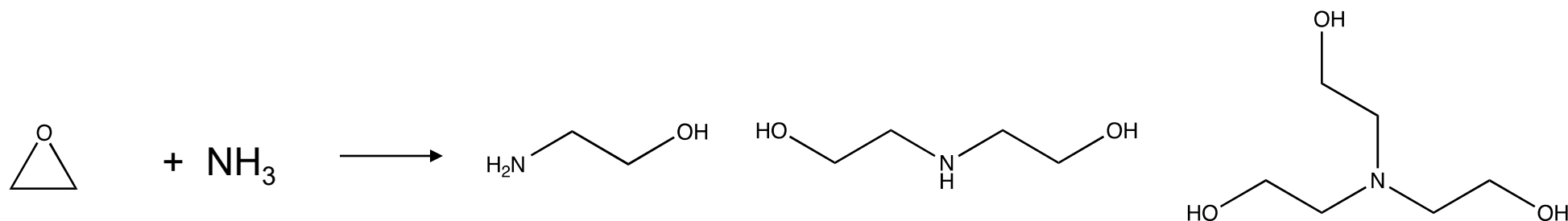
- In general multiple EO addition to compounds with movable H
 - Alkyl phenols
Octyl-, nonyl-, dodecyl-
 - Fatty alcohols, fatty acids, fatty amines
- 10-30 mol EO per mol starting compound
- Hydrophilic character of product regulated with EO concentration
- Process:
 - 120 - 220°C
 - increased pressure
 - alkaline catalysts (NaOH, NaOAc)

Polyethoxylate

- Non-ionic surface active compounds
- Detergents
- Wetting agent
- For Polyurethanes (plus iso-cyanates)
- 1990: 3,5 Mio t
- Important criterion:
biological degradability!
Alkylphenoethoxylates (difficult)



Ethanolamines



- EO reacts with an aqueous solution of NH₃
- Exothermal reaction, 60-150°C, 30-150 bar
- Mixture of all 3 possible ethanolamines
- Can be influenced with NH₃/EO ratio

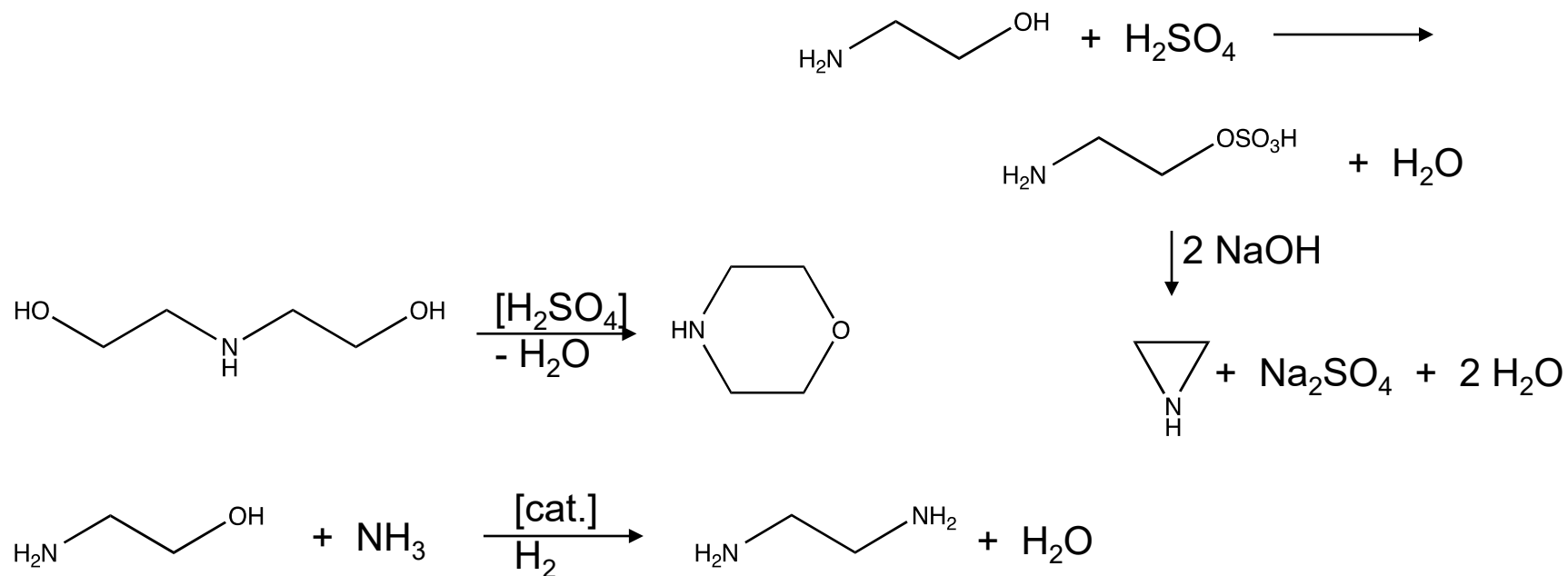
NH ₃ : EO	Mono-	Di-	Tri-
10:1	75	21	4
1:1	12	23	65

Ethanolamines

- Used for production of detergents (reaction with fatty acids)
 - Reaction of fatty acid with amino- or hydroxyl functionality
 - Fatty acid ethanol amide, fatty acid amino ethyl ester
- Roughly 1700 thousand t
- Or as weak base for gas scrubbing (H_2S , CO_2)
- Cosmetic industry (soaps, crèmes)
- Cement industry (grinding aid)
- Drilling grease
- In synthesis for heterocyclic build up

Derivatives of ethanolamines

1. Morpholin
2. Ethylene imin
3. Ethylene diamine



Acetaldehyde

- Molar mass: 44,1 g/mol
- Density: 0,784 g/cm³ @20°C
- T_m: -123°C
- T_b: 20,4°C
- pK_s: 13,57 (@ 25°C)
- Roughly 1,4 Mill t



Acetaldehyde

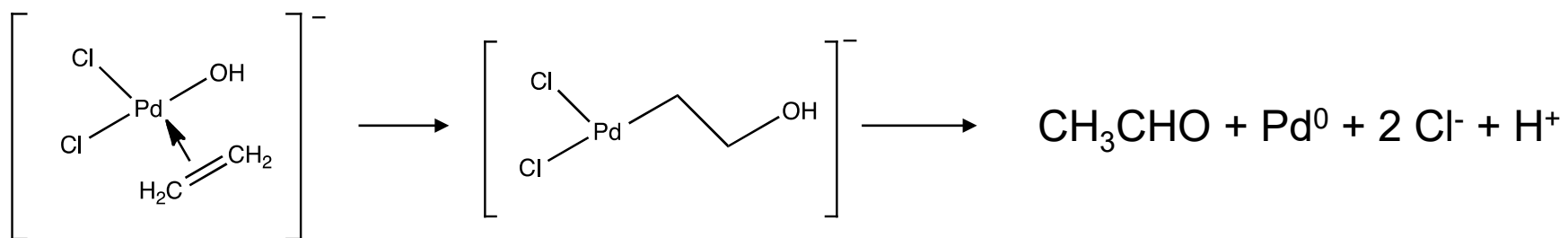
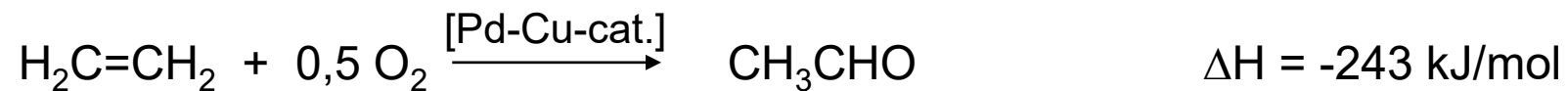
Classic processes

1. De- and oxo hydrogenation of ethanol
2. Hydration of acetylene
3. Oxidation of C₃/C₄ alkanes in the gas phase
4. Oxidation of ethylene

Acetaldehyde: oxidation of ethylene

- Principle of ethylene oxidation to acetaldehyde points back to F.C. Philips (1894)
- Pt metal salts can oxidize ethylene (stoichiometric mixture) to acetaldehyde and are reduced to the metal
- Process developed by Wacker and Hoechst 1957-1959
- Exothermic catalytic direct oxidation

Acetaldehyde: oxidation of ethylene

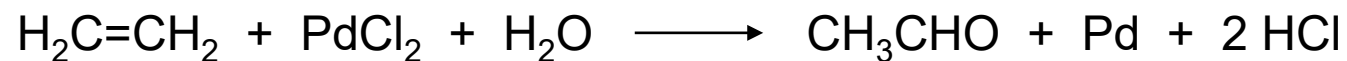


Total reaction



Acetaldehyde: oxidation of ethylene

Total reaction

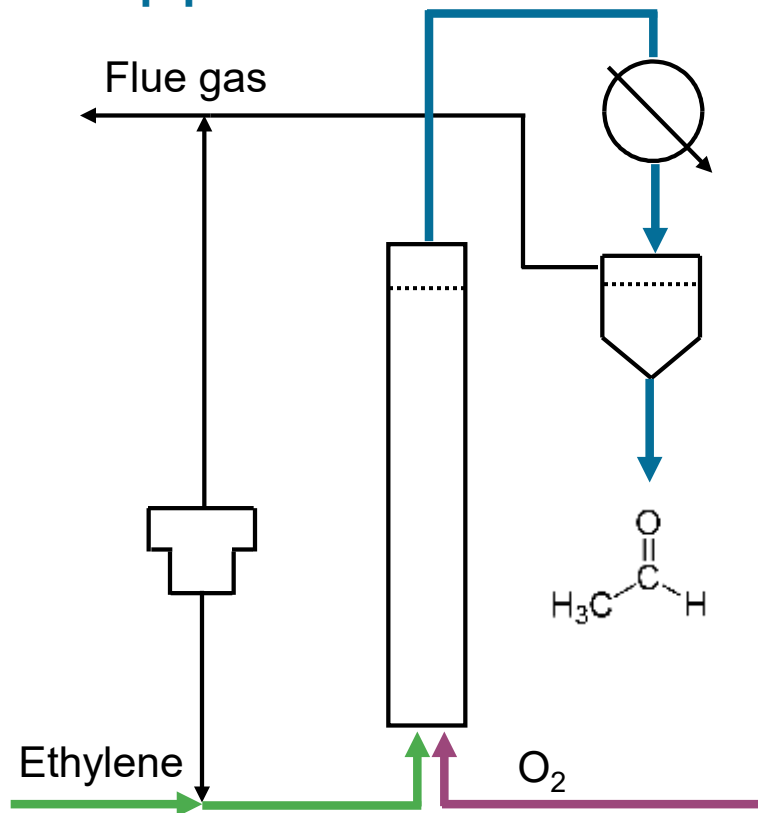


Actually a multiple step reaction with a fast oxidation and the reaction speed limiting regeneration step

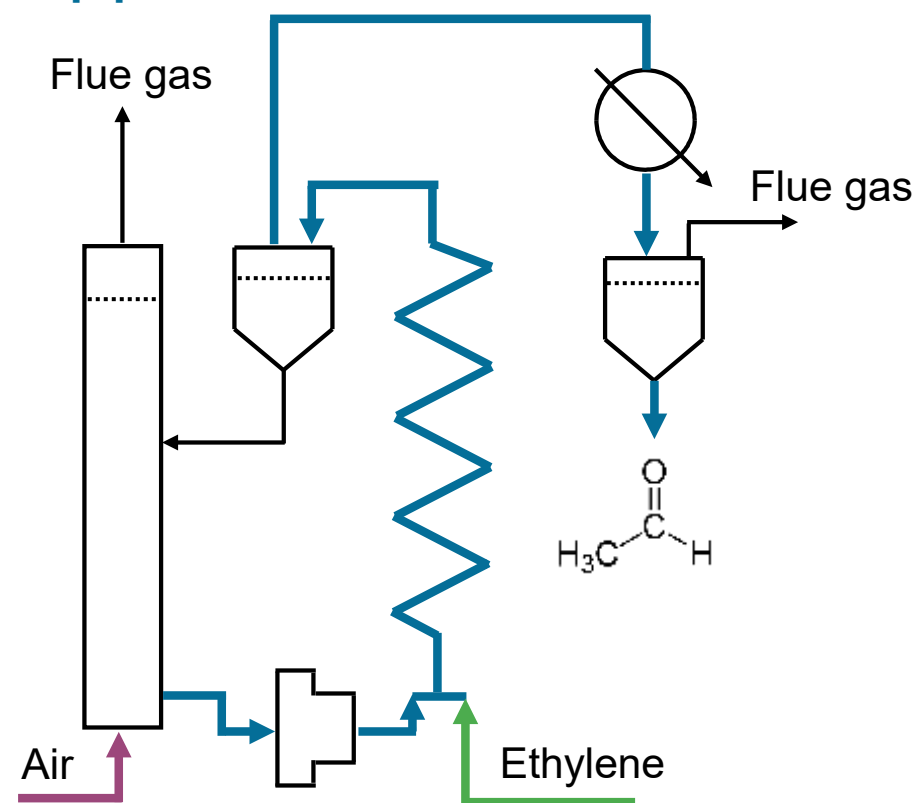


Acetaldehyde: process

1 step process:

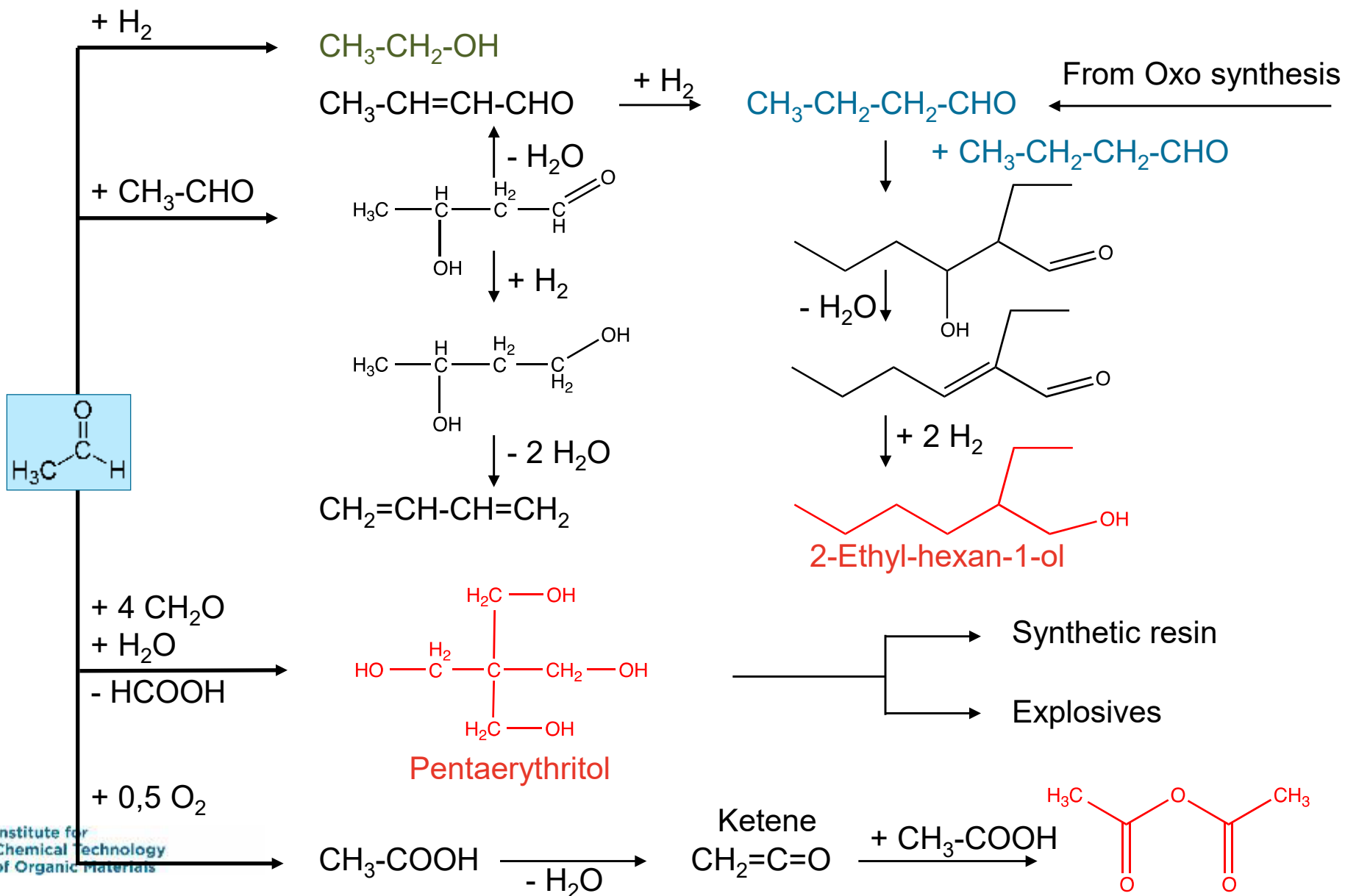


2 step process:



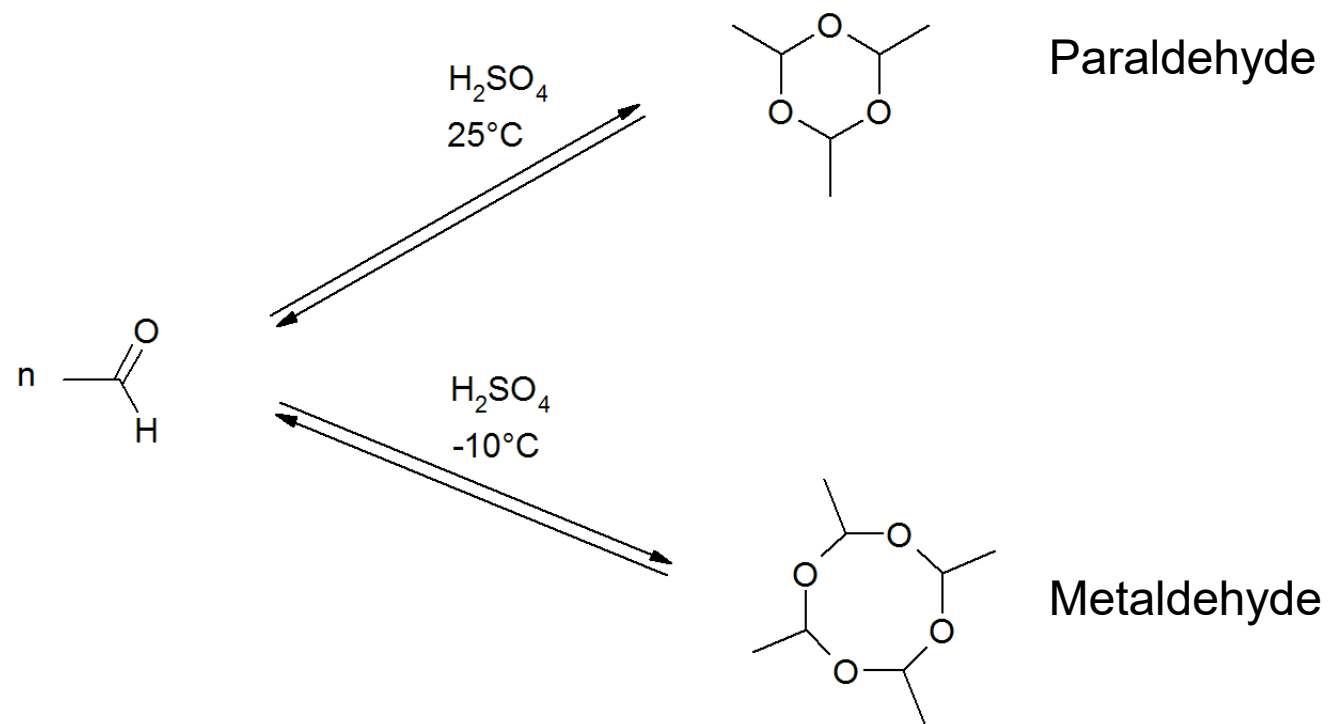
Process	Temp. [°C]	p [bar]	Ox agent	C ₂ H ₄ [turnover per cycle - %]
1-step	120-130	<5	O ₂	50
2-step	100-110	>9	Air	98

Acetaldehyde derivatives



Acetaldehyde

Para- and metaldehyde



Acetic acid

- Molar mass: 60,05 g/mol
- Density: 1,05 g/cm³
- T_m: 17°C
- T_b: 118°C
- pK_s: 4,76
- Production worldwide roughly 20 Mill t



E 260

Acetic acid

- Has been made over a long time from acetaldehyde
Beginning of 20th century oxidation processes from Wacker, Hoechst and Shawinigan
- Economic pressure to work up lower HC led to oxidation processes
Light paraffin's – BP, Celanese, Hüls, UCC
C₄-olefins – Bayer, Hüls
- In the 20ies carbonylation of CH₃OH – BASF
Got more attention through Monsanto Rh based process

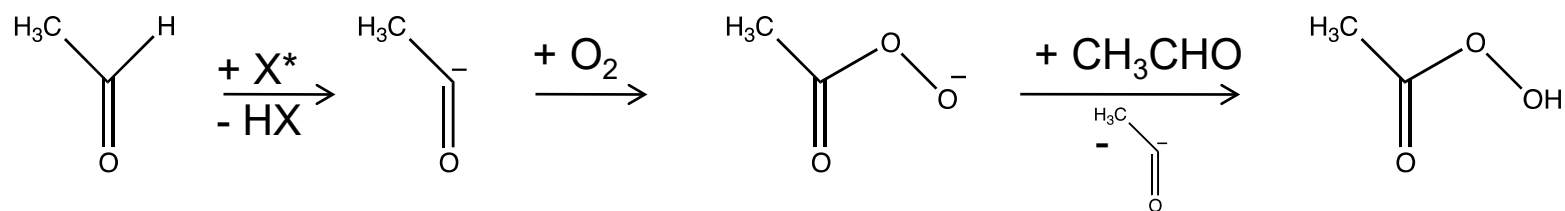
Acetic acid: change in educts

- Western Europe:
1979: 62% of acetic acid based on oxidation of acetaldehyde
1995: 28% - due to economic pressure of CH_3OH carbonylation (55%)
- USA:
No more based on acetaldehyde
1995: 80% of production based on CH_3OH

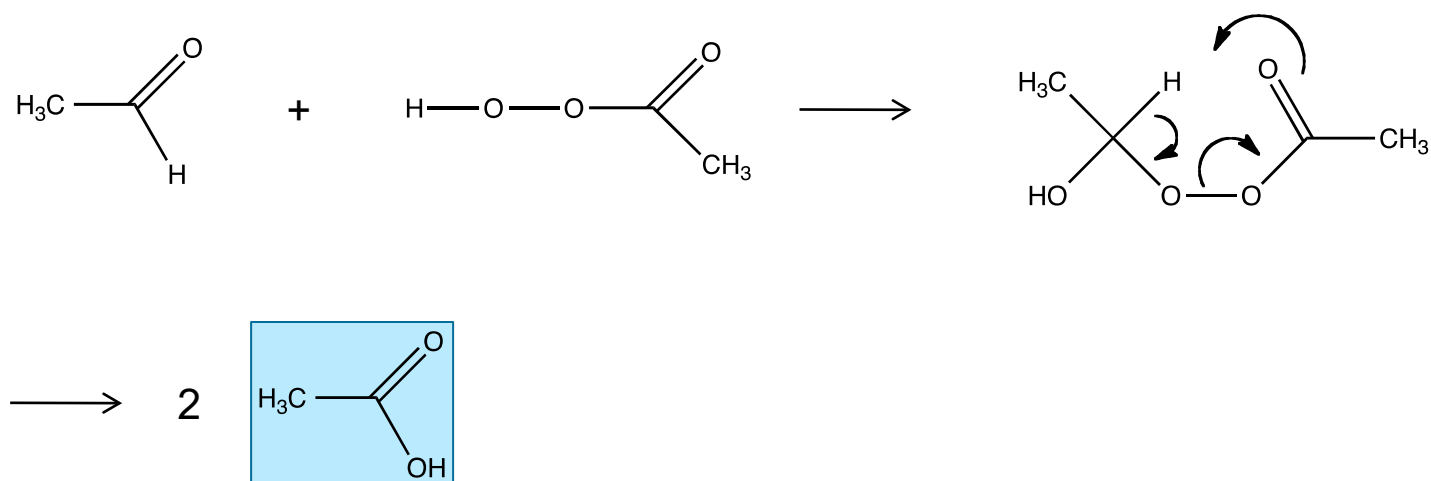
World [%-w.]	1988	1994	1996
CH_3OH carbonylation	47	58	60
CH_3CHO oxidation	27	23	20
$\text{C}_2\text{H}_5\text{OH}$ dehydration/oxidation	6	4	5
Butane-, naphtha- oxidation	7	9	9
Others	13	6	6

Oxidation of acetaldehyde

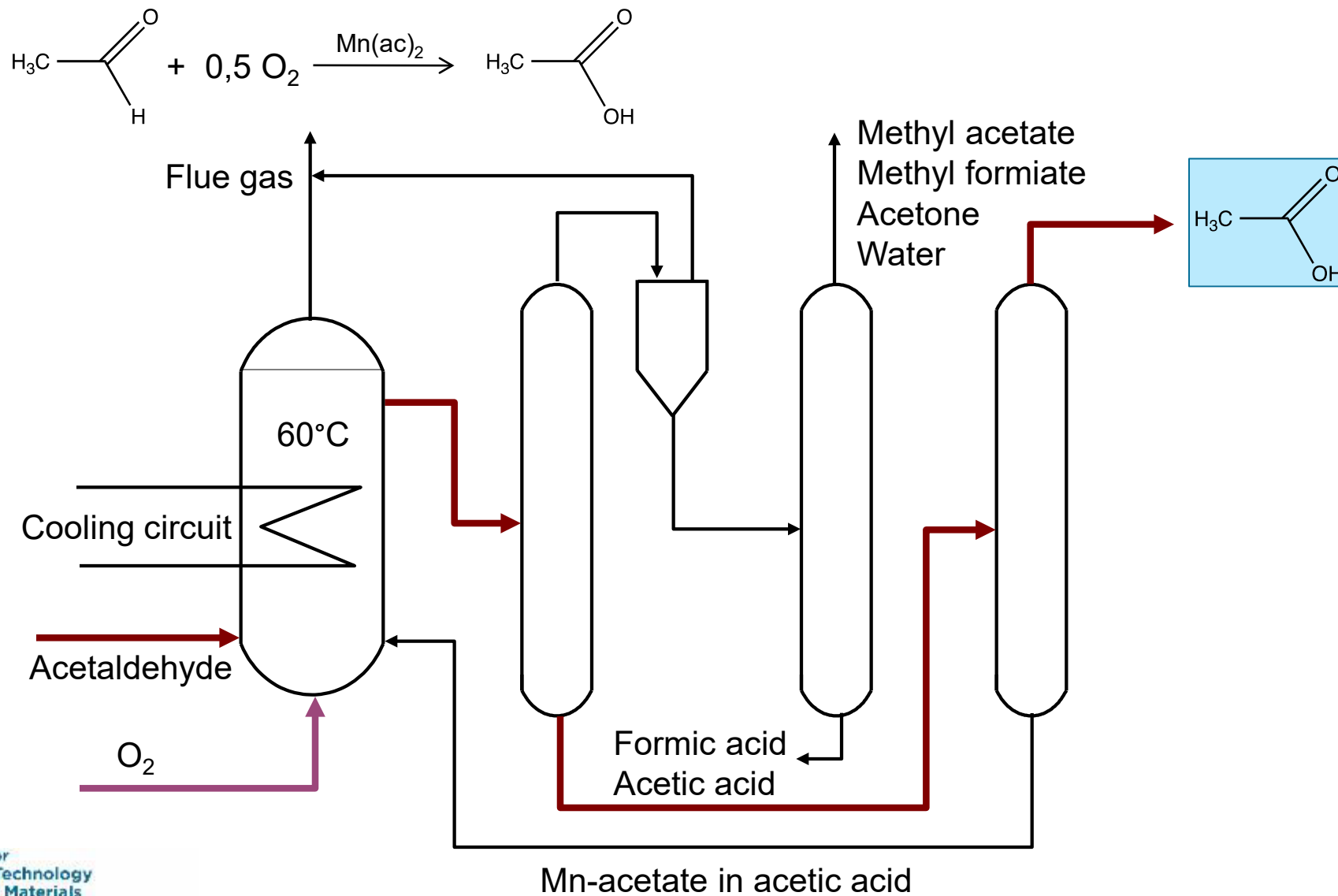
1: Peracetic acid



2: Acetaldehyde oxidation



Oxidation of acetaldehyde



Alkane / alkene oxidation

- Production of acetic acid through oxidative decomposition
- Preferably C₄ – C₈ hydrocarbons
- Highly exothermal radical processes
 1. Butane: Celanese, Hüls, UCC
 2. Butene: Bayer, Hüls
 3. Light gasoline: BP, Distillers

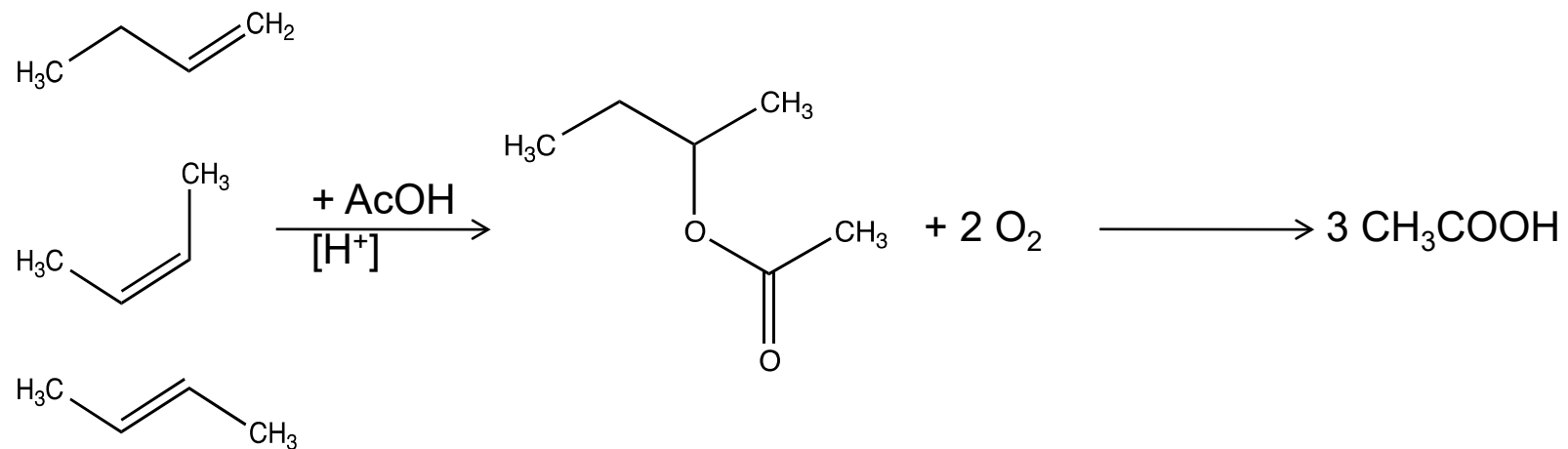
Butane oxidation

- Celanese LPO (liquid phase oxidation) process:
 - 175°C, 54 bar
 - Catalyst: Co-acetate
 - Byproducts: after separation of acetic acid fed back to process
- Hüls butane oxidation:
 - No catalyst
 - Oxidation at 60 – 80 bar and 170 – 200°C in air or O₂ enriched air (30%) in liquid phase
 - Product work up through distillation (byproducts)
- Celanese, Hüls, UCC

Butene oxidation

- Hüls butylene oxidation:
 - Liquid phase
 - Slight over pressure, 200°C
 - Catalyst: Ti or Sn-vanadate
 - Acetic acid is diluted (H_2O used in process) – high energy consumption
- Bayer butylene oxidation:
 - 2 step process
 - Mixture of 1-butene and cis/trans butylene
 - 1 step: 2-acetoxybutane
 - 2 step: oxidation to acetic acid

Bayer Butene oxidation

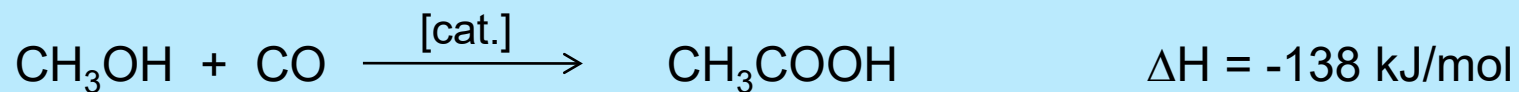


Alkane (light gasoline) oxidation

- Oxidation of crude oil with T_b 15 – 95°C
- Equivalent to light gasoline $C_4 - C_8$
- Oxidation with air in liquid phase
- 160 – 200°C
- 40 – 50 bar
- No catalyst
- Side products: formic acid, propanoic acid, acetone and small amounts of succinic acid
- BP, Distillers

Carbonylation of CH₃OH

- Points back to syngas (CH₃OH) works by BASF
- 1913 first carbonylation of CH₃OH
- In the beginning problems with corrosion
- With the 50ies access to Hastelloy (Mo-Ni alloys)
- 1960 first technical plant
 - 250°C, 680 bar
 - Catalyst: CoJ₂
- Further development through Monsanto

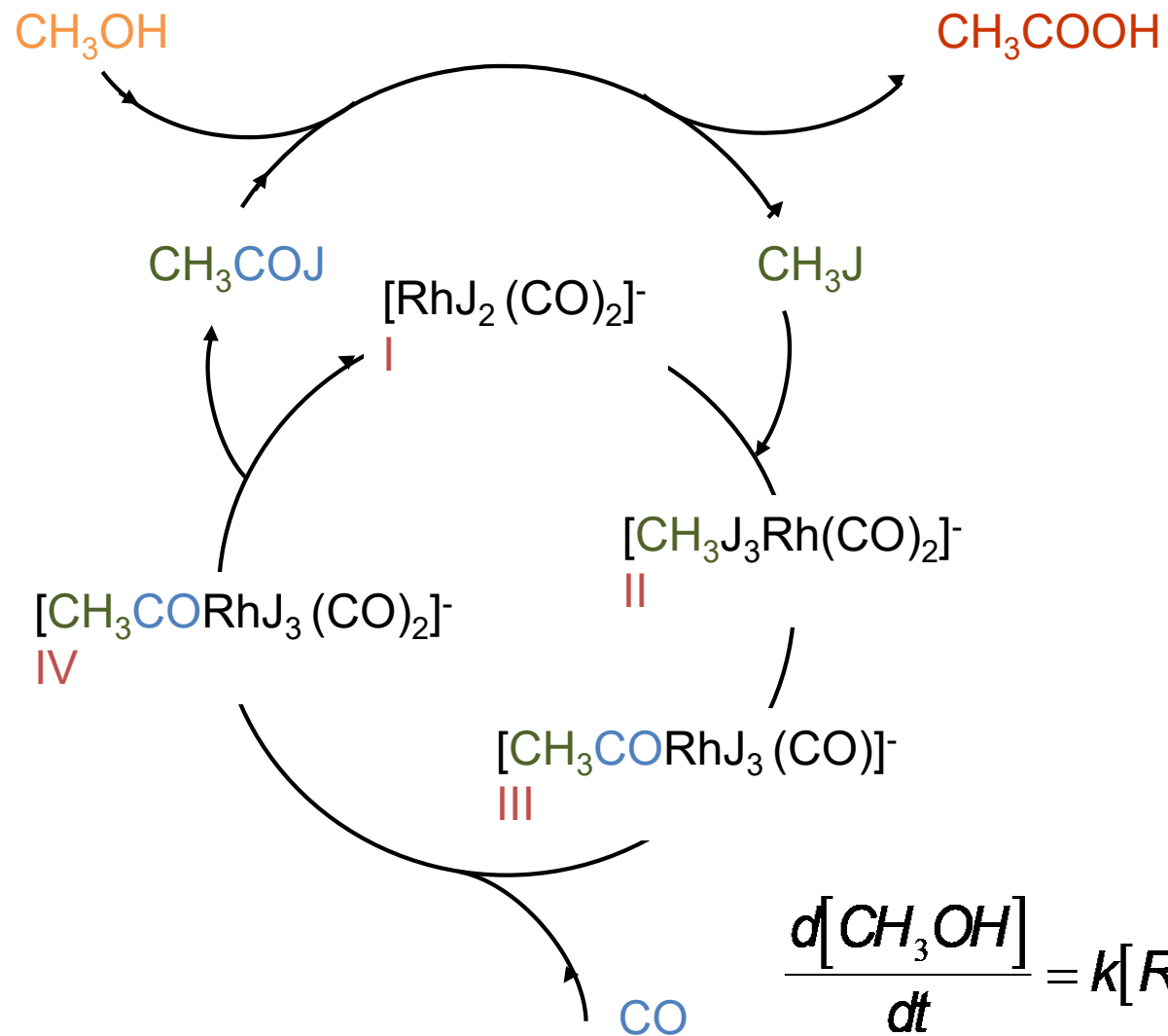


Hastelloy

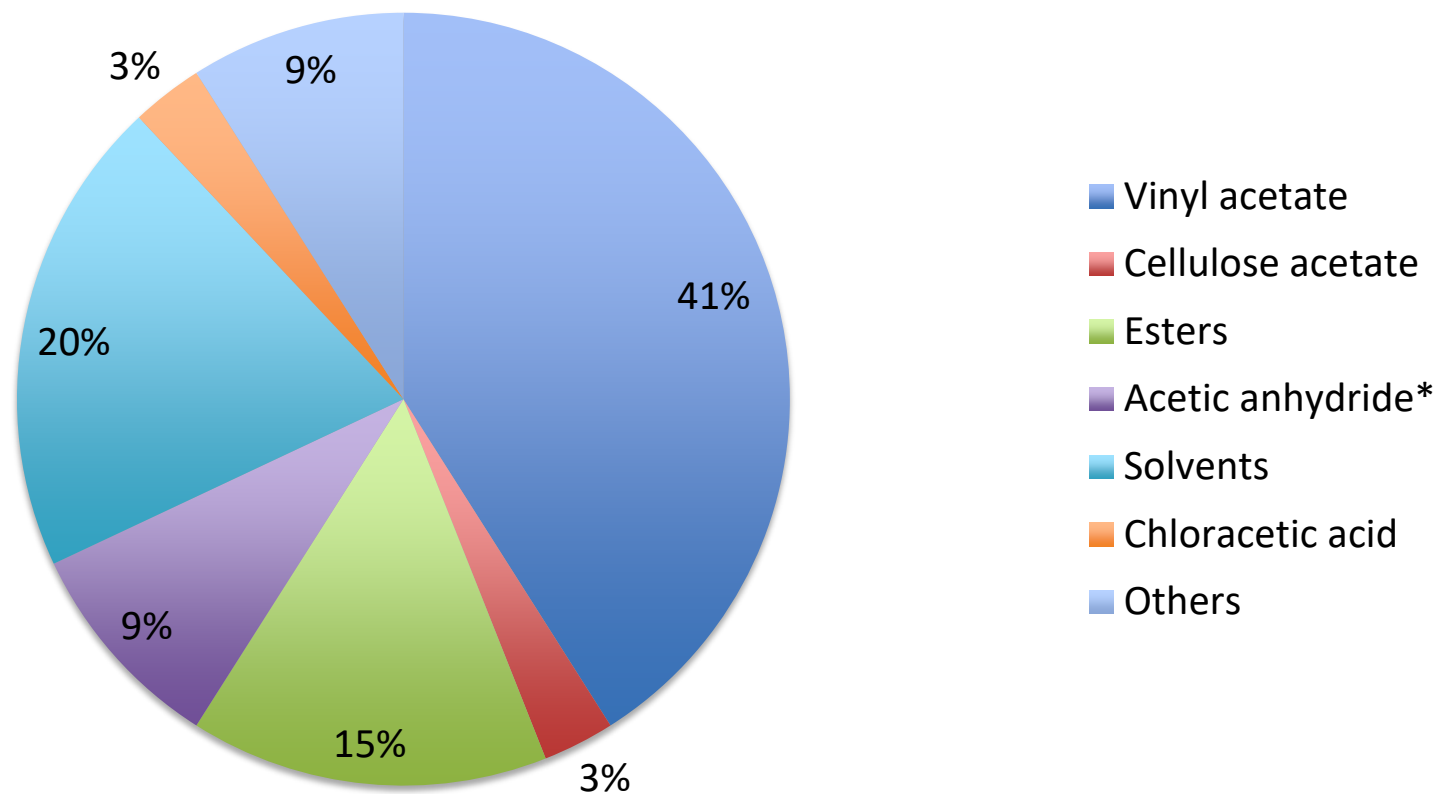
- Hastelloy Haynes International
- Range of twenty two different highly corrosion-resistant metal alloys loosely grouped under the term “superalloys” or “high-performance alloys”
- The predominant alloying ingredient: Ni
- Other alloying ingredients: Mo, Cr, Co, Fe, Cu, Mn, Ti, Zr, Al, C, W
- The primary function: survival under high-temperature, high-stress service in corrosive environment
- Pressure vessels of nuclear reactors, chemical reactors, distillation equipment and pipes and valves in chemical industry



Carbonylation of CH_3OH



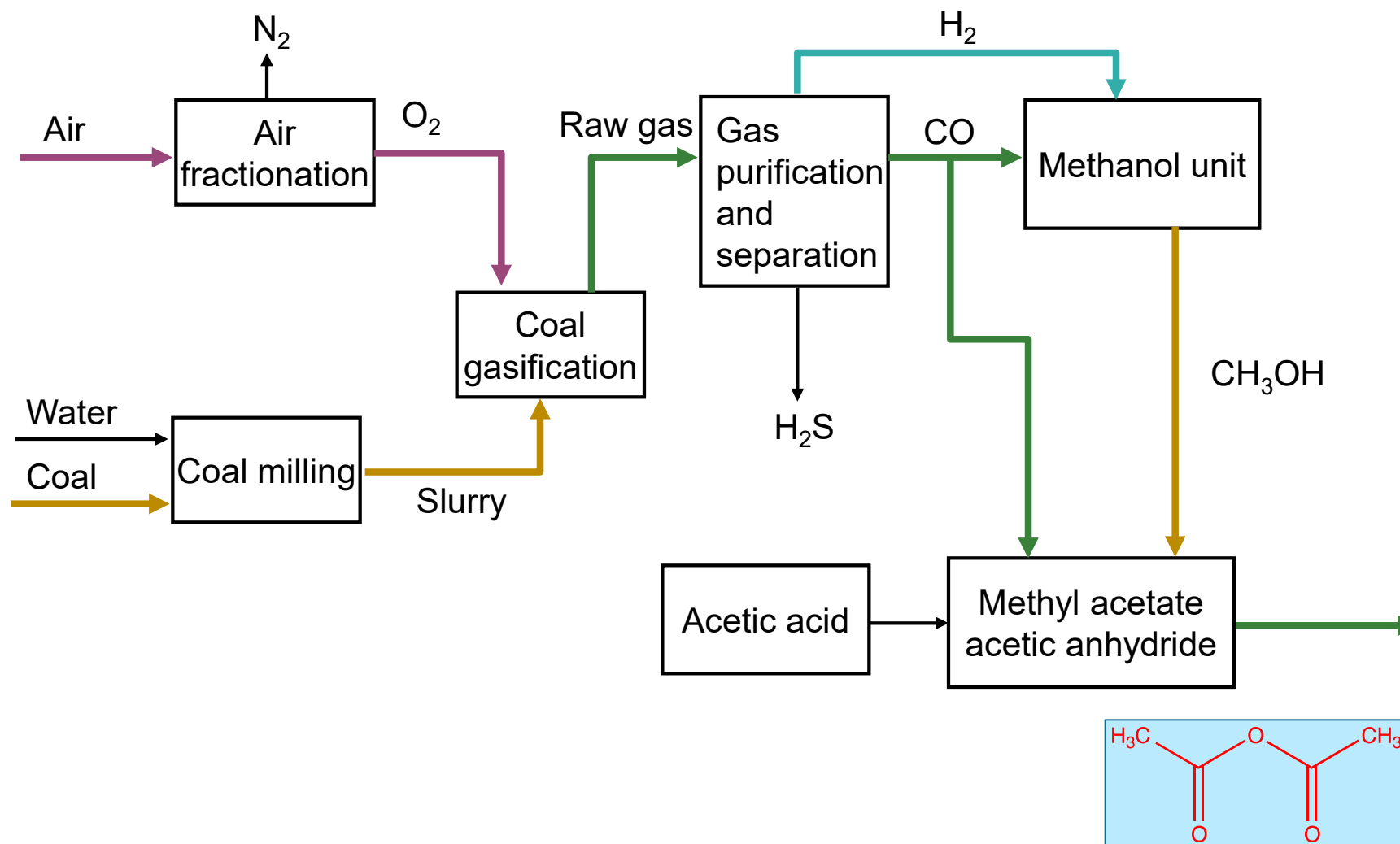
Use of acetic acid



Acetic anhydride

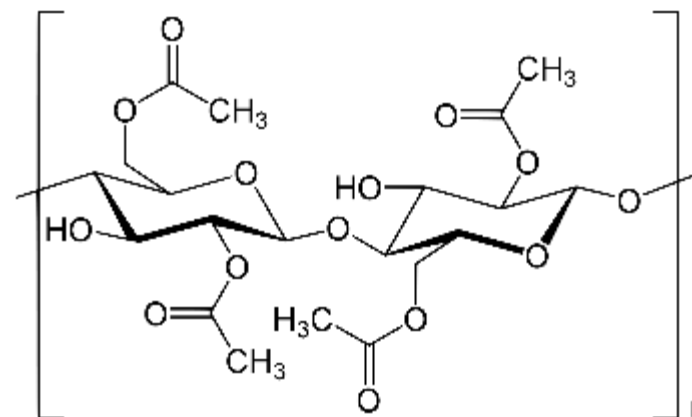
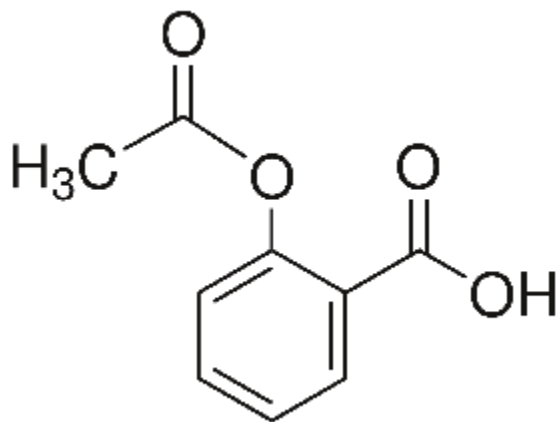
- Acetic acid can build 2 anhydrides
 - Acetic anhydride (important)
 - Keten (less important)
- Total volume roughly 2,3 Mio t
- Processes:
 - Oxidation of acetaldehyde
 - Carbonylation of methyl acetate
 - Carbonylation of di-methyl ether

Syngas - acetic anhydride



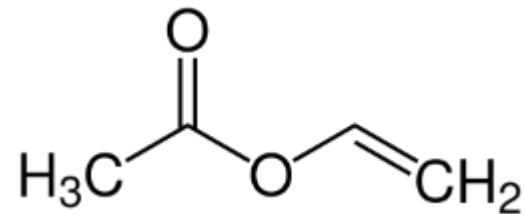
Acetic anhydride: use

- 1st place for acetylation
 - Cellulose acetate
 - Acetyl salicylic acid



Vinyl acetate

- Molar mass: 86,09 g/mol
- Density: 0,93 g/cm³
- T_m: -110,2°C
- T_b: 72°C
- Production worldwide roughly 8,5 Mill t



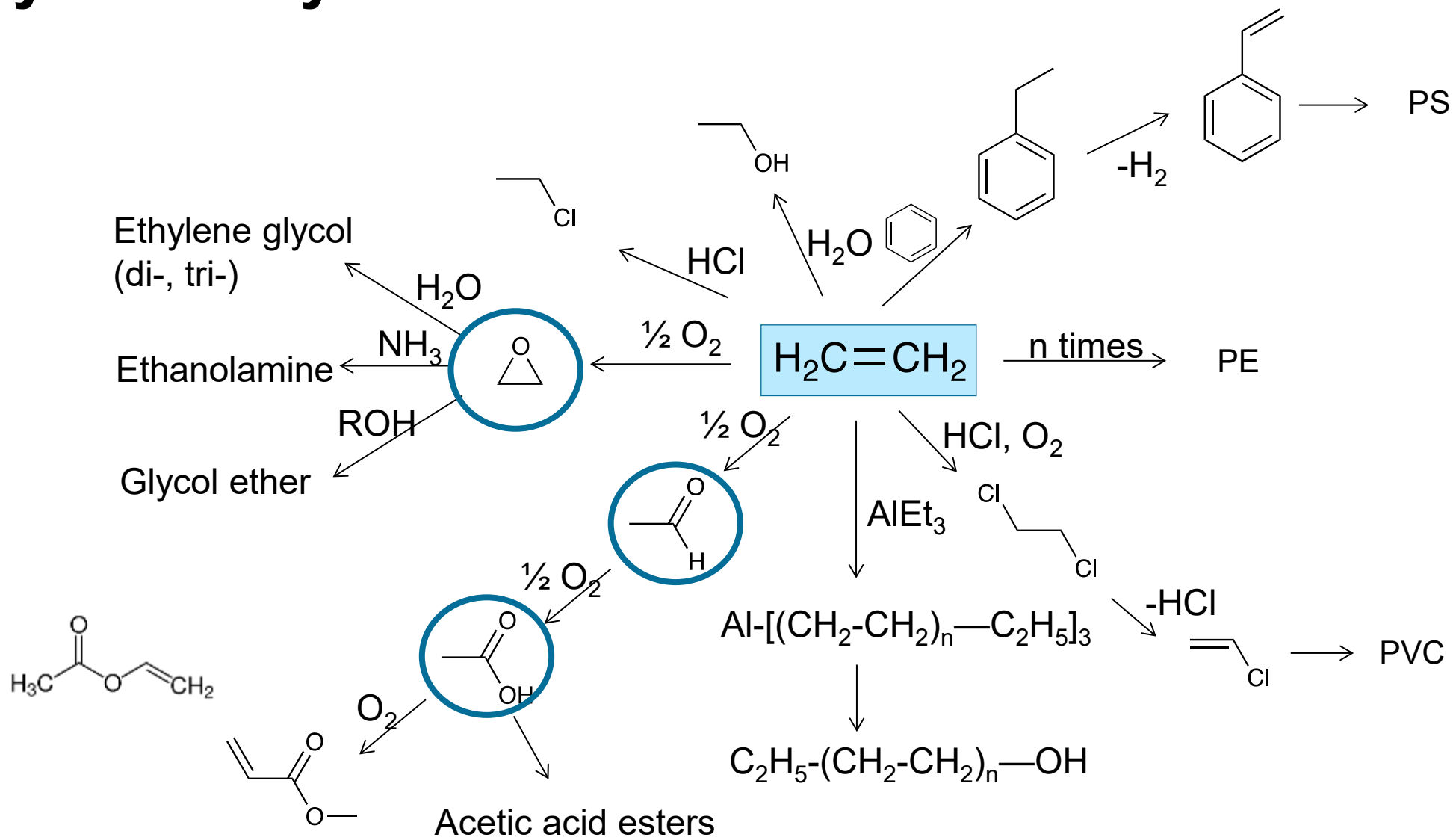
Vinyl acetate - processes

3 processes in use:

1. Addition of acetic acid to acetylene
2. Acetaldehyde with acetic anhydride via ethylene di-acetate
3. **Acetoxylation of ethylene**

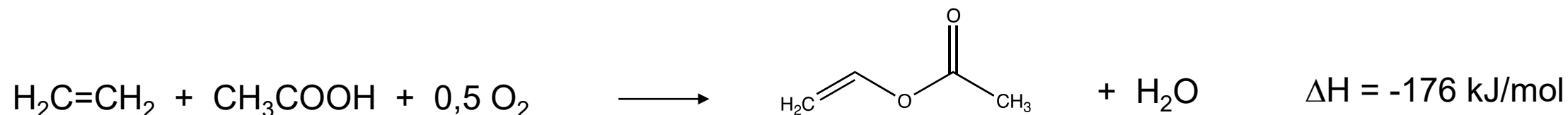
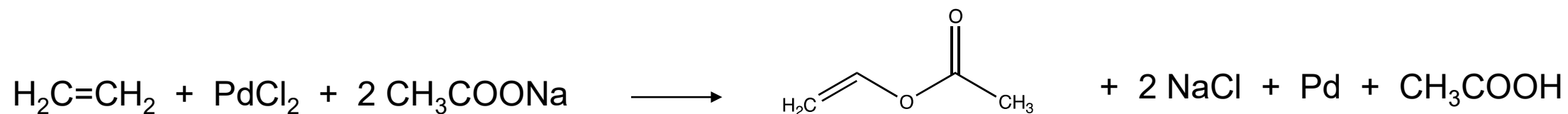
No 1 and 2 are still in use but replaced by ethylene acetoxylation

Ethylene family tree



Acetoxylation of ethylene

- Principle of discovered by J.J. Moiseev and co-workers
- Pd catalyzed oxidation of ethylene to acetaldehyde also leads to an acetoxylation of ethylene if done in acetic acid and in presence of Na-acetate



Use of vinyl acetate

Polyvinyl acetate (PVA)

- Color dispersion (paints) and laquers
- Base for chewing gum
- Co-monomer for vinyl chloride or ethylene
- PVA through saponification or transesterification to PVOH



Use of vinyl acetate

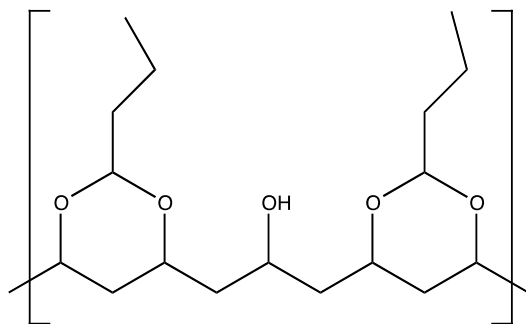
Poly vinyl alcohol (PVOH)

- 2002: 1,1 Mio t
- Glues (USA, WE), fibers (J), finishing of textiles, etc
- Reaction of free OH groups in PVOH with aldehydes leads to acetal plastics (formaldehyde, acetaldehyde, butyric aldehyde)



Polyvinylbutyral PVB

- Intermediate layer in safety glass



- Acetalisation with butanal
- E.g. Mowital (Kuraray)



Alcohols

- Methanol, ethanol, iso-propanol and butanols are the most important lower alcohols, allyl alcohol are gaining more attraction
- Higher alcohols C_6 to C_{18} used for plasticizer and for detergents
- The notation fatty alcohols points to $> C_8$ alcohols

Technically important alcohols

Alcohol	Process
Methanol	Low pressure methanol synthesis
Ethanol	Hydration of ethylene, hydrogenation of acetaldehyde
n-Propanol	Hydrogenation of propionaldehyde
i-Propanol	Hydration of propylene
n-Butanol	Hydrogenation of n-butyraldehyde
i-Butanol	Hydrogenation of i-butyraldehyde
t-Butanol	Hydration of i-butylene, side product of propylene oxide production via Halcon process
2-Ethylhexanol	Hydrogenation of 2-ethyl hexanal
Cyclohexanol	Oxidation of cyclohexane, hydrogenation of phenol
Ethylene glycol	Saponification of ethylene oxide
Allyl alcohol	Saponification of allyl chloride, relocation of propylene oxide
Higher alcohols	Hydrogenation of fatty acid esters, Alfol process

Capacities

Alcohol	1000 t/a
Methanol	3.500
Cyclohexanol	1.800
Ethylene glycol	1.400
i-Propanol	750
2-Ethylhexanol	700
Ethanol	600
Higher alcohols C7 – C16	500
n-Butanol	425
i-Butanol	200
Propylene glycol	200
Glycerin	190
1,4-Butane diol	140
Pentaerythrite	100

Alcohols

- Ethanol
- Isopropanol
- Butanol
- Higher alcohols
- Alfol process
- Polyols
- Butanediol

Ethanol

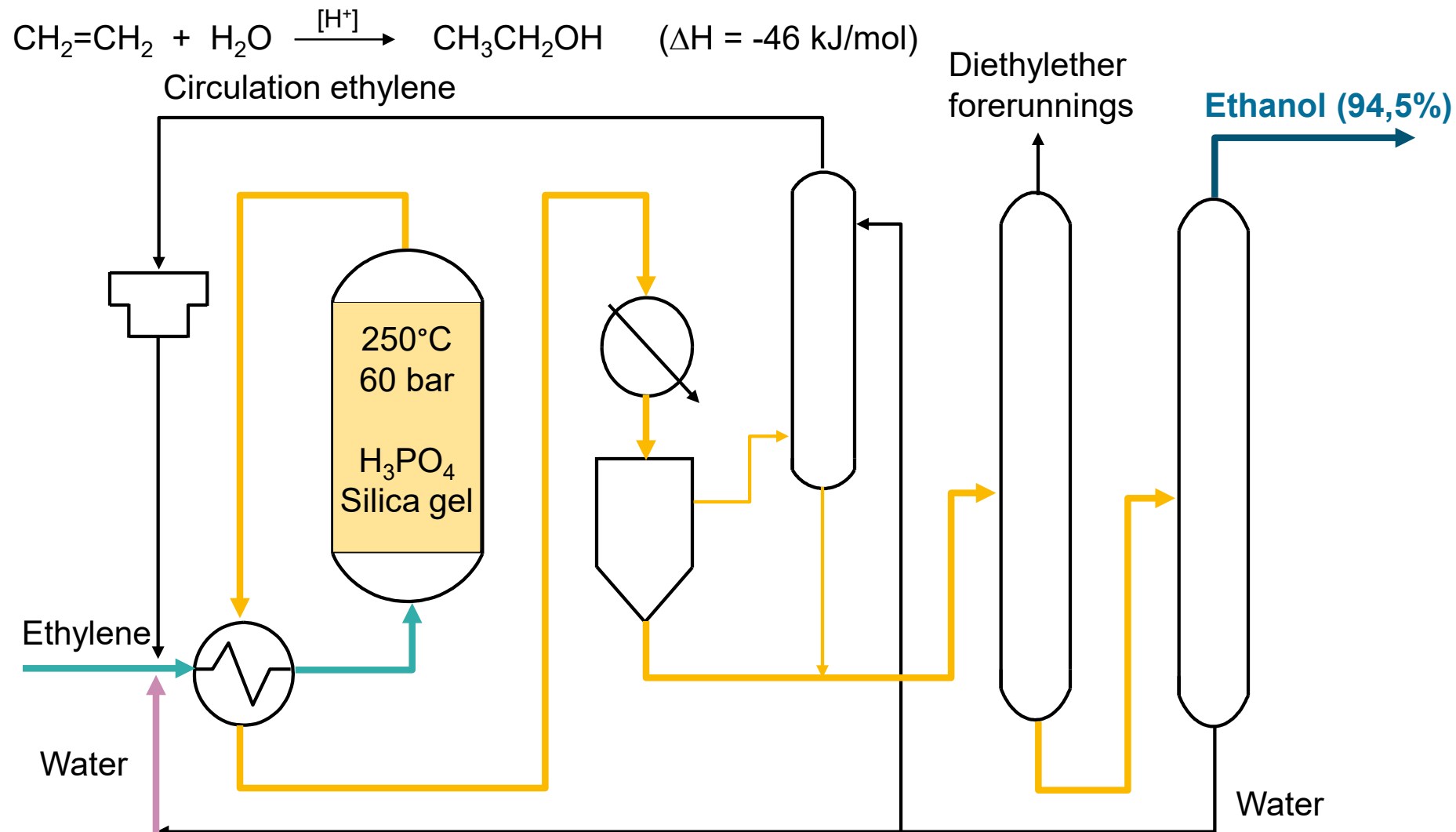
- Molar mass: 46,07 g/mol
- Density: 0,79 g/cm³
- T_m: -114°C
- T_b: 78°C
- Roughly 112 Bill liters (about 90 Mill t)



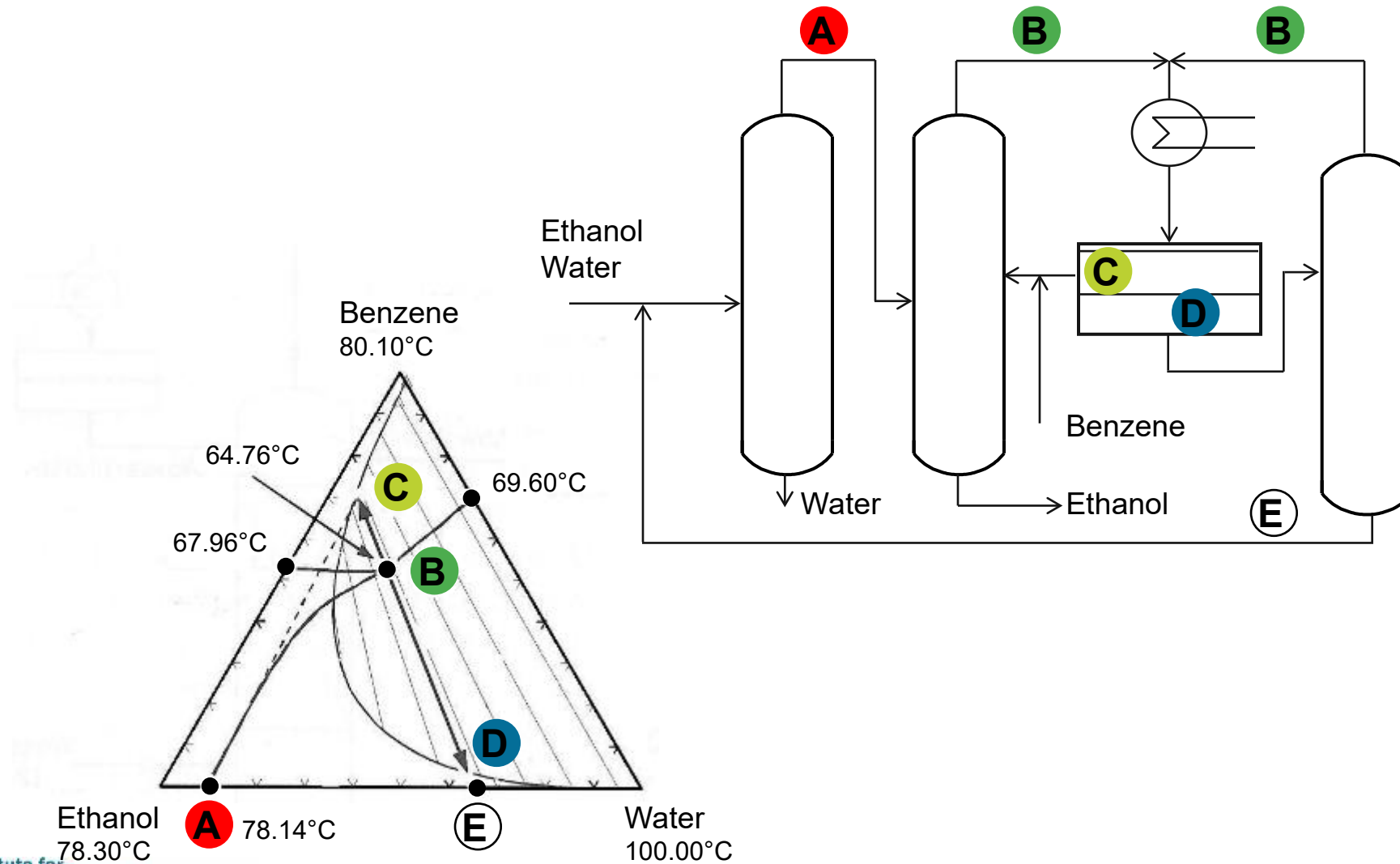
Ethanol - processes

1. Indirect hydration:
Since 1930 in an industrial scale
Ethylene rich gases (35 – 95%) are run over several absorption towers (55 – 80°C, 10 – 35 bar) with H_2SO_4 (94 – 98%)
In an exothermic reaction mono- and diethylsulfate are formed, and later hydrolysed to ethanol
2. Direct catalytic hydration:
First process 1947 by Shell
3. Routes based on syngas
Methanol with CO/H_2 in presence of Rh/Co catalysts in liquid or gas phase
Technical realization pending

Ethanol: catalytic hydration



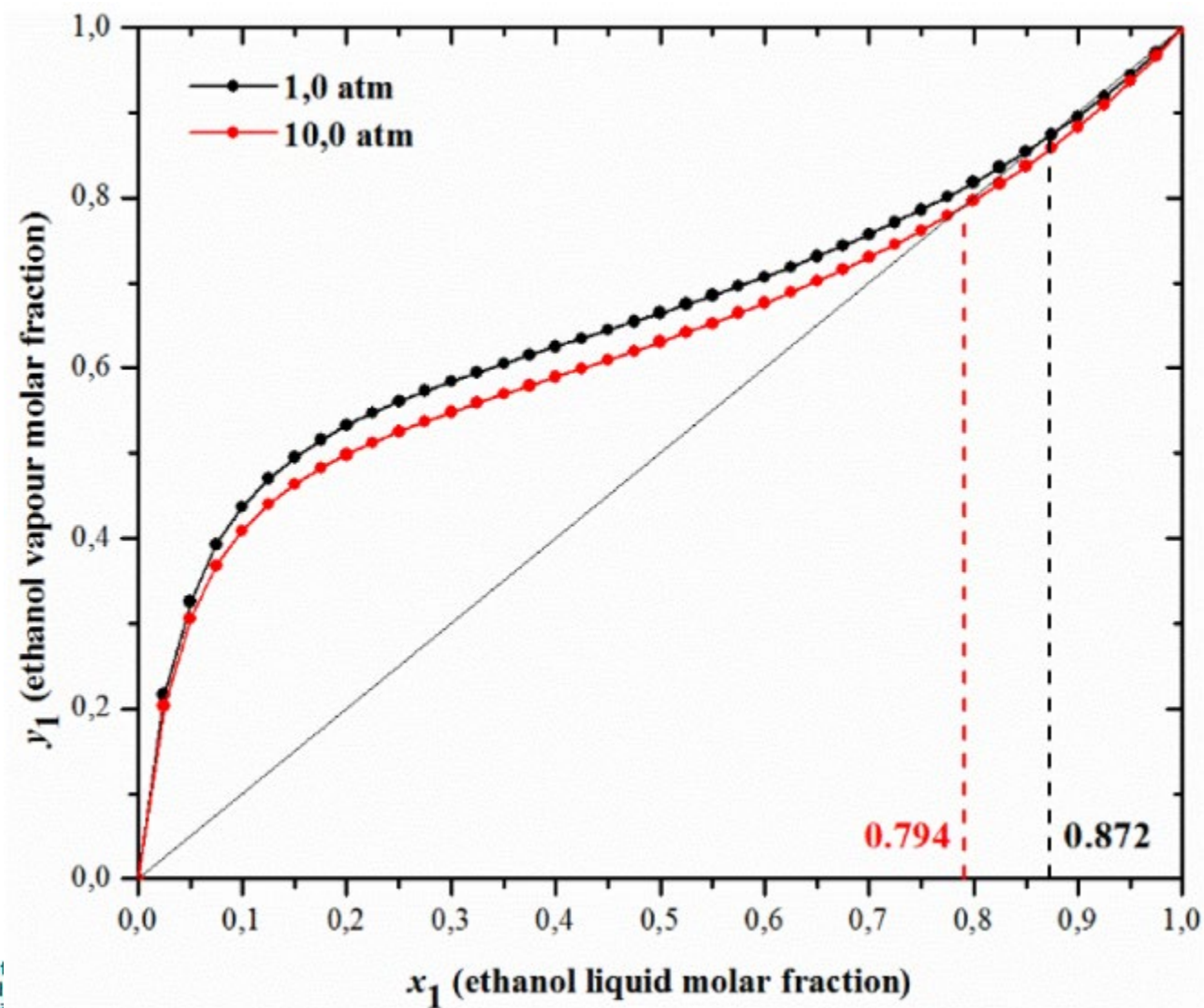
Azeotropic distillation



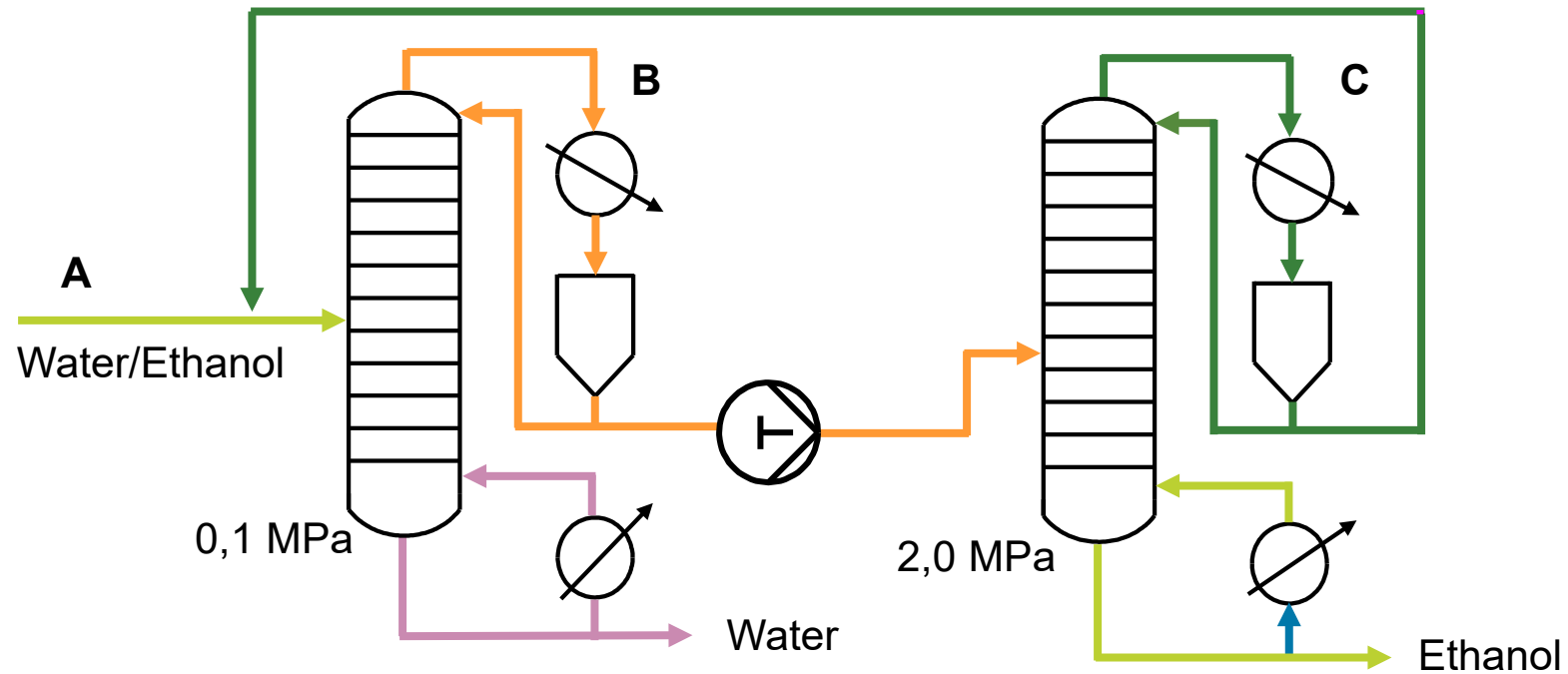
Pressure dependence of azeotrope

Pressure [MPa]	Temp [°C]	Ethanol [%-w]
0,013	34,3	99,3
0,033	53,0	97,6
0,066	68,0	96,0
0,100	78,3	95,6
0,340	112,6	95,0
0,680	136,7	94,7
1,360	164,2	93,9
2,040	182,6	93,6

Separation at 2 different pressures

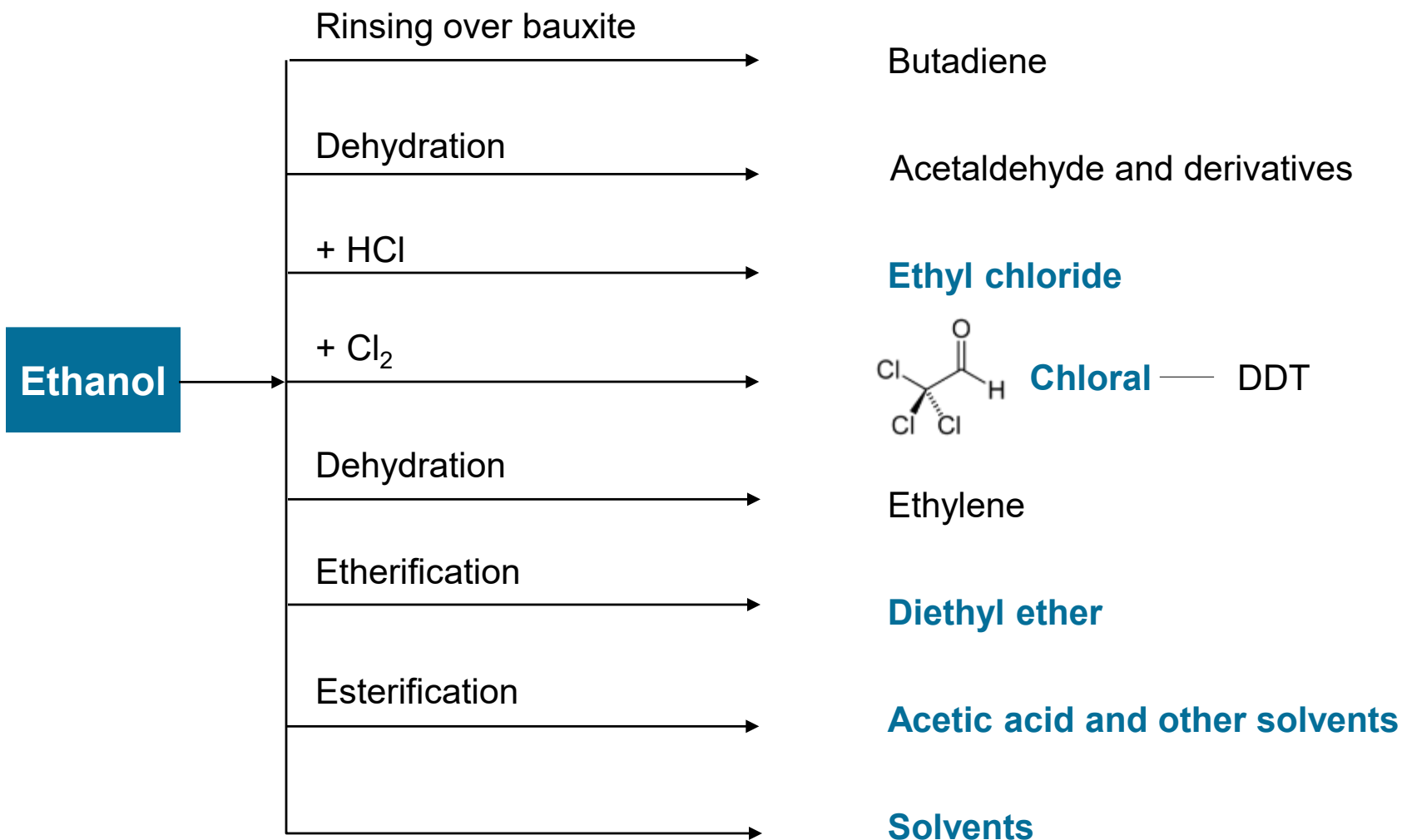


Separation at 2 different pressures



	Ethanol [%]	Water [%]	Tb [°C]	p [MPa]
A	20,0	80	-	-
B	95,6	4,4	78,2	0,1
C	93,6	6,4	182,6	2,0

Use of ethanol



Use of ethanol

- As carrier fluid for odorous substances (perfumes etc.)
- Cleaning (glass, plastics, windscreen washer)
- Antifreeze
- Additive in food (port wine – to stop fermentation process)
- Energy source (65 % of total EtOH volume used for biofuels)

Isopropanol

- Molar mass: 60,10 g/mol
- Density: 0,78 g/cm³
- T_m: -88°C
- T_b: 82°C
- Roughly 2,3 Mill t produced worldwide



Isopropanol

Production processes:

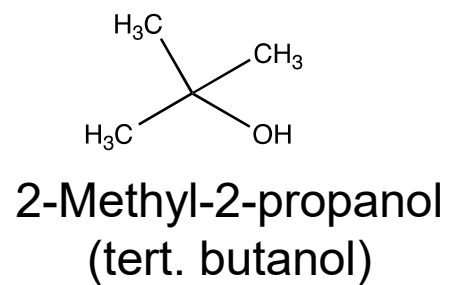
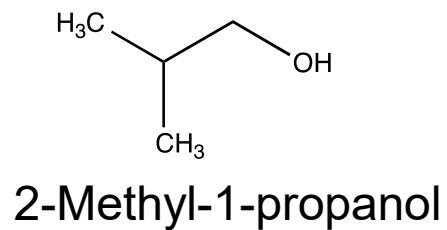
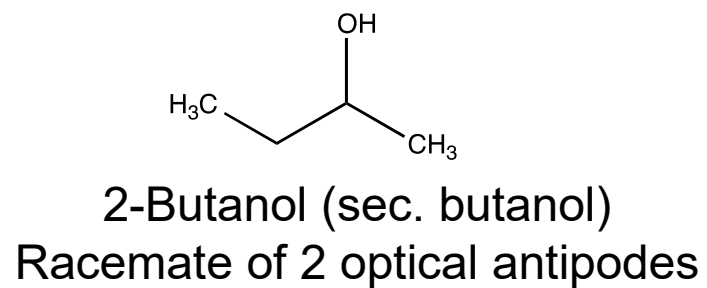
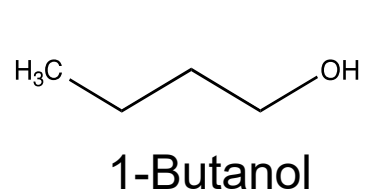
- Historically propylene hydration in liquid phase via H_2SO_4
- Catalytic hydration in gas or liquid phase, single step
- Acidic catalysts
- Mineral acids
- Acid ion exchanger
- Water soluble systems (tungsten)

Isopropanol - use

- Production of acetone
- Fuel additive (avoid icing of carburetor)
- Solvent
- Replacing ethanol in cosmetics and pharmaceuticals
- Iso-propylacetate

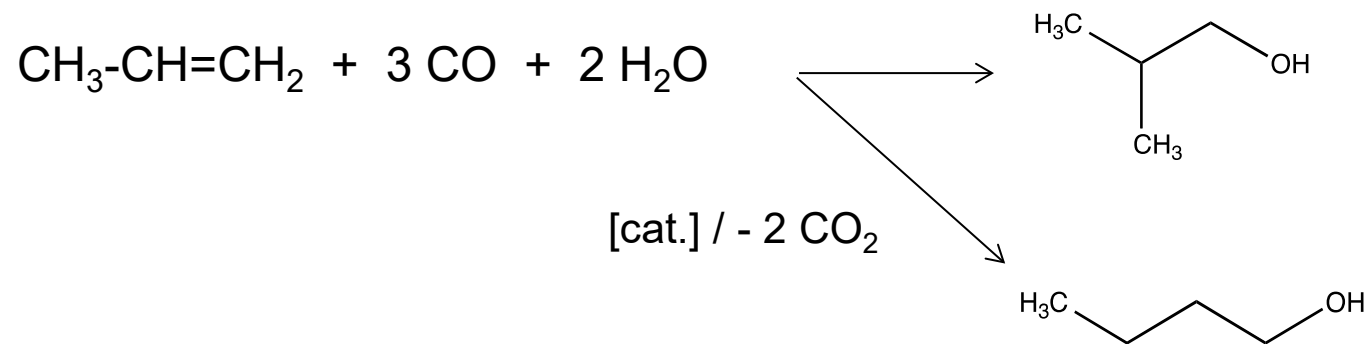
Butanol

- C₄ alcohols
- 4 different isomer structures

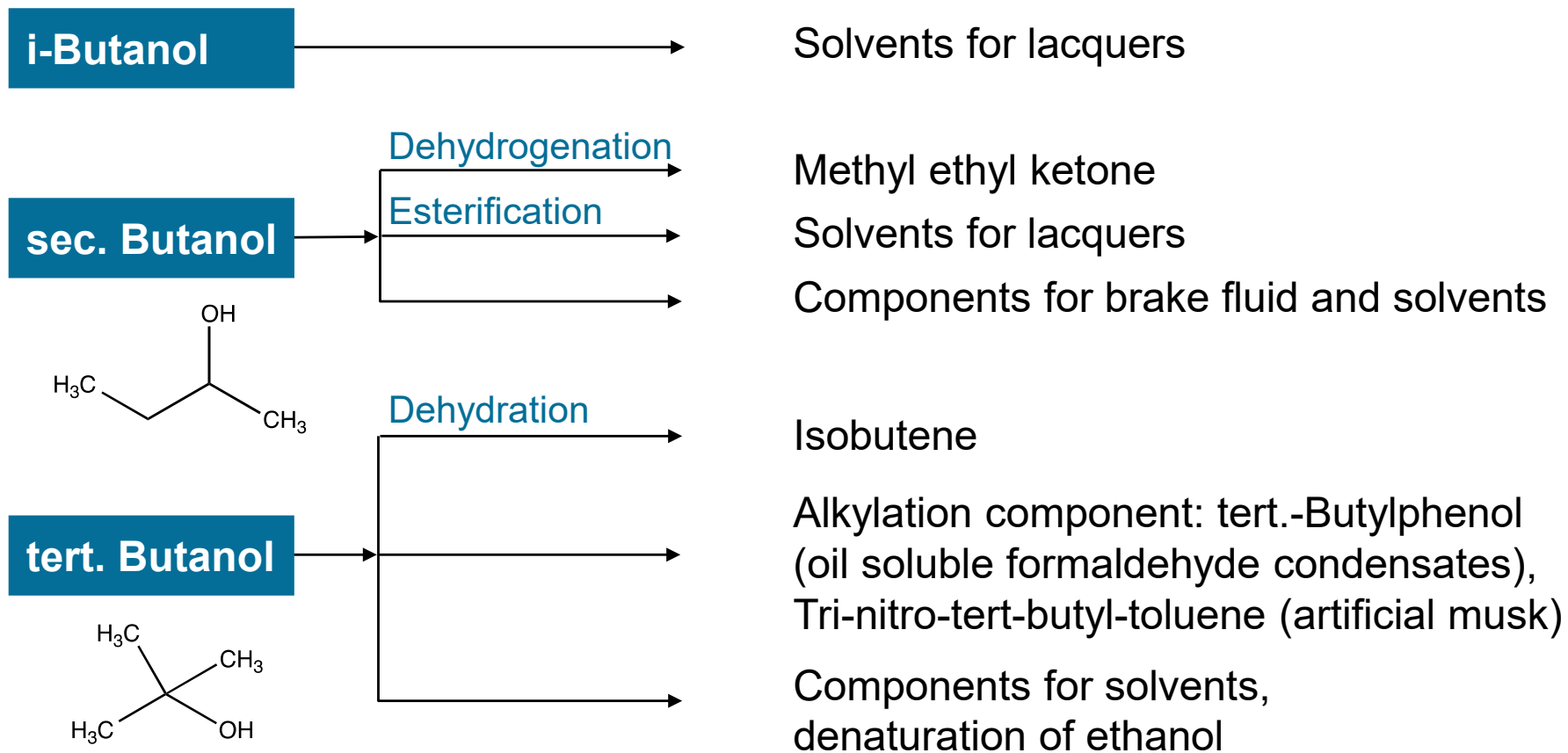


Butanol - processes

- Hydroformylation of propylene with following hydration (see our chapter on oxo synthesis)
- Aldol condensation of acetaldehyde (see acetaldehyde)
- Reppe process:
Propylene plus CO and water in presence of $\text{Fe}(\text{CO})_5$ catalyst



Butanol - use

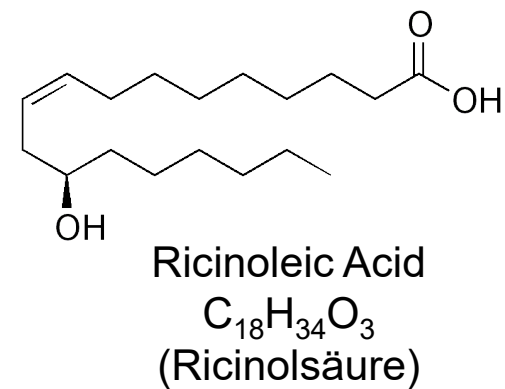
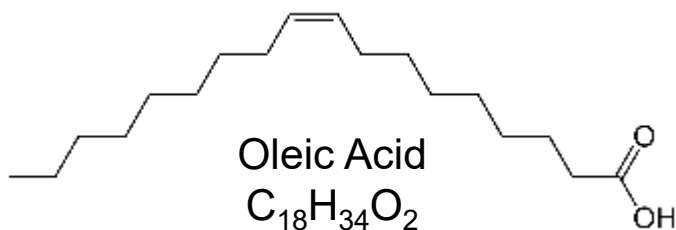


Higher alcohols

For the production of higher alcohols 4 routes are used

1. Hydrogenation of fatty acids

- Native fatty acids
- Catalytic oxidation of paraffin's
- Alkaline oxidative cleavage of ricinoleic acid
- Ozonolysis of oleic acid

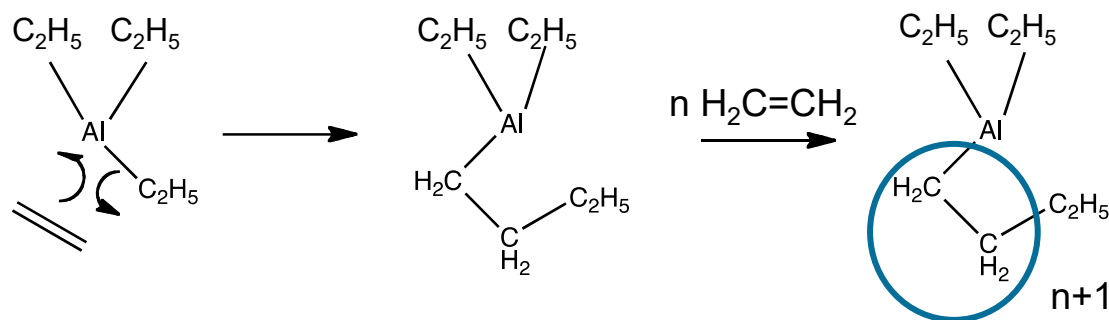


Higher alcohols

2. Hydroformylation of olefins with successive hydrogenation of formed aldehydes (mixtures of linear and branched primary alcohols)
3. Partial oxidation of paraffin's leads to linear secondary alcohols
4. Alfol process

Alfol process

- Al organic reaction based on K. Ziegler
- **Synthesis of higher α -olefins based on Ziegler (Alfen) process**
- Leads to primary, linear and even alcohols
- 4 stage process
- Build-up reaction: 90-120°C, 100 bar insertion of ethylene in TEAL and growth of alkyl chain



- Displacement reaction: 200-300°C, 50 bar release of α -olefins (even number of carbons)

Alfol process

1. Manufacturing of TEAL



fine Al powder hydrogenated with TEAL

@ 110-140°C and 50-200 bar in presence of H₂ to Diethylaluminiumhydride

Next conversion to TEAL with ethylene

@ 100°C and 25 bar

Alfol process

2. Chain build up

Exothermic reaction in plug flow reactor, 120°C and 100 – 140 bar



3. Oxidation of Trialkylaluminium

with dried air @ 50 – 100°C (exothermic reaction)

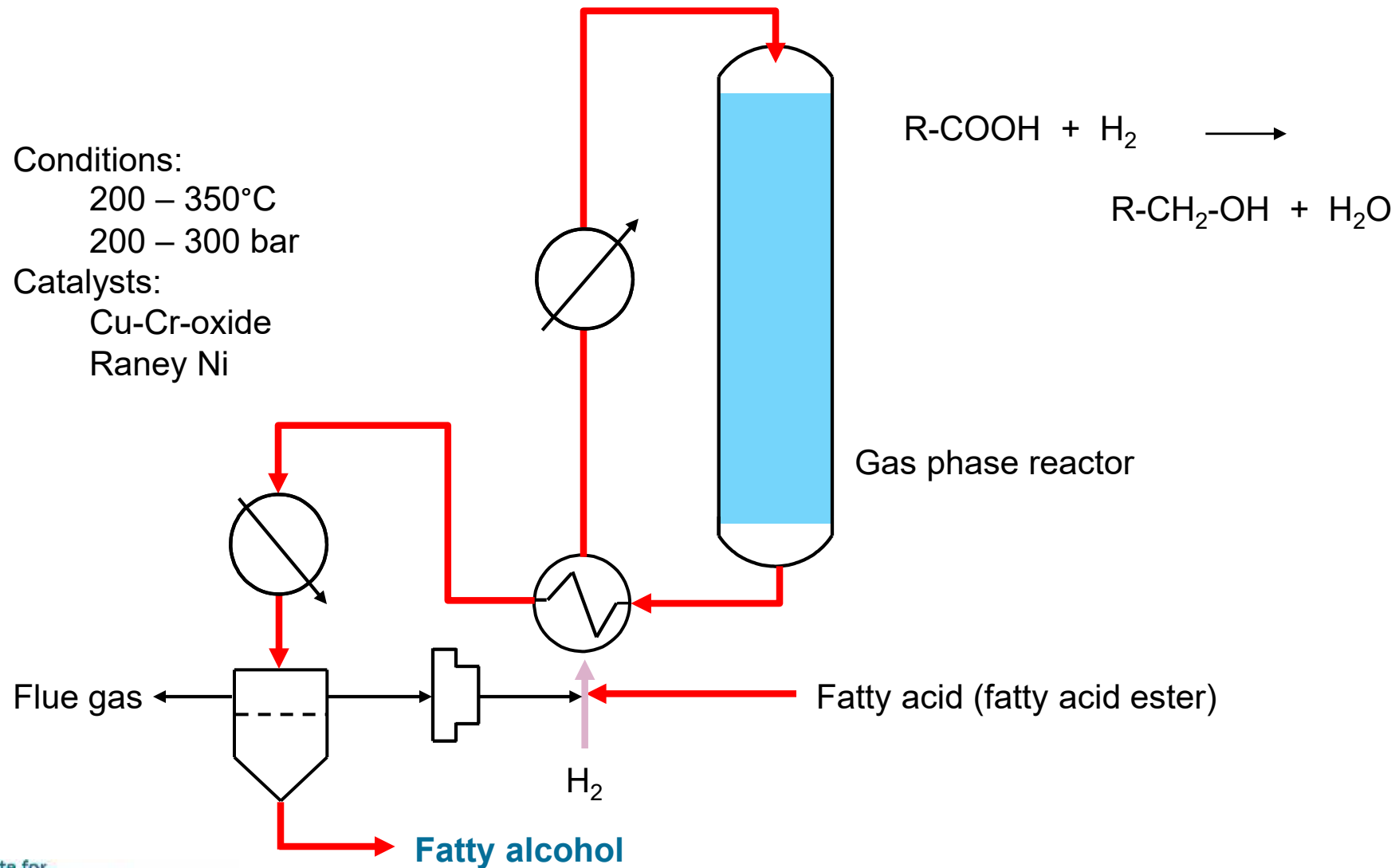


4. Saponification of Al-Alkoxids

with water or diluted H_2SO_4 to the alcohol and a $\text{Al}_2(\text{SO}_4)_3$ solution



Higher alcohols: hydrogenation of fatty acids



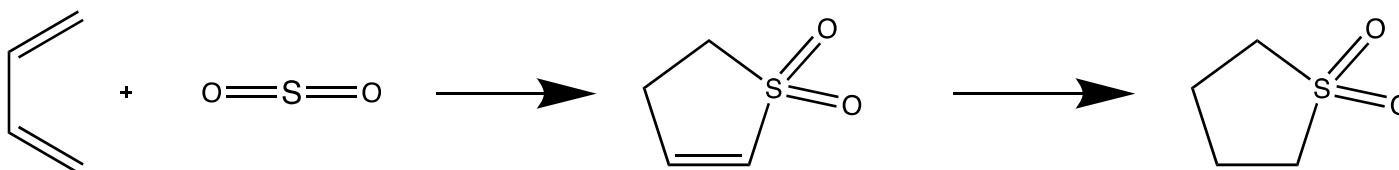
Raney Ni

- Particularly active form of Ni
- Pyrophoric heterogeneous catalyst formed by etching the inactive aluminum component from a nickel – aluminum alloy with alkali (NaOH) to afford a sponge like catalyst with large surface
- Used for catalytic hydrogenation or directly as a reducing agent
- Invented by Murray Raney in 1926 (Chattanooga, Tennessee)
- Initially used to turn liquid vegetable oils (cottonseed oil) into solid (hardening of fats/oils)
- By 1950 it had become one of the most widely utilized catalysts
- Cheaper than precious metals (Pd, Pt) based alternatives
- Can be tuned
 - Parameters of the digestion process, addition of small amounts of metals like Cr, Fe, Cu or Mo
 - Determine where hydrogen will be added to molecules with more than one potential insertion site

Raney Ni

Some examples:

- Petroleum-derived toluene is nitrated to dinitrotoluene followed by hydrogenation to make diaminotoluene → used for PU
- Butanediol is made by hydrogenating butynediol (made from acetylene) → used for Polyesters
- Sulfolane solvent used to refine petroleum and to remove unwanted gases, made from sulfolene (in turn, is made from butadiene)



- Non-petroleum example is sorbitol: this sugar substitute is produced by hydrogenation of glucose
- And many more

Higher alcohols – native oils

Fatty alcohols from coco oil

	%-w
$\text{CH}_3(\text{CH}_2)_6\text{CH}_2\text{OH}$	8-9
$\text{CH}_3(\text{CH}_2)_8\text{CH}_2\text{OH}$	7-10
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_2\text{OH}$	54-51
$\text{CH}_3(\text{CH}_2)_{12}\text{CH}_2\text{OH}$	16-18
$\text{CH}_3(\text{CH}_2)_{14}\text{CH}_2\text{OH}$	7-10
$\text{CH}_3(\text{CH}_2)_{16}\text{CH}_2\text{OH}$	1-3
$\text{CH}_3(\text{CH}_2)_{18}\text{CH}_2\text{OH}$	6-11

Fatty alcohol capacities [2000]

	Total [1000 t]	%-w native oils
WE	550	59
USA	531	22
J	108	20
World	1.300	51

Higher alcohols: oxidation of paraffin's

- Production of linear secondary alcohols
- Developed in the 1930ies in D
from the 40ies technically used Hydrocarbons from the Fischer-Tropsch synthesis were oxidized to a mixture of alcohols, ketones, esters and acids
- Selectivity was drastically increased by the use of Boron acid
- Bashkirov oxidation

Higher alcohols: Bashkirov oxidation

Conditions:

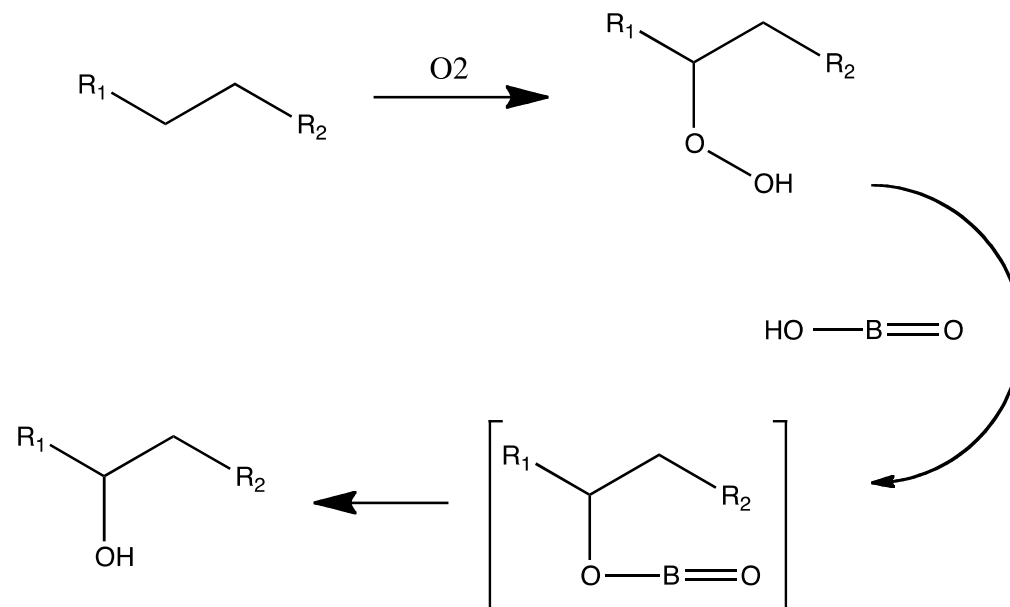
140-190°C

4-5 %-w H_3BO_4

(0,1 %-w KMnO_4)

Turnover 15-25%

Normal pressure



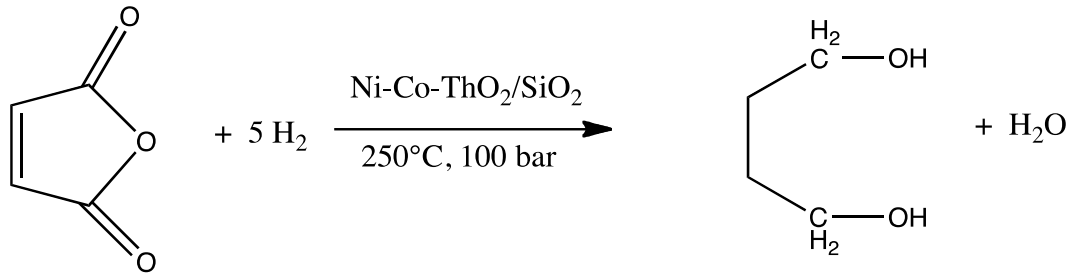
Polyols

Technically of importance:

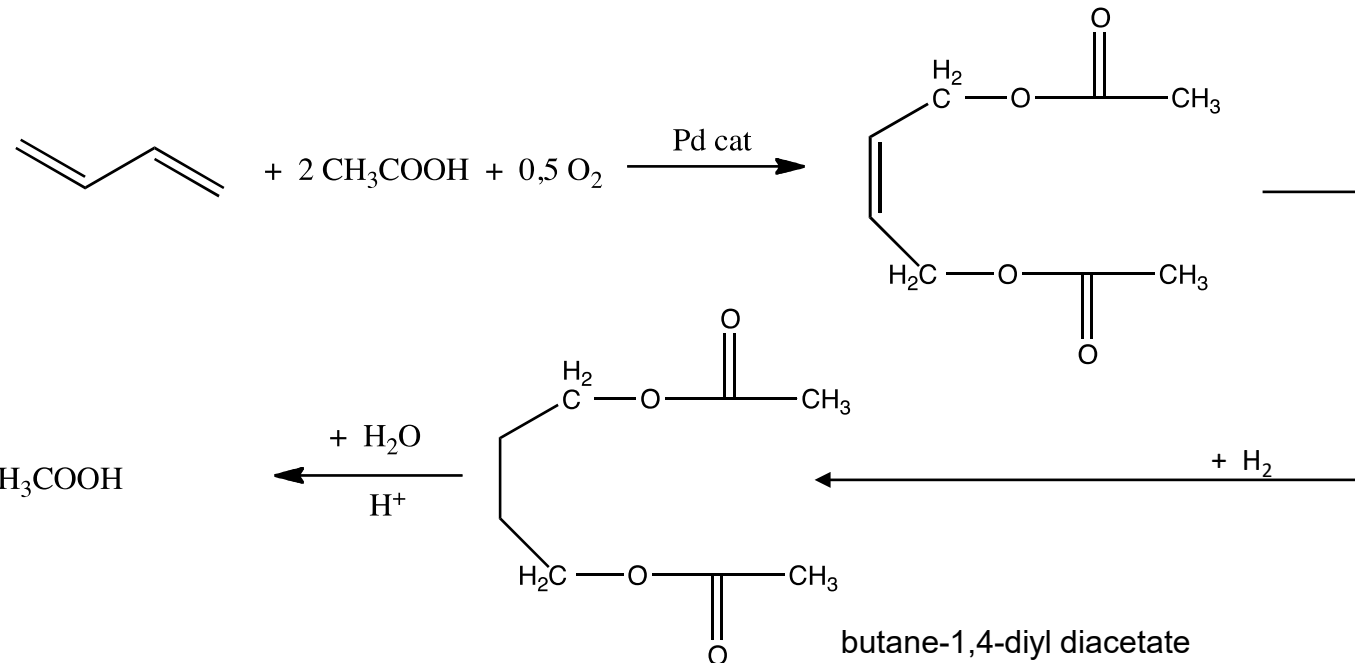
- 1,2 diols: ethylene-, propylene glycol
- 1,3 diols: propane diol, dimethyl propane diol, butane diol
- 1,4 diols: butine-, butene-, butane diol
- 1,6 diols: hexane diol
- Triols: glycerin, trimethylol propane
- Tetraols: Pentaerythrite

Butanediol

- MSA (ICI) process:

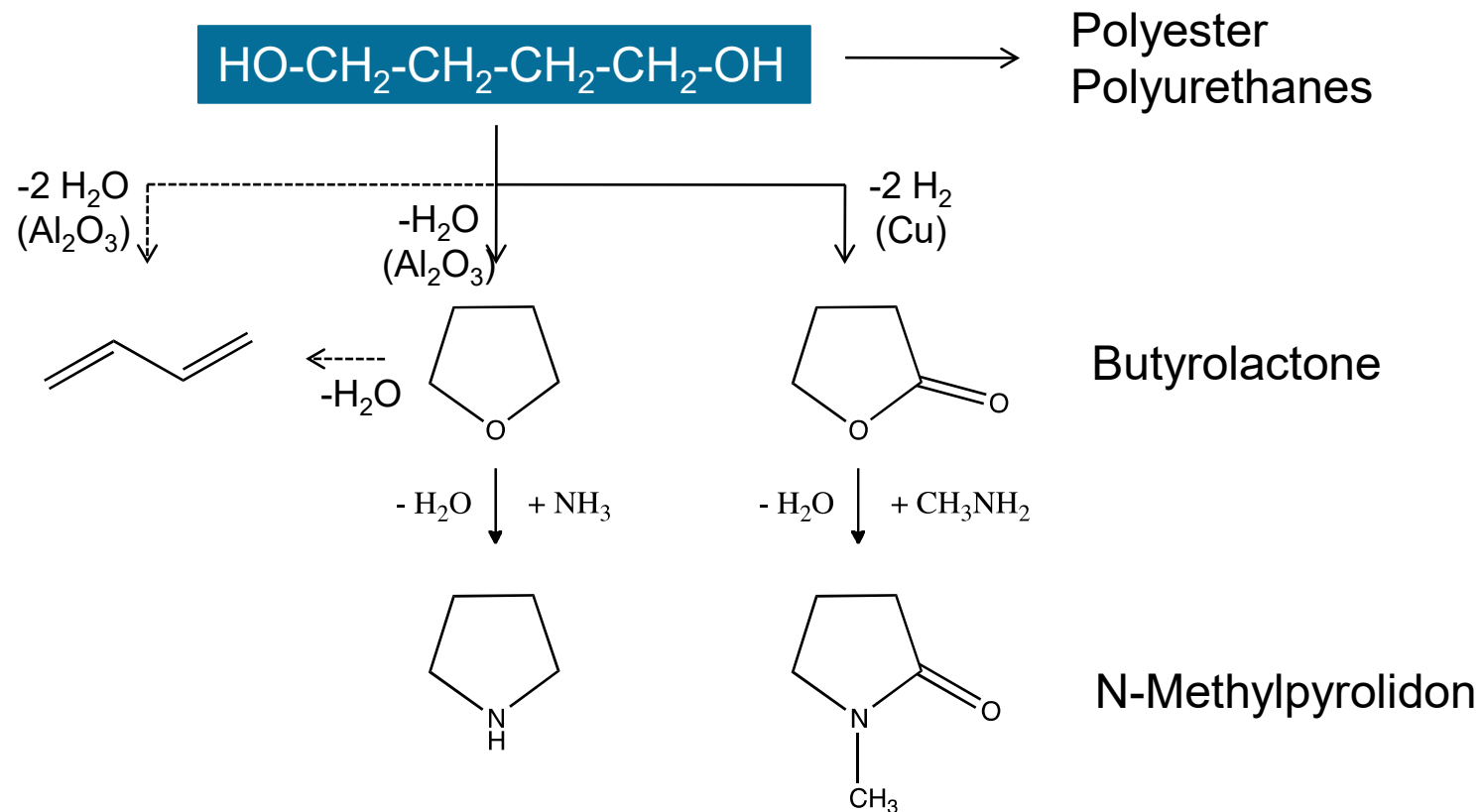


- Butadiene:

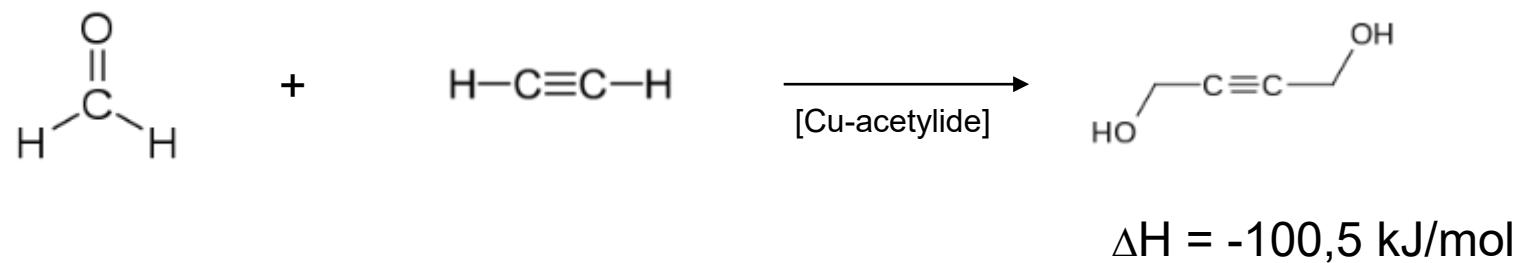


Butanediol

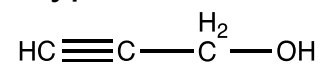
Technically important reactions



Butynediol

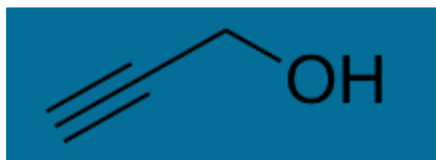


Byproduct:

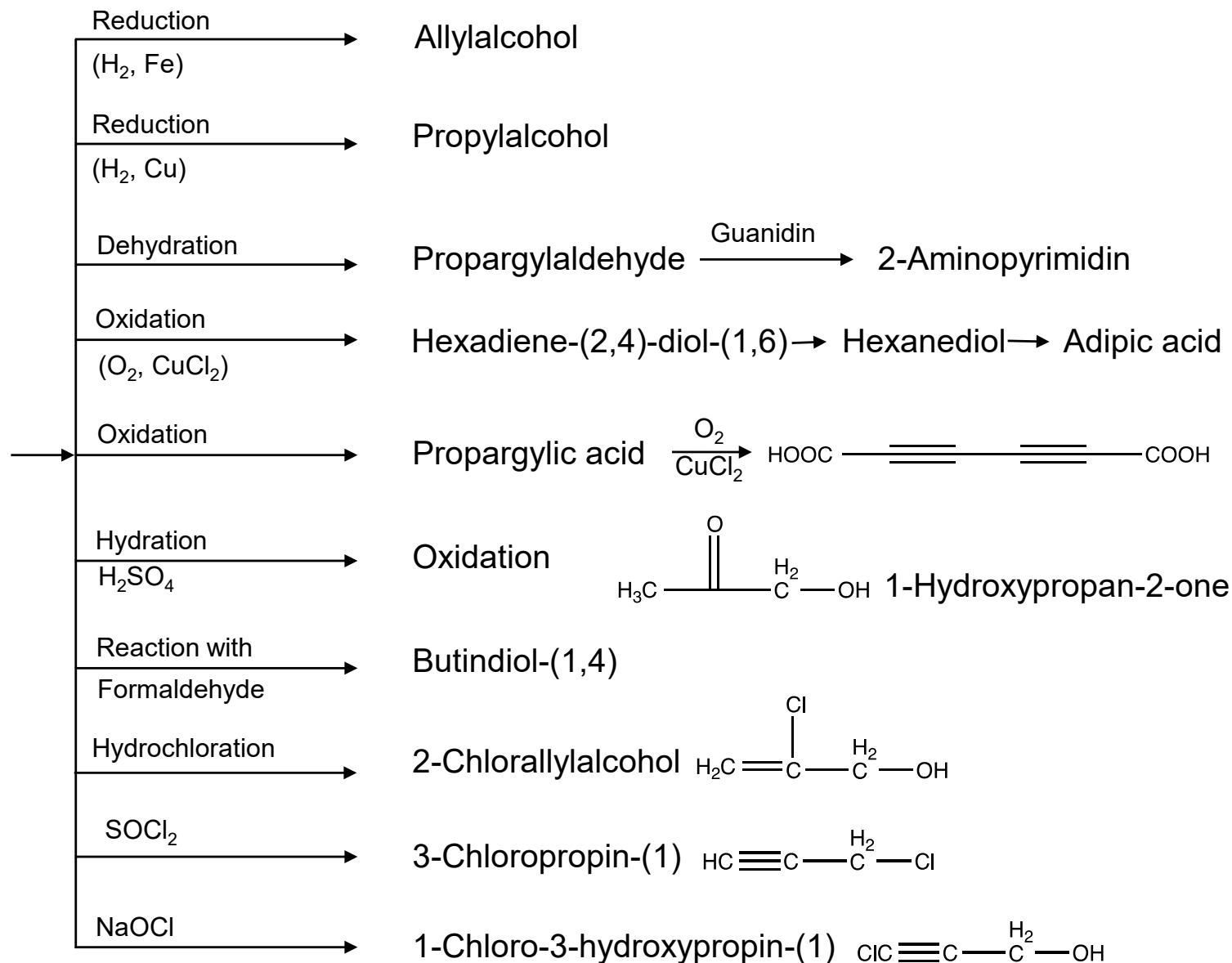


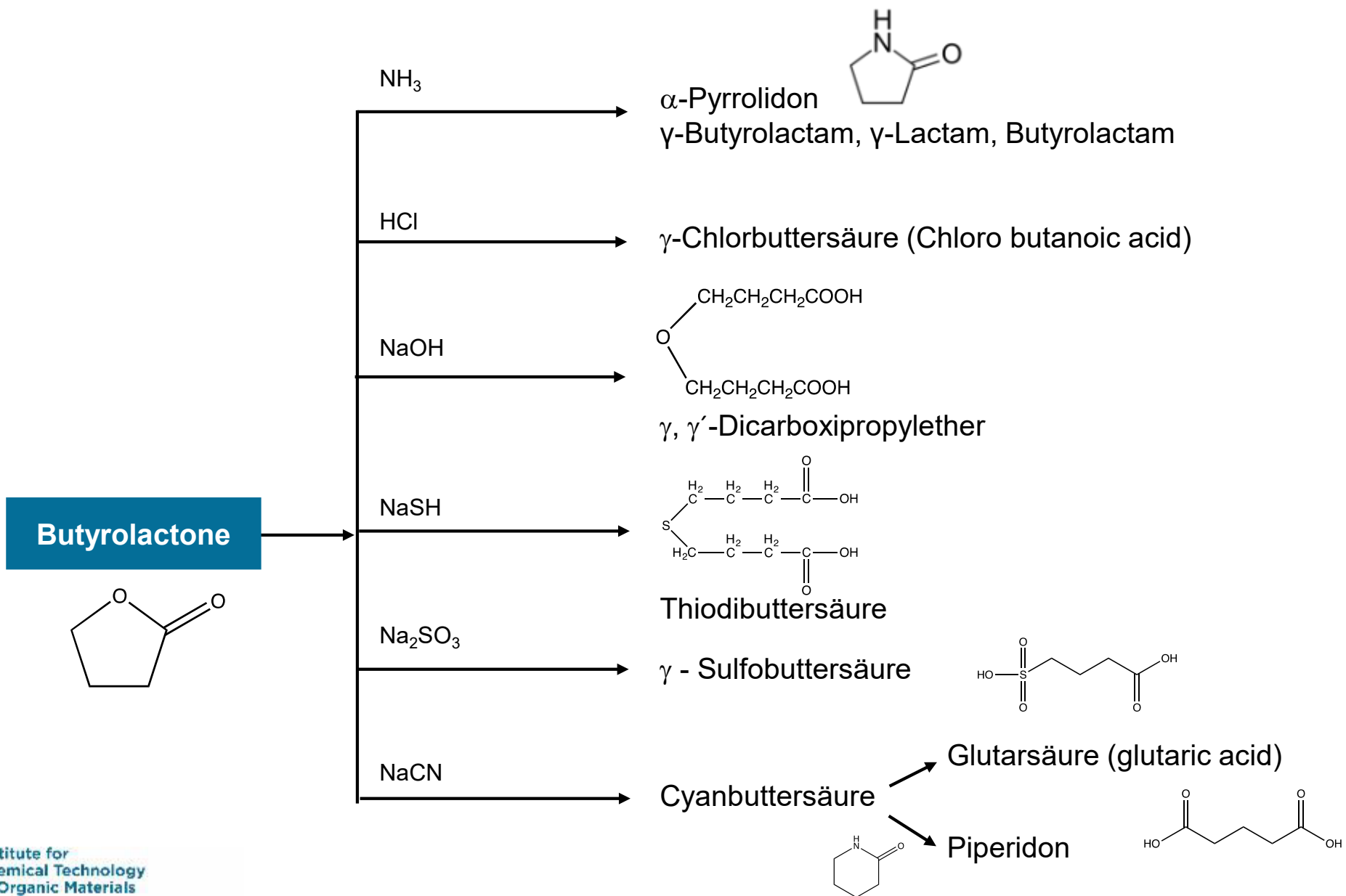
Propargylalcohol

Prop-2-yn-1-ol



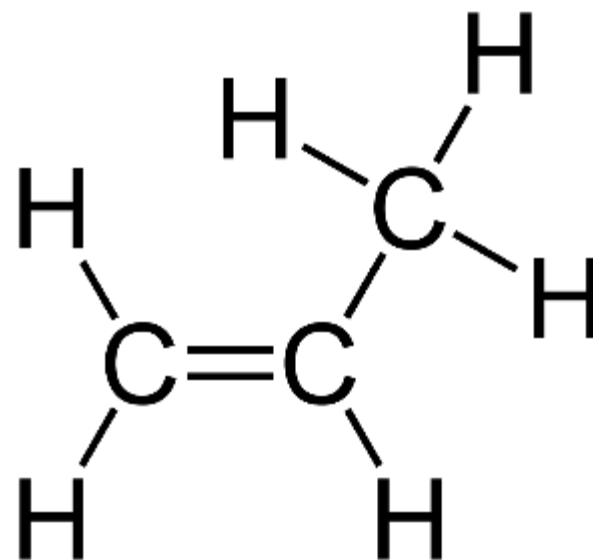
Propargylalcohol
Prop-2-yn-1-ol



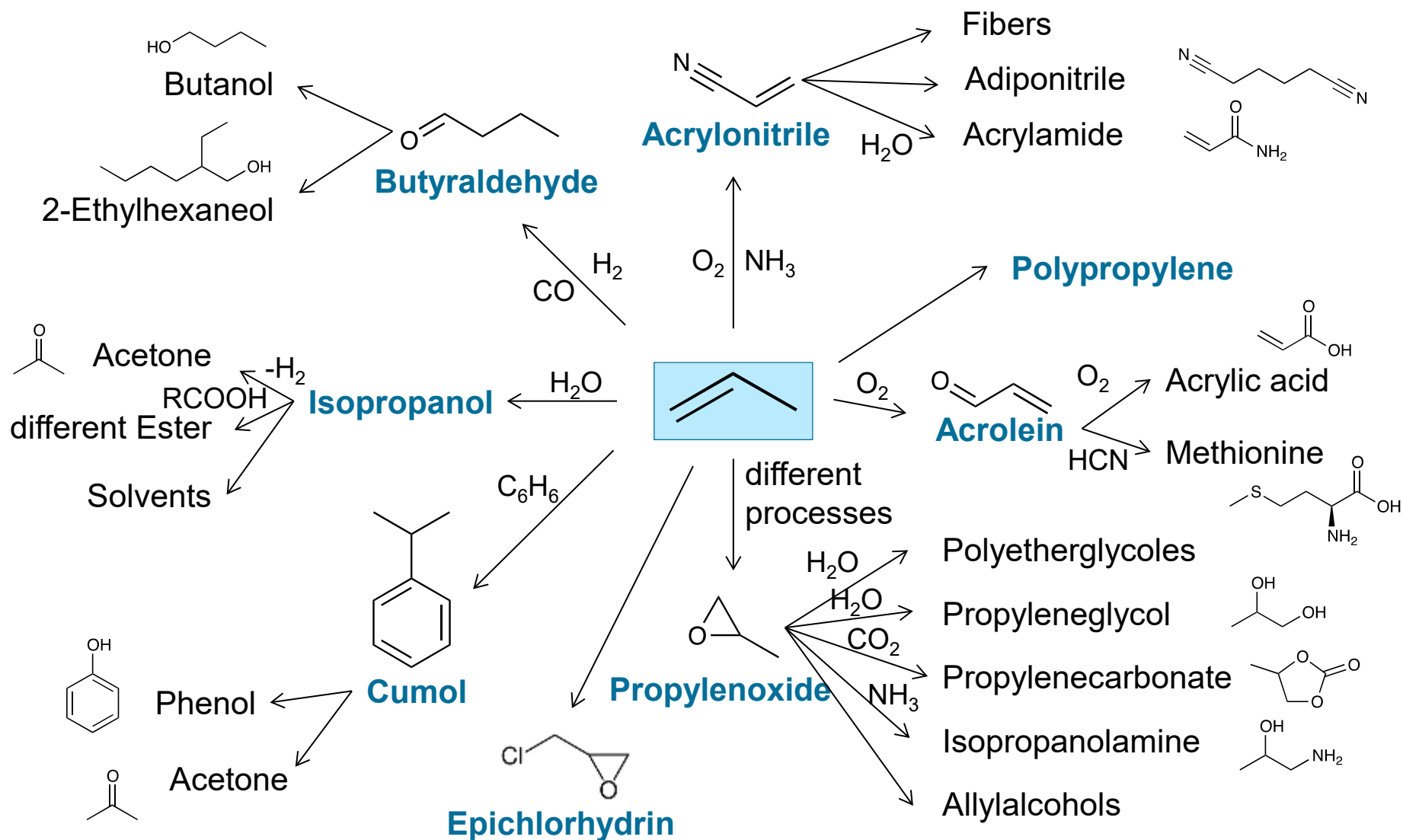


Propylene

- Molar mass: 42,08 g/mol
- Density: 1,91 kg/m³ (0°C)
- T_m: -185,3°C
- T_b: -47,7°C
- Flash point: -108°C

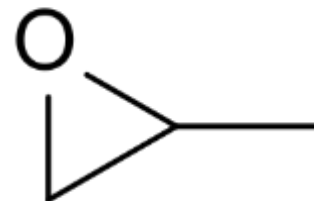


Propylene family tree



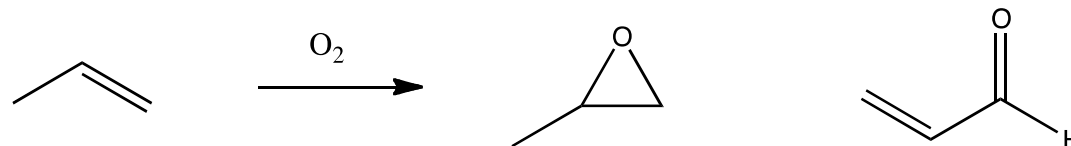
Propylene oxide

- Molar mass: 58,08 g/mol
- Density: 0,83 kg/m³ (20°C)
- T_m: -112°C
- T_b: 34°C
- Worldwide production > 6 Mill tons



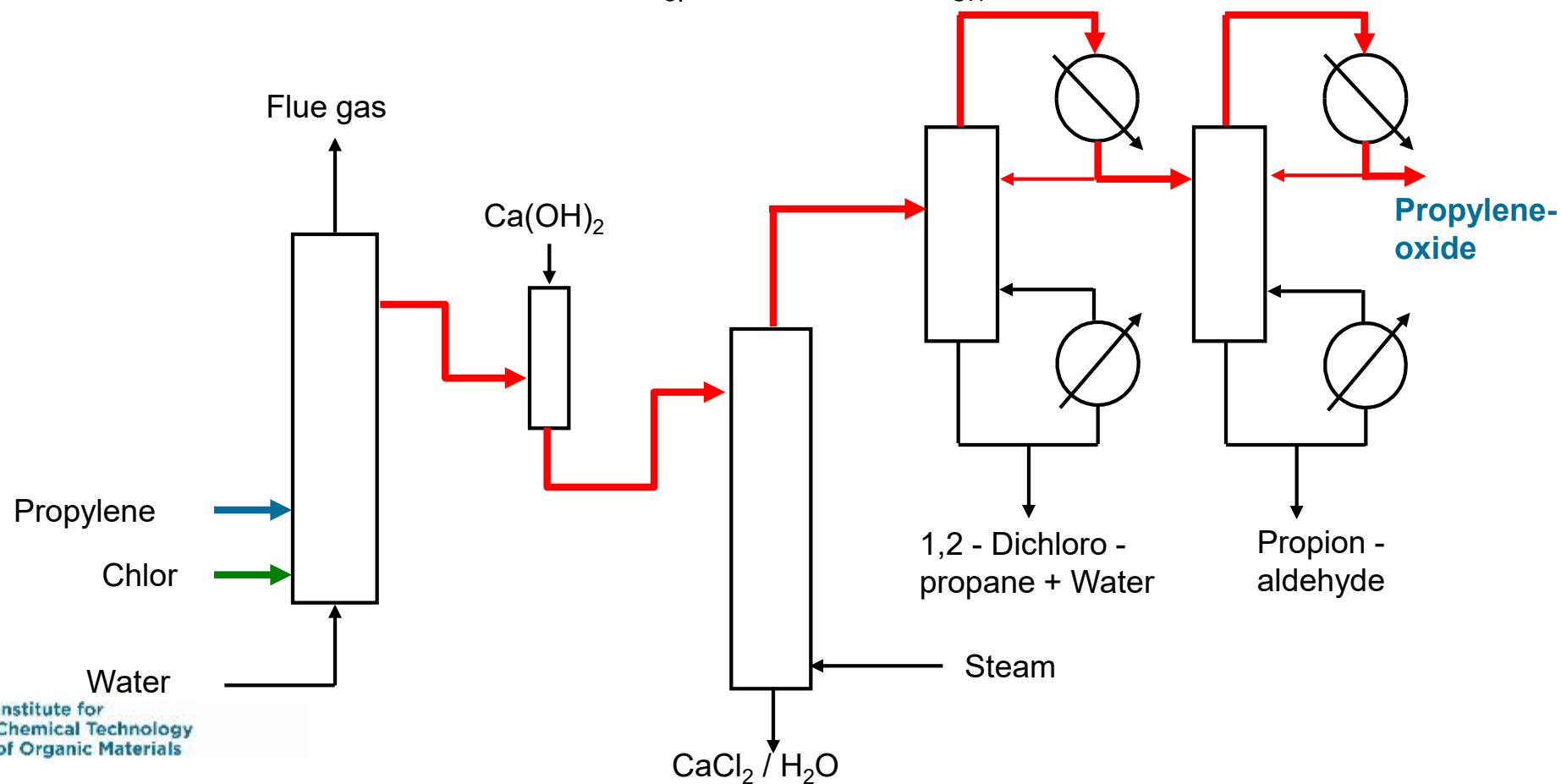
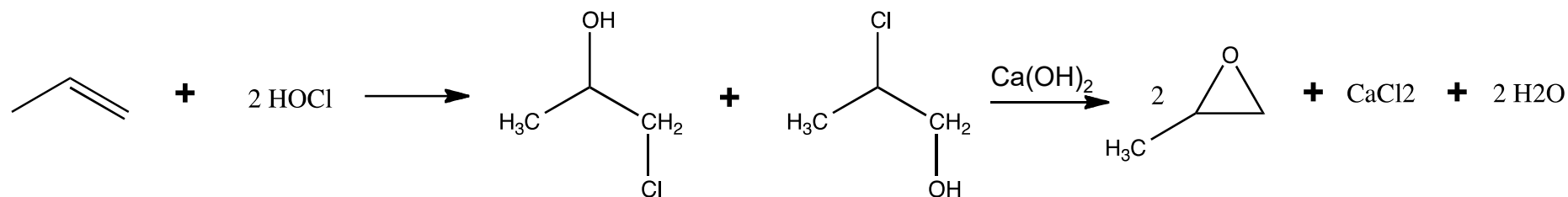
Propylene oxide

- Direct oxidation like $C_2=$ to EO does not work

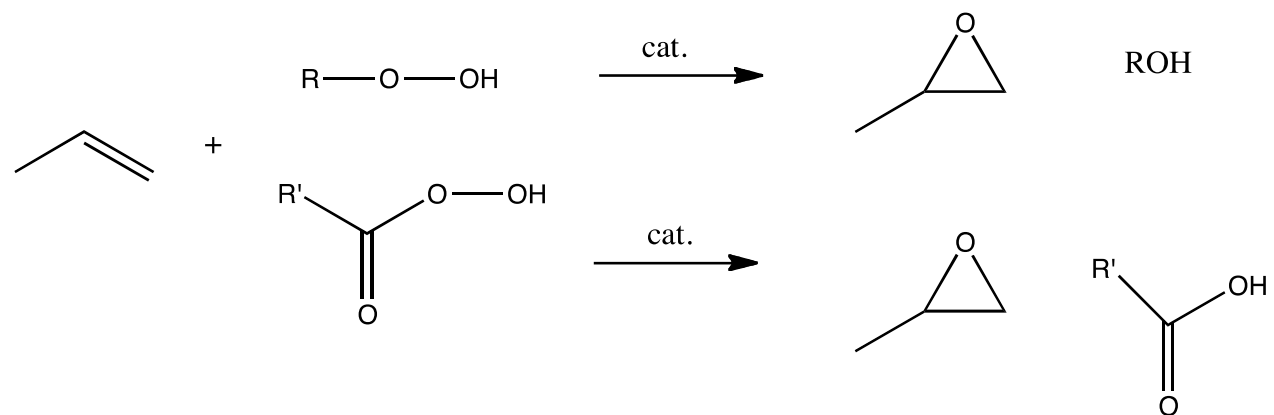


- All efforts so far negative
- H of the allyl methyl group gets oxidized – acrolein
- Only little epoxy selectivity
- Byproducts (C1-, C2-, C3- materials) make work up more difficult

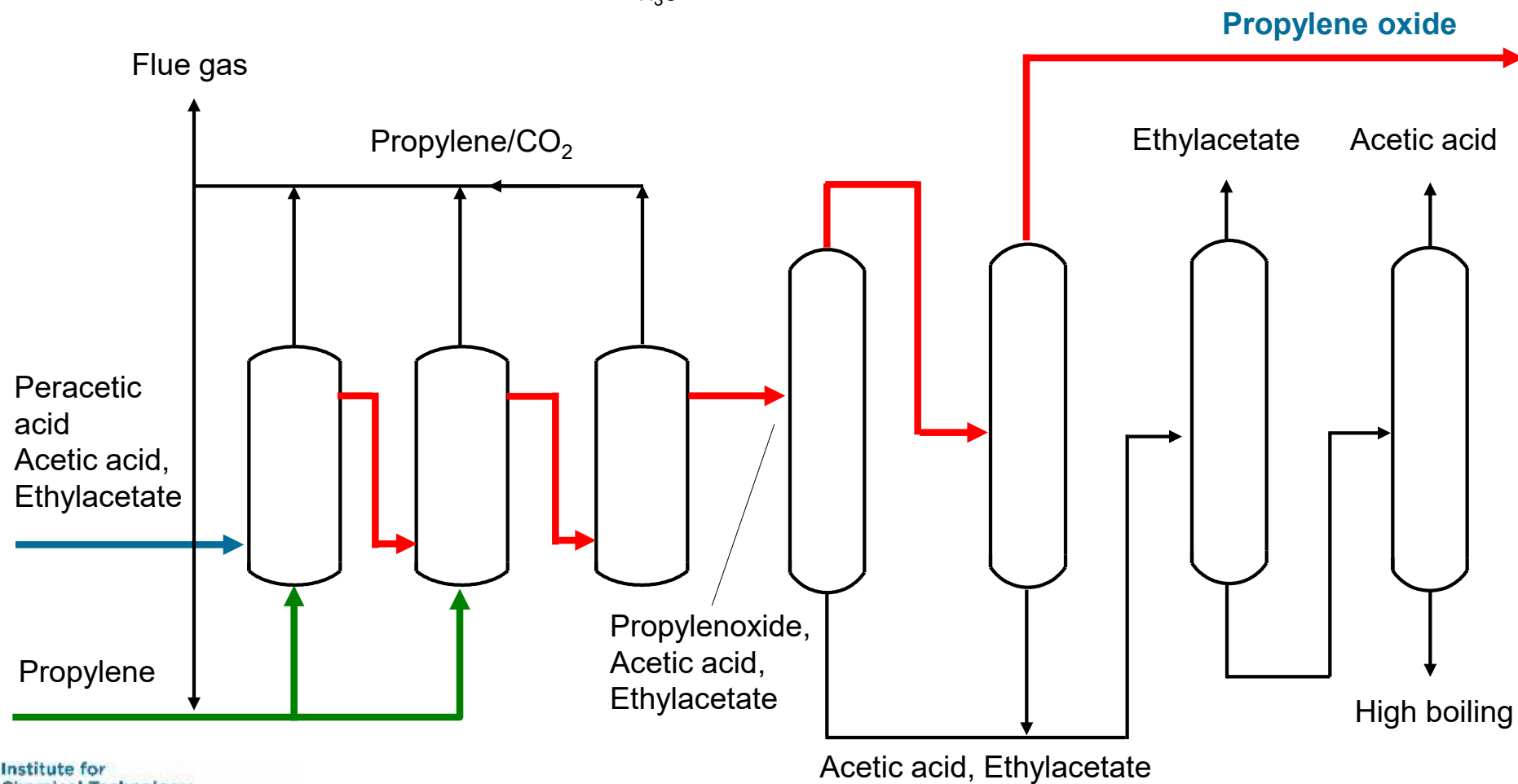
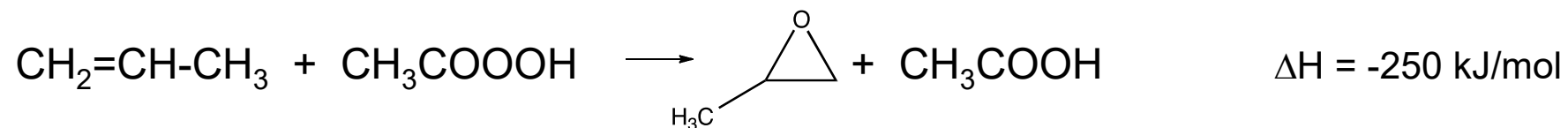
PO synthesis - Chlorohydrin



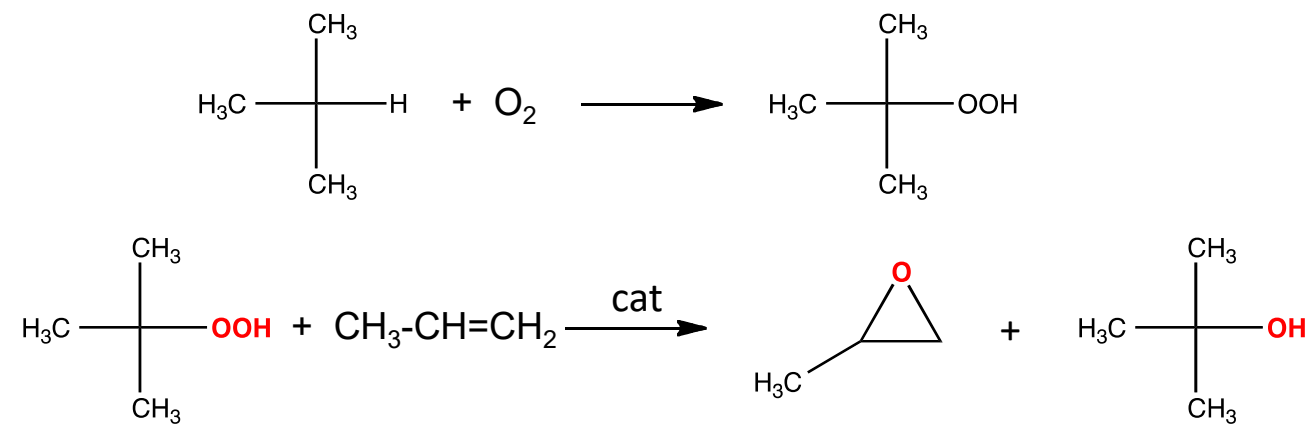
PO synthesis - indirect



PO synthesis



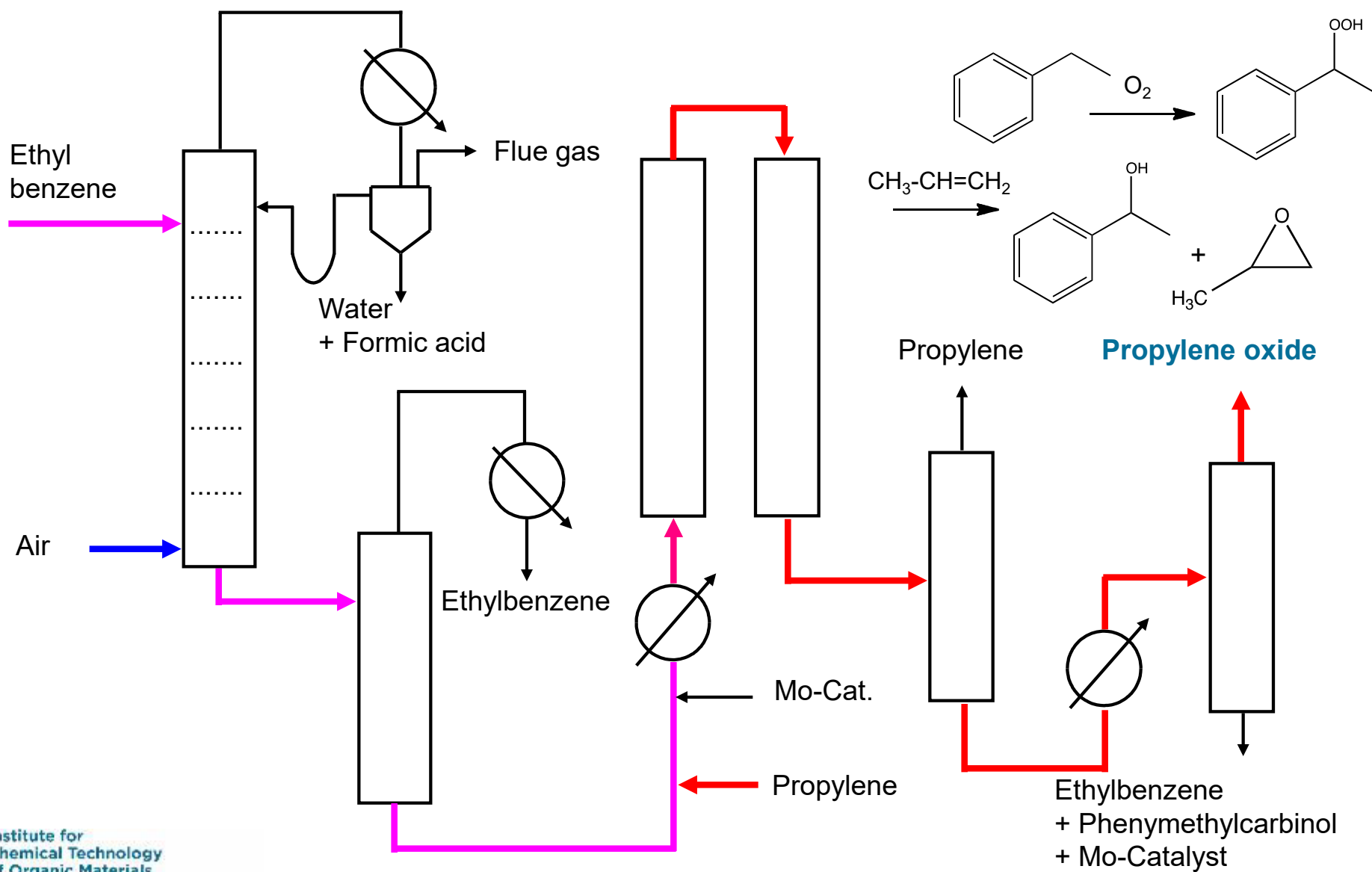
Oxirane (Halcon) process



PO synthesis

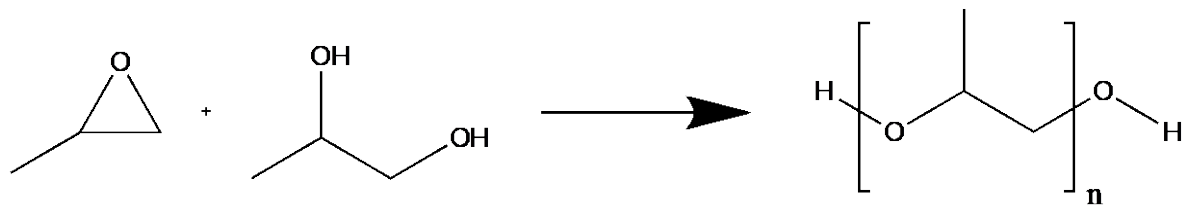
Educt	Co-oxidate	Byproduct
Acetaldehyde	Acetic acid, acetic anhydride	
Iso butane	Tert. butanol	Iso butene, MTBE
Iso pentane	Iso pentanol	Isopentene, isoprene
Cyclohexane	Cyclohexanol/on	
Ethylbenzene	Methyl phenyl carbinol	Sytrene
Cumol	Di methyl phenyl carbinol	a-Methyl styrene

PO synthesis



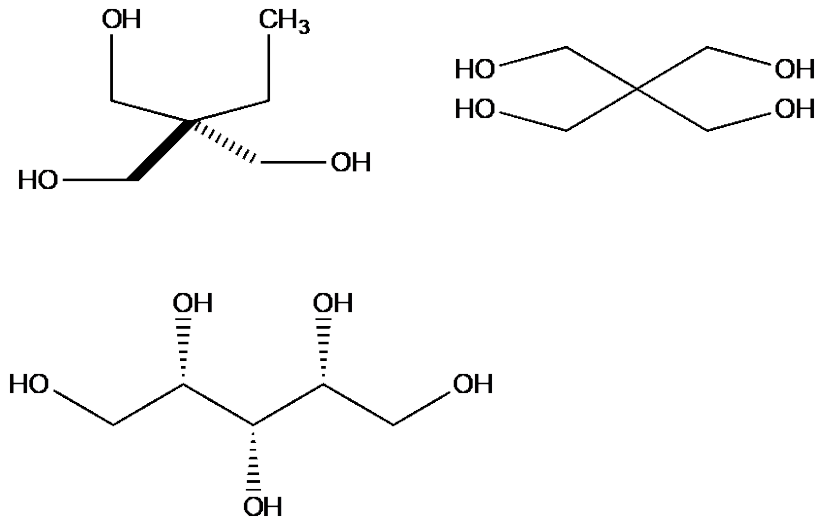
Propylene glycol

- Volume wise biggest derivative of PO
- Made through the reaction of PO with propylene glycol (1,2 propylene glycol)
- Mw > 1000 not soluble in water – via addition of EO these ethers can be made soluble again
 - Surface active products used for detergents and as emulsifiers



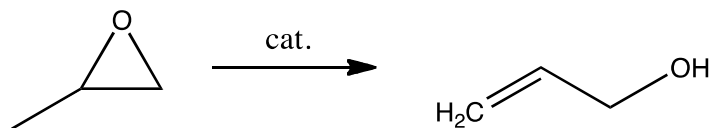
Propylene glycol: monomer for PU

- PO plus higher alcohols
(glycerin, trimethylolpropane, pentaerythrite)
to adjust PU properties



Allyl alcohol

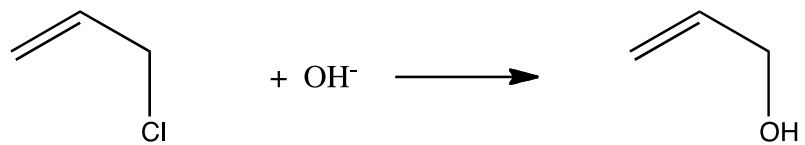
- Can be made from allyl chloride, PO, acrolein or allyl acetate
- Isomerization of PO:



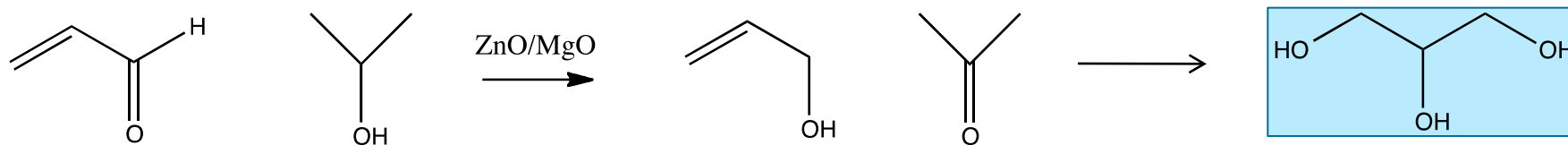
- Liquid or gas phase
 - Catalyst: Li_3PO_4
 - 275 - 280°C, 10 bar
 - Selectivity roughly 92%
- Used for the **production of glycerin** (mainly)
 - Historically byproduct in the saponification of fats
 - Used for synthetic detergents

Glycerin

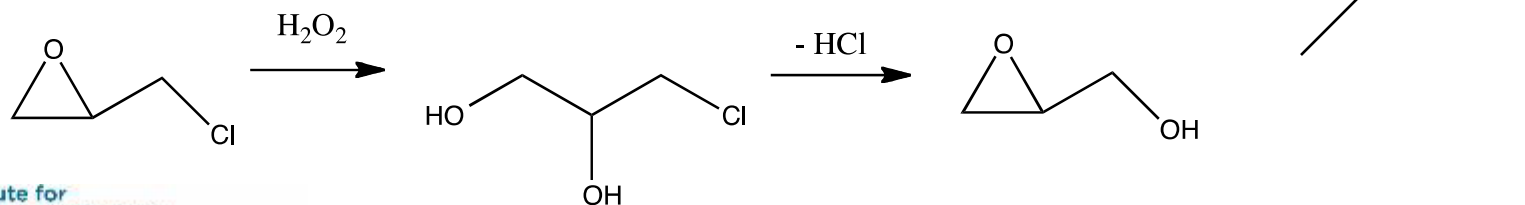
Saponification of allyl chloride:



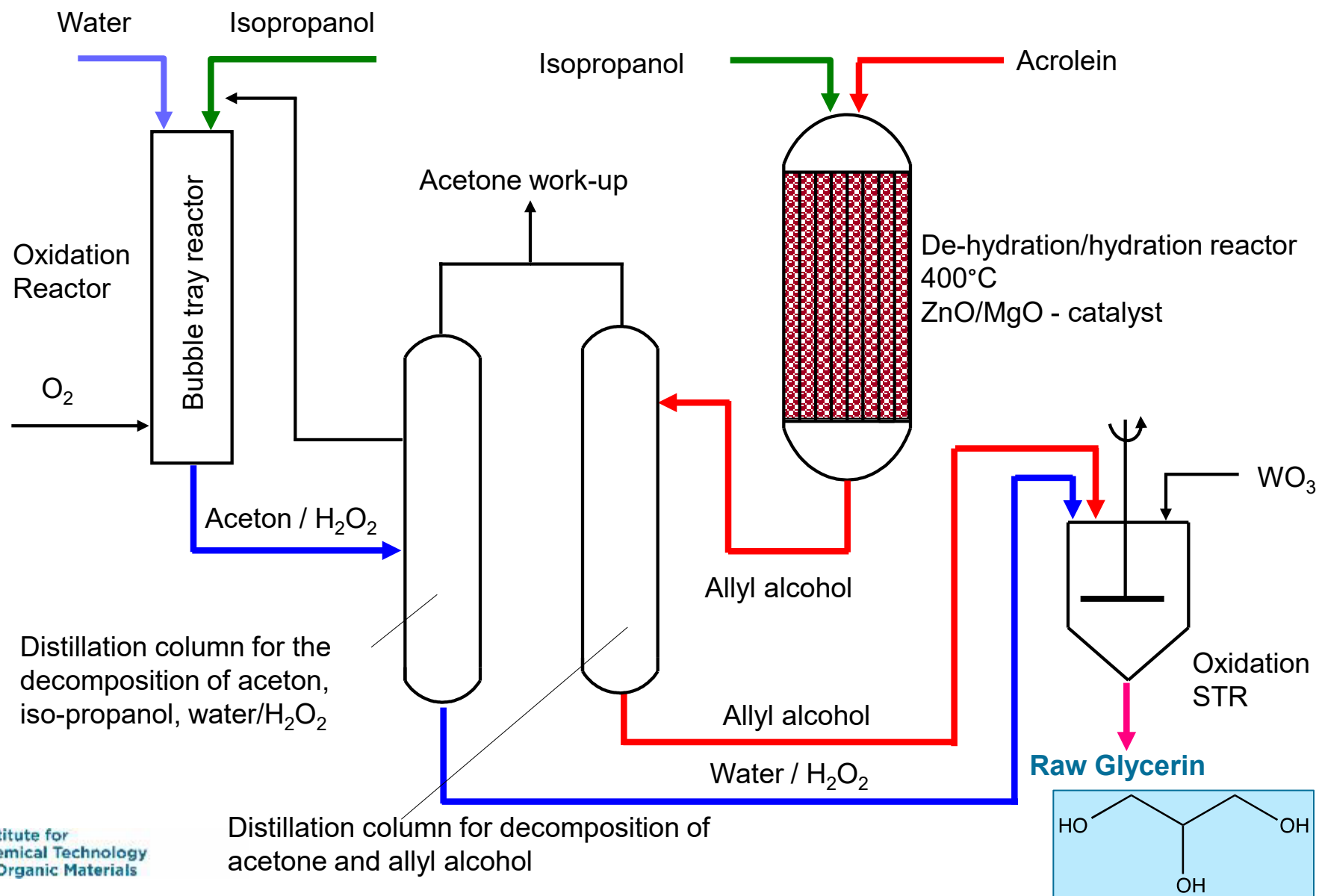
Acrolein:



Epichlorohydrin:



Glycerin from acrolein



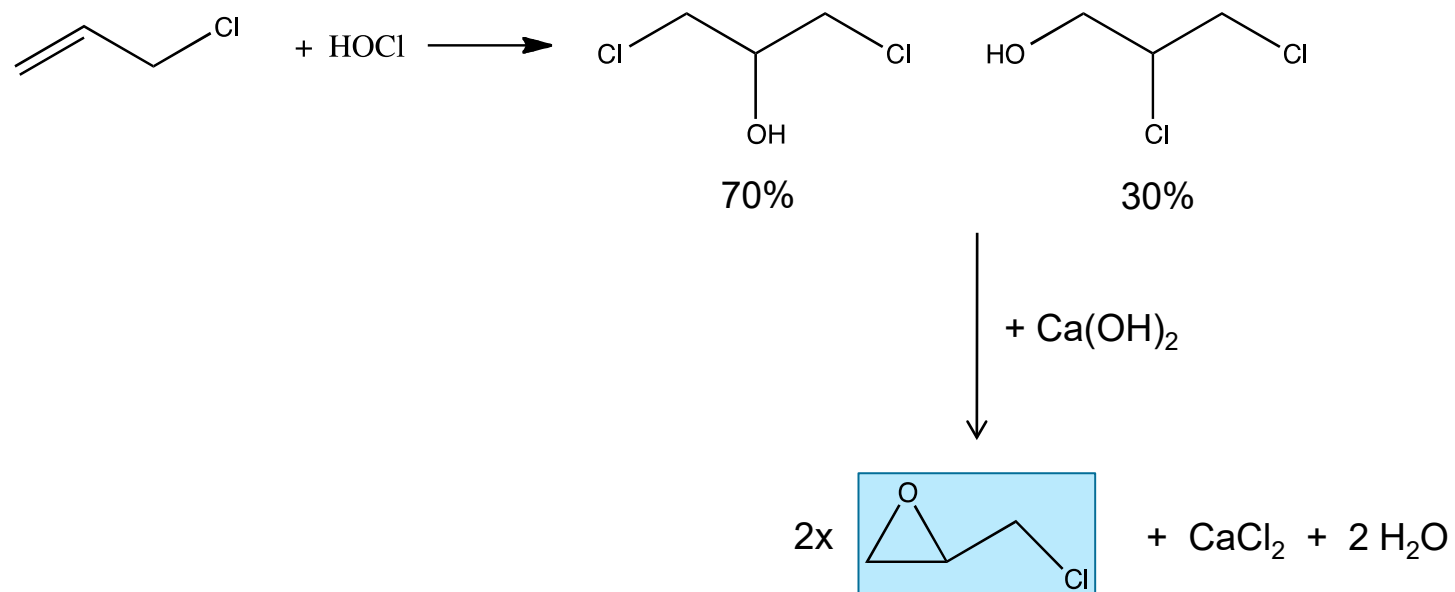
Allyl chloride

- Cl adds easier on propylene than on ethylene
- Dichloropropane has no technical importance



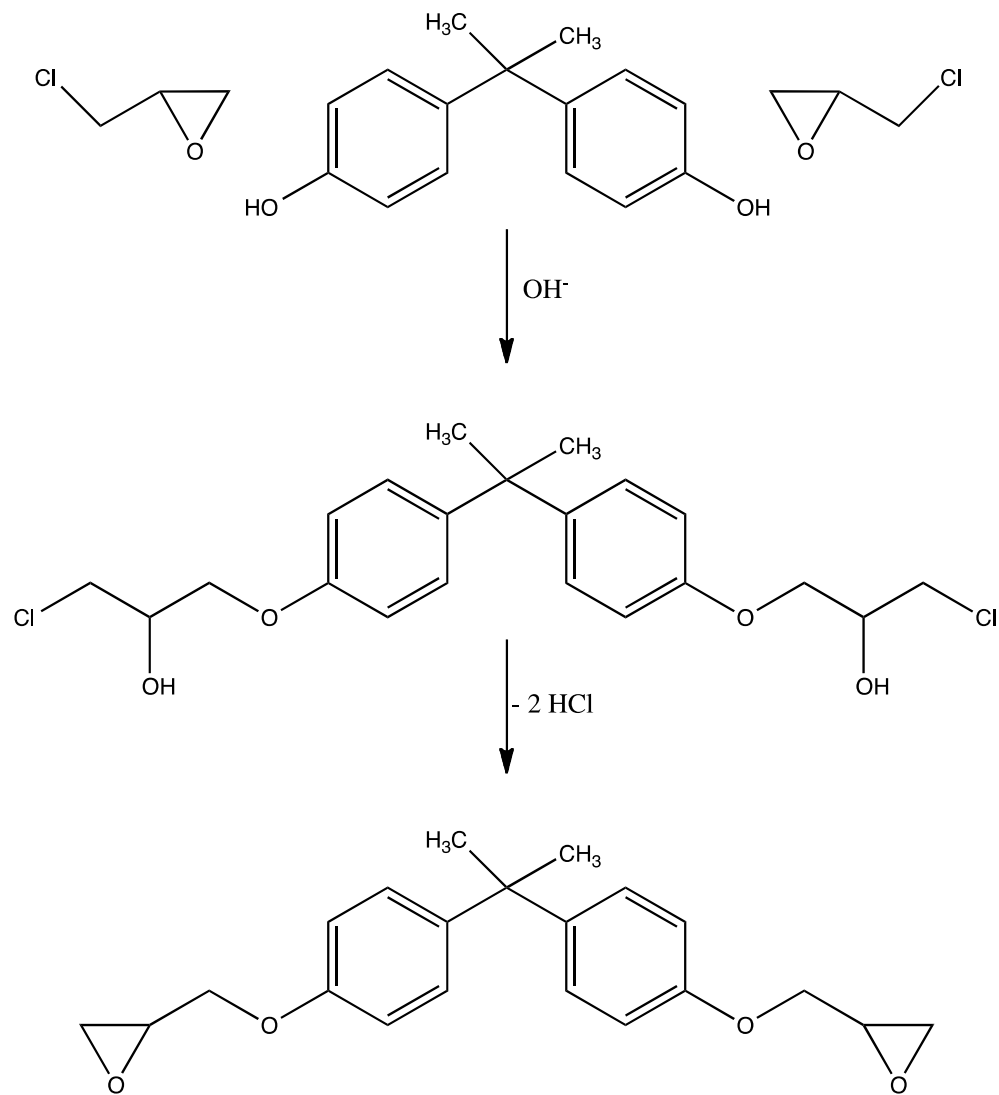
- Propylene in excess
 - Byproducts: 1,3 di-chloro-1-propene
- 1996: 0,41 Mill. t/a
- Mainly used for the production of **epichlorohydrine**

Epichlorohydrin



Other route:
Allyl chloride plus per propionic acid

Plus Bisphenol-A to epoxy resins

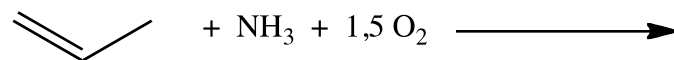
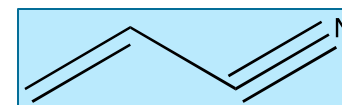
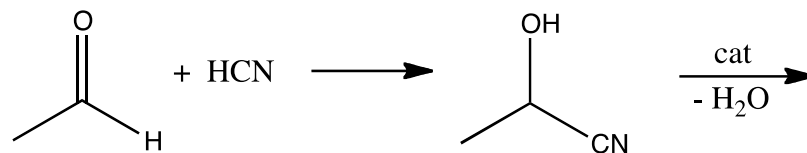
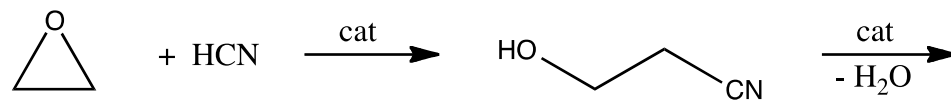


Acrylonitrile

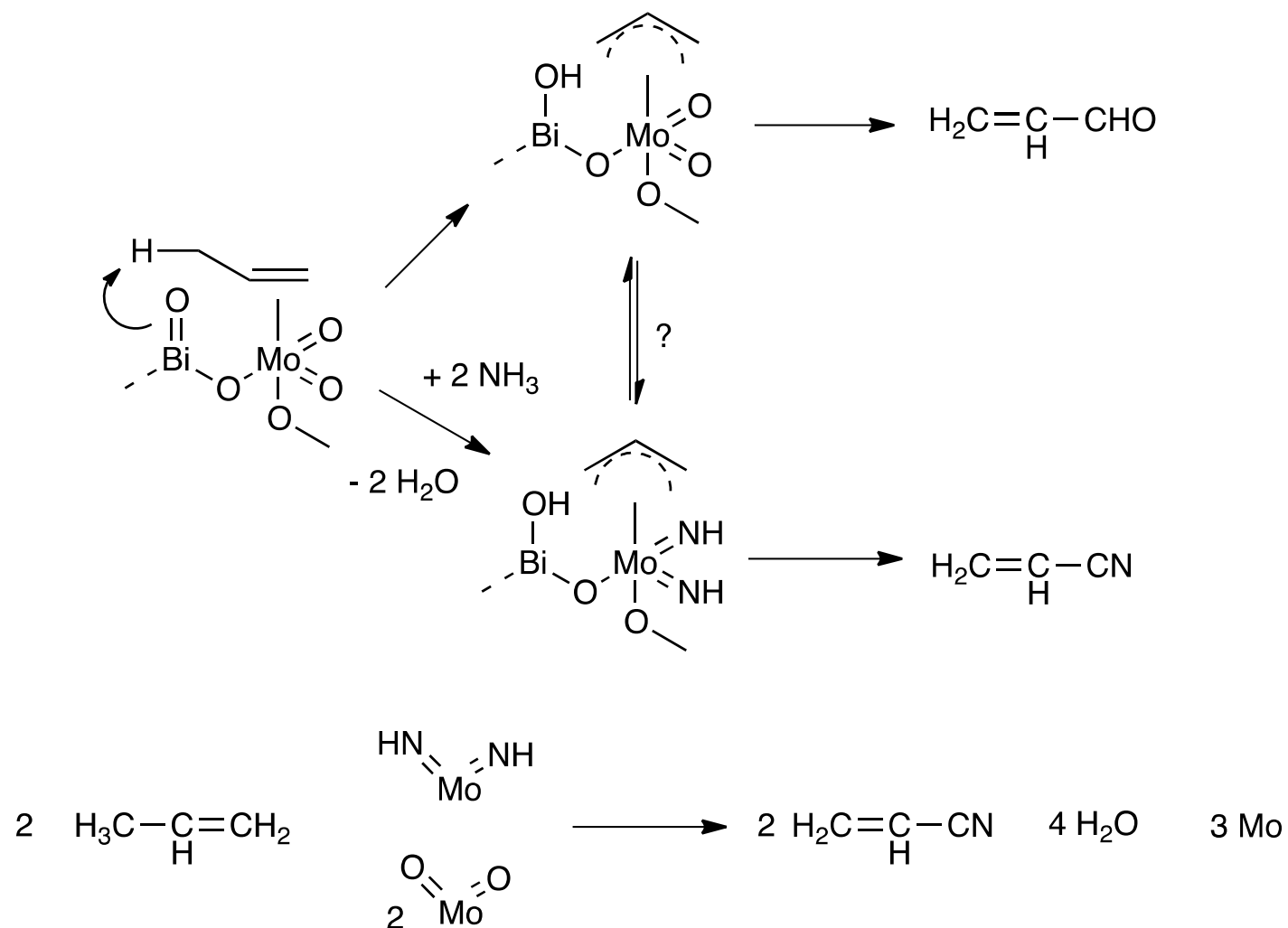
- Copolymerization with butadiene to synthetic rubber Perbunan
- Perbunan was developed by Bayer in the 1930s as the first-ever oil-resistant rubber
- 2022: 8,6 Mill. t worldwide



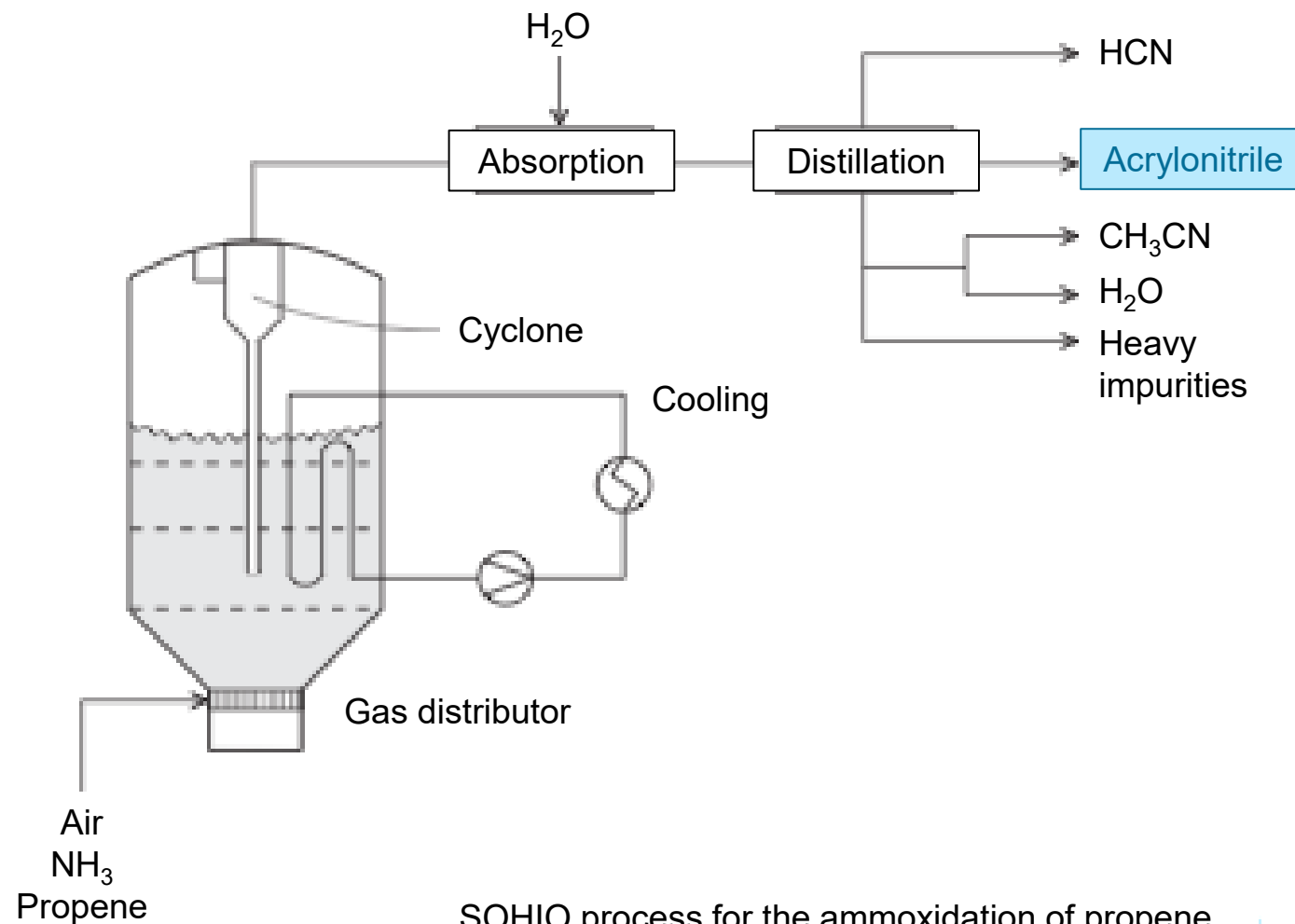
Production of acrylonitrile



Ammonoxidation of propylene



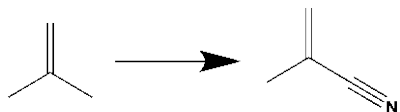
SOHIO acrylonitrile process



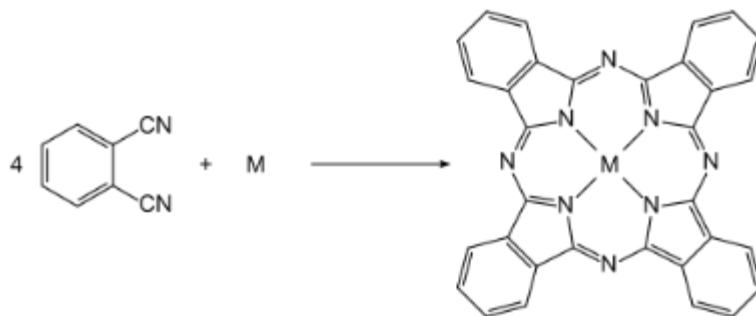
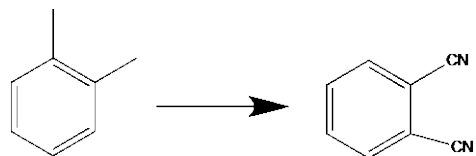
SOHIO process for the ammoxidation of propene

Ammoxidation

- Most important process: ammoxidation of propene
 - > 4 Mill.t/a
 - Plant capacities up to 180.000 t/a
- Other products
 - Methacrylonitrile from isobutene

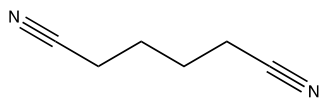


- HCN from methane
- Phthalodinitrile from o-xylene



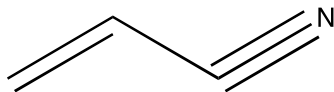
Acrylonitrile - use

- Mostly used as monomer for the production of homo- and copolymers for fibers, plastics and synthetic rubbers
- Also used as intermediate, e.g. adiponitrile



- Because of the reactive double bond used in many reactions with alcohols, amines, aldehydes and ketons – in combination with a basic catalyst – cyanethylation
- Pharmaceuticals and colors
- Acrylamide
Hydrolysis in the presence of stoichiometric amounts of H_2SO_4

Acrylonitrile



Polymerization

PAN-Fiber

Copolymerization
with butadiene

Buna N

Copolymerization
with vinyl chloride

Vinyon N, Dynel

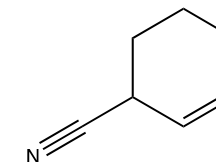
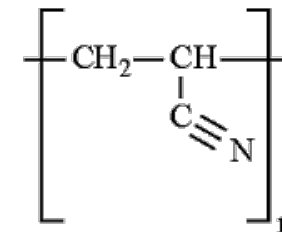
Copolymerization
with vinyl acetate

Acrilan

Butadiene

DIELS-ALDER-reaction

1,2,5,6,-Tetrahydrobenzonitrile



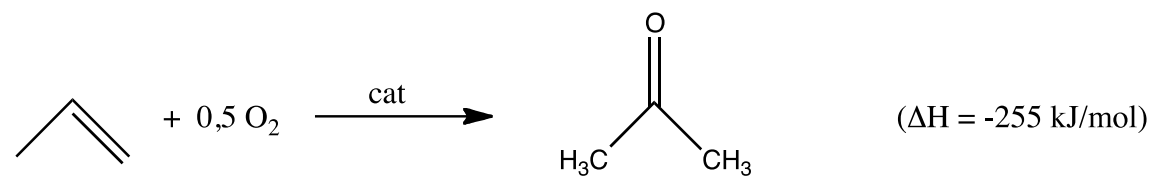
Acetone

- Molar mass: 58,08 g/mol
- Density: 0,79 kg/m³ (20°C)
- T_m: -95°C
- T_b: 56°C
- Worldwide production roughly 7 Mill tons

Processes:

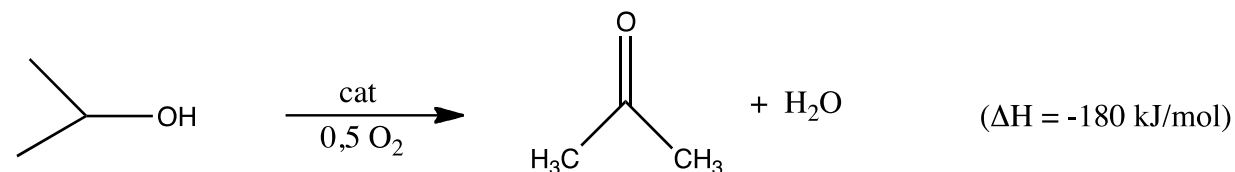
- Propylene direct oxidation
Wacker-Hoechst
- Isopropanol dehydrogenation
- Co-product in phenol
production – Hock cumene process

Direct oxidation of propylene



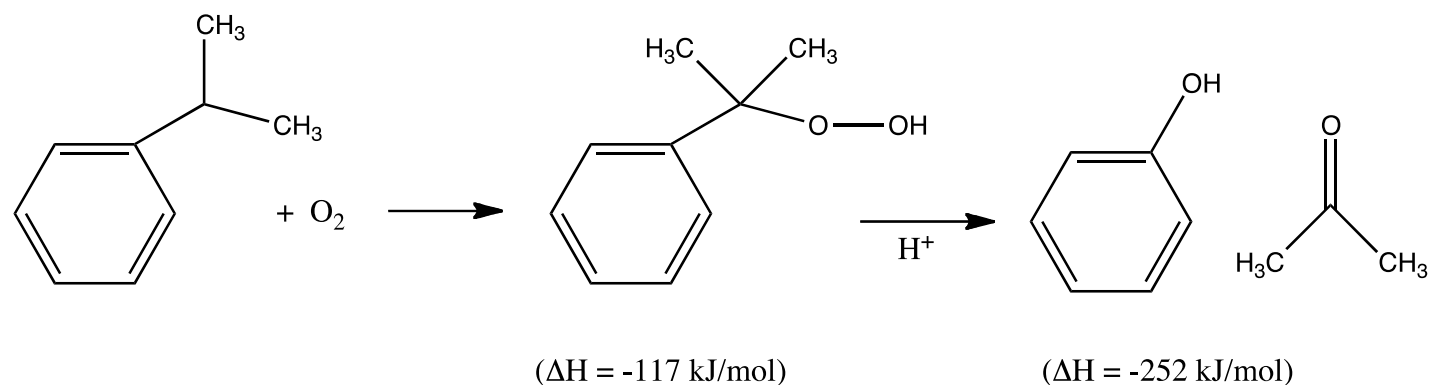
- Wacker-Hoechst process since 1964
- Catalyst: PdCl₂
- Liquid phase, 110-120°C and 10-14 bar
- 92% selectivity, side product: propion aldehyde
- Propylene yield > 99%

Isopropanol dehydrogenation



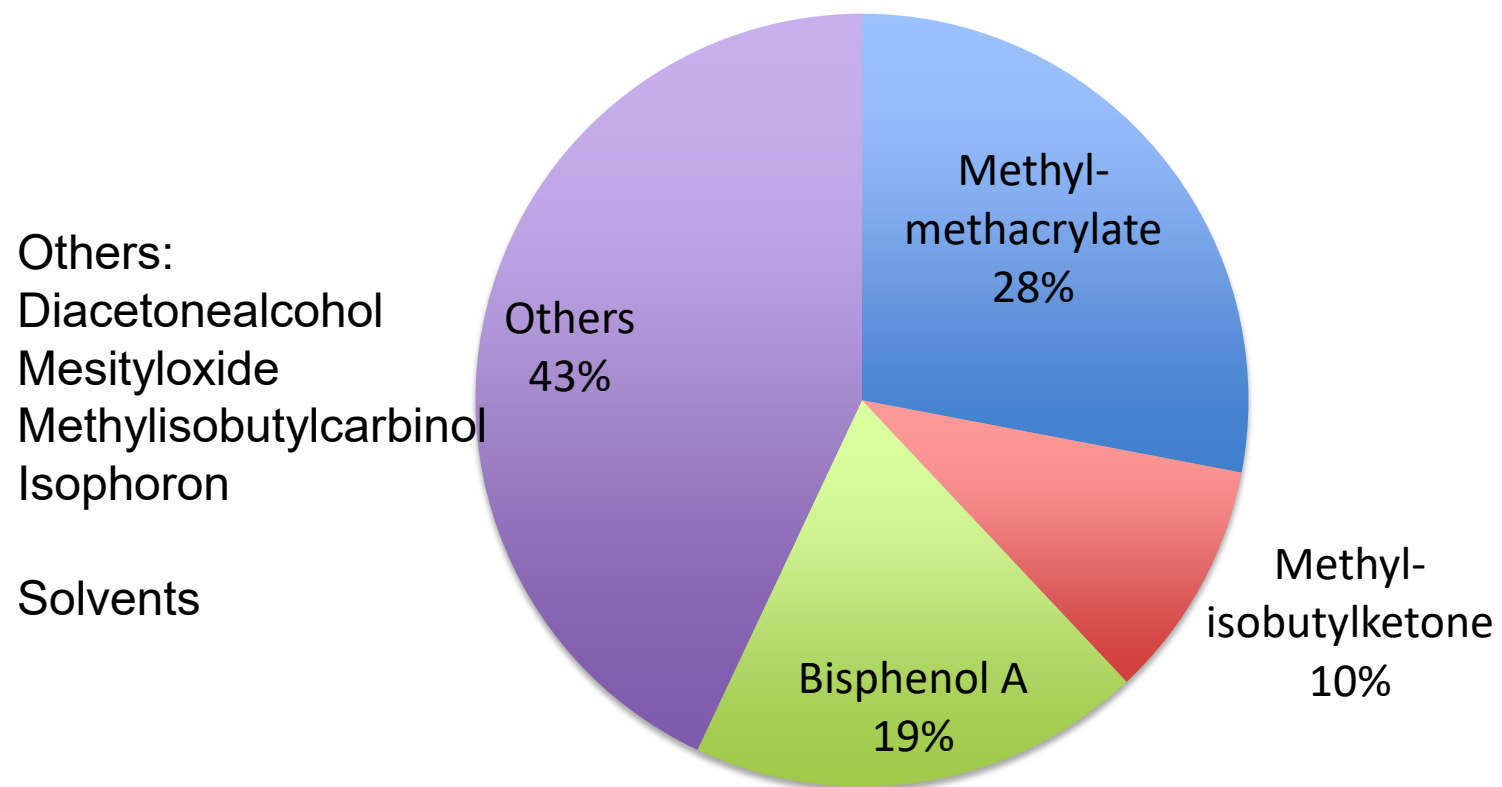
- Oxidative dehydrogenation
 - 400 - 600°C // Ag or Cu catalysts
- Dehydrogenation
 - 300 - 400°C // ZnO catalyst
 - 500°C, 3 bar // Cu or brass catalyst
 - 98% isopropanol conversion, 90% selectivity
- In US 1970 still 50-60% of acetone based on isopropanol
- Today cumene process more important

Cumene process

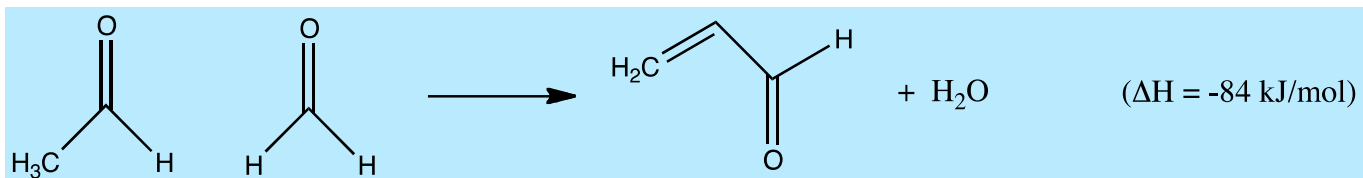


- Cleavage of cumene to phenol and acetone
- 1944 developed by Heck and Lang
- We will touch this process when we talk about aromatics

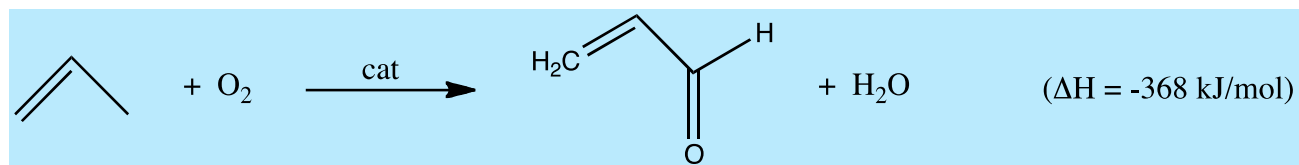
Use of acetone



Acroleine

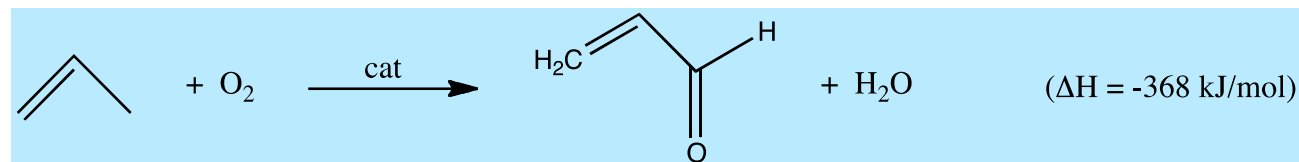


- In the 1930ies developed by Degussa
- 300-320°C in the gas phase on N-silicate/SiO₂

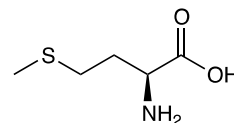


- Gas phase oxidation on Cu₂/SiC catalysts with J₂, 350 - 400°C
 - Propylene conversion 20%, selectivity 70-85%
- SOHIO process brought here a breakthrough

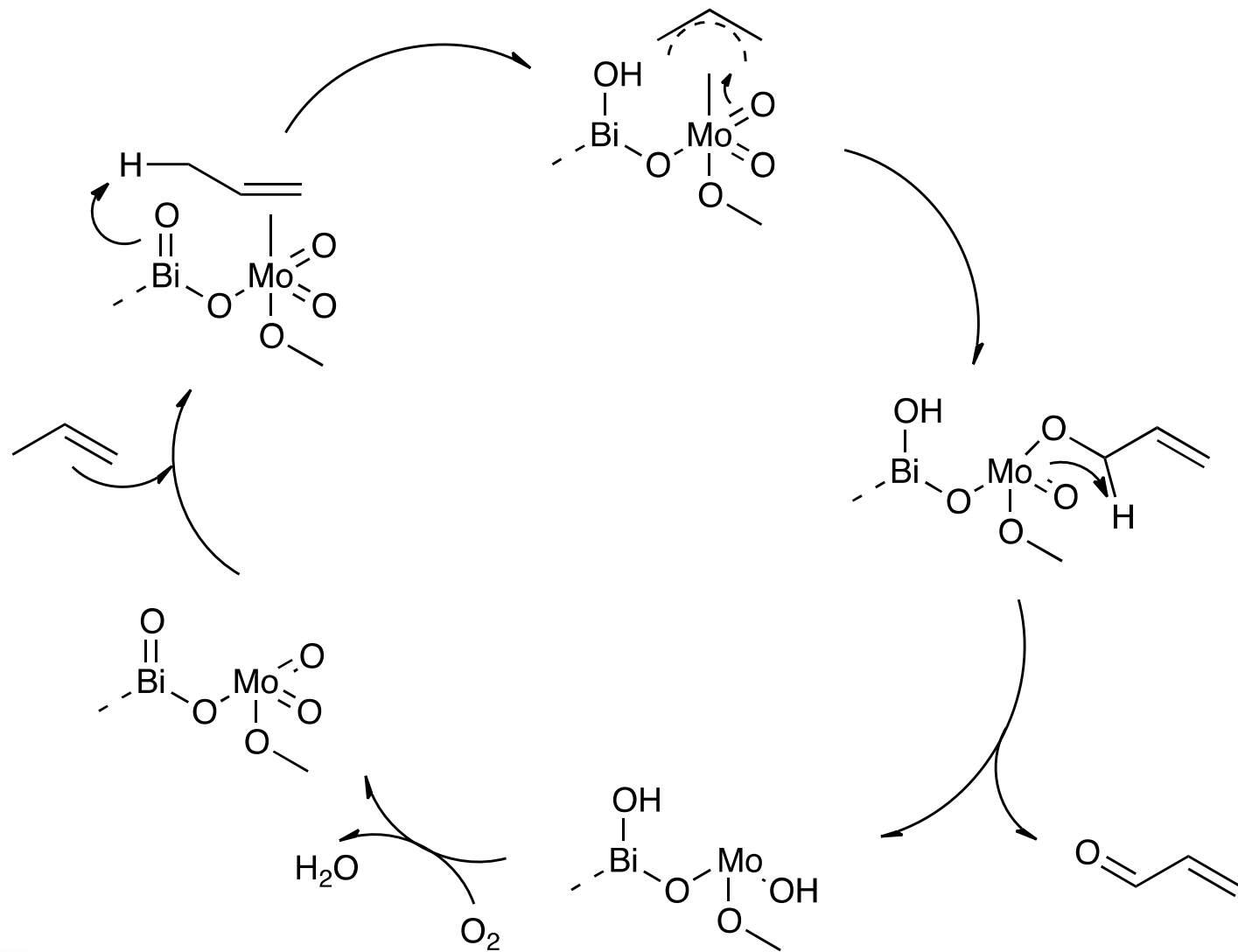
SOHIO process for acrolein



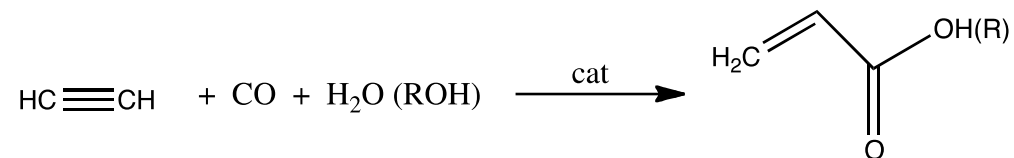
- Oxidation in air, 300 - 360°C, 1-2 bar
- $\text{Bi}_2\text{O}_3/\text{MoO}_3$ catalyst in solid bed / pipe bundle reactor
- Propylene is oxidized in excess of air and presence of steam, under high propylene conversion (96%)
- Big air/propylene ratio important to keep oxidation state of catalyst above a threshold (avoiding decrease of selectivity - 90%)
- Byproducts: acetic aldehyde, acrylic acid, acetic acid
- Used for the synthesis of Methionin (animal feed)



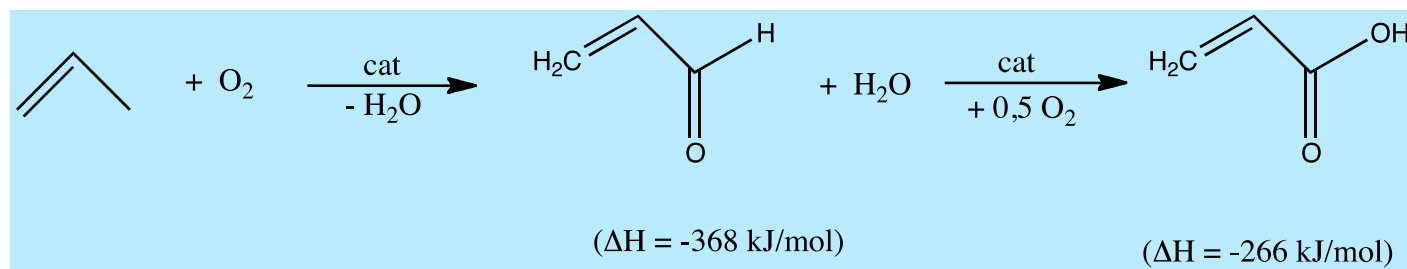
SOHIO process for acrolein



Acrylic acid

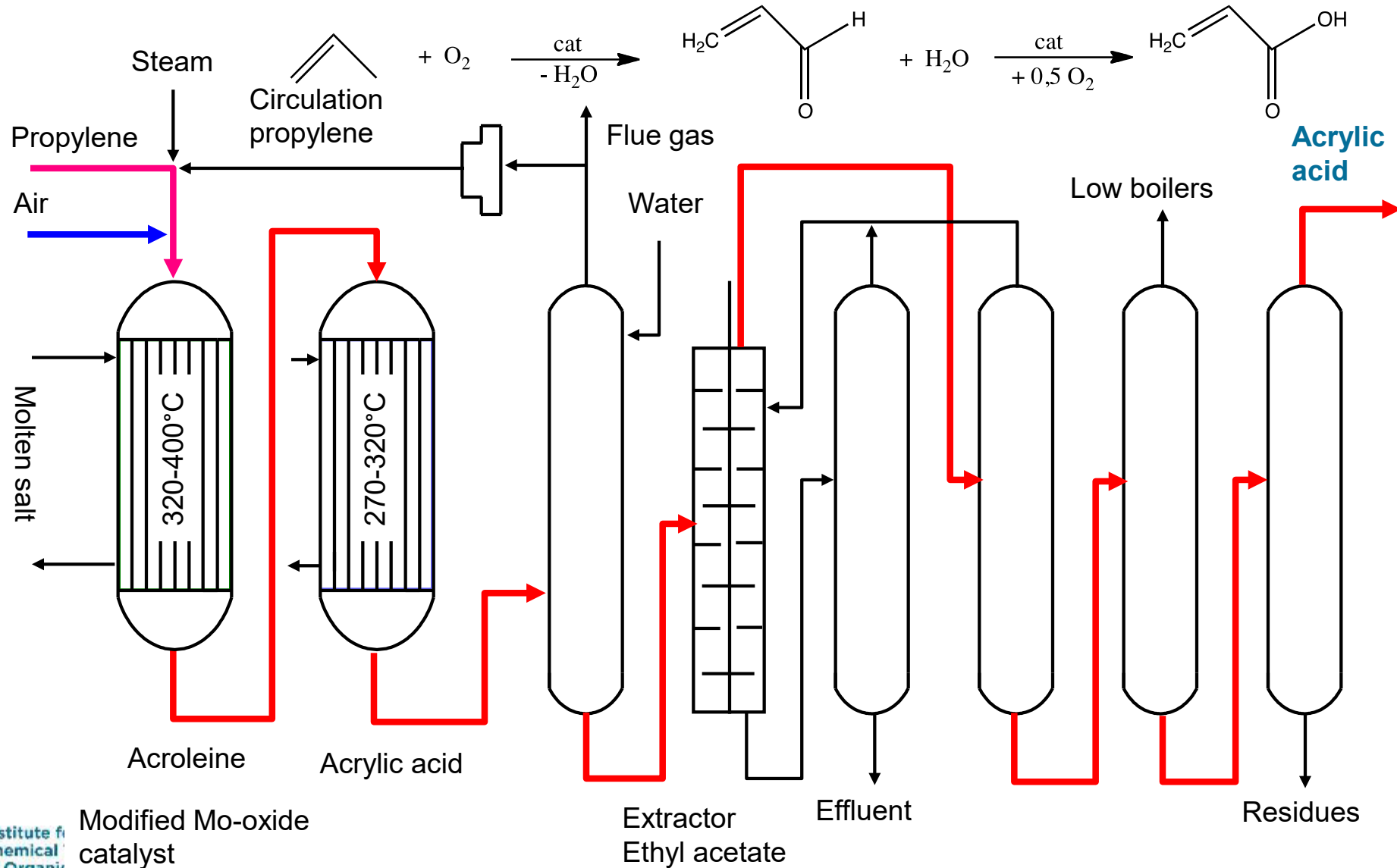


- Historically Reppe process
- Carbonylation of acetylene in presence of water or alcohols

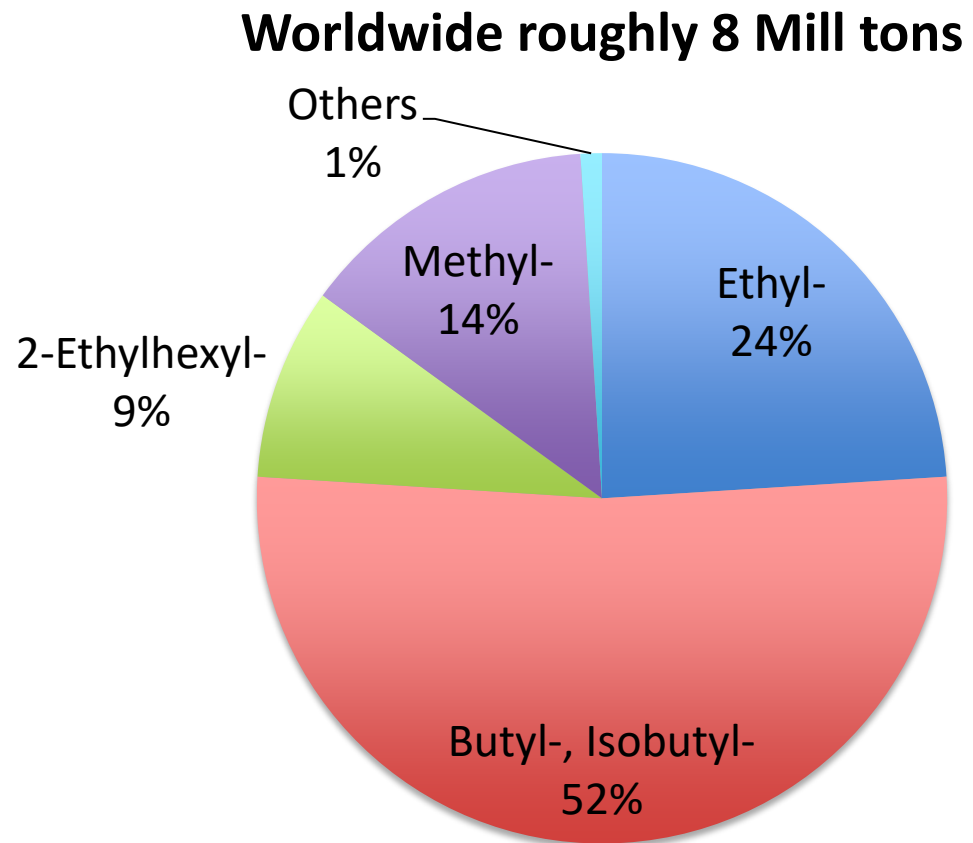


- In a one step oxidation process propylene, air (O₂) are reacted at 200 - 500°C and up to 10 bar with a multi component catalyst system (Mo as main metal)
- Because of different kinetics a 2 step process is preferred

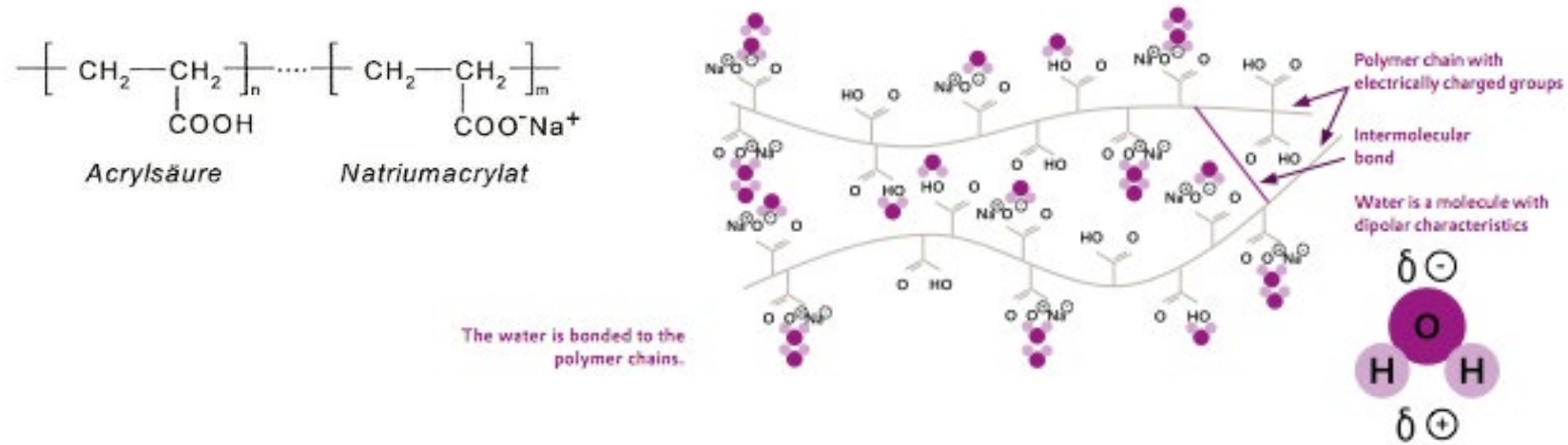
Acrylic acid – 2 step oxidation



Use of acrylic acid

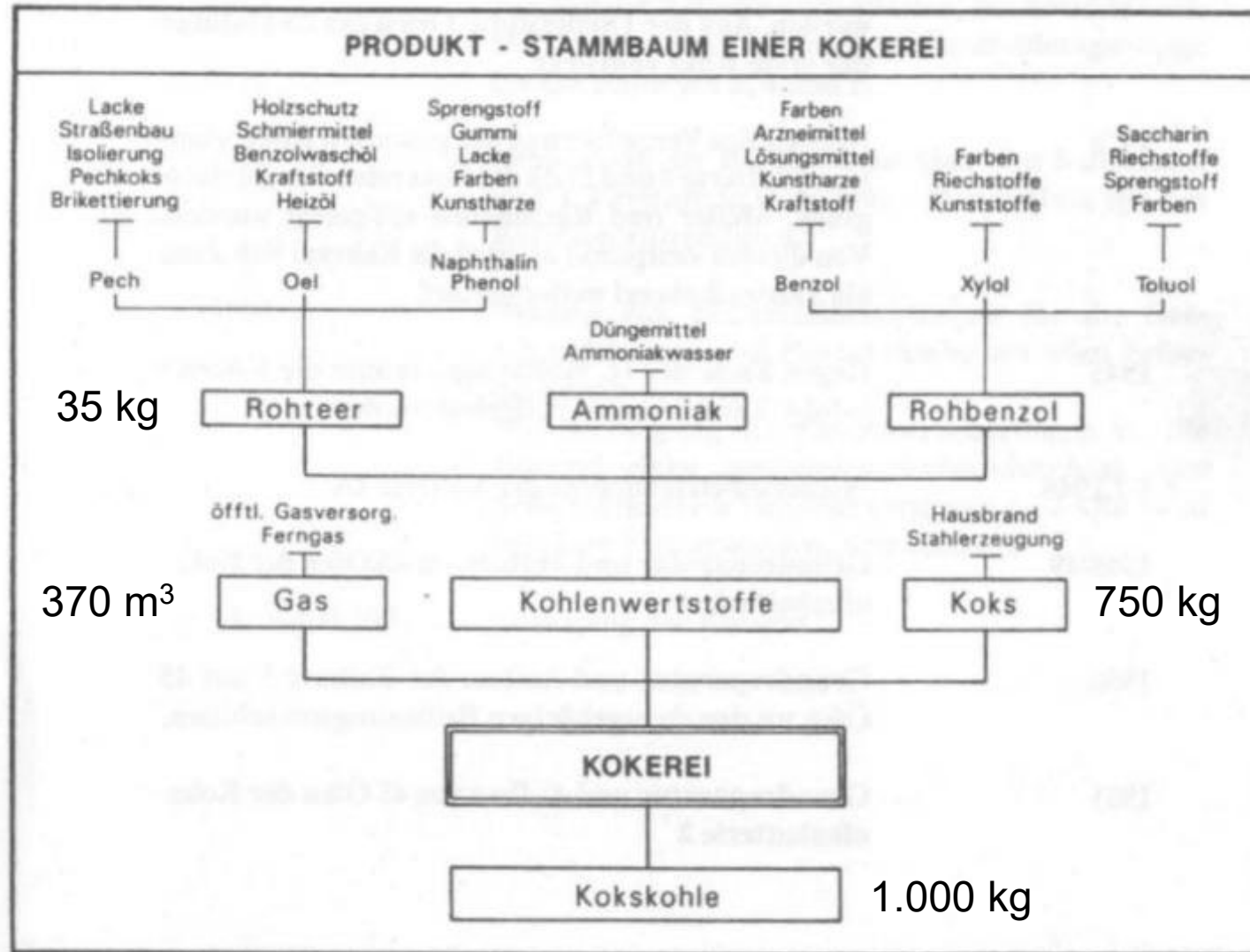


Superabsorber

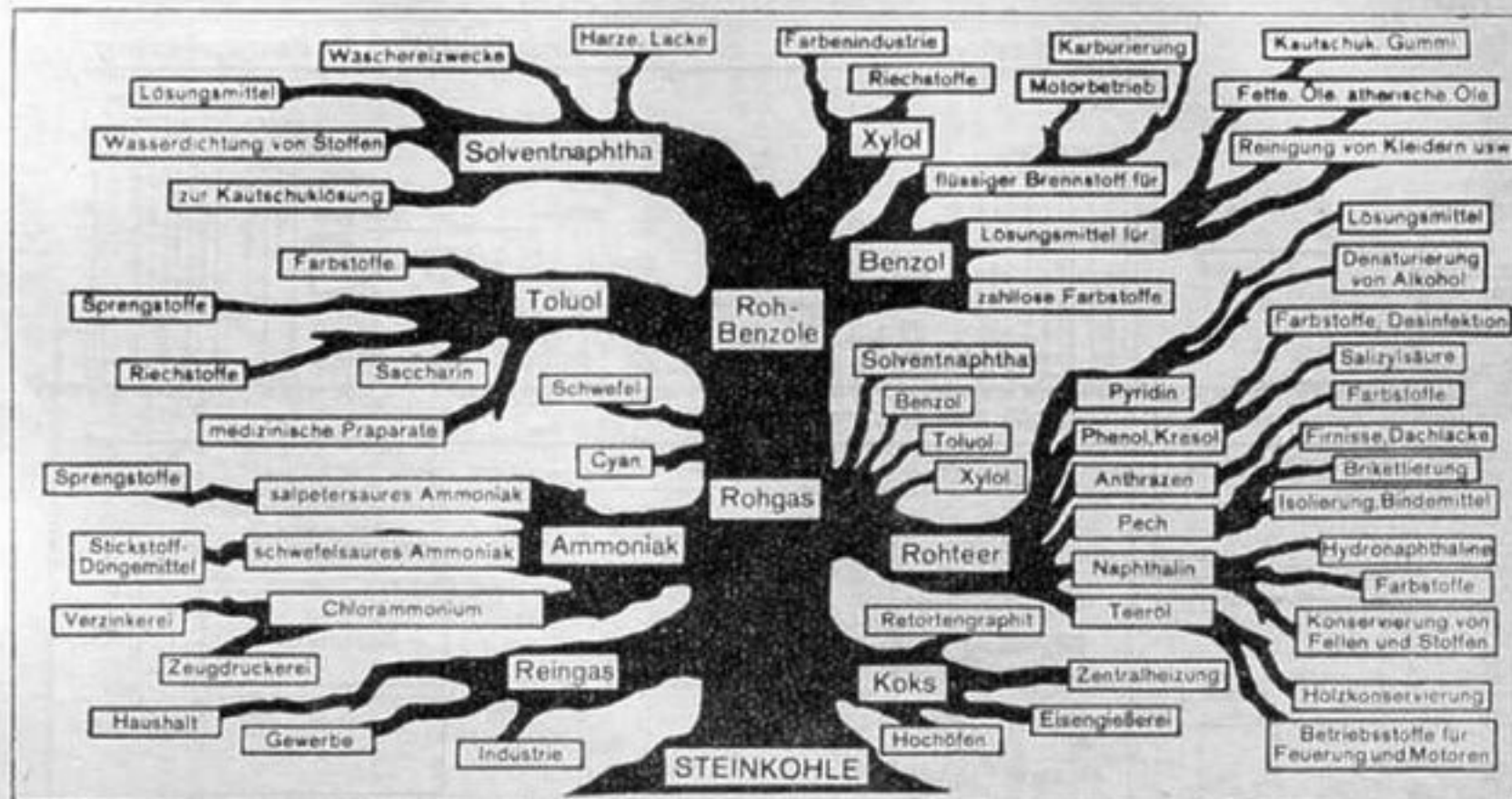


Aromatics

- Main products:
 - Benzene, Toluene, Xylene, Ethyl benzene (BTX or BTXE)
 - Naphthalene or Anthracene are less important
- Basically aromatics belong to the most important raw materials in the chemical industry
- Plastics, synthetic rubber, synthetic fibers
- Practically all aromatics are derived from fossil raw materials
oil and coal
(through thermal / catalytic conversion in refineries or coking plants)
- European production more than 12 Mill tons



Kokerei und Kohleveredlung



Die bei der Verkokung der Steinkohle anfallenden Kohlenwertstoffe,
ihre Veredelungserzeugnisse und deren Verwendung

Aromatics content of different oils

%	Libya	Texas	Nigeria	Iran
Benzene	0,37	0,18	0,11	0,19
Toluene	0,37	0,51	0,92	0,56
C8 aromatics	0,56	1,10	1,47	1,05

Raw materials: production of BTX

	Reformate	Pyrolysis fuel	Coke
Benzene	4	38	65
Toluene	17	20	18
Xylene	18	5	6
Ethylbenzene / styrene	5	4	2
Higher aromatics	11	5	7
Non aromatics	45	28	2

- Raw fuel (coking process) before distillation through H_2 raffination removal of thiophenes and olefin impurities
- Pyrolysis fuel: before extractive distillation selective hydrogenation of olefins and acetylene

Isolation of aromatics

- The isolation of aromatics basically consists of
 - Separation of non-aromatics
 - Separation of the different aromatic components
- Fractionating distillation not feasible (economically)
 - Cyclohexane, heptane and other alkanes build with benzene or toluene azeotropic systems
- Special separation methods

Data of aromatics

	Tb [°C]	Tm [°C]
Benzene	80,2	-5,6
Toluene	110,6	-95,2
Ethylbenzene	136,2	-95,0
p-Xylene	138,3	13,3
m-Xylene	139,1	-47,9
o-Xylene	144,4	-25,2
Styrene	145,3	-30,7

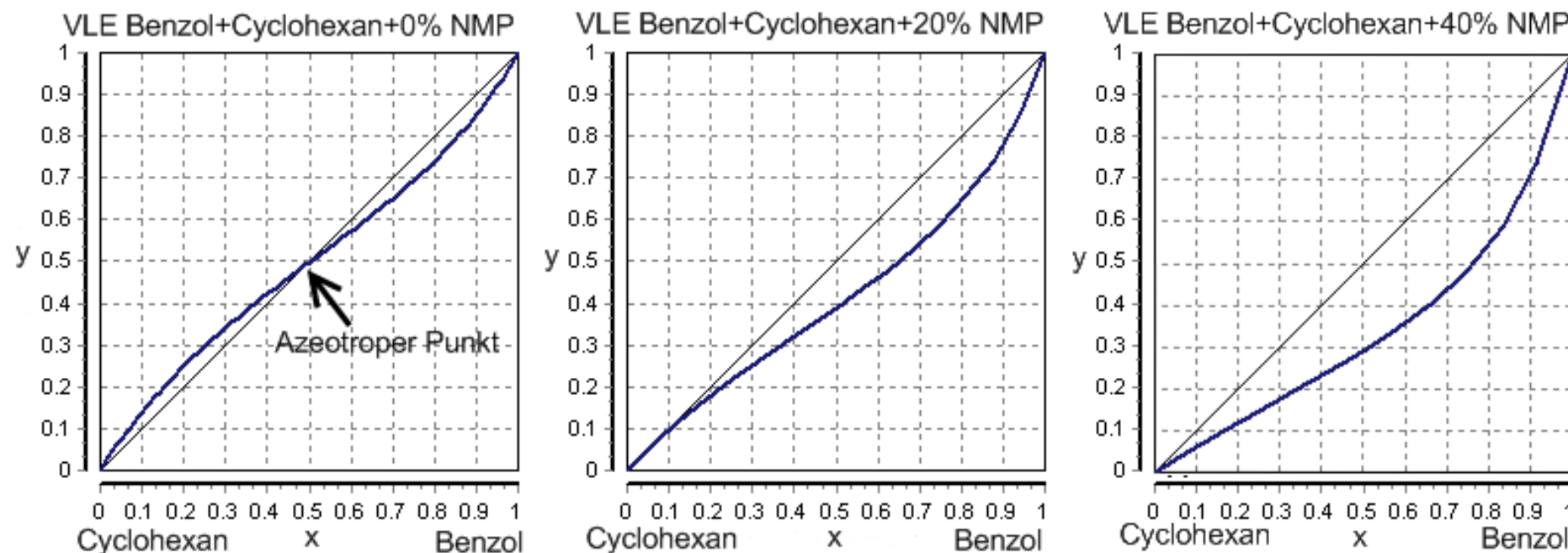
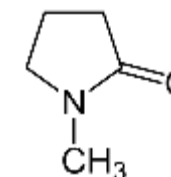
Aromatics

- **Separation methods**
- Benzene & benzene products
- Phenol, Cyclohexanone, ...
- Toluene
- C8 aromatics
- Polycyclic aromatics
- Heterocyclics

Separation methods

Process	Problem	Requirement
1. Azeotropic distillation	BTX separation Pyrolysis fuel	High aromatic content >90%
2. Extractive distillation	BTX separation Pyrolysis fuel	Middle aromatic content 65-90%
3. Liquid-liquid extraction	BTX separation Reformat fuel	Low aromatic content 20-65%
4. Freeze out crystallization	Isolation of p-xylene from m/p mixtures	Pre separation of o-xylene (distillation) and ethyl-benzene from C8 aromatic fractions
5. Adsorption on solids	Isolation of p-xylene from C8 aromatic fractions	Continuous, reversible and selective adsorption

Equilibrium diagram: benzene / cyclohexane

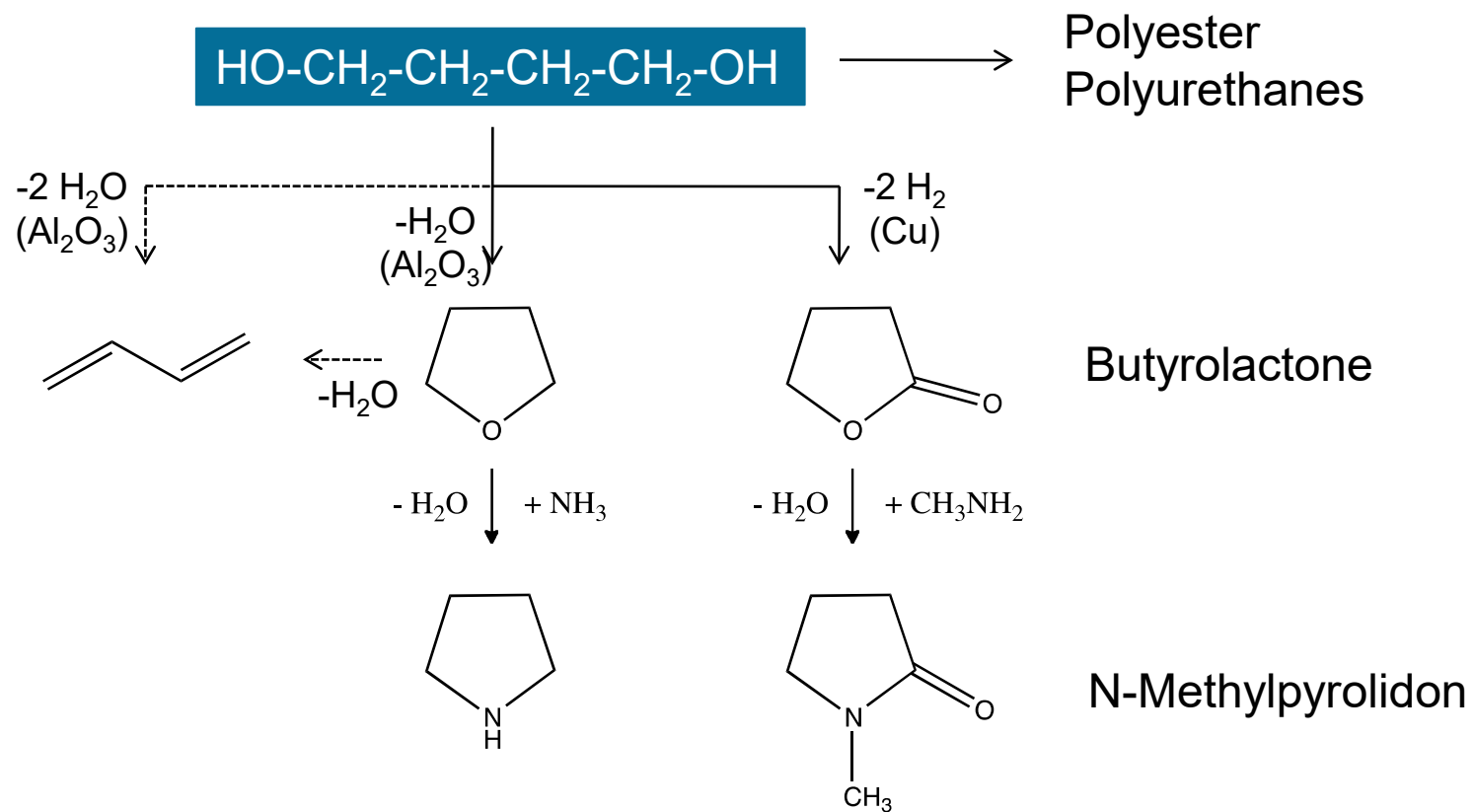


T_b: 202°C

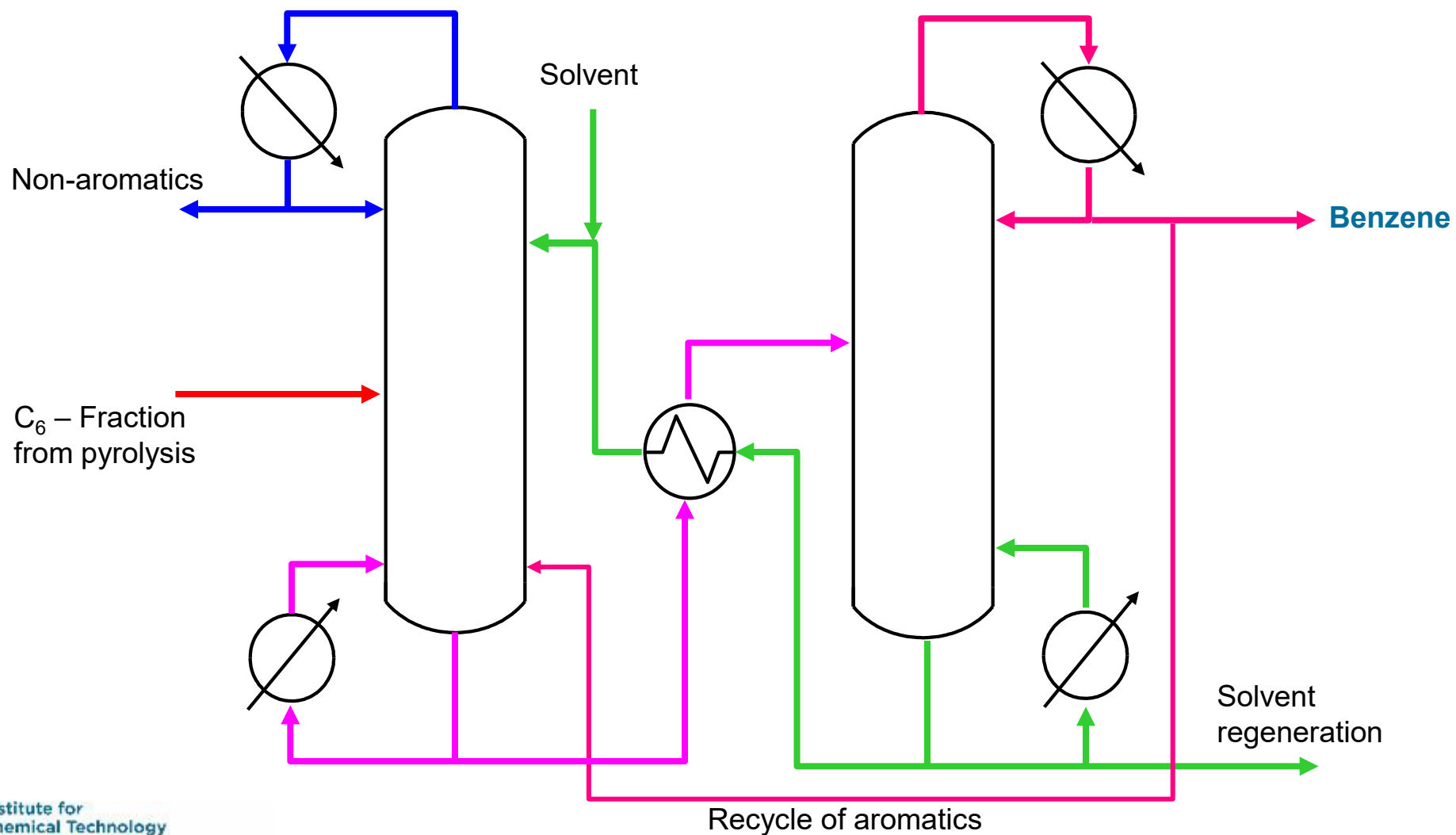
T_m: -24°C

N-Methylpyrrolidon

Butanediol



Extractive distillation of benzene



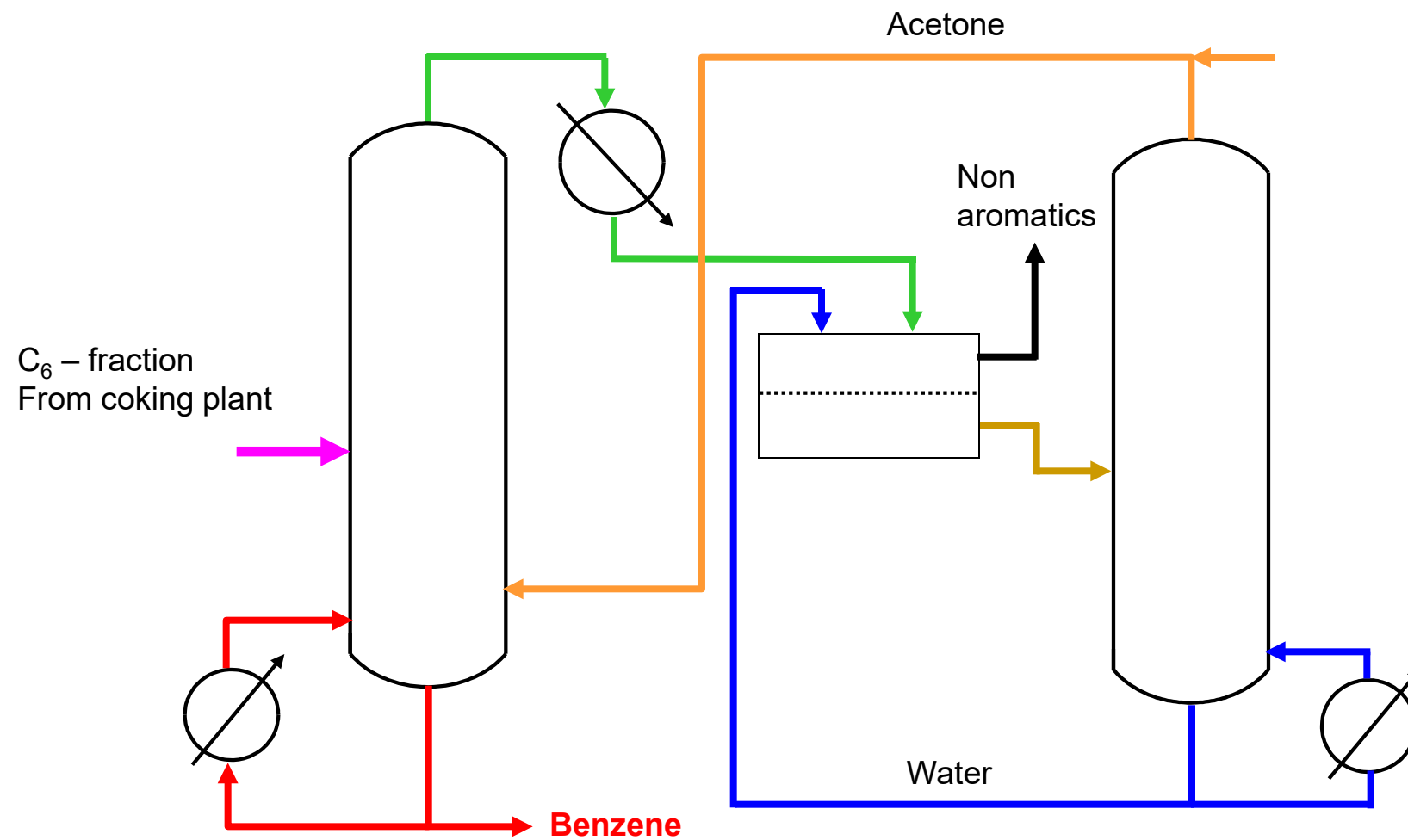
Activity coefficients for benzene / cyclohexane

Solvent	T [°C]	f^0_c	f^0_b	$f^0_c:f^0_b$
Sulfolan	40	50,1	4,3	11,7
Dimethylsulfoxid	40	41,7	3,8	11,0
Ethylenglykol	40	281	32	8,7
N-Methylpyrrolidon	50	11,3	1,7	6,6
N,N-Dimethylacetamid	50	20,7	3,7	5,5
Diethylsulfoxid	40	9,9	1,8	5,5
N-Methylpiperidon	50	5,7	1,1	5,1
N,N-Dimethylformamid	50	24,1	5,1	4,7
N-Methylcaprolactam	50	4,8	1,2	4,1
N,N-Diethylacetamid	50	6,4	1,7	3,7
N-Methylformamid	50	17,4	4,8	3,6
N-Methylacetamid	50	13,4	4,0	3,4

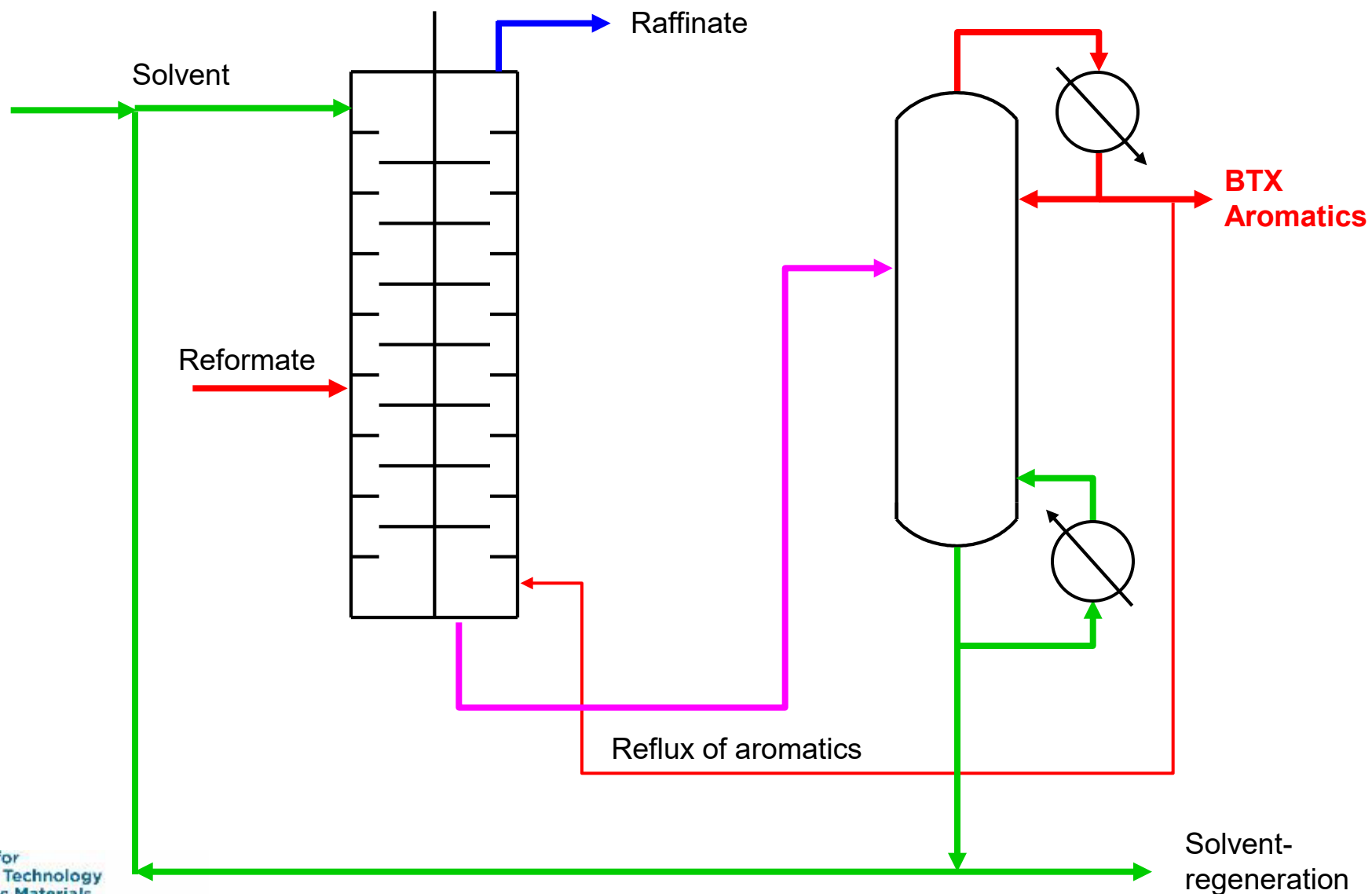
Technically important extractive distillation processes

Educt	Selective solvent	Product	Process
C ₄ from fuel pyrolysis	Dimethylformamide N-Methylpyrrolidone Furfurol Acetonitrile Dimethylacetamide Dimethylsulfoxide	Butadiene	Difex
C ₅ from fuel pyrolysis	Dimethylformamide N-Methylpyrrolidone	Isoprene	
C ₆ from fuel pyrolysis	Phenole N-Methylpyrrolidone N-Formylmorpholine	Benzene	Distex Distapex Morphylan
Isopropanol/Water	CaCl ₂ /Water	(Water)	

Azeotropic distillation of benzene



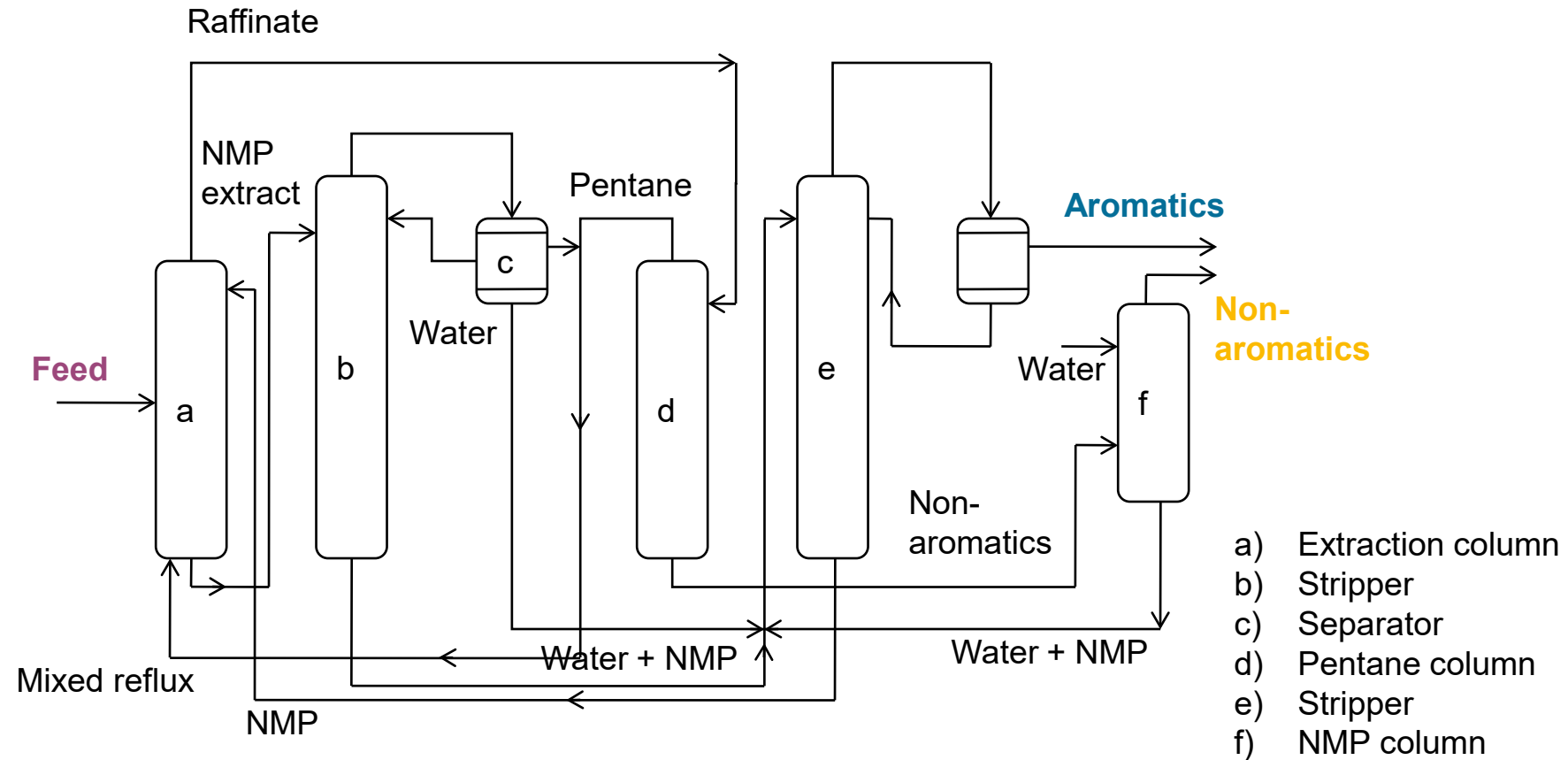
Liquid-liquid extraction: BTX



BTX aromatics through liquid liquid extraction processes

Process	Solvent	Extractor	Company
UDEX	Diethylene glycol	Sieve bottom column	Union Oil
SULFOLAN	Tetramethylenesulfon	Sieve bottom column	Shell
AROSOLVAN	N-Methylpyrrolidone + ethylene glycol	Mixer-settler	Lurgi
AREX	N-Methylcarpolactam + ethylene glycol	Swivel plate column	Leuna

Arosolvan process

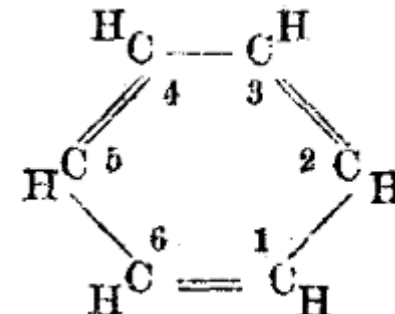
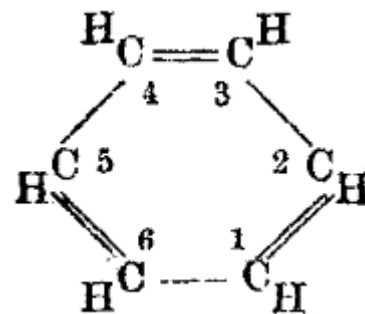


Aromatics

- Separation methods
- **Benzene & benzene products**
- Phenol, Cyclohexanone, ...
- Toluene
- C8 aromatics
- Polycyclic aromatics
- Heterocyclics

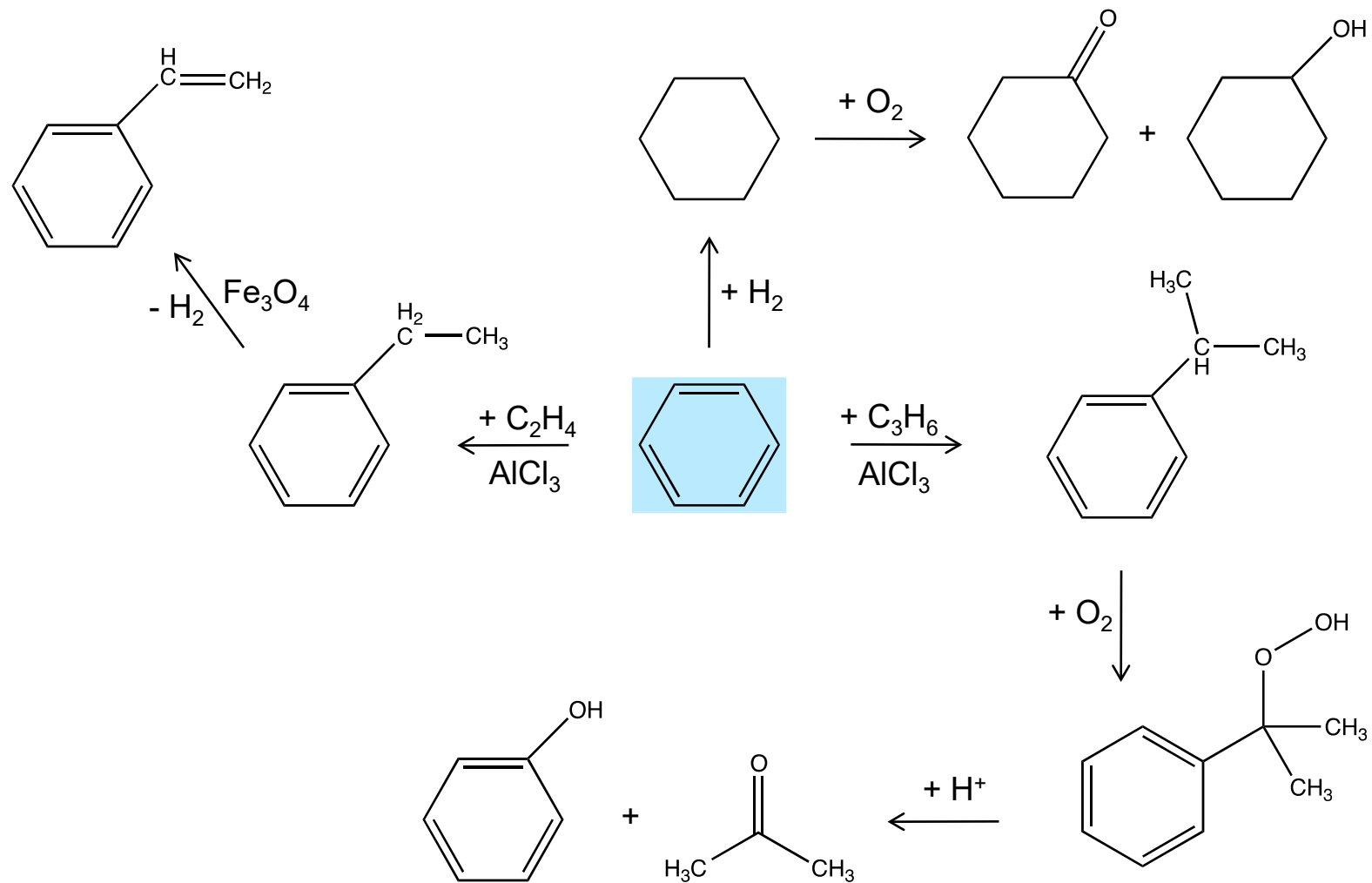
Benzene

- Molar mass: 78,11 g/mol
- Density: 0,88 kg/m³ (20°C)
- T_m: 5,5°C
- T_b: 80,1°C

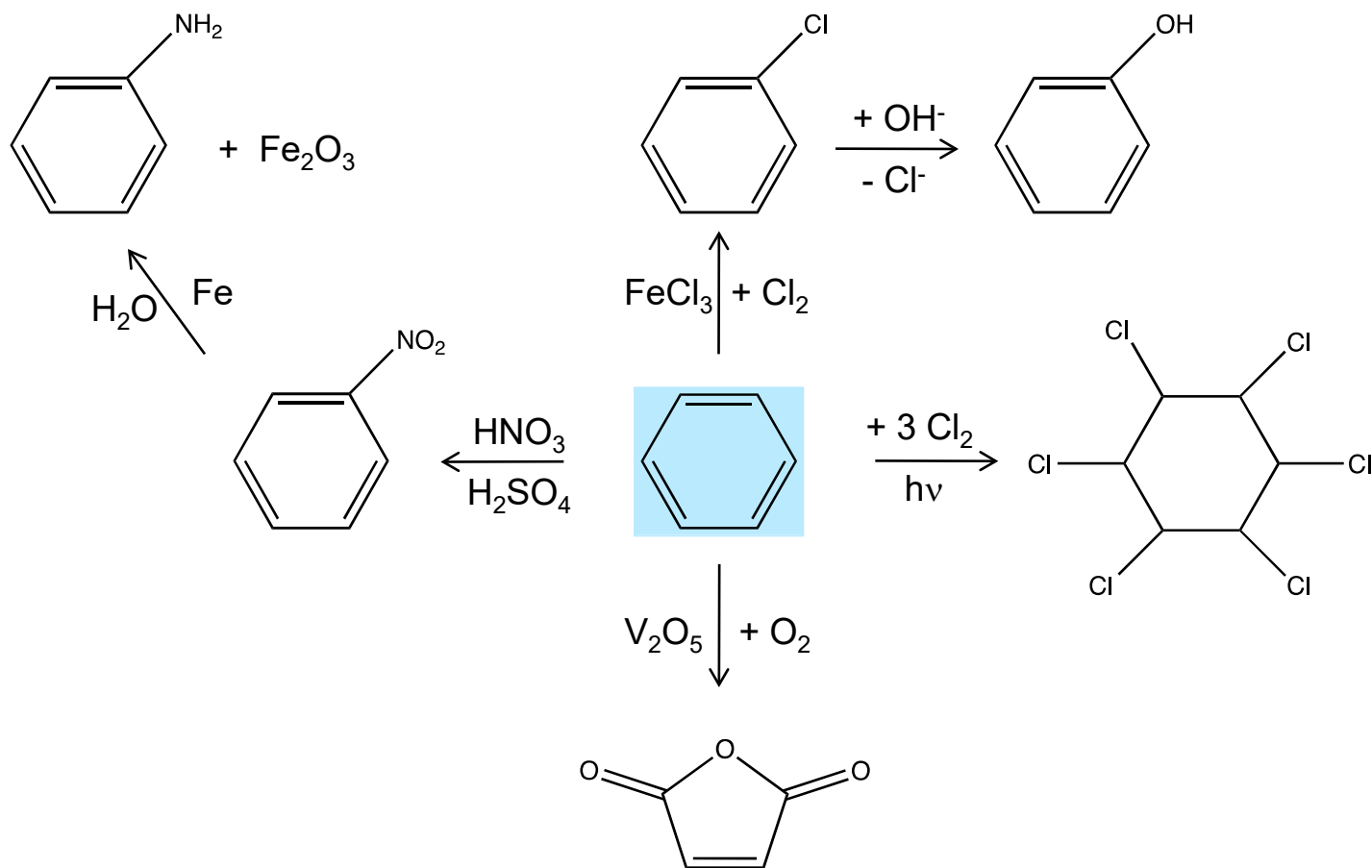


Use [%]	USA (2004)	WE (1999)	J (1999)	D (1999)
Styrene	51	49	59	43
Cumene (phenol)	25	24	19	27
Cyclohexane	14	12	15	14
Nitro benzene	6	11	3	13
Maleic anhydride	-	1	2	-
Alkyl benzene	2	2	1	1
Others	2	1	1	2
Total [Mill. t]	8,7	7,5	4,4	2,1

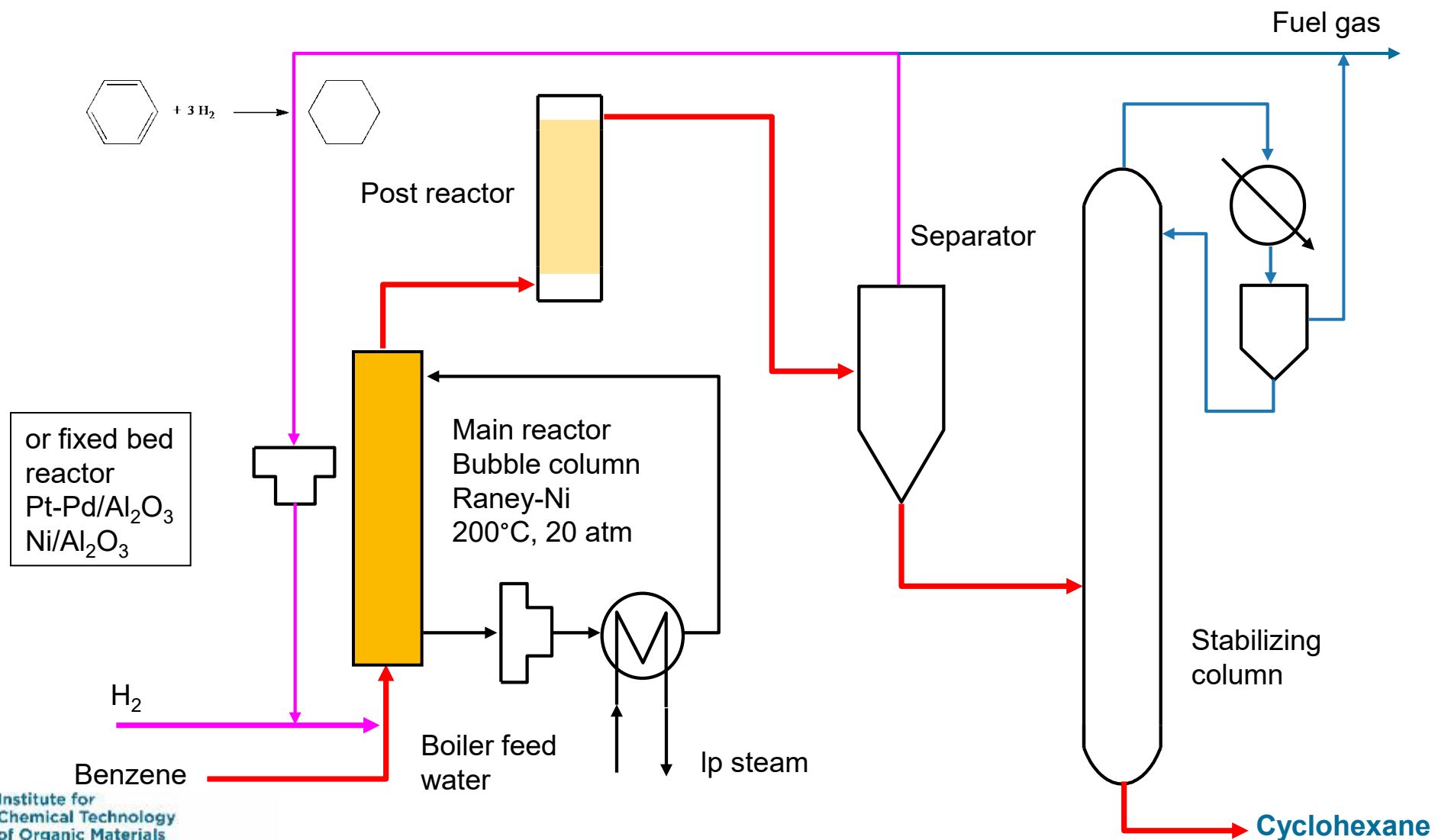
Benzene family tree



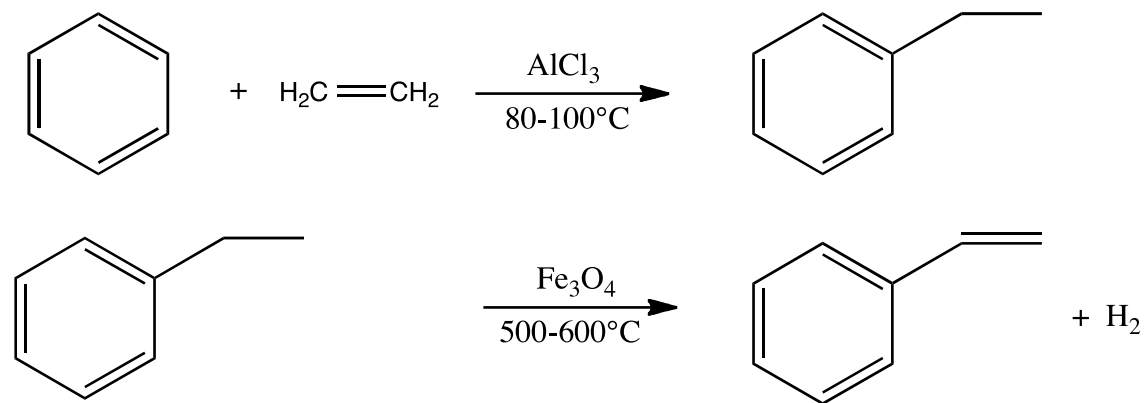
Benzene family tree (ctd.)



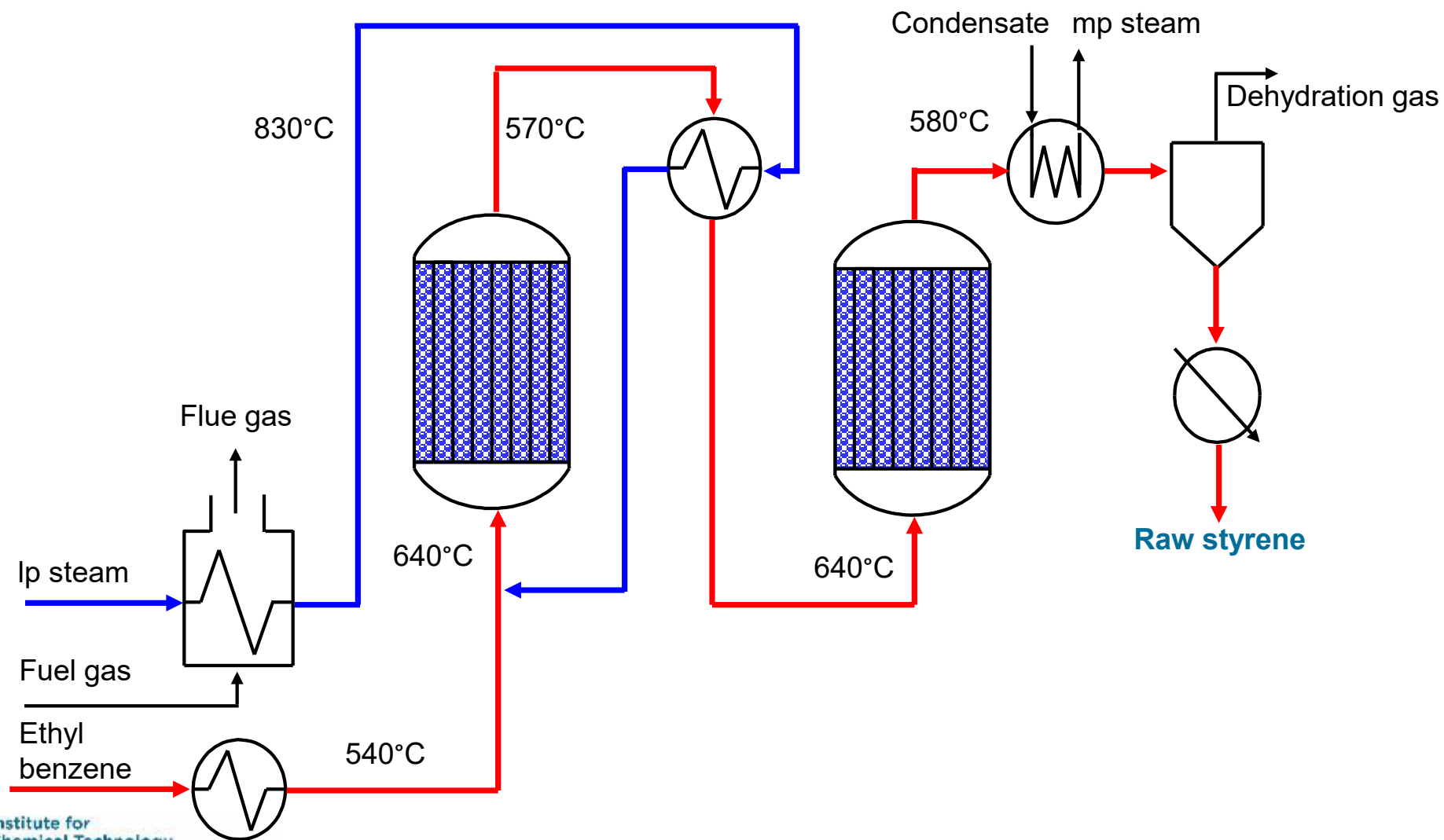
Cyclohexane hydrogenation of benzene in liquid phase



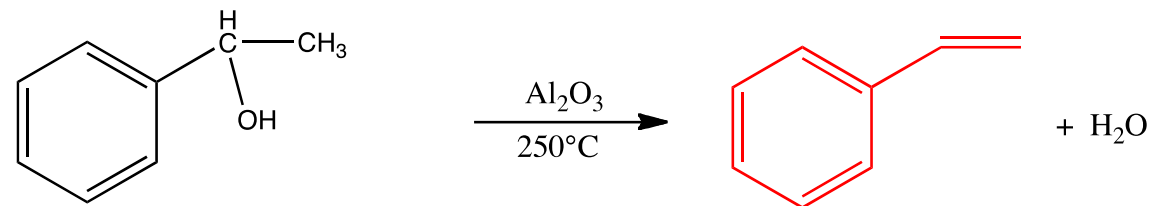
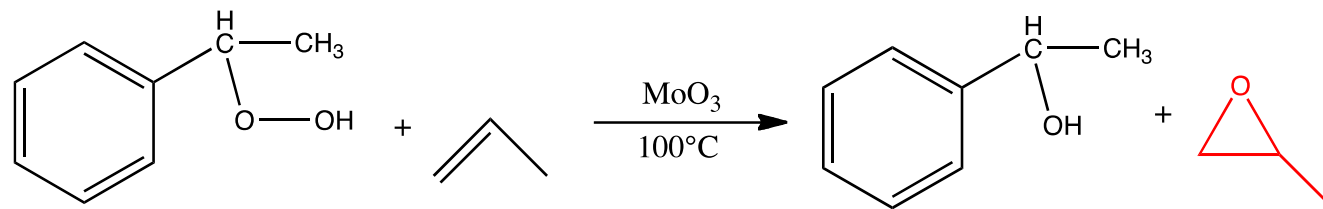
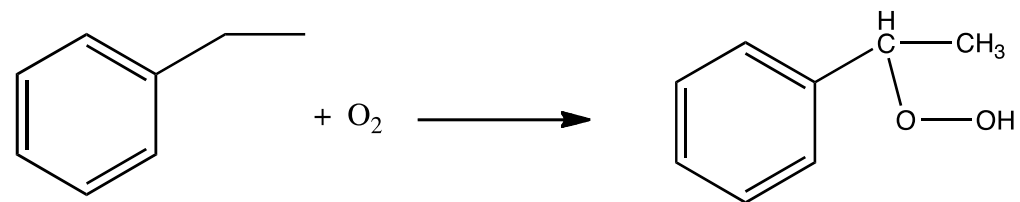
Styrene



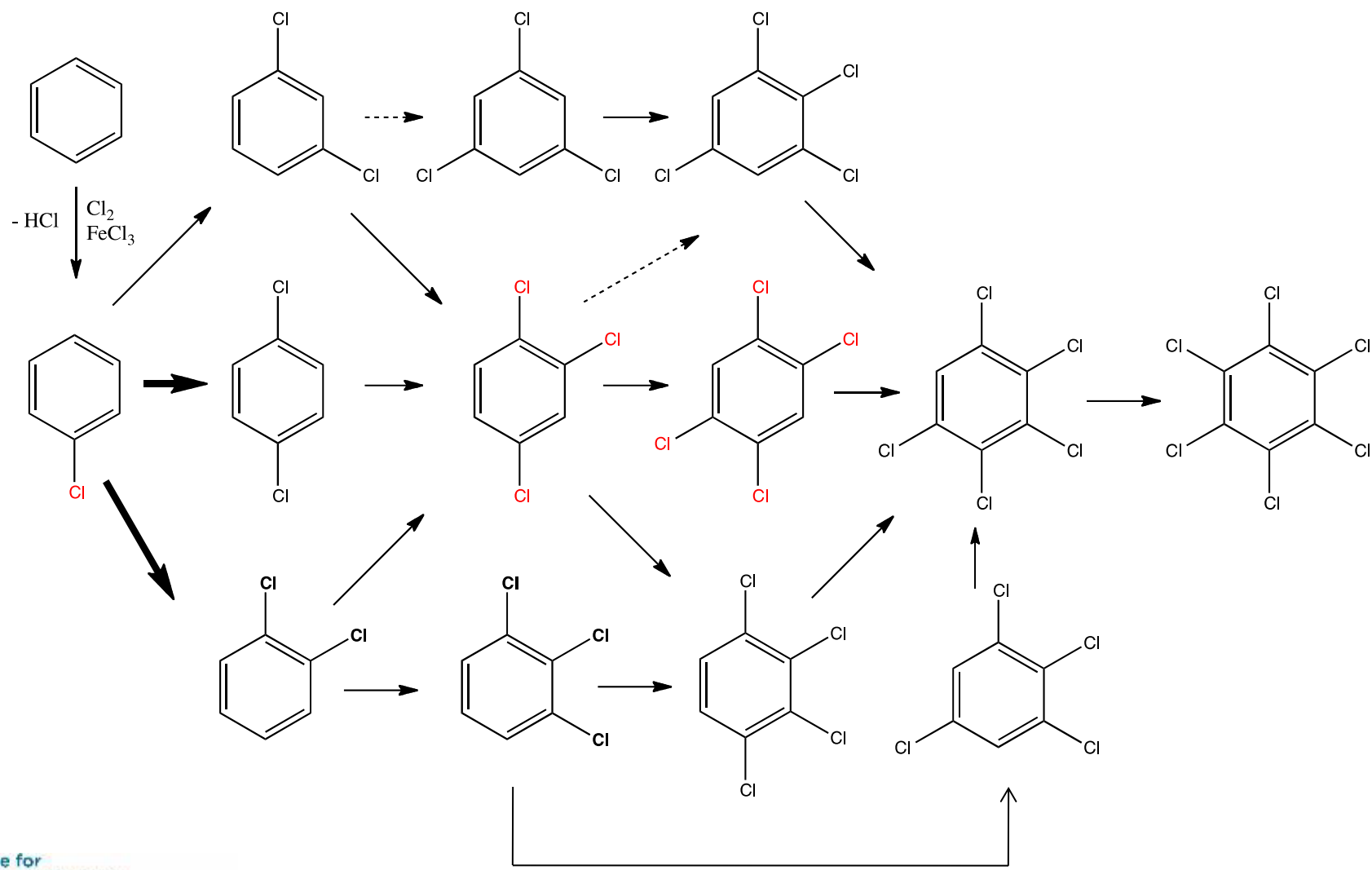
Dehydration of ethyl benzene



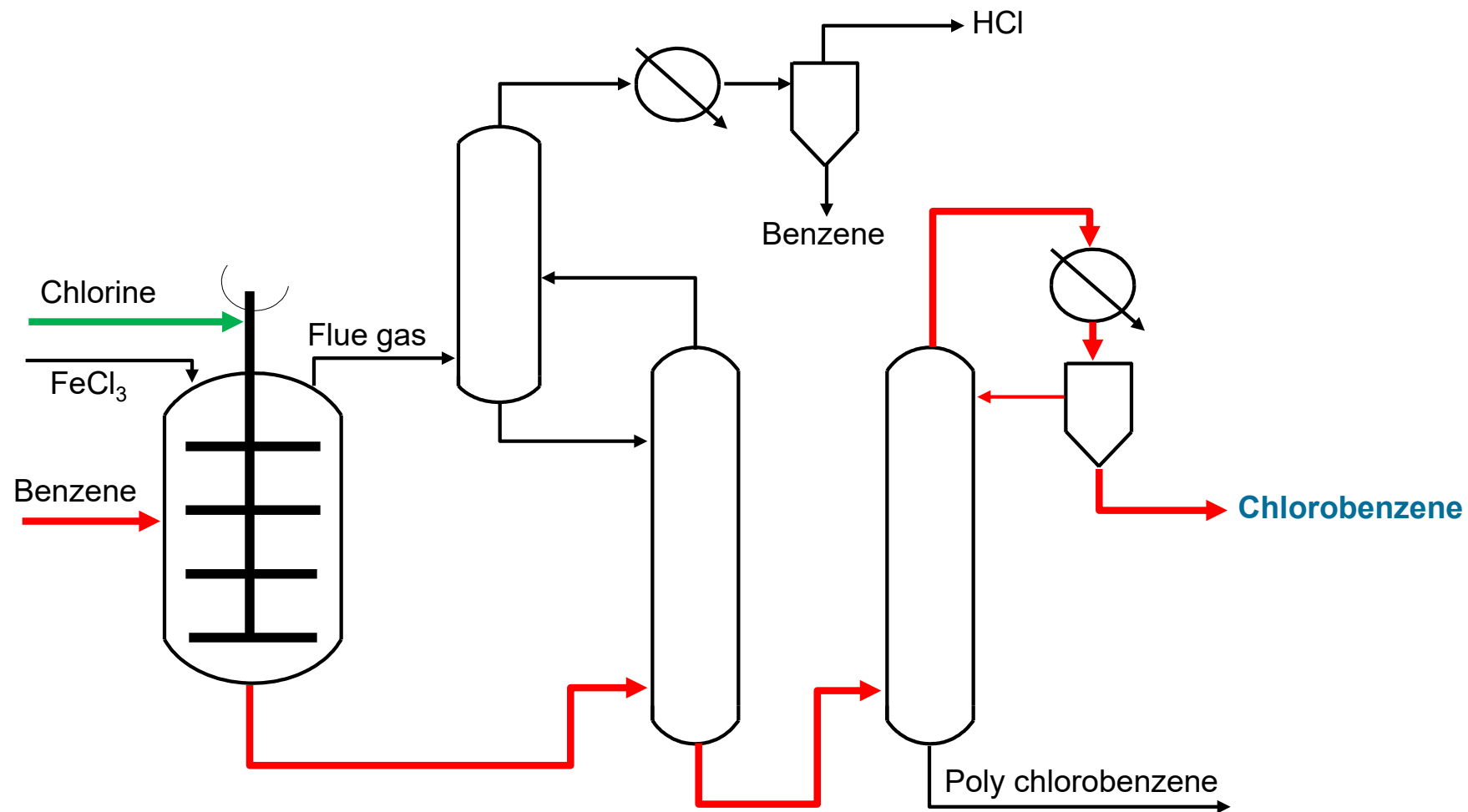
Styrene: byproduct in Halcon process



Chlorination of benzene



Chlorobenzene process



Saponification of chlorobenzene - Dow process

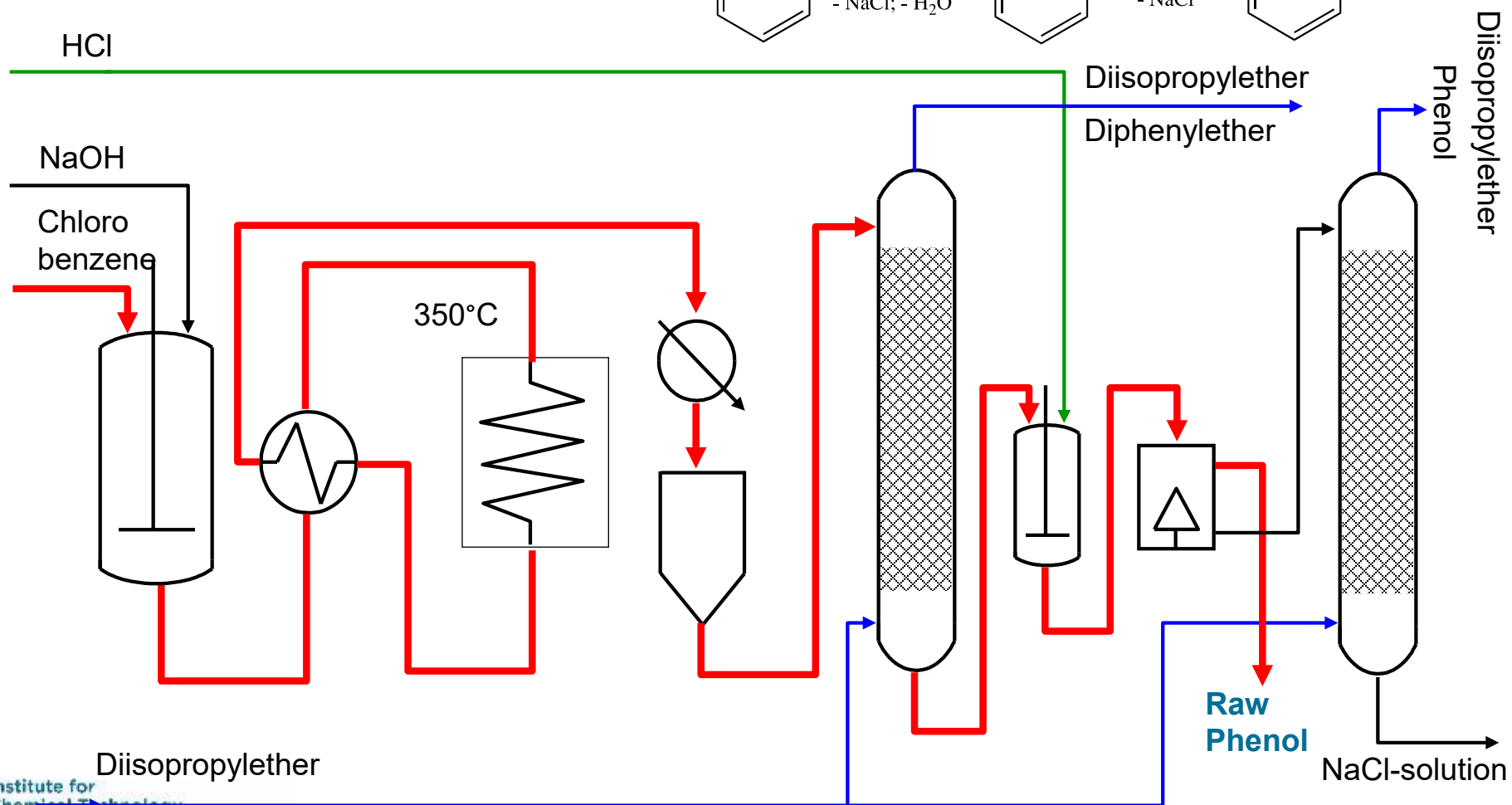
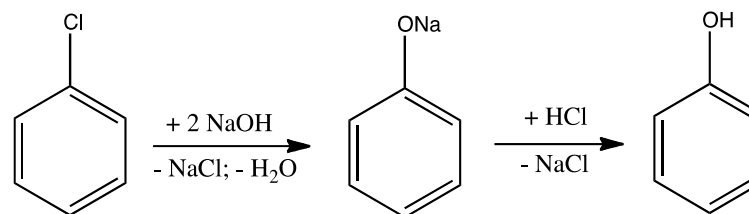
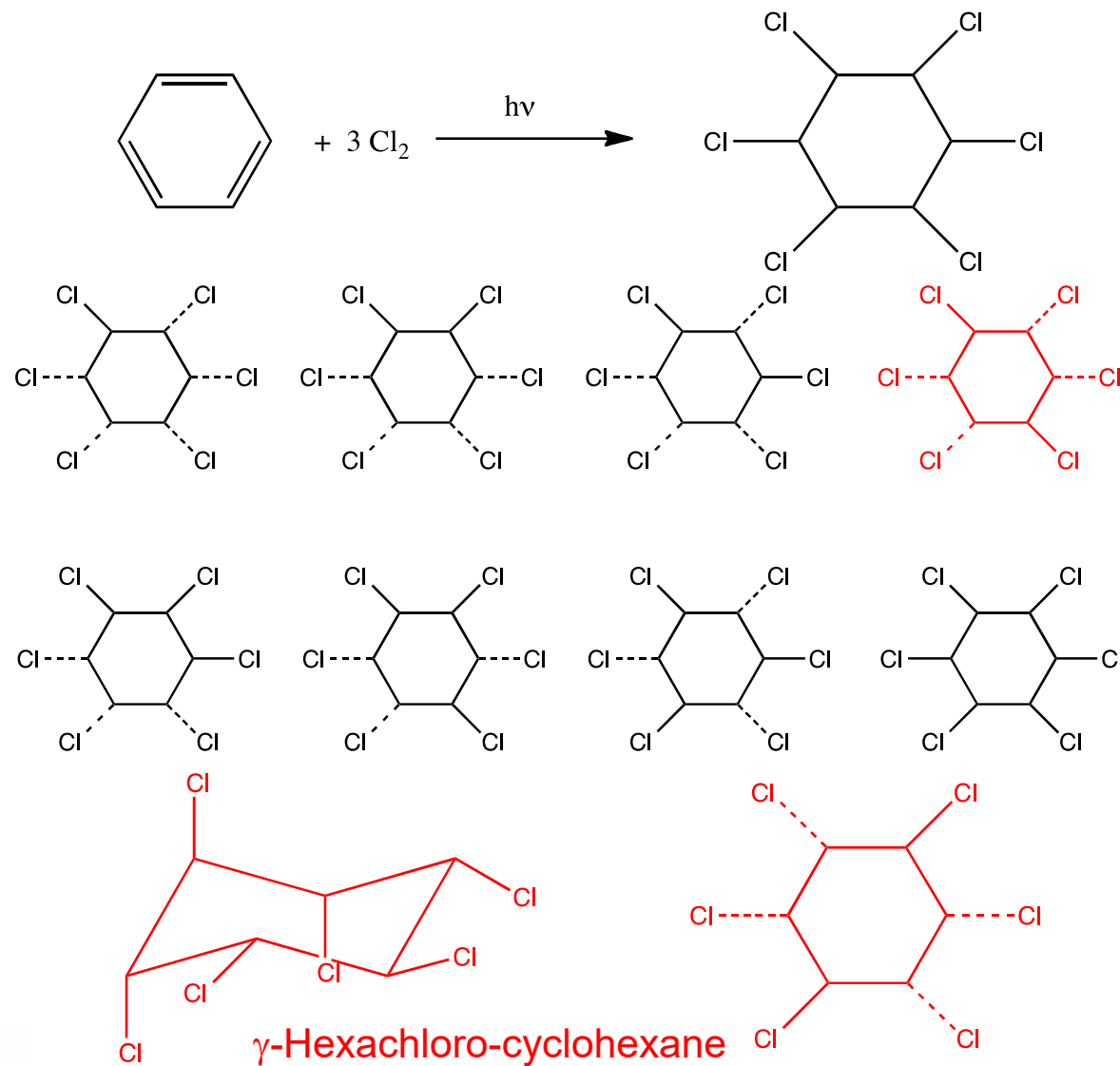
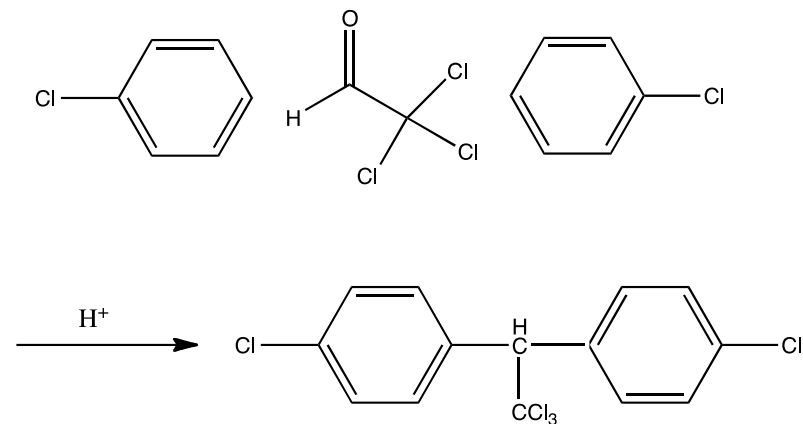


Photo chlorination of benzene

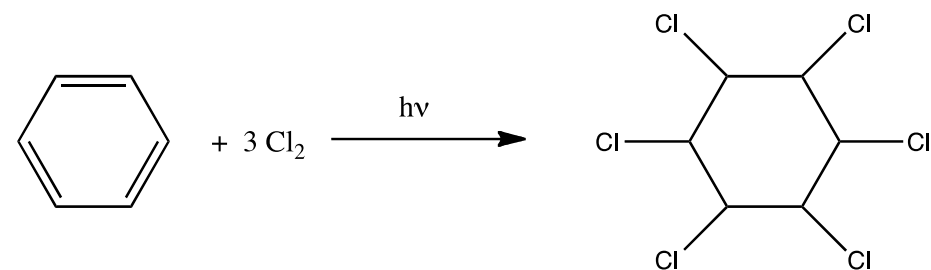


Chlorine insecticides based on benzene

Dichloro-diphenyl-trichloromethyl-methane (DDT)



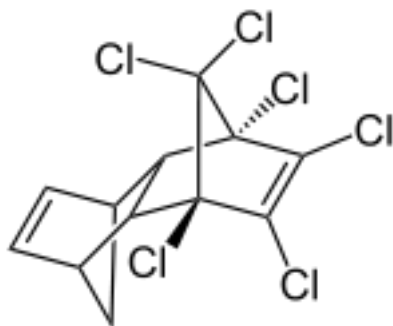
Hexachlorocyclohexane (only the γ isomer is active) - Lindan



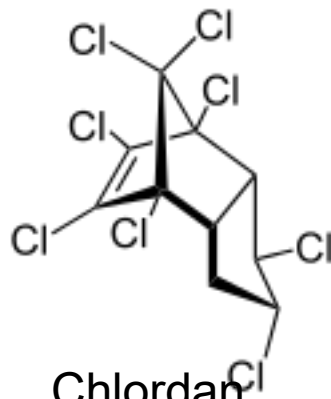
“Dirty Dozen”

Stockholmer convention on persistent organic pollutants – 2001

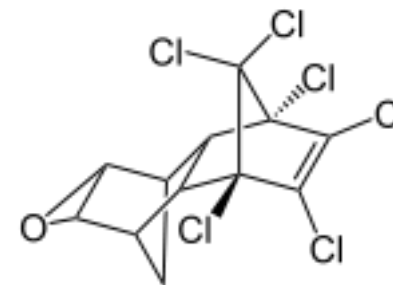
Since May 2004 effective



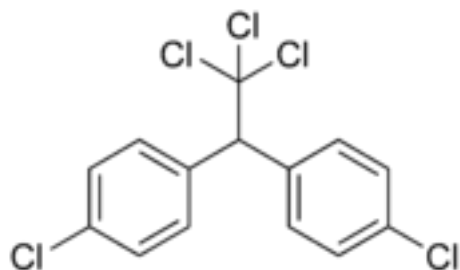
Aldrin



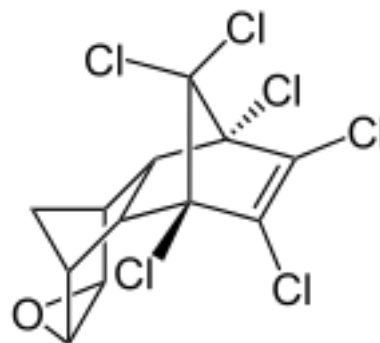
Chlordane



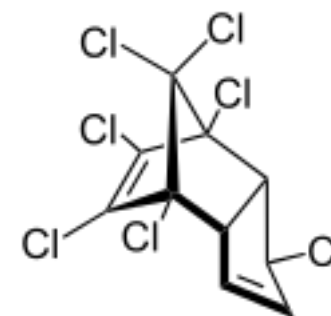
Dieldrin



DDT

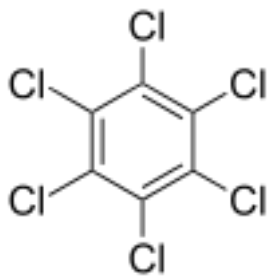


Endrin

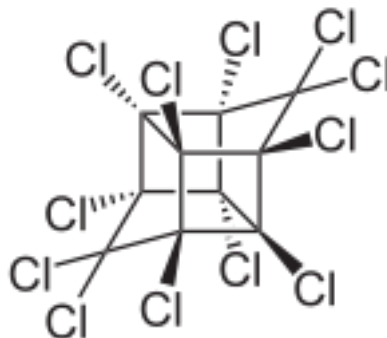


Heptachlor

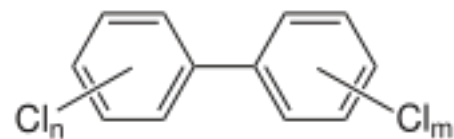
“Dirty Dozen”



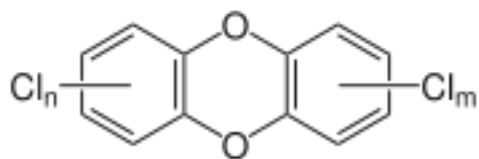
HCB



Mirex

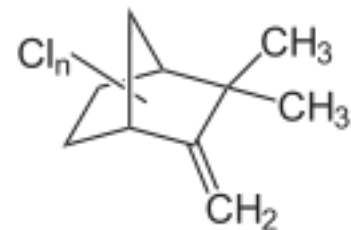
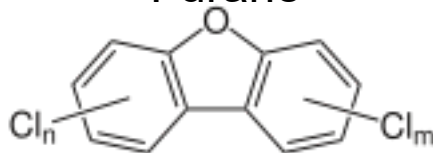


PCB's



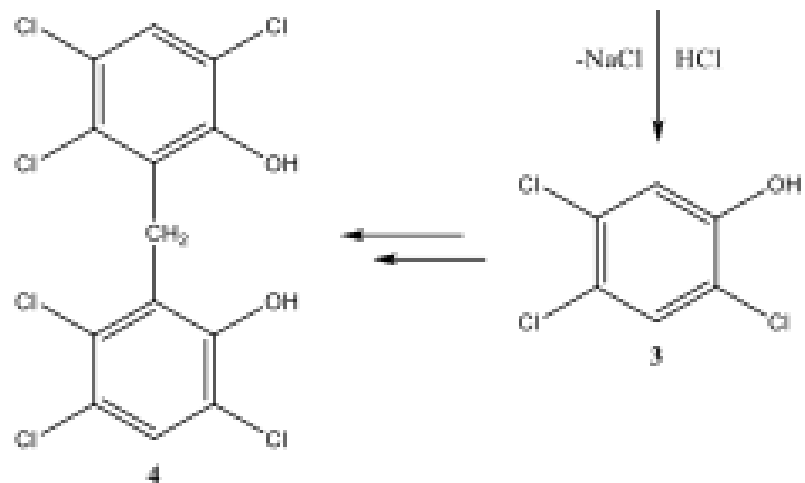
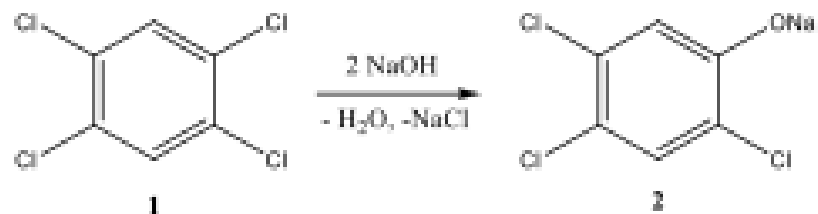
Polychlorierte Dibenzodioxine
“Dioxine”

Polychlorierte Dibenzofurane
“Furane”



Toxaphen

Hexachlorophen

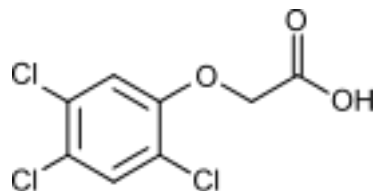


TCP

Icimesa (Givaudan (Hoffmann La-Roche))

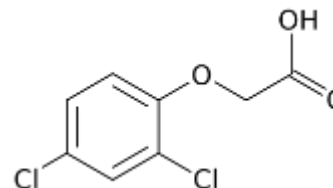
Disinfectant Hexachlorophene

Agent orange

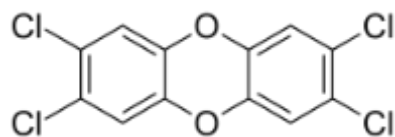


2,4,5-trichlorophenoxyacetic acid (2,4,5-T)

In iso-octyl ester form

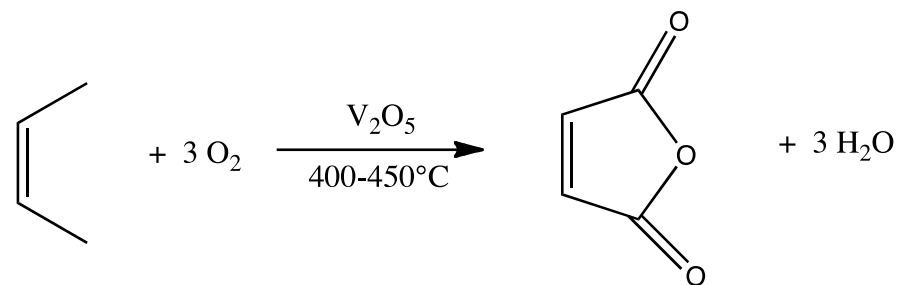
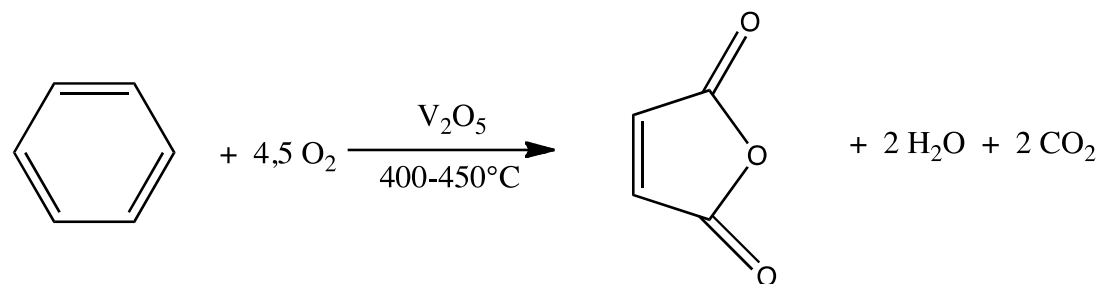


2,4-Dichlorophenoxyacetic acid (2,4-D)

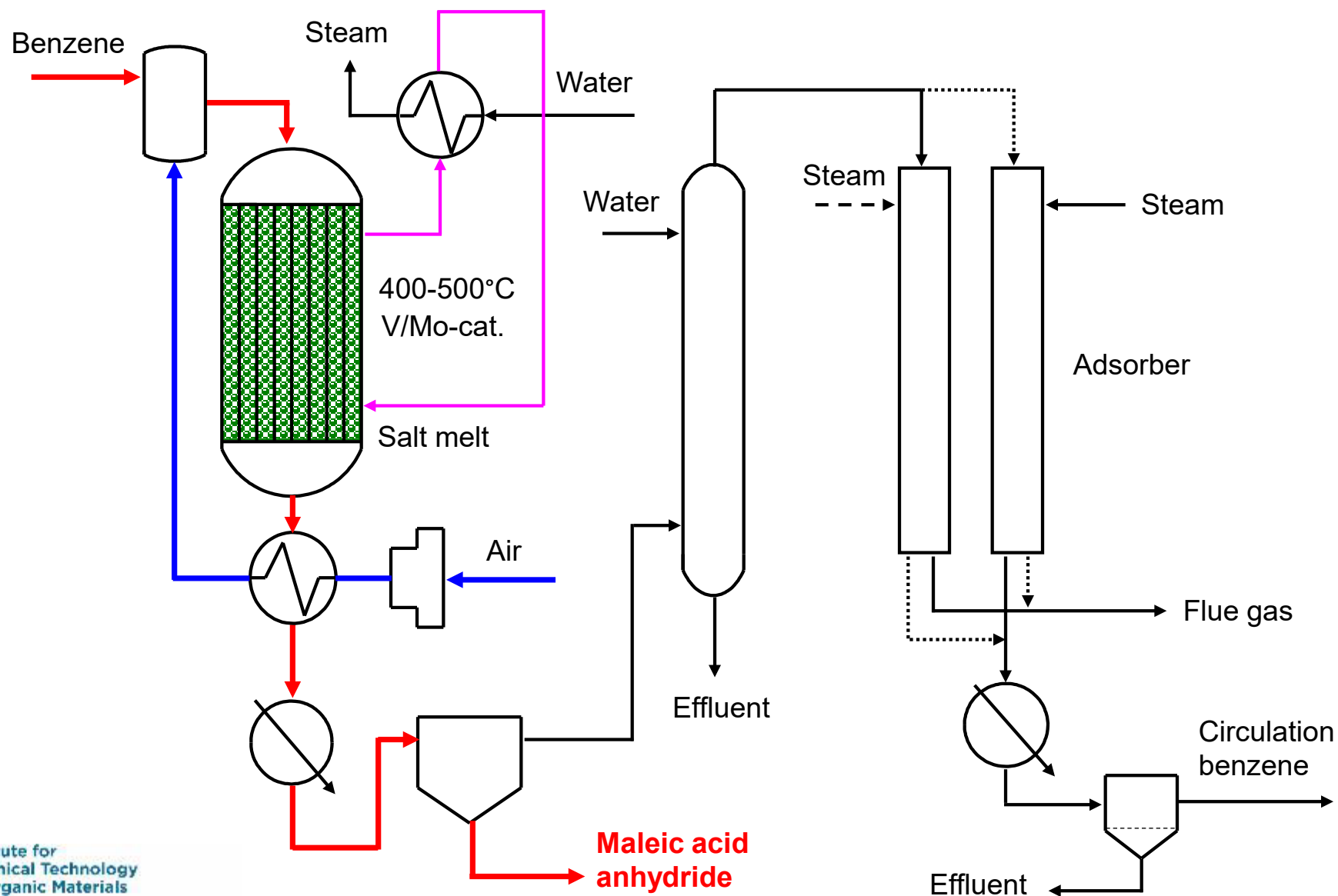


In the mixture as by product

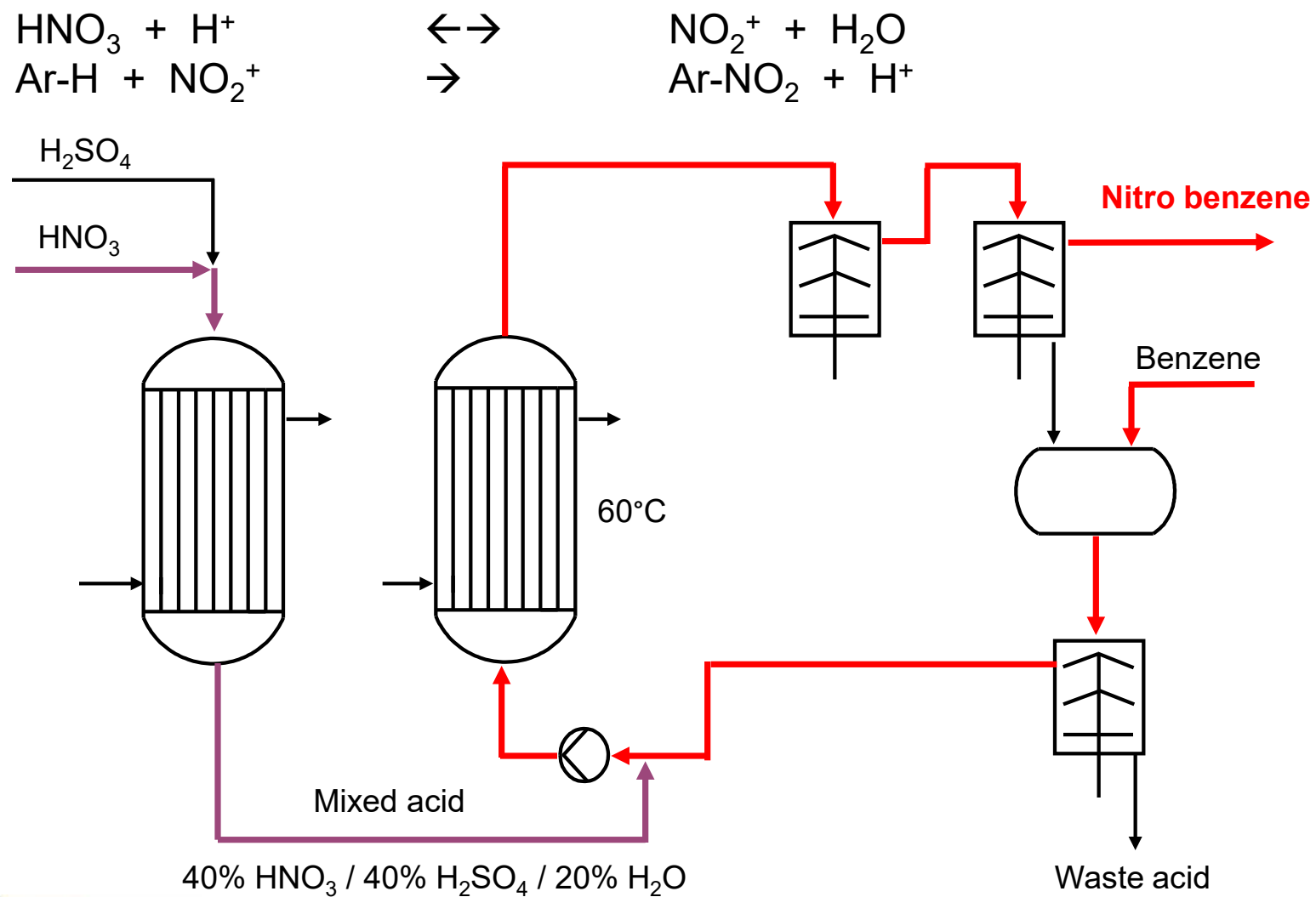
Maleic acid anhydride



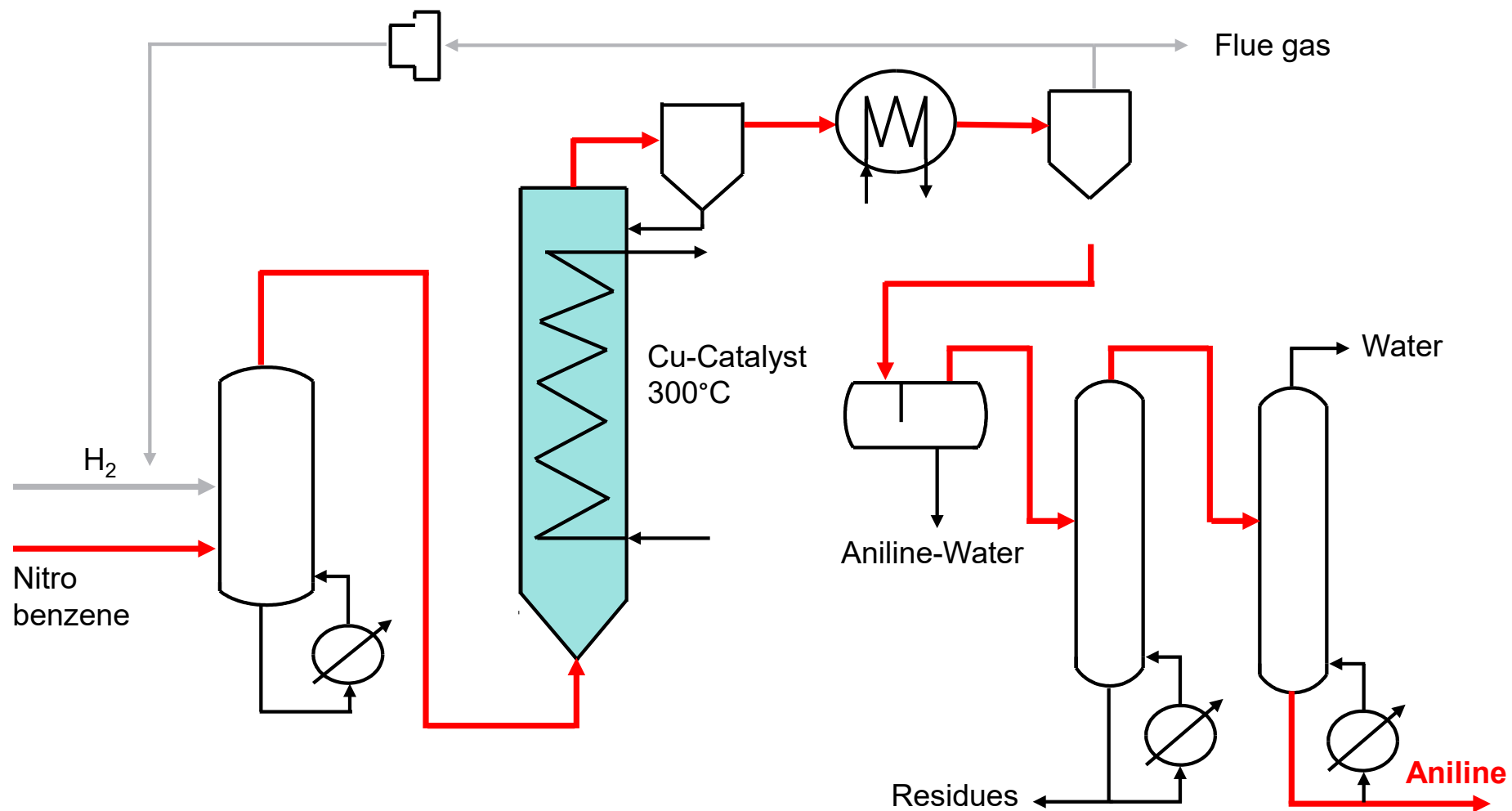
Selective oxidation of benzene



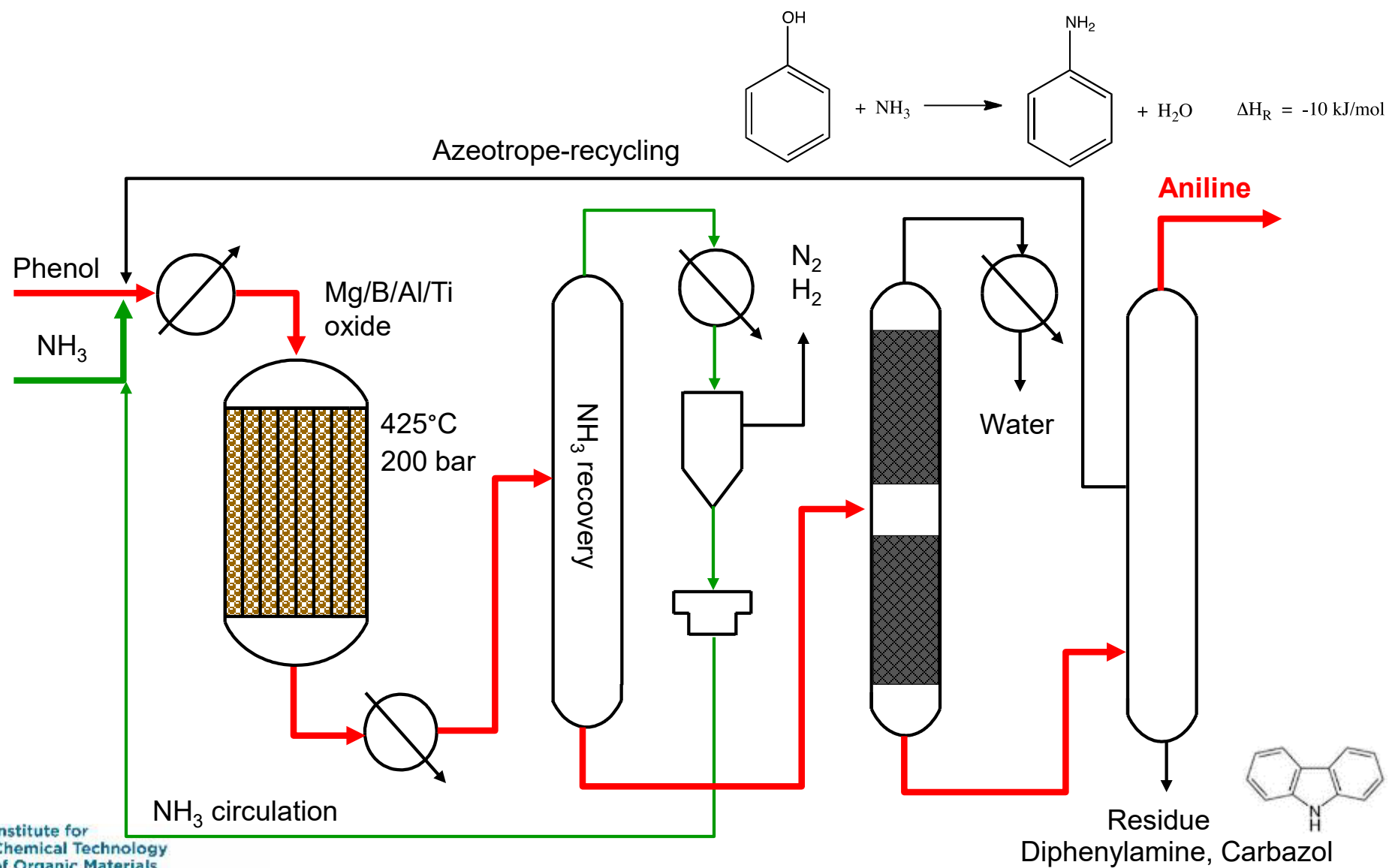
Nitro benzene



Aniline: hydrogenation of nitro benzene

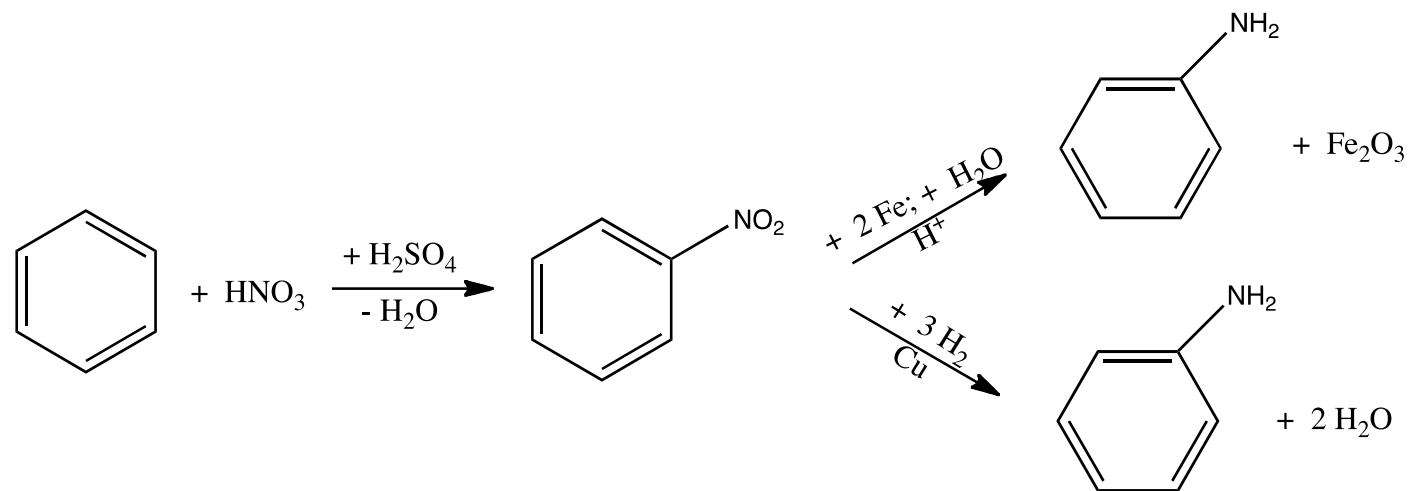


Aniline: Halcon process

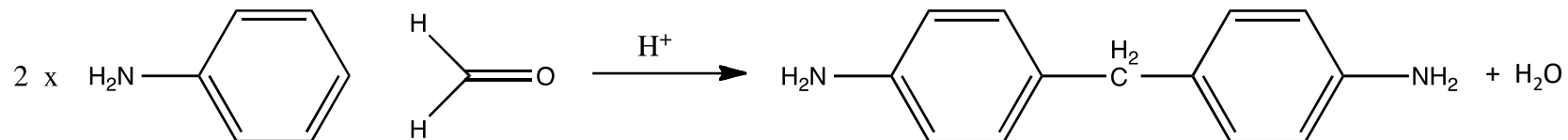


Aniline and derivatives

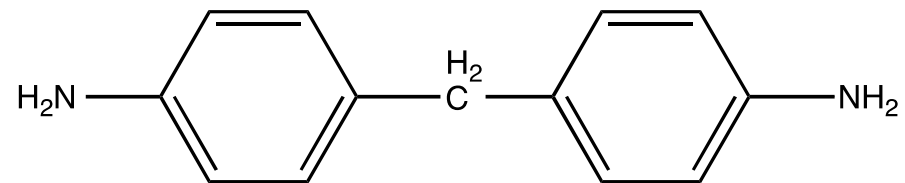
Aniline from benzene



4,4'-Diaminodiphenylmethane



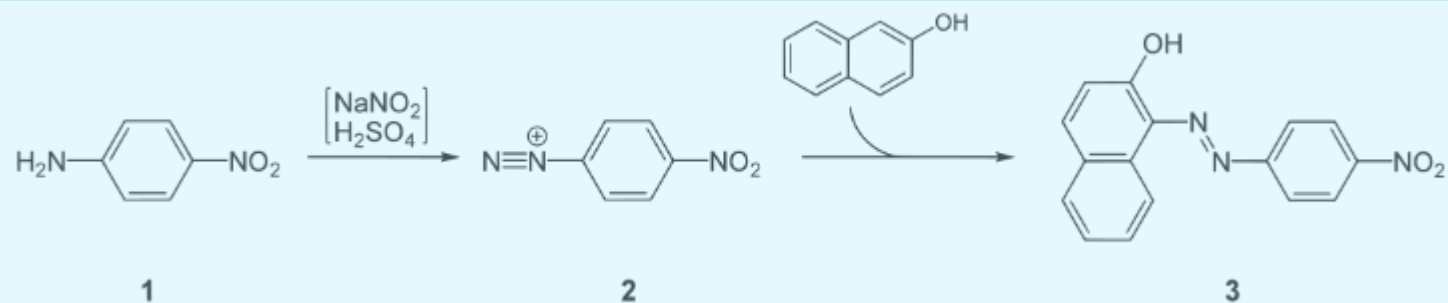
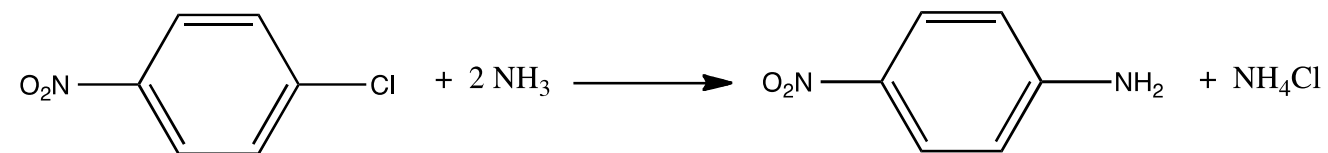
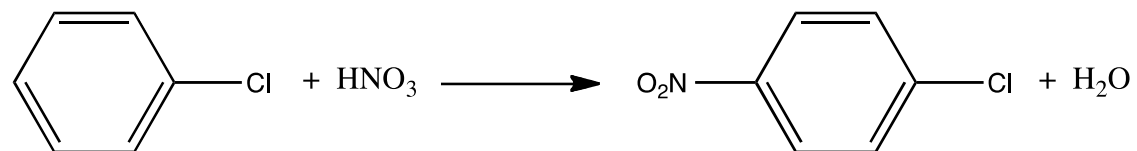
4,4'-Diaminodiphenylmethane



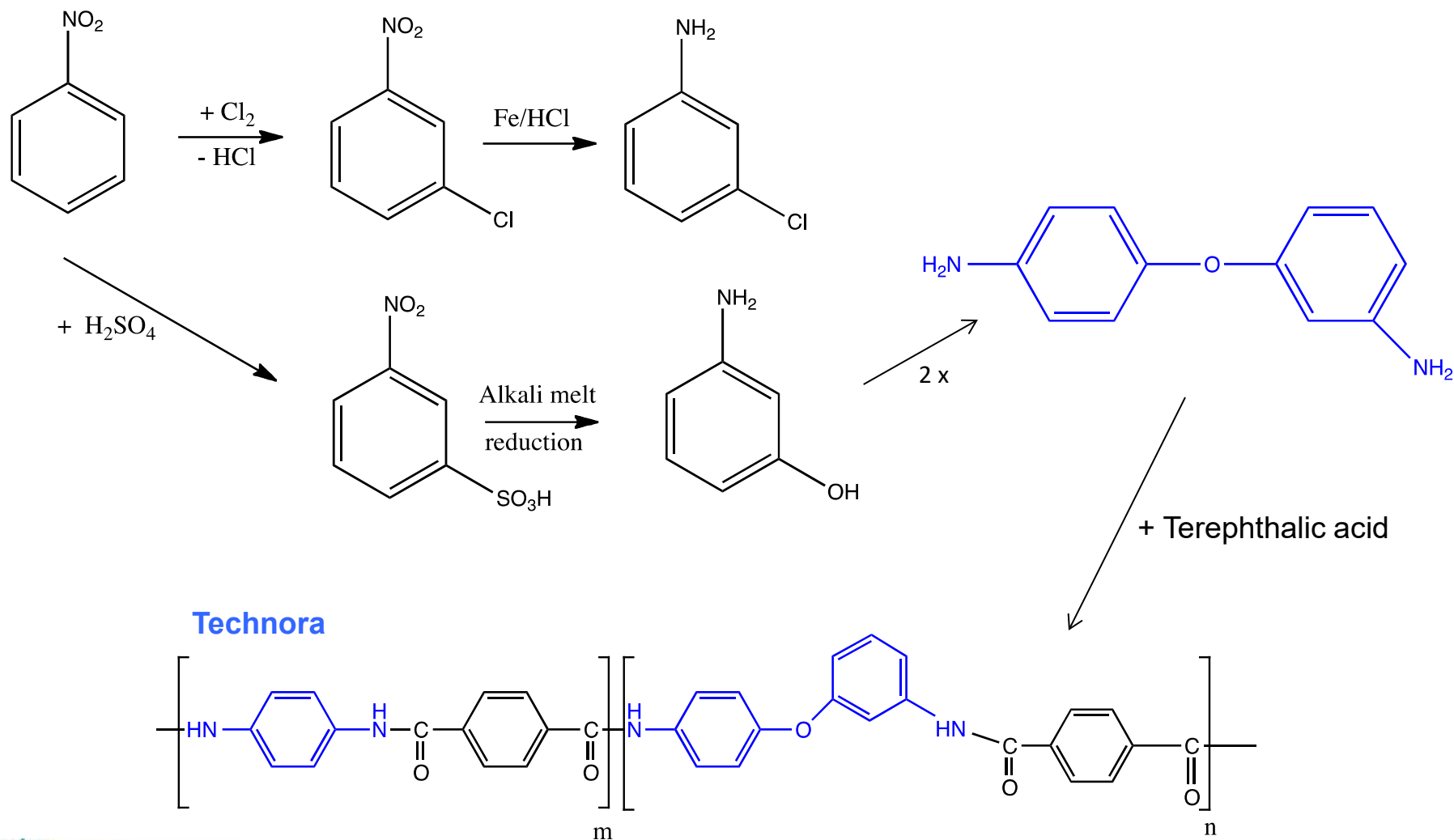
- Epoxy hardening agent
- Raw material in the production of polyurethane elastomers
- In the rubber industry - a curative for neoprene
- Anti-frosting agent (anti-oxidant) in footwear
- Raw material in the production of Qiana Nylon
- Raw material in the preparation of poly(amide-imide) resins (used in magnet wire enamels)

Aniline and derivatives

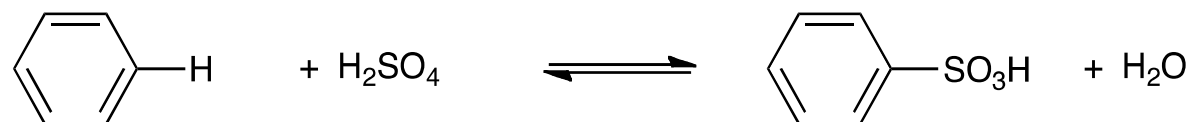
p-Nitroaniline



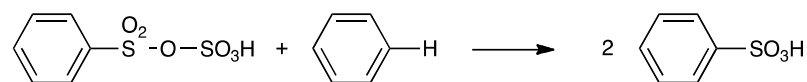
Nitro benzene - Aramide



Aromatic sulfonic acids



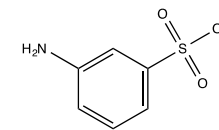
1. Azeotropic sulfonation in a reactor cascade
2. Sulfonation with oleum
3. Sulfonation with SO₃ in a bubble column reactor or in a **thin-film evaporator** (heat dissipation $\Delta H = -170$ kJ/mol)
Formed pyrosulfonic acids are split in a second reactor



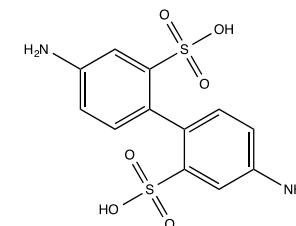
4. Sulfonation with chloro sulfonic acid and successive hydrolysis to the sulfonic acid
In excess of chloro sulfonic acid one gets aromatic sulfo chlorides

Sulfonation of nitro- and Cl aromatics

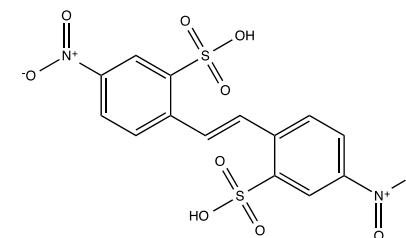
Nitro- or chloro aromatic	Use
Nitrobenzene	Metanilic acid (1) Benzidine-2,2'-disulfonic acid (2)
o-Nitro toluene	Colors
p-Nitro toluene	4,4'-Di nitro stilebene -2,2'-di sulfonic acid (3)
o-Chloro toluene	2-Chloro-4-nitro toluene-5-sulfonic acid (4) 2-Chloro-4-amino toluene-5-sulfonic acid (5)
o-Chloro nitro benzene	Colors
p-Chloro nitro benzene	Colors



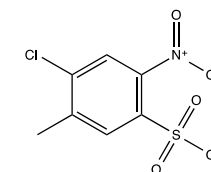
(1)



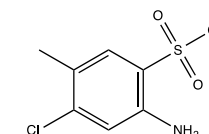
(2)



(3)

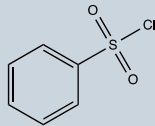


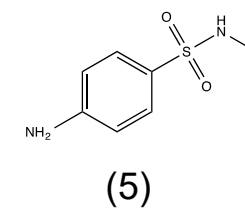
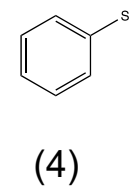
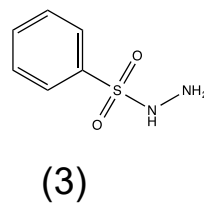
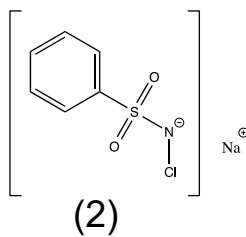
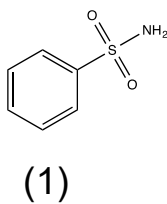
(4)



(5)

Aromatic sulfo chlorides

Aromatic	Sulfo chloride	Use
Benzene	 Benzene sulfochloride	Benzene sulfonamide (1) Chloroamine B (2) Benzolsulfonohydrazid (foaming agent) (3) Thiophenole (4) Sulfonamides (5)
Toluene	o-Toluene sulfochloride p-Toluene sulfochloride o-,p-Toluene sulfochloride	Saccharine p-Toluene sulfonamide o-,p-Toluene sulfonamide
Nitro benzene	m-Nitro benzene sulfochloride	Sulfinic acid



Aromatics

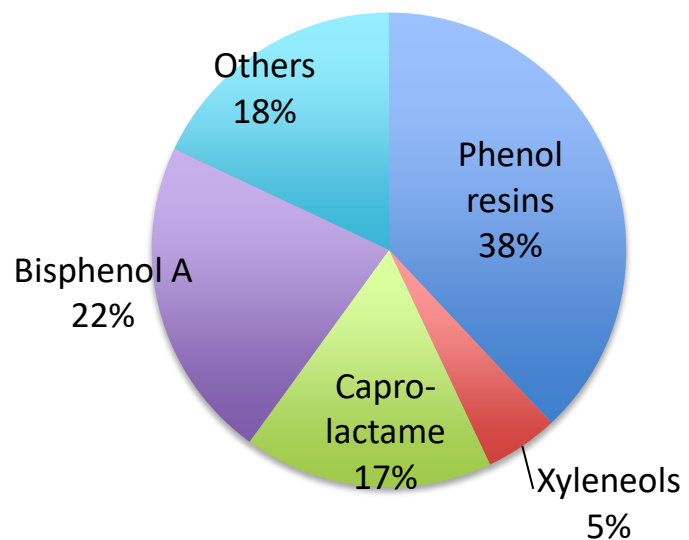
- Separation methods
- Benzene & benzene products
- **Phenol, Cyclohexanone, ...**
- Toluene
- C8 aromatics
- Polycyclic aromatics
- Heterocyclics

Phenol

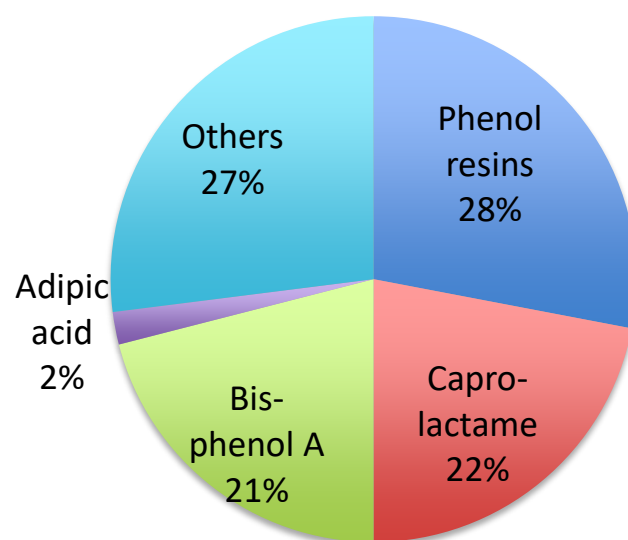
- Molar mass: 94,11 g/mol
- Density: 1,07 kg/m³ (20°C)
- T_m: 41°C
- T_b: 182°C
- pK_s: 9,99
- Production volume roughly 12 Mill tons

Phenol - use

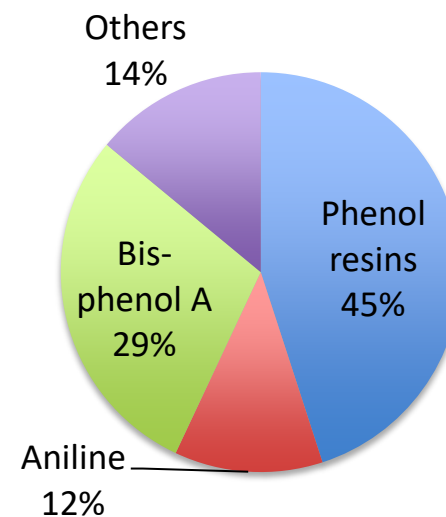
USA



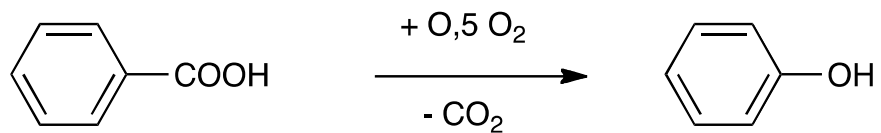
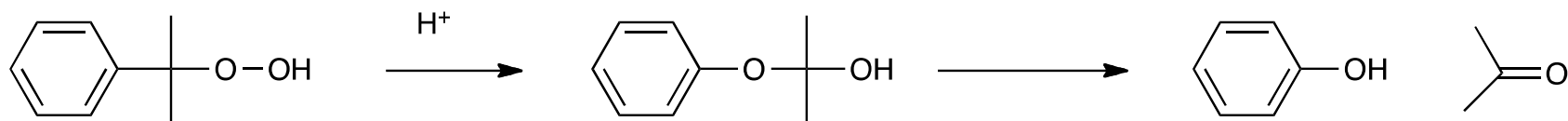
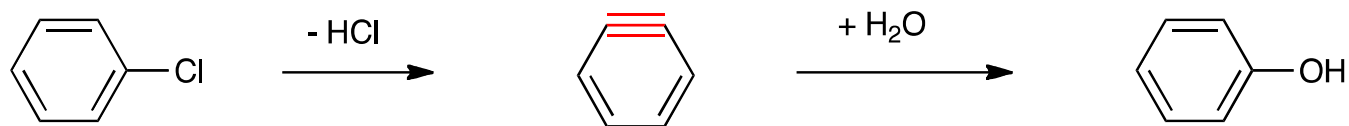
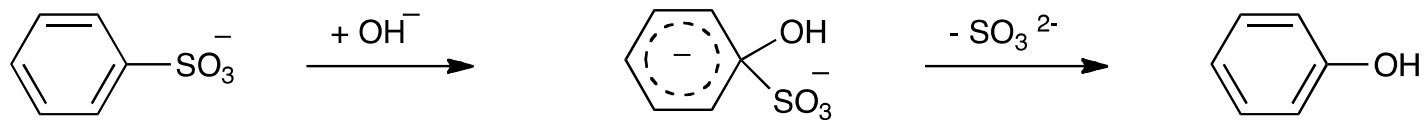
WE



J

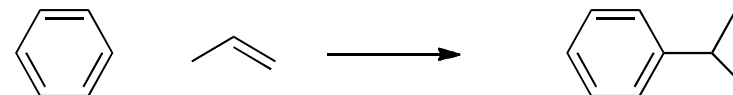


Phenol - synthesis

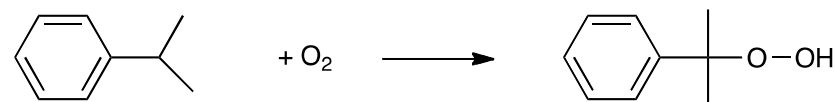


Phenol cumene process

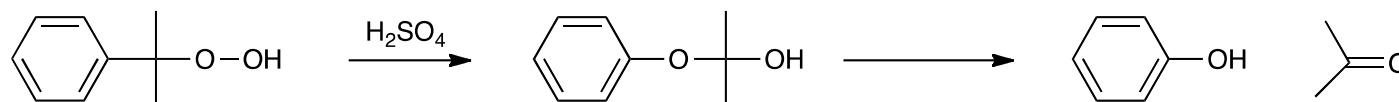
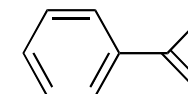
1. Benzene + propylene: cumene



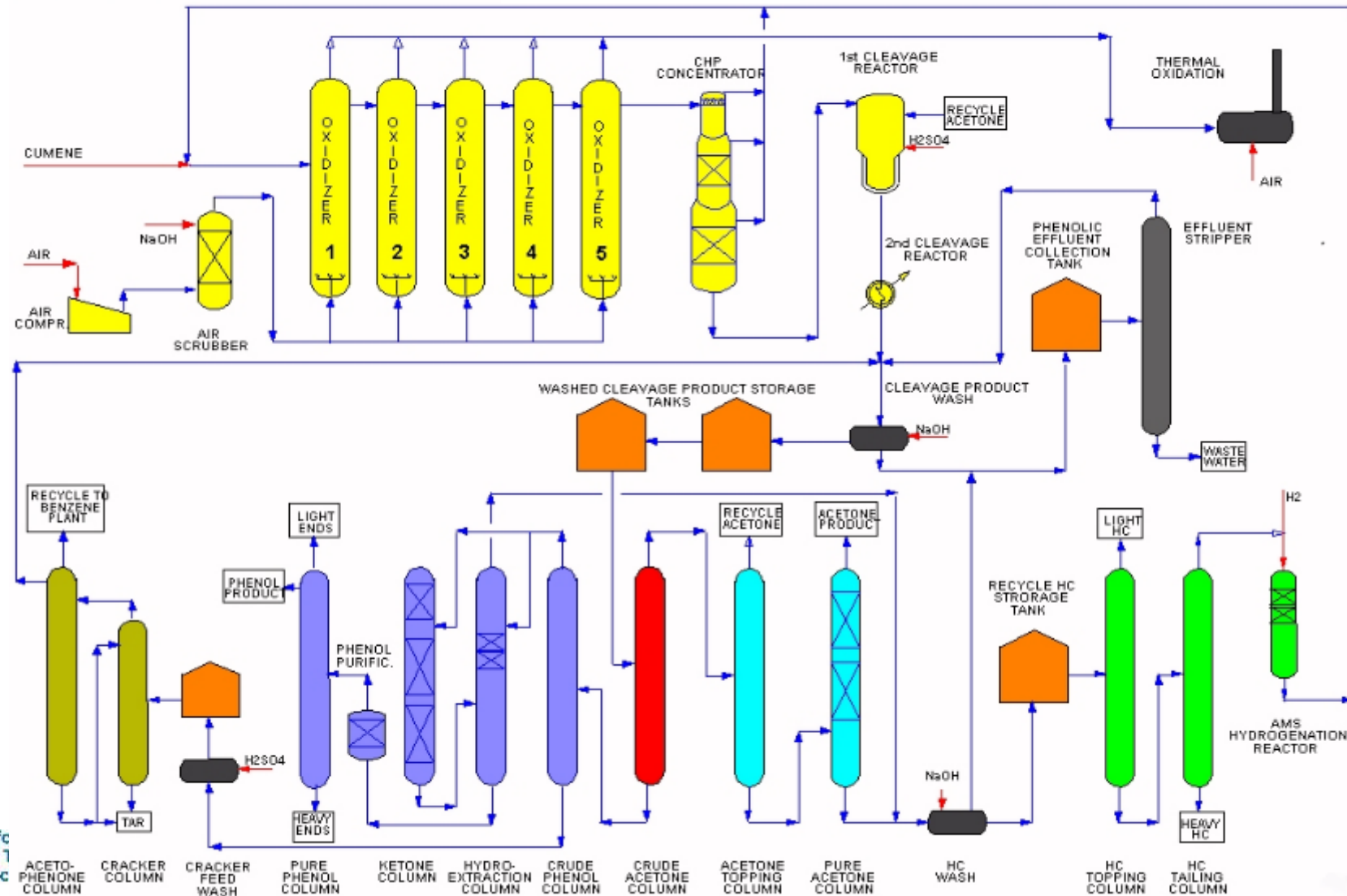
2. Cumene oxidation:
Cumene hydro peroxide (CHP)
 $\Delta H_f = -110 \text{ kJ/mol}$



3. Cleavage to phenol and acetone
 $\Delta H_f = -270 \text{ kJ/mol}$

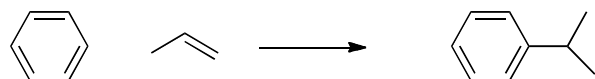


Phenol cumene process

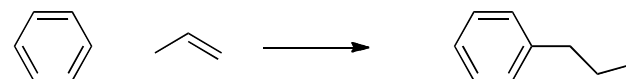


Cumene reactions

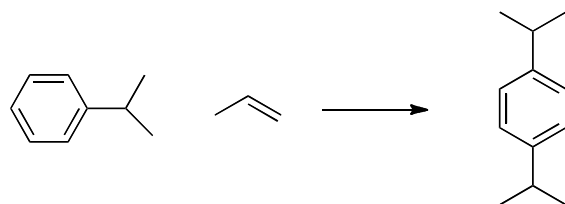
Alkylation



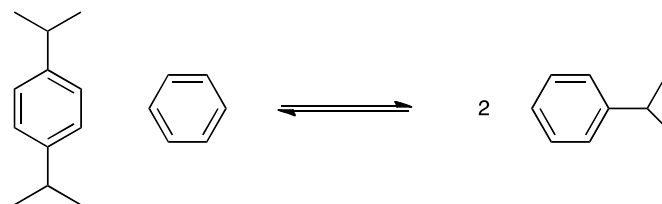
Formation of propylbenzene



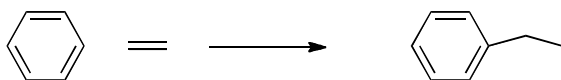
“Over” alkylation (*p*-DIPB)



Transalkylation

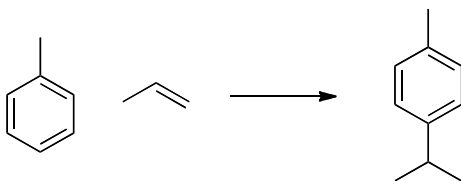


Cumene: side reactions & impurities



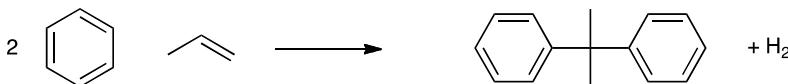
Impurities in the propylene feed such as ethylene, cyclopropane and butene react with benzene to form alkylbenzenes:

Ethylbenzene, n-Propylbenzene and Butylbenzene

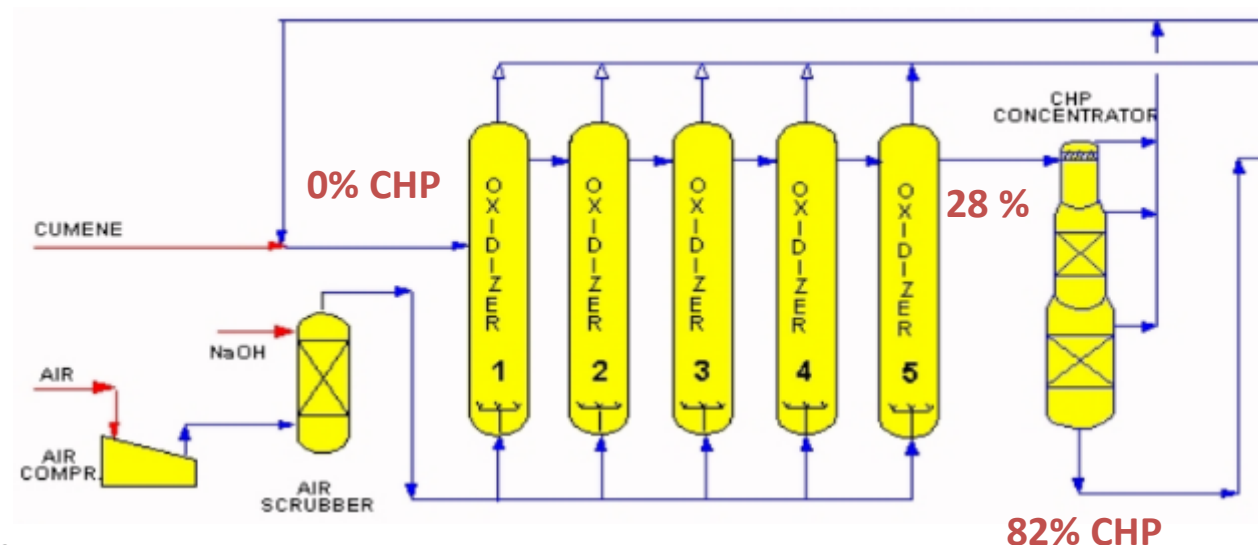


Formation of Cymenes:

~ 73 % of the toluene reacts to cymenes (p-cymene)

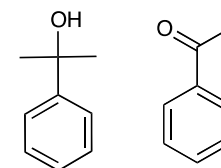


Cumene: oxidation



Oxidation & concentration reaction:

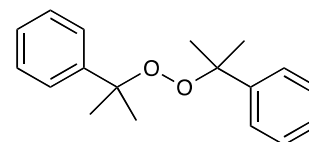
Main reaction: cumene + O₂ → CHP (Cumenehydroperoxide)



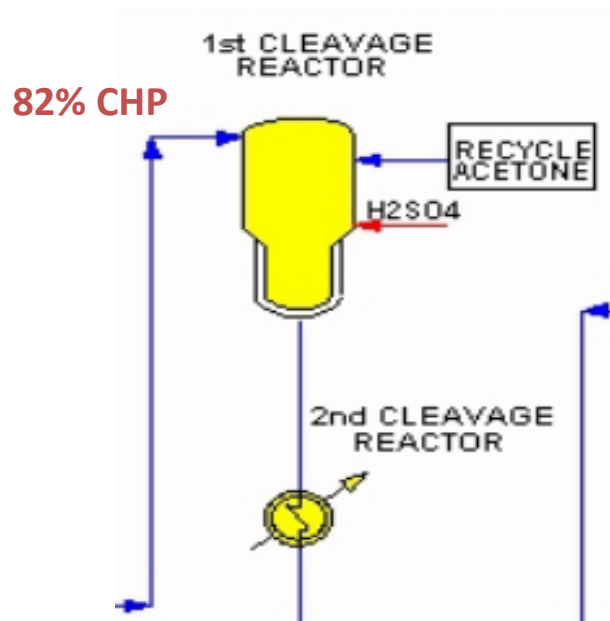
Side reactions (major components):

Cumene + O₂ → DMBA, AP (**D**i **M**ethyl **B**enzene **A**lcohol, **A**ceto **P**henone)

CHP + DMBA → DCP (**D**i **C**umyl **P**eroxide) and water

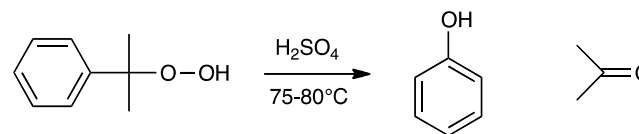


Cumene - cleavage

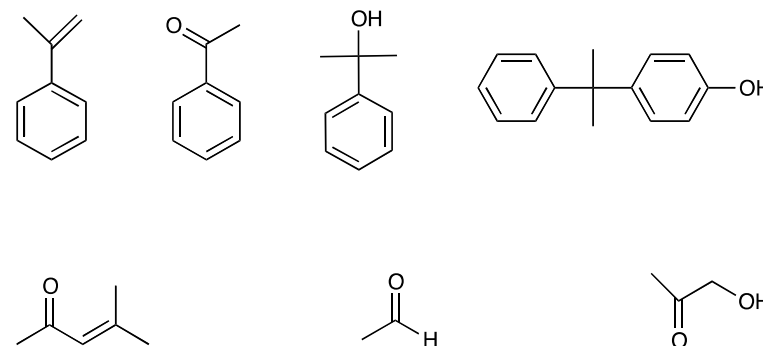


Cleavage product:
Mixture of phenol, acetone,
cumene, water, byproducts

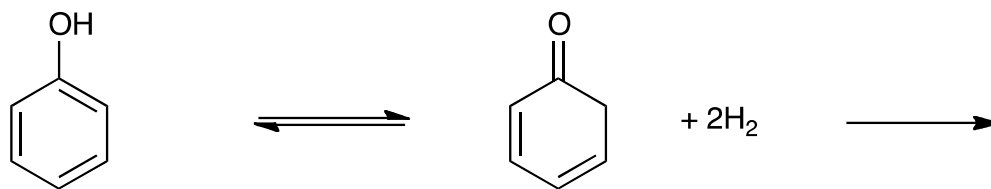
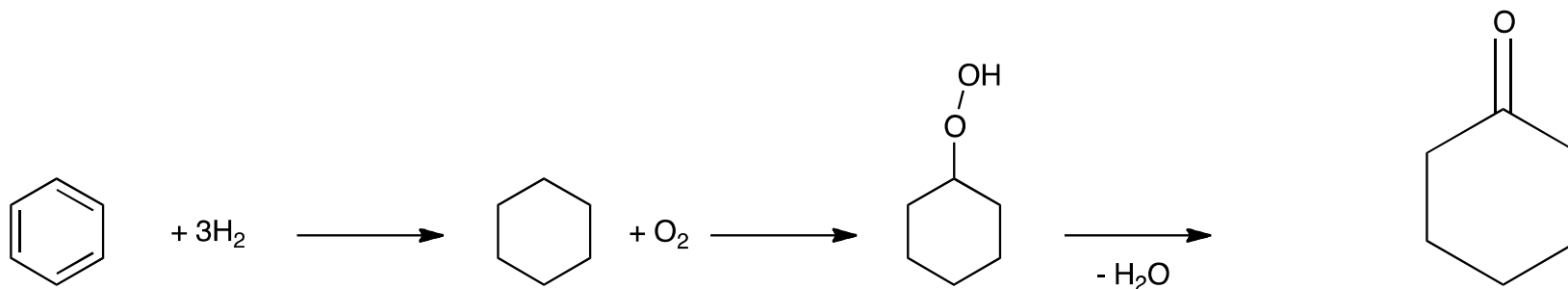
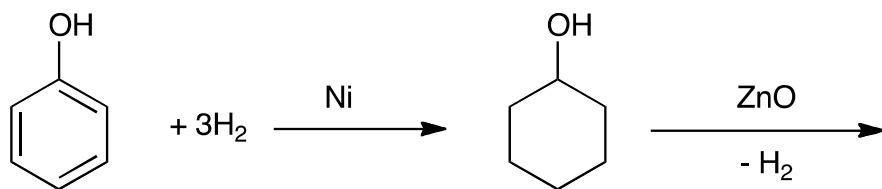
Main reaction:



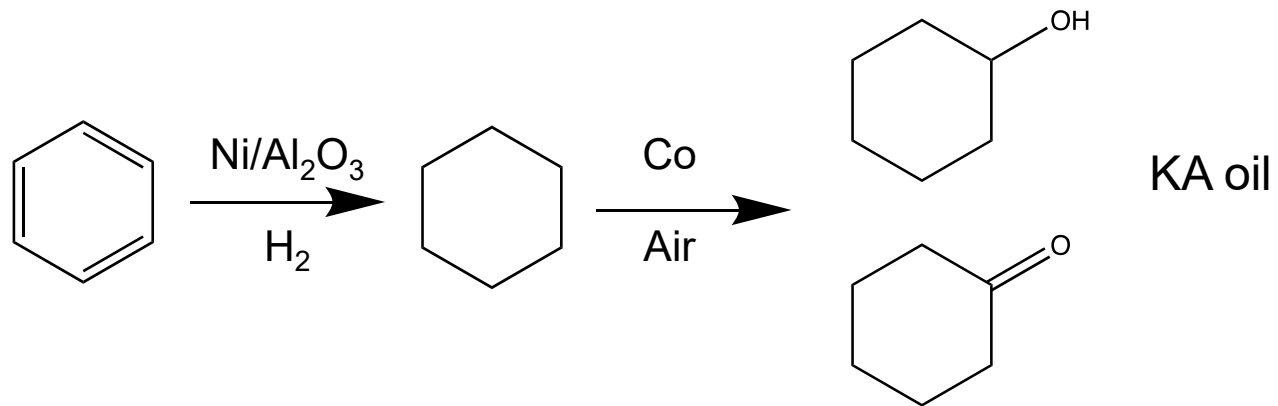
Byproduct formation:



Cyclohexanon synthesis

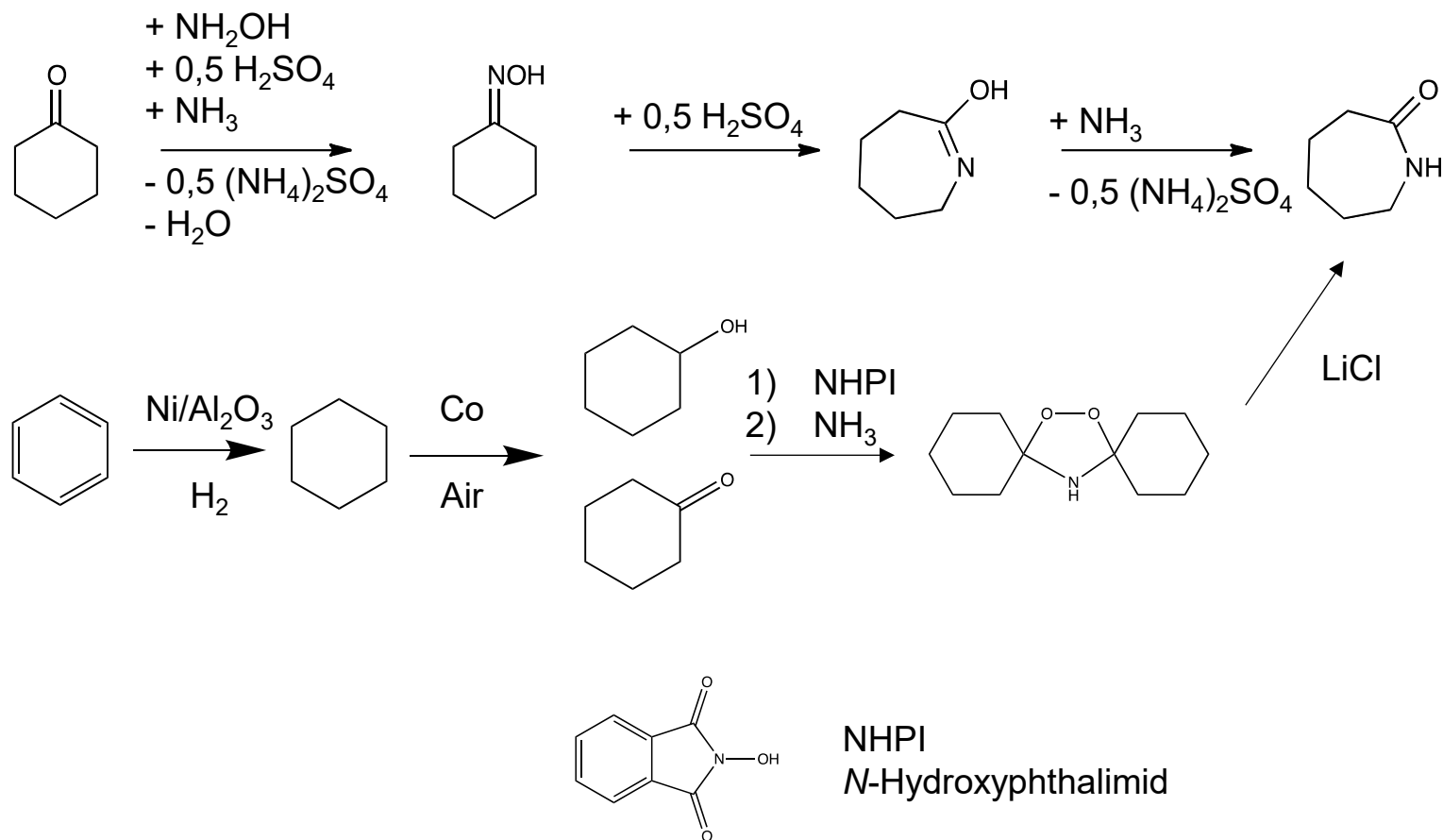


"KA" oil



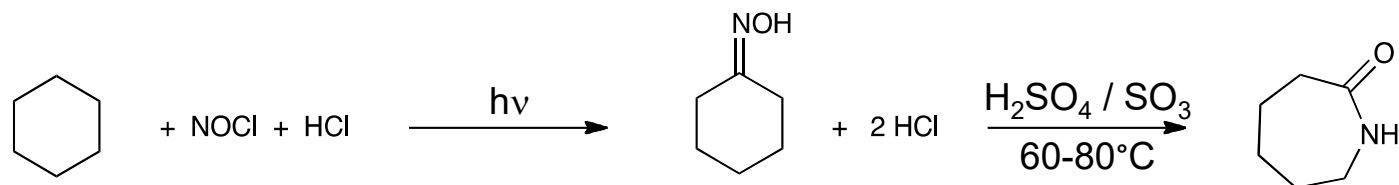
KA Keton/Alcohol

Caprolactam: synthesis

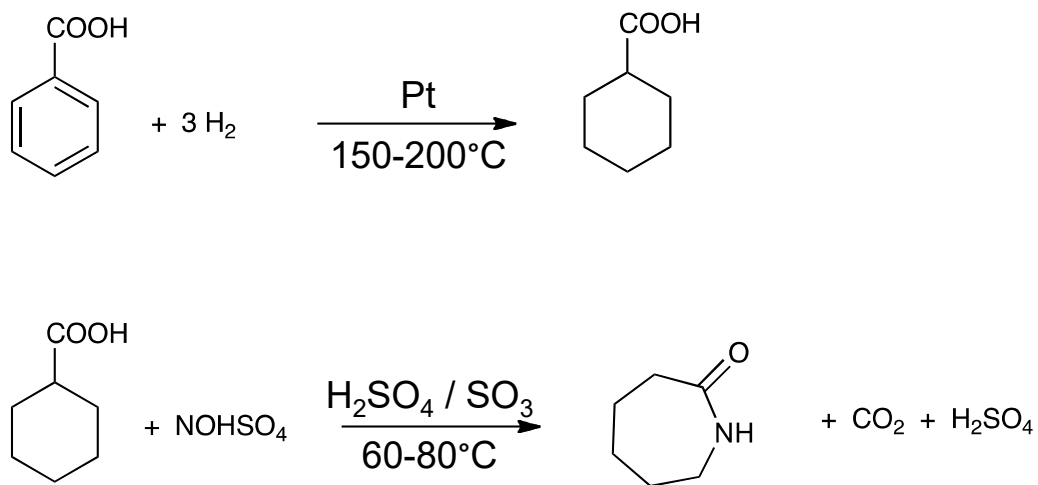


Caprolactam: synthesis

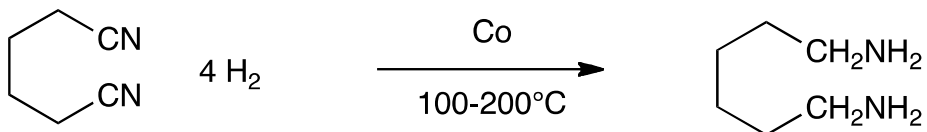
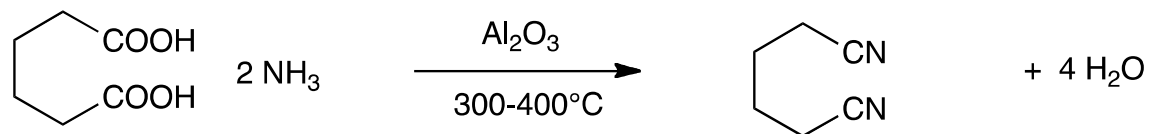
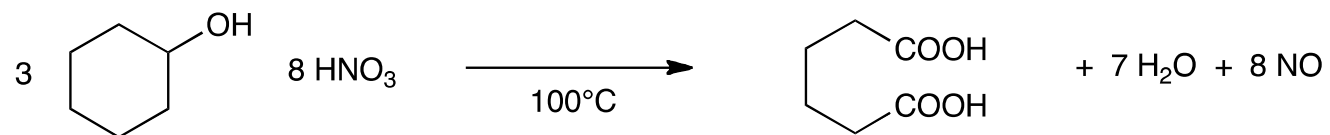
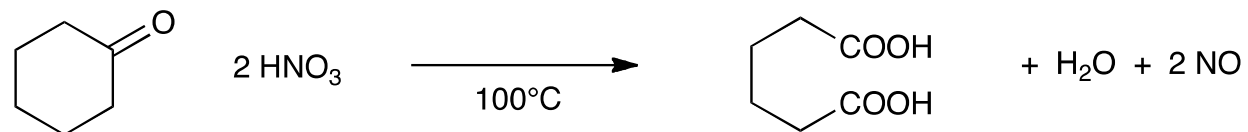
Caprolactam through photo nitrosation of cyclohexane



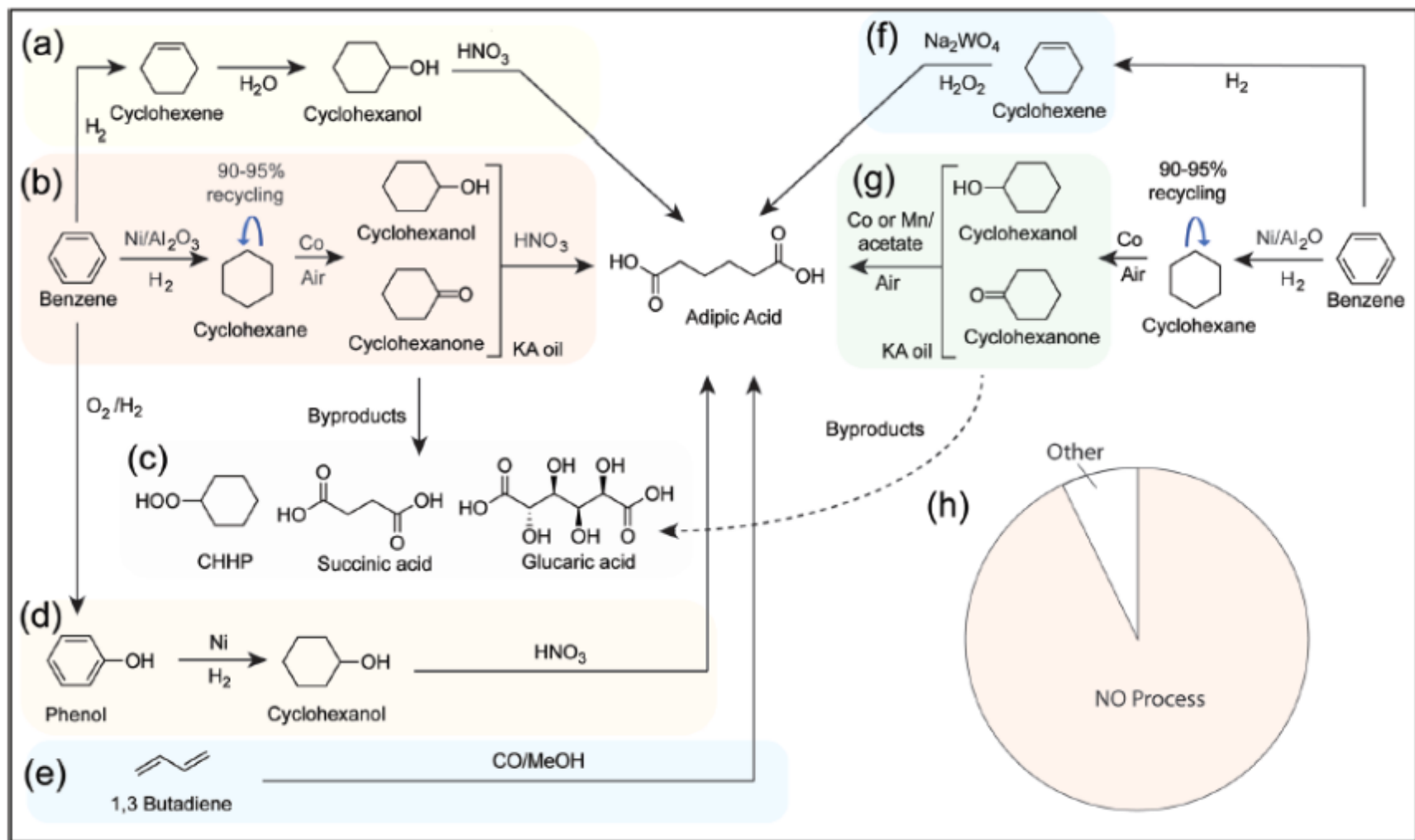
Benzoic acid (nitroso decarboxylation of hexahydroxy benzoic acid)



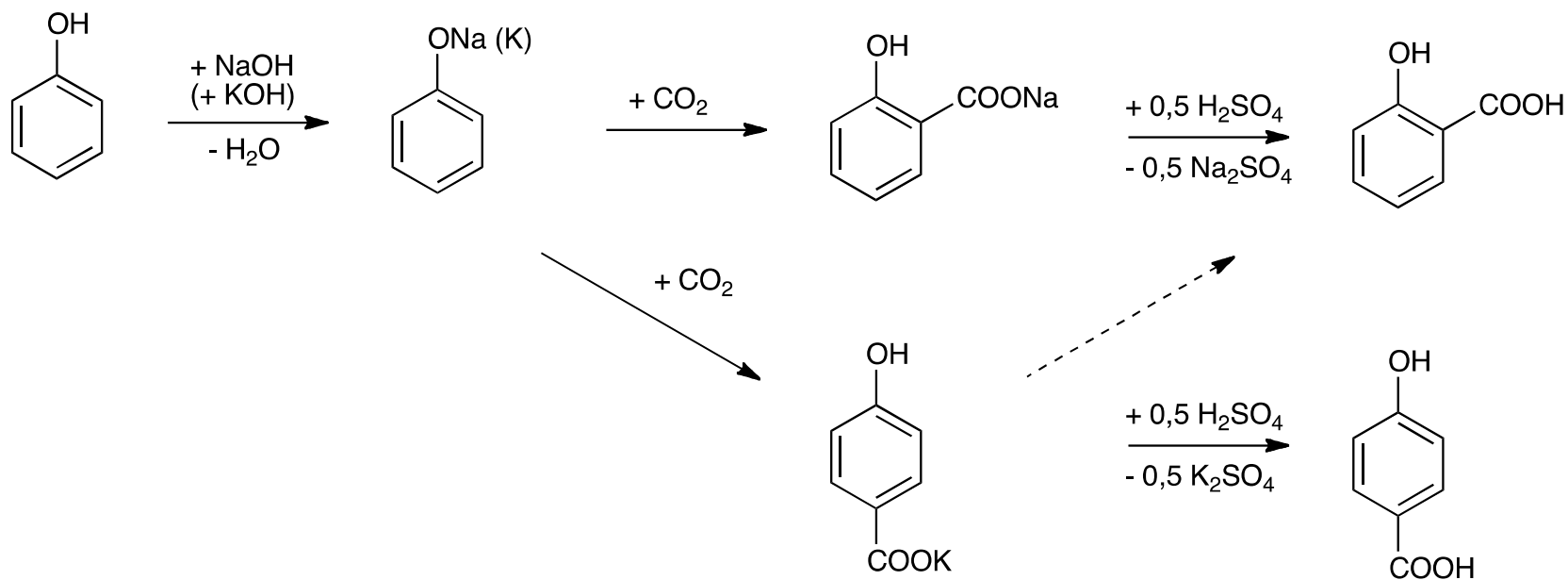
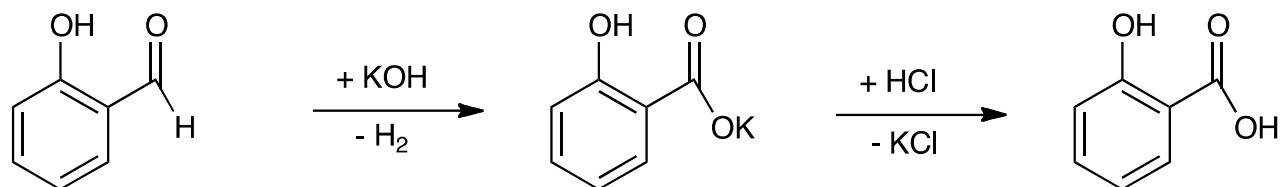
Adipic acid and hexamethylene diamine



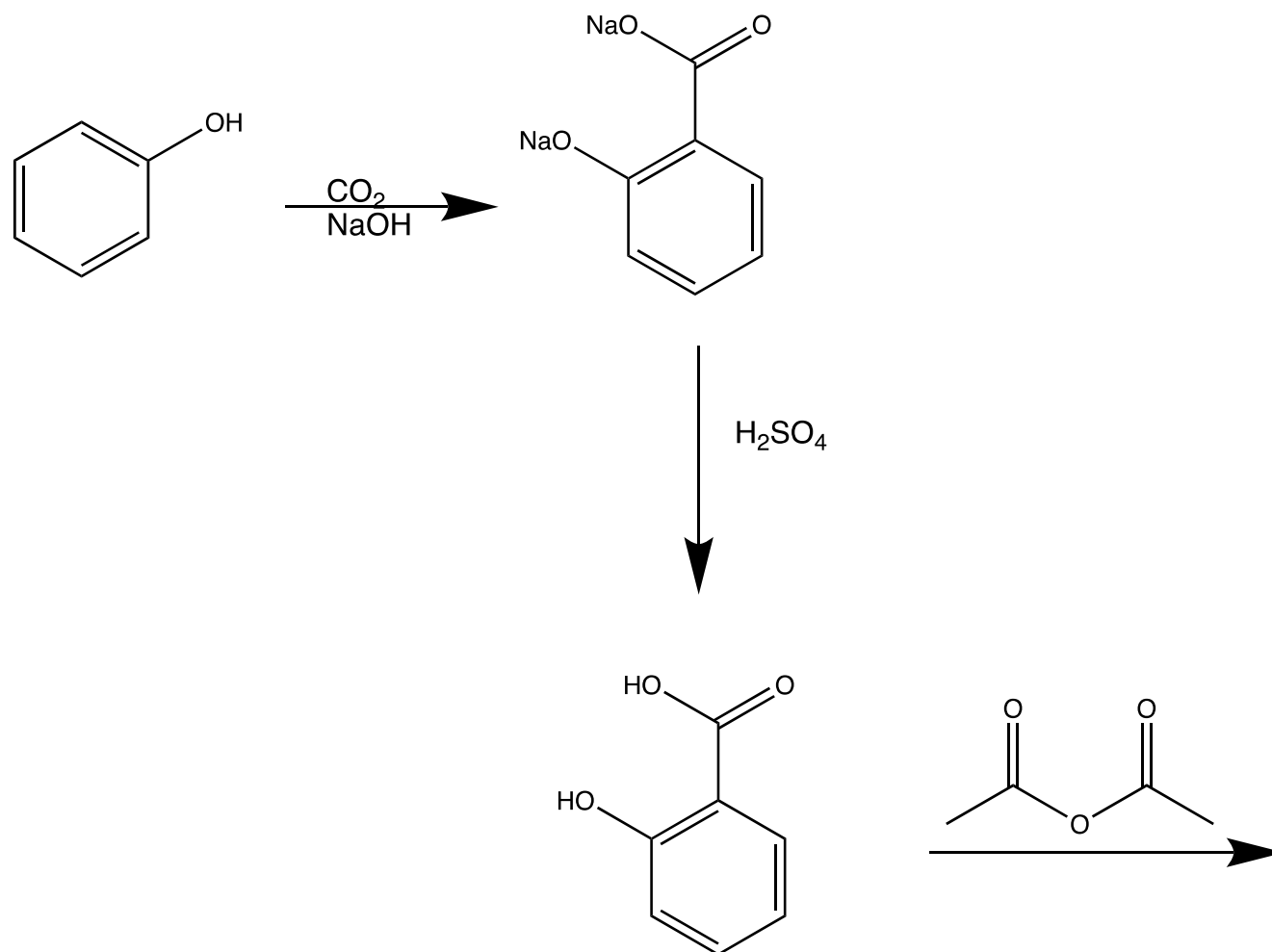
Adipic acid



Salicylic acid and *p*-hydroxy benzoic acid



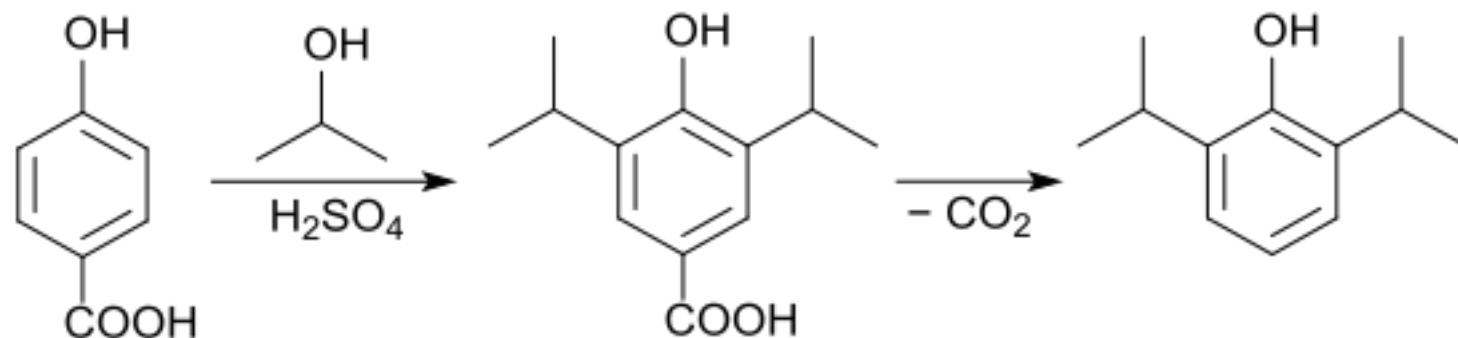
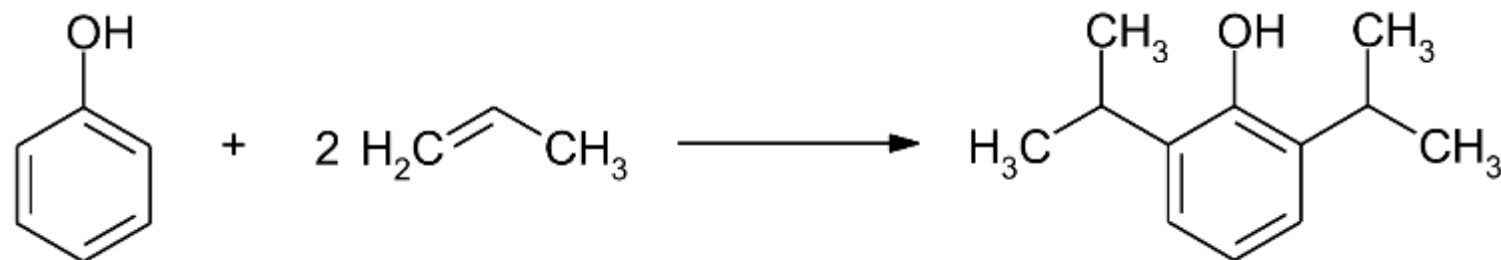
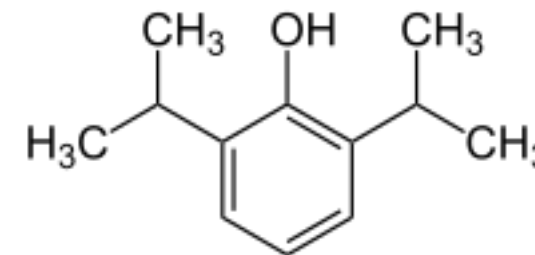
Aspirin



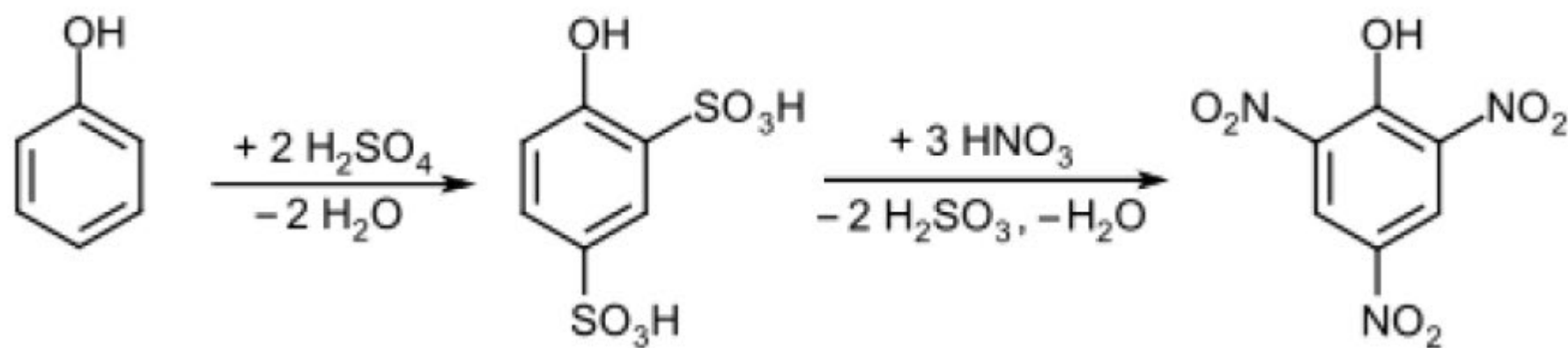
Roughly 41.000 t/a (2023)

Propofol

active component of an intravenous anesthetic formulation



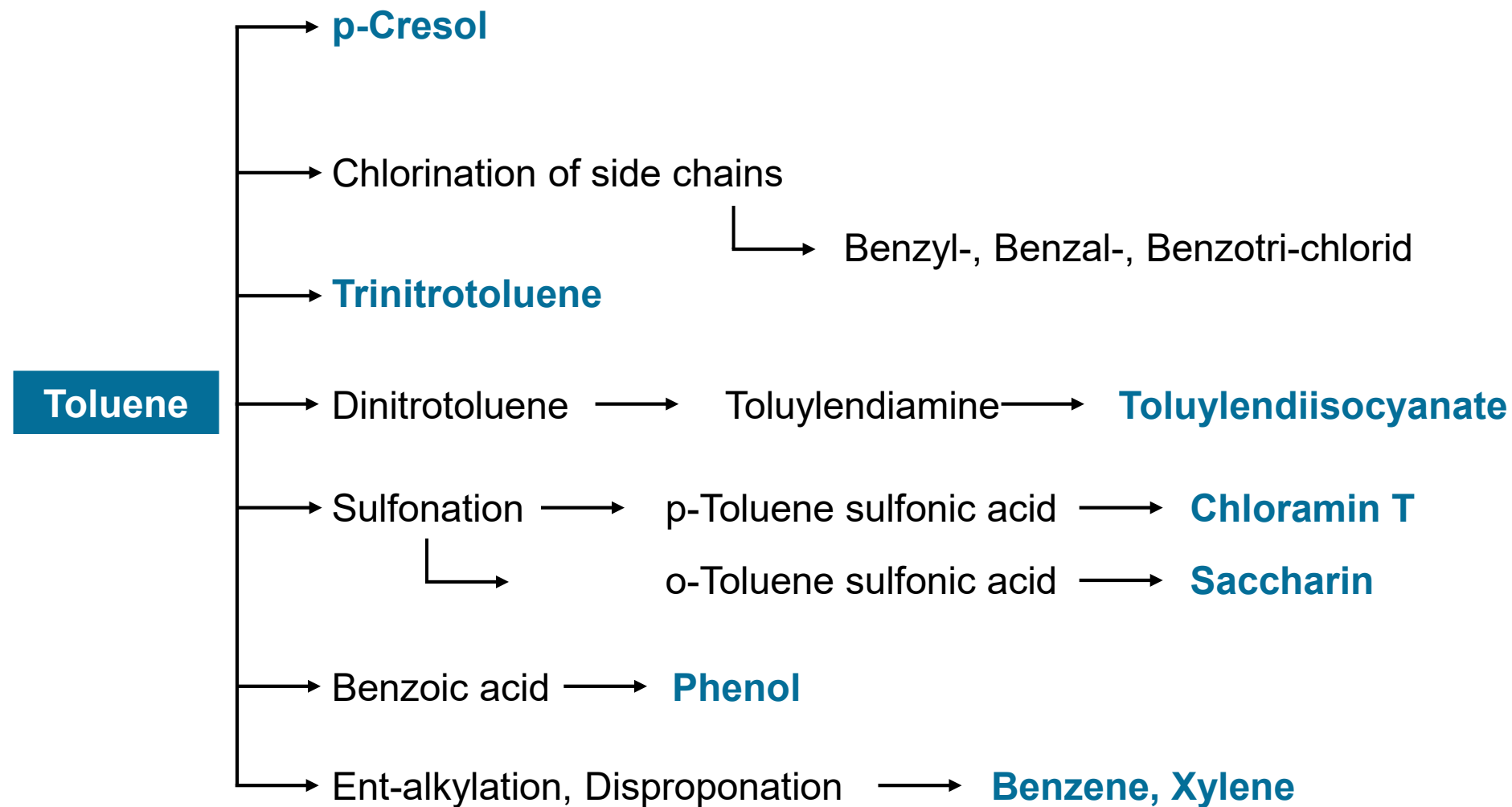
Picric acid



Aromatics

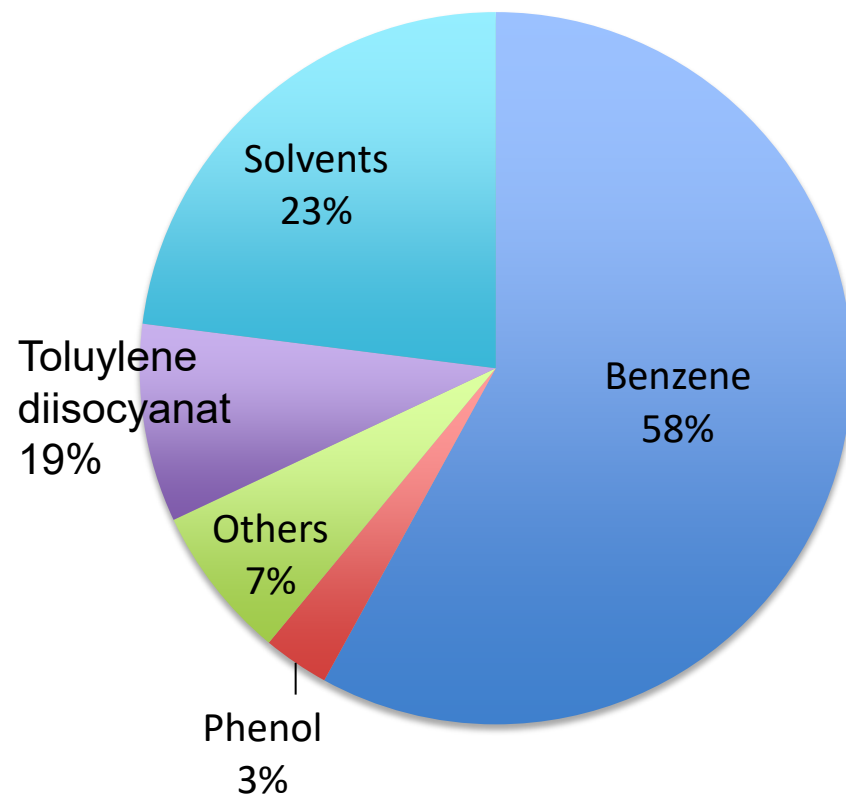
- Separation methods
- Benzene & benzene products
- Phenol, Cyclohexanone, ...
- **Toluene**
- C8 aromatics
- Polycyclic aromatics
- Heterocyclics

Toluene

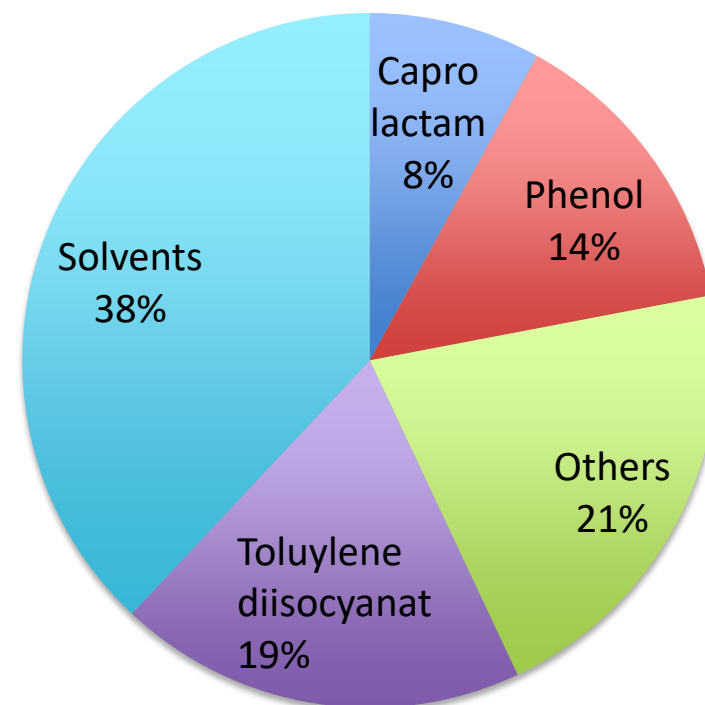


Toluene - use

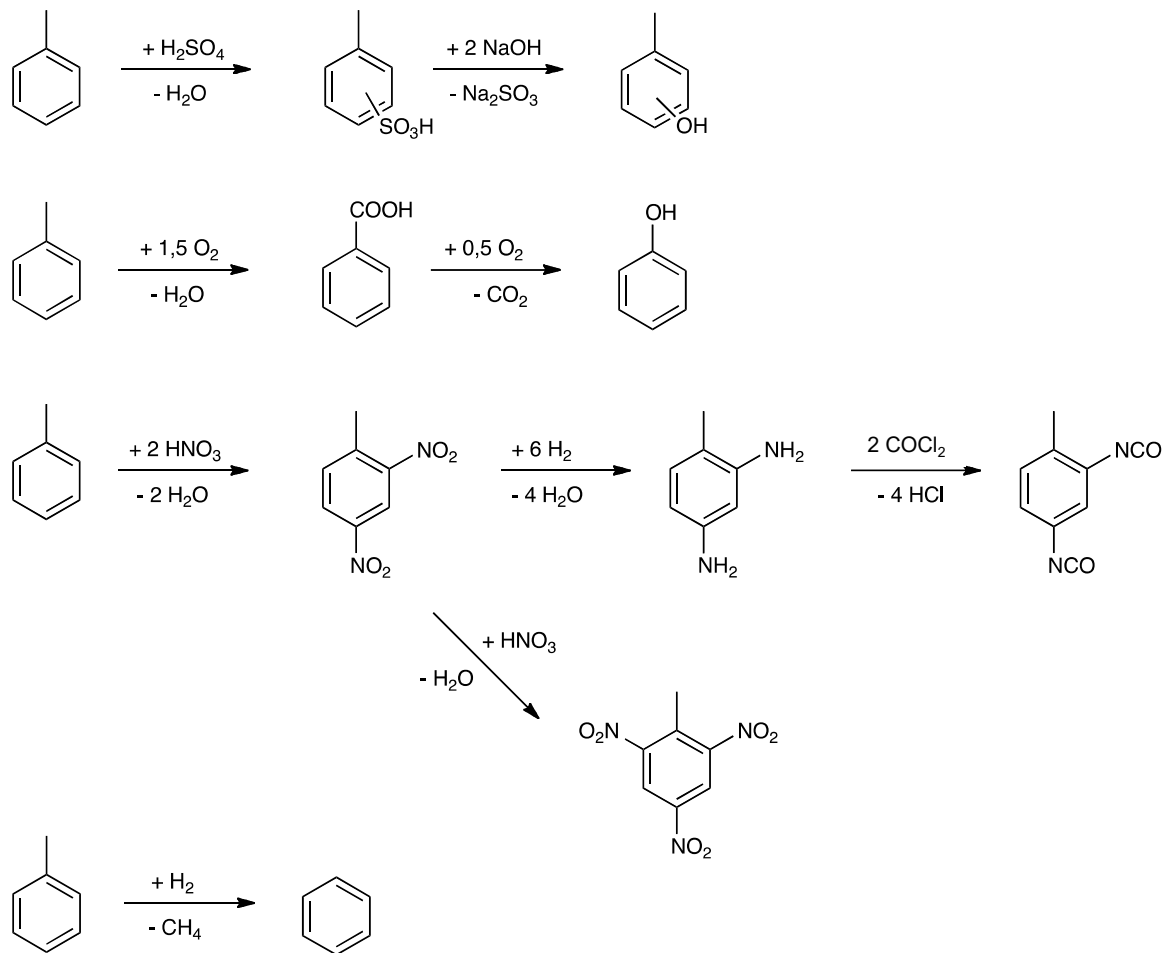
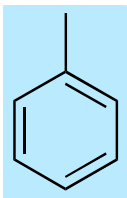
USA



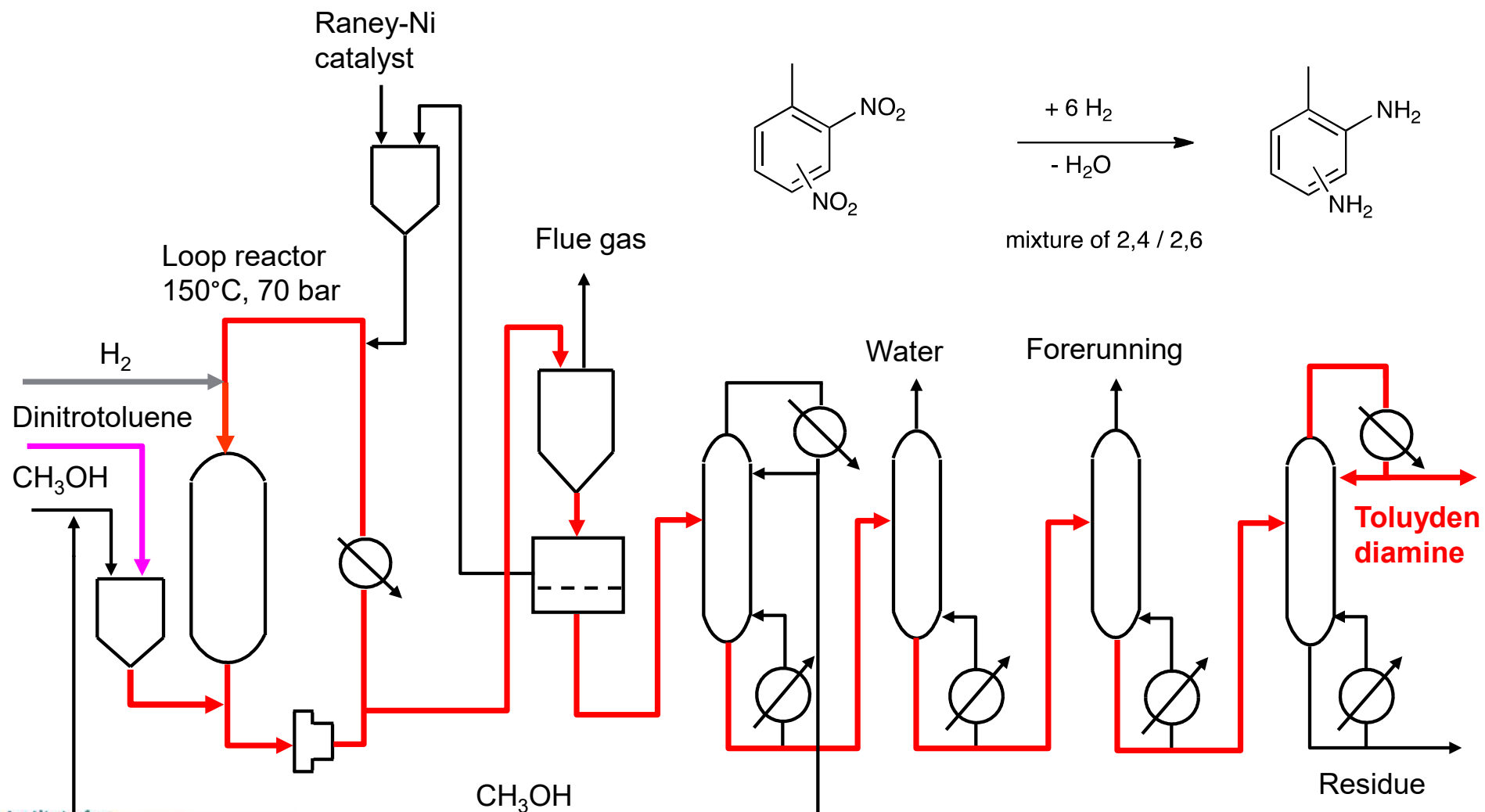
WE



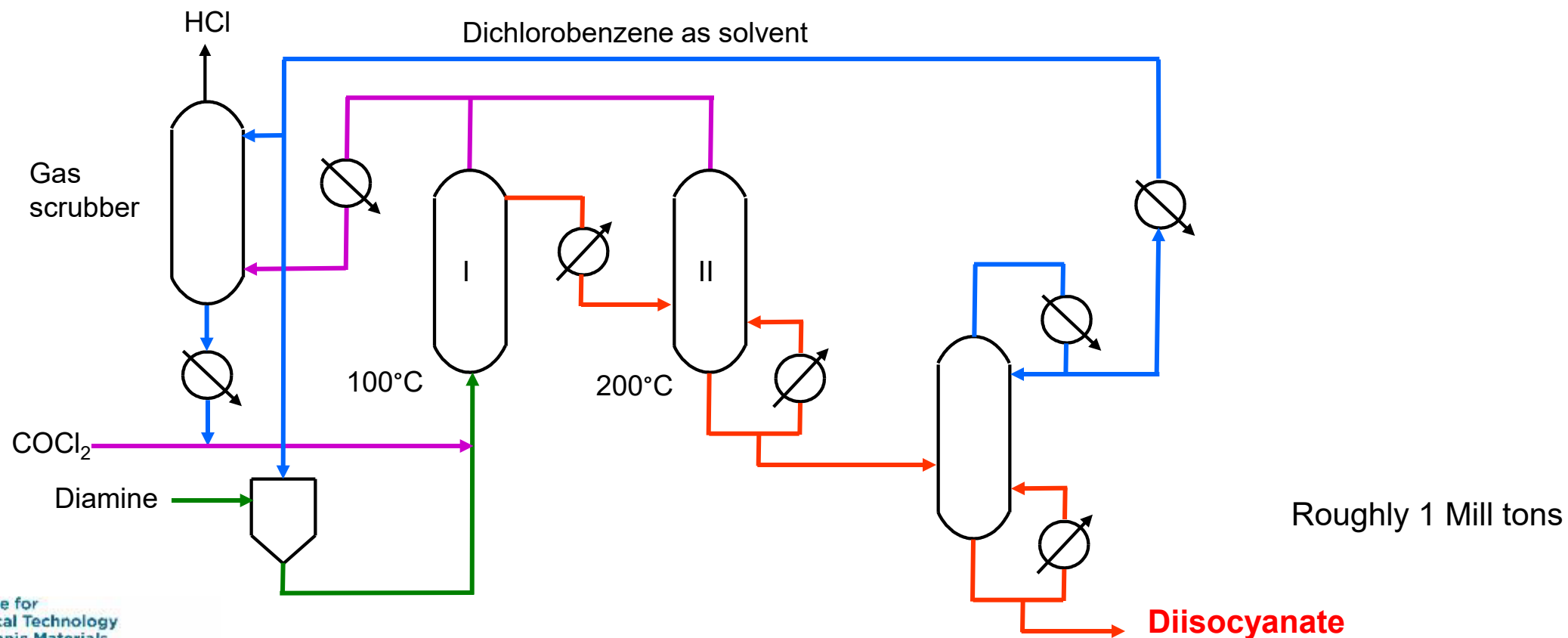
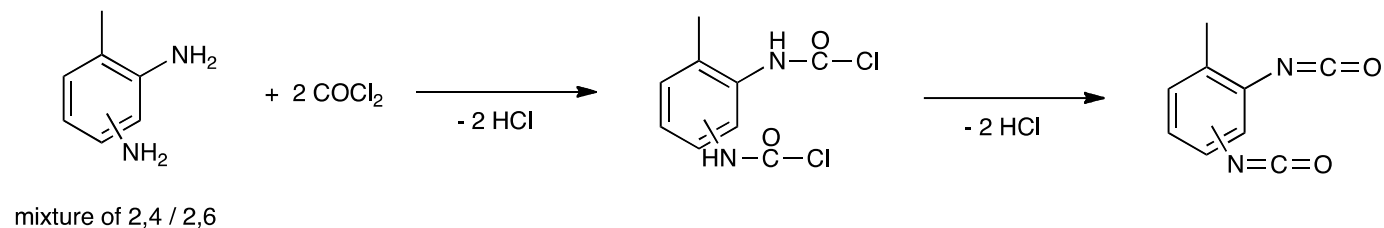
Toluene



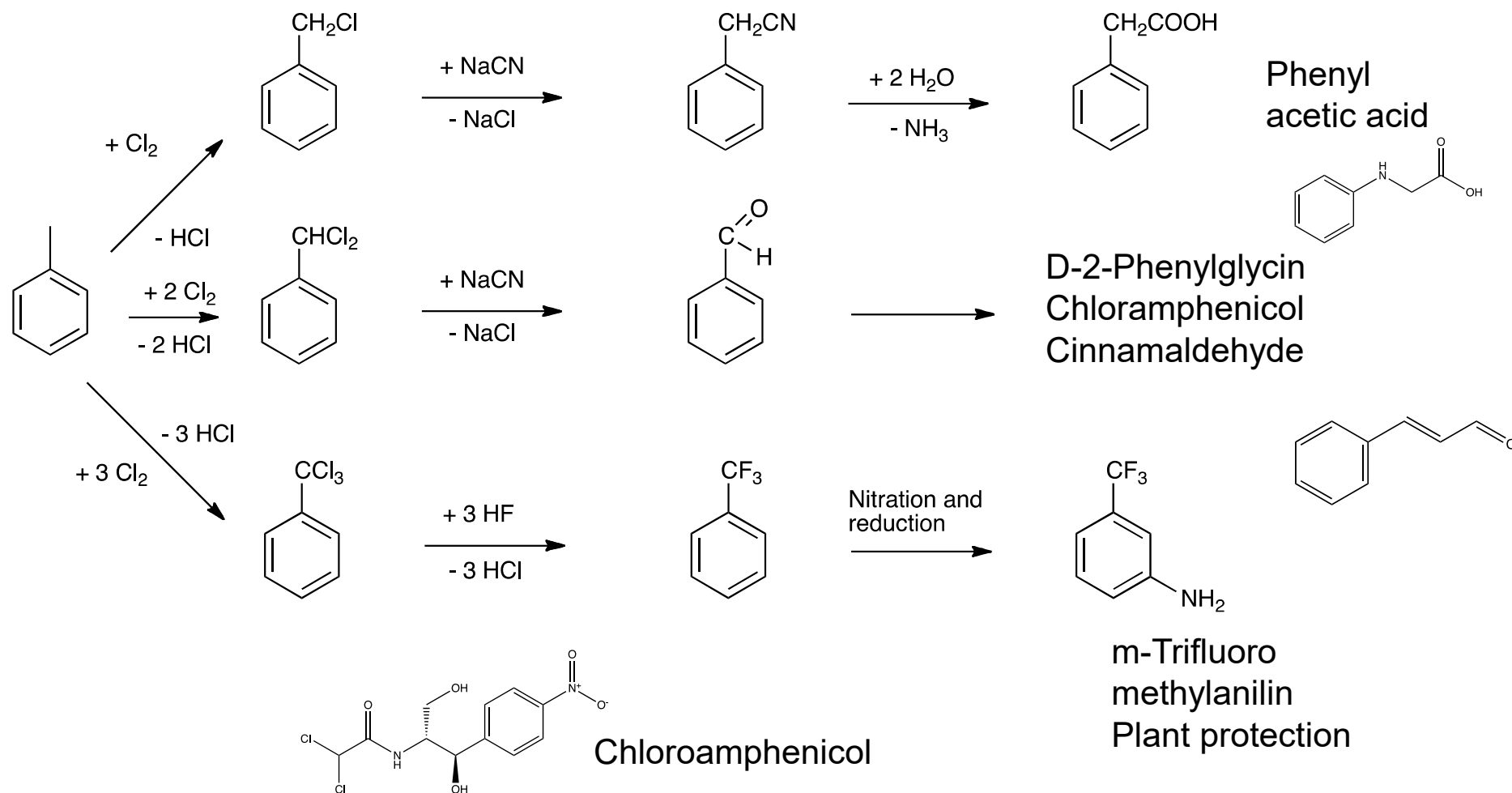
Hydrogenation of di-nitro toluene to toluyden diamine



Toluydene diisocyanate

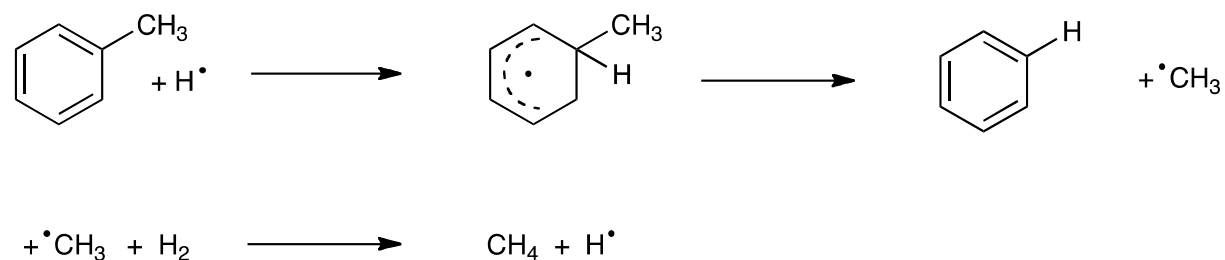


Side chain chlorination

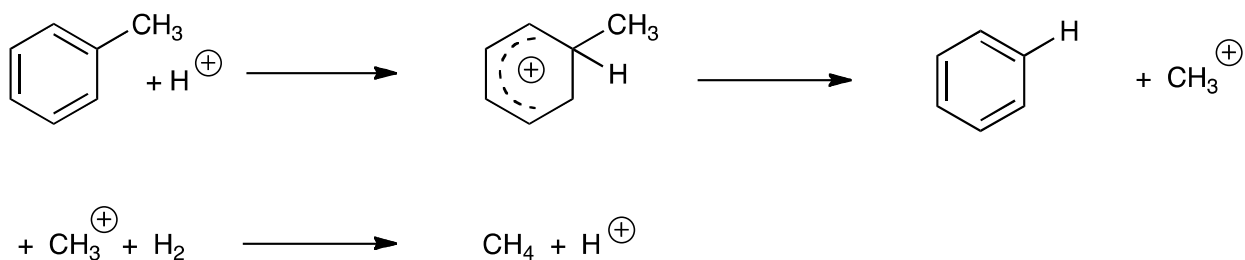


Toluene - dealkylation, disproportionation, alkylation

Thermal dealkylation (700-900°C, 10-30 bar)



Catalytic dealkylation (500-600°C, 20-40 bar)

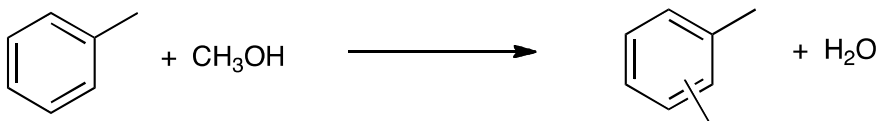


Toluene - dealkylation, disproportionation, alkylation

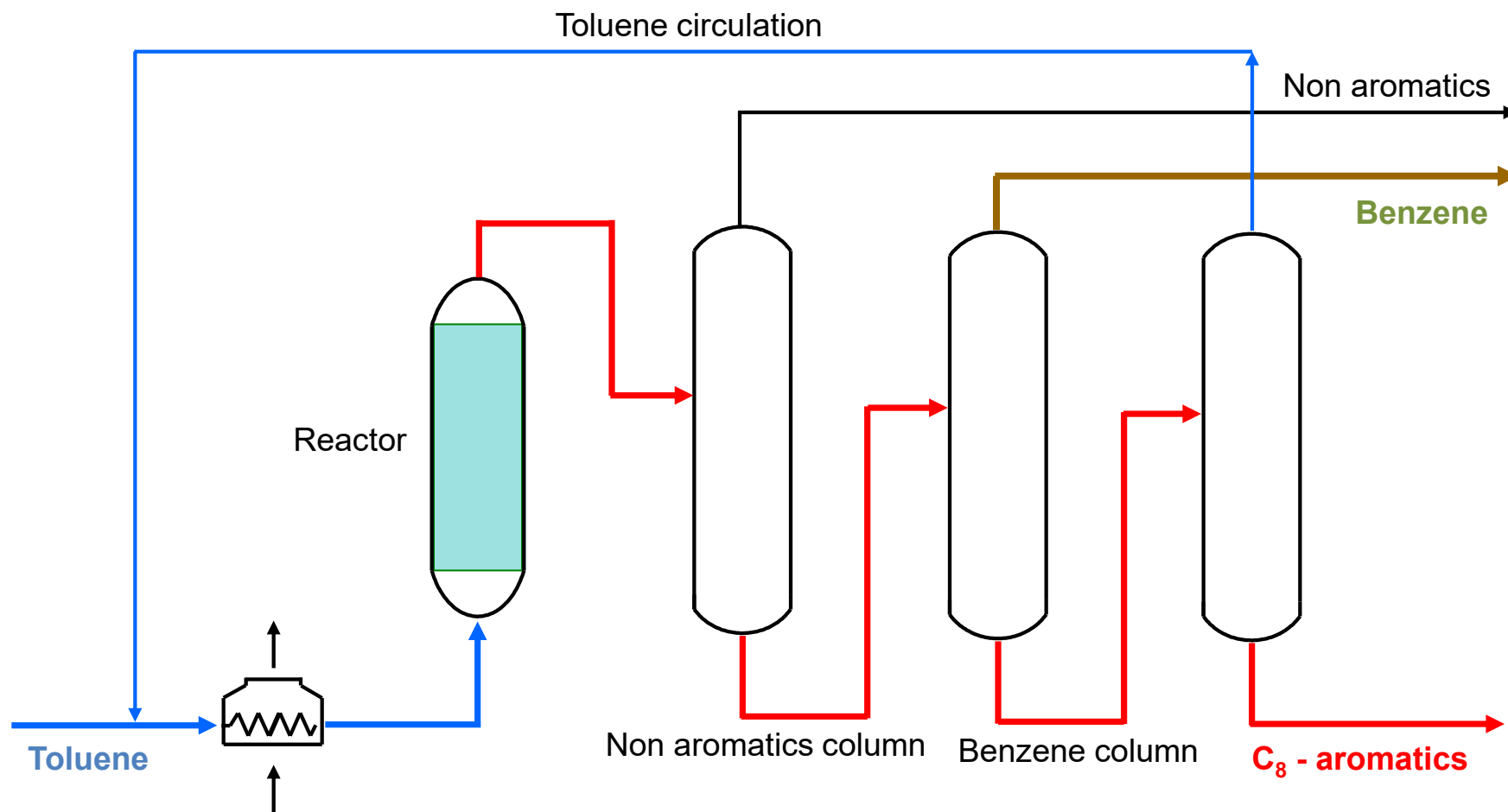
Catalytic disproportionation (400-500°C, 5-30 bar)



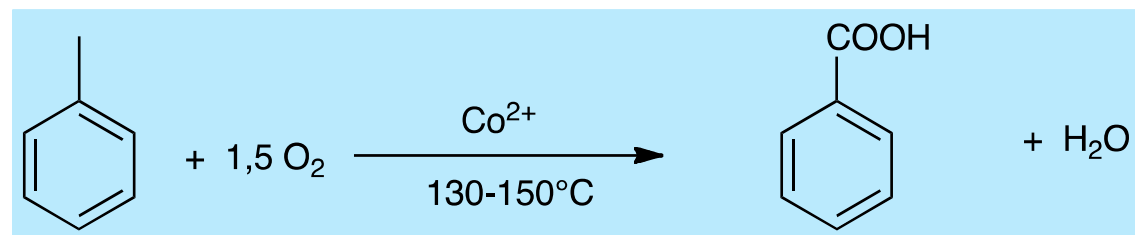
Catalytic alkylation with methanol (250-350°C, 10-43 bar)



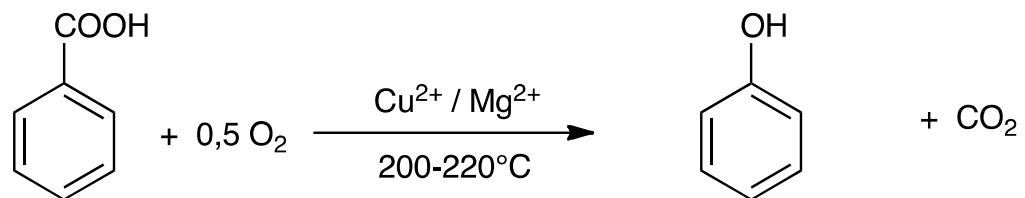
Dismutation of toluene



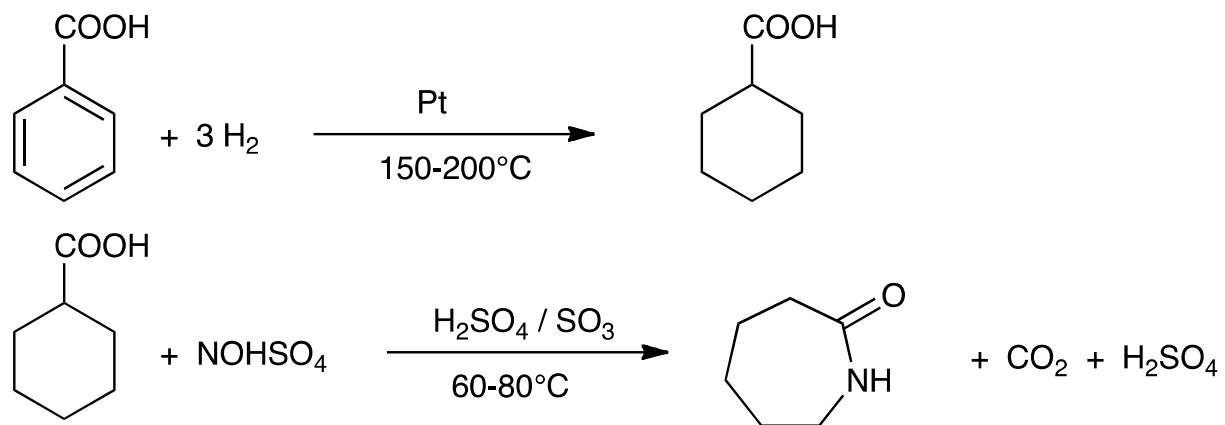
Benzoic acid



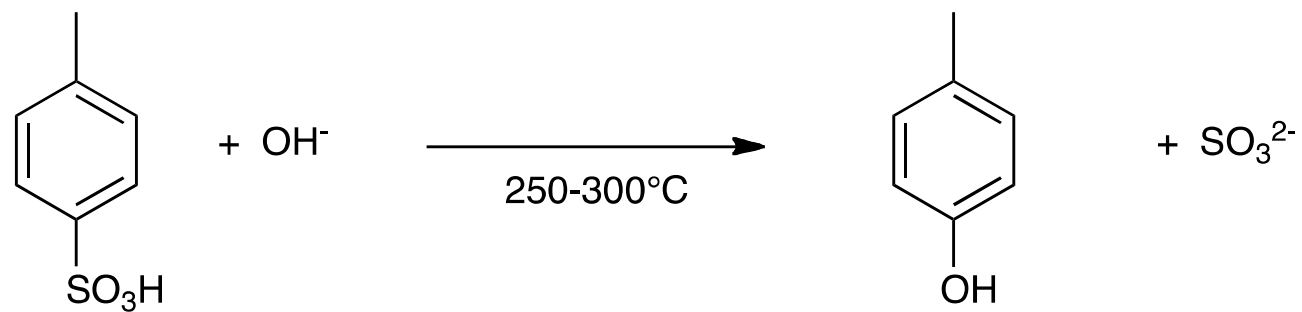
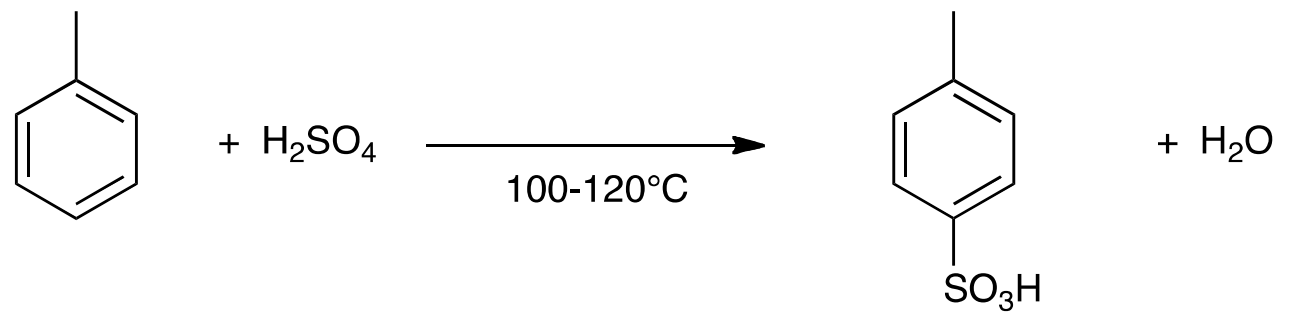
Phenol



Caprolactam



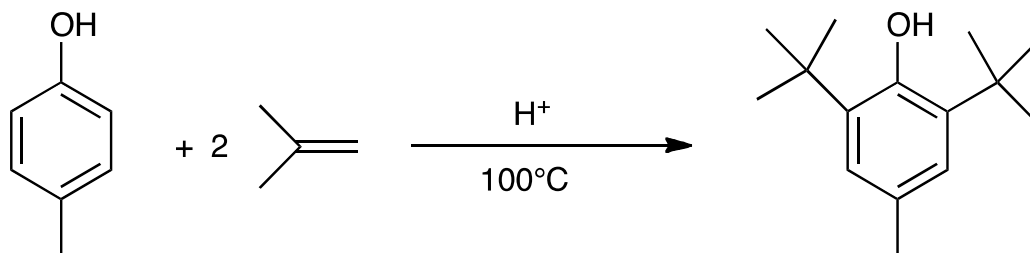
***p*-Cresol**



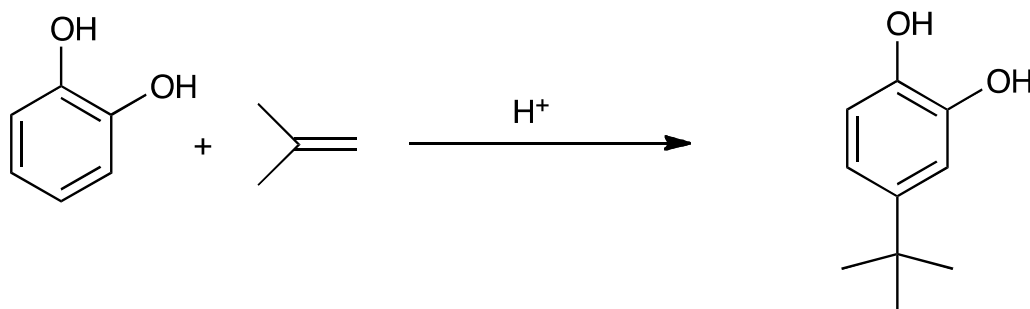
Antioxidants

Stabilization of plastics and oils

Di-tert.-butyl-*p*-cresol (BHT)



Tert.-butyl-brenzcatechine (TBC)



Irganox 1010

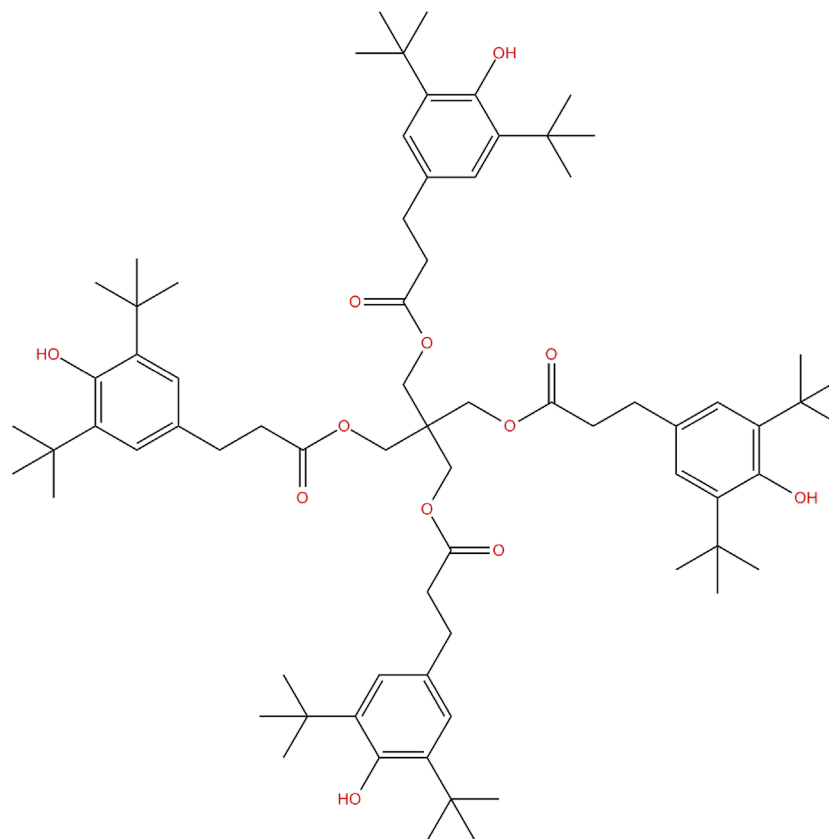
Pentaerythritol tetrakis(3-(3,5-di-tert-butyl-4 hydroxyphenyl)propionate)

Formula: $\text{C}_{73}\text{H}_{108}\text{O}_{12}$

Molar Mass: 1177,655

Most common antioxidant

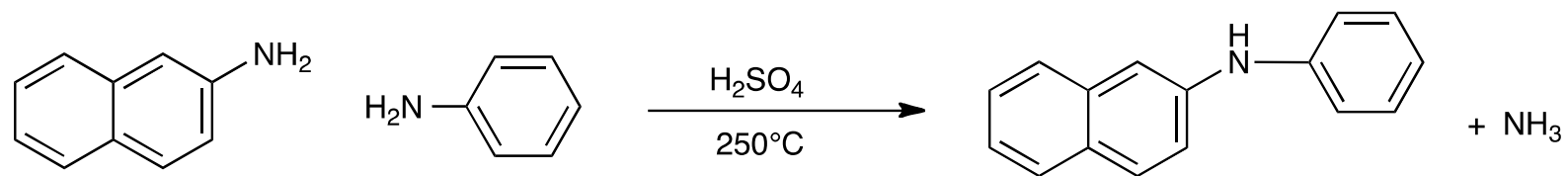
Invented by Ciby (BASF)



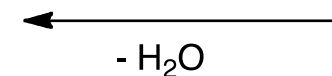
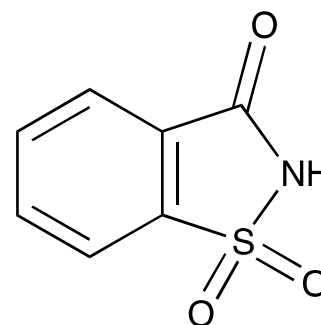
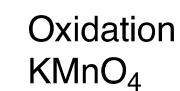
Antioxidants

Stabilization of plastics and oils

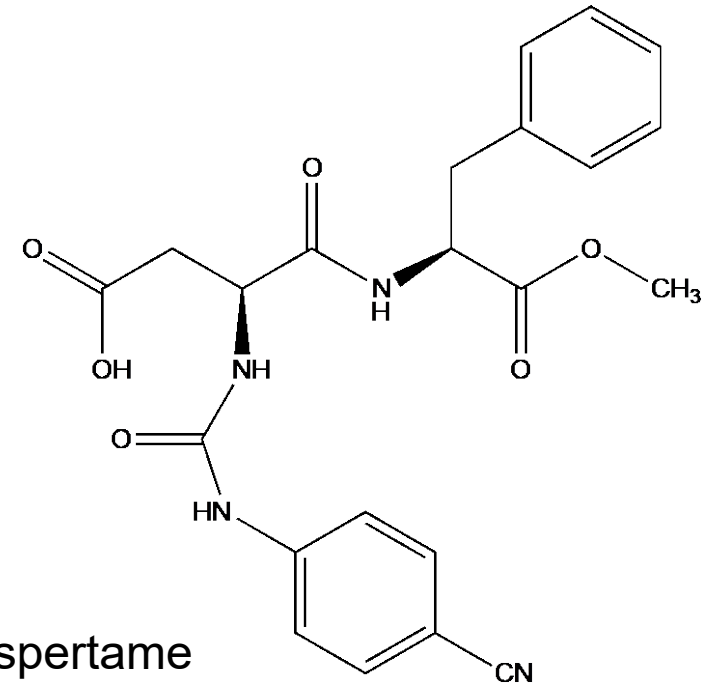
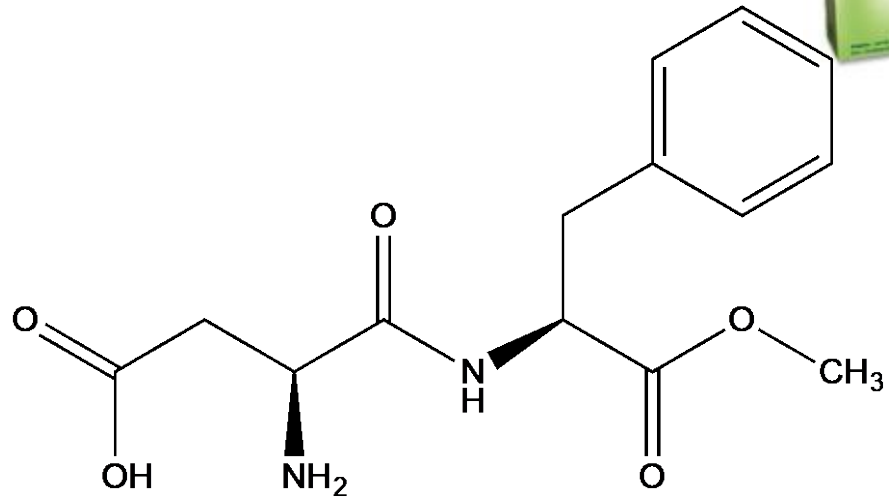
Phenyl- β -naphthylamin (PBN)



Serendipity

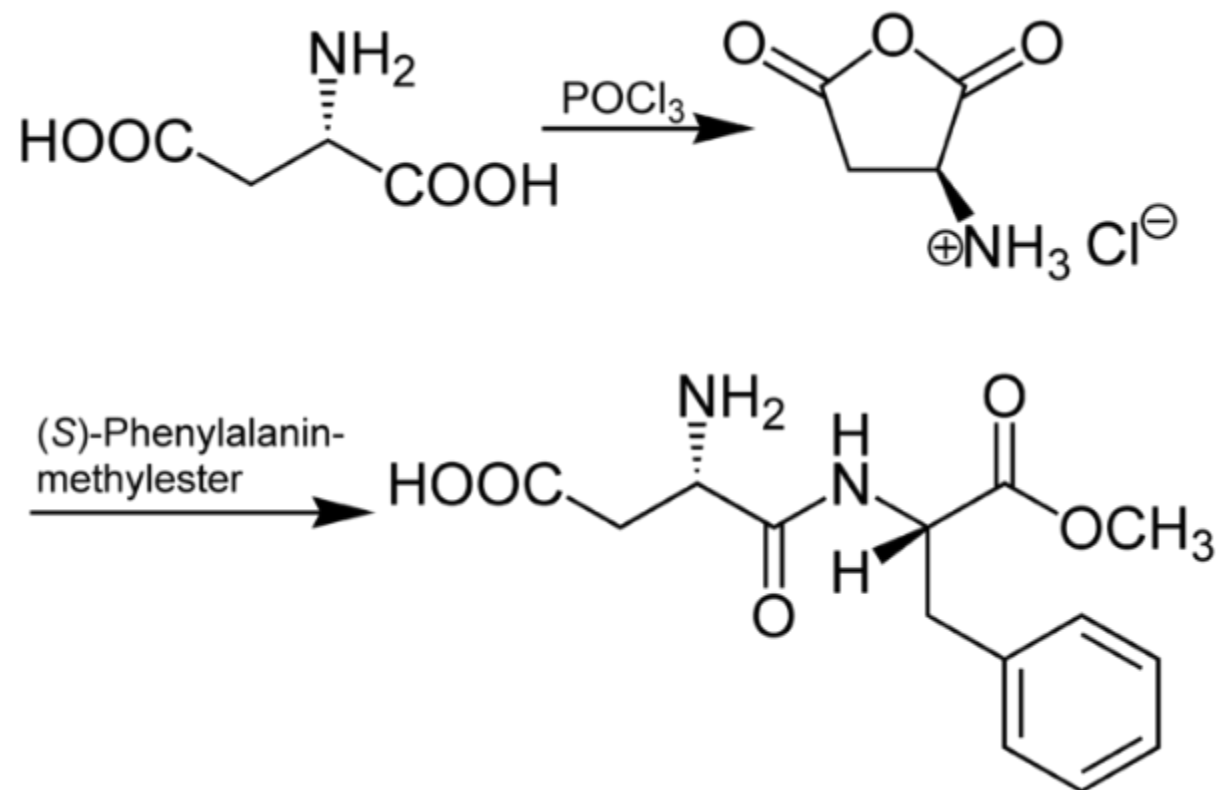


Aspartame

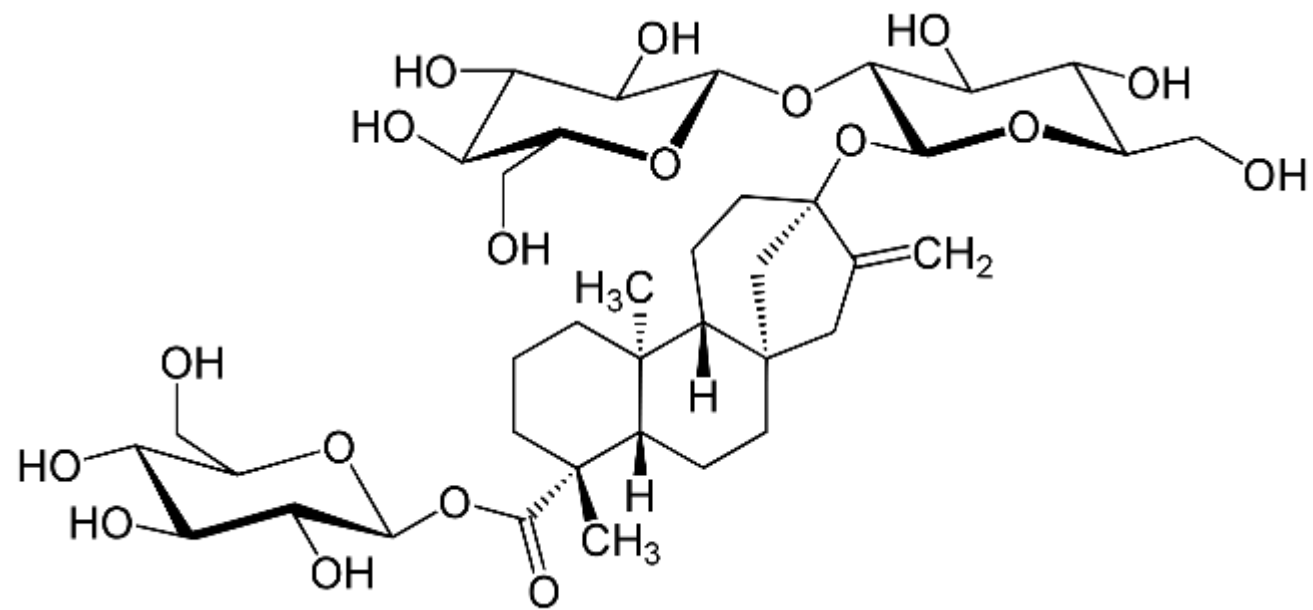


Super aspartame

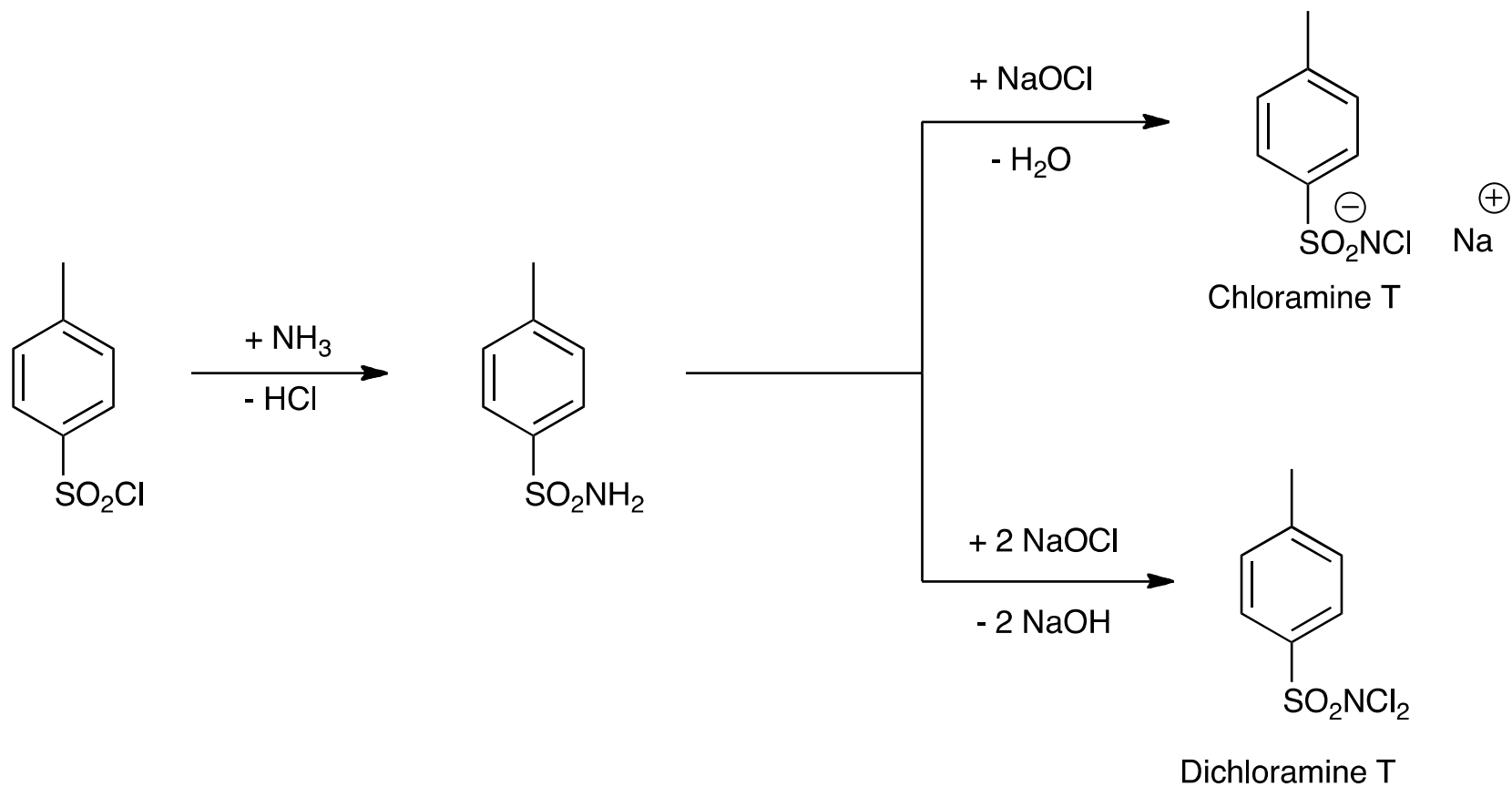
Synthesis of aspartame



Stevia



Chloramine T: synthesis



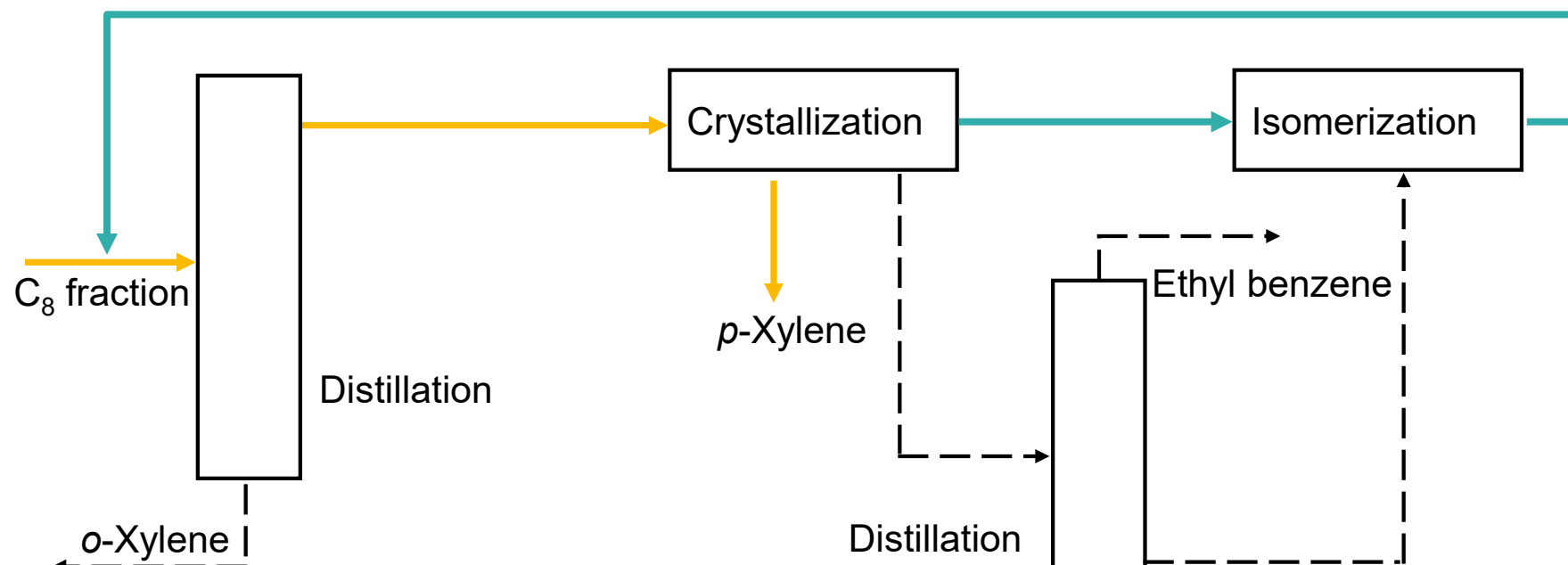
Aromatics

- Separation methods
- Benzene & benzene products
- Phenol, Cyclohexanone, ...
- Toluene
- **C8 aromatics**
- Polycyclic aromatics
- Heterocyclics

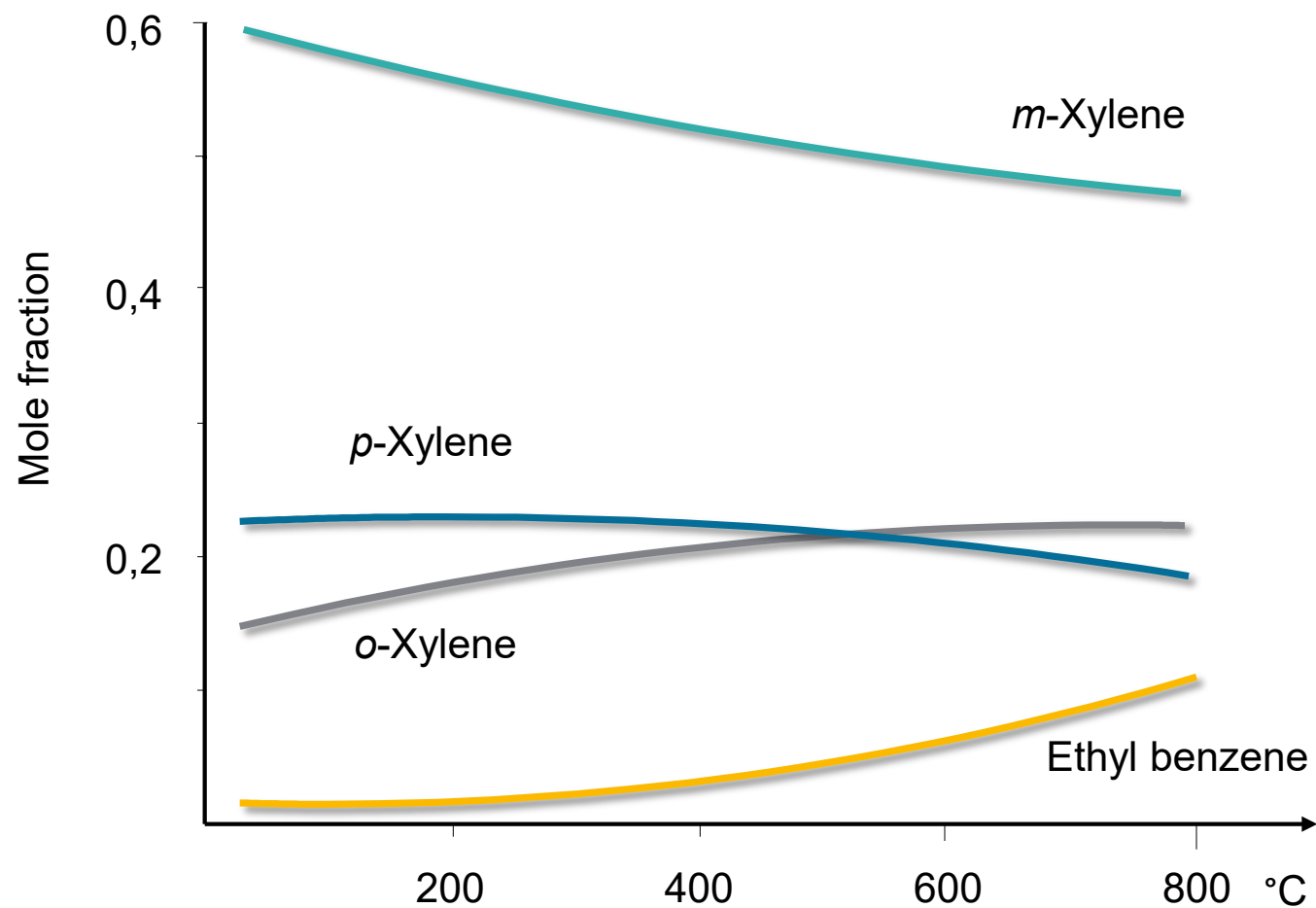
C₈ aromatics

From reforming
stripper work up

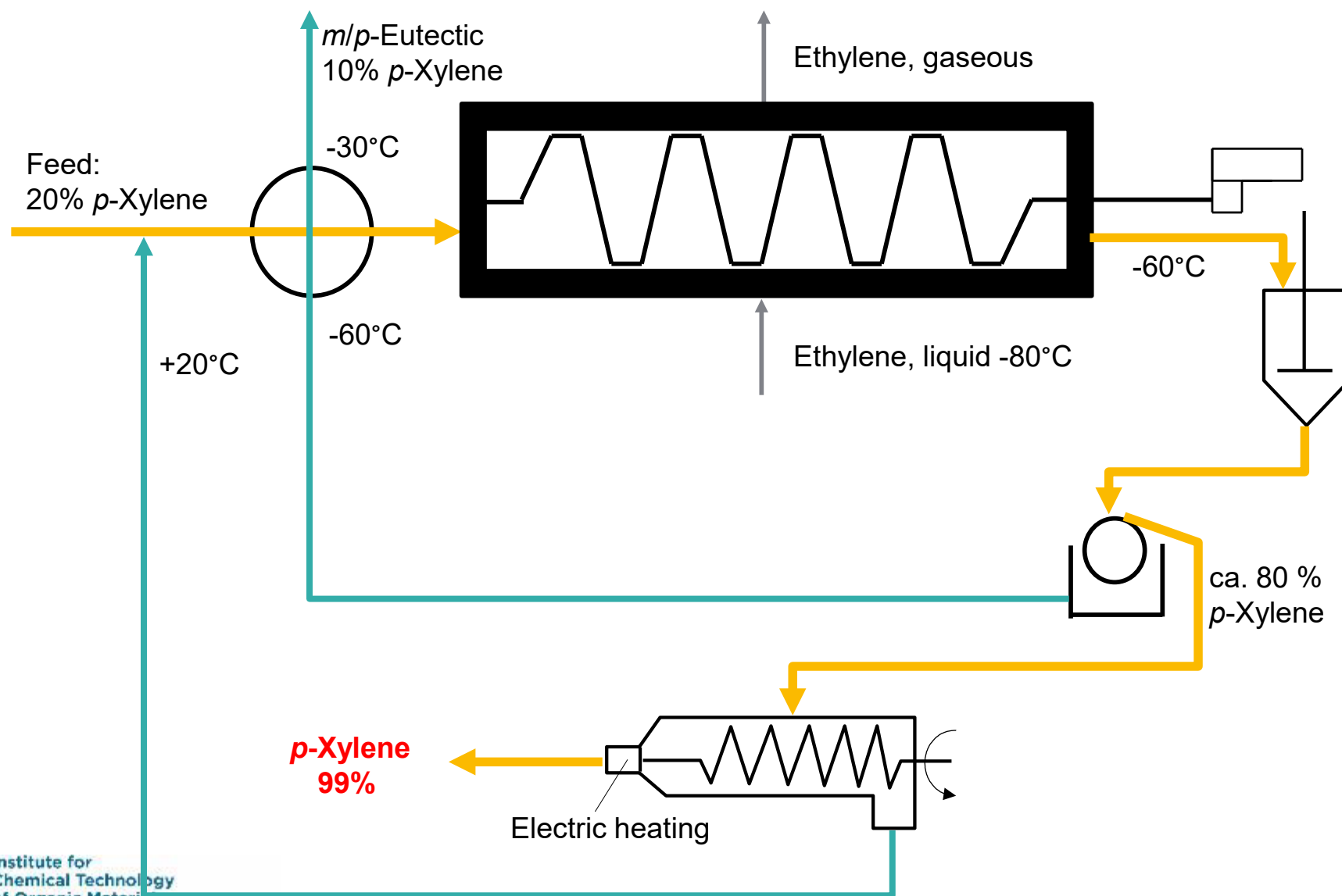
Hydrocarbon	T _b [°C]	T _c [°C]	Content [%] C ₈ -feed	Equ.
Ethyl benzene	136,2	-95,0	22	10
p-Xylene	138,3	+13,3	18	21
m-Xylene	139,1	-47,9	40	46
o-Xylene	144,4	-25,2	20	23



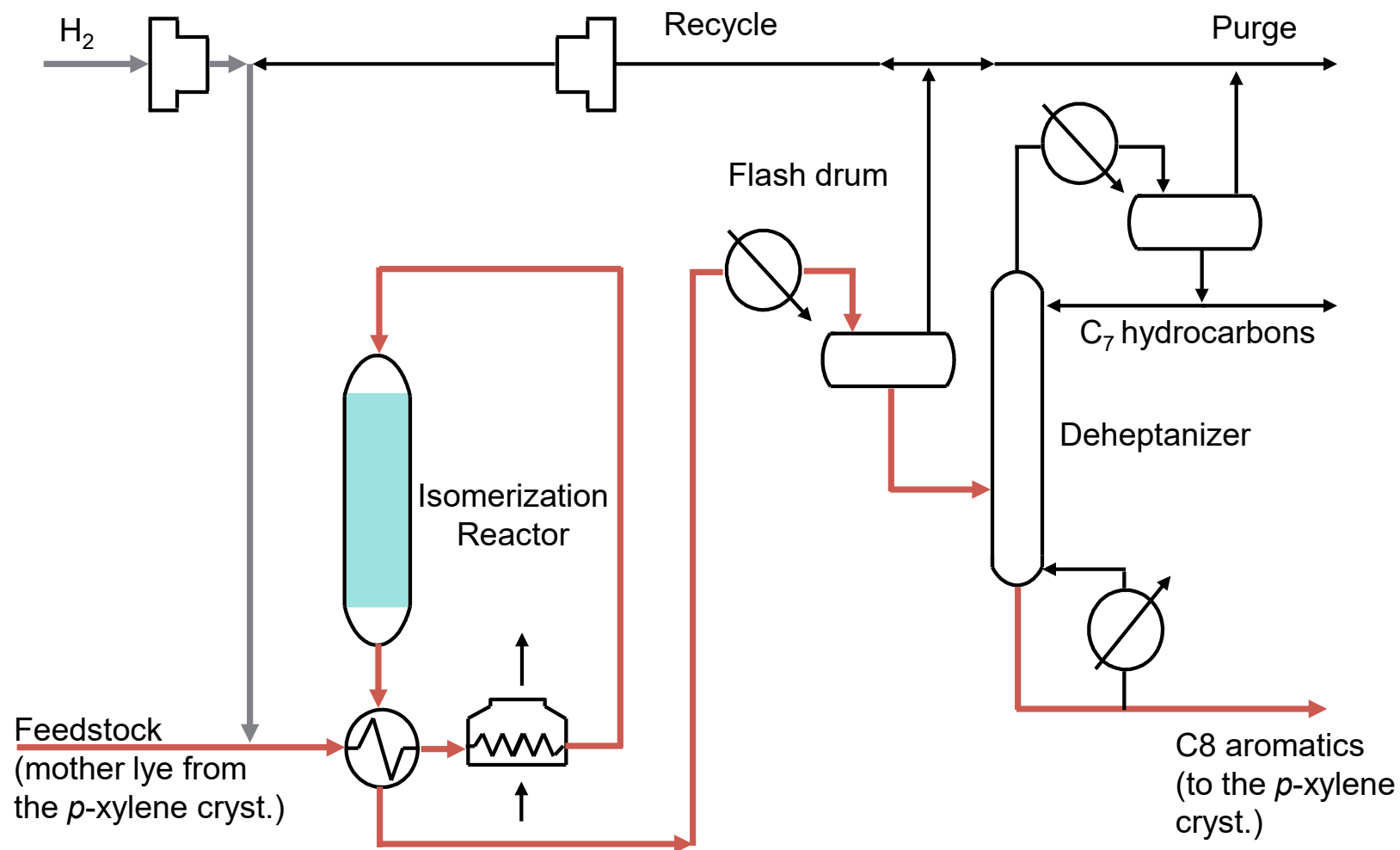
Equilibrium conc. of C₈ aromatics



p-Xylene crystallization

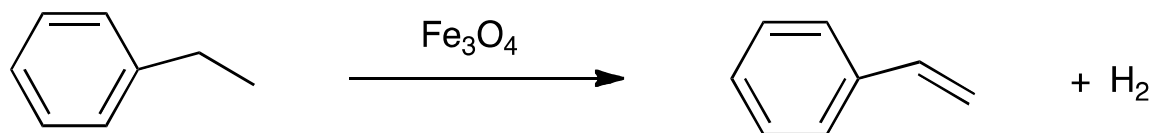


Octafining process

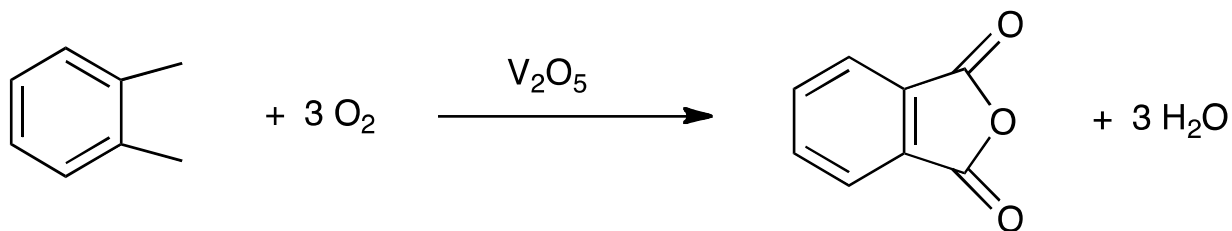


Processing C₈ aromatics

Dehydrogenation of ethyl benzene to styrene

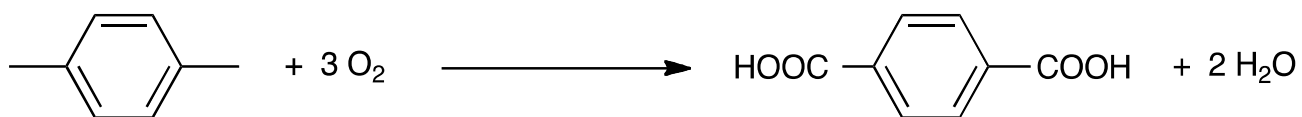


Oxidation of *o*-xylene to phthalic acid anhydride in the gas phase

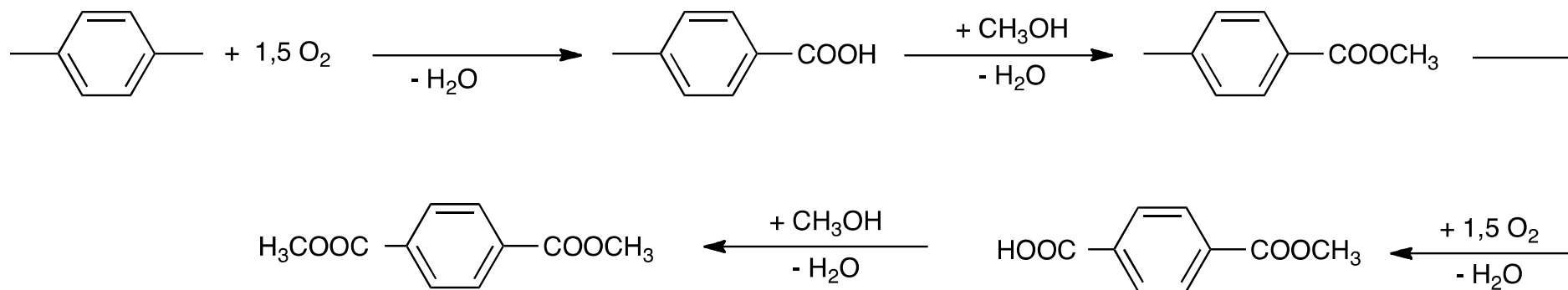


Processing C₈ aromatics

Oxidation of *p*-xylene to terephthalic acid in liquid phase



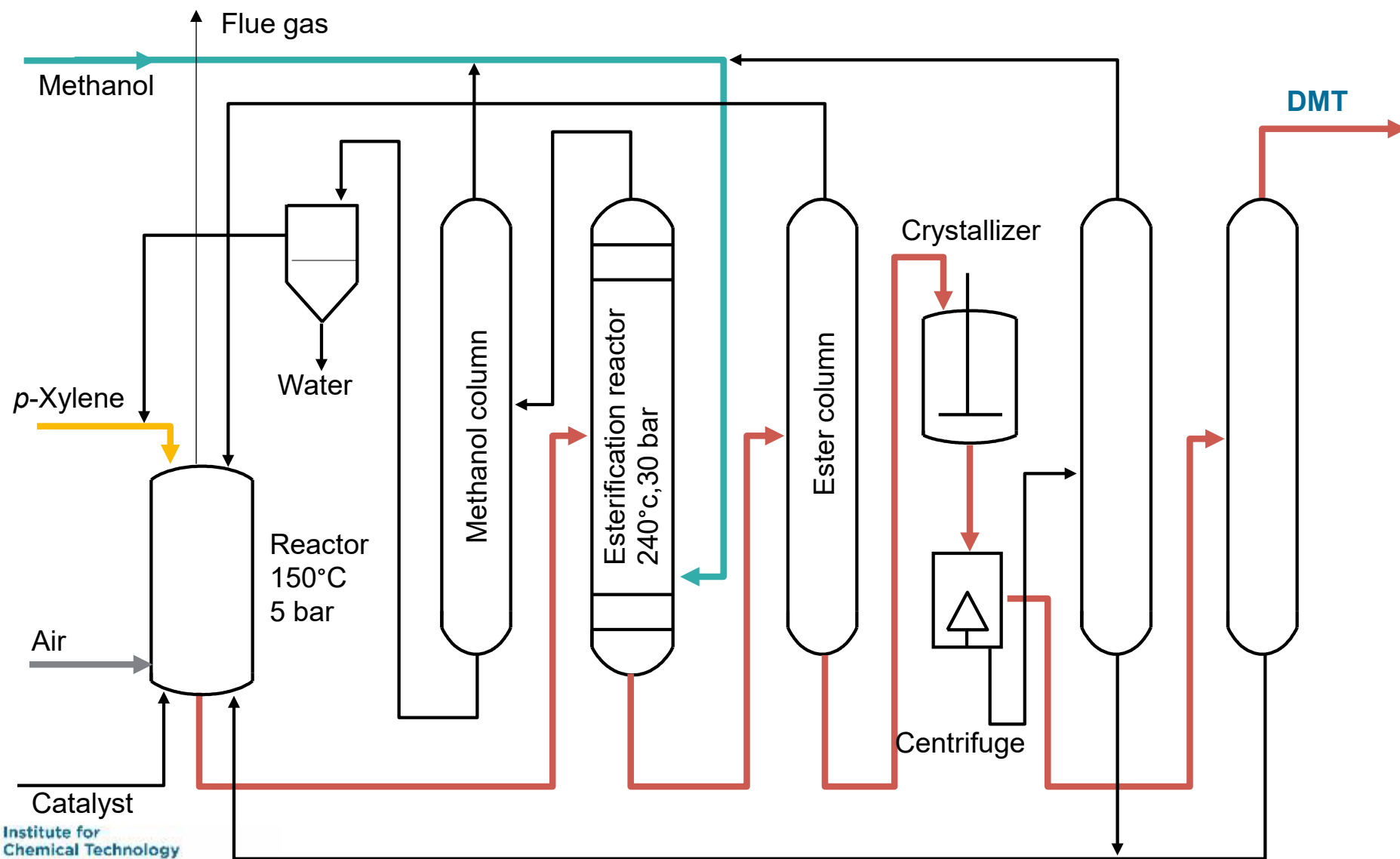
Amoco process



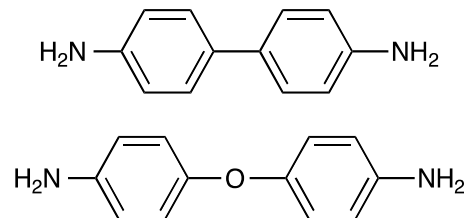
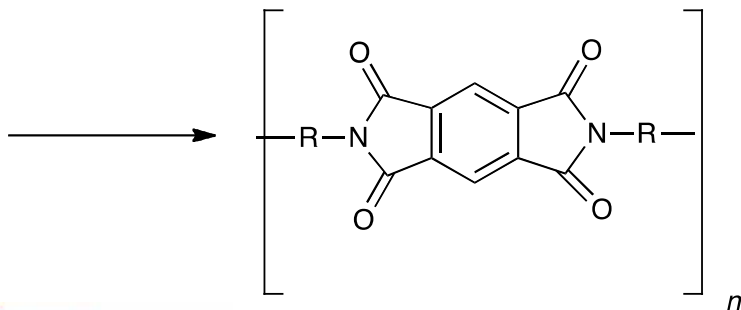
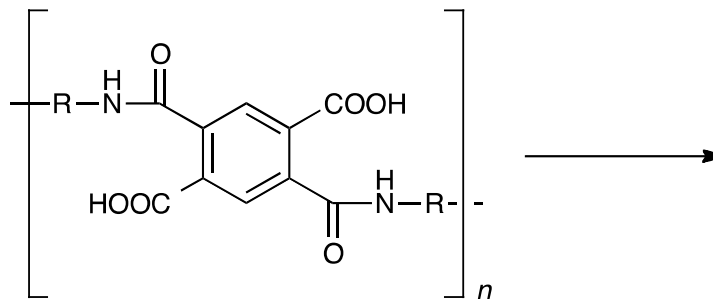
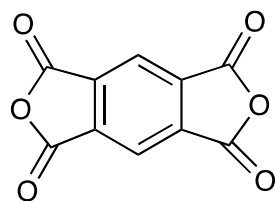
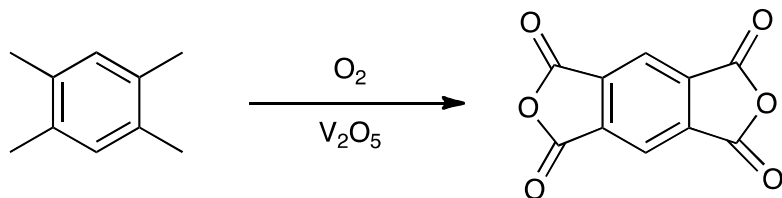
Innhausen process

2022: 72 Mio t terephthalic acid produced

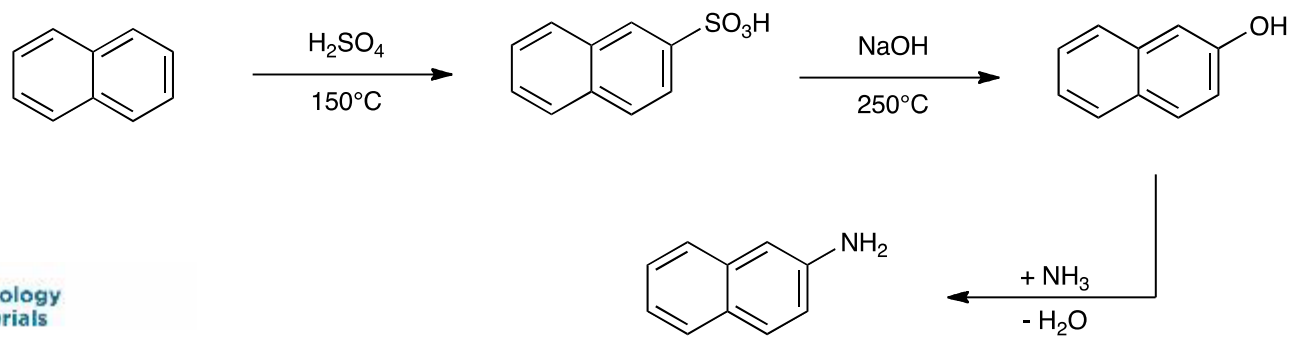
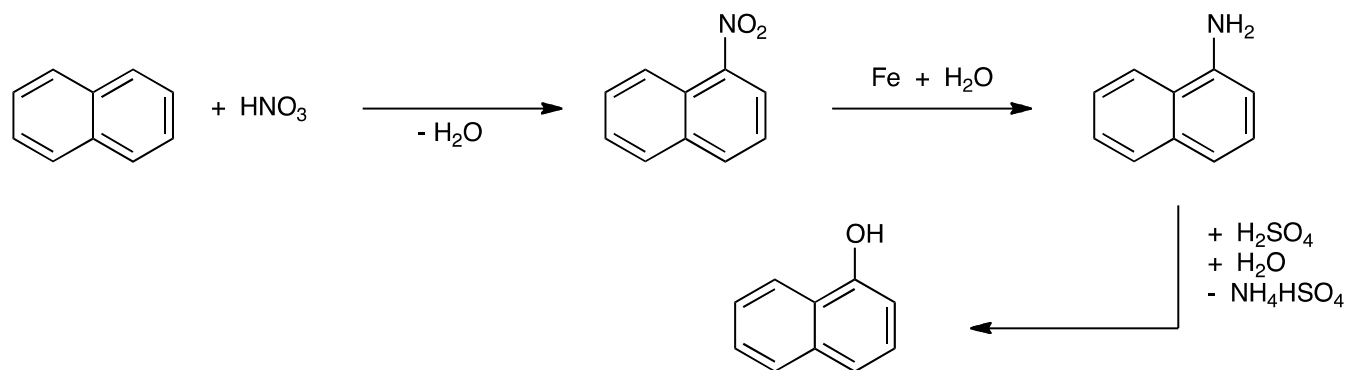
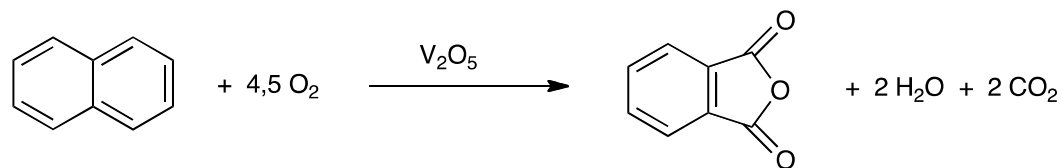
Dimethyl terephthalate



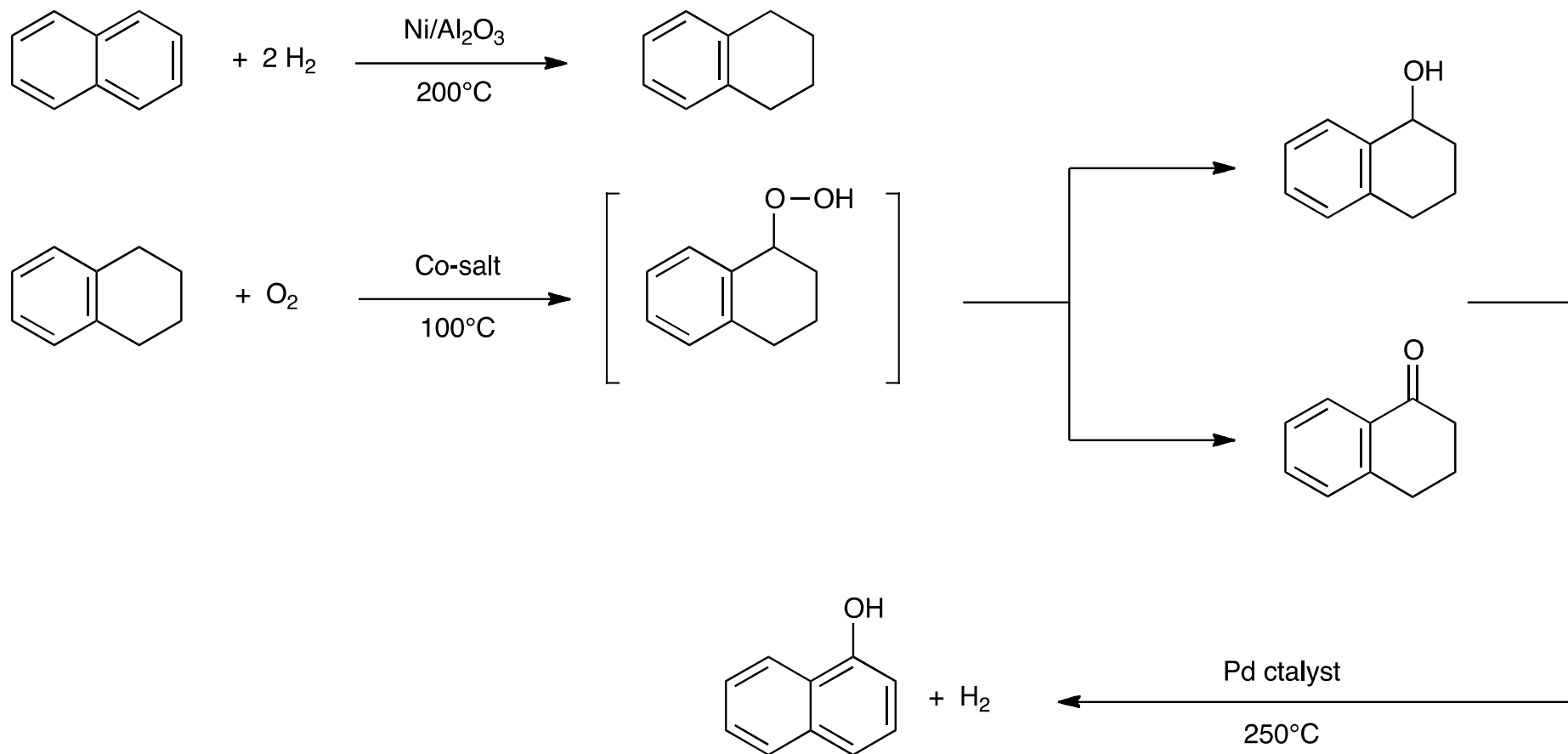
Tetramethylbenzene Durene – Polyimide



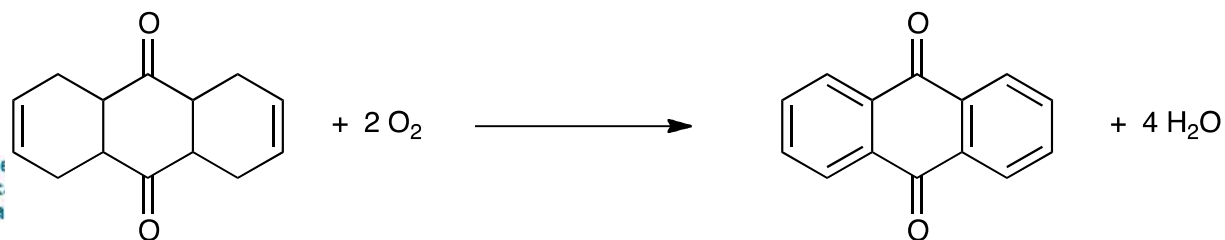
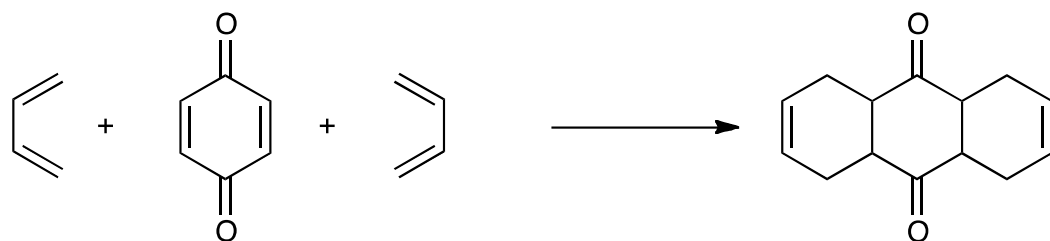
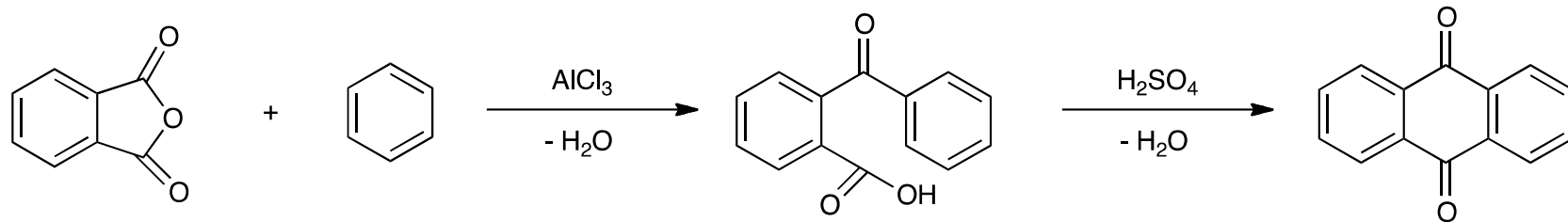
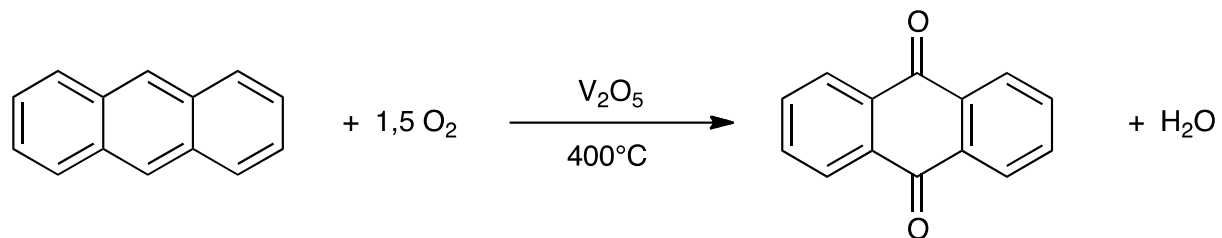
Naphthalene



α -Naphthol: oxidation of naphthalene



Anthracene - anthraquinone



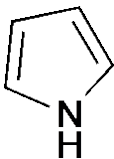
Aromatics

- Separation methods
- Benzene & benzene products
- Phenol, Cyclohexanone, ...
- Toluene
- C8 aromatics
- Polycyclic aromatics
- **Heterocyclics**

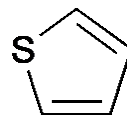
Heterocyclics



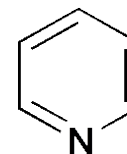
Furan



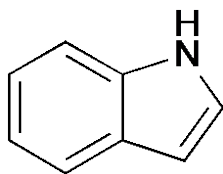
Pyrrol



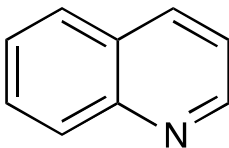
Thiophene



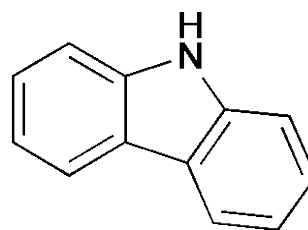
Pyridine



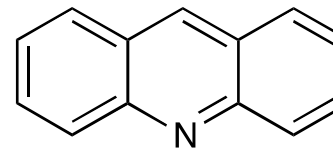
Indole



Quinoline



Carbazole



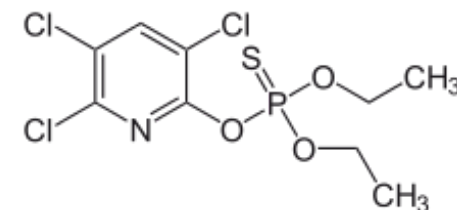
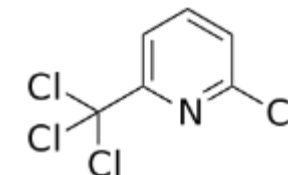
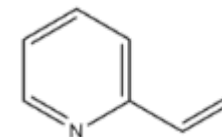
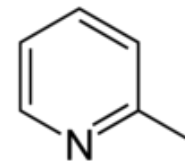
Acridine

Typical composition of bases from light oil refining of black coal tar

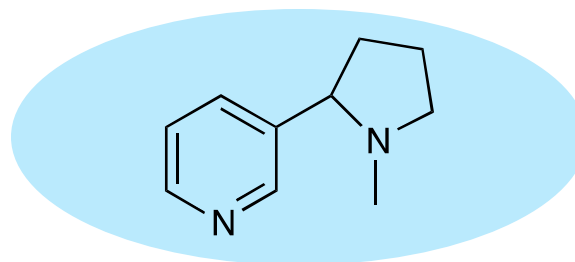
Compound	[%]	Compound	[%]
Pyridine	12	2,3,6-Collidine 2,4,6-Collidine	9
α -Picoline	10		
β -Picoline	6		
γ -Picoline	8	Aniline	12
2-Ethyl pyridine	2	o-Toluidine	4
2,3-Lutidine	2	m- and p-Toluidine	8
2,4-Lutidine	11		
2,5-Lutidine	4	higher boiling bases	6
2,6-Lutidine	6		

2-Picoline (α -Picoline)

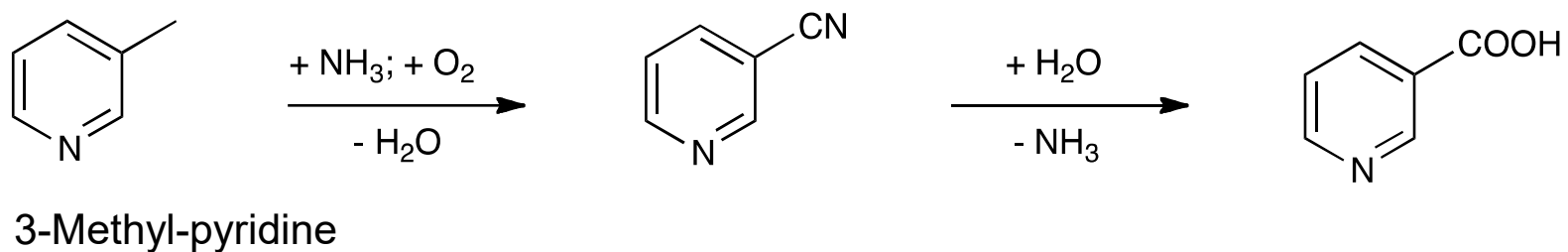
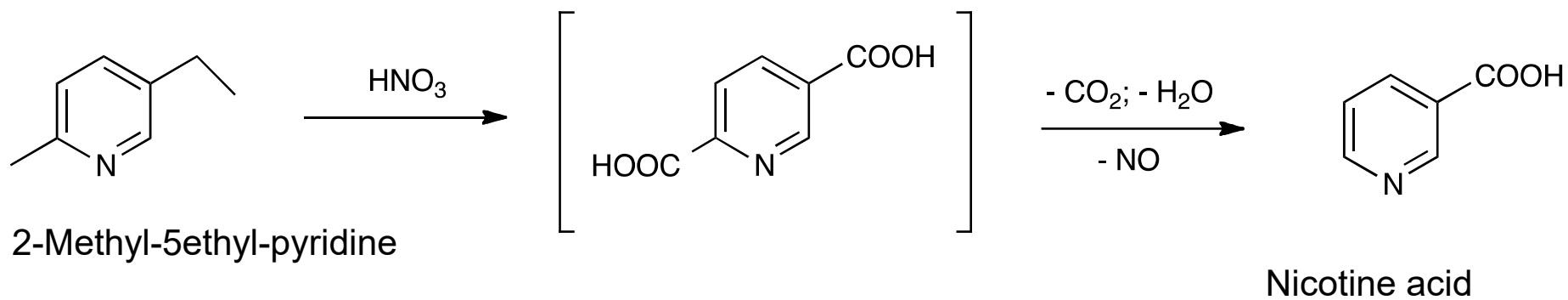
- 2-Vinylpyridine: polymer, tire cord binder – terpolymer of 2-vinylpyridine, styrene and butadiene
- Nitrapyrin: is a nitrification inhibitor in agriculture as well as a soil bactericide
Produced via photochlorination
- Chlorpyrifos: is an organophosphate pesticide used on crops, animals, and buildings to kill insects and worms

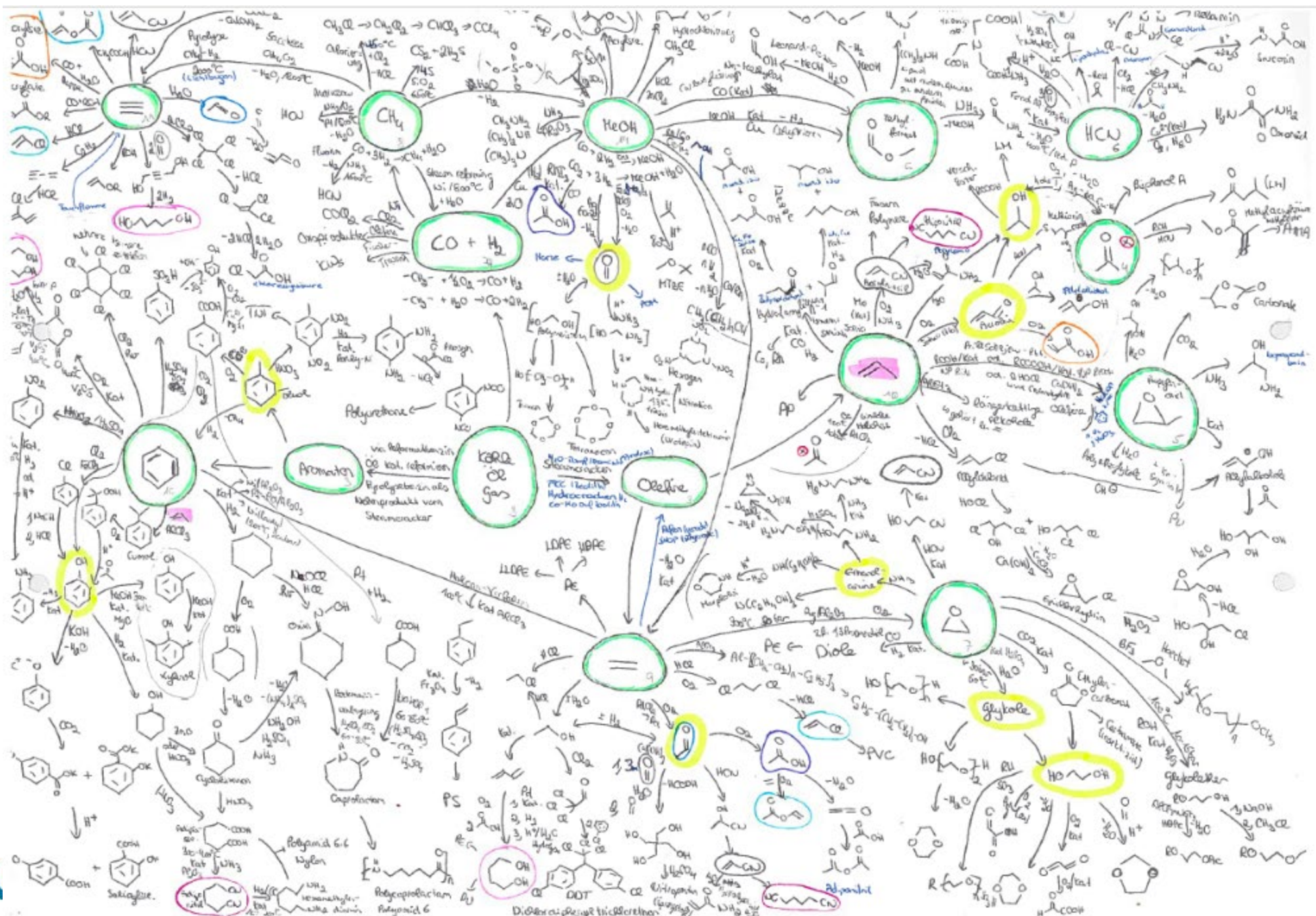


Nicotine & nicotine acid

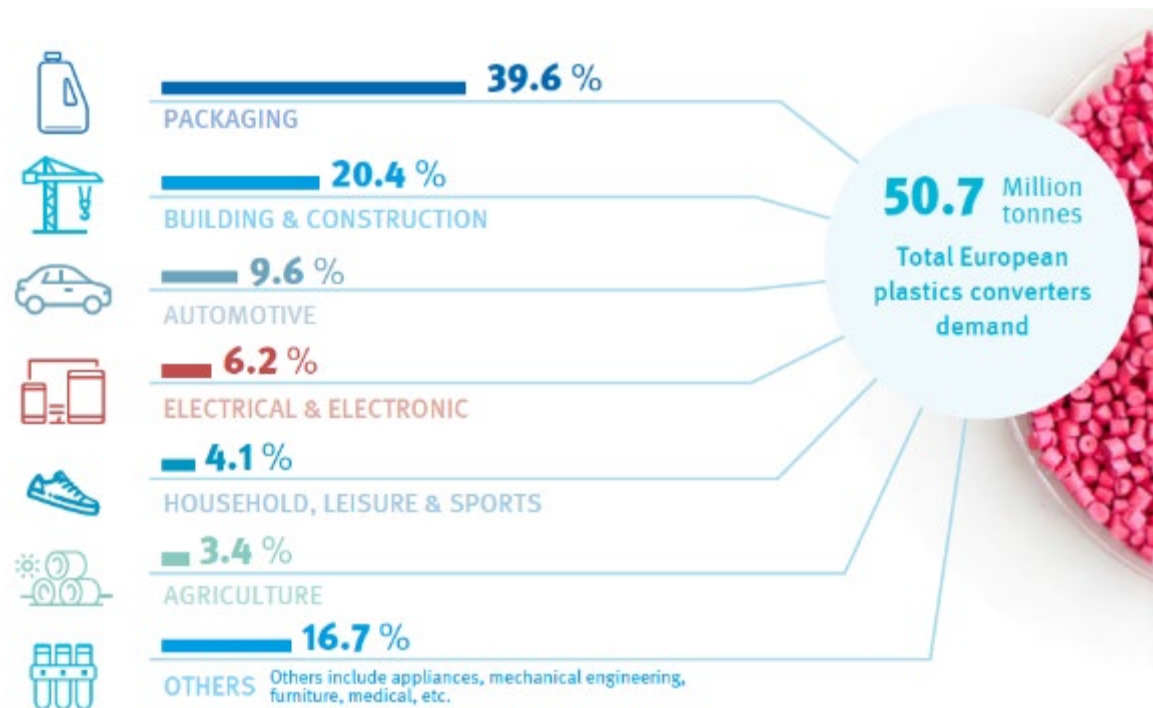


Nicotine





World plastic production is continuously growing

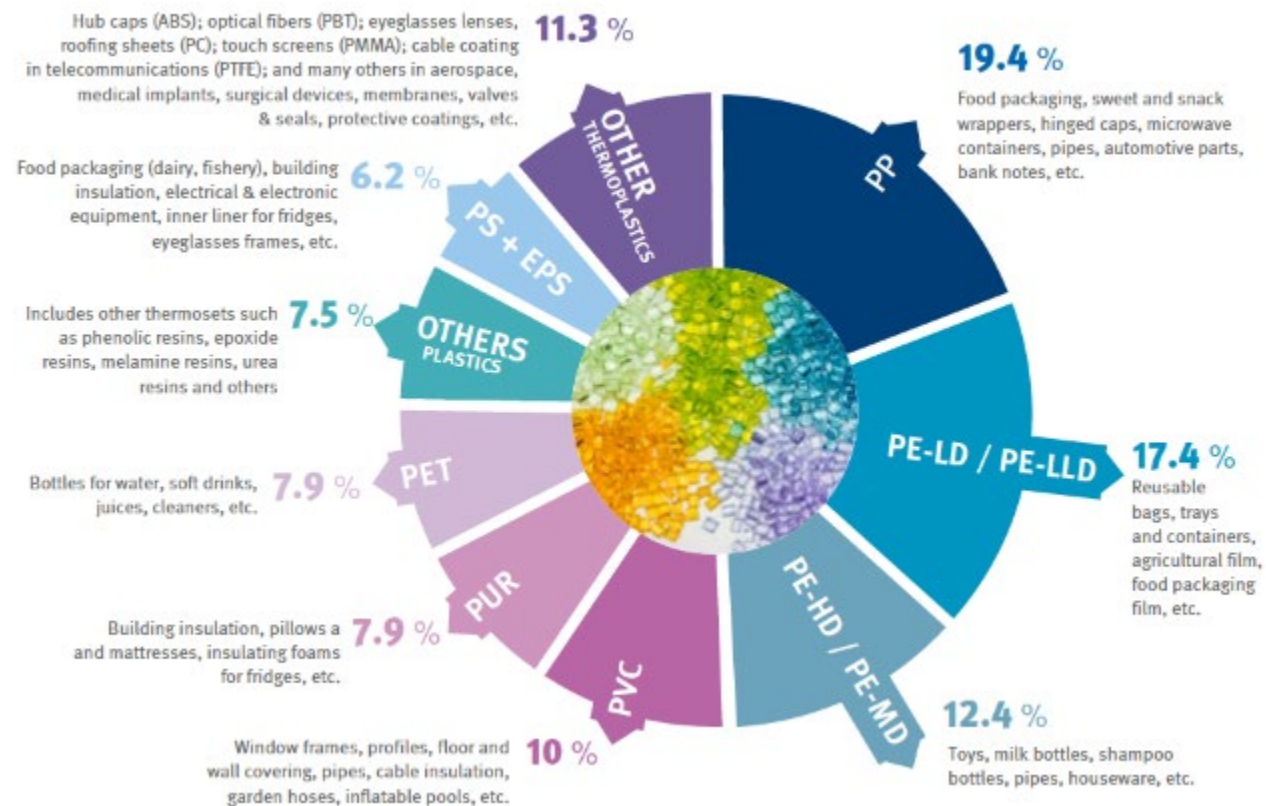


Source: PlasticsEurope 2020

A handful of polymers dominate the market

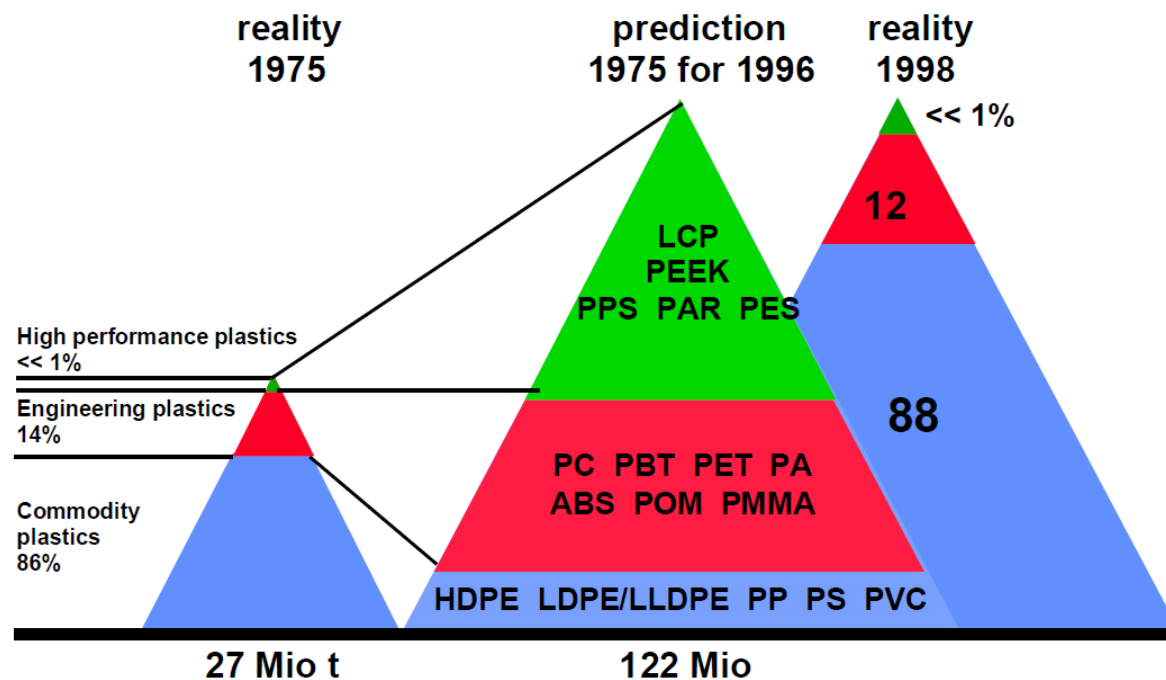
European plastic demand by resin type 2019:

Data for EU28+NO/CH.



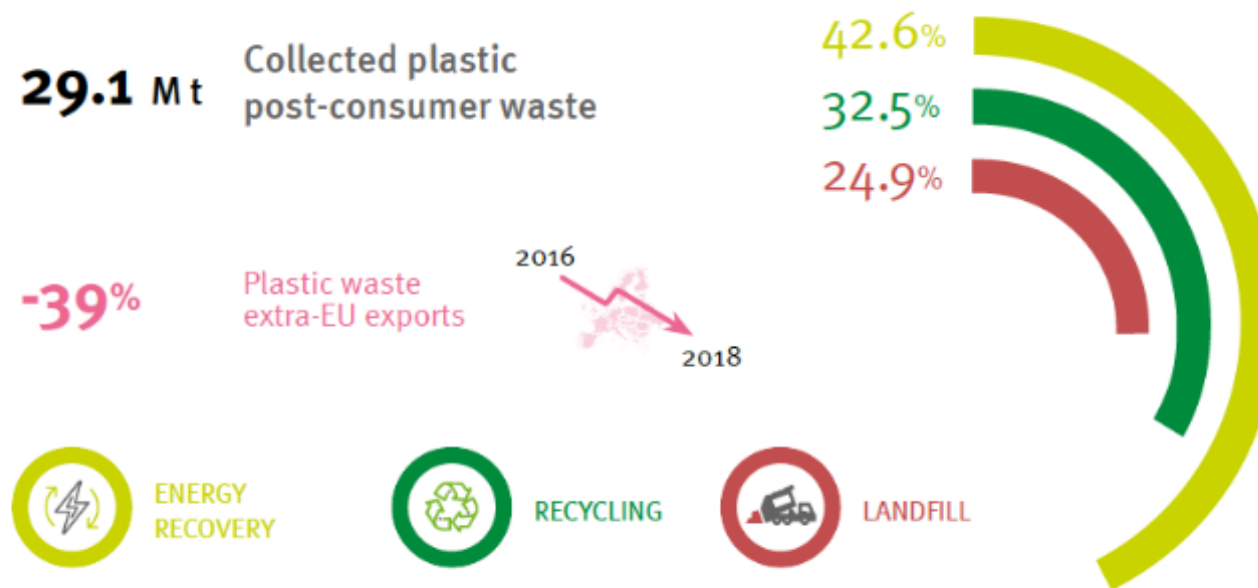
Source: PlasticsEurope 2020

Fiction and reality



Plastic Post-consumer waste treatment 2018

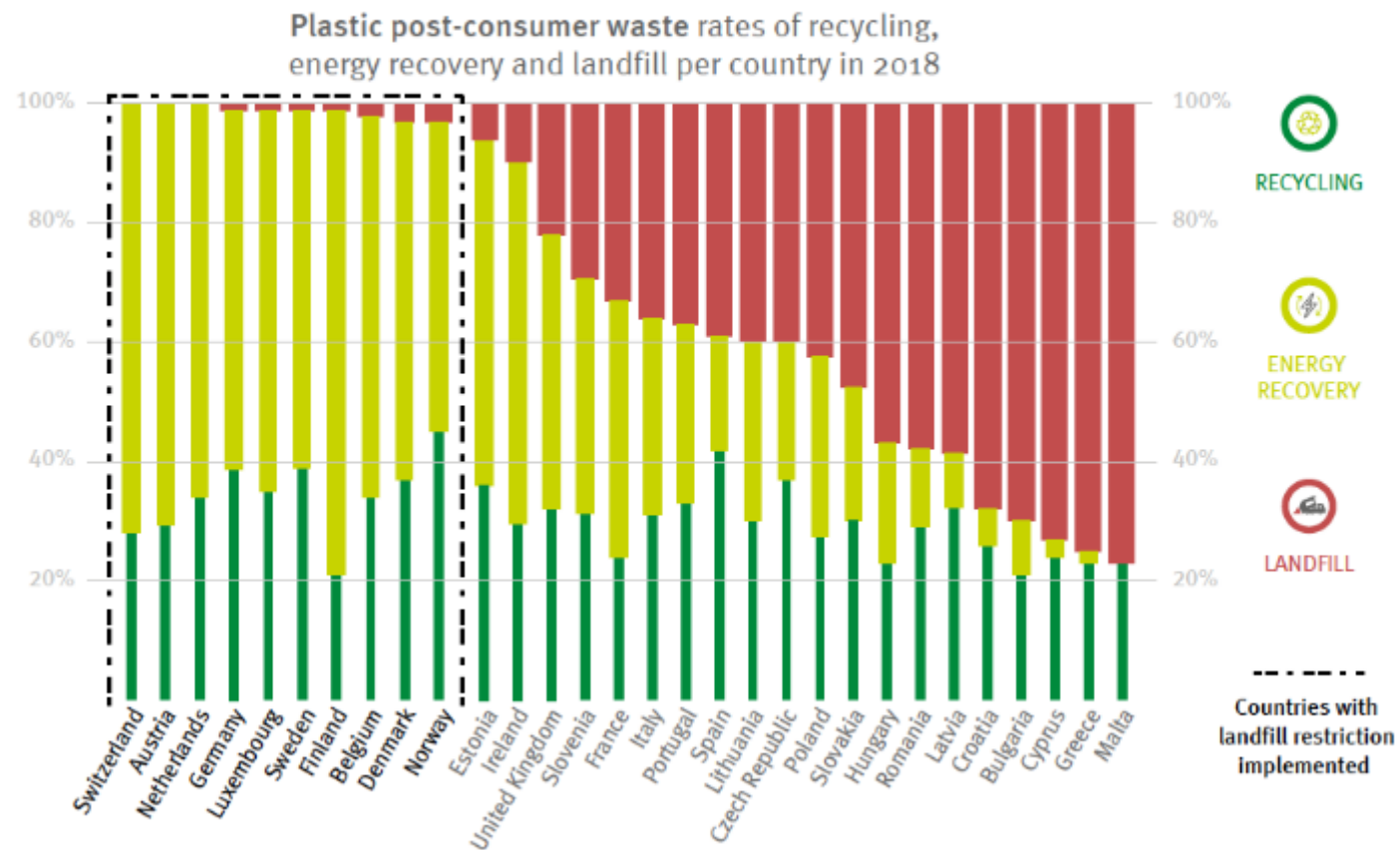
In 2018, 29.1 million tonnes of plastic waste were collected in the EU28+NO/CH in order to be treated. Plastic waste exports outside the EU have decreased by 39% from 2016 to 2018.



Source: PlasticsEurope 2020

Recycling of Plastic Waste in Europe

Countries with landfill restrictions of recyclable and recoverable waste have, on average, higher recycling rates of plastic post-consumer waste.



Source: PlasticsEurope 2020

History of polymers

16th century: artificial horn (Schobinger)

“Alchemists dreams: durable, easy to produce and process materials which can be colored as well” (casein)

1839: vulcanization of natural rubber with sulphur

Charles Goodyear

1872: J.W. Hyatt: commercial exploitation of celluloid
(replacing ivory)

History of polymers

1872: A. Baeyer, polycondensation of phenol and formaldehyde

1884: G. Eastman, photographic films based on cellulosedinitrate

1891: de Chardonnet, artificial silk

1892: artificial silk based on viscose

History of polymers

1897: W. Kricheldorf, A. Spitteler: Galalith (casein plus formaldehyde)

1907: L.H. Baekeland: Bakelite (phenol plus formaldehyde)

1920: H. Staudinger: organic polymers

Nobel prize 1953 – long chains not aggregates

H.F. Mark: X-ray diffraction of polymers

1927: O. Röhm: Polymethacrylate (organic glass)

1930: IG-Farben: starts radical polymerization of styrene (1936 start of production)

1931: DuPont: polymerization of chloroprene (Dupren - Neoprene)

History of polymers

1933: ICI discovers LDPE: “Polyethylene is good for inert laboratory beakers and very little else”

1935: W.H. Carothers: Polyamide 6,6 (Nylon)

1937: O. Bayer – Polyurethane

1935: Start of PVC production (IG Farben)

1937: Production of LDPE at ICI starts (later BASF)

1938: P. Schlack – Polyamide 6 (Perlon)

History of polymers

1938: R.J. Plunkett - PTFE

1940: E. Müller, E.G. Rochow: polycondensation of chloro silanes – Silicons

1941: Bayer – Polyacrylonitrile fibers (PAN), Dralon

1941: ICI – PET fibers

1947: DuPont – Polyformaldehyde

1953: H. Schnell (Bayer): PC

History of polymers

1953: K. Ziegler – low pressure PE

1954: G. Natta – i-PP

1960: First PP plant started up in Austria

1963: Nobel prize for Ziegler & Natta

1955: Development of PU foams

1956: Bayer – production of PC (Makrolon)

1960: First production of LLDPE by DuPont (Canada)

1965: Union Carbide – Polysulfon (PSU)

History of polymers

1967: ICI develops Poly-4-methylpenten

1971: Philips Petroleum – Polyphenylenesulfide

1975: W. Kaminsky – metallocene chemistry

1982: General Electric – Polyetherimide

2000: Nobel prize for A. Heeger, A. Mac Dermid, H. Shirakawa for their works on electrical conductive polymers

Classification of polymers

- Chemical structure
- Synthesis
- Physical properties
- Use
- Origin

Classification of polymers

1. Synthesis

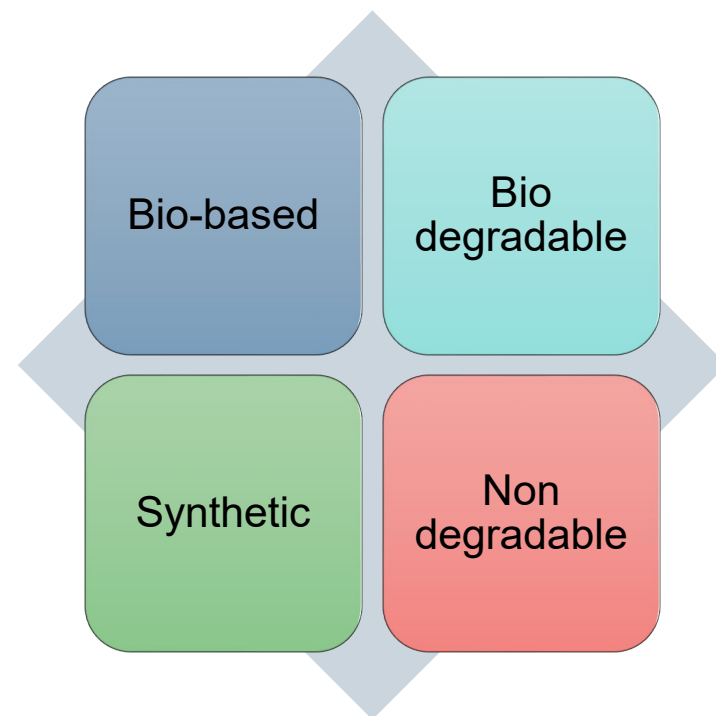
- Polymerization
- Poly condensation
- Poly addition
- Chemical modified polymers

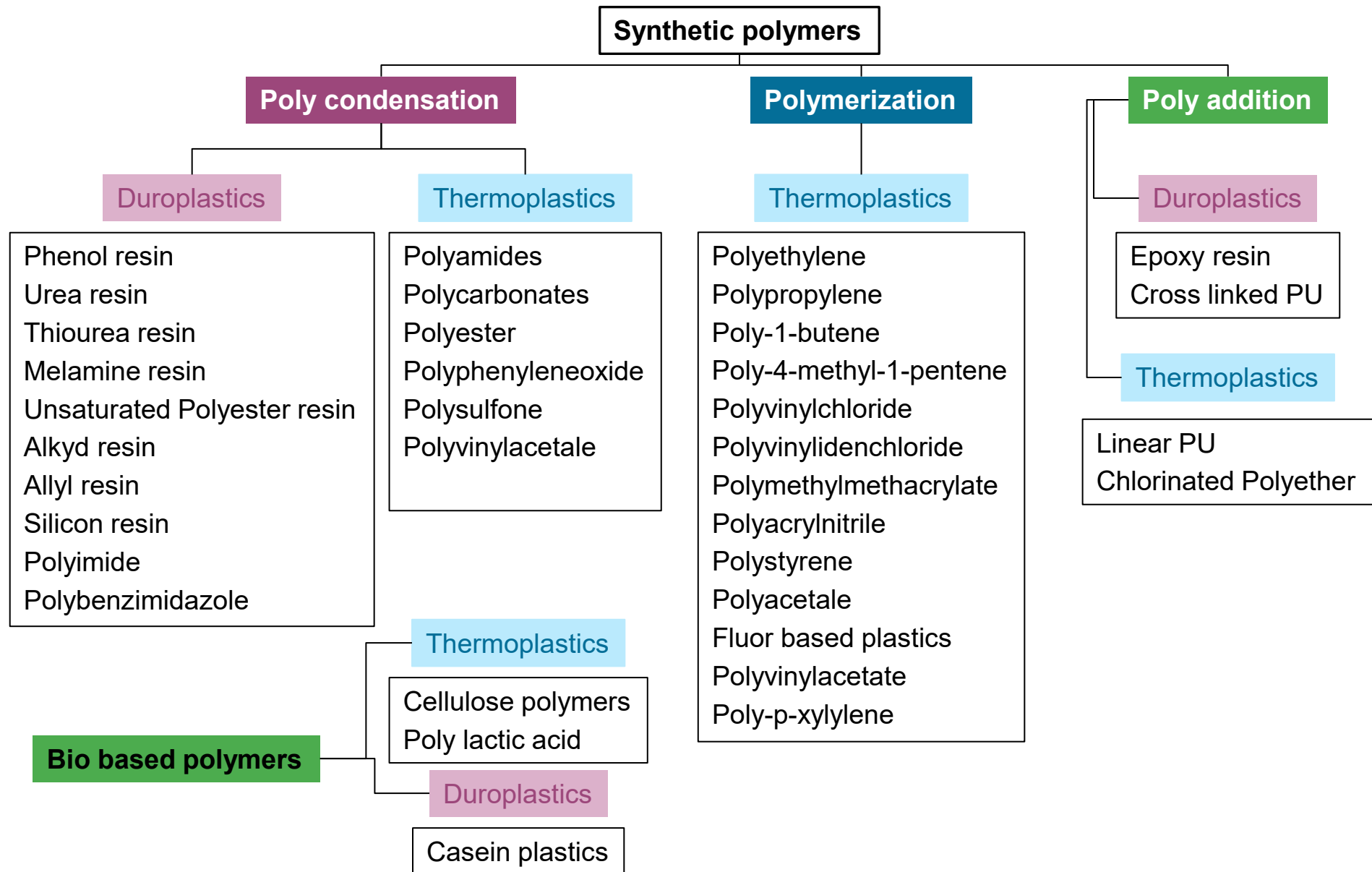
2. Molecular structure

- Polyolefins
- Vinyl polymers
- Polyester
- Polyamides
- Polyurethanes
- Cellulose ether
- Etc.

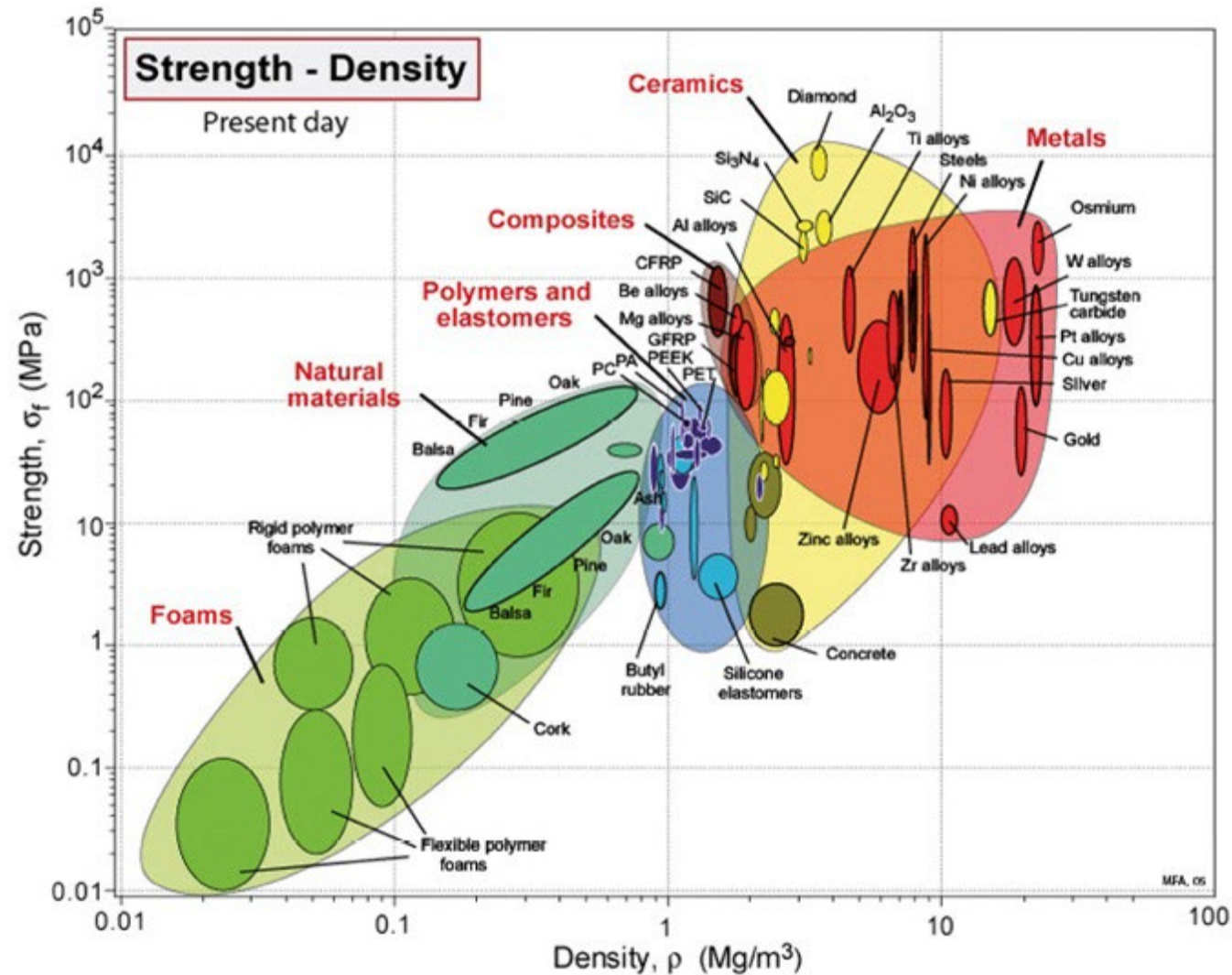
3. Use and processing

- Construction
- Fibers
- Elastomers
- Foams
- Etc.



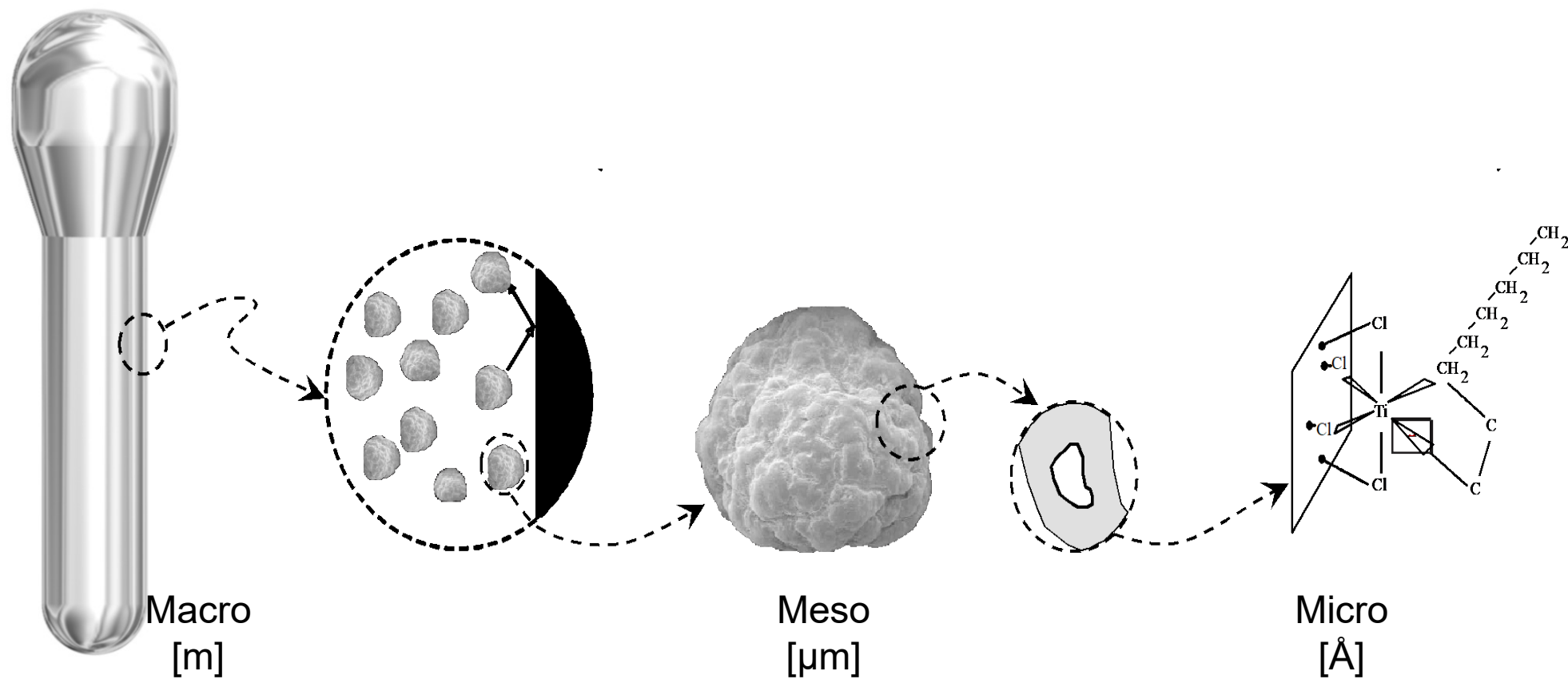


Excellent performance ratio



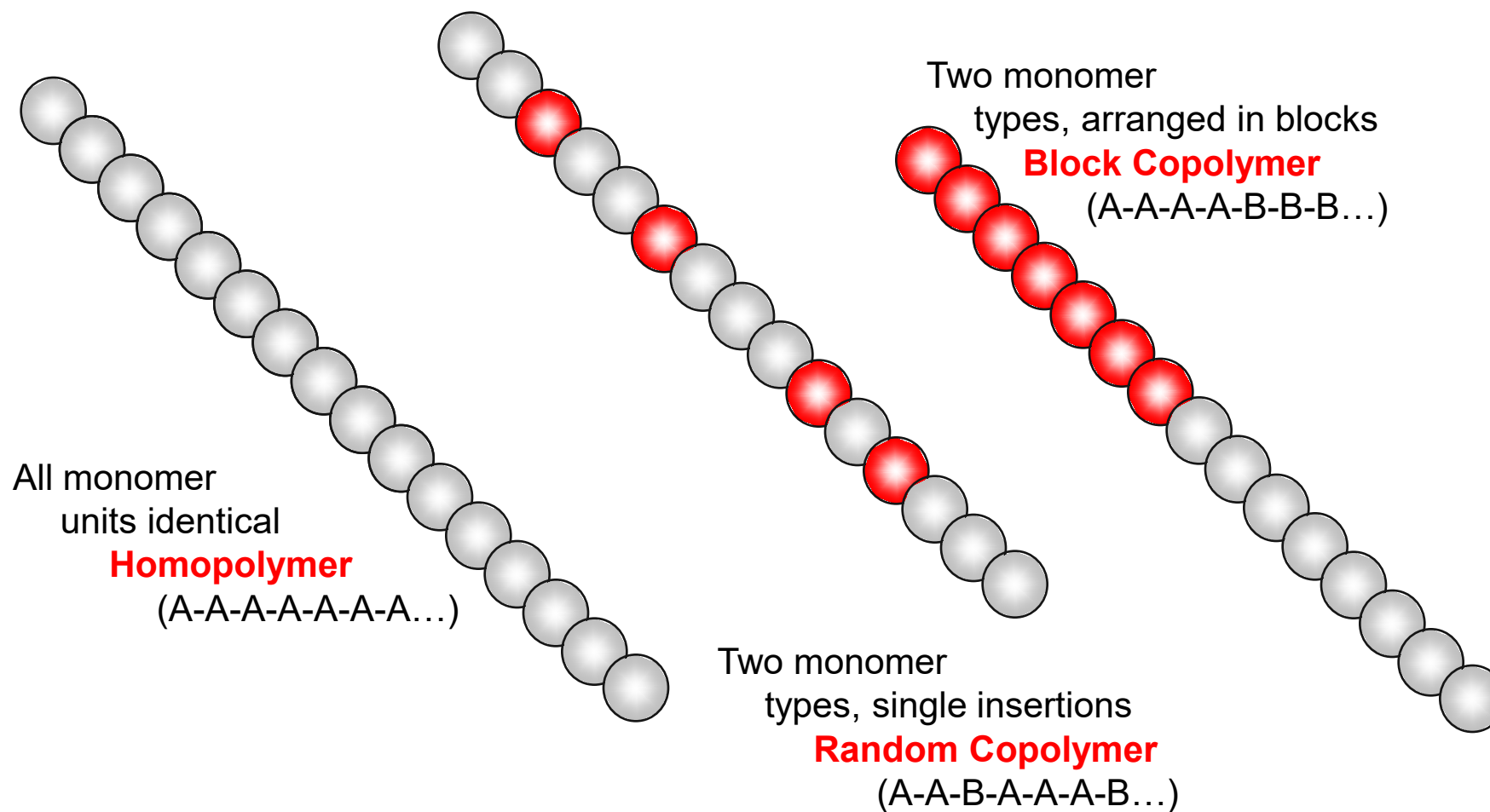
Plus	Minus
Specific weight	Ageing
Tailoring of properties	Mechanical properties esp. at elevated temperatures
Long lasting	Flammability
Electrical isolation	Recycling Repair
Easy processing	Thermal expansion Post crystallization
Coloration	

Scale in polymer research

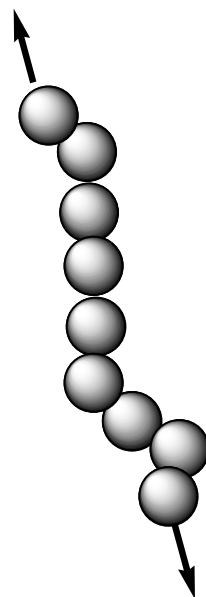


Picture created by Tim McKenna

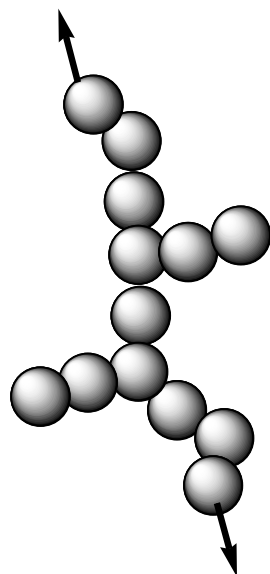
Polymer chains



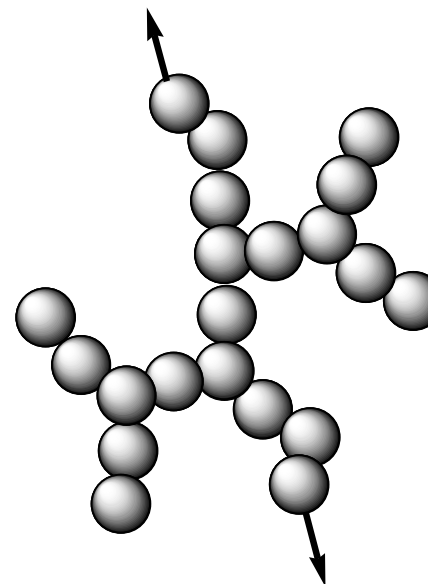
Chain topology



unbranched



branched



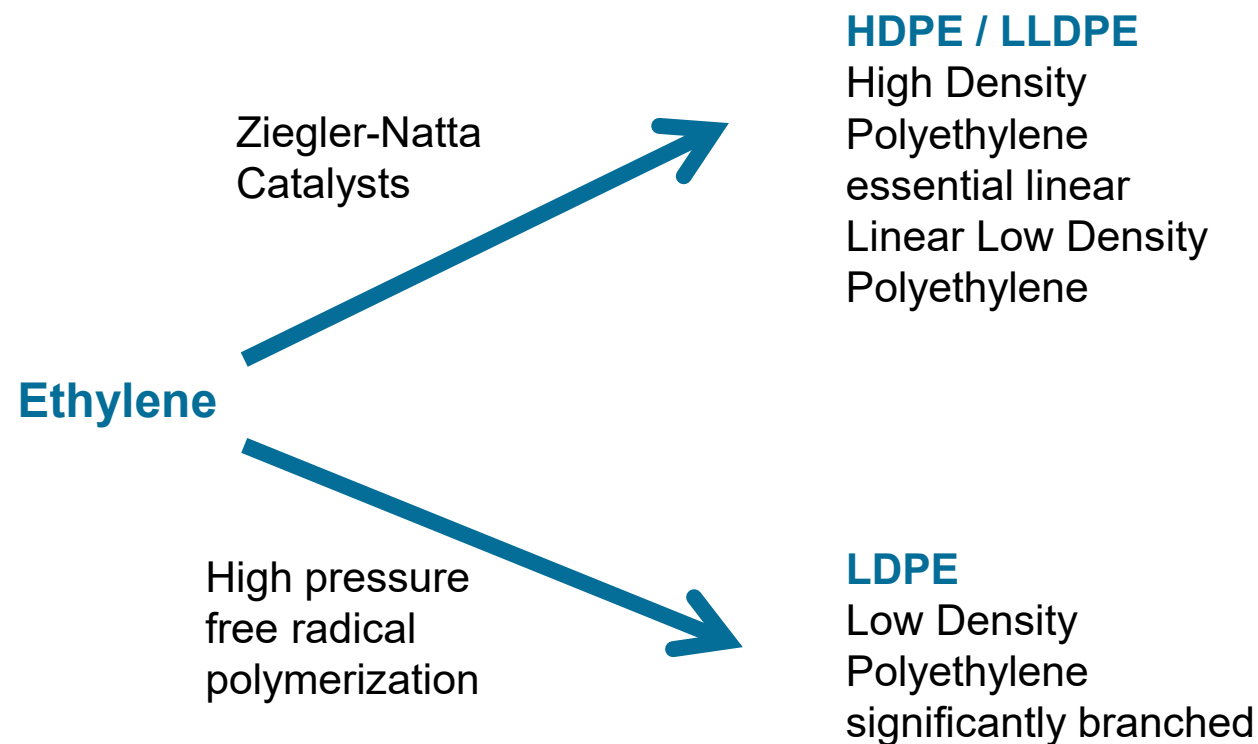
branches on branches

Linear or branched:

Short chain branched = long comonomers (e.g. LLDPE)

Long chain branched = side chains from polymerization (e.g. LDPE)

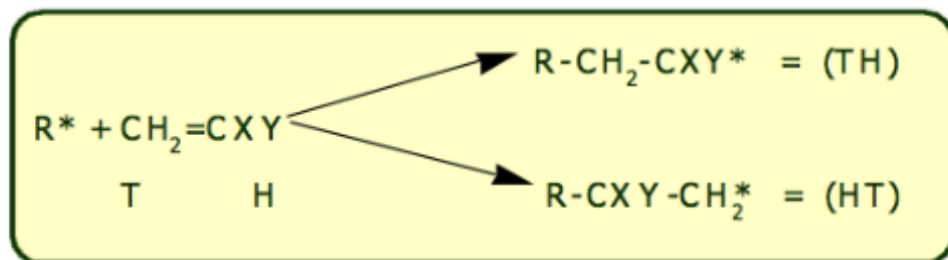
One monomer different structures



Affects ability to crystallize

Isomerism

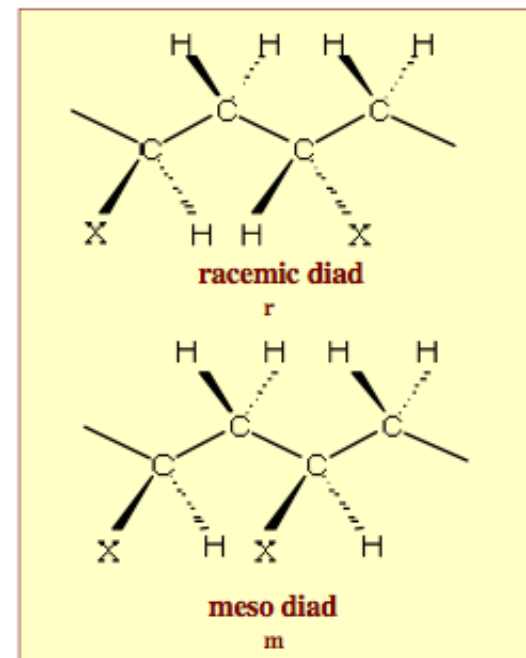
- Sequence isomerism
 - When a monomer unit adds to a growing chain it usually does so in a preferred direction
 - Polystyrene, poly(methyl methacrylate) and poly(vinyl chloride) are only a few examples of common polymers where addition is almost exclusively head-to-tail (HT)



- Stereoisomerism (in vinyl polymers)
- Structural isomerism (in diene polymers)

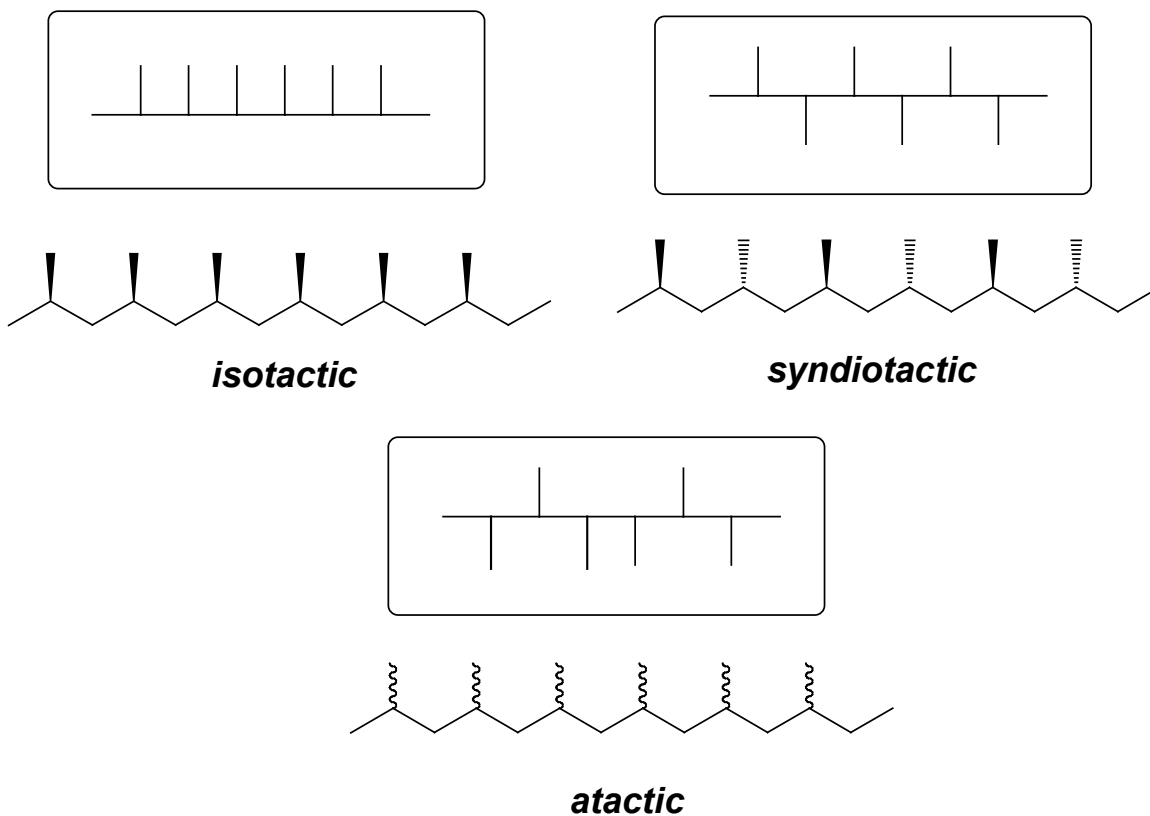
Isomerism

- Stereoisomerism (vinyl polymers)
 - Polymerization of a vinyl monomer, $\text{CH}_2=\text{CHX}$, where X may be a halogen, alkyl or other chemical group, leads to polymers with microstructures that are described in terms of tacticity
 - The substituent placed on every other carbon atom has two possible arrangements relative to the chain and the next X group along the chain
 - These arrangements are called racemic diad (r) or meso diad (m)



Isomerism

Stereoisomerism (vinyl polymers)



Regular structure:
able to crystallize
PP, PB1, P4MP1, ...

Irregular structure:
amorphous
PS, PMMA, ...

**PVC (largely atactic,
some syndiotactic
sequences)**

Isomerism

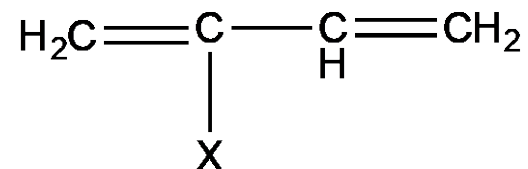
Structural isomerism

- Seen in the synthesis of polydienes from conjugated dienes: where if

X = H we have butadiene

X = CH₃ we have isoprene

X = Cl we have chloroprene



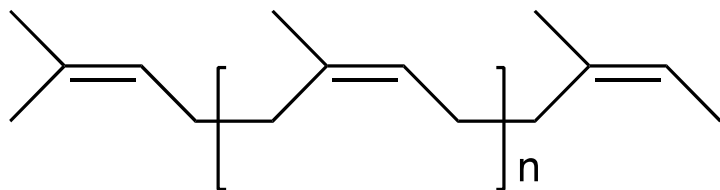
The polymers made from these monomers are elastomers or rubbers

Poly(isoprene): natural rubber

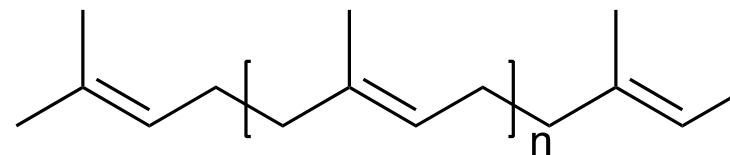
Poly(chloroprene): neoprene

Isomerism

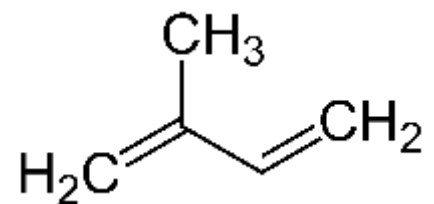
Structural isomerism



Cis 1,4 polyisoprene
Natural rubber
higher M_w



Trans 1,4 polyisoprene
Guttapercha



2 types of 1,4 units: cis and trans

Refers to the arrangement of carbon atoms around the double bond

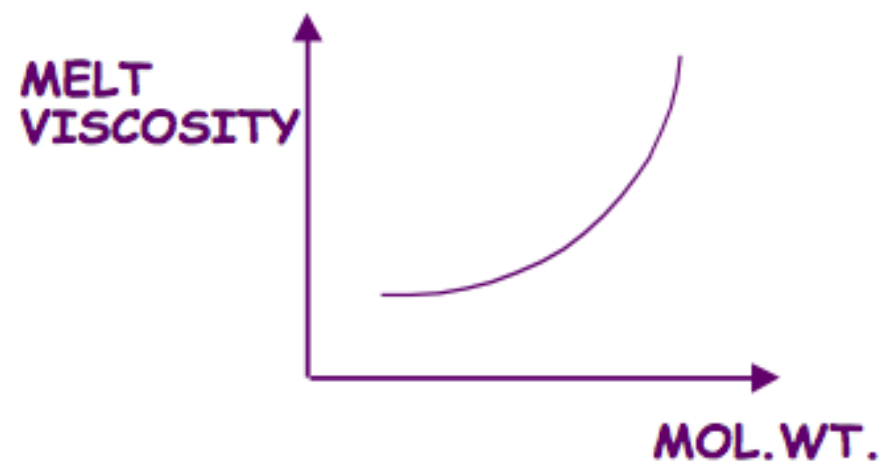
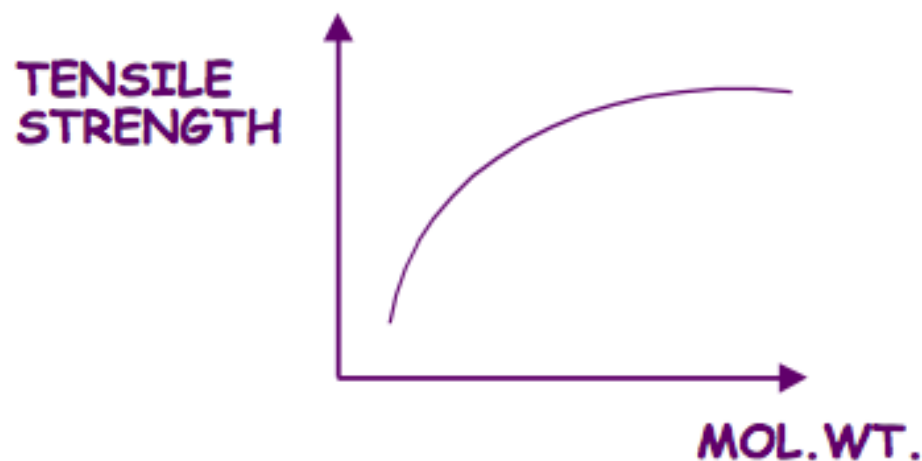
Polymers – properties: molecular weight

CH_4	-----	16
$\text{CH}_3 - \text{CH}_3$	-----	30
$\text{CH}_3 - \text{CH}_2 - \text{CH}_3$	-----	44
$\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$	-----	58
Gases		
$\text{CH}_3 - (\text{CH}_2)_6 - \text{CH}_3$	-----	114
Liquids		
$\text{CH}_3 - (\text{CH}_2)_{30} - \text{CH}_3$	-----	450
"Semi-solid"		
$\text{CH}_3 - (\text{CH}_2)_{30000} - \text{CH}_3$	-----	420030
Solids		

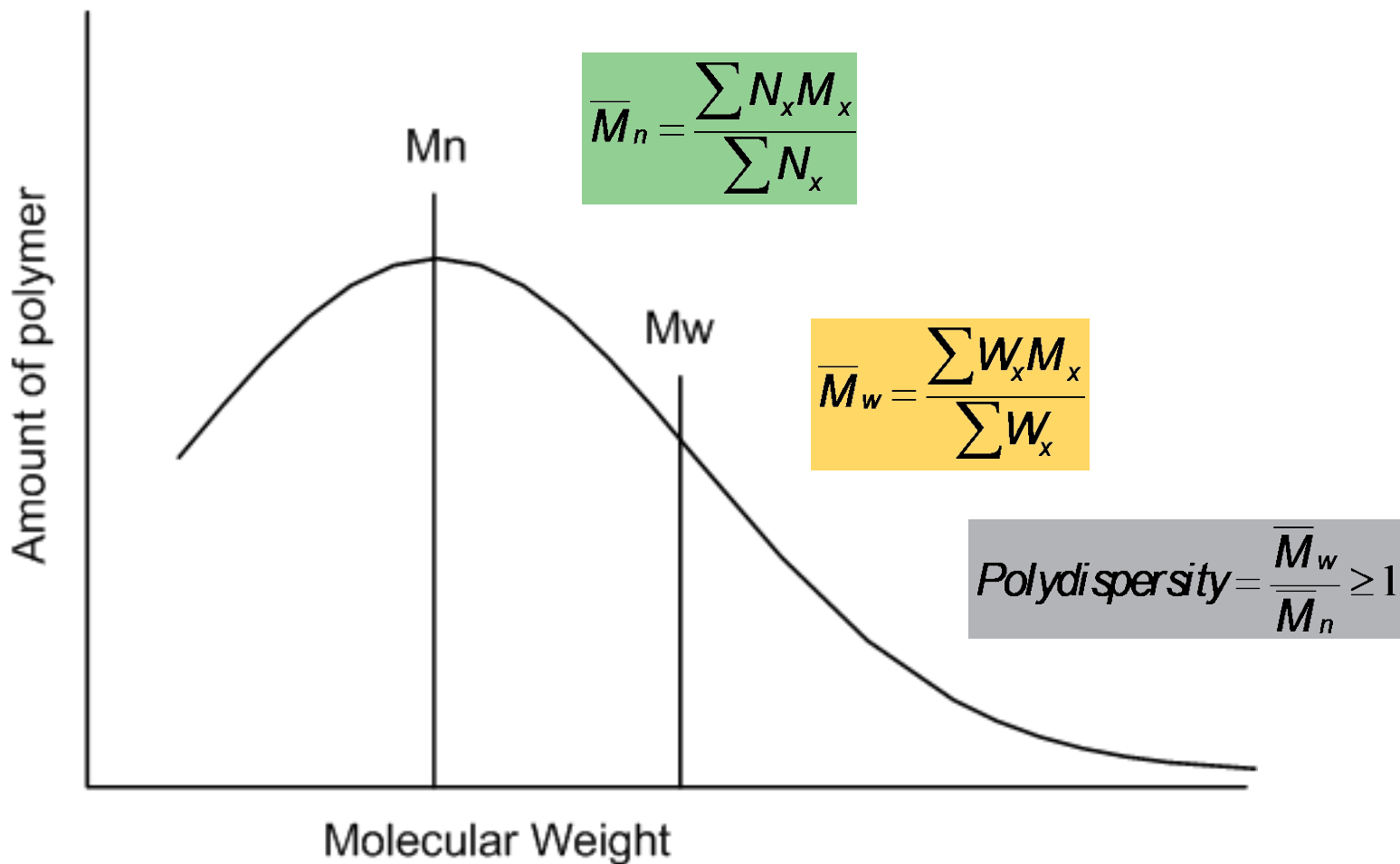
↓
**Increasing
Molecular Weight**

Polymers – properties: molecular weight

- Commercially produced polyethylene's often have molecular weights in the hundreds of thousands
- All synthetic polymers have a distribution of chain lengths
- As a result, we usually have to define an average molecular weight - there are different types of averages



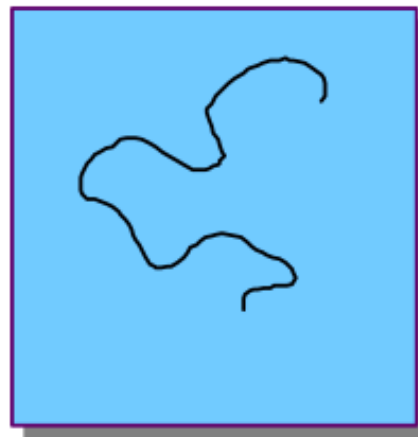
Polymers – properties: molecular weight



Polymers – properties: crystallinity

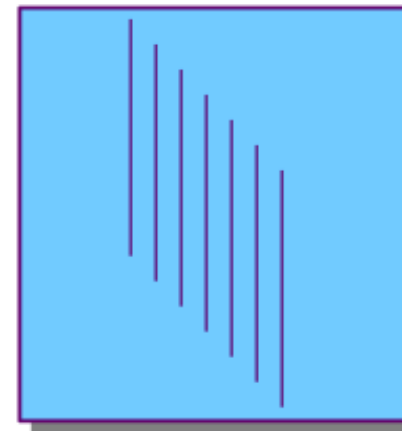
- Strength
- Stiffness
- Toughness
- Barrier Properties
- Solubility
- Transparency
- Thermal Properties
- Etc.

- However, in reality much more complicated!



RANDOM COIL

Like "cooked spaghetti"



CRYSTALLINE

Like "uncooked spaghetti "

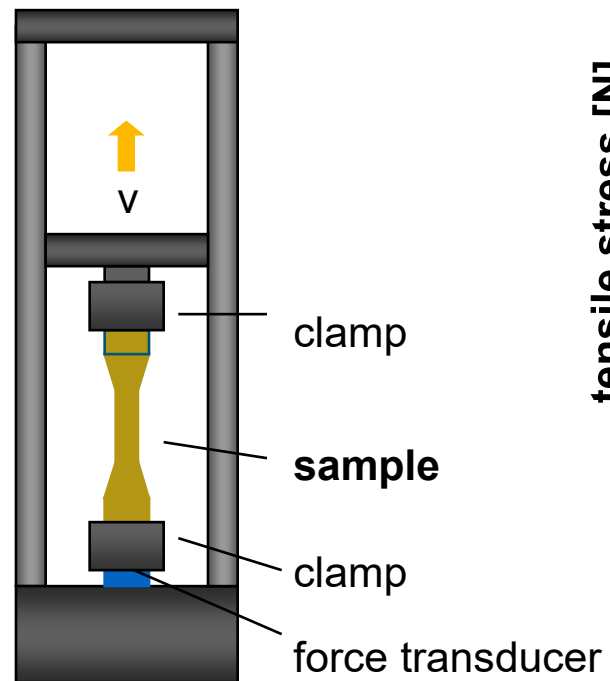
Polymers – properties: stiffness

Stiffness is the resistance of an elastic body to deflection or deformation by an applied force

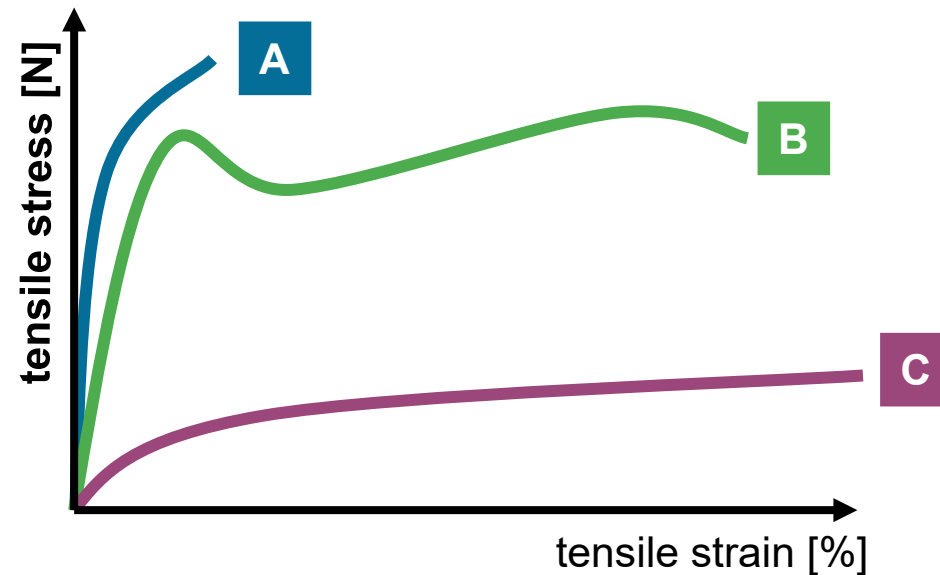
It is an extensive material property

Polymers – properties: tensile testing

Scheme of Tensile tester



Stress-strain diagram



A ... typical for thermosets

B ... typical for thermoplastics (e.g. PE & PP)

C ... typical for elastomers

ISO 527

Polymers – properties: stress strain curve

- **E-Modulus (Elastic modulus) /MPa**

0,05 – 1,05 % secant modulus
the higher the E-modulus,
the higher the stiffness

- **I = Yield area**

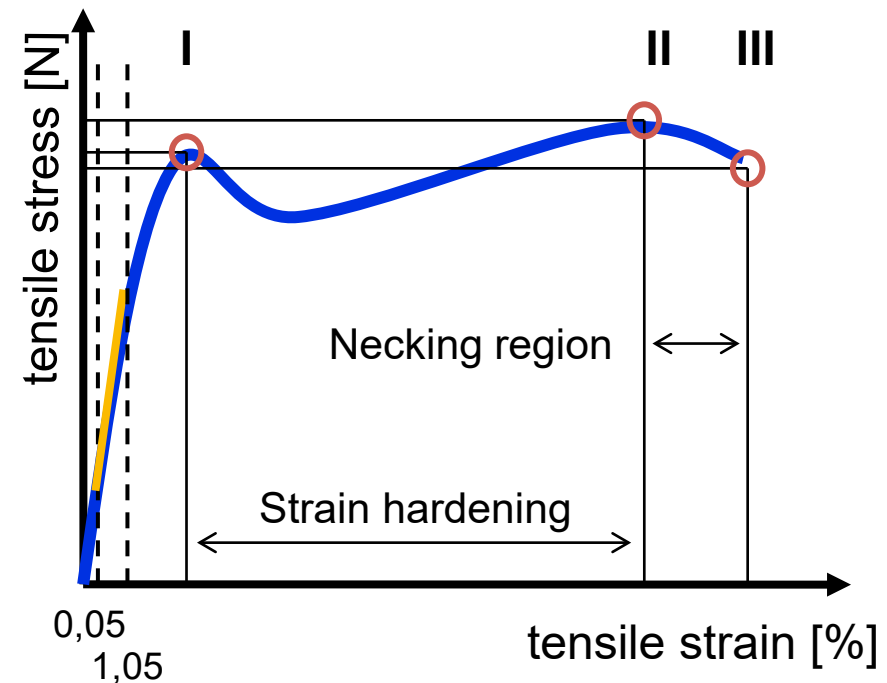
Yield stress /MPa
Yield strain /%

- **II = Tensile area**

Tensile (ultimate) strength /MPa
Tensile strain /%

- **III = Break area**

Stress /MPa
Strain /% at break



Polymers – properties: tensile strength of materials

Comparison of organic and inorganic materials

Force F needed to tear apart a bar with area $A \rightarrow Z = F/A$ [N/m²]

Material	N/m ²
Steel (depends on quality)	$7,0 \cdot 10^8$
Copper	$3,9 \cdot 10^8$
Alumina	$2,9 \cdot 10^8$
Glass	$8,8 \cdot 10^7$
Wood (spruce, perpendicular to fiber)	$2,6 \cdot 10^6$
Rubber (cross linked)	$1,9 \cdot 10^7$
Thermoplastics	$1,0 \dots 6,0 \cdot 10^7$
Foams	$2,0 \dots 7,0 \cdot 10^5$

Polymers – properties: tensile strength

- Dyneema (UHMWPE by DSM): up to 15 times stronger than steel!
- Ropes, fishing nets, medical applications, protective gloves, protective arming



Material	Strength / N/m ²
Dyneema	$3-4 \cdot 10^9$
Kevlar	$3-4 \cdot 10^9$
Spider silk	$1 \cdot 10^9$

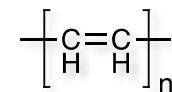
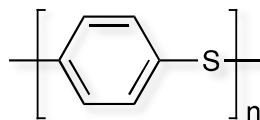
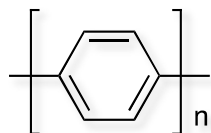
Polymers – properties: E - modulus

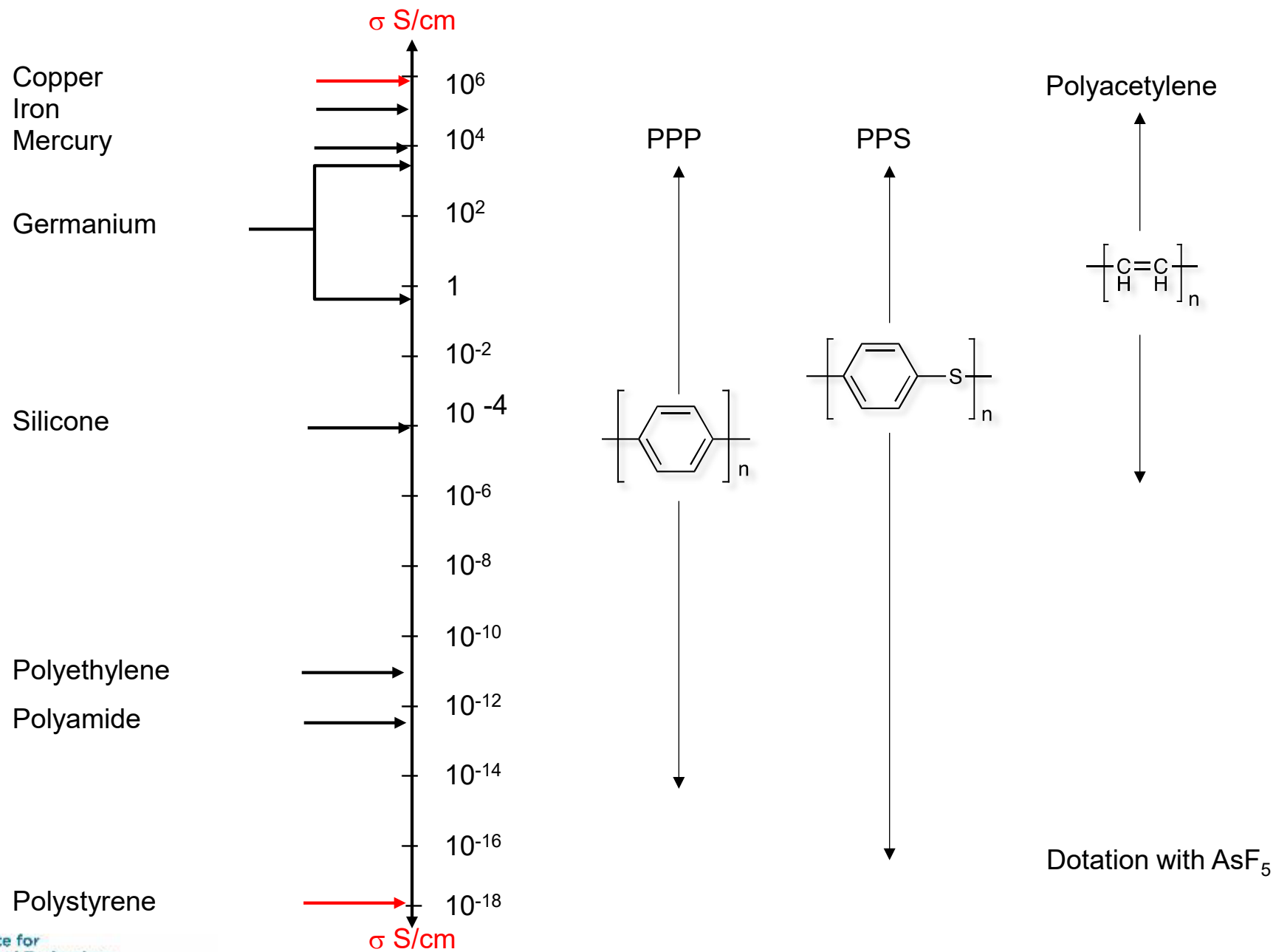
Material	N/m ²
Steel	2,1*10 ¹¹
Copper	1,2*10 ¹¹
Alumina	7,6*10 ¹⁰
Glass	6,9*10 ¹⁰
Wood (spruce, perpendicular to fiber)	1,1*10 ¹⁰
Glass fiber reinforced polyester resin	2,0*10 ¹⁰
PVC (hard)	3,3*10 ⁹
PS	3,3*10 ⁹
HDPE	1,0*10 ⁹
LDPE	2,0*10 ⁸

Hook's law $\frac{\Delta l}{l} = \frac{1}{E} \frac{F}{A}$

Polymers – properties: electrical conductivity

Most plastics are insulators, some are semiconductors, and only a few are conductors



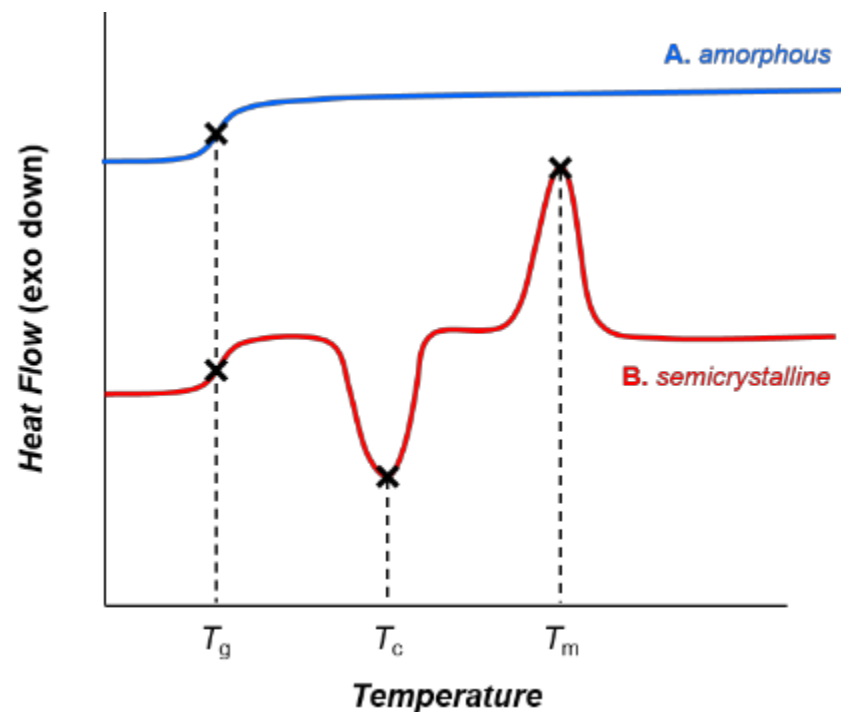


Polymers – properties: thermal behavior

- Glass transition temperature (T_G)
The liquid-glass transition (or glass transition) is the reversible transition in amorphous materials (or in amorphous regions within semicrystalline materials) from a hard and relatively brittle state into a molten or rubber-like state
- $T > T_G$
 - Increase of thermal expansion coefficient
 - Change in mechanical properties, above T_G plastic deformation possible (for non-cross linked materials) or rubber elasticity (for partial cross linked plastics)
 - Specific heat reaches maximum

Polymers – properties: T_G and T_M

DSC thermogram



T_g
 T_c
 T_m

Glass transition temperature
Crystallization temperature
Melting temperature

Polymer	T_M /°C	T_G /°C
PE	105 - 135	-70
PP	165	-1
PB-1	142	-24
PVC	150	77
1,4-cis-Polyisoprene	40	-67
POM	181	-82
PA6	229	57
PS	240	100
PET	265	69

Polymer build up reactions: chain growth

Polymerization of C=C double bonds

- Polyolefins from ethylene, propylene or butene-1
- Polystyrene, Polybutadiene, Polyisoprene, butadiene/styrene copolymers
- Polyacrylnitrile, Polyacrylate, Polymethacrylate
- Polyvinylchloride, Polyvinylacetate

Polymerization of C=O double bonds

- Polyformaldehyde from formaldehyde

Ring opening polymerization

- Polyformaldehyde from trioxane
- Polypentamer from cyclopentene
- Polyamide 6 from caprolactame or from C₁₂ lactam

Polymer build up reactions: **step growth**

Polycondensation

- Amino plastics from urea, melamine and formaldehyde
- Phenoplastics from phenol and formaldehyde
- Polyamide from dicarboxylic acids and diamins
- Polyamide from α -amino acids
- Polyester from dicarboxylic acids and diols
- Polyester from dicarboxylic ester and diols

Polyaddition

- Polyurethanes from diisocyanates and diols
- Epoxy resins from bis-epoxies and diamines
- Epoxy resins from bis-epoxies and dicarboxylic acid anhydrides

Ceiling temperature

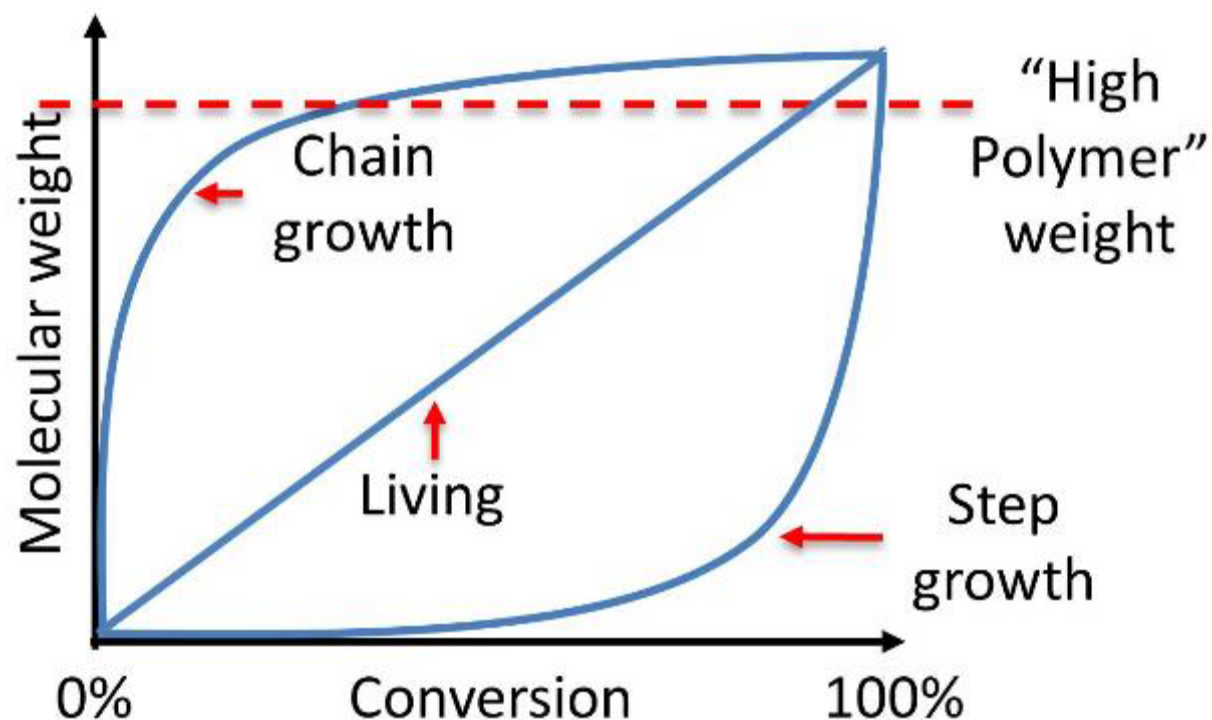
- Growth in polymerization reactions is reversible, one always finds polymerization and de-polymerization.
- At the ceiling temperature both reactions are equally fast.

$$\Delta_R G = \Delta_R H - T \Delta_R S = 0$$

$$T_C = \frac{\Delta_R H}{\Delta_R S}$$

Styrene (PS): 395°C

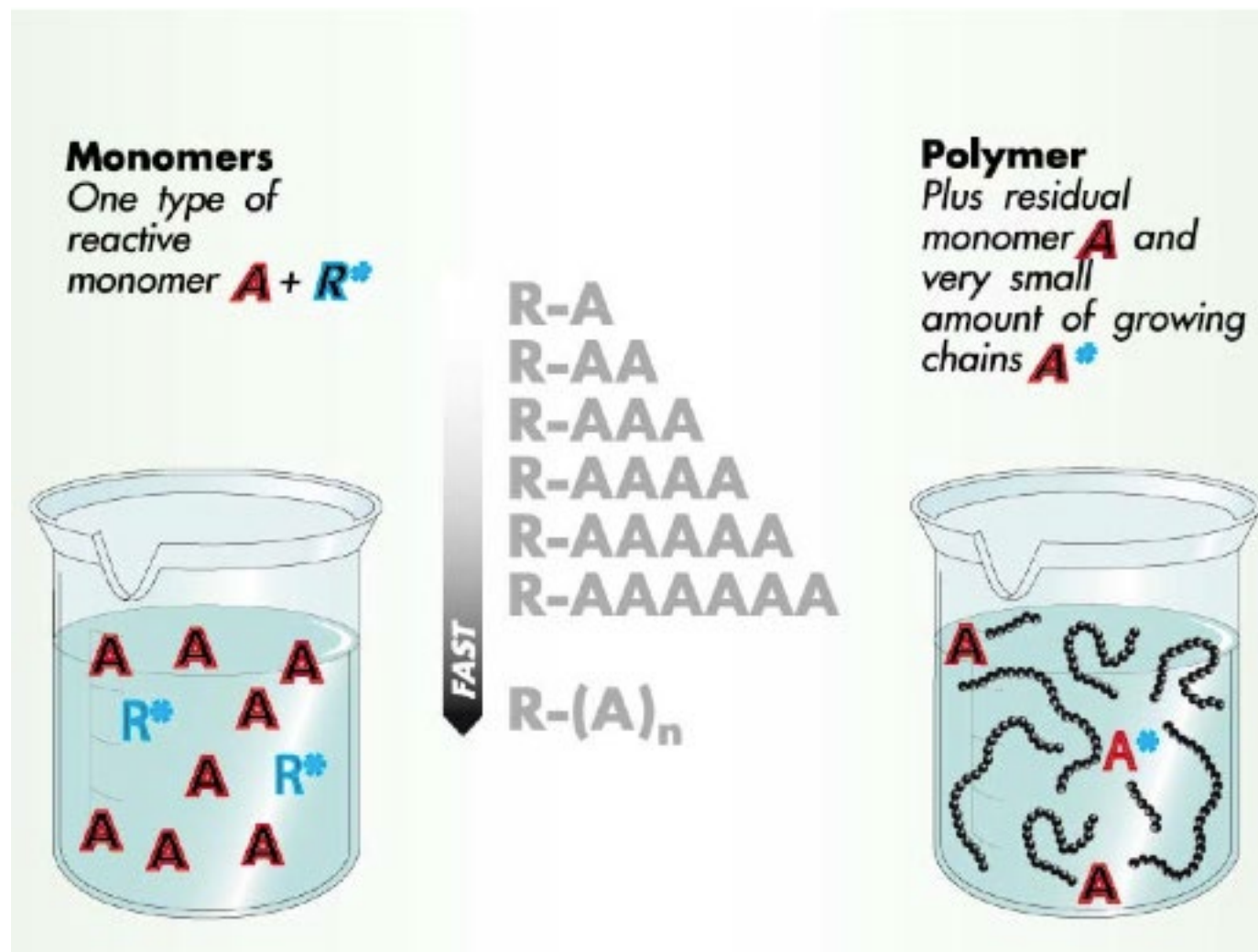
α -Methyl Styrene: 66°C



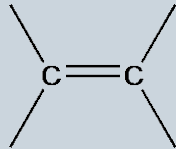
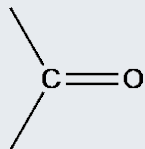
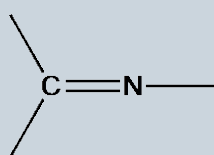
Chain growth polymers

- Chain propagation
- Polymerization processes
- Polyethylene (PE)
- Polypropylene (PP)
- Polyvinylchloride (PVC)
- Fluor polymers
- Styrene polymers
- Polyvinyl -ester, -alcohol, - ether
- Polyacrylates

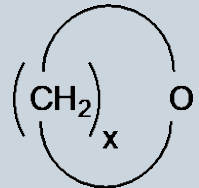
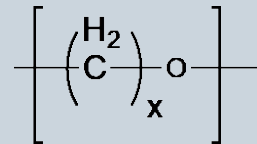
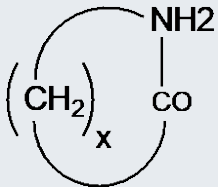
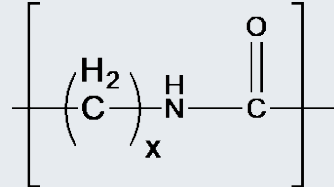
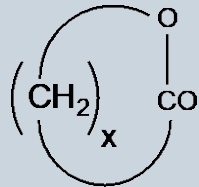
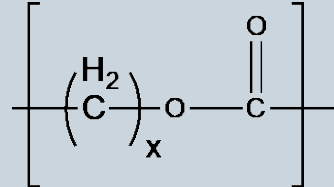
Chain propagation



Chain propagation: type of monomer

Bond	Monomer	Polymer
	$\begin{array}{c} \text{H}_2\text{C}=\text{CH} \\ \\ \text{X} \end{array}$	$\left[\begin{array}{c} \text{H}_2 \\ \\ \text{---C---} \end{array} \begin{array}{c} \text{H} \\ \\ \text{---C---} \\ \\ \text{X} \end{array} \right]$
	$\begin{array}{c} \text{R---C=O} \\ \\ \text{H} \end{array}$	$\left[\begin{array}{c} \text{R} \\ \\ \text{---C---} \\ \\ \text{H} \end{array} \text{---O---} \right]$
	R---N=C=O	$\left[\begin{array}{c} \text{R} \\ \\ \text{---N---} \end{array} \begin{array}{c} \text{O} \\ \\ \text{---C---} \end{array} \right]$
$\text{---C}\equiv\text{N}$	$\text{R---C}\equiv\text{N}$	$\left[\begin{array}{c} \text{---C=N---} \\ \\ \text{R} \end{array} \right]$

Chain propagation: type of monomer

Bond	Monomer	Polymer
Cyclic ether		
Lactame		
Lactone		

Chain propagation

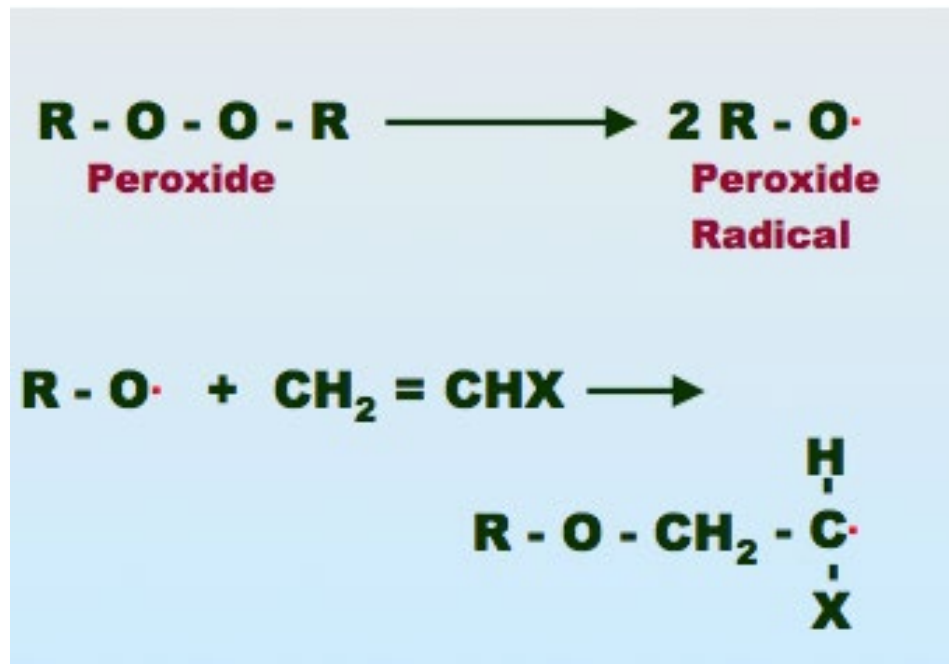
- Radical polymerization
- Cat-ionic / an-ionic polymerization
- Ziegler-Natta (co-ordination) polymerization

Need to consider:

1. Initiation
2. Propagation
3. Termination
4. Chain Transfer

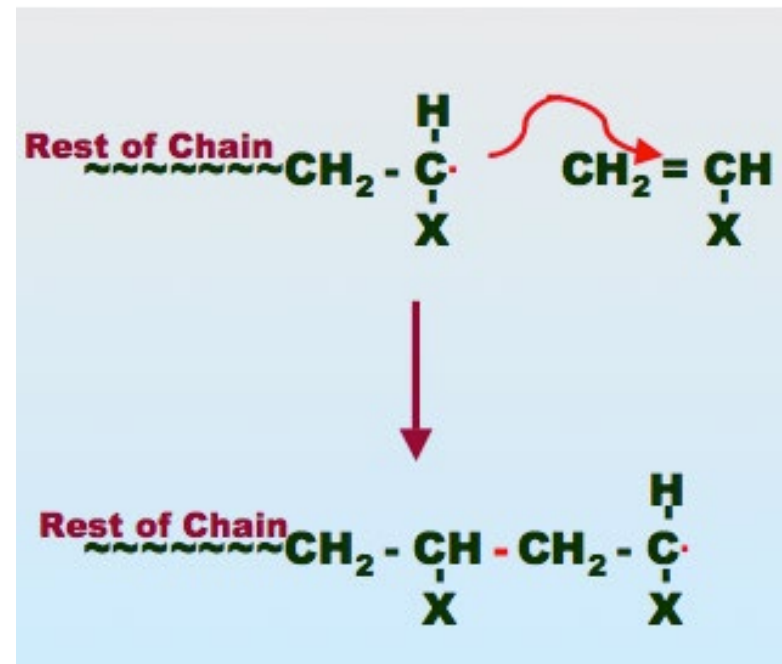
Radical polymerization

1. Initiation



$$v_i = \frac{d[R^*]}{dt} = 2fk_i[I]$$

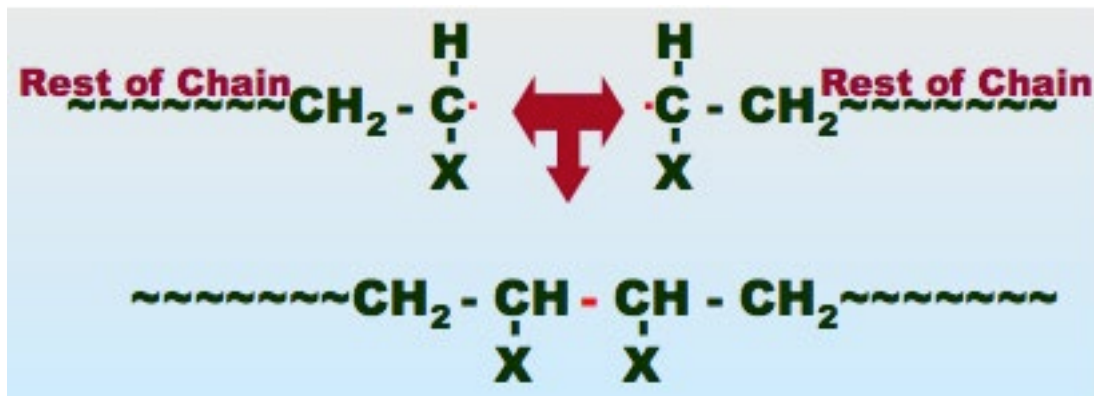
2. Propagation



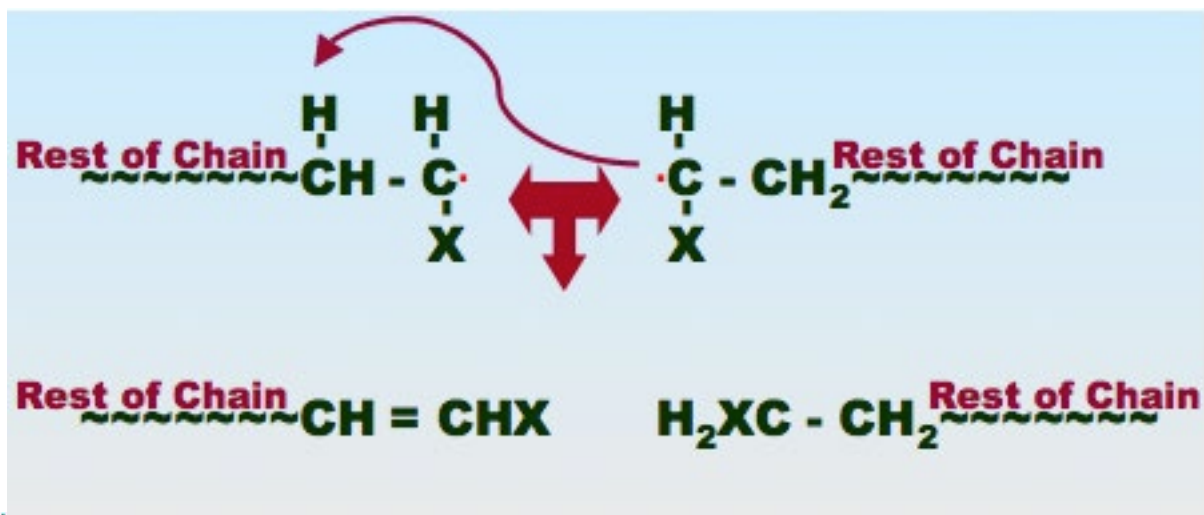
$$v_p = -\frac{d[M]}{dt} = k_p[M][P^*]$$

Radical polymerization

3. Termination / recombination



$$v_T = -\frac{d[P^*]}{dt} = k_T [P^*]^2$$



Radical polymerization

4. Chain transfer

$$v_{CT} = k_{CT} [P^*] [Transfer]$$

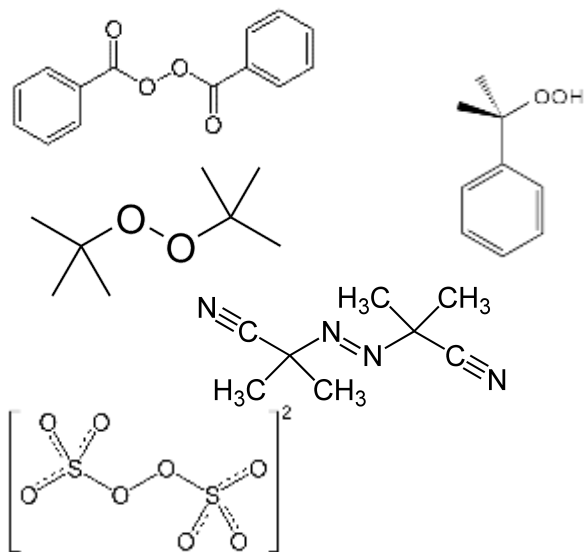
Overall reaction speed

$$v_{Total} = k_P \sqrt{2f \frac{k_I}{k_T}} [M] \sqrt{[I]}$$

Average degree of polymerization

$$\bar{P} = \frac{v_P}{v_T + v_{CT}}$$

Radical polymerization: initiators



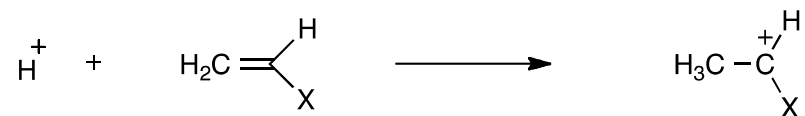
Initiator	Temp. range [°C]
Benzoyl peroxide	40-90
Cumene hydroperoxide	50-100
Di tert butyl peroxide (DTBP)	80-150
Azobisisobutyronitrile (AIBN)	40-90
Potassium persulfate	25-80

Important polymers made through radical polymerization:

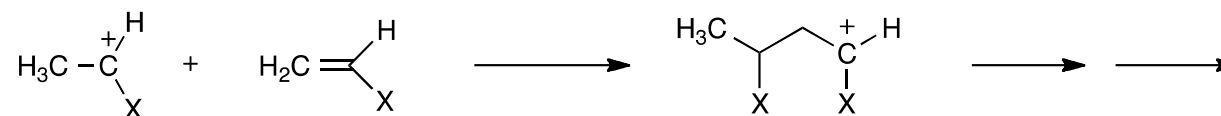
- LDPE (hp PE)
- PS, PVC
- Polyacrylonitrile, Polyacrylate
- Styrene-butadiene-copolymers

Cationic polymerization

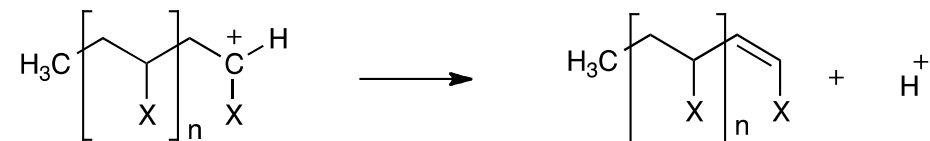
1. Initiation



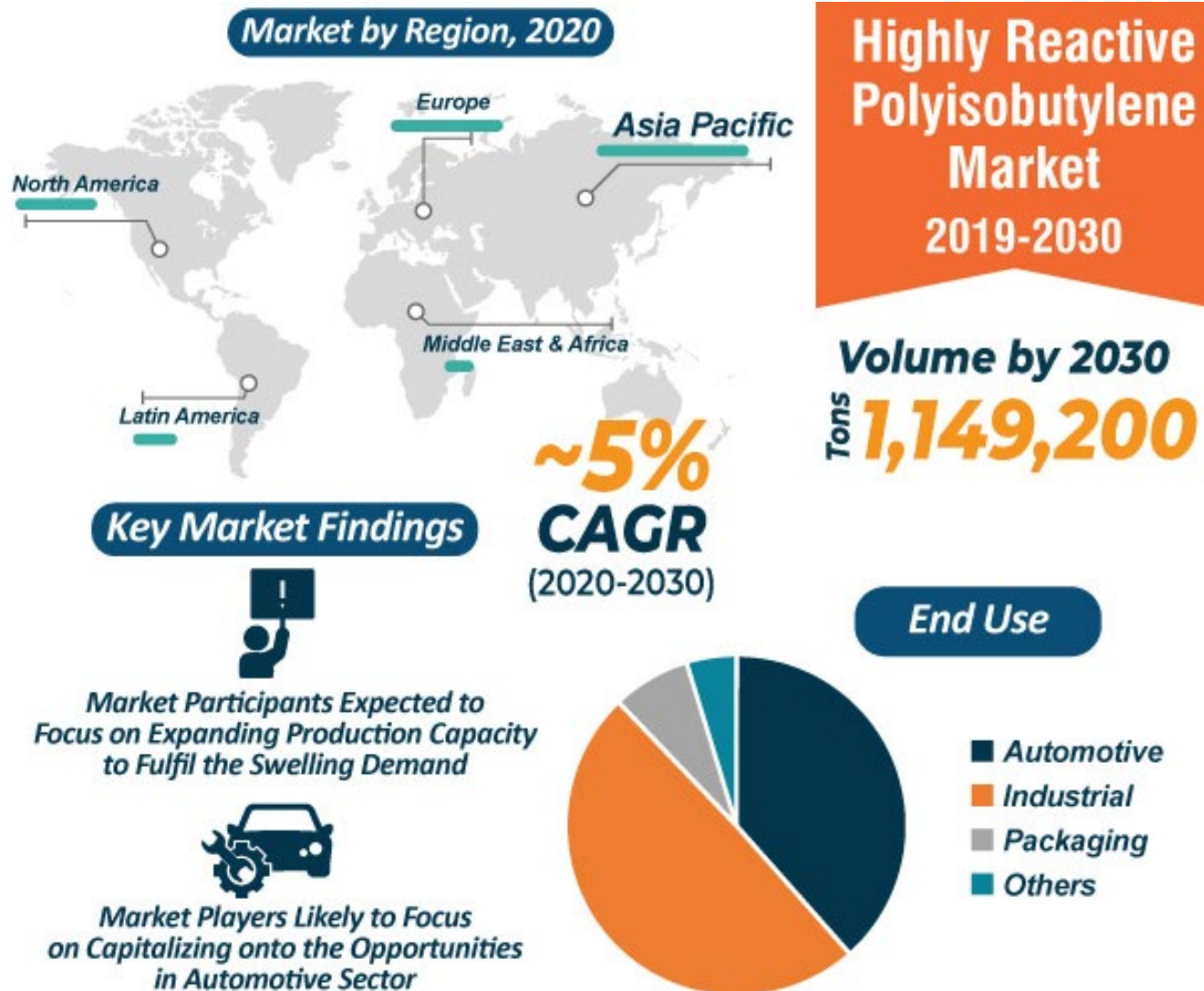
2. Propagation



3. Termination



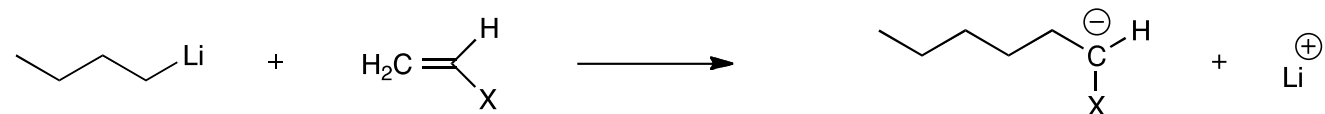
e.g. Polyisobutylene, Polyformaldehyde, Polyvinyl ether



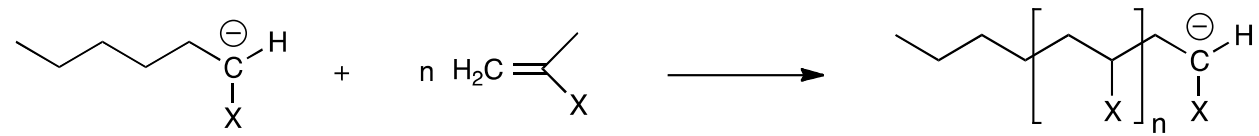
HR-PIB is a polyisobutene that has more than 85% of its double bonds located in the terminal position of the molecule, allowing it to react more quickly when producing derivatives, especially detergents and dispersants.

Anionic polymerization

1. Initiation



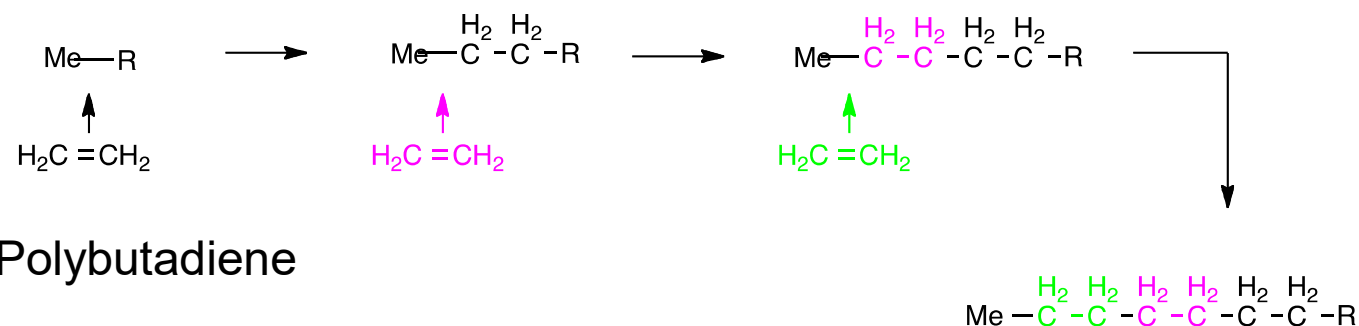
2. Propagation



e.g. cis-1,4-Polybutadiene, cis-1,4-Polyisoprene

Coordination polymerization: Ziegler-Natta

- Insertion mechanism:



e.g. lp PE, PP, PB-1, all-cis-1,4-Polybutadiene

- Ring-opening polymerization / metathesis

Polypentenamer (n=5)

Polynorbornenamer

Differences: radical // ionic, coordination

Parameter	Radical	Ionic / Co-ordination
Activation energy [kJ/mol]	75 - 125	8 - 25
Polymerization temperature [°C]	-5 – 300	-78 - 80
Purity (monomer/solvent)	Standard	High
Polymer structure	Atactic	Tacticity tuneable
Molecular weight increase	Trommsdorf effect	“living” polymers

Chain growth polymers

- Chain propagation
- **Polymerization processes**
- Polyethylene (PE)
- Polypropylene (PP)
- Polyvinylchloride (PVC)
- Fluor polymers
- Styrene polymers
- Polyvinyl -ester, -alcohol, - ether
- Polyacrylates

Polymerization processes

1. Mass polymerization (bulk polymerization)

- High speed and partial conversion
High pressure polymerization of ethylene
Bulk polymerization of vinyl chloride
- Low speed and complete conversion
Polymerization of styrene
Polymerization of methyl methacrylate
- Removal of reaction heat difficult

Polymerization processes

2. Solution polymerization

- Monomer soluble in the solvent – the formed polymer is soluble as well
- Removal of reaction heat is no problem
Polymerization of acrylonitrile in dimethyl formamide or dimethyl sulfoxide
- Polymerization of butadiene or isoprene in n-hexane

Polymerization processes

3. Precipitation polymerization

- Monomer soluble in the solvent – the formed polymer not
Polymerization of acrylonitrile in water
- Low pressure polymerization of ethylene or propylene in n-hexane

4. Gas phase polymerization

- Polyethylene, Polypropylene

Polymerization processes

5. Emulsion polymerization

- Monomer and polymer not soluble in water
initiator soluble in water
- Use of emulsifiers (surfactants)
- Copolymerization of butadiene and styrene
- Polymerization of chloroprene, vinyl acetate or vinyl chloride

Polymerization processes

6. Suspension polymerization

- Monomer and polymer in water not soluble
initiator soluble in the monomer
- Use of dispersants e.g. polyvinyl alcohol or carboxymethylcellulose (CMC)
- Polymerization of styrene, vinyl chloride
- Copolymerization of styrene with divinylbenzene

Reactor		Media	Application			
			Mass-, solution-, precipitation	Suspension	Emulsion	Gas phase
Stirred tank (STR)						
batch	liquid		LDPE, HDPE, PS, PB, PMMA, PC, resins	PVC, PTFE, PMMA, EPS	PVC, PTFE, SBR, co- polymers	PVC
continuous	liquid		LDPE, PS, PMMA		PVC	
STR cascade						
	liquid		PE, PS, HDPE, PP	LDPE	SBR	
Loop reactor						
	liquid			HDPE, PP		
Fluidized bed						
	solid/gas					HDPE, PP, LLDPE
Tower-, plug flow reactor						
	liquid/liquid		PS, PA, LDPE, EVA, PA			

Chain growth polymers

- Chain propagation
- Polymerization processes
- **Polyethylene (PE)**
- Polypropylene (PP)
- Polyvinylchloride (PVC)
- Fluor polymers
- Styrene polymers
- Polyvinyl -ester, -alcohol, - ether
- Polyacrylates

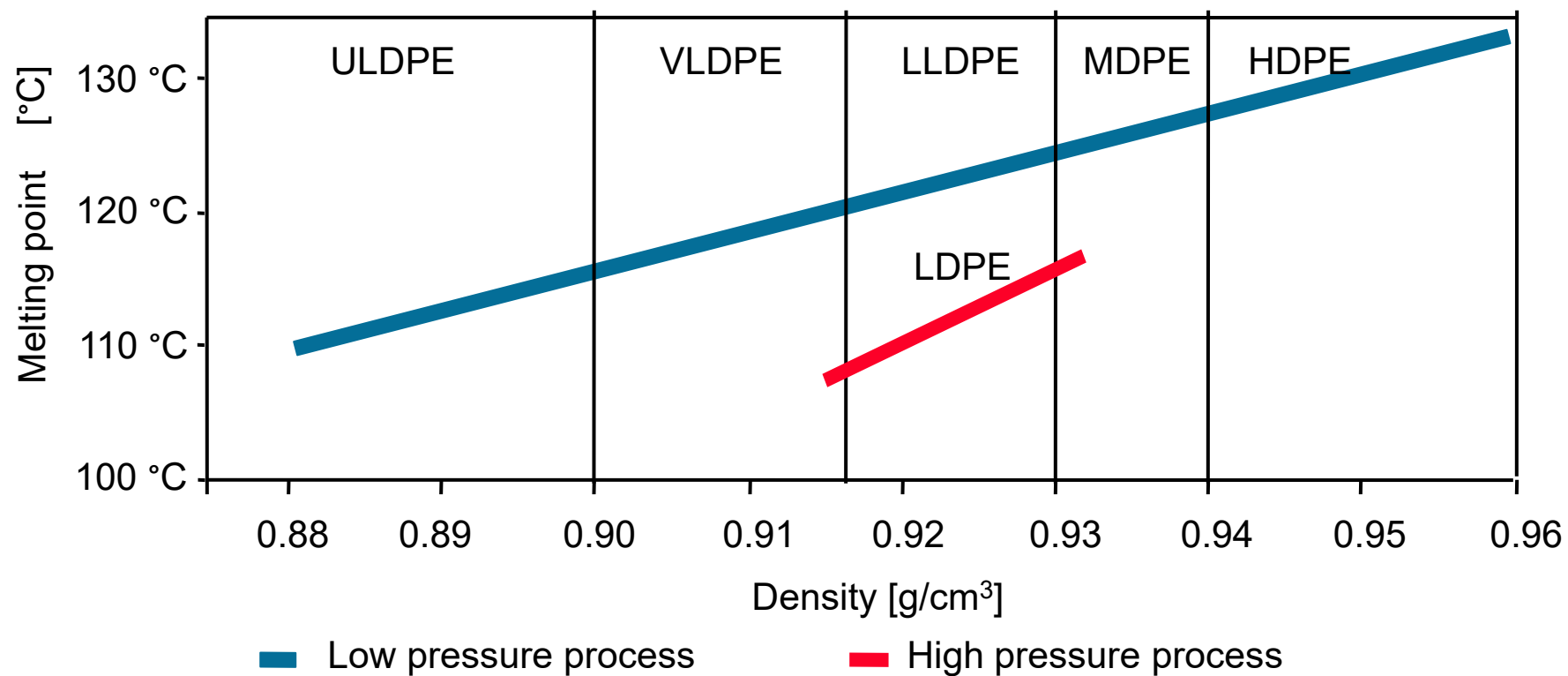
Polyethylene (PE)

Low Density Polyethylene (LDPE)

- Radical polymerization of ethylene
- 130-300°C // 200-300 MPa
- Initiators: oxygen and/or peroxides
- Conversion: 18 to 35% in CSTR or plug flow reactors
- Short and long chain branching
- Low crystallinity
- Lower stiffness / higher flexibility

Polyethylene (PE)

Density versus T_M – for linear and branched PE's



Polyethylene (PE)

Linear Low Density Polyethylene (LLDPE)

- Co-ordination polymerization
Ziegler or Chromium catalysts, mostly heterogenized
- Polymerization conditions / processes as for HDPE
- Co-monomer: α -olefins (butene, hexene, octene)
- Adjusting properties with amount of co-monomer:
density, T_M , stiffness/toughness
- Short chain branching

Branching in LDPE

Long chain branching

- Intermolecular transfer of growing polymer chains to already existing chains
- Transfer of a H atom from a secondary C-H bond to the primary C-radical of the growing chain - further growth at the newly formed secondary radical
- Branches on branches

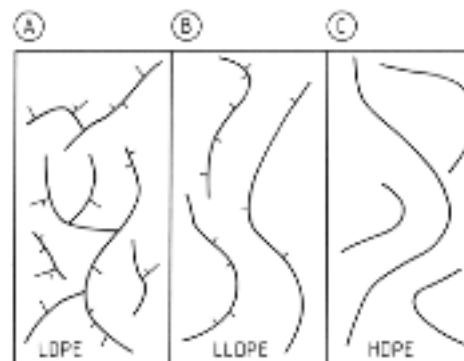
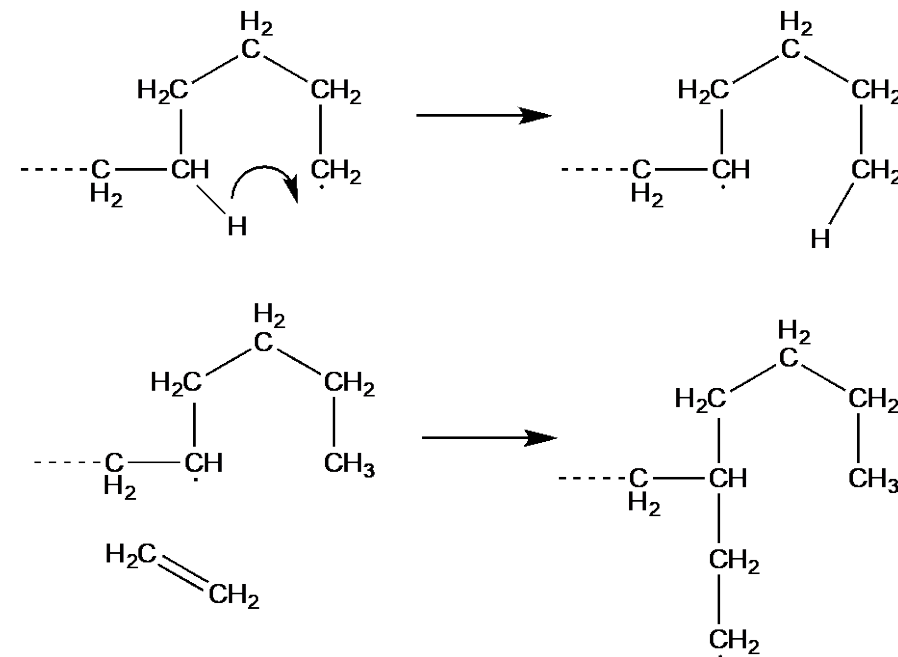


Figure 1. Schematic molecular structure
A) Low-density polyethylene; B) Linear low-density polyethylene; C) High-density polyethylene

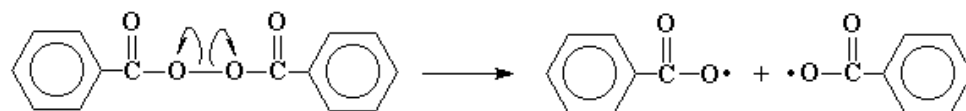
Branching in LDPE

Short chain branching

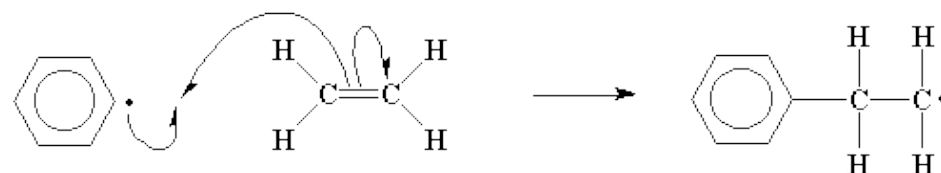
- Intramolecular transfer in growing polymer radical
- Transfer of a H atom from a secondary C-H bond to the primary C radical of growing chain
- n-Butyl side chain



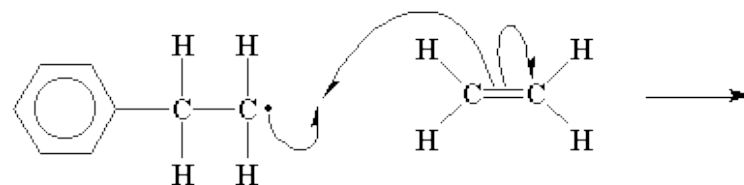
LDPE - high pressure process



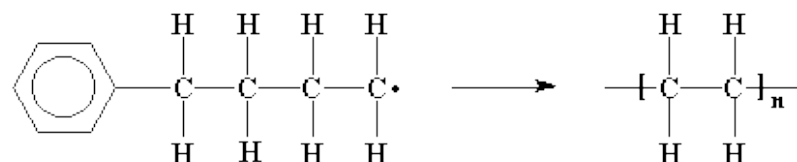
Decomposition of initiator (peroxide),
formation of radicals



Start reaction with first monomer unit

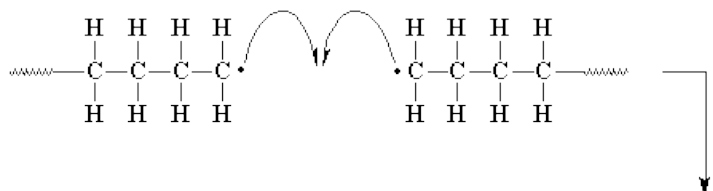


Polymerization reaction
(build-up reaction)

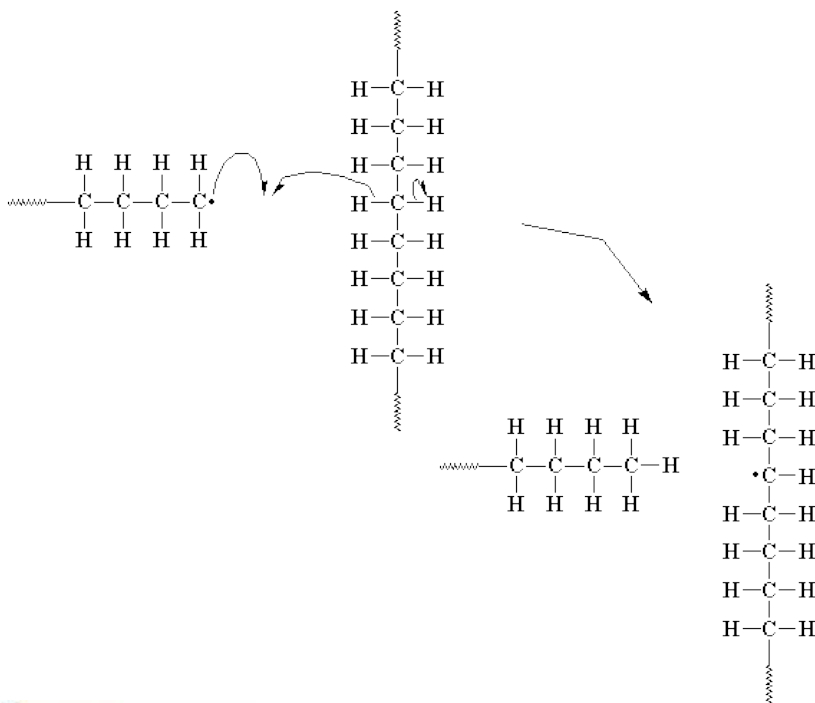
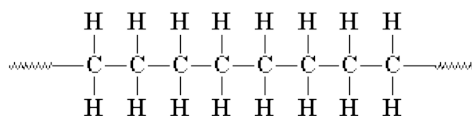


T = 130-300°C
p = 1000-3000 bar

LDPE - high pressure process



Termination reaction by recombination



Branch formation by hydrogen transfer

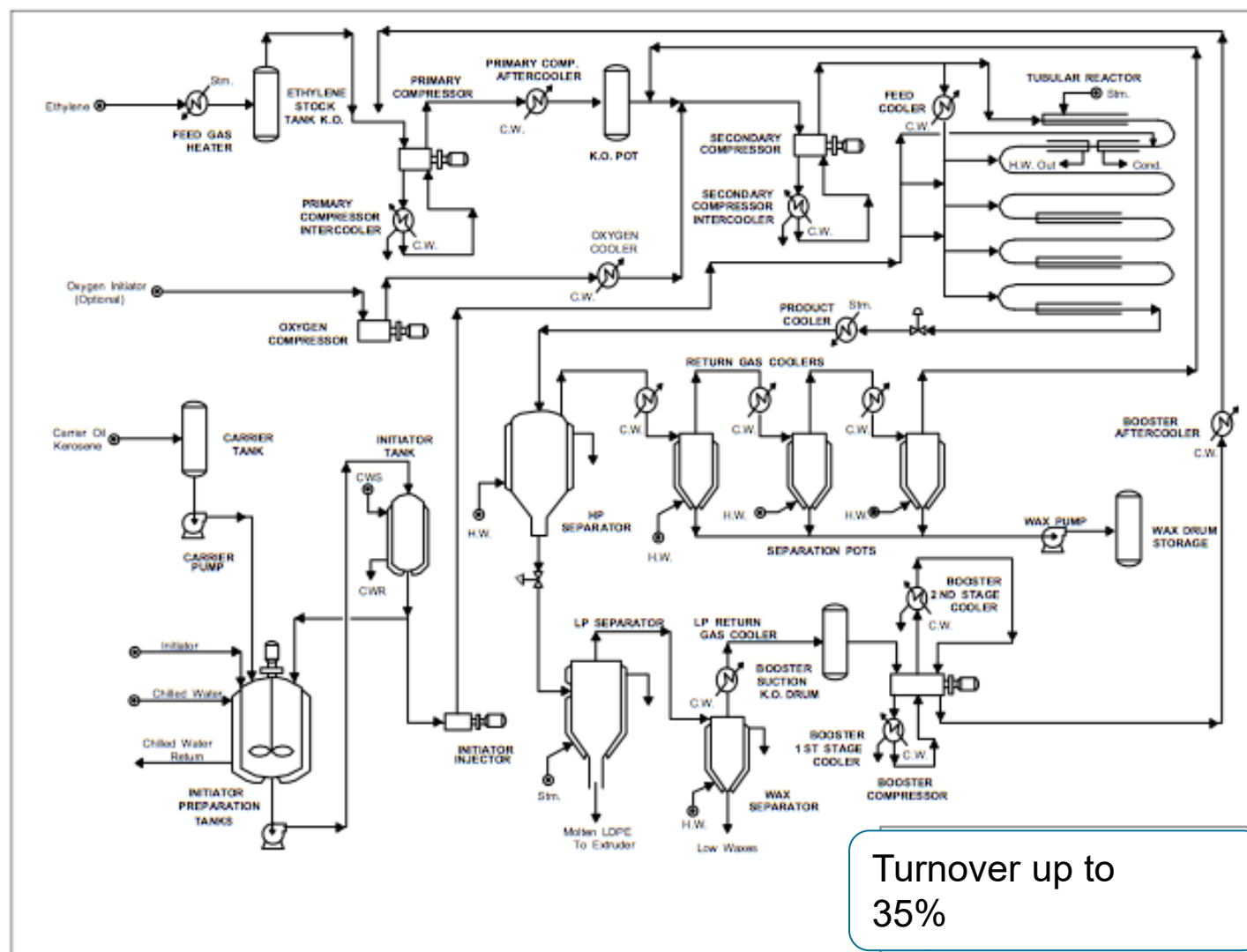
T = 130-300°C
p = 1000-3000 bar

Two LDPE processes

Autoclave or tubular - basic principles

- Necessity of a 2 stage compression of the ethylene to reach the polymerization pressure of 1.200 – 3.000 bar
- Extraction of the polymer in molten stage, this means one can use short extruders for the granulation (dispersion of small amounts of additives)
- Copolymerization with vinyl acetate or methyl acrylate possible

LDPE Tubular process



LDPE tubular process

- Initiator (organic Peroxide) and monomer can be injected at different places of the reactor
- Examples: Novolen G and the Equistar process
- The whole reactor pipe has a heat exchange jacket, reactor loops are placed in a bunker (open at the top)



LDPE autoclave process

- Polymerization in a stirred high pressure reactor (possible to set them in series)
- Different systems – rough differentiation in “broad” and “slim” reactors
- In comparison to tubular reactors the technical problems for autoclave systems are bigger (use of a stirrer, heat removal)
- However, the homogeneity of copolymers is better, EVA copolymers with up to 40 % co-monomer possible

LDPE property control

With increasing value of ...	Average relative molecular weight	Frequency of short chain branching	Frequency of long chain branching
Temperature	--	++	+
Pressure	+	-	-
Initiator concentration	-	+	+
Regulator conc.	--	-	-
Polymer concentration	+	0	++

Regulator: aliphatic or olefinic hydrocarbons

Low pressure PE: HDPE and LLDPE

- Process closely related to catalyst design
- Ziegler (Natta) systems
 - By far the most important class of catalysts for olefin polymerization
 - The actual catalyst is made via a reaction of an Al alkyl and a Ti compound
 - The polymerization reaction takes place on the surface of a solid catalyst particle
 - H_2 is terminating the reaction (regulating the MW)
- Chromium
- Metallocene

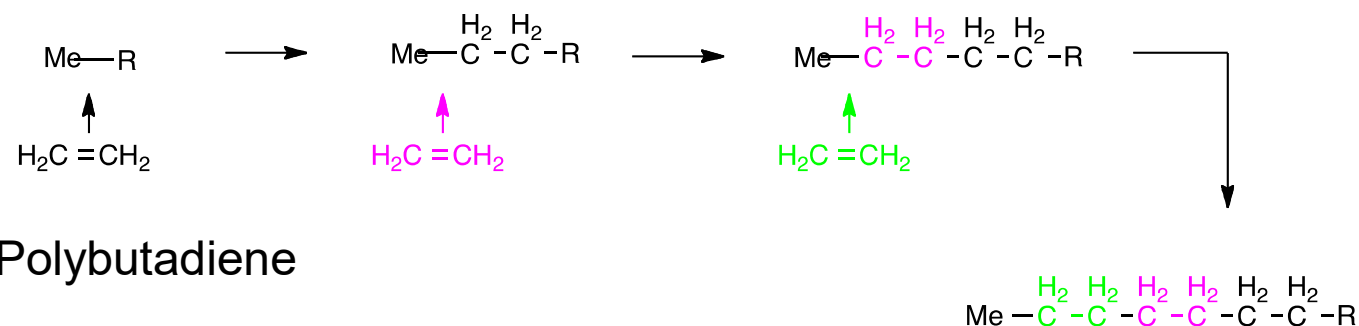
Polyethylene (PE)

High Density Polyethylene (HDPE)

- Co-ordination polymerization
Ziegler or Chromium catalysts, mostly heterogenized
- 30-130°C // 0,1-6,0 MPa
- Solution polymerization in hexane or cyclohexane (>100°C)
- Precipitation polymerization in propane, hexane or cyclohexane (<100°C)
- Gas phase polymerization in fluidized beds (80-100°C)
- Only few branches, high density (crystallinity), stiff

Coordination polymerization: Ziegler-Natta

- Insertion mechanism:



e.g. lp PE, PP, PB-1, all-cis-1,4-Polybutadiene

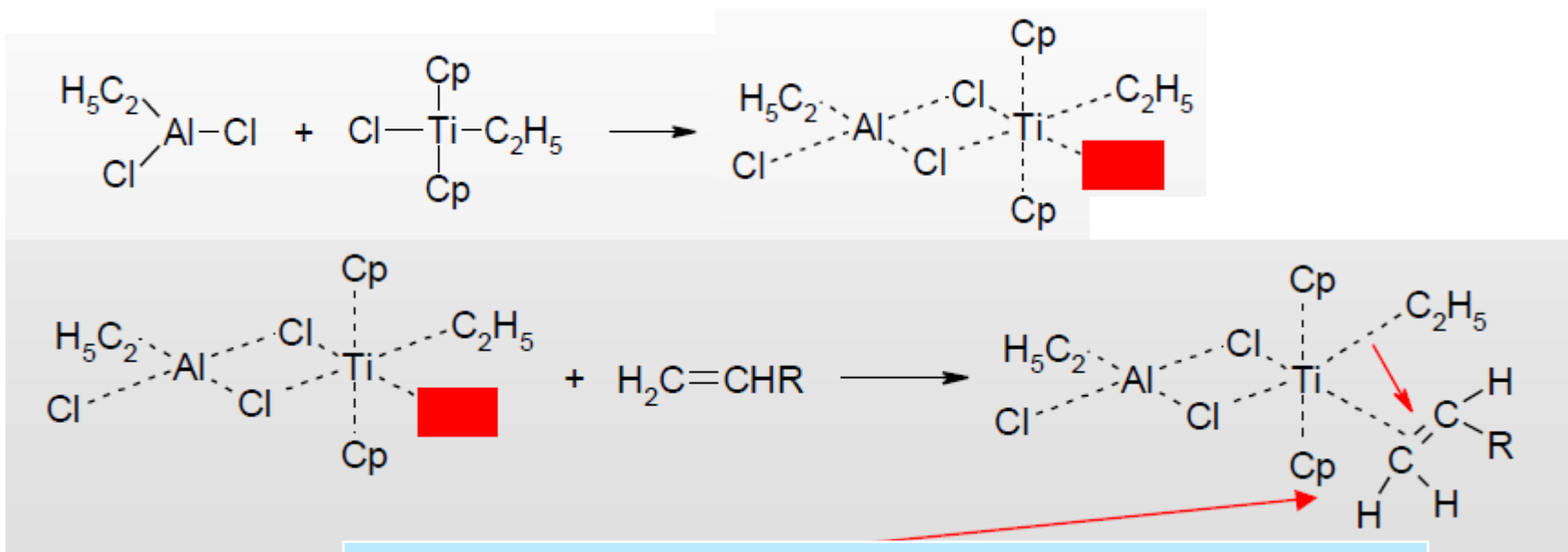
- Ring-opening polymerization / metathesis

Polypentenamer (n=5)

Polynorbornenamer

Ziegler Natta PE

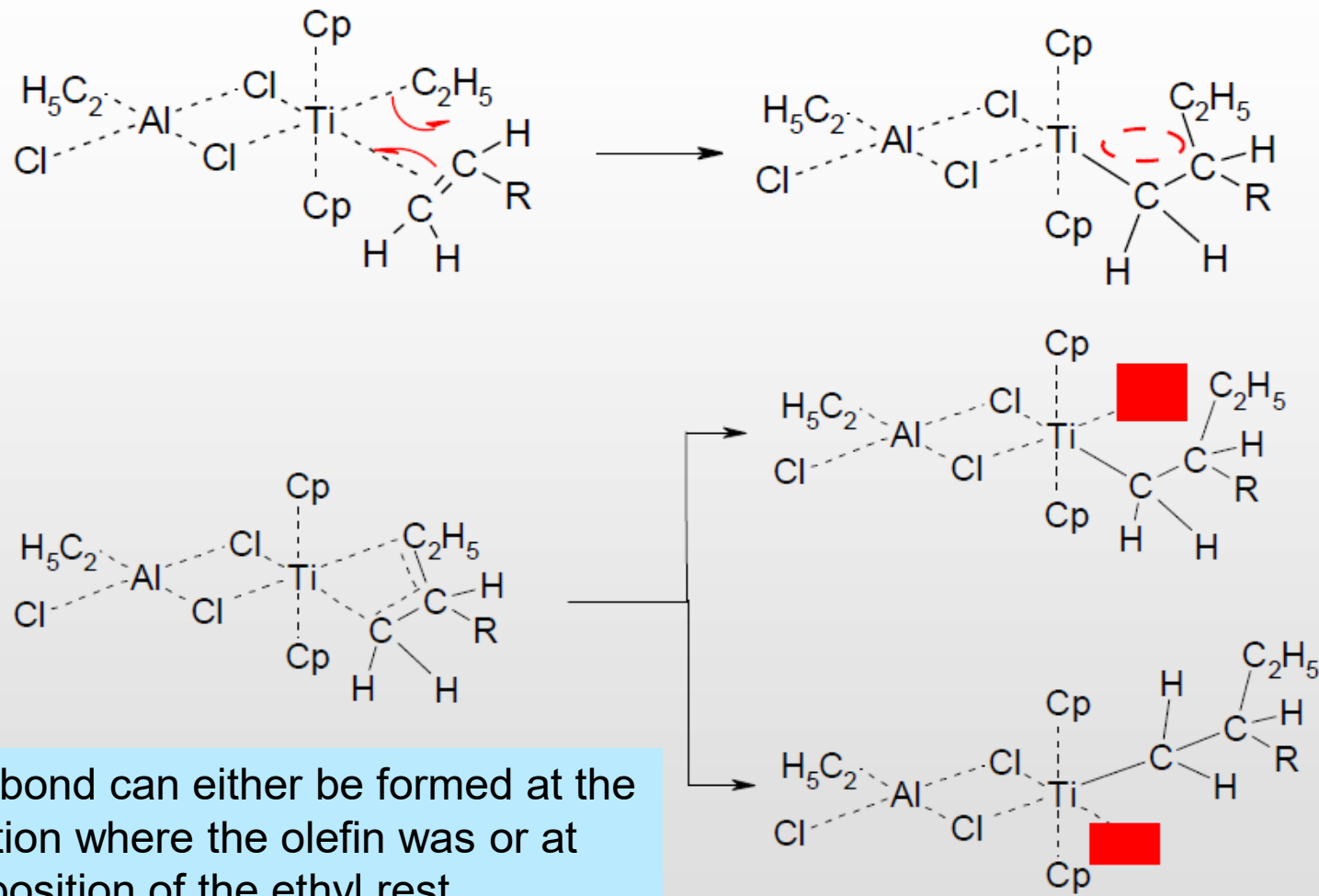
Reaction of metal alkyl (ethylaluminadichloride) with soluble transition metal compound (TiCl_4 , dicyclopentadienylethyl-titaniumchloride) leads to an octaedric Ti complex with an open coordination site as active center



Because of the coordination the $\text{Ti}-\text{C}_2\text{H}_5$ bond is destabilized and it can react with the double bond

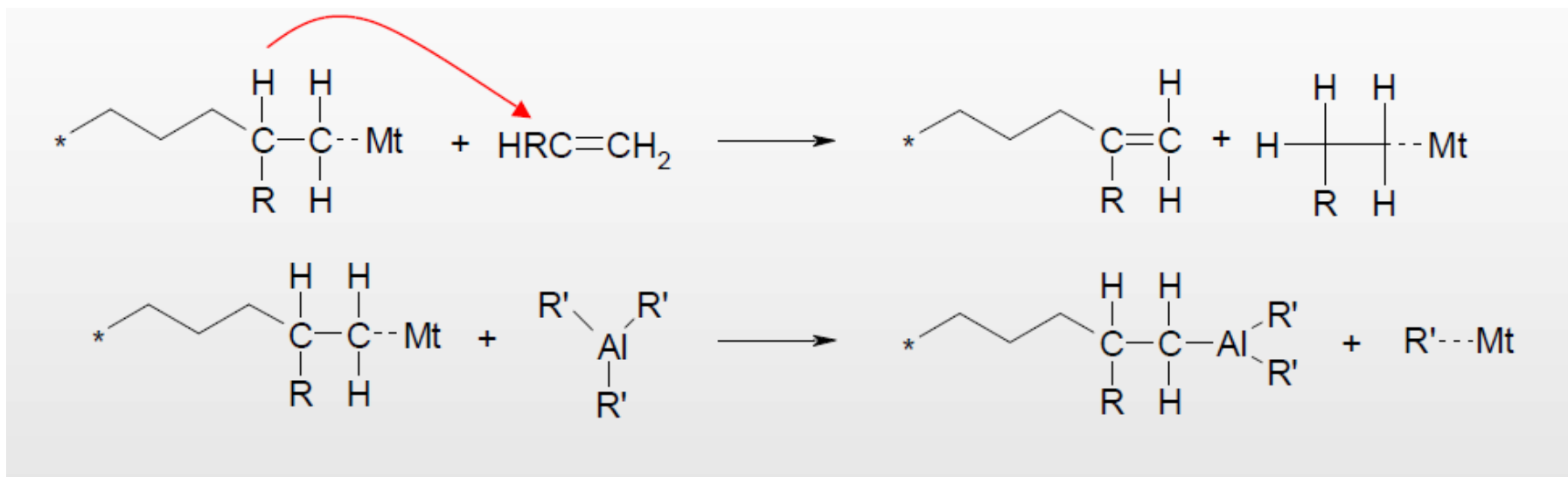
Ziegler Natta PE

Monomer insertion and formation of a new Ti-C bond



Ziegler Natta PE

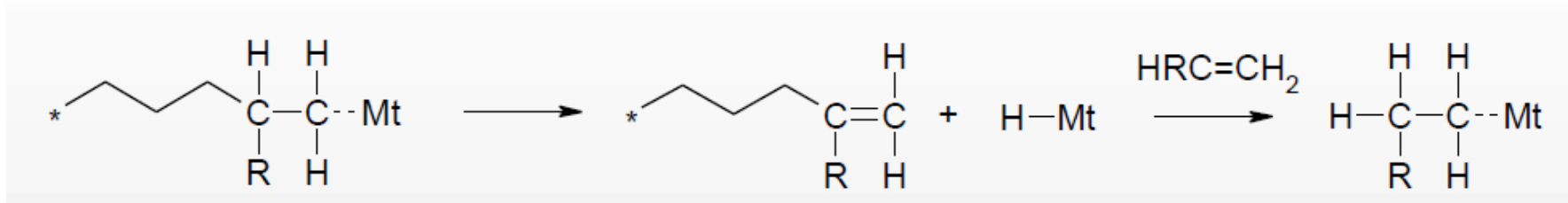
Chain transfer either to the monomer or to the Alkyl-alumina



Stable and soluble ZN catalysts (homogenous) lead to living polymerization (insertion) – no termination

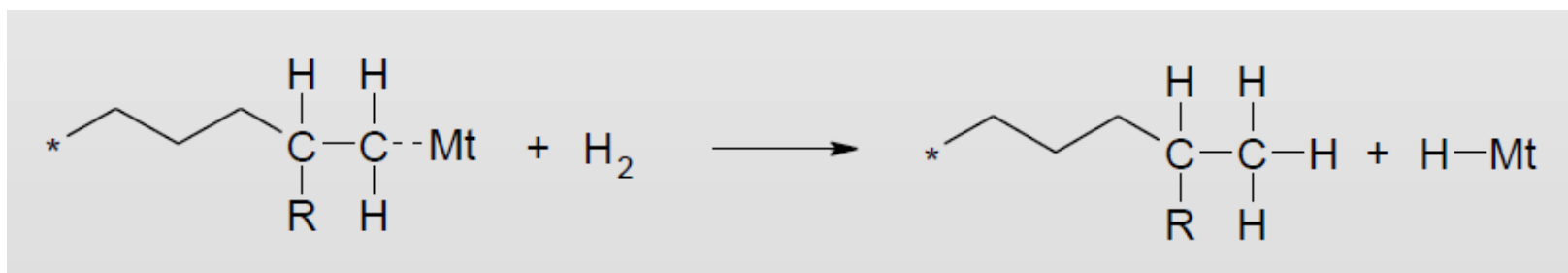
Ziegler Natta PE

β -elimination at higher temp under formation of transition metal hydrides



Re-alkylation of Mt
hydride through
monomer - transfer

Termination only be addition of H_2



Reactivity

Alkene	Formula	Relative reactivity
Ethylene	$\text{CH}_2=\text{CH}_2$	100 (reference)
Propylene	$\text{CH}_2=\text{CH}-\text{CH}_3$	$\sim 5-10$
1-Butene	$\text{CH}_2=\text{CH}-\text{C}_2\text{H}_5$	2.0
1-Pentene	$\text{CH}_2=\text{CH}-\text{C}_3\text{H}_7$	1.8
1-Hexene	$\text{CH}_2=\text{CH}-\text{C}_4\text{H}_9$	1.3
1-Decene	$\text{CH}_2=\text{CH}-\text{C}_8\text{H}_{17}$	0.8
1-Octadecene	$\text{CH}_2=\text{CH}-\text{C}_{16}\text{H}_{33}$	0.6
3-Methyl-1-butene	$\text{CH}_2=\text{CH}-\text{CH}(\text{CH}_3)_2$	0.2
3-Methyl-1-pentene	$\text{CH}_2=\text{CH}-\text{CH}-\text{C}(\text{CH}_3)_2-\text{C}_2\text{H}_5$	0.2
4-Methyl-1-pentene	$\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}(\text{CH}_3)_2$	0.8
4-Methyl-1-hexene	$\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{C}_2\text{H}_5$	0.6
Vinylcyclohexane	$\text{CH}_2=\text{CH}-\text{cyclo}-\text{C}_6\text{H}_{11}$	0.05
Styrene	$\text{CH}_2=\text{CH}-\text{C}_6\text{H}_5$	~ 1
Vinylnaphthalene	$\text{CH}_2=\text{CH}-\text{C}_{10}\text{H}_7$	~ 0.5

Metallocenes or Single Site catalysts

- Organo-metallic substances (mostly Zr, Hf as central atom)
- Metallocenes are most important class
- Only one catalytic center
- Basically homogenous catalyst, however, supporting for commercial polymerization processes necessary
- High catalyst activity
- Narrow molecular weight distribution
- Catalytic effect of Dicyclopentadienyltitan dichloride known since 1957
Activation with Methylaluminoxane (MAO) discovered by Kaminski (Hamburg) in 1980

Branching in HDPE / LLDPE

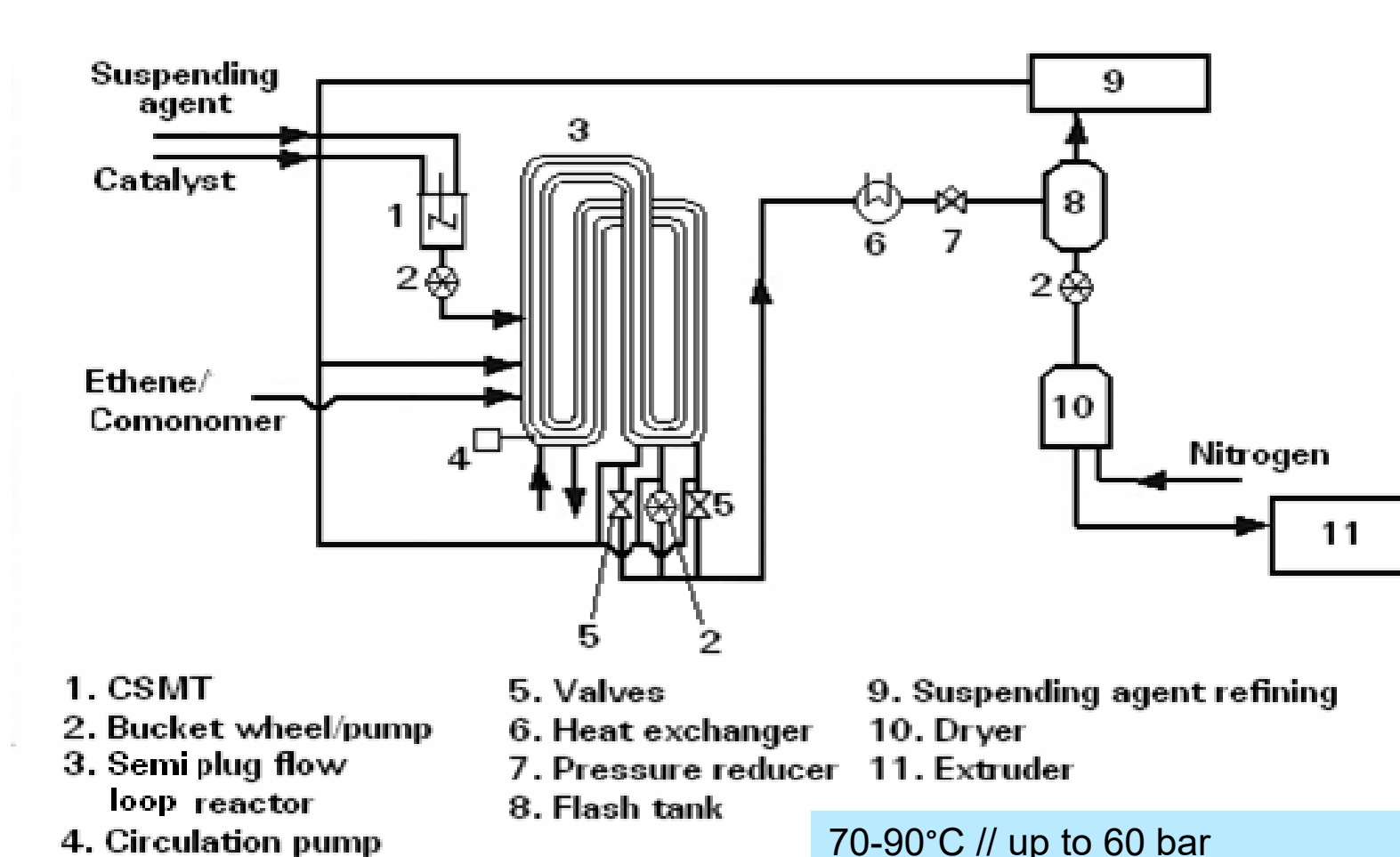
HDPE

- A small part of the ethylene is dimerized at the Z/N catalyst to 1-butene, then co-polymerized with the growing chain
- Side chains: ethyl

LLDPE

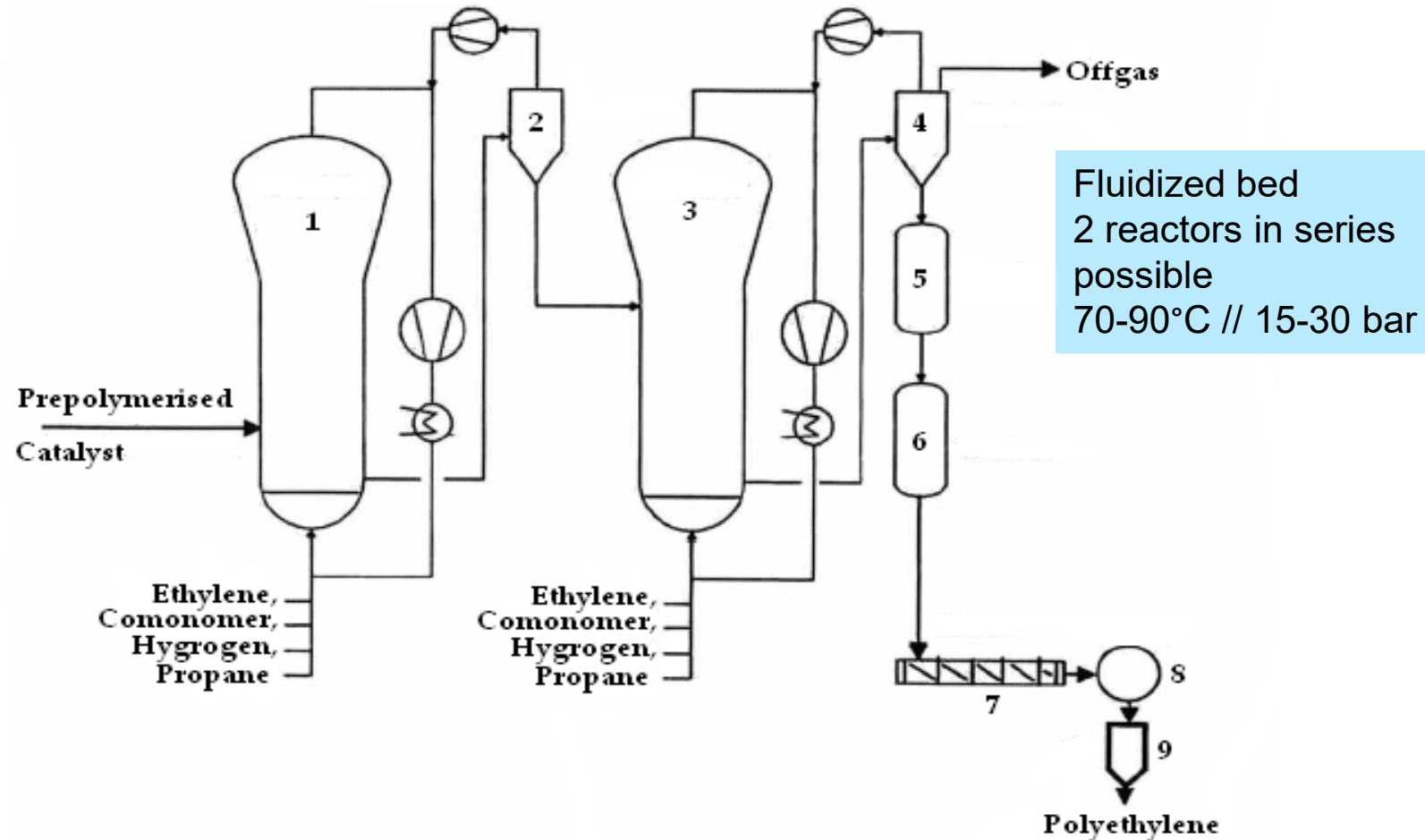
- Copolymerization of ethylene in presence of co-monomer
- Side chains: ethyl, butyl or hexyl (depending on co-monomer)

PE slurry loop



70-90°C // up to 60 bar
Lower density of loop product 918
Combination with gas phase

PE gas phase process

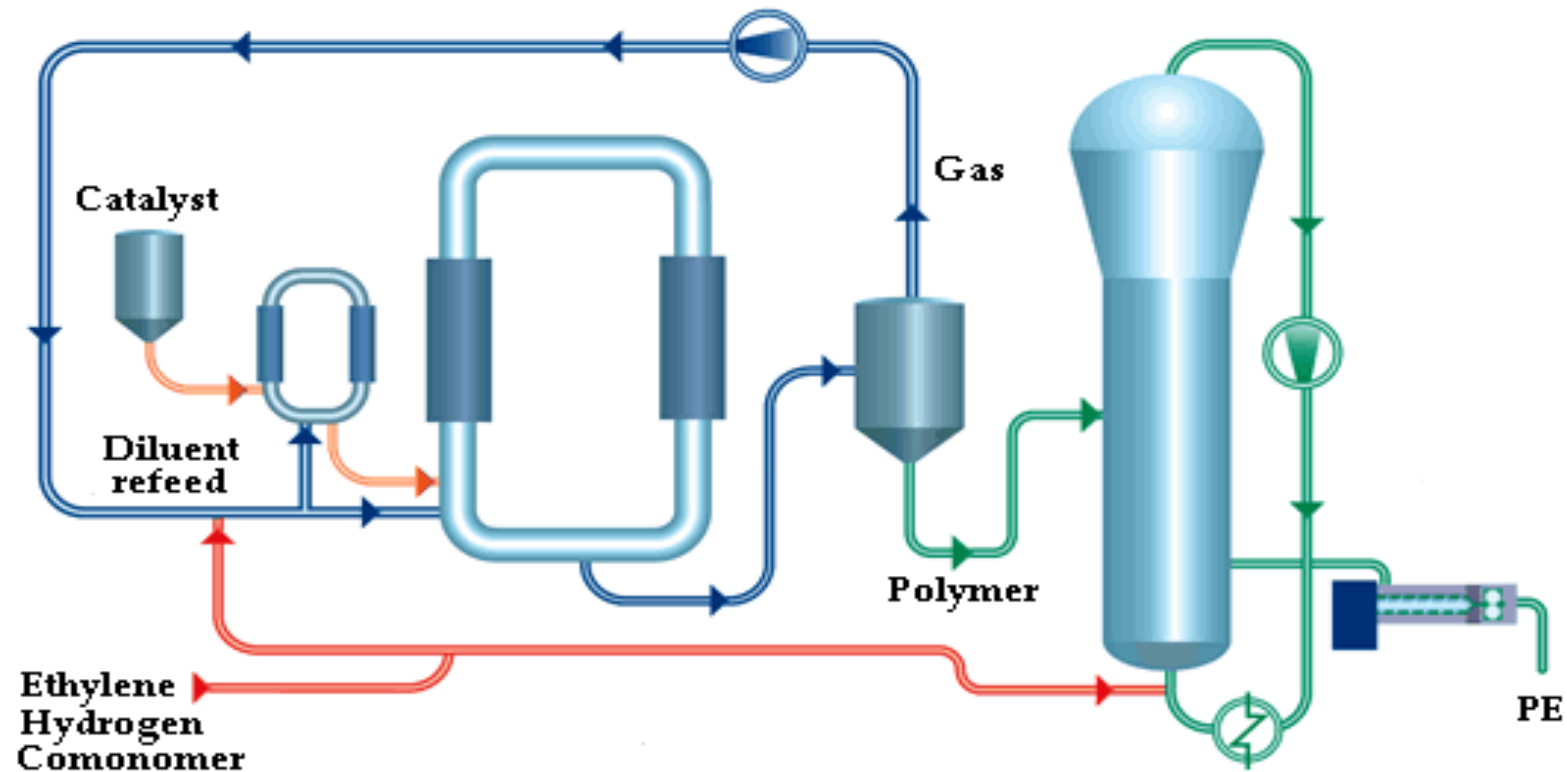


- 1. CSTR cascade 1
- 2. Stripper
- 3. CSTR cascade 2

- 4. Stripper
- 5. Treatment with steam
- 6. Dryer

- 7. Extruder
- 8. Dryer
- 9. Hopper

Borealis Borstar process



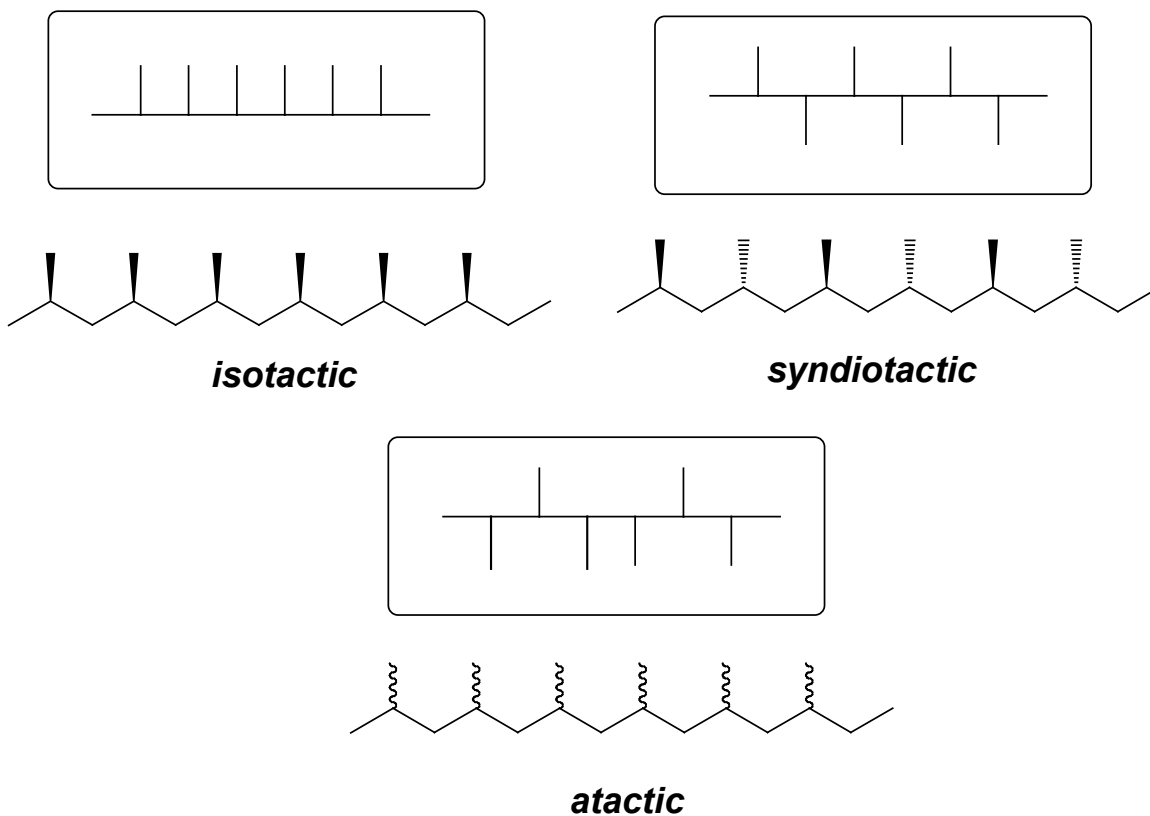
1. Prepolymerisation CSTR
2. Loop semi plug flow reactor
3. Flash
4. Gas phase CSTR cascade
5. Polymer degassing/pelletising

Polypropylene PP

- Economical material (monomer cost!)
- Low density (0,9 g/cm³)
- High melting point (homopolymer: 165°C)
- High solvent resistance
- Surface hardness and gloss
- Good processability and recyclability
- Modification in wide range possible
- Ecologically advantageous

Isomerism

Stereoisomerism (vinyl polymers)



Regular structure:
able to crystallize
PP, PB1, P4MP1, ...

Irregular structure:
amorphous
PS, PMMA, ...

**PVC (largely atactic,
some syndiotactic
sequences)**

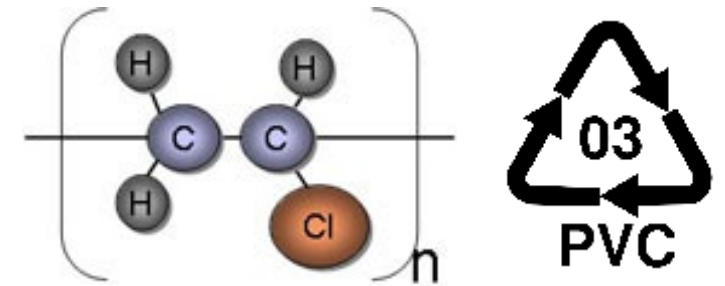
Development of PP catalysts

Generation (year)	Catalyst composition	Productivity (kg _{pp} /g _{cat})	I.I. ^b %	mmmm %	M _w /M _n
First (1954)	δ-TiCl ₃ ·0.33AlCl ₃ + AlEt ₂ Cl	2–4	90–94		
Second (1970)	δ-TiCl ₃ + AlEt ₂ Cl	10–15	94–97		
(1968)	MgCl ₂ /TiCl ₄ + AlR ₃	15	40	50–60	
Third (1971)	MgCl ₂ /TiCl ₄ /benzoate + AlR ₃ /benzoate	15–30	95–97	90–94	8–10
Fourth (1980)	MgCl ₂ /TiCl ₄ /phthalate + AlR ₃ /silane	40–70	95–99	94–99	6.5–8
Fifth (1988)	MgCl ₂ /TiCl ₄ /diether + AlR ₃	100–130	95–98	95–97	5–5.5
	MgCl ₂ /TiCl ₄ /diether + AlR ₃ /silane	70–100	98–99	97–99	4.5–5
“Next” (1999)	MgCl ₂ /TiCl ₄ /succinate + AlR ₃ /silane	40–70	95–99	95–99	10–15

^a(Polymerization conditions: liquid propylene, 70°C, H₂).

^bIsotactic Index.

PVC

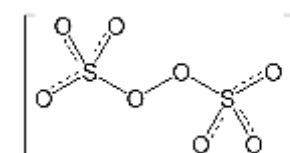


- Widely used thermoplastic
- Over 50% is used in construction
- Hard PVC: magnetic stripe cards, window profiles, gramophone records, pipe, ...
- Soft PVC: by addition of plasticizers it can be made more flexible and soft – clothing, hoses, flooring, roofing membranes, cable
- 1935 start of PVC production at IG Farben



PVC

- **Bulk or mass polymerization** (precipitation)
Discontinuous polymerization in STR (2 steps)
Heat removal difficult
- **Emulsion polymerization**
Continuous polymerization at 40-60°C and 1 to 2 MPa in presence of potassium persulfate in STR (no cascade, yield 50-80%)
Work-up: spray drying, product contains rests of initiator and emulsifier



Emulsion polymerization

- Monomer and polymer not soluble in water, initiator soluble in water
- The monomers are dispersed in water in the presence of surfactants
- The surfactants adsorb on the surface of the monomer droplets, stabilizing them
- Ionic surfactant stabilizes the droplets by electrostatic repulsion, whereas non-ionic surfactants provide steric stabilization
- Most initiators are water-soluble, therefore the radicals are formed in the aqueous phase
- These radicals are often too hydrophilic to directly enter into the organic phases

PVC

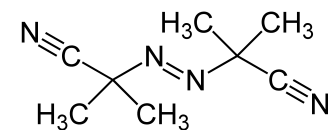
- **Suspension polymerization**

Discontinuous polymerization at 40-75°C and 1 to 2 MPa in presence of organic peroxides or AIBN in large STR (100-400 m³)

Yield: 90-95%

Work-up: filtration

Products are very clean

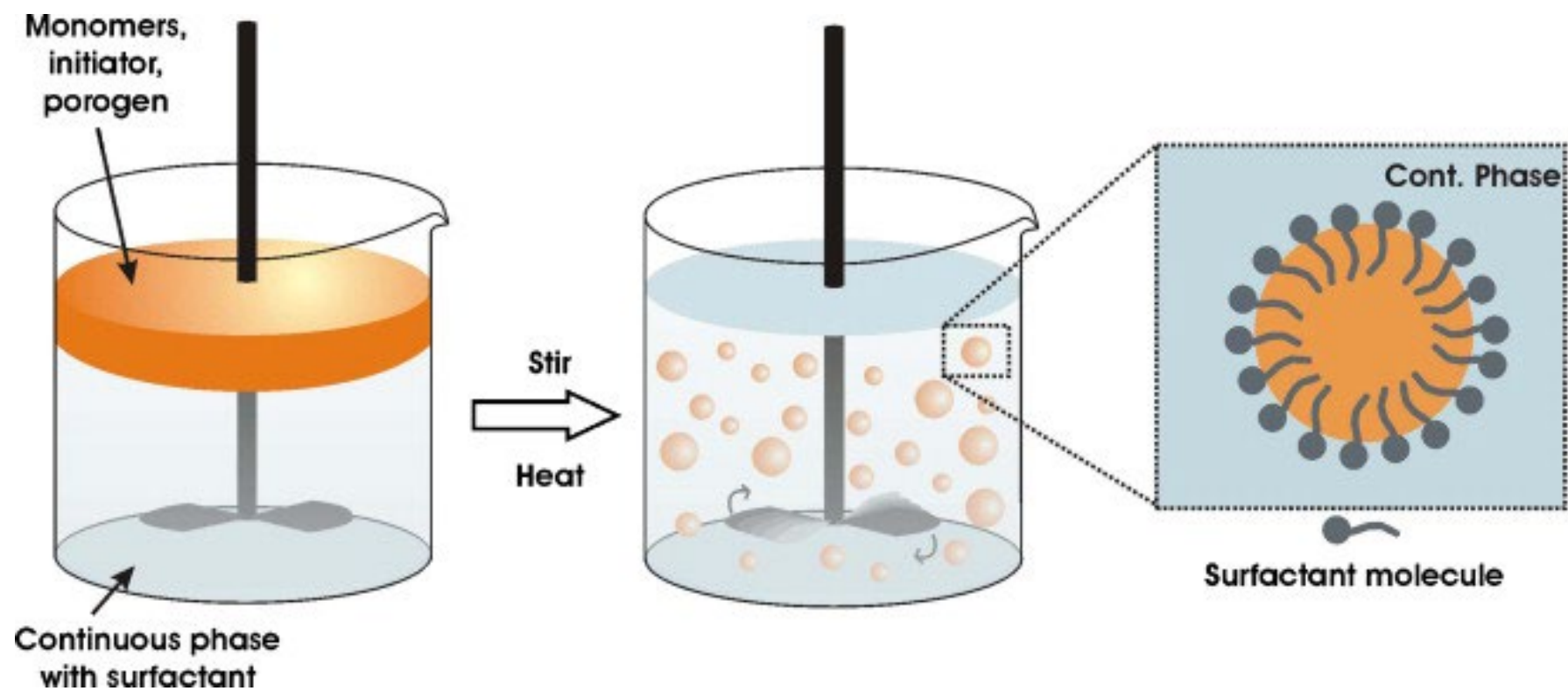


Azobisisobutyronitrile (AIBN)

Suspension polymerization

- **Monomer and polymer not soluble in water, initiator soluble in the monomer**
- The polymerization occurs in the monomer droplets that are progressively transformed into sticky, viscoelastic monomer–polymer droplets and finally into rigid, spherical polymer particles in the size range of 50–500 μm
- The polymer solids' content in the fully converted suspension is typically 30–50% w/w
- In the inverse suspension polymerization, the hydrophilic monomer(s) (e.g., acrylamide, acrylic acid) and initiator are dispersed in the hydrophobic continuous organic phase (e.g., hexane, paraffin oil)

Suspension polymerization



Initiators for PVC

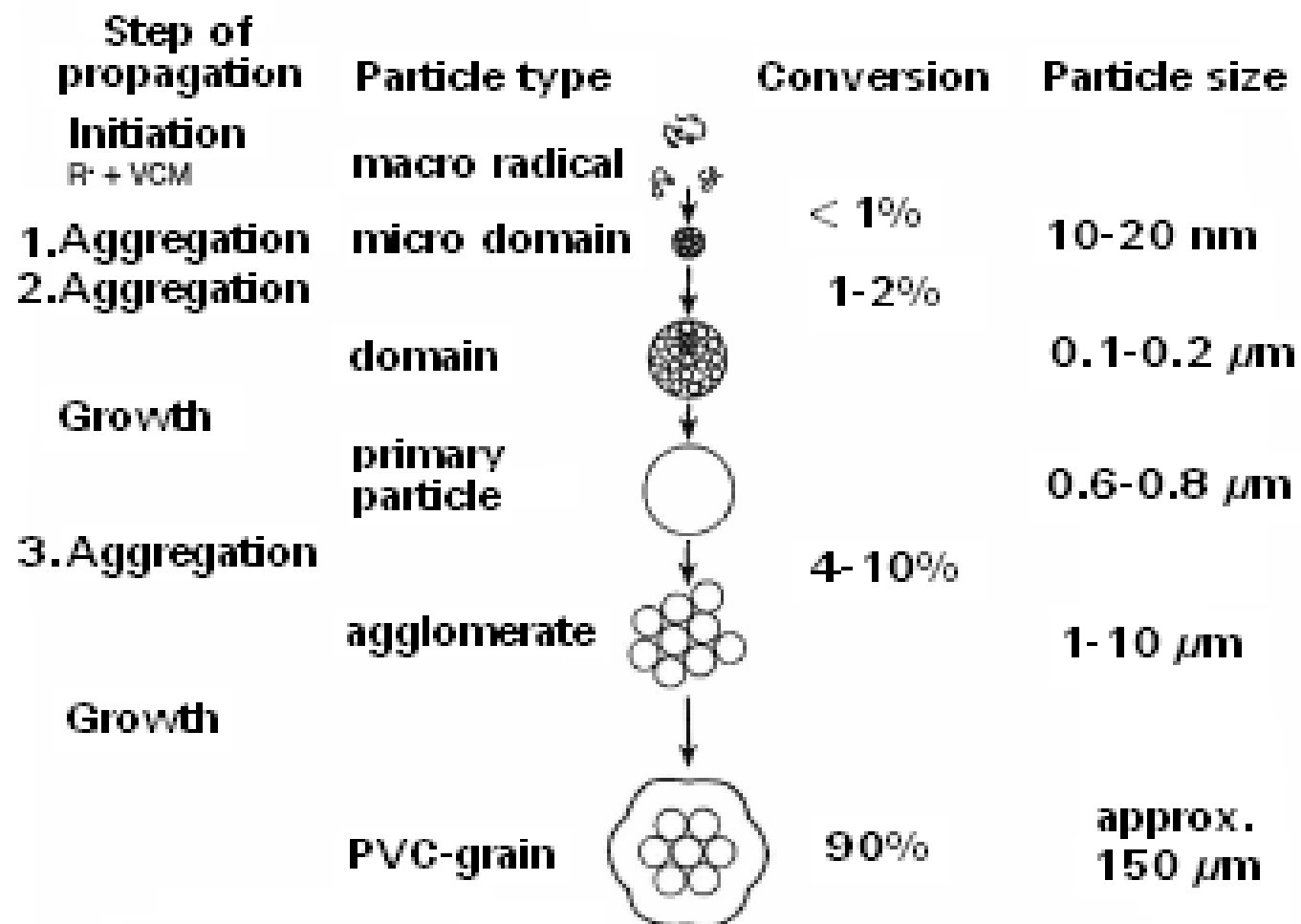
No.	Name	Storage temperature ^a , °C	<i>M</i> _r	Supplied form	<i>T</i> _{1/2} , °C ^b			<i>t</i> _{1/2} , min ^c		
					10 h	1 h	6 min	50 °C	60 °C	70 °C
<i>Diacyl peroxides</i>										
1	acetylcyclohexane sulfonyl peroxide		— 15 222	solution	37	51	66	70	15	
2	dioctanoyl peroxide		10 286	solid	61	79	100		700	190
3	didecanoyl peroxide	25	343	solid	60	79	99		700	190
4	dilauroyl peroxide	30	398	solid	62	80	99		850	220
<i>Peroxydicarbonates</i>										
5	bis(2-ethylhexyl) peroxydicarbonate	— 15	346	solution	44	61	80	250	70	21
6	dicyclohexyl peroxydicarbonate	10	286	solid	44	59	76	250	58	16
7	dicetyl peroxydicarbonate	25	571	solid or dispersion	45	61	79	320	70	20
8	bis(4- <i>tert</i> -butyl cyclohexyl) peroxydicarbonate	30	398	solid or dispersion	43	60	78	250	60	18
9	dimyristyl peroxydicarbonate	20	515	solid	46	62	79	320	70	20
<i>Alkyl peroxyesters</i>										
10	<i>tert</i> -butyl peroxyneodecanoate	— 5	246	solution	42	62	84	250	70	25
11	<i>tert</i> -amyl peroxyneodecanoate	— 5	260	solution	30	59	83	560	180	55
12	cumyl peroxyneodecanoate	— 20	306	solution	36	54	74	100	30	9
13	<i>tert</i> -butyl peroxy-pivalate	— 5	174	solution	51	71	94	700	200	70
<i>Azo initiator</i>										
14	Azobisisobutyronitrile (AIBN)	20	164	solution	59	79	101		500	170

^a Maximum.

^b Temperature at which $t_{1/2}$ = 10 h, 1 h, or 6 min, as indicated.

^c Approximate.

PVC particle growth in bulk polymerization



Use of PVC

- **Hard PVC**

Extrusion at temperatures of 160-180°C to avoid decomposition

Stabilizer: earth alkali soaps, organic tin compounds, epoxidized vegetable oils

T_G : about 80°C

- **Soft PVC**

Compounding with plasticizer (20-35%) and stabilizers at 160-180°C, then extrusion at about 130-150°C

T_G : about -10°C

PVC physical properties

Property	Hard-PVC	Soft-PVC
Density [g/cm]	1,38-1,50	1,16-1,30
Tensile modulus [N/mm ²]	50-65	10-25
Elongation at break [%]	20-100	170-400
T _G [°C]	80	-10
Electr. Resistance [Ω]	10 ¹⁵	10 ¹¹

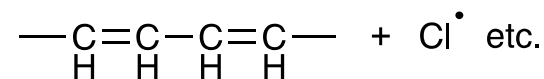
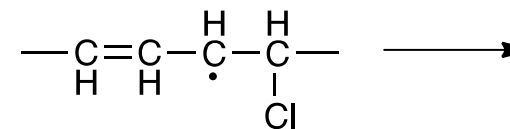
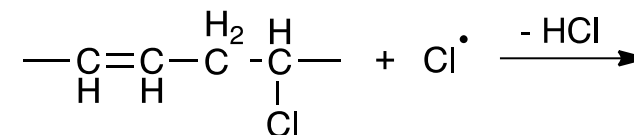
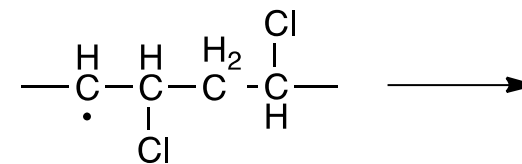
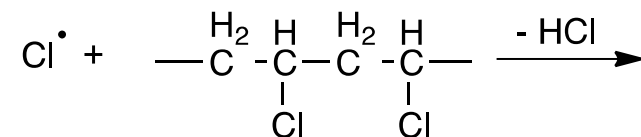
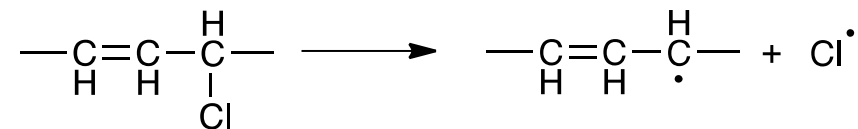
Plasticizer for PVC

- Phthalic acid ester
- Alkyl sulfonic acid aryl ester
- Phosphoric acid aryl ester
- EVA copolymers
- Chlorinated PE

Stabilization of PVC

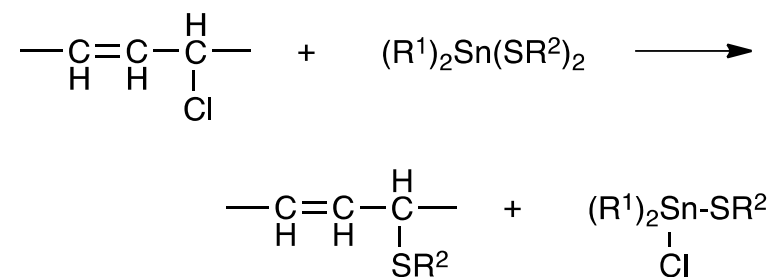
PVC is thermally unstable

- Heating results in elimination of HCl, the formation of polyene sequences, and rapid discoloration
- This autocatalytic reaction begins at ca. 100°C
- Special heat stabilizers are therefore required

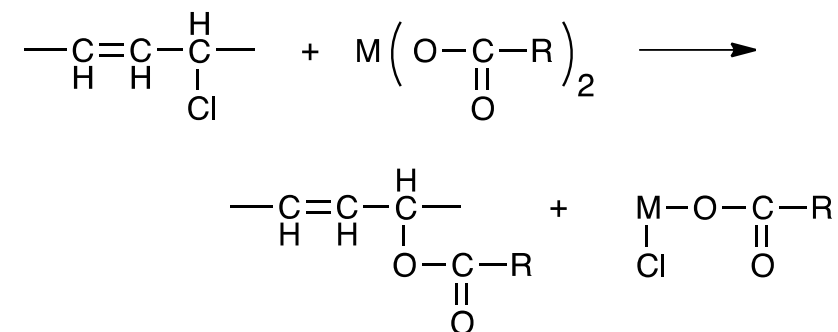


Mechanism of PVC stabilization

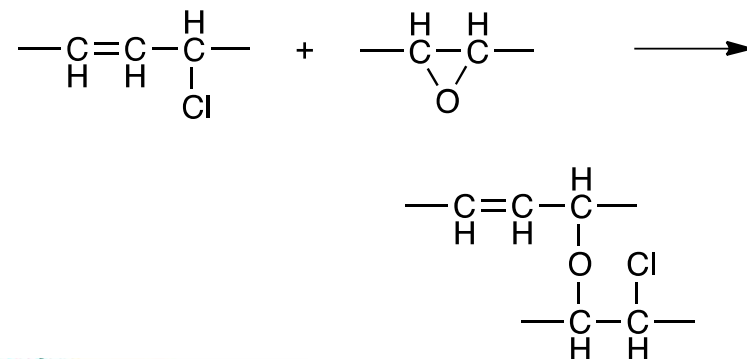
Tin mercaptide



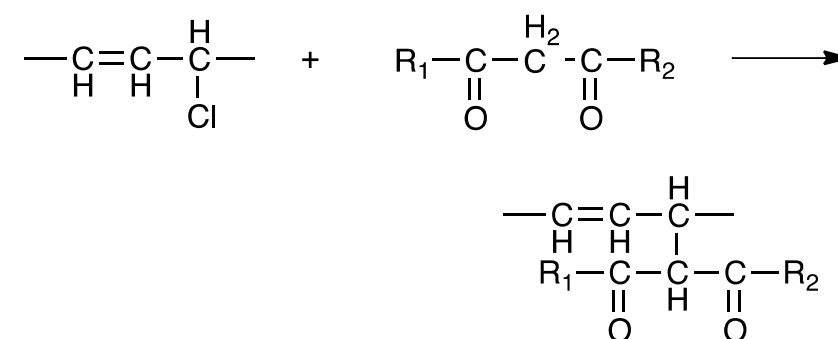
Metal carboxylate



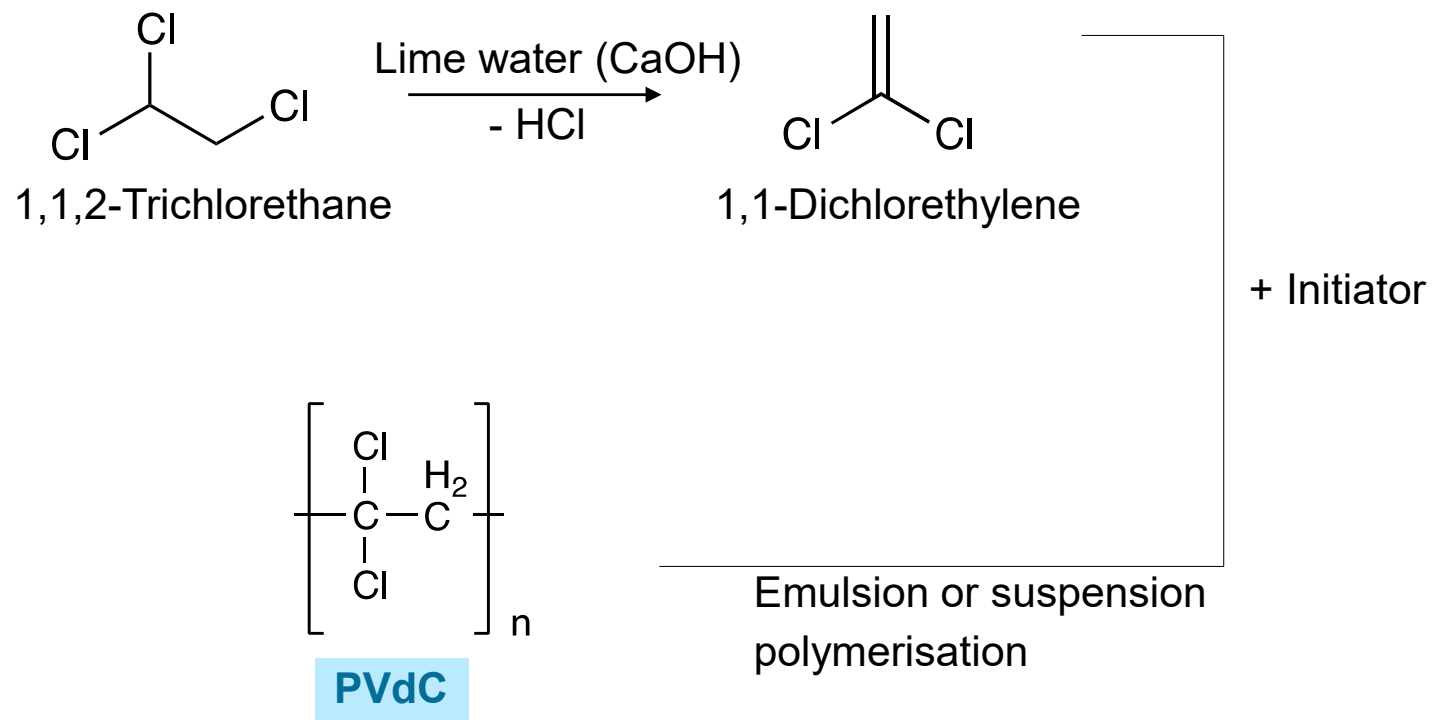
Epoxy compound



1,3-Diketone



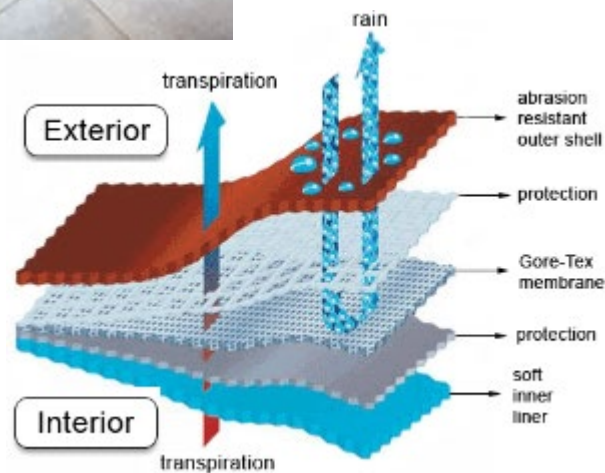
Polyvinylidenechloride (PVdC)



Copolymers with: vinyl acetate, vinyl chloride, acrylic acid ester, acrylonitrile

PTFE

1938: R.J. Plunkett – PTFE



Fluor polymers

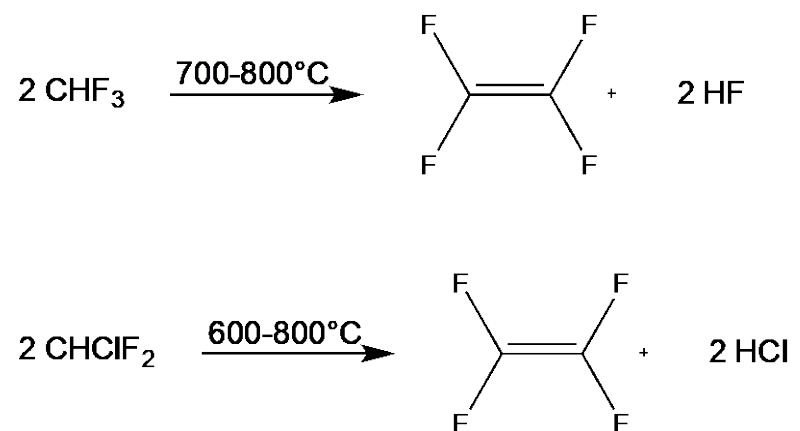
Polymer	Monomer	Production	Properties / use
PTFE	$\text{CF}_2=\text{CF}_2$	Suspension- or emulsion polymerization at 30°C and redox initiators	T_M : 320-345°C Use: 200-260°C Isolation, coating, fibers
Polyvinylfluoride	$\text{CH}_2=\text{CHF}$	In aqueous systems at 50-150°C and elevated pressures	T_M : 185-190°C Use: -70-110°C films and laquers
PVDF Polyvinylidene fluoride	$\text{CH}_2=\text{CF}_2$	As PTFE	T_M : 168-172°C Use: -30-150°C thermoplastic conversion, films and profiles
Polytrifluorochloroethylene	$\text{CF}_2=\text{CFCl}$	Suspension- or emulsion polymerization	T_M : 210-212°C Use: -200-180°C thermoplastic conversion, films, boards, pressure resistant gaskets

Polytetrafluoroethylene

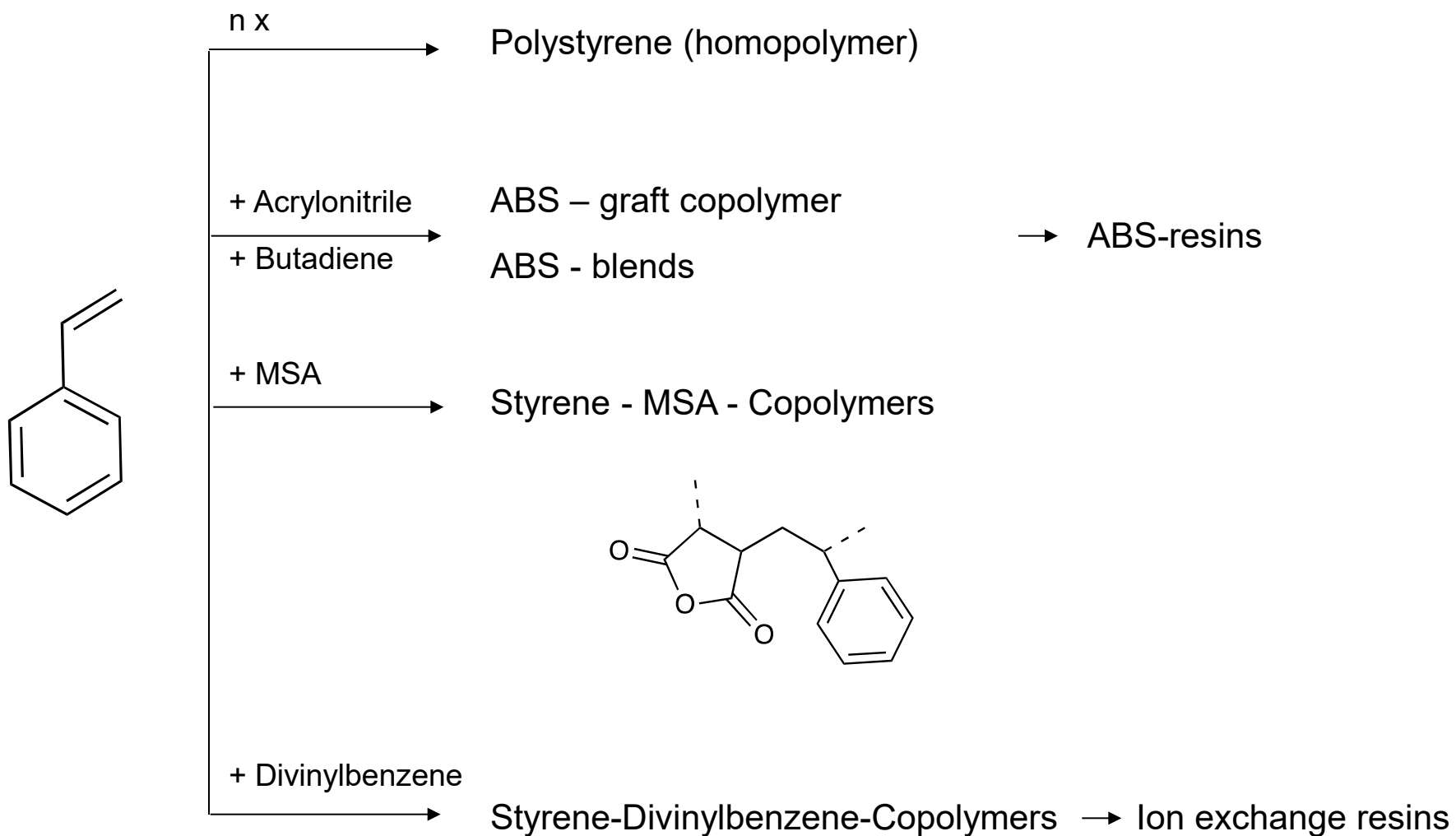
- **Granular PTFE** is prepared by **suspension polymerization**. The coarse product is cut to uniform particle size and processed by powder metallurgy techniques.
- **Fine powder resins** are prepared by **dispersion polymerization** and coagulation by high-speed agitation. The fine powder resins consisting of uniform particles of 0.2 μm diameter are mixed with a lubricant such as kerosene and processed by a cold extrusion method called paste extrusion.
- The third form of PTFE is an **aqueous dispersion**. The dispersion products are processed by latex processing technique (dip coating or spray coating) followed by baking at high temperature which causes coalescence into a continuous film.

Production of tetrafluoroethylen

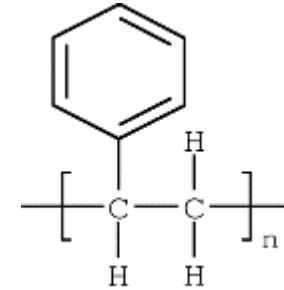
The two principal commercial methods are **pyrolysis of trifluoromethane and chlorodifluoromethane**



Styrene polymers



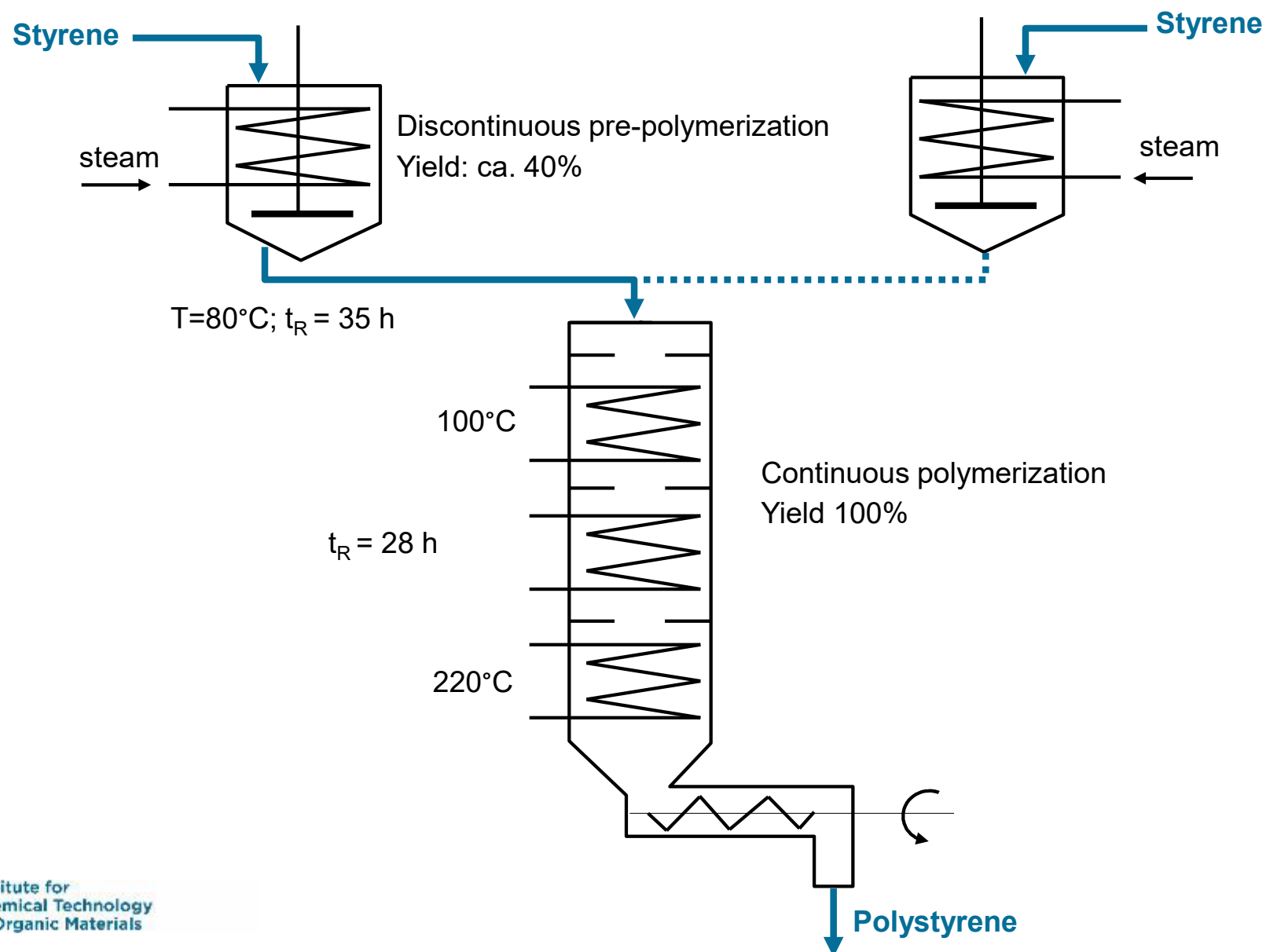
Polystyrene



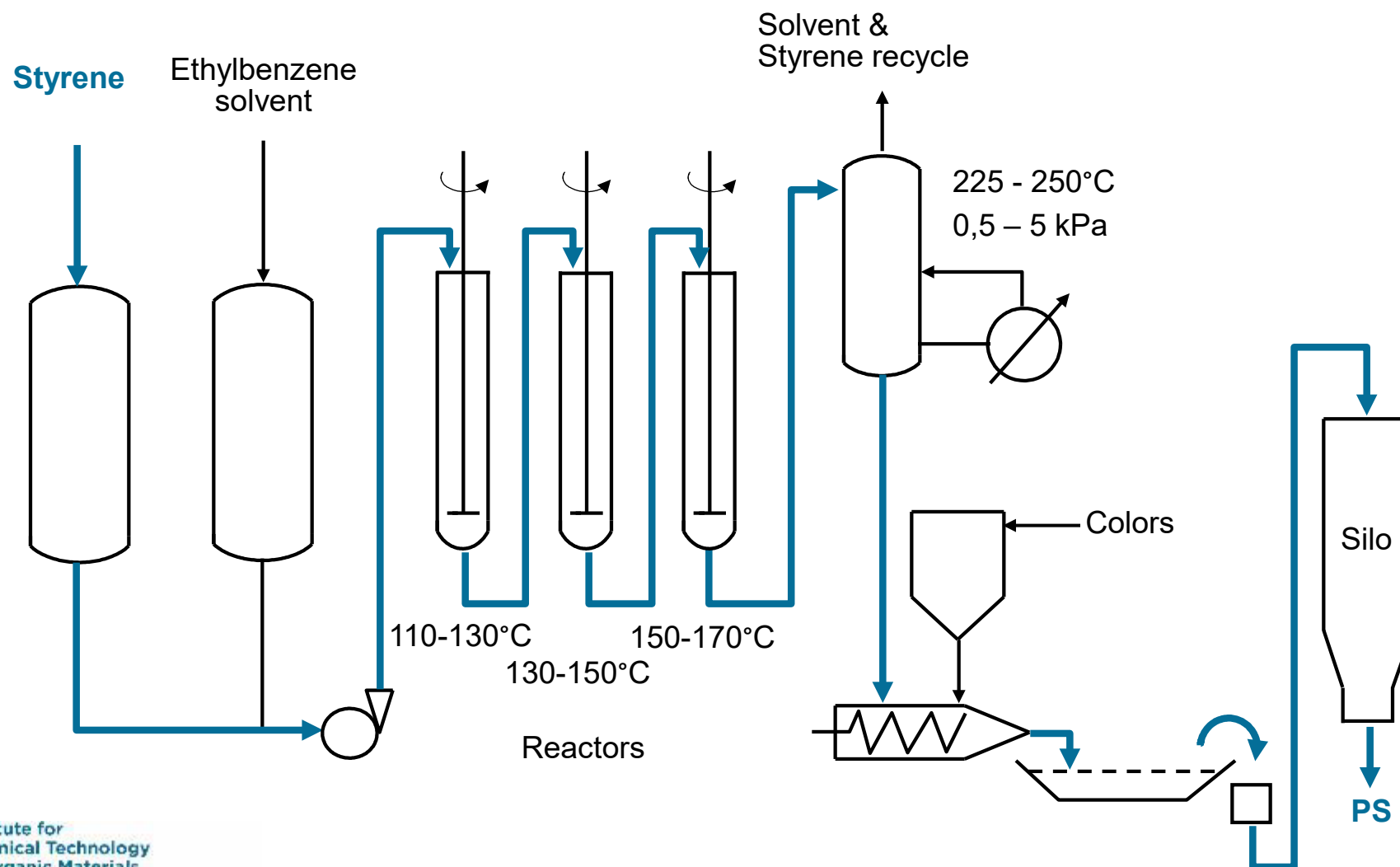
- PS is a transparent, hard plastic with limited flexibility
- Commonly used as expanded PS (EPS), Styrofoam
- Co-polymerization with butadiene and mixing with Poly-butadiene results in HIPS (high impact PS)
- Density: 1050 kg/m³
- Young's modulus: 3000 – 3600 MPa
- T_G: 95°C
- T_M: 240°C



PS thermal process

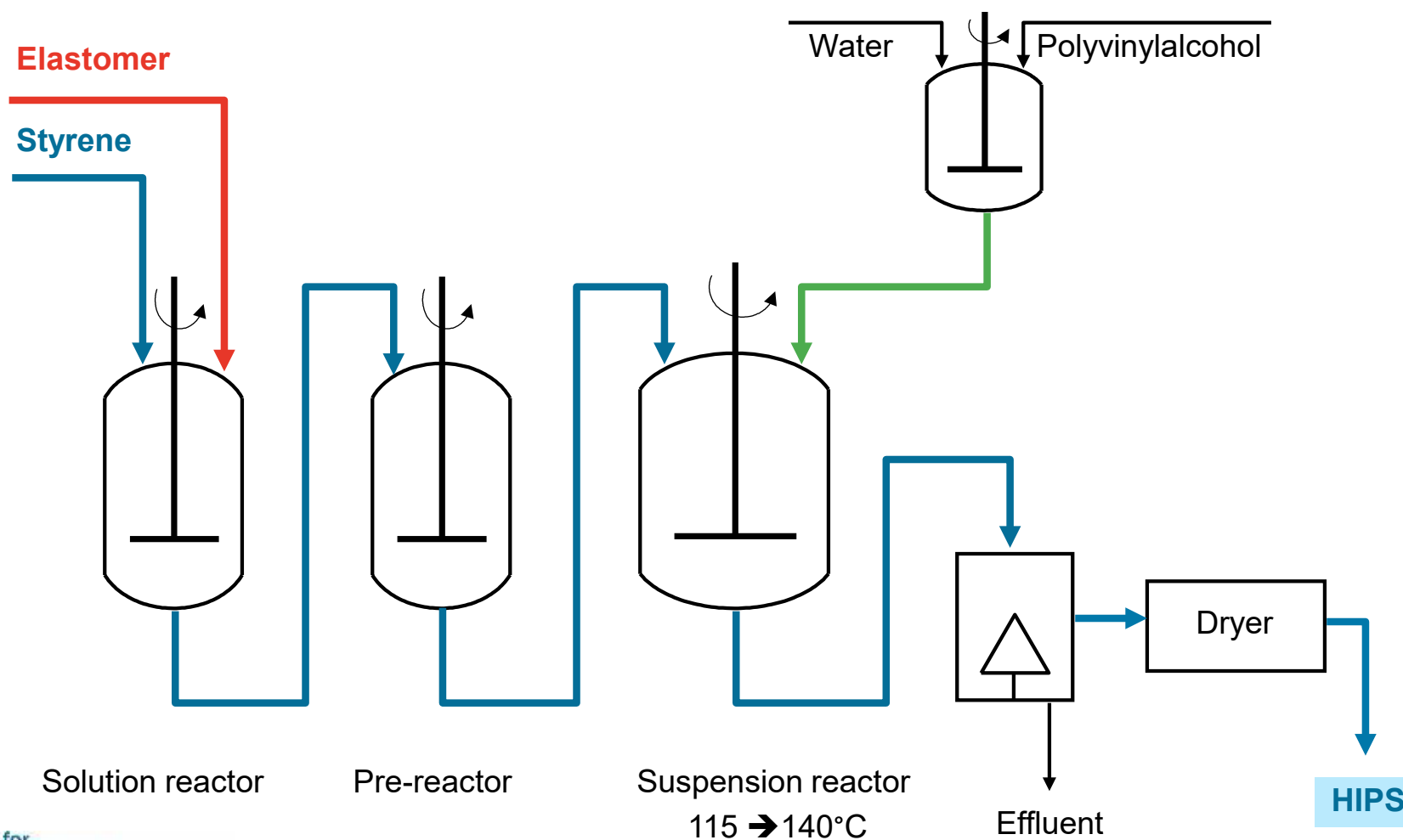


PS continuous solution process



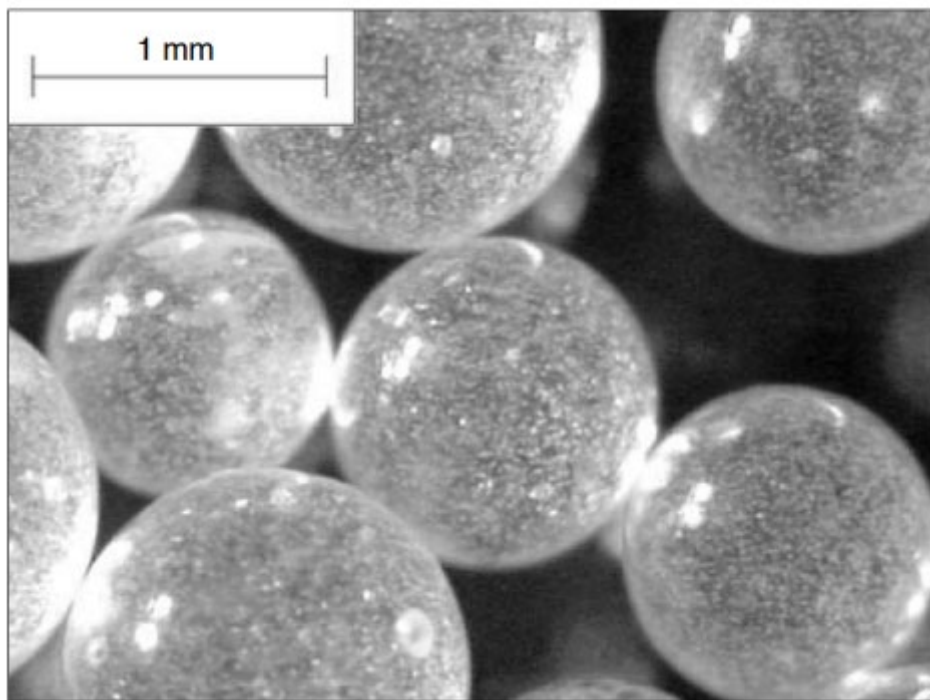
PS suspension polymerization

Elastomer modified PS (HIPS) – mass suspension polymerization



Expanded PS (EPS)

- Suspension polymerization in the presence of a blowing agent (e.g., pentane)
- It is also possible to introduce the blowing agent to the polymer after polymerization and allowing it to diffuse into the beads



- Once the beads are hard, the reaction mixture is heated to a temperature above the glass transition temperature of the PS, 100°C
- During heating, the reactor is pressurized with a blowing agent (usually n-pentane) at 5–8% w/w
- Subsequently, the reactor is pressurized with nitrogen at 7–9 bars and the so-called impregnation stage starts
- n-pentane diffuses into the “beads”
- The system is cooled down to 20–30°C, so that no bead expansion can take place during the discharge
- In the next stage, the excess of stabilizer is chemically removed
- In the final processing, the EPS beads are warmed up to 80–110°C, generally with steam that causes the beads to expand by foaming and their volume to increase by a factor of 30–50

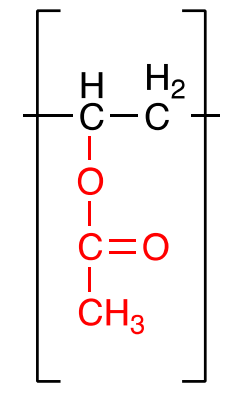
Polyvinylester

- **Poly vinyl acetate**

Emulsion polymerization, w/o work-up useable as dispersion

Glues, paintings, base material for Poly vinyl alcohol

Low hydrolysis resistance

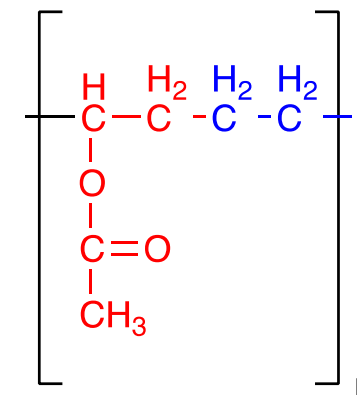


- **Ethylene vinyl acetate copolymers**

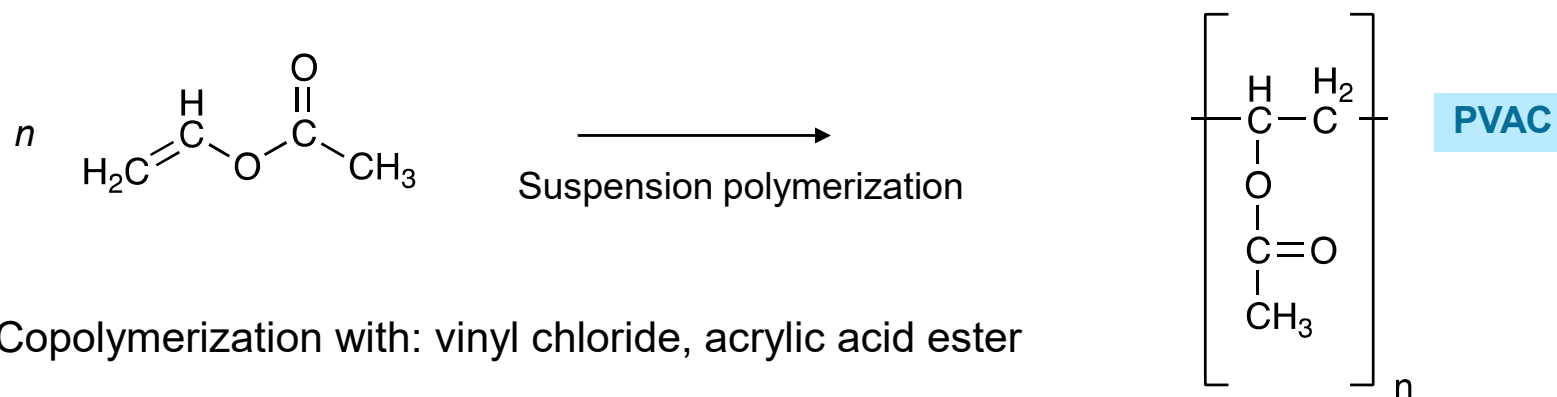
Vinyl chloride – vinyl acetate copolymers

Ethylene content > 60% high pressure process

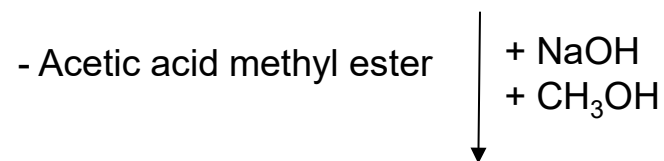
Ethylene content < 40% solution polymerization



Polyvinylalcohol

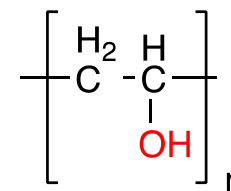
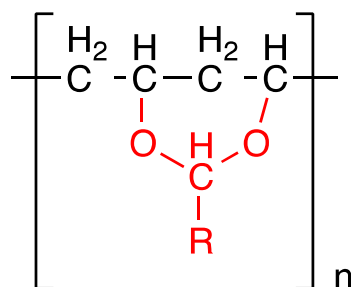


Copolymerization with: vinyl chloride, acrylic acid ester



Polyvinyl acetale

Polyvinyl formiate
Polyvinyl butyrate

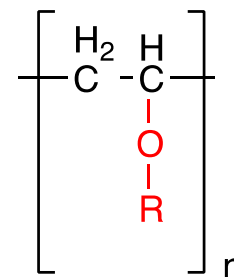


Polyvinyl alcohol

Emulsifier, coatings, sizing
for textiles, water soluble films,
glues

Polyvinylether

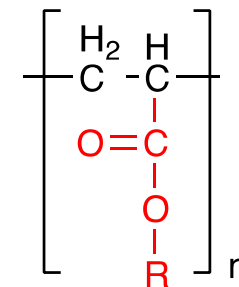
- Cationic polymerization:
BF₃, AlCl₃ in propane
- Use:
paintings, glues / adhesives, plasticizer
resistant against diluted acids and bases



R: Methyl-, Ethyl-, Isobutyl-

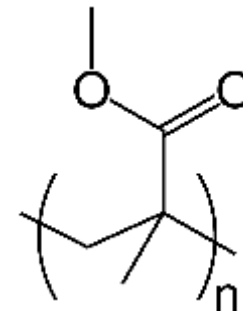
Polyacrylates

- Radical polymerization
- Emulsion- and solution processes
- Excellent UV and hydrolysis resistance
- Resistant against diluted acids and bases
- Use: paintings, glues, adhesives, plasticizer



R: Methyl-, Ethyl-,
n-Butyl-, 2-Ethylhexyl-

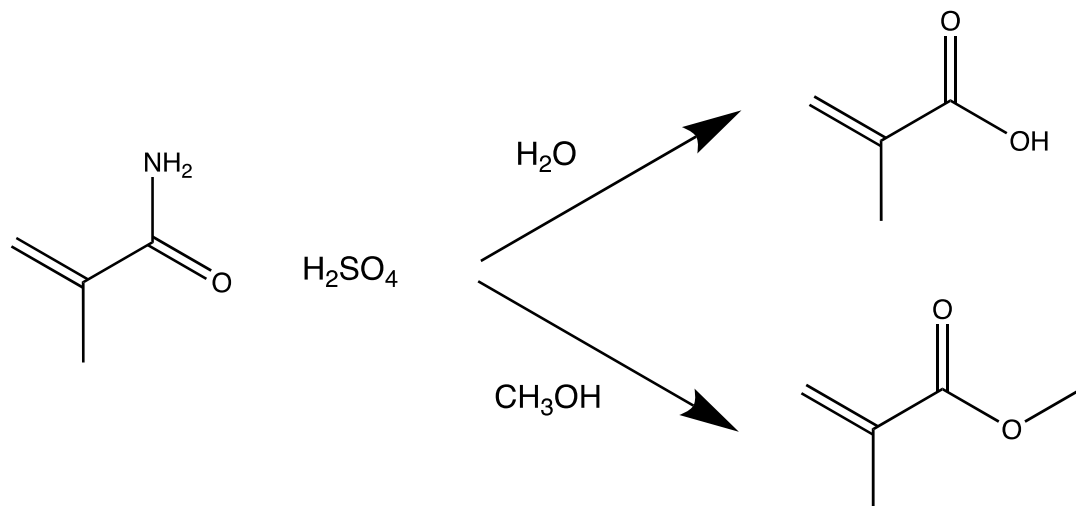
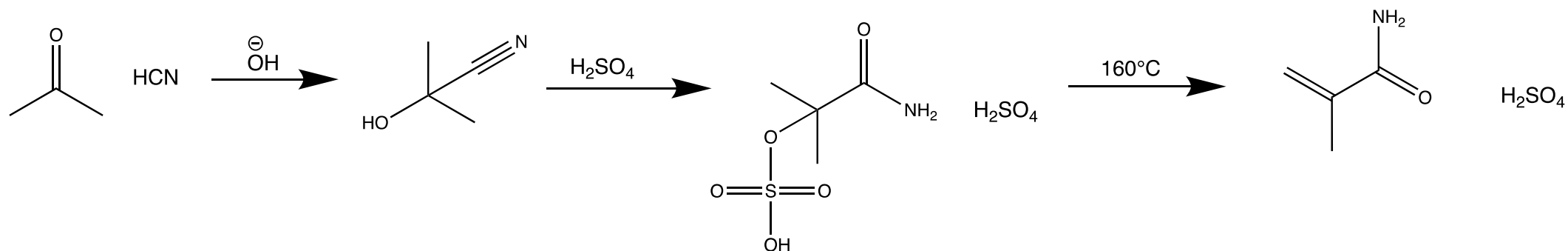
Polymethylmethacrylate (PMMA)



- Radical polymerization, mass polymerization
- Can be initiated by light, heat or radical forming agents
- Density: 1150 - 1190 kg/m³
- T_M: 130 - 140°C
- Highly transparent material, often used as an alternative to glass
- In competition with PC it is better for outdoor applications (cheaper)
- Use: artificial glass, dispersions, dental



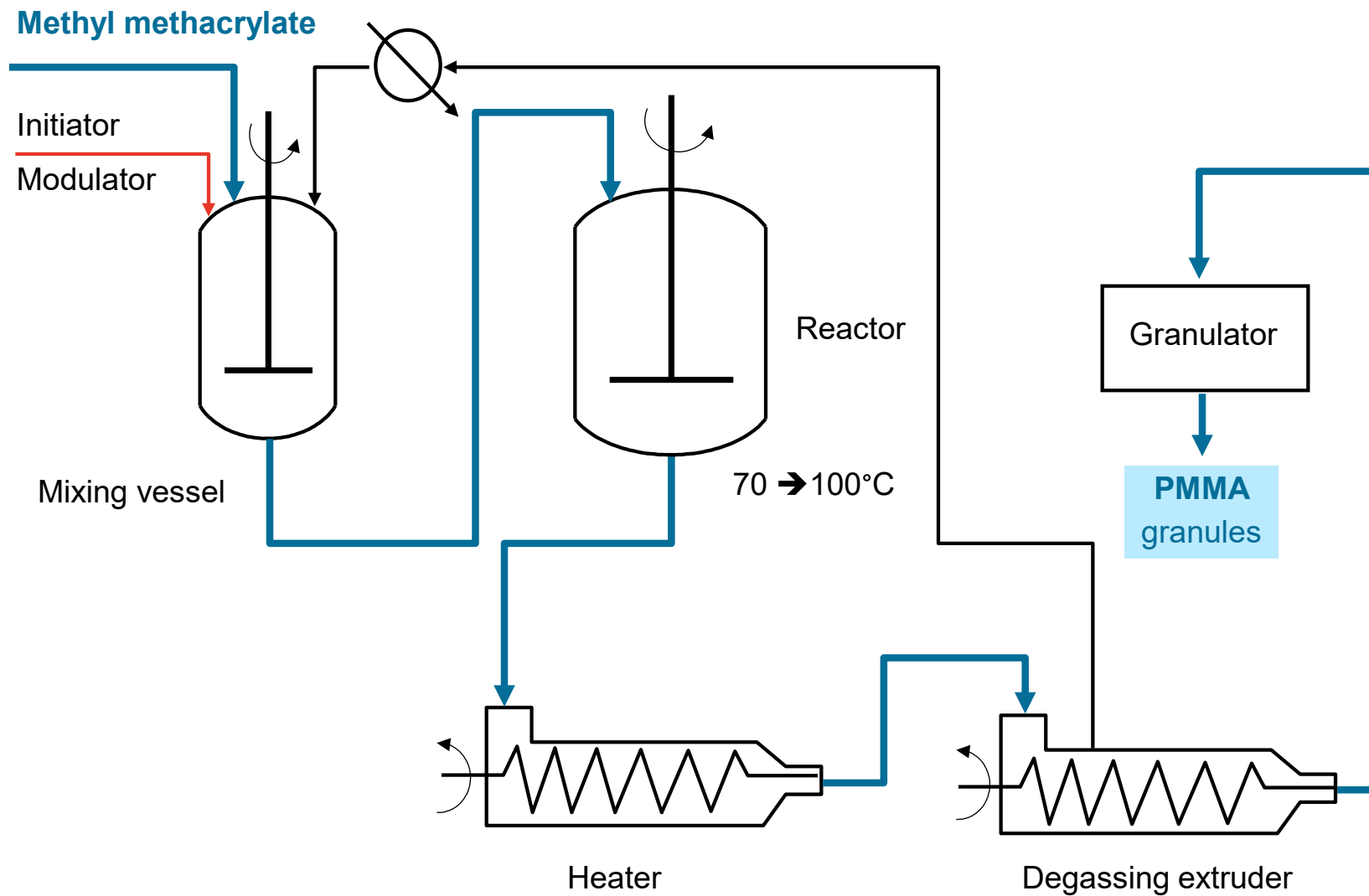
Methacrylic acid / Methylmethacrylate via cyanhydrin



Methacrylic acid / Methylmethacrylate: other routes

- From isobutene: oxidized to methacrolein and in a second stage to methacrylic acid
- From isobutyric acid:
 - Carbonylation of propene to isobutyric acid
 - Oxidative dehydrogenation to methacrylic acid
- From ethylene:
 - Ethylene to propanal (hydroformylation)
 - Condensation of propanal with formaldehyde to give methacrolein
 - Oxidation of methacrolein to methacrylic acid
- However, the most important route is still the methanolysis of methacrylamide sulfate

PMMA for moldings



Rubbers & elastomers

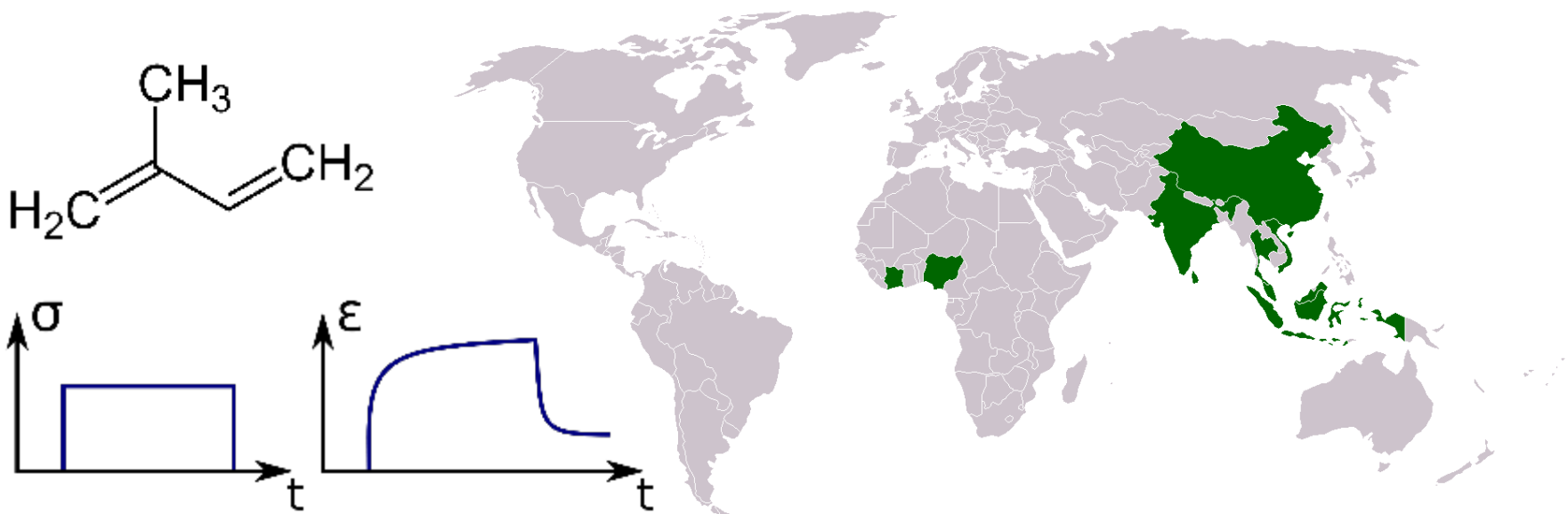
- 1600 BC: 1st artifacts made from rubber in Middle America
- 1615: first rubber materials in Europe
- 1770: Priestley uses rubber as eraser
- 1783: 1st balloon flight, the silk was coated with rubber latex
- 1824: Johann Nepomuk Reithoffer gets a patent on waterproof clothing
- 1839: Goodyear invents vulcanization
- 1845: Thompson invents the pneumatic tire
- 1852: 1st European production for rubber materials founded in Wimpassing (Semperit)
- 1888: Dunlop invents it a second time
- 1909: Hoffmann, files first patent for the production of artificial rubber



Ancient rubber ball from Mexico

Natural rubber

- 5 most important producers: Thailand, Indonesia, Malaysia, India and China
In Africa: Cote Ivoire, Nigeria and Lybia
- World production: 7,6 Mill. t (dry, 2003)
- Main converters in USA, J, China, D and F



Composition of fresh rubber latex

Component	%. Wt.
Total solids	41,5
Rubber	36,0
Amino acids and N-bases	0,3
Neutral lipids	1,0
Proteins	1,6
Phospholipides	0,6
Carbohydrates	1,5
Salts (K, P and Mg)	0,5
Water	58,5

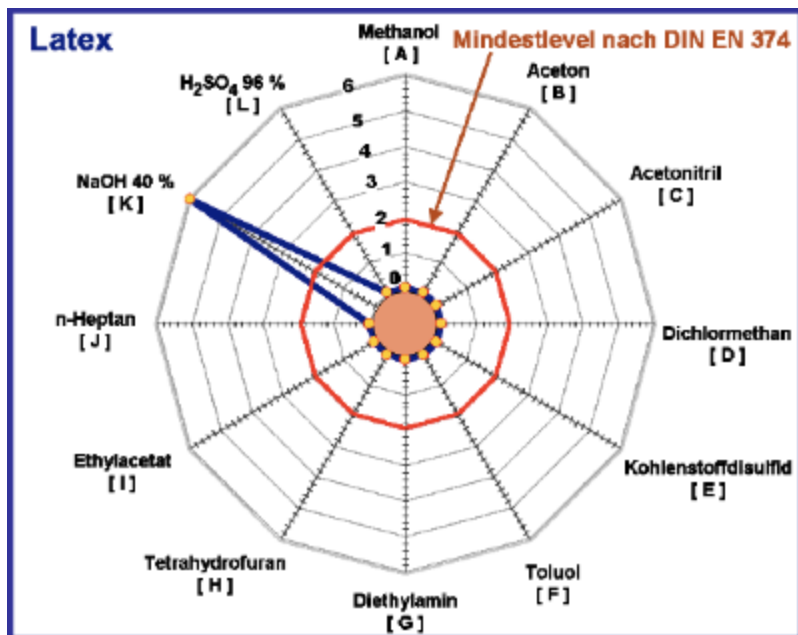
- About 90% of the latex are processed directly at the farms to natural rubbers
- By adding acetic acid the pH is lowered, the carboxy groups at the surface are protonated, it gets electrically neutral, the latex particles coagulate
- These rubber particles swim at the surface, are separated from the water and rolled to plates

Technical elastomers

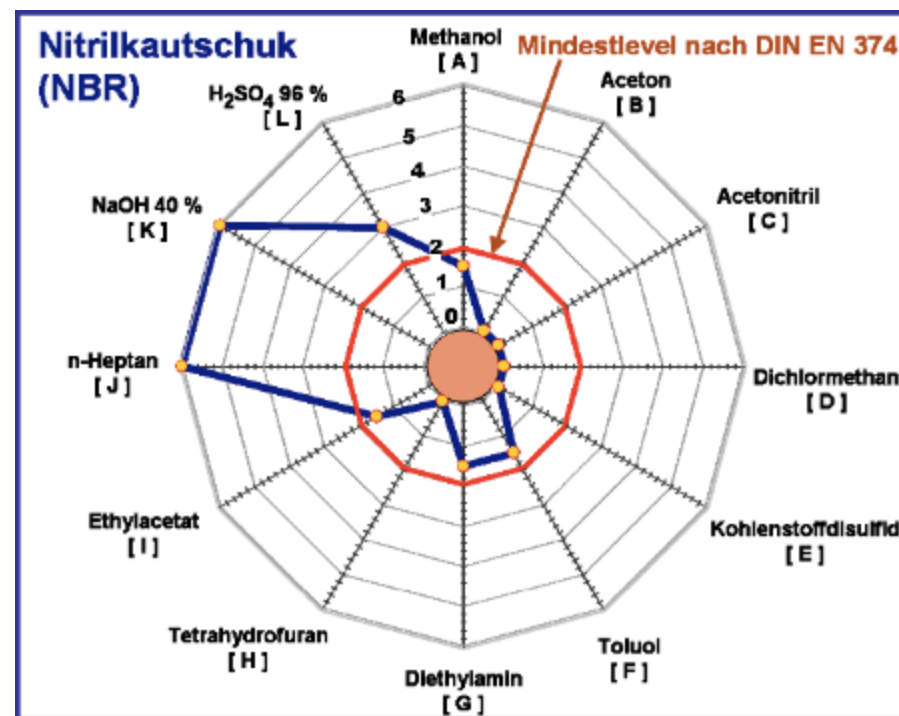
- Production: ca. 18 Mill t / year
- 70% synthetic rubbers and 30% natural rubbers

Elastomer	Abbreviation	% of production
Butadiene styrene copolymers	SBR	56
cis-1,4-Poly butadiene	BR	15
cis-1,4-Poly isoprene	IR	10
Poly chloroprene	CR	5
Ethylene-propylene-co and terpolymers	EPM, EPDM	4
Poly isobutylene (butyl rubber)	IIR	3
Butadiene-acrylonitrile-copolymers	NBR	3
Others	-	2

Properties of gloves



How to make gloves?



Schutzindex	Durchbruchzeit [min]
Klasse 1	> 10
Klasse 2	> 30
Klasse 3	> 60
Klasse 4	> 120
Klasse 5	> 240
Klasse 6	> 480

Special elastomers

Synthesis	Synonym
Polymerization	Ethylene-vinyl acetate rubber Acrylate rubber Epoxide rubber Fluor carbon rubber
Poly condensation	Silicone rubber Poly ester rubber Poly sulfide rubber Poly urethane rubber
Poly addition (reactions on polymers)	Chloro polyethylene Chloro butyl rubber

Thermoplastic elastomers

Block copolymers with multi (2) phase structure (hard- and soft segments), or polymer mixtures, no vulcanizes

Type	Soft	Hard
Poly urethane	Poly ester alcohols Poly ether alcohols	Diisocyanate + Butanediol-1,4
Styrene-diene-block copolymers	Poly butadiene Poly isoprene	Polystyrene
Poly ether esters	Poly ether alcohols	Terephthalic acid + Butanediol-1,4
Polyolefin blends	Ethylene-propylene rubber (EPDM)	Polyethylene Polypropylene

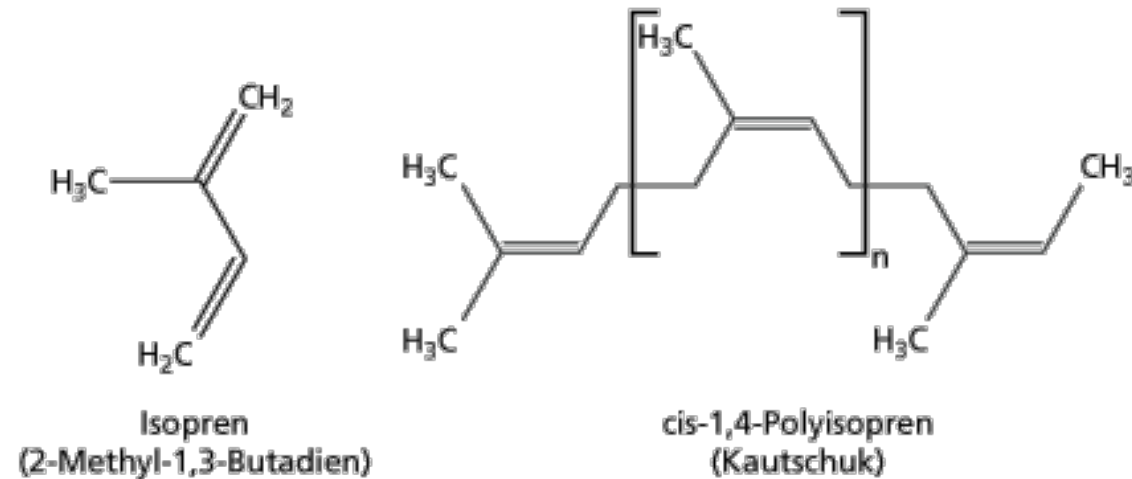
Rubbers

- Polyisoprene (IR)
- Polybutadiene (BR)
- Polychloropren (CR)
- Polyisobutylene (IIR)
- Styrene butadiene rubber (SBR)
- Acrylonitrile butadiene rubber (NBR)
- *Styrene Acrylonitrile Copolymer (SAN)*
- *Rubber modified SAN (ABS)*
- Other elastomers - Silicones

Polyisoprene (IR)

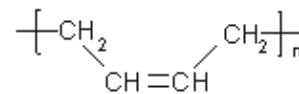
- As natural rubber, but more even, cleaner and expensive, transparent
 - Main uses: special tires, rubbers in contact with food
- Synonyms: Natsyn Cariflex IR

Chemische Struktur von Isopreneinheiten und Kautschuk

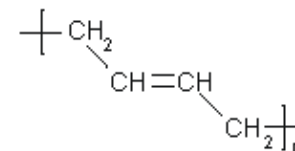


Polybutadiene (BR)

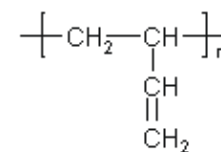
- High elasticity, abrasion resistant, flexible at low temperatures (-80 to +90°C), mostly compounded with NR or SBR
- Main uses: tires, transportation belts, heavy used articles
- Synonyms:
Buna CB 11 Titan type,
Buna CB 10 Cobalt type,
Buna CB 30, oil mixed



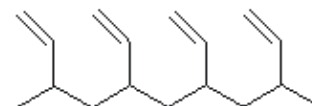
cis -1,4-Poly(butadien)



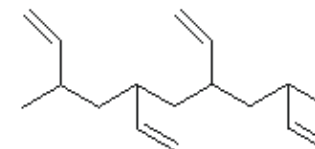
trans -1,4-Poly(butadien)



1,2-Poly(butadien)

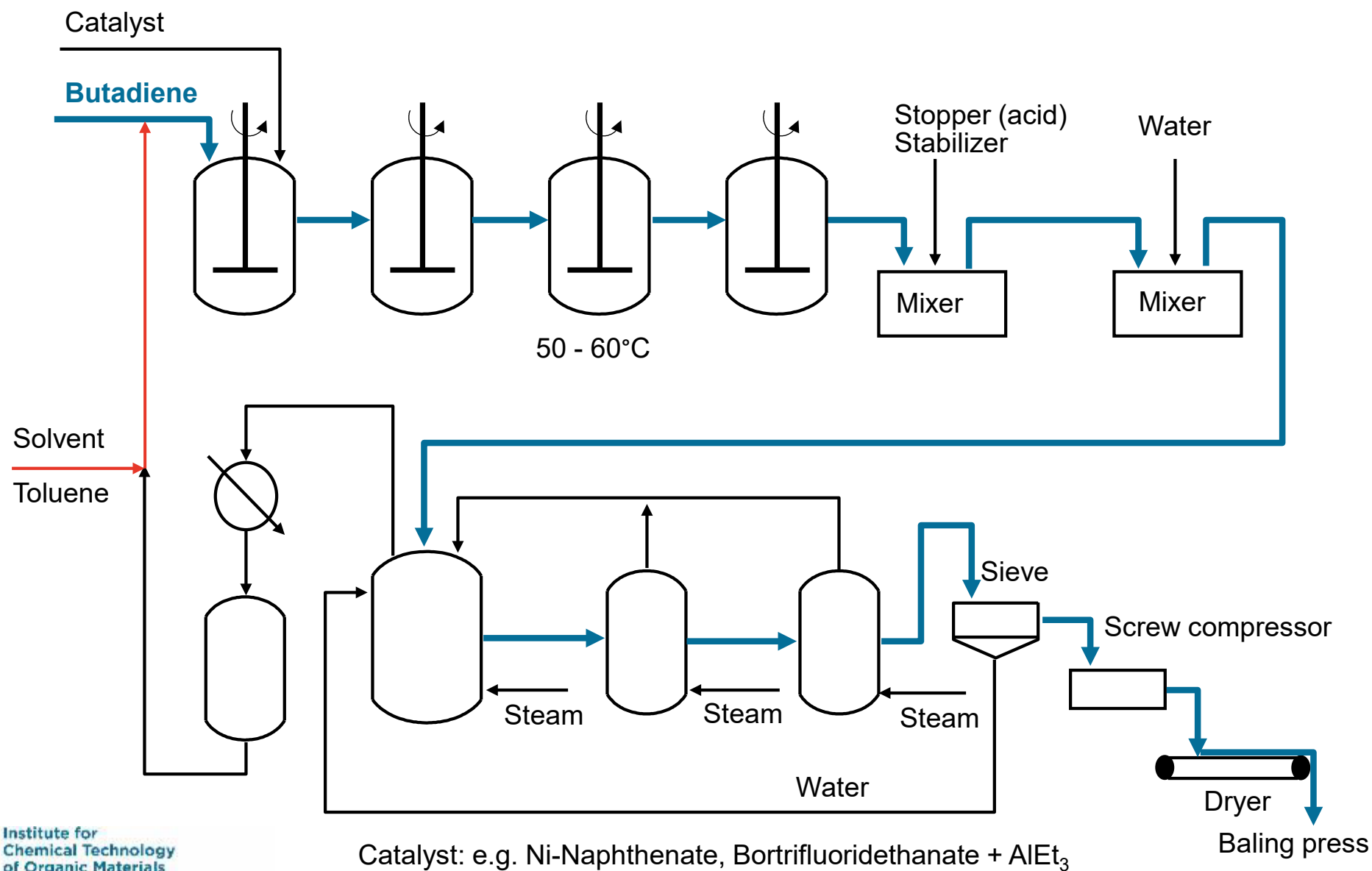


isotaktisches 1,2-Poly(butadien)



syndiotaktisches 1,2-Poly(butadien)

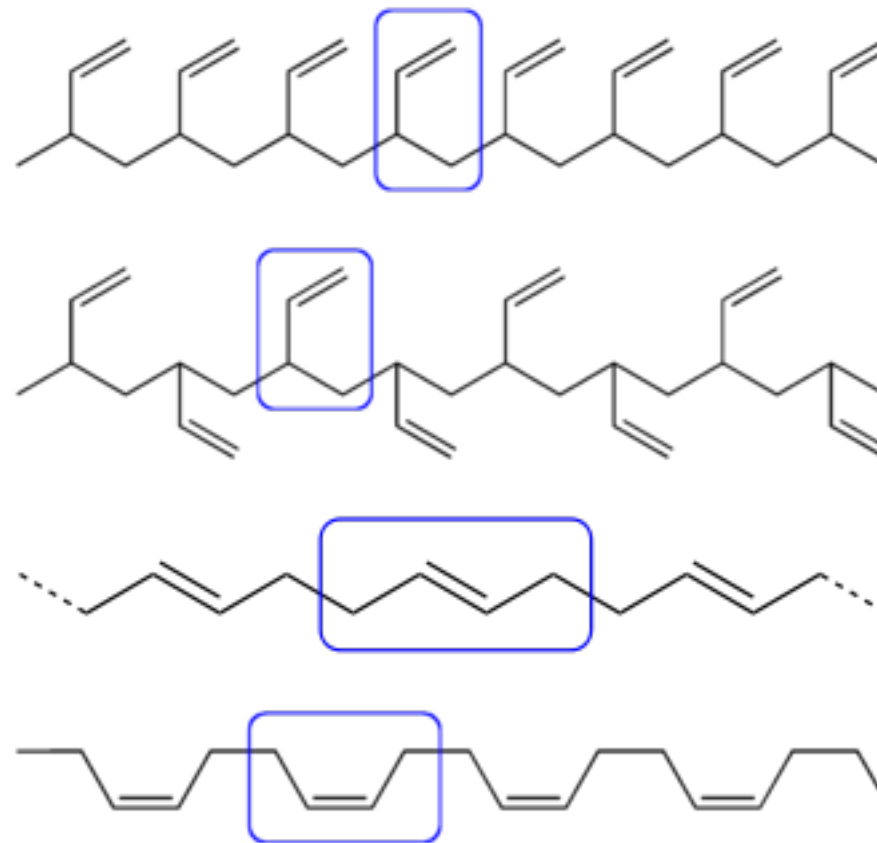
Polybutadiene



Polybutadiene

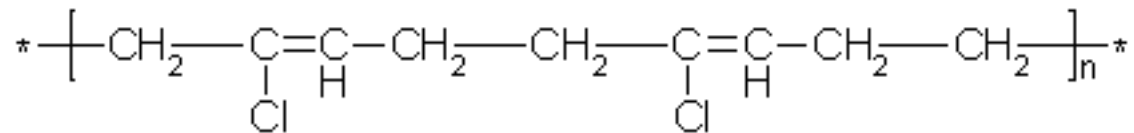
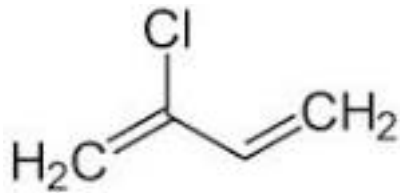
Microstructure of poly butadiene with respect to catalyst
(Ti, Co, Ni as component of Ziegler/Natta systems, Li as n-butyl-lithium)

Catalyst	cis-1,4 [%]	trans-1,4 [%]	1,2-content [%]
Ti	93	3	4
Co	96	2	2
Ni	97	2	1
Li	35	55	10



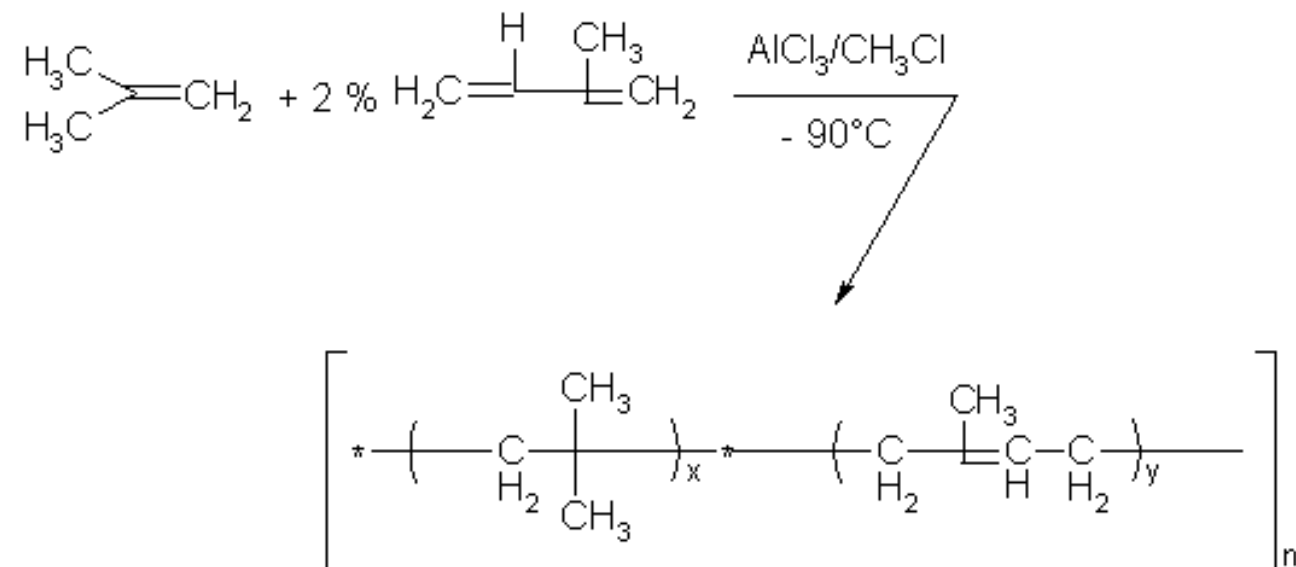
Polychloropren (CR)

- Working temperature: -40 to +110°C, low swelling in oil, hardly flammable, dynamic
- Main uses: transportation & driving belts, cables
- Synonyms: Baypren (number code), Neoprene (character code)

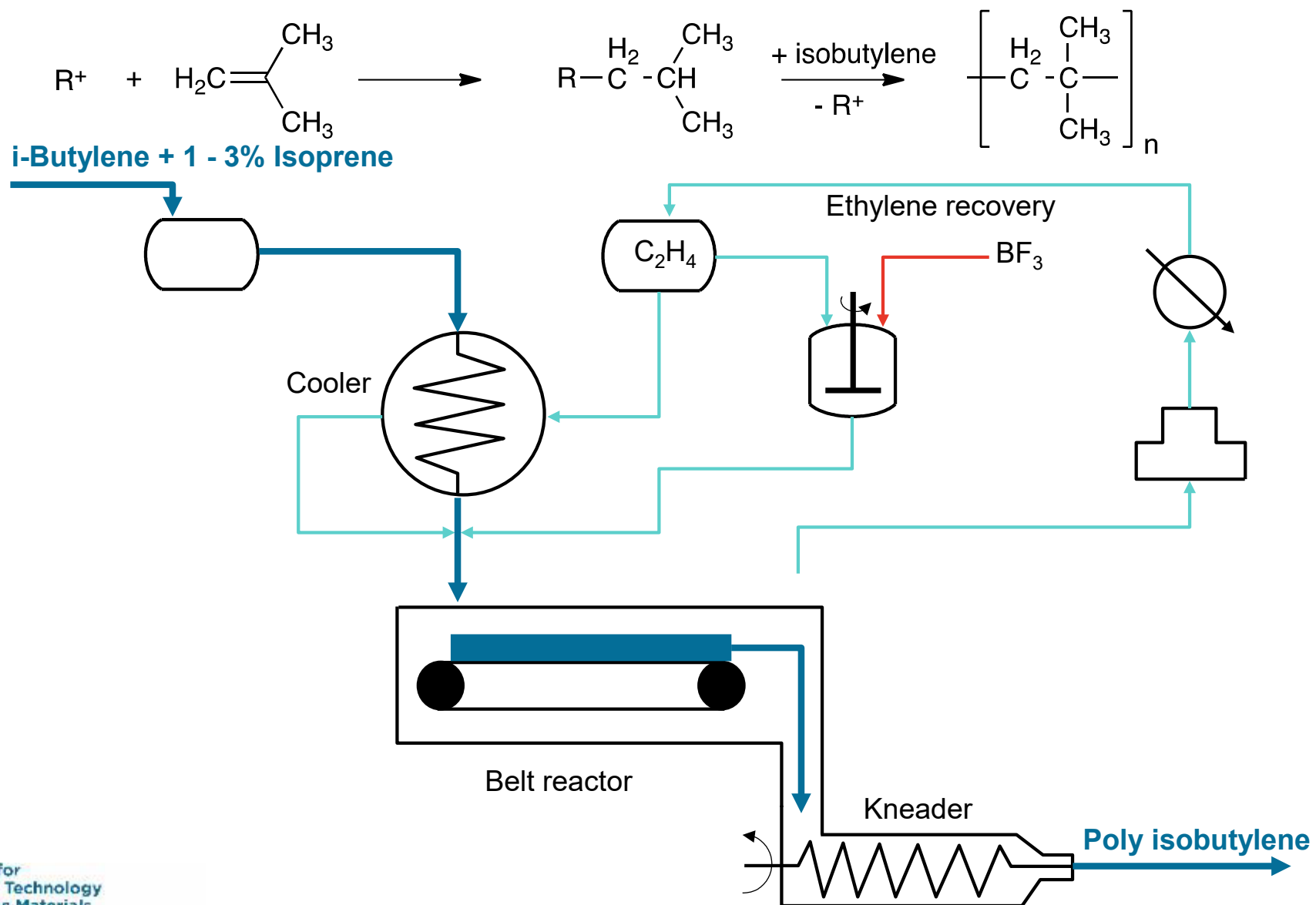


Butylrubber (IIR)

- Butyl rubber (poly isobutylene), IIR
- Useable up to 130°C, good damping properties, low gas barrier
- Main uses: tire-tubes, vibration damping
- Synonyms: special coding with respect to isoprene content

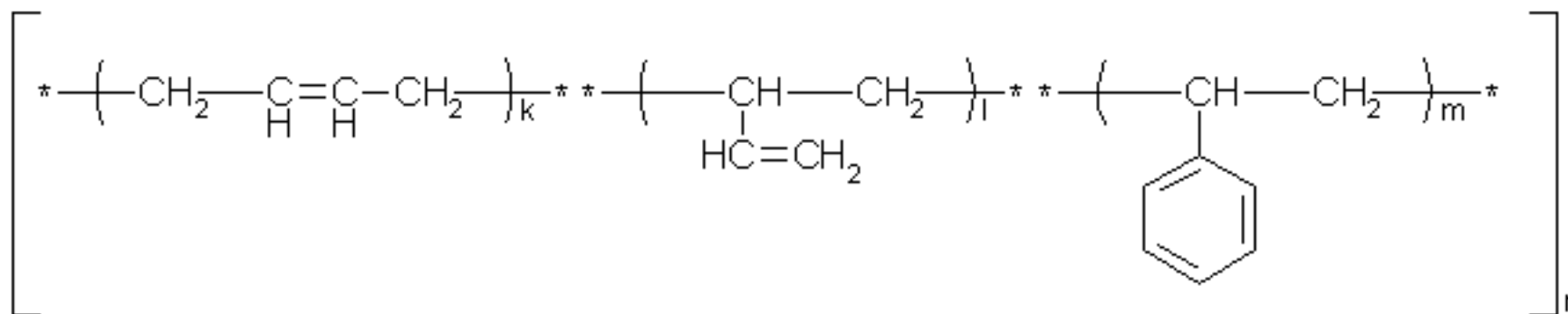


Polyisobutylene

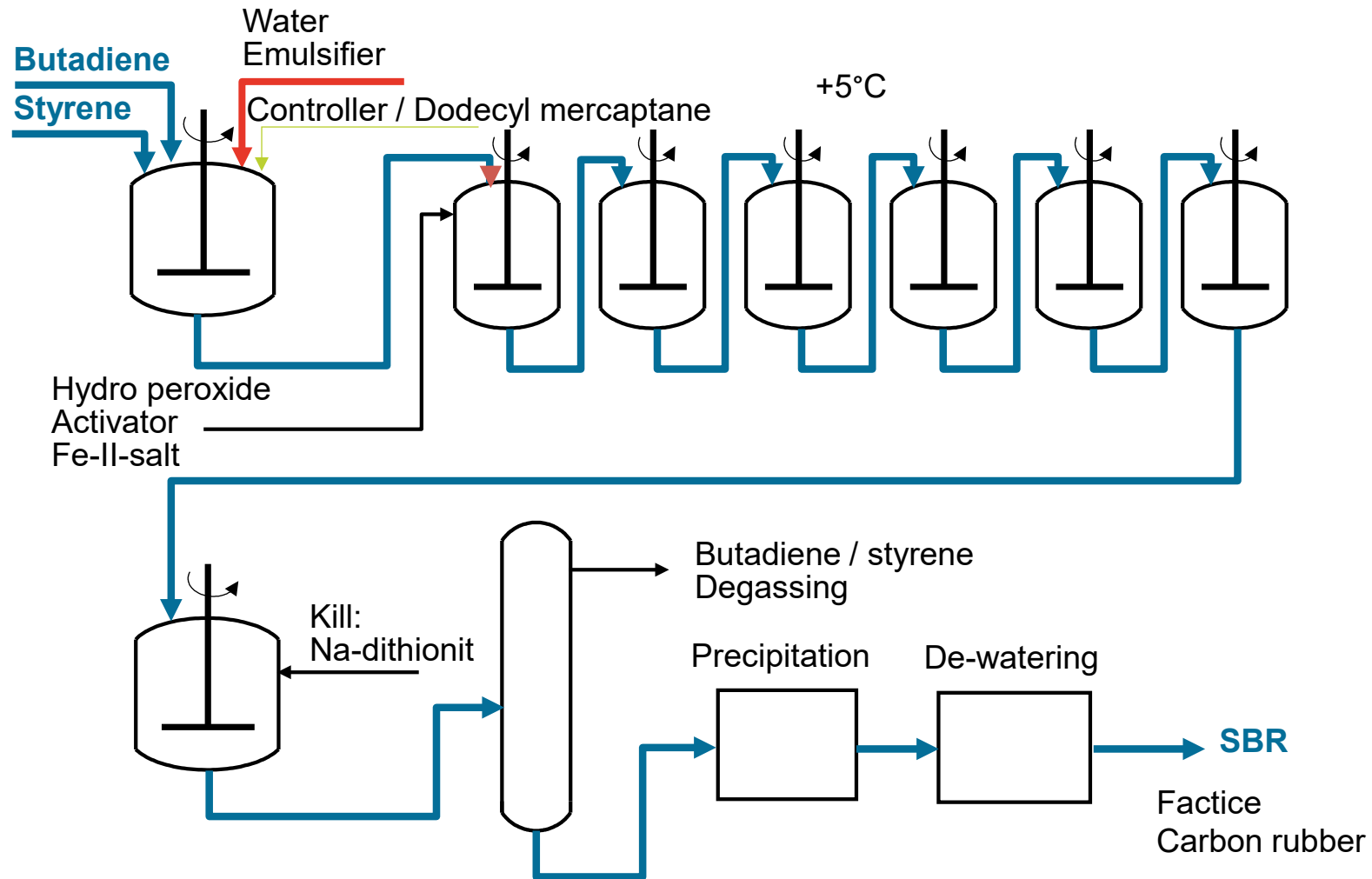


Styrene butadiene rubber (SBR)

- Cheap, better aging than NR, rubber for general use, good resistance against chemicals
- Main uses: tires, transportation belts, technical rubber articles
- Synonyms: Buna SB, Krylene, Cariflex S, (many different grades)

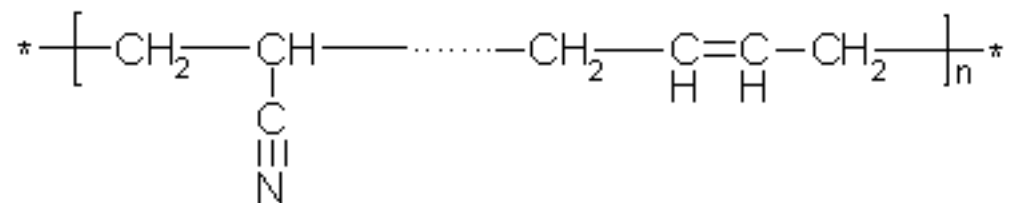


SBR: emulsion polymerization

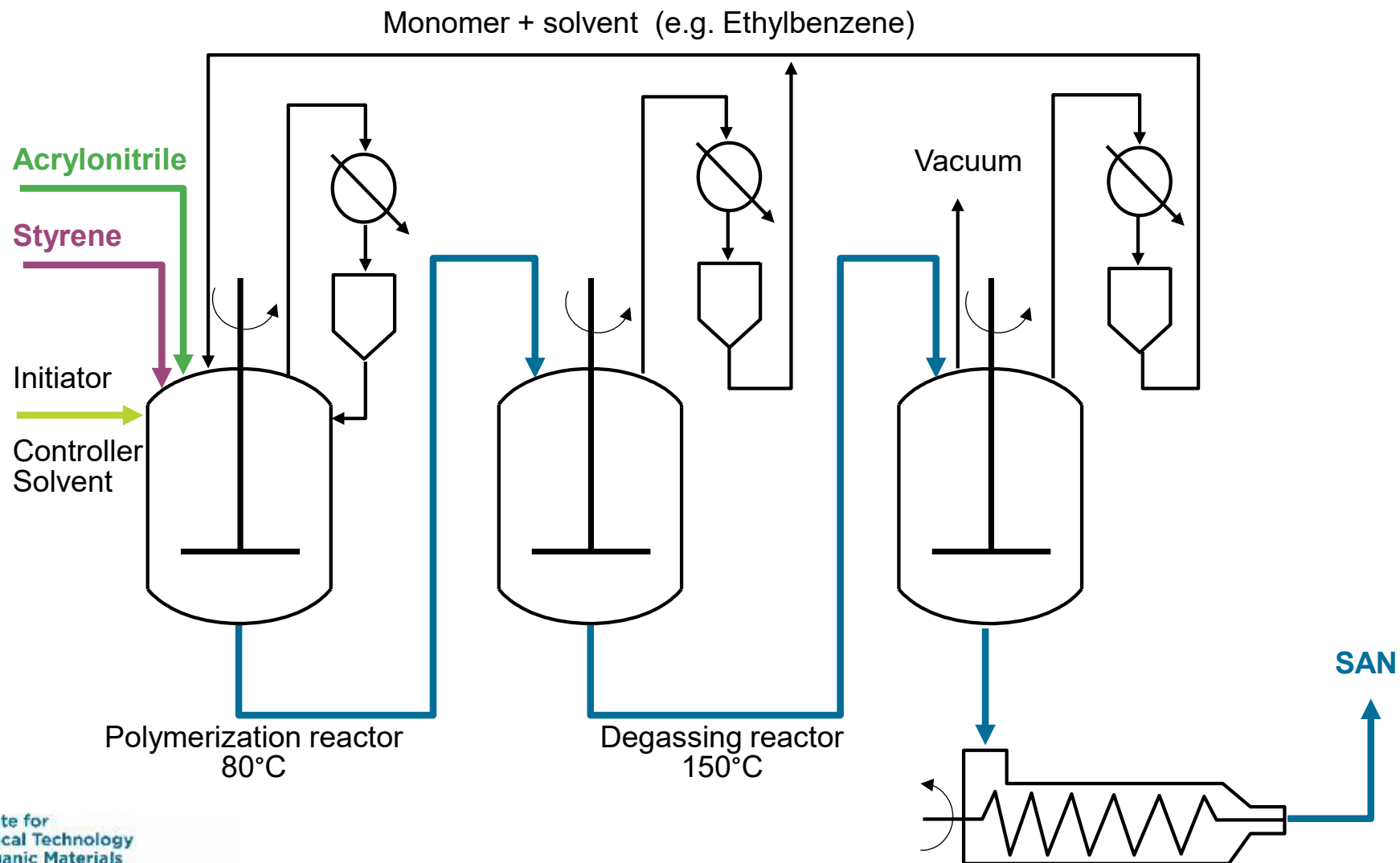


Acrylonitrile butadiene rubber (NBR)

- Acrylonitrile butadiene rubber, NBR
- Hydrogenated acrylonitrile butadiene rubber, HNBR
- Working temperature up to ca. 110°C, low oil/gasoline swelling, low gas diffusion, good chemical resistance
- Main uses: car parts, oil and gasoline hoses, gaskets, articles in contact with food
- Synonyms: special codes by manufacturers, dependent on acrylonitrile content

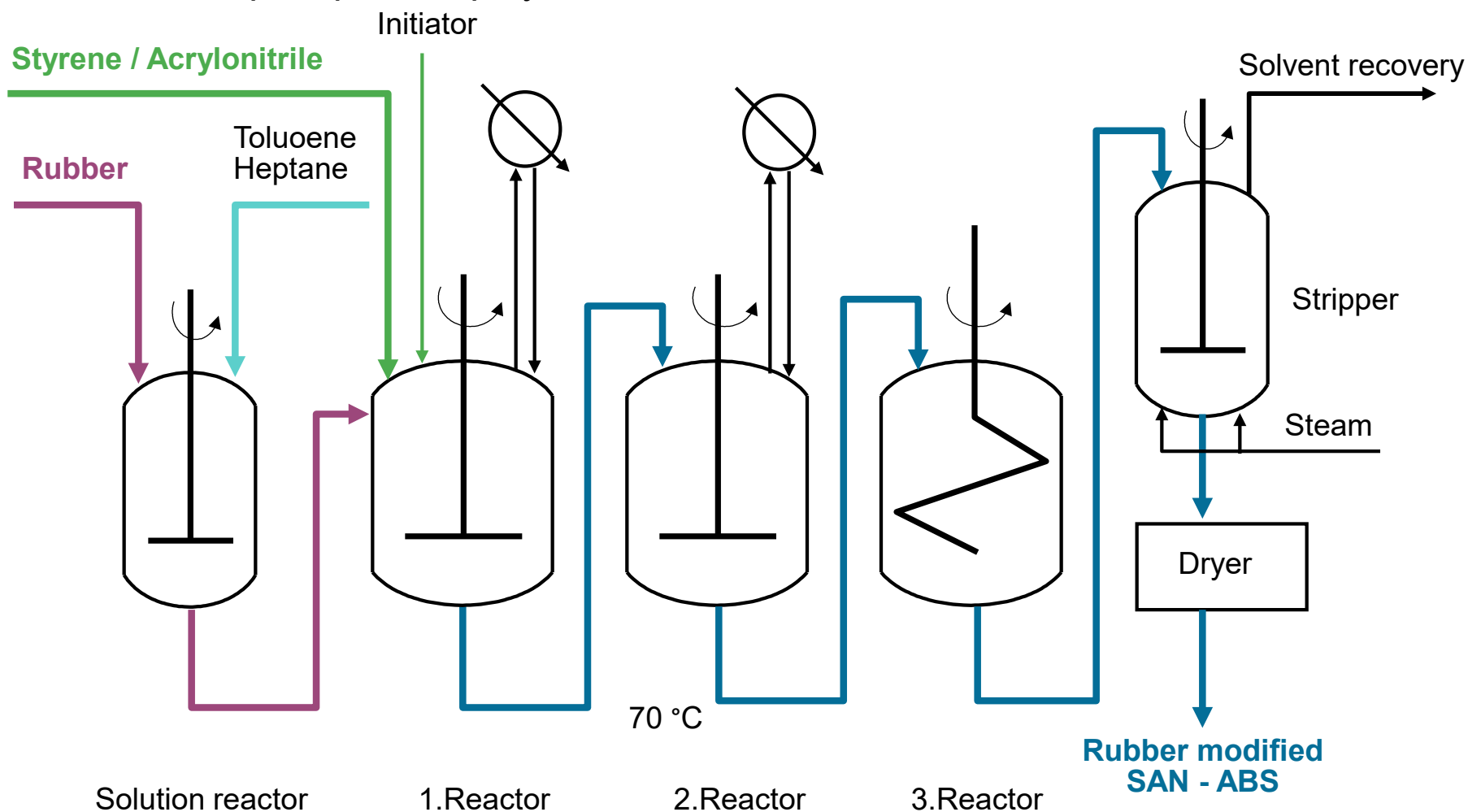


Styrene acrylonitrile copolymer (SAN)



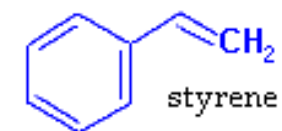
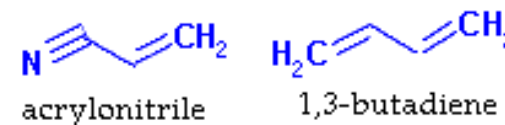
Rubber modified SAN → ABS

Solution / precipitation polymerization

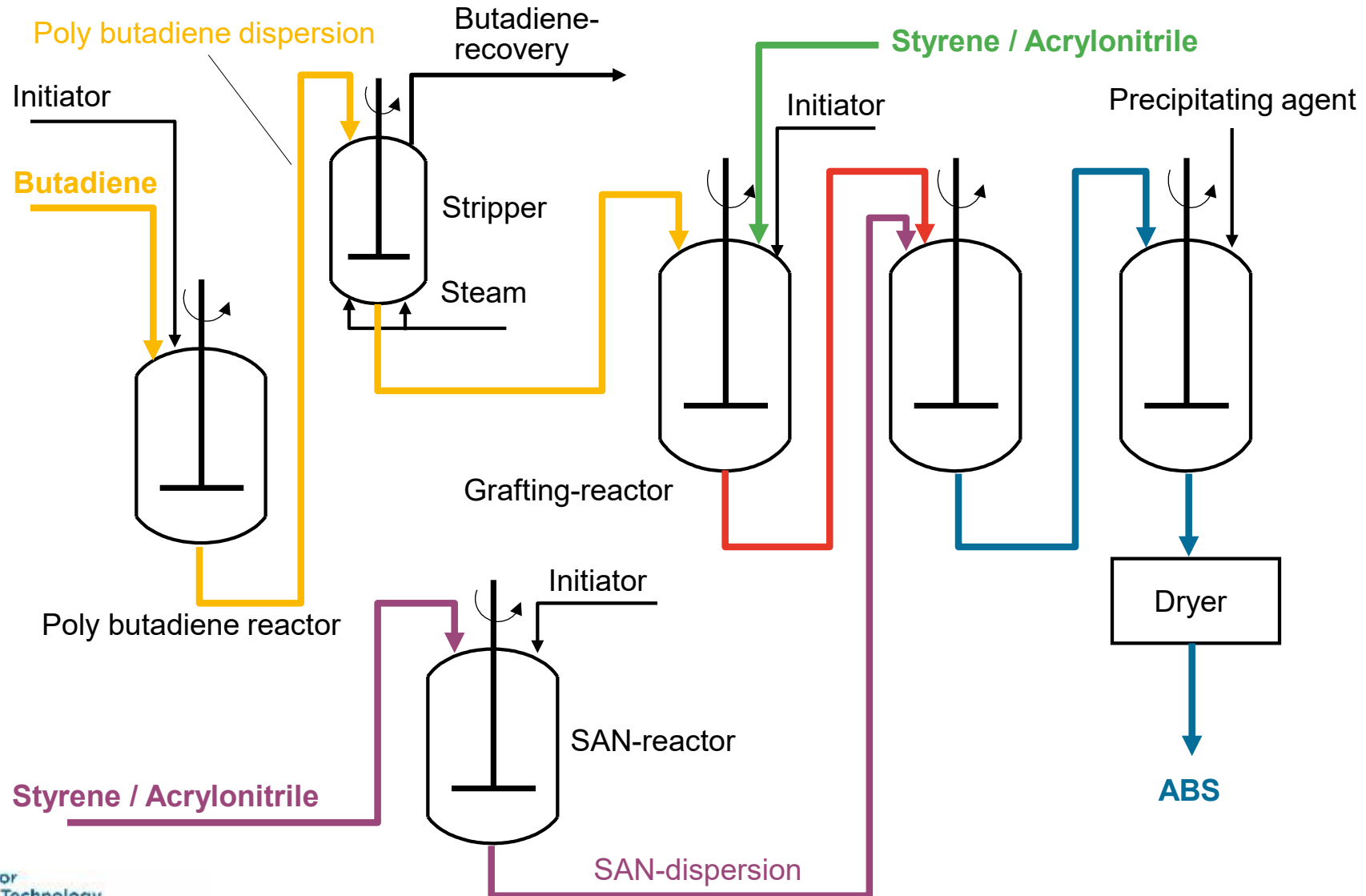


ABS

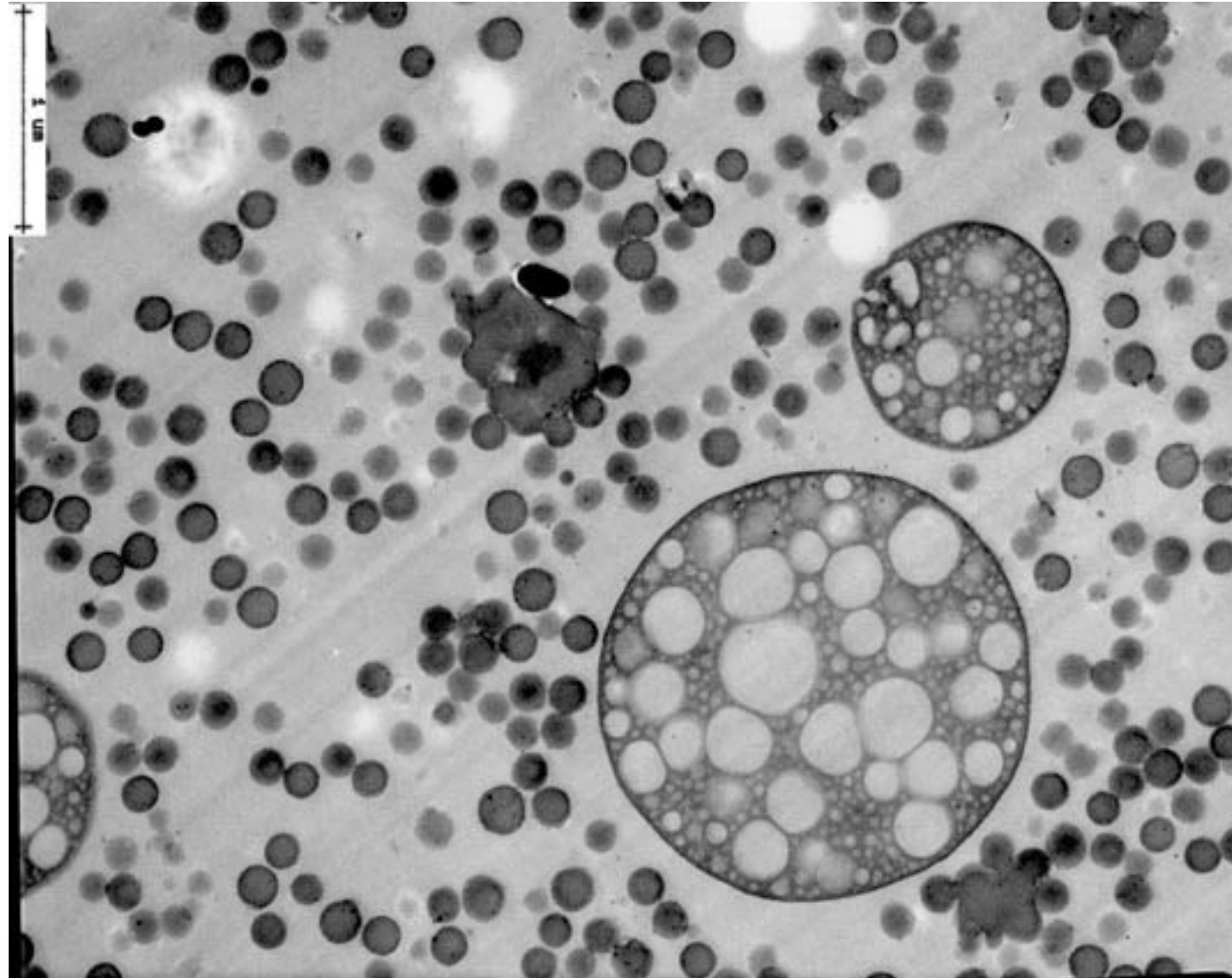
- Engineering plastic for rigid molded products (piping, musical instruments, automotive parts, toys and of course Lego bricks)
- Composition
 - Acrylonitrile: 15 – 35%
 - Butadiene: 5 – 30%
 - Styrene: 40 – 60%
- ABS combines the strength and rigidity from acrylonitrile and styrene polymers and the toughness of the poly butadiene rubber



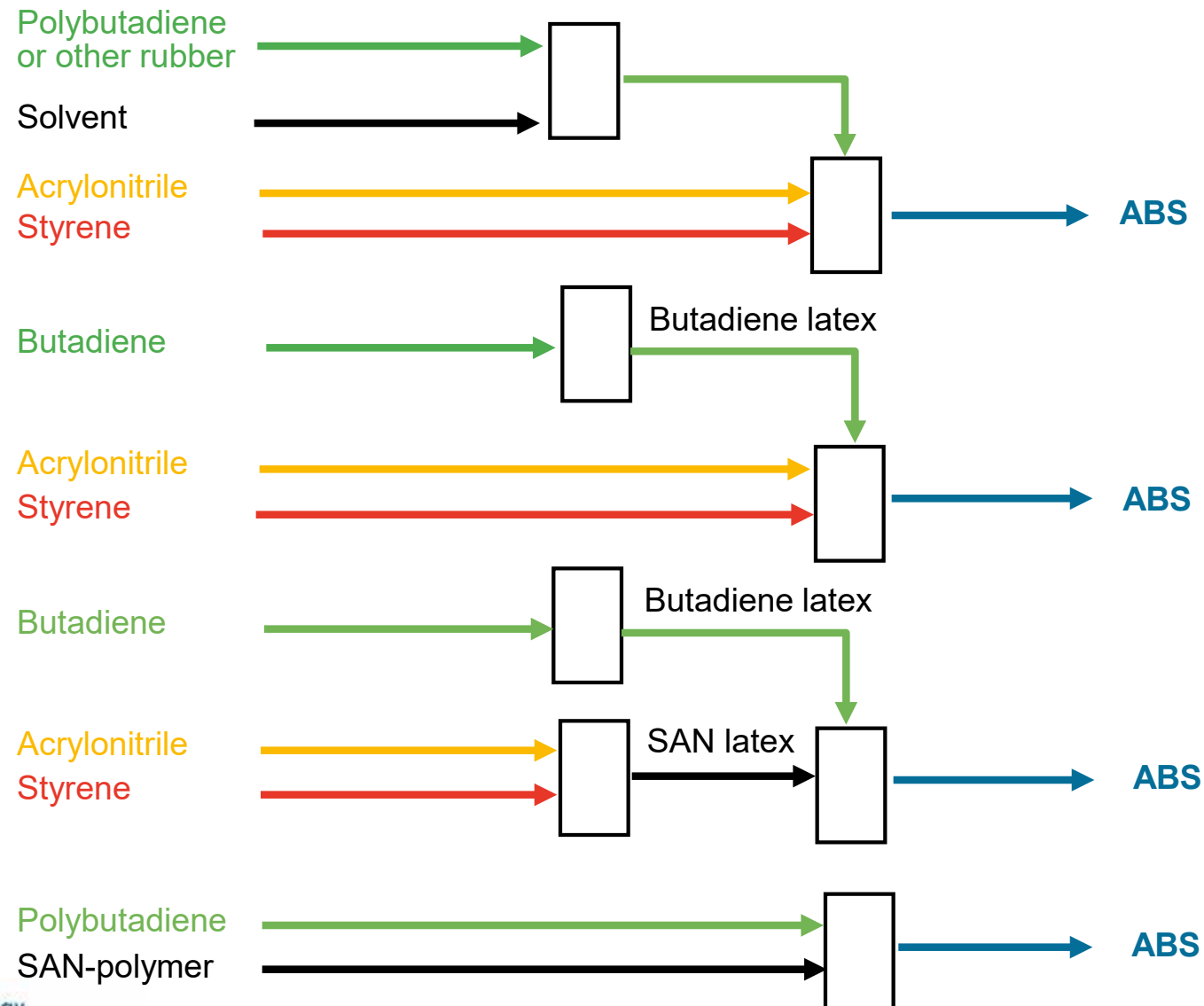
ABS through grafting



ABS morphology

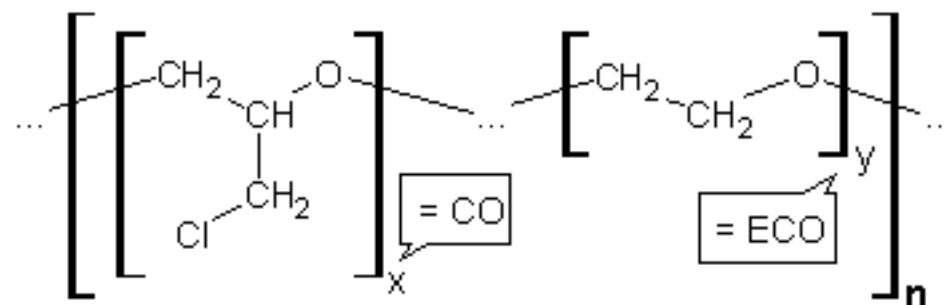


ABS production



Other rubbers / elastomers

- Ethylene propylene rubber, copolymer EPM, terpolymer EPDM
- Acrylate rubber, ACM
- Epichlorohydrine rubber, CO (homo-), ECOCO (copolymer)
- Silicon rubbers

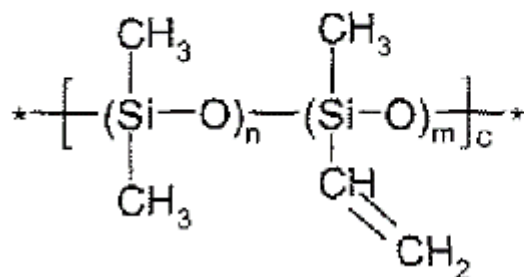
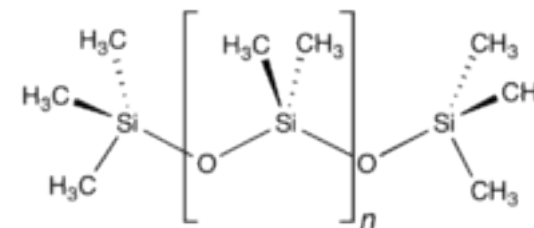


Silicon rubbers

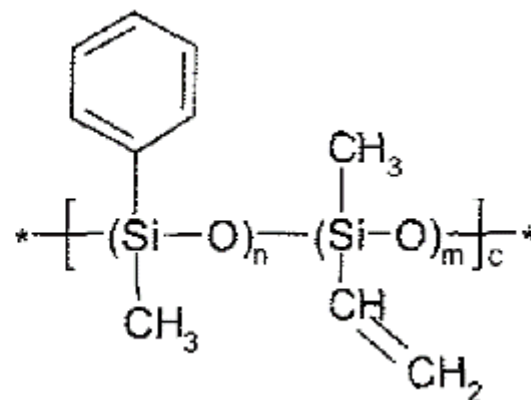
- 3 dimensional cross-linked polydiorganosiloxanes (PDMS)
- Temperature resistance to 200°C
- Low temperature flexibility to -100°C; high electrical resistance
- Main uses: cables, hoses, films, gaskets
- Special nomenclature: M-V-P-F-Q
 - Q: Polysiloxanes
 - M: Dimethylsiloxy
 - V: Vinylmethylsiloxy
 - P: Phenylmethylsiloxy
 - F: Trifluoropropylmethylsiloxy

Silicon rubbers

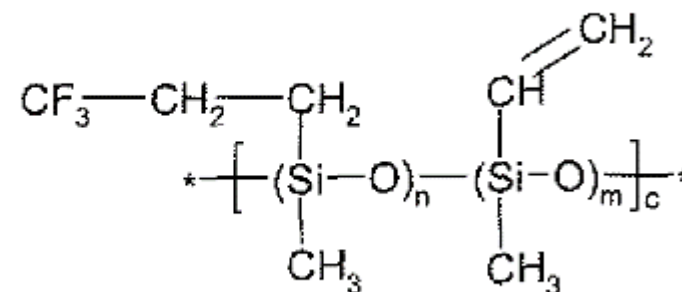
PDMS



MVQ

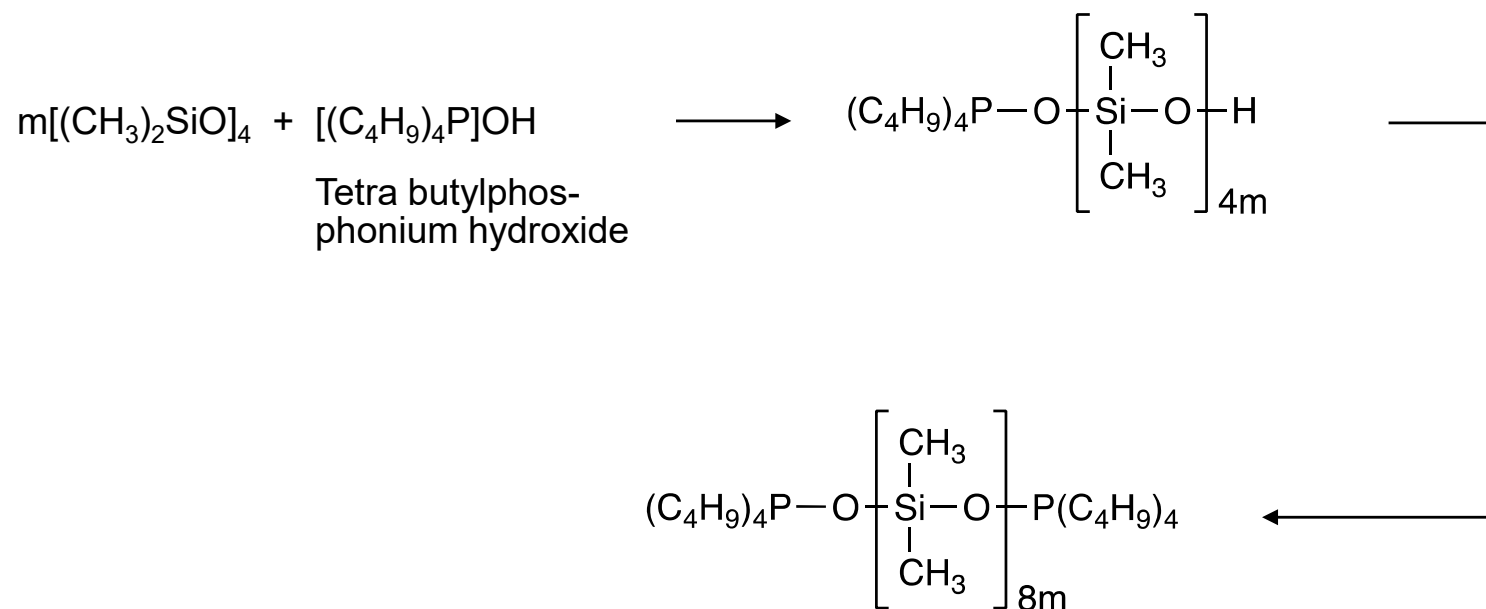
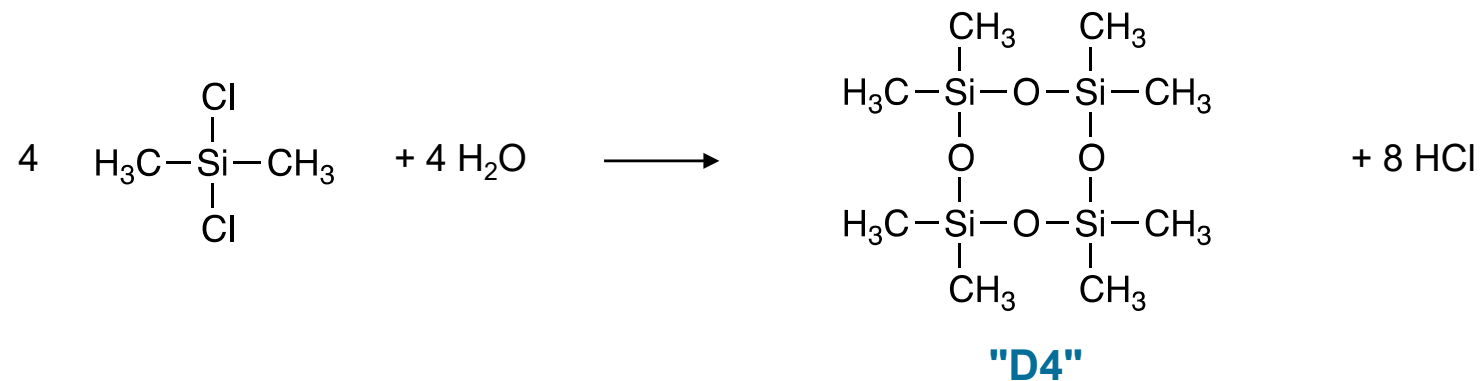


PVQ

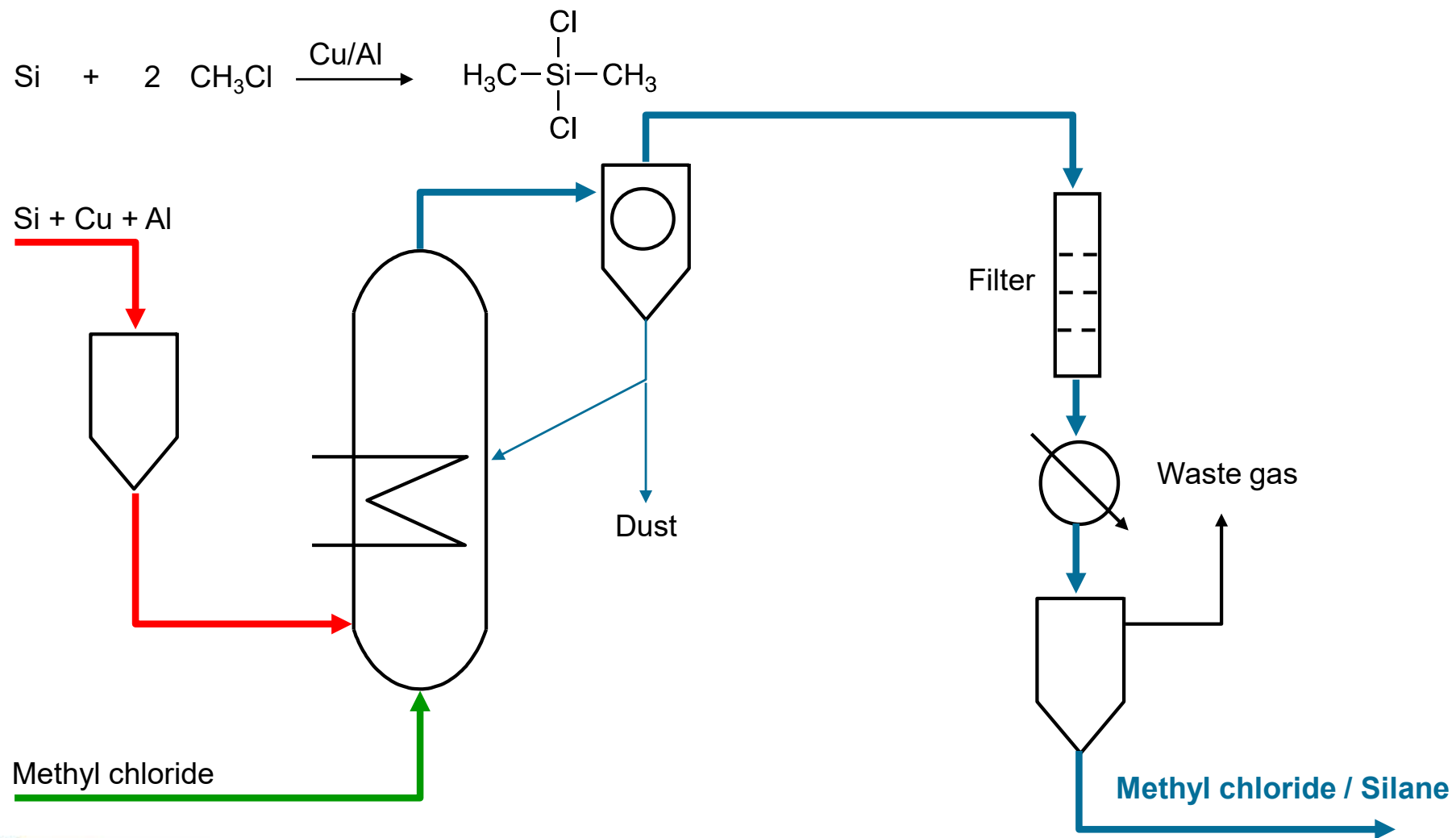


FVQ

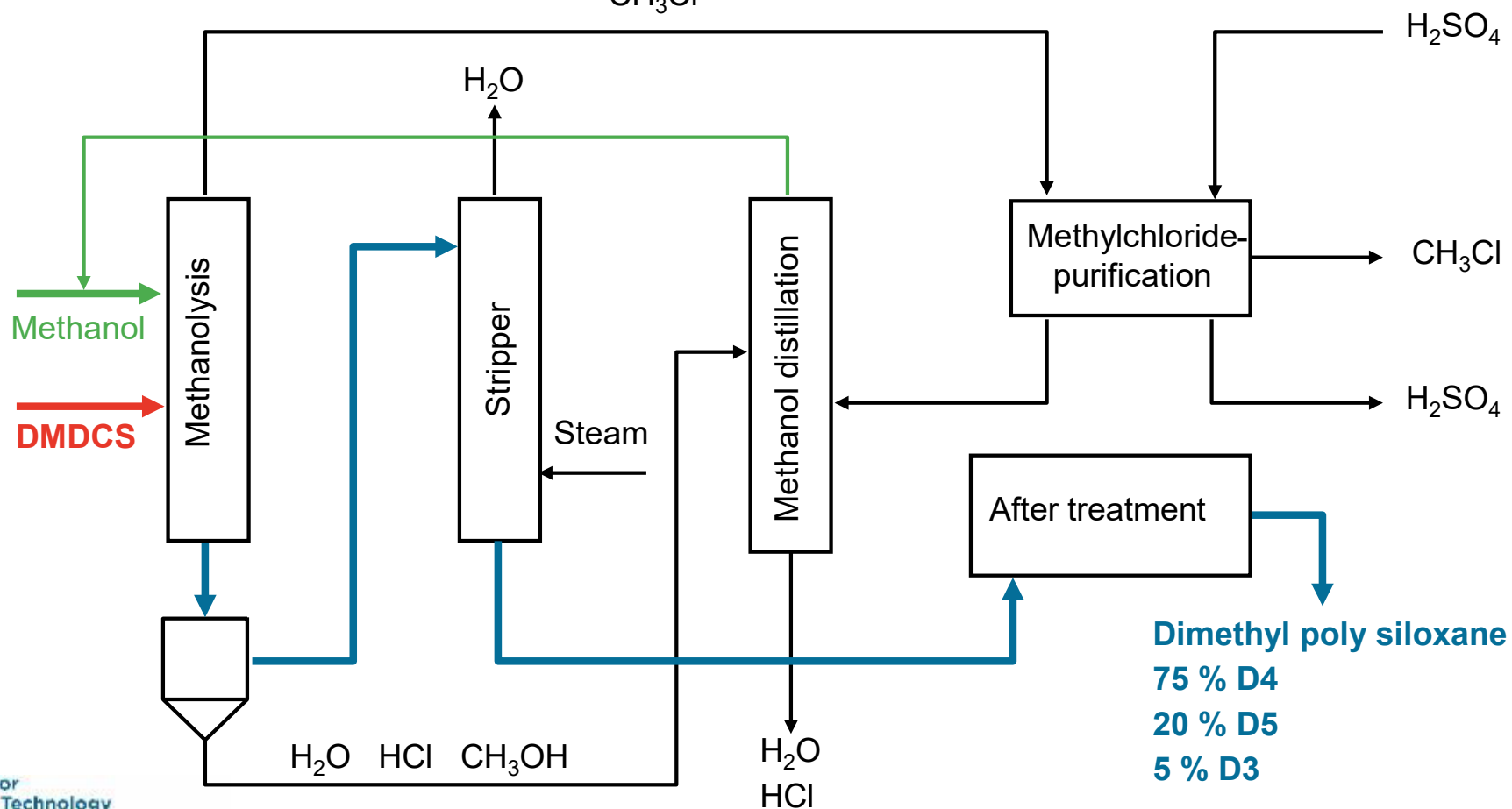
Silicon rubbers



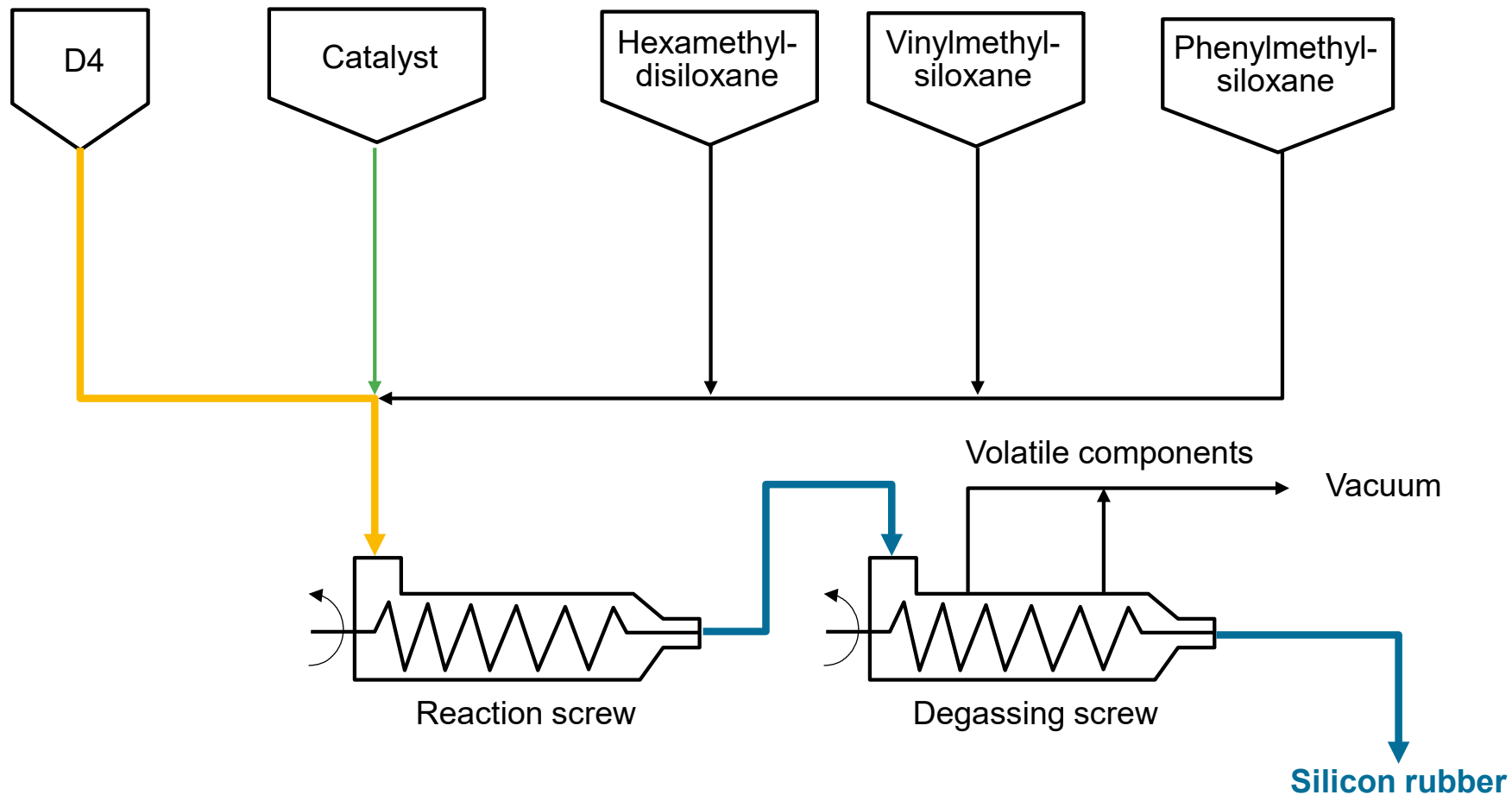
Production of chloro silanes



Dimethyl polysiloxanes



Production of silicone rubber



Synthetic rubbers

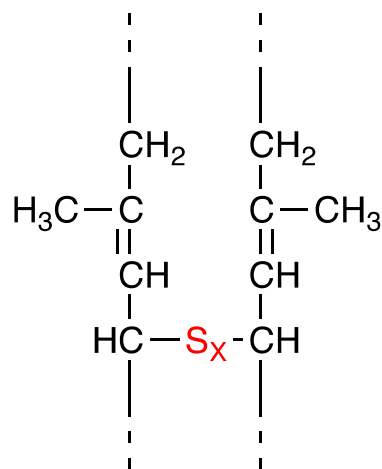
S vulcanizable

Monomers	Polymerization	Labeling
Butadiene + styrene	Radical (EP)	Buna S, SB-rubber
Butadiene + acrylonitrile	Radical (EP)	Buna N, Nitrile-rubber
Chloroprene	Radical (EP)	Chloroprene rubber
Butadiene	Ziegler Natta (SP)	all-cis-1,4-Poly butadiene
Isoprene	Ziegler Natta (SP) or anionic (Li-butyl)	all-cis-1,4-Poly isoprene
Cyclo pentadiene	Ziegler Natta (SP)	Poly pentenamer
Cyclo octadiene	Ziegler Natta (SP)	Poly octenamer
Isobutene + isoprene	Cationic (SP)	Butyl rubber
Ethylene + propylene + termonomer	Ziegler Natta	EPDM rubber

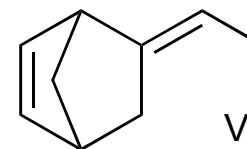
EP: emulsion polymerization; SP: solution polymerization

Vulcanization

Intermolecular sulfide bridges

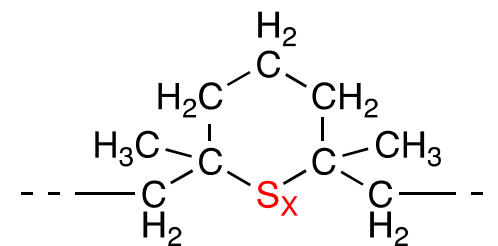
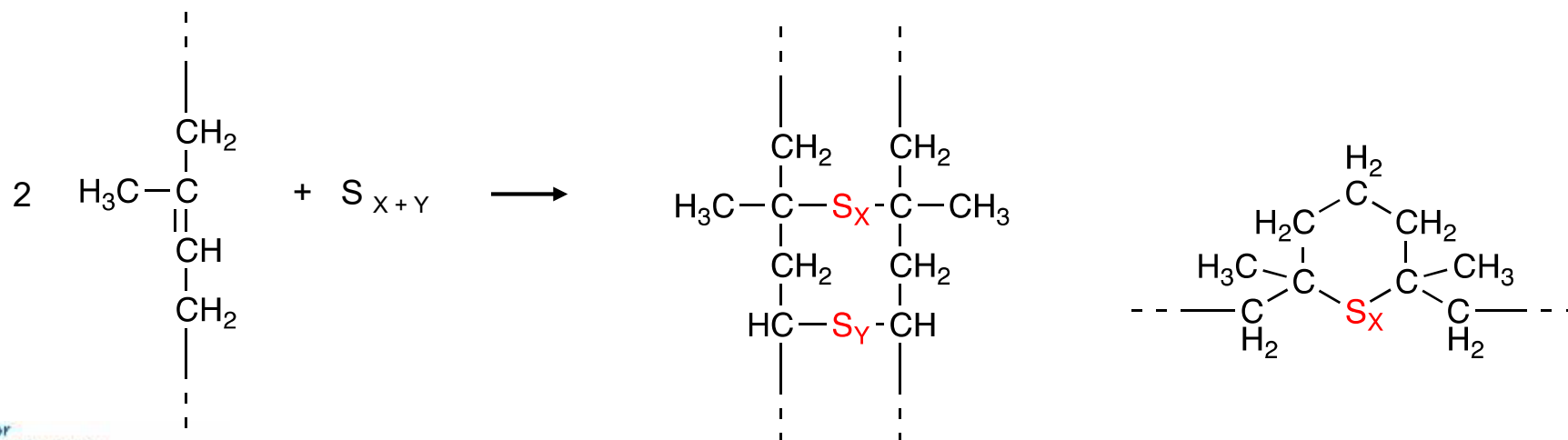


Polymerization



Vulcanization

Formation of cyclic structures

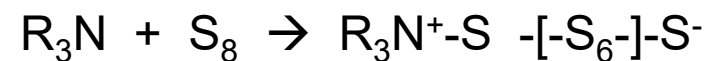


Vulcanization

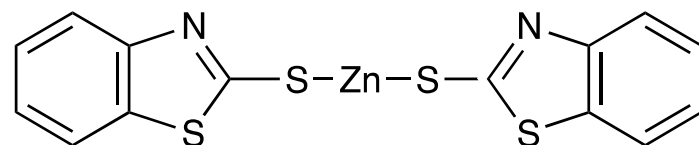
- The amount of sulfur linkages defines the hardness of the product
- > 35%: strongly crosslinked and hard rubber
- < 5%: soft rubber
- Toy balloon: < 1%

Vulcanization promoters

Amines

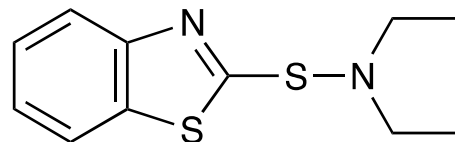


Zink salt of Mercaptobenzthiazol



(R-S-Zn-S-R)

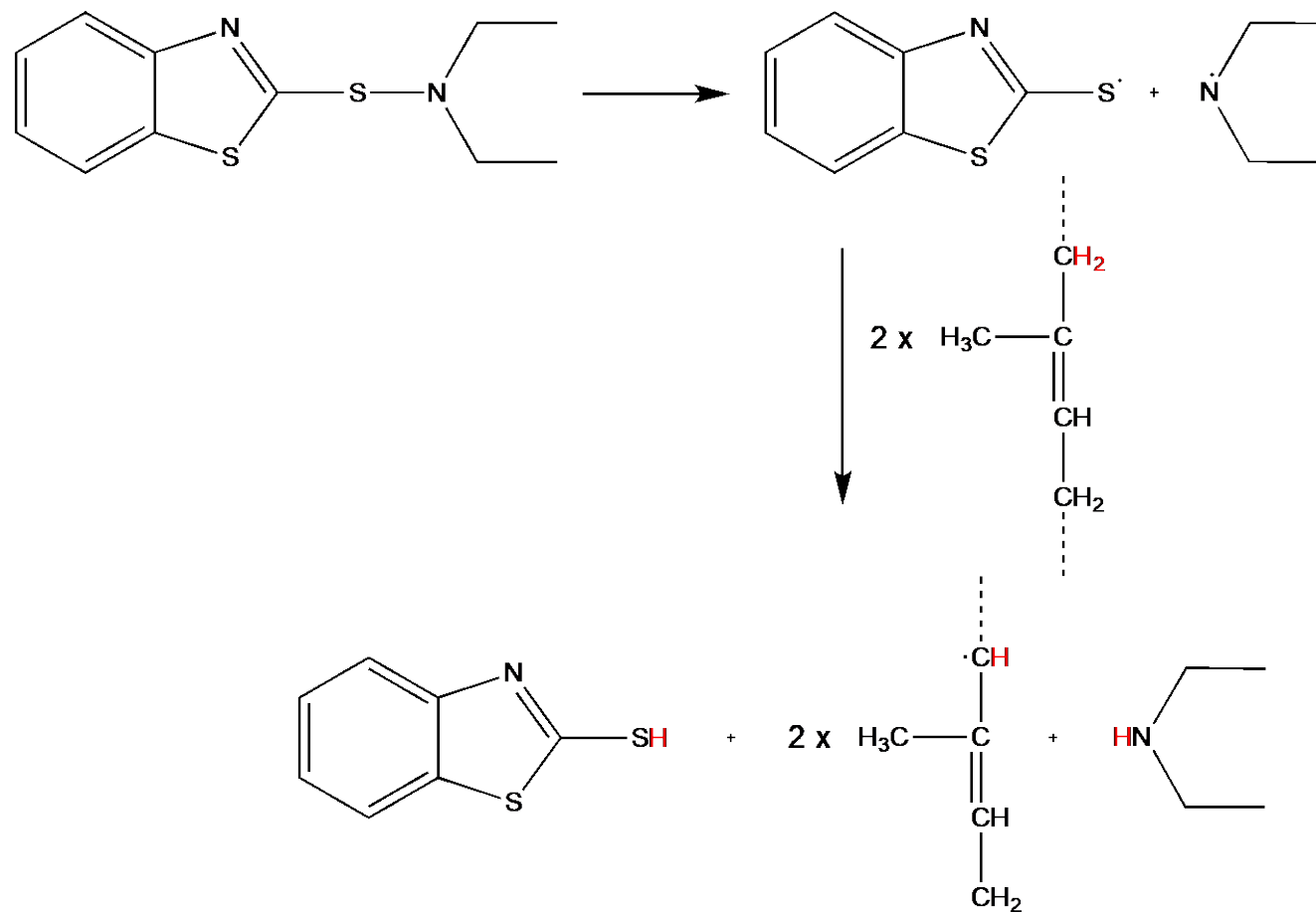
Sulfenamide



→ Radical decomposition

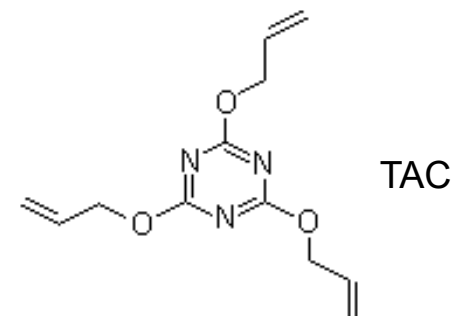
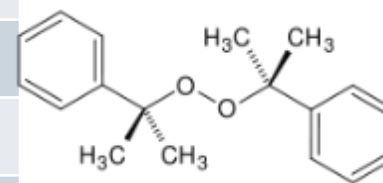
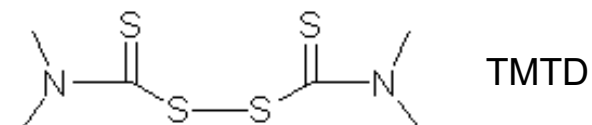
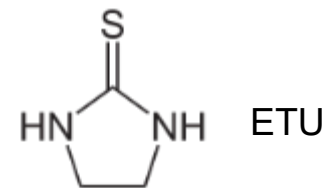
Vulcanization promoters

Acceleration through Sulfenamide

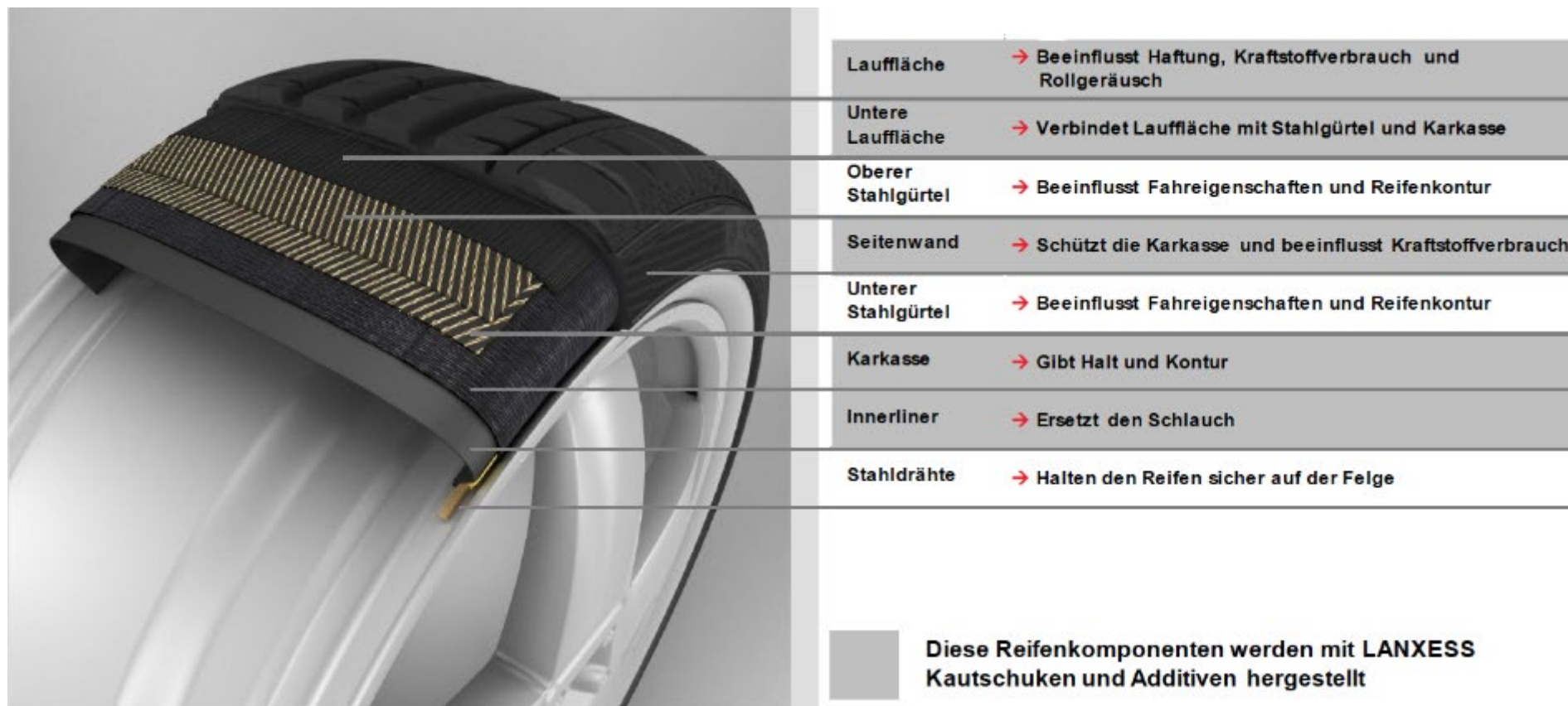


S free vulcanization

Rubber	X-linking agent	Promoter / inhibitor
CR	Zink oxide Peroxides	Ethylene thiocarbamide (ETU), MgO
CSM	Pb oxides, MgO Epoxy resins Peroxides	ETU, Diamines, Tetramethylthiuramdisulfid (TMTD) Pentaerythrite
ECO	PbO ₂ , Pb ₂ O ₃ , ZnO, MgO	ETU, Pentaerythrite
ACM	Metal soaps, Diamines, Epoxy resins	ZnO, PbO
EPM	Peroxides	Dicumylperoxide
SI	Peroxides	Dicumylperoxide
IIR	Methylol phenol resins	
PUR	Diisocyanates	Triallylcyanurate (TAC)



Tire



Tire



Merkmale eines Hochleistungsreifens: LANXESS Produkte

Optimierte Mikrostruktur und Füllstoff-Interaktion

→ Kraftstoffeffizienz, Nass- und Trockenrutschen, Abriebfähigkeit

Lange Ketten (physische Anordnung)

→ Rissfestigkeit

Flexible Ketten

→ Elastizität

Hoher Härtegrad

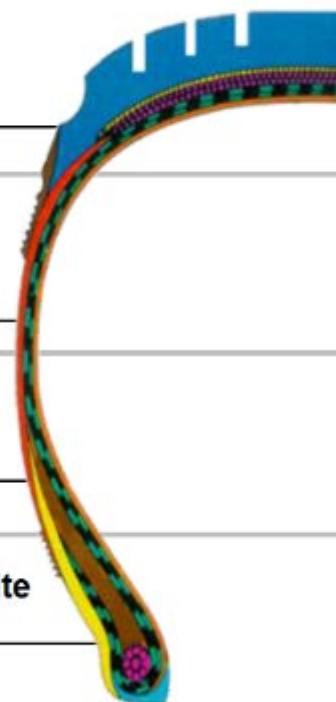
→ Abriebfestigkeit

S-SBR
Nd-BR

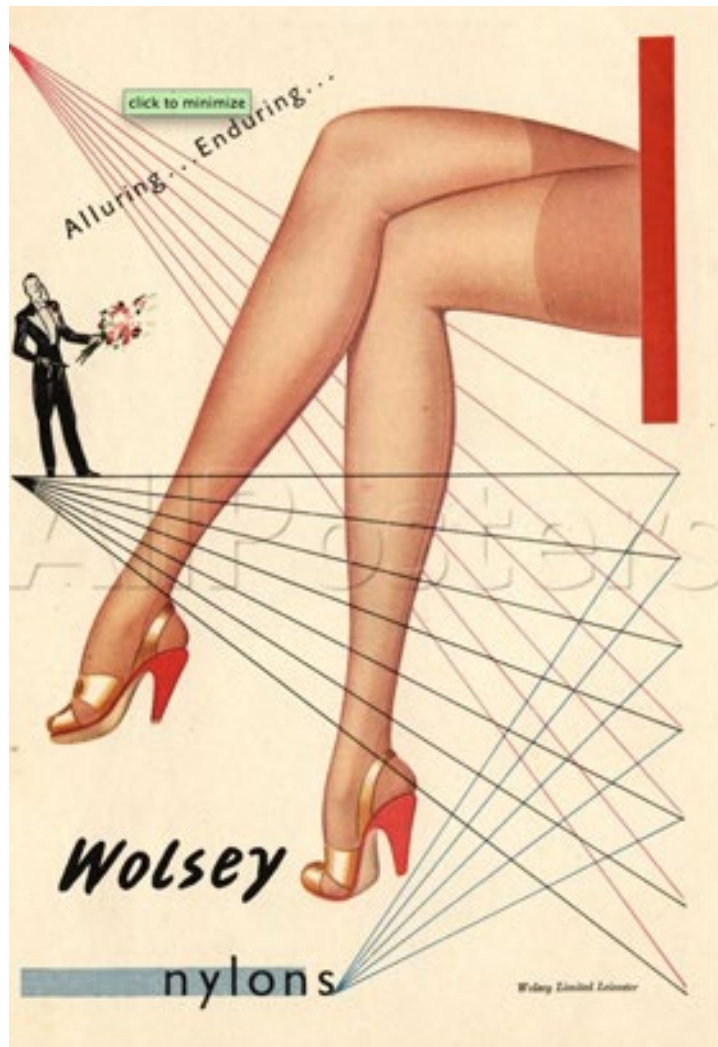
Nd-BR

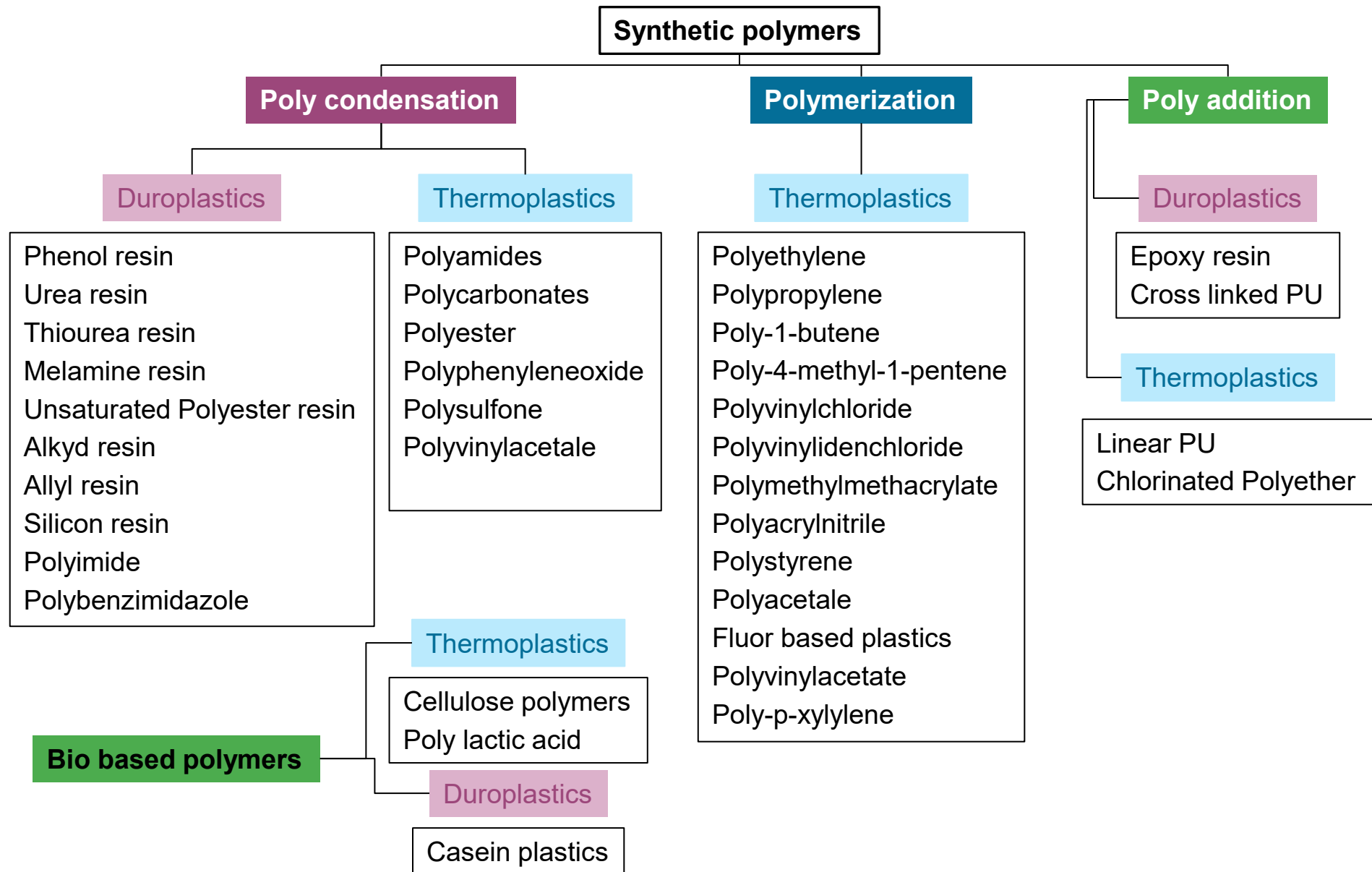
Nd-BR
Co-BR

Gekoppelte
Li-BR



Step growth polymerizations: Polycondensation & Polyaddition





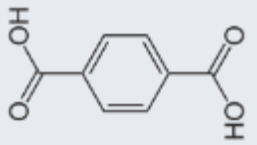
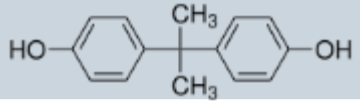
Polycondensation

- Monomers with 2 different functional groups in the same molecule
ω-Amino acids
→ Polyamides
- Two different monomers with different functional groups each
Terephthalic acid and ethylene glycol
→ Polyester
- Reversible poly condensation → equilibrium reaction
→ linear polyester
- Irreversible poly condensation → reaction of formaldehyde with phenol
→ Duromer (thermoset)

Balance of ester formation:

$$K = \frac{C_{Ester} * C_{Water}}{C_{Acid} * C_{Alcohol}}$$

Important monomers for Polycondensation

Monomers	Formula	Polymer
Adipic acid Hexamethylene diamine	$\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$ $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$	Polyamide 6,6
Terephthalic acid Ethylene glycol	 $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$	Polyethylene terephthalate
Phosgene 4,4'-Dihydroxy-diphenyl-propane (Bisphenol A)	COCl_2 	Polycarbonate
Dimethyl dichlorsilane	$\text{Si}(\text{CH}_3)_2\text{Cl}_2$	Silicones
ω -Amino-undecanic acid	$\text{HOOC}-(\text{CH}_2)_9-\text{CH}_2-\text{NH}_2$	Polyamide 11

Polymers

- Polyester
- Polycarbonate
- Phenolic resins
- Amino- and urea resins
- Polyamides & -imides

Polyester

- **Polyester for fibers & bottles:**

Terephthalic acid or dimethyl terephthalate and ethylene glycol

Catalysts: salts of divalent transition metals

- **Polyester alcohols:**

(diol component for the production of poly urethanes)

Aliphatic dicarboxylic acids, phthalic anhydride and/or maleic anhydride and ethylene glycol, diethylene glycol or propylene glycol

Catalysts: no catalyst or sulfuric acid

- **Alkyd resin:**

(binder for paint coatings)

Reaction of natural fats with glycerol, TMP and/or pentaerythrite, followed by cross linking of the partially esterified polyols with phthalic anhydride

Catalysts: sulfuric acid

Polyester

- **Unsaturated polyester resins:**

Phthalic anhydride or maleic anhydride plus ethylene glycol or diethylene glycol

Catalysts: sulfuric acid

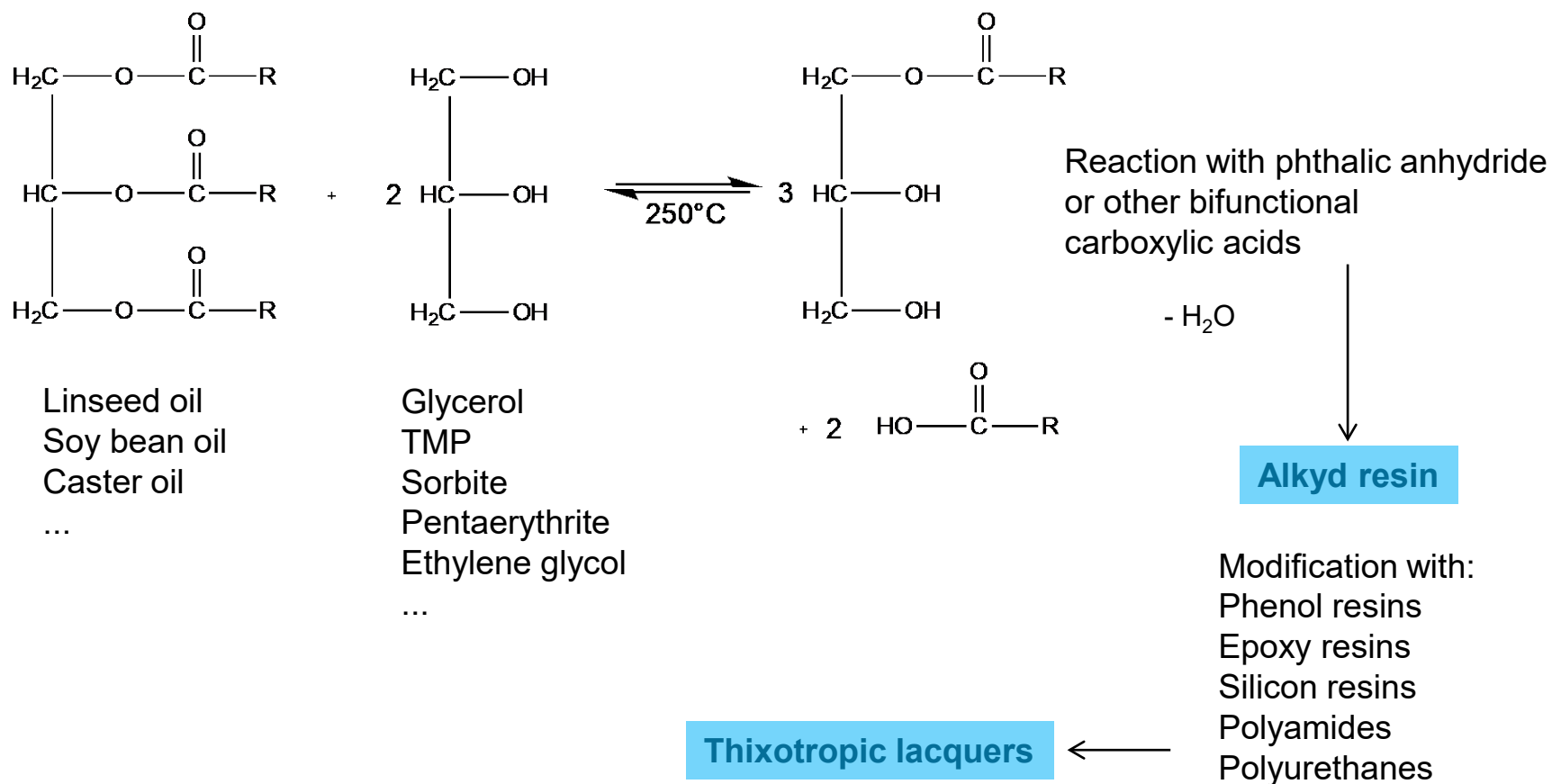
Followed by mixing with styrene and curing (cross linking) through polymerization – peroxides as initiators

- **Polycarbonates:**

Diphenylolpropane (Bisphenol A) or similar bis-phenols and phosgene or diphenylcarbonate

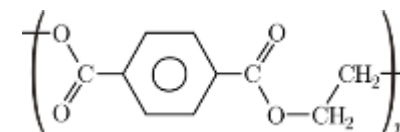
Catalyst: base substances

Alkyd resins



Polyethylene terephthalate PET

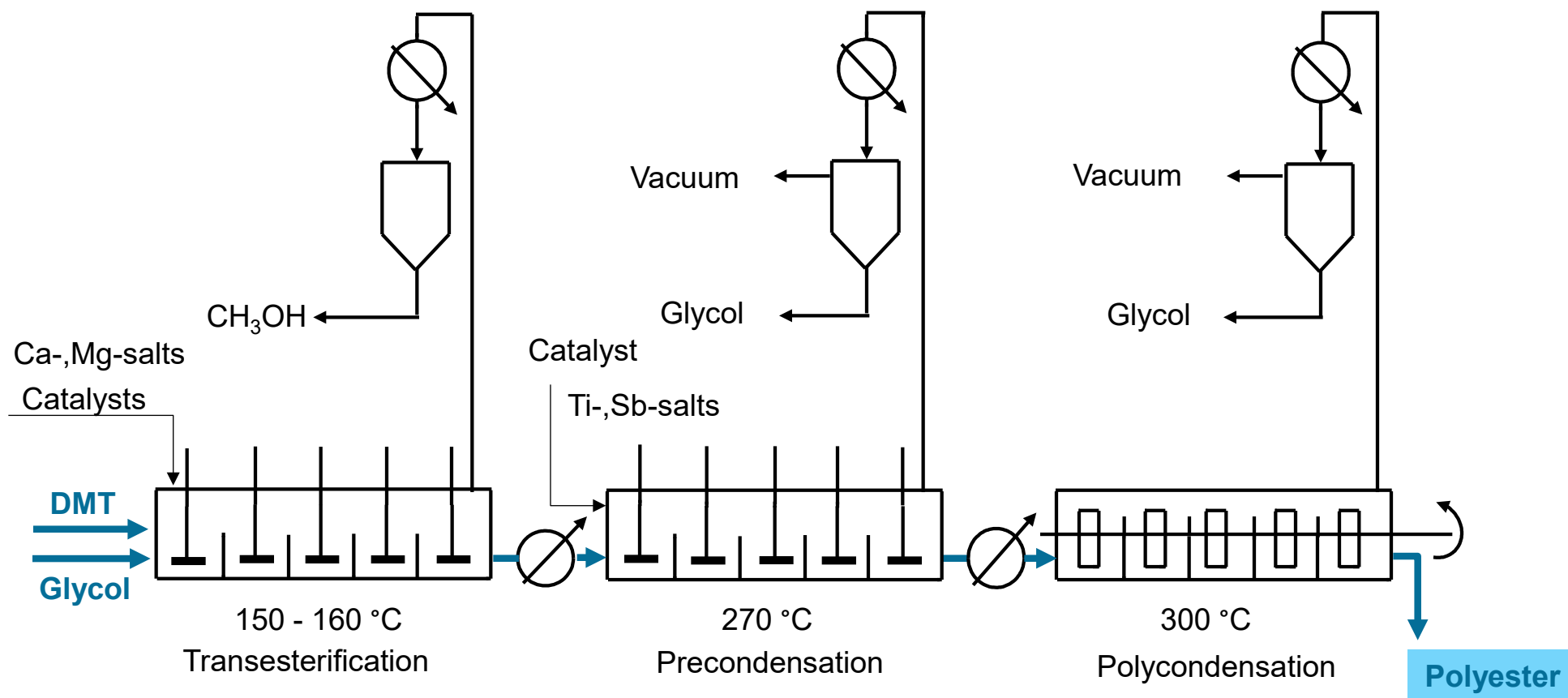
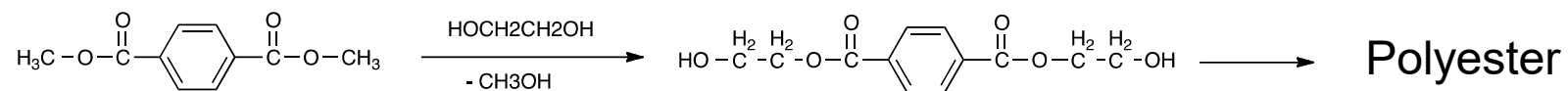
- Widely used thermoplastic of the Polyester family
- Use
 - Fibers 60%
 - Bottles 30% (carbonated drinks)
 - and other containers
- Polycondensation of terephthalic acid with ethylene glycol



- Density: 1370 kg/m³
- Young's modulus: 2800-3100 MPa
- T_G: 75°C
- T_M: 260°C

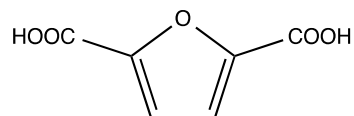
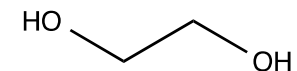


PET production

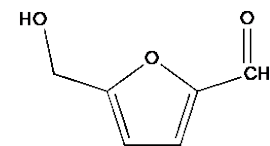
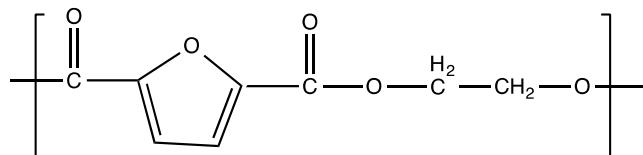


Coca Cola plant bottle

- Ethylene glycol is made from bio ethanol
- Bio based polyester with promising properties



Furandicarboxylic acid



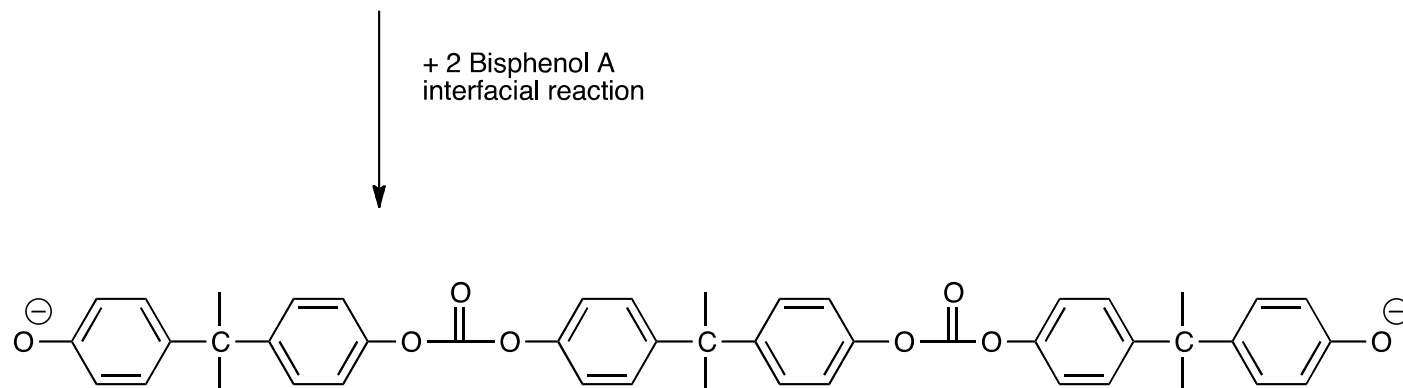
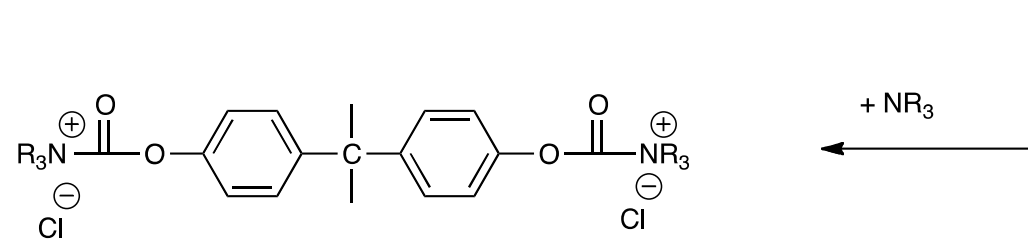
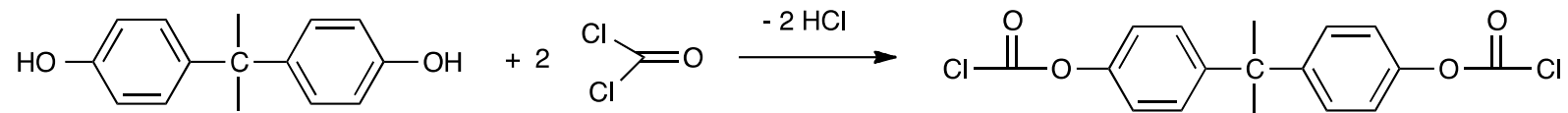
Hydroxymethylfurfural (HMF)

Polycarbonate PC

- Highly transparent material
- Use: eye-wear lenses, automotive headlamps, CD's, DVD's, lab equipment, iPOD player cases, ...
- Poly condensation of bisphenol A and phosgene
- Density: 1200 – 1220 kg/m³
- Young's modulus: 2000 – 2400 MPa
- T_G : 150°C
- T_M : 267°C



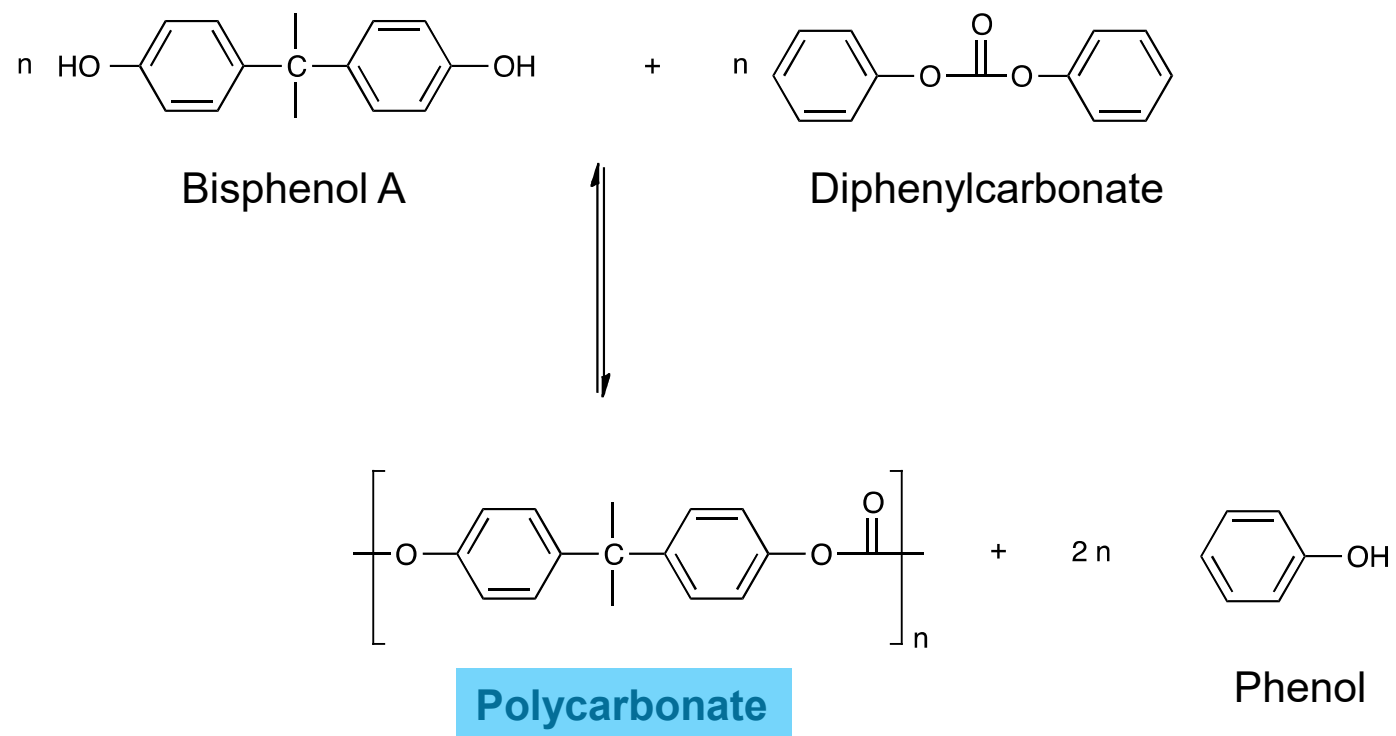
PC interfacial Polycondensation



(in Dichloromethane + NaOH)

Polycarbonate

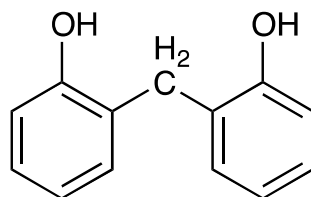
PC trans esterification



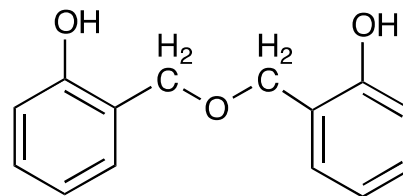
Phenolic resins

Condensation of phenol, cresol and/or xyleneols with formaldehyde

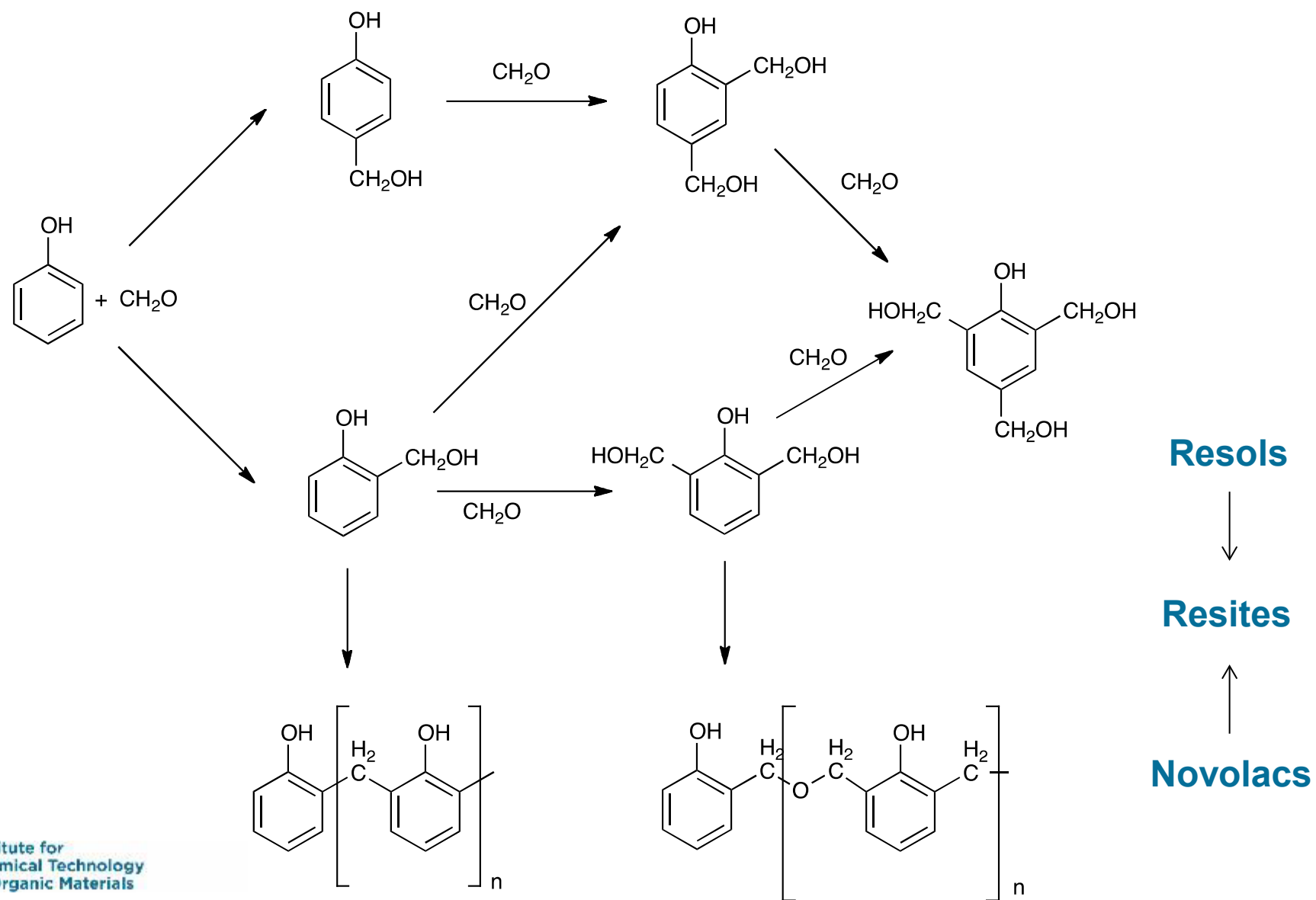
Linkage of aryl rests via bridges:



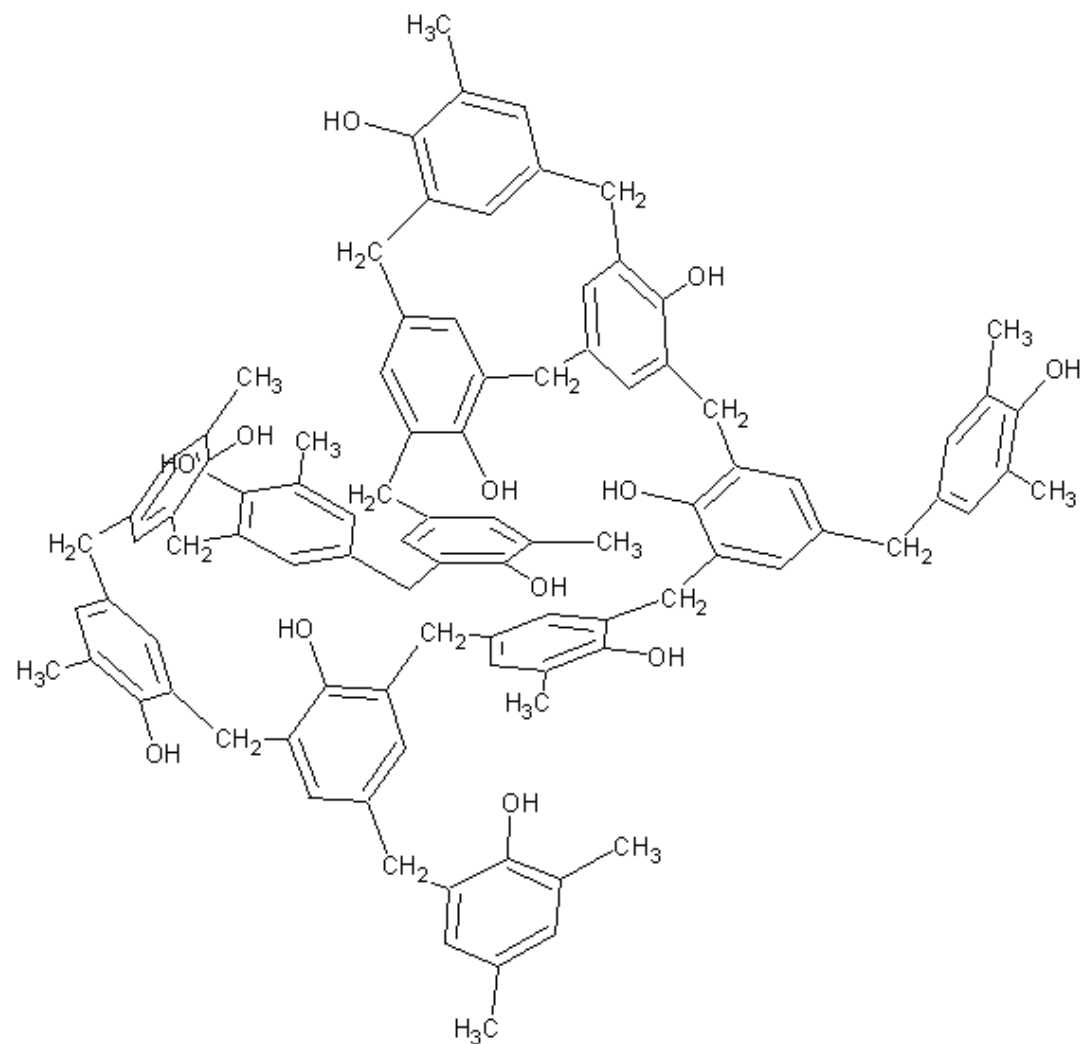
o or p to OH group



Phenolic resins

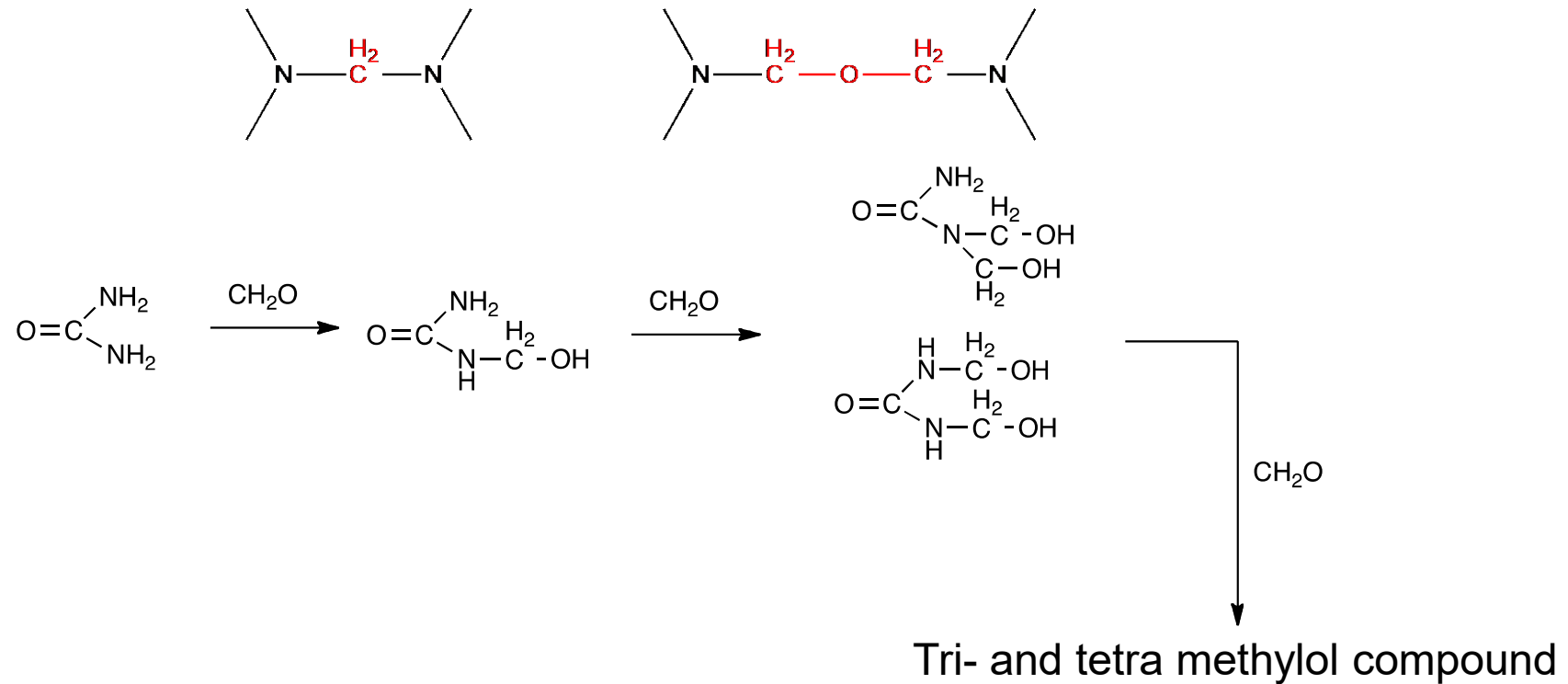


Phenolic resins



Amino resins

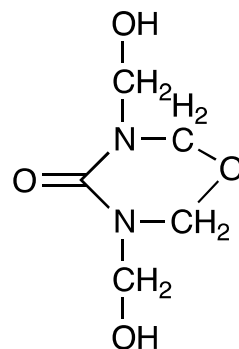
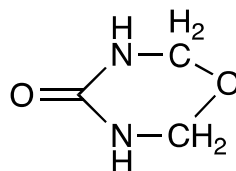
Condensation of urea and/or melamine with formaldehyde
Linkage of NH_2 groups via bridges:



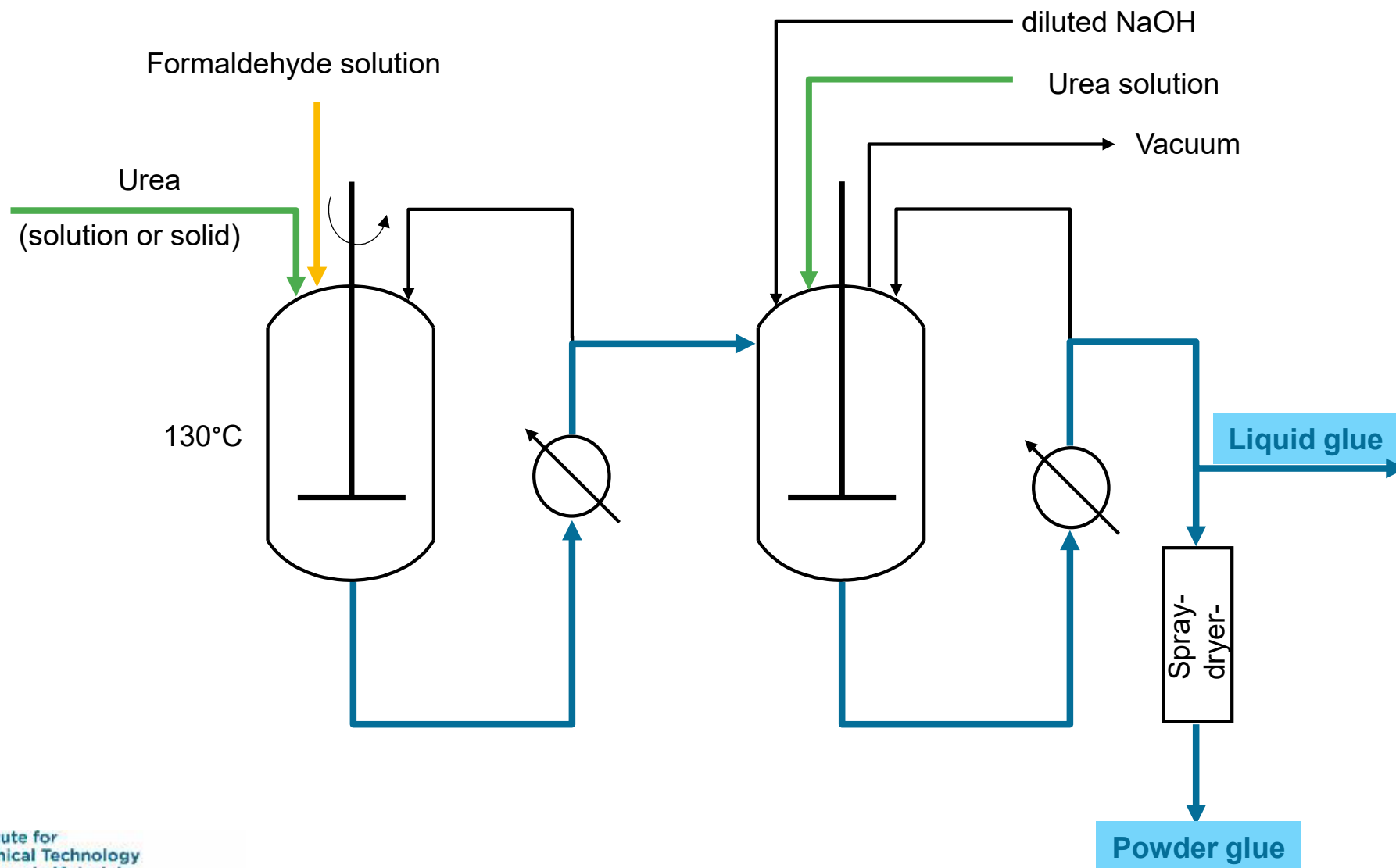
Amino resins

- Poly condensation at low temperatures via ether:
 $\text{H}_2\text{N}-\text{CO}-\text{NH}-\text{CH}_2-\text{O}-\text{CH}_2-\text{NH}-\text{CO}-\text{NH}_2$
- Poly condensation at high temperature via methylene bis urea:
 $\text{H}_2\text{N}-\text{CO}-\text{NH}-\text{CH}_2-\text{NH}-\text{CO}-\text{NH}_2$

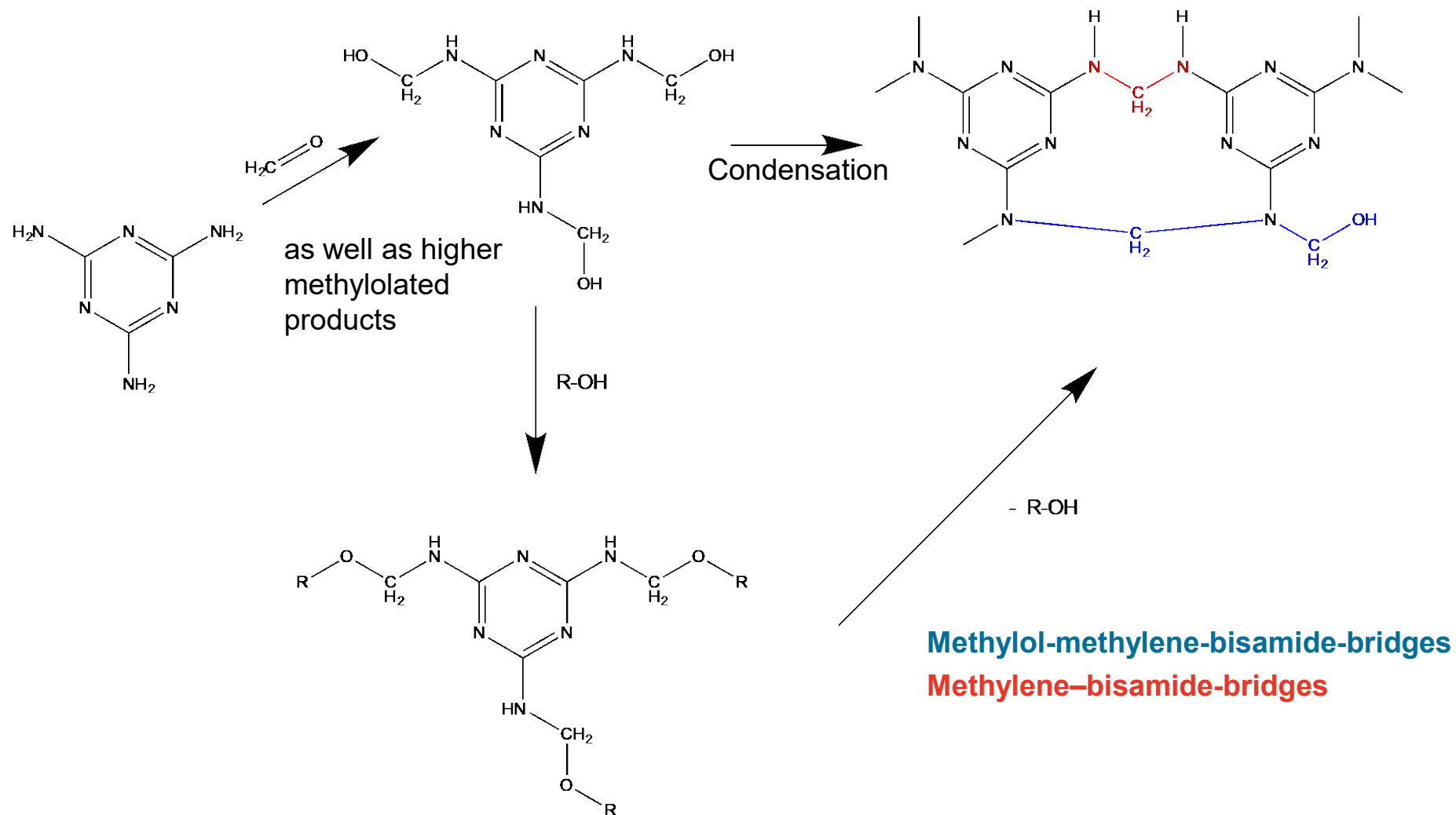
Cyclic structures:



Production of urea resins



Melamine resins



Polyamides

PA can be spun to fibers (melt spinning) or they can be processed like thermoplastics (used for construction purposes due to their high stiffness and stability)

Abbreviation	Components
PA 6,6	Adipic acid and hexamethylene diamine
PA 6	ϵ -Caprolactame
PA 11	Amino undecane acid
PA 12	C ₁₂ -Lactame



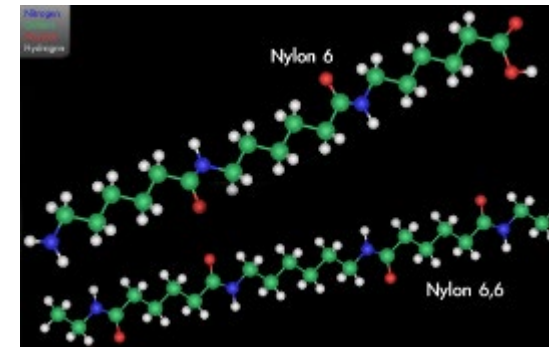
Polyamide

Nylon 6

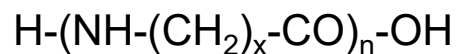
- ring-opening polymerization of caprolactam
- T_G : 40°C / T_M : 215°C
- bristles for toothbrush, knitted garments, threads, ropes, filaments, nets, tire cords, etc.

Nylon 6,6

- condensation polymerization of hexamethylene diamine and adipic acid
- T_G : 50°C / T_M : 280°C
- carpet fibers, airbags, tires, ropes, conveyor belts, hoses



Important PA resins



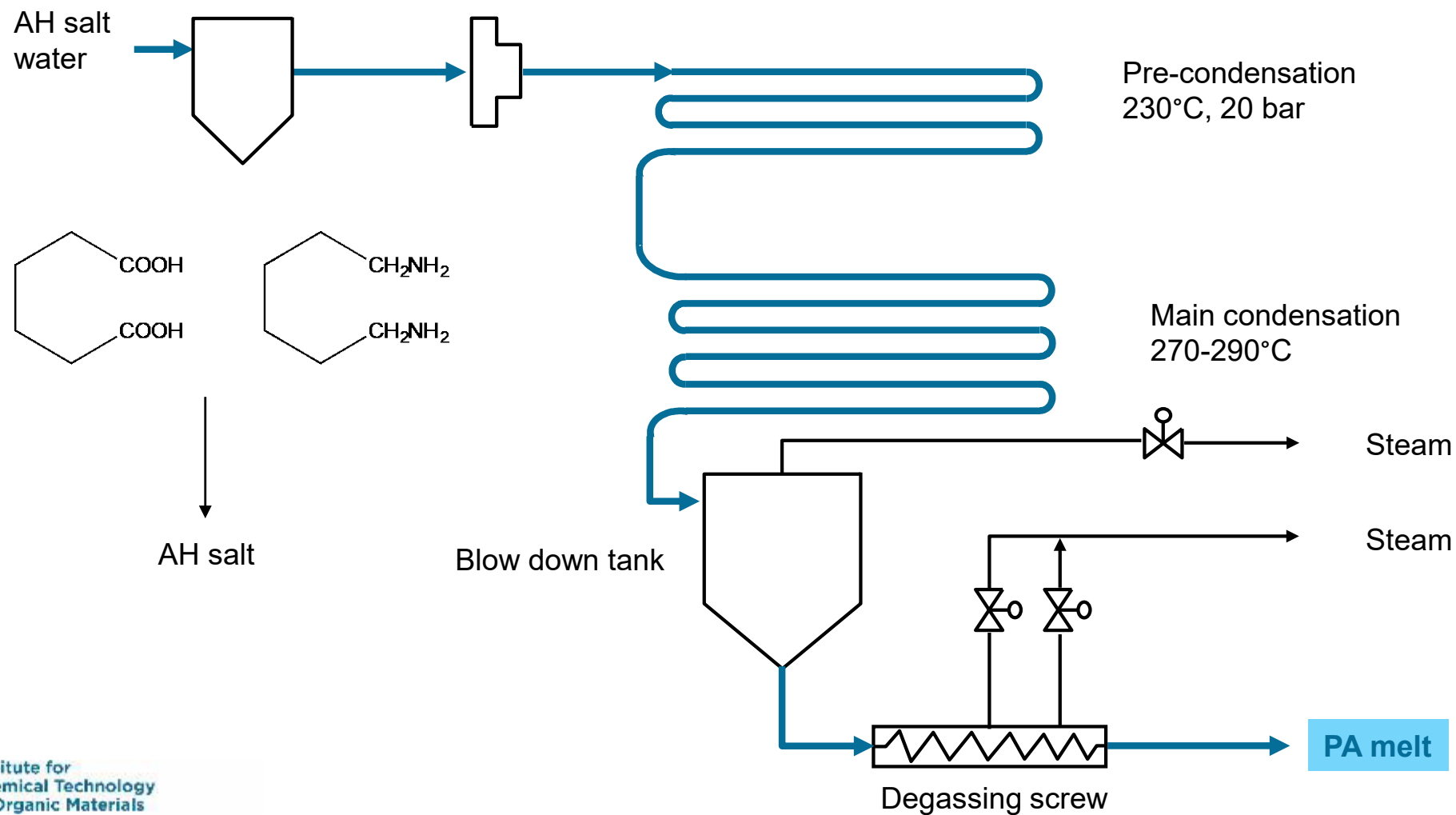
PA	Synthesis	Technical importance	Fiber properties
2 (x=1)	Poly condensation of amino acetic acid	No	(comparable to natural silk)
4 (x=3)	Anionic polymerization of g-Butyrolactame	No, T_M and T_D too close	Cotton
6 (x=5)	Polymerization of e-Caprolactame	High, raw materials are cheap	Excellent
7 (x=6)	Poly condensation of w-Amino heptane acid, hydrolytic polym. of the lactame (Önanthlactame)	Low, Synthesis of monomer too difficult	Excellent textile properties
11 (x=7)	Poly condensation of w-Amino undecane acid	Low, raw material (castor oil) too expensive	Low water up-take
12 (x=11)	hydrolytic polym. of the lactame (Laurinlactame)	For textiles low but for construction materials important	Low water up-take

Properties of PA resins

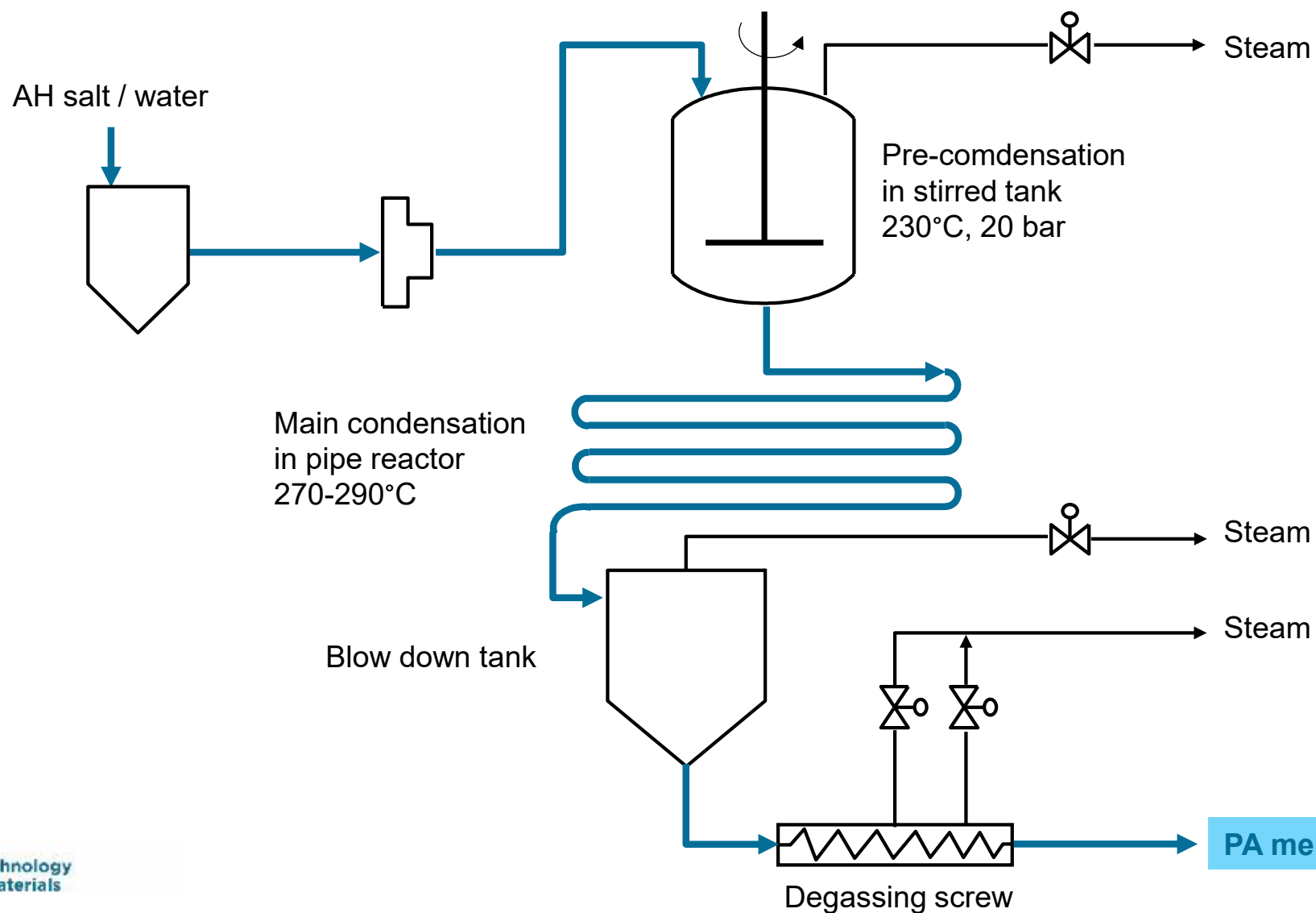
PA	T _M [°C]	Water up-take [%]*	Density [g/cm ³]
4	265	9,1	-
6	223	4,3	1,14
7	233	2,8	1,10
11	190	1,3	1,04
12	179	1,3	1,04

* @ 65% relative humidity

Production of PA 6,6 in a reactor cascade



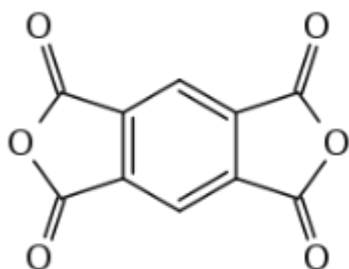
Production of PA 6,6 reactor cascade STR/pipe



Polyimides

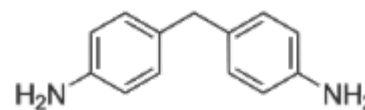
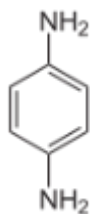
Extreme rigid polymers with high TM, stiffness and form stability

Very stable against O₂ at high temperatures → construction materials for aviation and space industry



Pyromellithic dianhydride

Plus elongated aromatic diamines



p-Phenylenediamine, 4,4'-Diamino-diphenylmethane
4,4'-Diamino-diphenylether, 4,4'-Diamino-diphenylsulfon

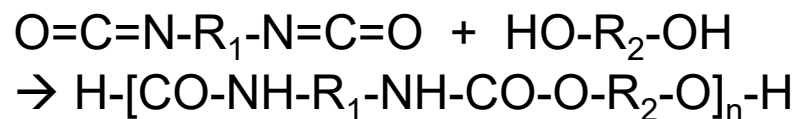
Polyaddition

- Reaction of 2 different functional groups without elimination of low-molecular compounds, in contrast to poly condensation
- Build up of chain is step-wise via addition of monomers to the growing chain

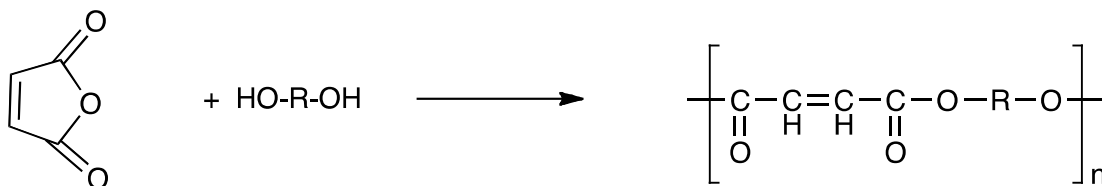
- Polyurethanes
- Non-saturated Polyester
- Epoxy resins

Polyaddition

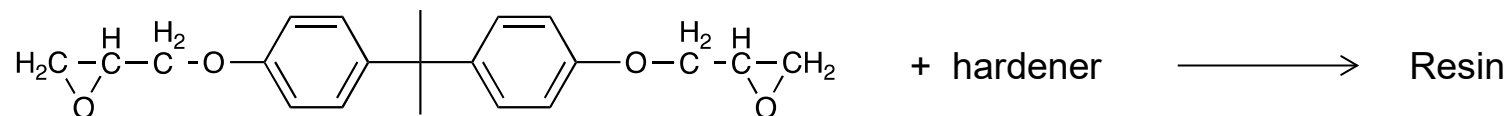
- Polyurethanes



- Non-saturated Polyester

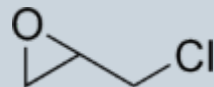
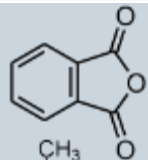
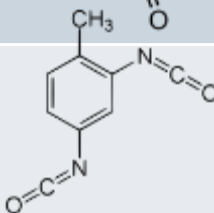
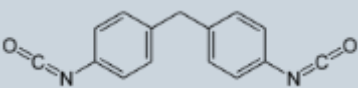
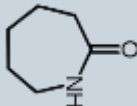


- Epoxy resins



2,2'-(((propane-2,2-diylbis(4,1-phenylene))bis(oxy))bis(methylene))bis(oxirane) or
DGEBA: Di-Glycidyl Ether of Bisphenol A

Technically important monomers

Monomer	Formula	Polymer
Epichlorohydrin		Epoxy resin
Aliphatic di- and tri- amines	$\begin{array}{c} \text{H}_2\text{N}-\text{R}-\text{NH}_2 \\ \text{H}_2\text{N}-\text{R}-\text{NH}-\text{R}-\text{NH}_2 \end{array}$	Epoxy resin
Phthalic anhydride		Epoxy resin
Toluylenediisocyanate (TDI)		Polyurethane
4,4'-Diisocyanate-diphenyl-methane		Polyurethane
Diols	$\text{HO}-(\text{CH}_2)_4-\text{OH}$	Polyurethane
e-Caprolactame		PA 6

Synthesis of iso-cyanates

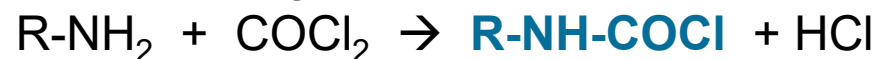
Phosgenation of primary amines

- Production of phosgene:



50-100°C and 0,2-0,5 MPa; catalyst: activated carbon

1. Step of phosgenation: formation of urea acid chloride



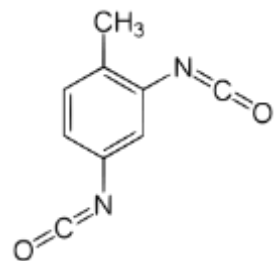
10-100°C and 0,1-5 MPa, Cl-benzene or o-Di-chloro-benzene as solvent, phosgene excess, exothermic reaction

2. Step of phosgenation: formation of iso-cyanate

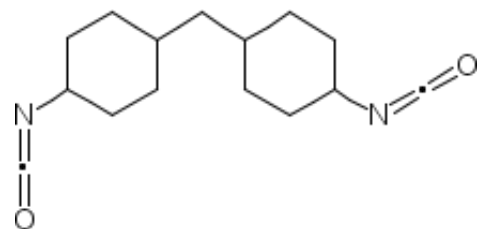


10-100°C and 0,1-5 MPa, endothermic reaction

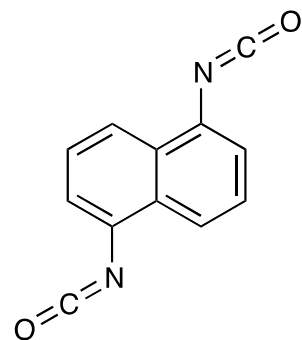
Technically important diisocyanates



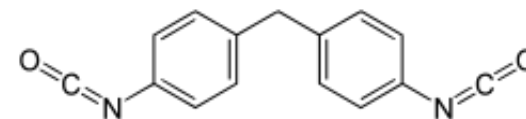
2,4-Toluylenedi-isocyanate (TDI)



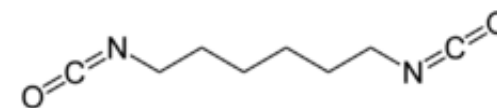
4,4'-Dicyclohexyl-methane-diisocyanate



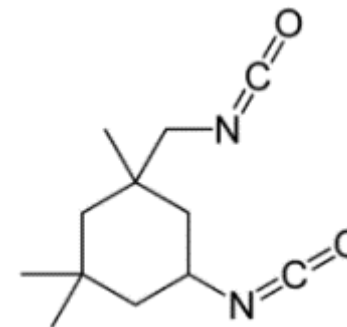
1,5-Naphthylene-diisocyanate (NDI)



4,4'-Diphenyl-methane-diisocyanate (MDI)



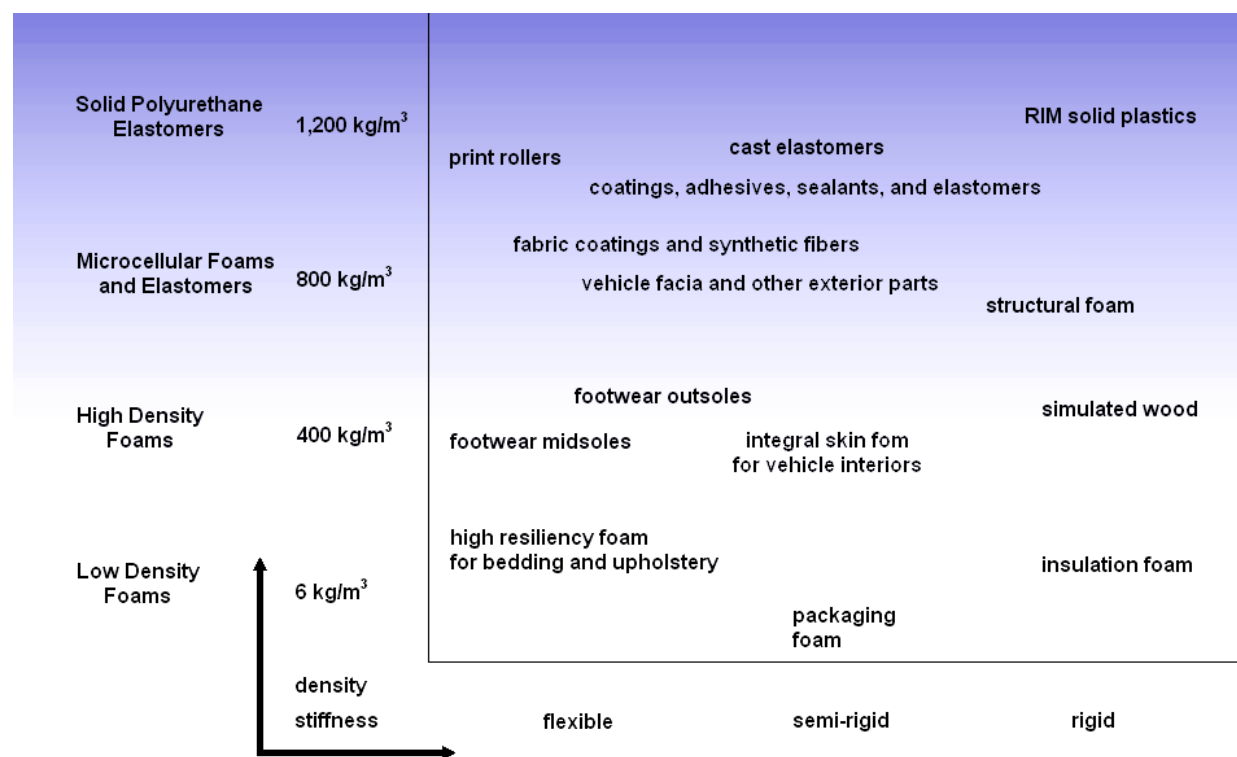
1,6-Hexa-methylene-diisocyanate (HDI)



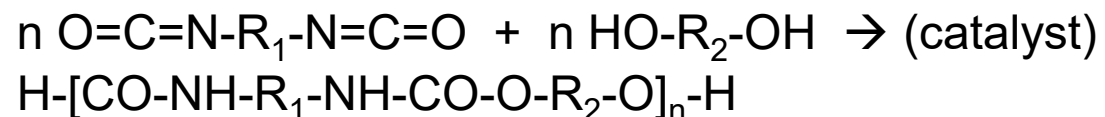
Isophorondiisocyanate (IPDI)

Polyurethanes PU

PU covers an extremely wide range of stiffness, hardness and densities

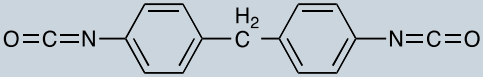
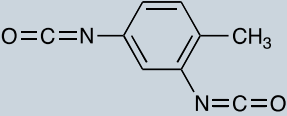
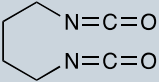


Production of PU



Addition of 3 or 4 functional alcohols → cross linking

Addition of water → release of CO₂ → foaming

$\text{O}=\text{C}=\text{N}-\text{R}_1-\text{N}=\text{C}=\text{O}$	$\text{HO}-\text{R}_2-\text{OH}$
 4,4'-Diphenyl-methane-diisocyanate (MDI)	
 2,4-Toluylenedi-isocyanate (TDI)	
 1,6-Hexa-methylene-diisocyanate (HDI)	Polyester alcohols Polyether alcohols Hexane diol Butane diol

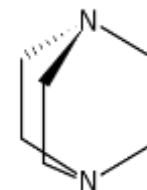
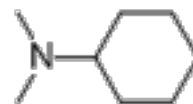
Production of PU

- **Catalysts:**

tert. Amines: e.g. N,N-Dimethyl-cyclohexyl-amine, 1,4-Diaza-bicyclo-octane
Alkali salts of organic acids and phenols, Sn organic compounds

- **Stabilizers:**

Oxidation- and UV stabilizer, e.g. Phosphites
Foam stabilizers, e.g. Polyethersiloxanes



DABCO

TEDA Triethylenediamine

Technically important PU

PU foams:

- Soft foams

Based on TDI and MDI plus linear polyols
padding material, filter materials

- Hard foams

Based on MDI and highly branched polyols (e.g. polyether from sorbitol)
Insulation material for construction

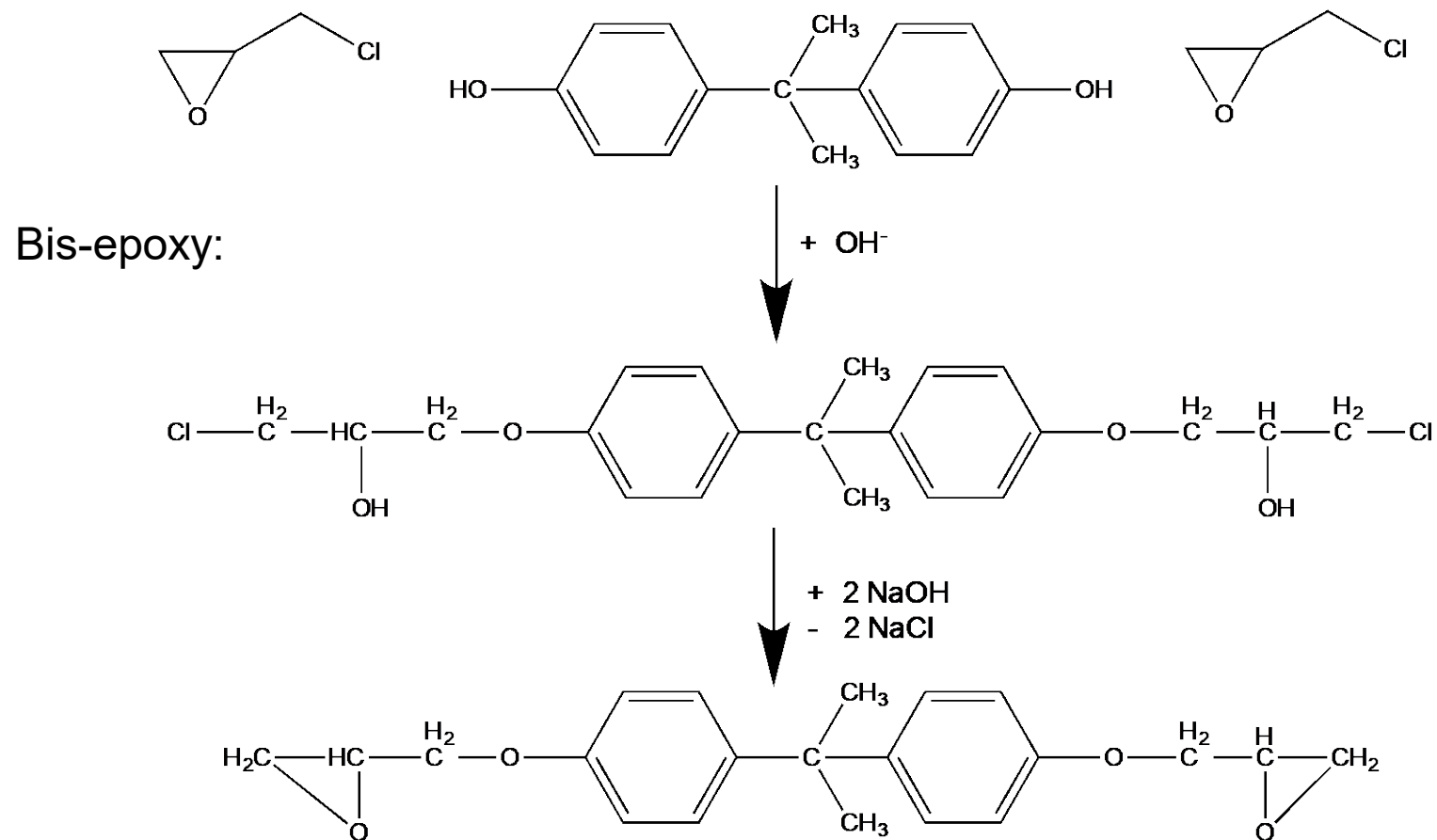
- Integral foams

Core has a foam structure, shells has a solid surface
Automotive industry, shoe industry, construction industry

Technically important PU

- PU casting resin
Coatings, sealing compounds, floorings
- PU coatings for textiles
Artificial leather
- PU elastomers
- PU lacquers
- PU glues
- PU fibers

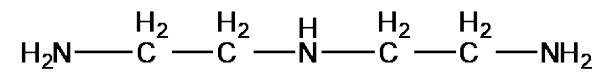
Epoxy resins



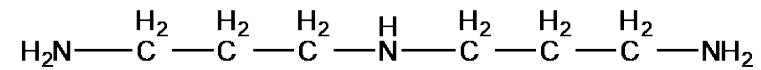
Epoxy resins

Hardener:

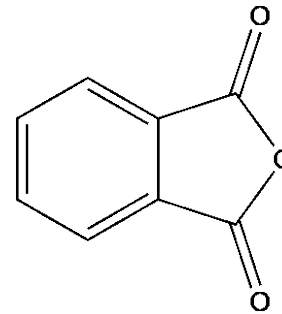
Diethylentriamin



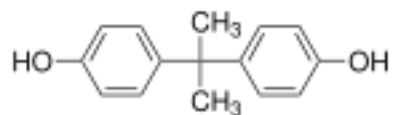
Dipropylentriamin



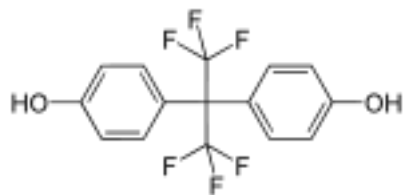
Phthalsäureanhydrid



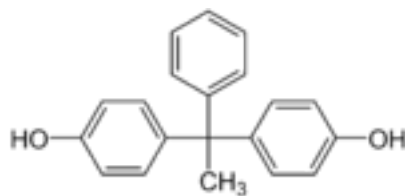
Other Bisphenoles



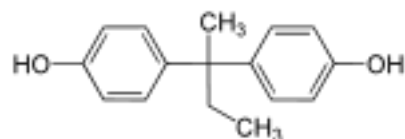
Bisphenol A



Bisphenol AF



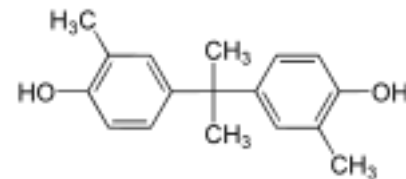
Bisphenol AP



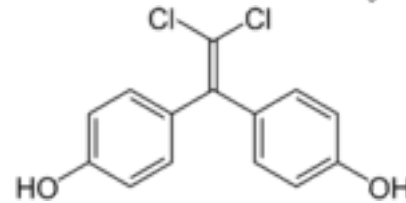
Bisphenol B



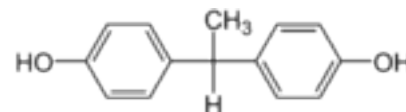
Bisphenol BP



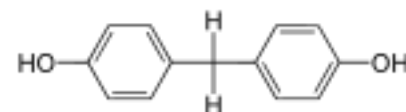
Bisphenol C



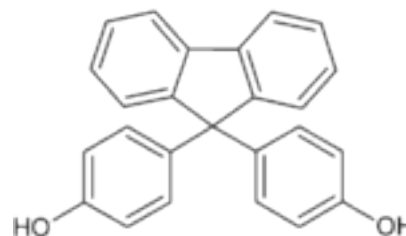
Bisphenol C II



Bisphenol E

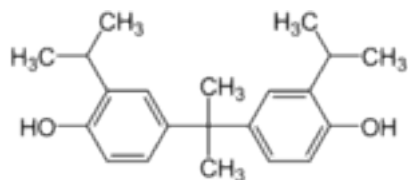


Bisphenol F

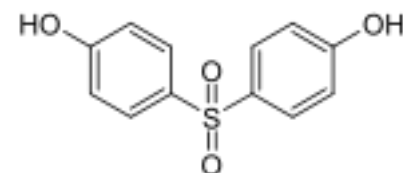


Bisphenol FL

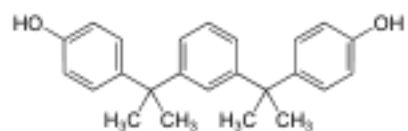
Other Bisphenoles



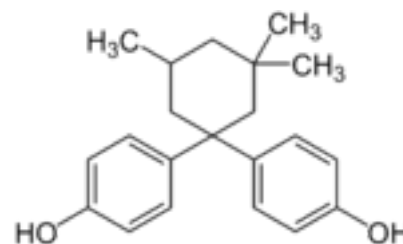
Bisphenol G



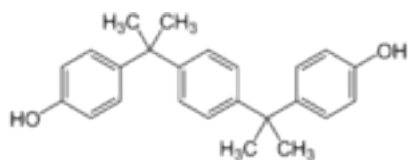
Bisphenol S



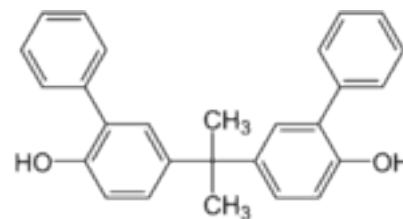
Bisphenol M



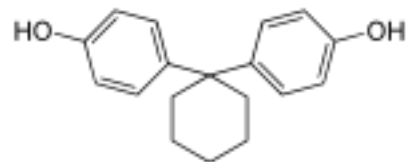
Bisphenol TMC



Bisphenol P

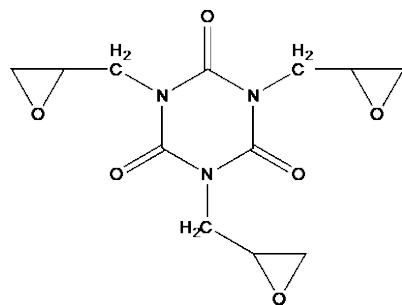


Bisphenol PH

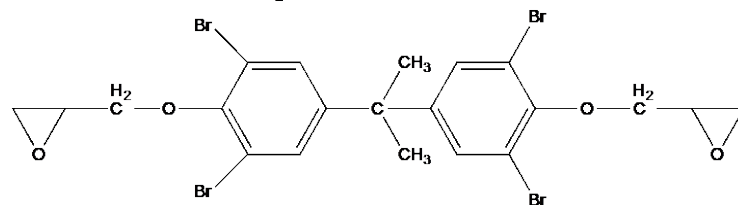


Bisphenol Z

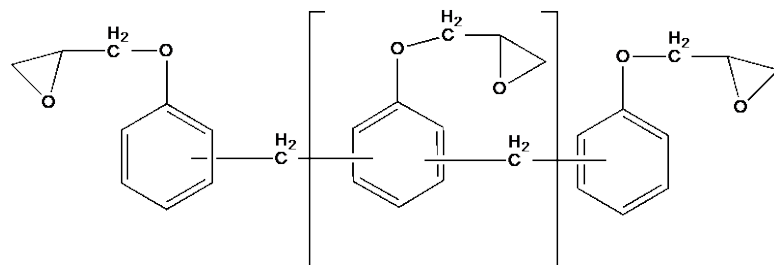
Other Bisepoxies



Isocyanuric resins

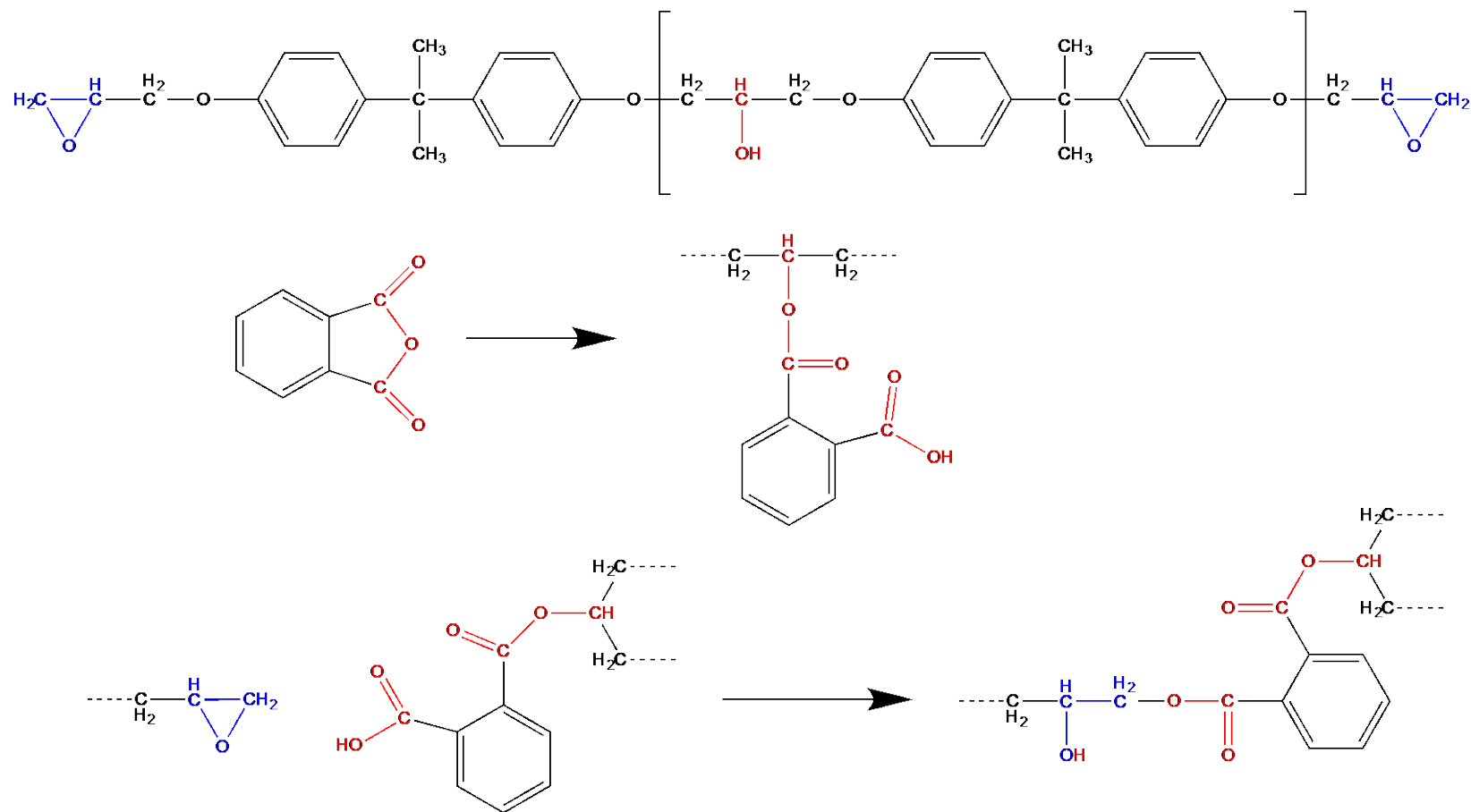


Brominated epoxy resins

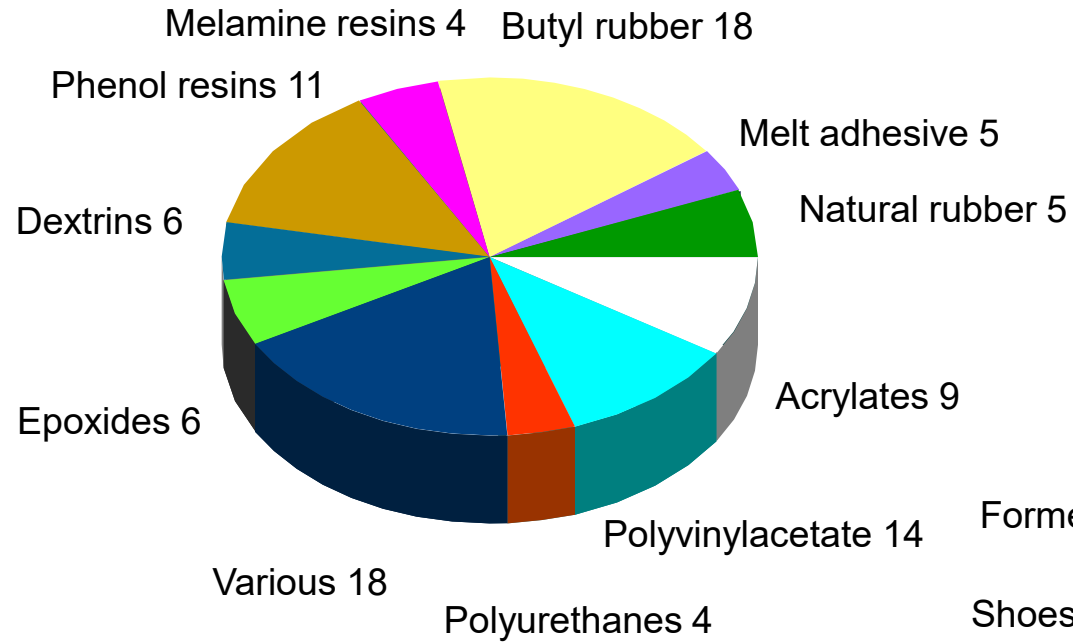


Novolak epoxy resins

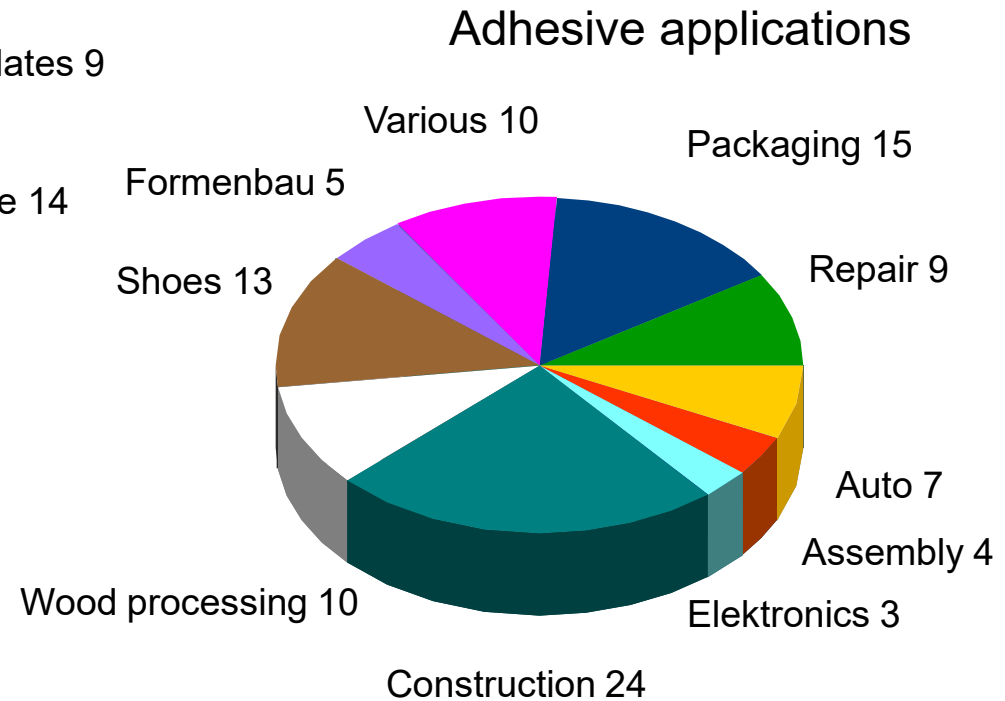
Curing of epoxies with acid anhydrides



Adhesives



Adhesives

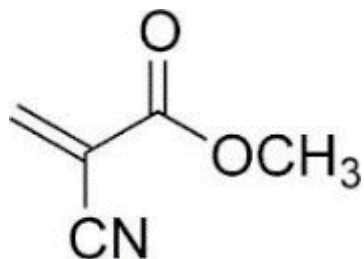


Thermoplastic adhesives

- **Polyvinyl acetate** is mainly used as a dispersion adhesive. Because of its relatively high glass transition temperature (28°C) it is used as a copolymer with ethylene, vinyl propionate or acrylates. The main application is the bonding of paper and furniture industries.
- **Ethylene-vinyl acetate copolymers** (EVA) with 25-50% vinyl acetate are particularly used as hot melt adhesives. Applications include bookbinding, carpet production, furniture and footwear industries.

Thermoplastic adhesives

- **Cyanoacrylates** are derived from α -cyanoacrylic esters which readily polymerize anionically (ambient moisture is enough). They adhere very well to almost all surfaces and bind them very quickly. Their use in vascular and bone surgery is controversial.



Thermoplastic adhesives

- **Polyvinyl chloride pastes** are almost always copolymers with 5-20% VAc and a few percent maleic anhydride, they are designed for bonding of PVC with a multitude of other materials (paper, metal, leather, etc.).
- **Polyamides** are used as hot melt adhesives or solutions. They have a very wide range of applications.
- **Acrylate copolymers** are usually copolymers with methacrylates. They are used as dispersions and solutions, as well as powder. Applications include wood/plastic composites, foams, metal bonding.

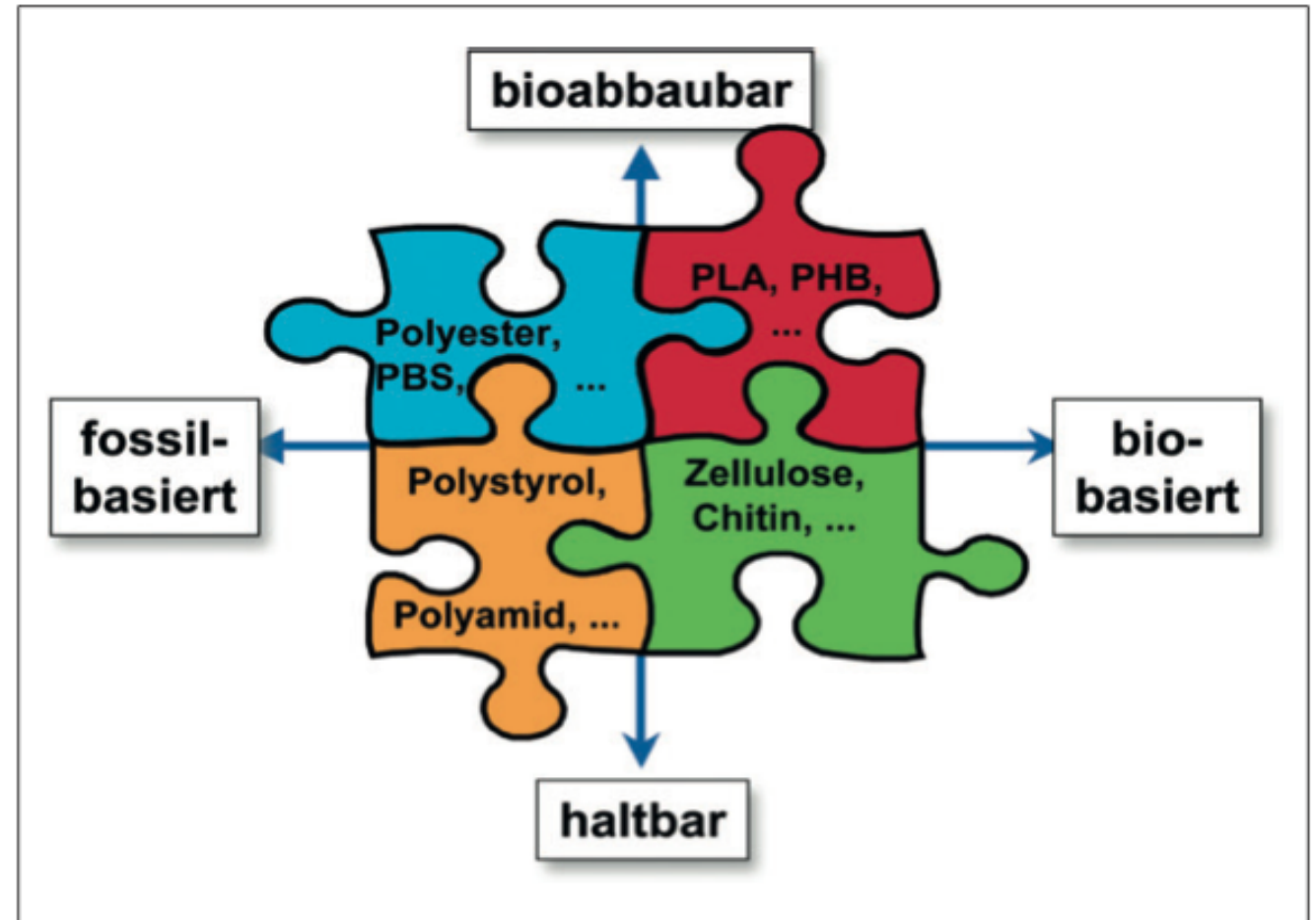
Thermoset adhesives

- **Urea-formaldehyde resins (UF)** resins are widely used as adhesives and glues in chipboard, plywood and furniture industries.
- **Melamine-formaldehyde resins (MF)** resins give better weather resistant and whiter bondings, but more expensive. They are used like UF resins also for mixed condensates.
- **Phenol-formaldehyde resins (PF)** resins are widely used both as novolac and resole. Resols are cured directly. The main applications are chipboard.

Bio plastics

What does bio really mean?

- Bio plastics: polymeric materials partially or totally derived from annually grown resources by opposition to petroleum based resources
- Does not imply that the material is bio degradable



Bio degradable polymers

- Are around on an industrial scale since the end of the 1990s
- Not defined in terms of their raw material basis
- Can be completely converted into water, CO₂ and biomass through the action of microorganisms (fungi and bacteria)
- Biodegradability is a special functionality – gives the product an additional value
 - Disposing packaging material along with organic waste or tilling agri films into the ground

Bio degradability

- Degradability: mechanical disintegration
- Bio degradability: metabolism
fulfilling norms, e.g. EN 13432: the polymer has to be converted to CO₂ by over 90% within 180 days under defined conditions of humidity, temperature and oxygen level

Bio degradability

- Aerobic process:
organic matter is transformed in the presence of oxygen
water, CO₂, energy and biomass
in soil, or in compost
- Anaerobic process:
organic matter is transformed in the absence of oxygen
water, CO₂, energy, biomass and methane

Bio degradability

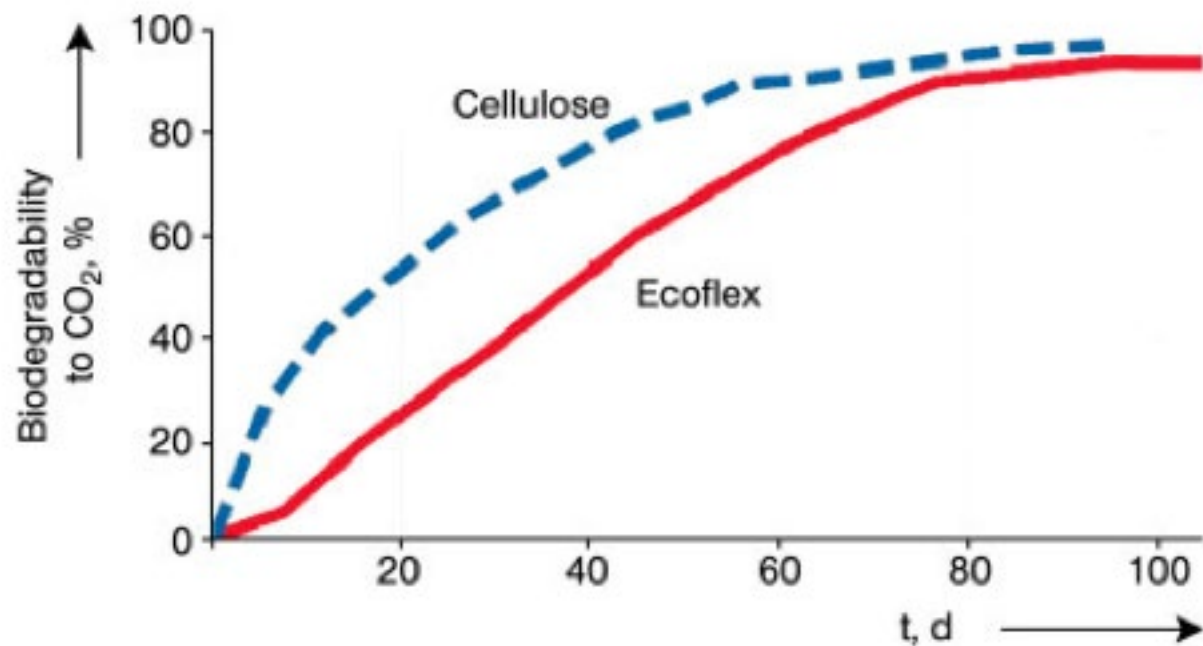
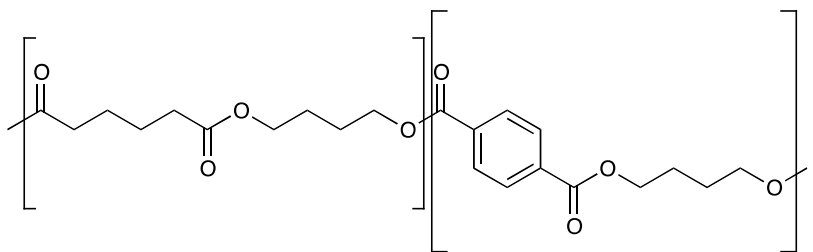
Process

- Most of the biodegradable polymers are insoluble in water
- Too big to pass the cellular membrane
- Biodegradation begins with chain cleavage by enzymes (lipases and esterases) to oligomers which can be metabolized
- Surface process (enzymes are too large to diffuse into the polymer)
- Influencing factors: humidity, oxygen level, temperature

Bio degradability

An example: aliphatic aromatic polyester

Poly-(butylene adipate)-co-(butylene terephthalate), PBAT, Ecoflex



Raw materials

Fossil based:

- Butanediol and dicarboxylic acids (adipic-, succinic- and terephthalic acid)
 - Butanediol: butadiene
 - Succinic acid: maleic anhydride
 - Adipic acid: cyclohexane
 - Terephthalic acid: p-xylene

Renewables:

- Feedstock for monomer production (e.g. lactic acid) or polymer production (e.g. poly hydroxy alkanoates, PHA)
- Structural material (e.g. cellulose) and compound ingredient (e.g. starch) for bio degradable polyester

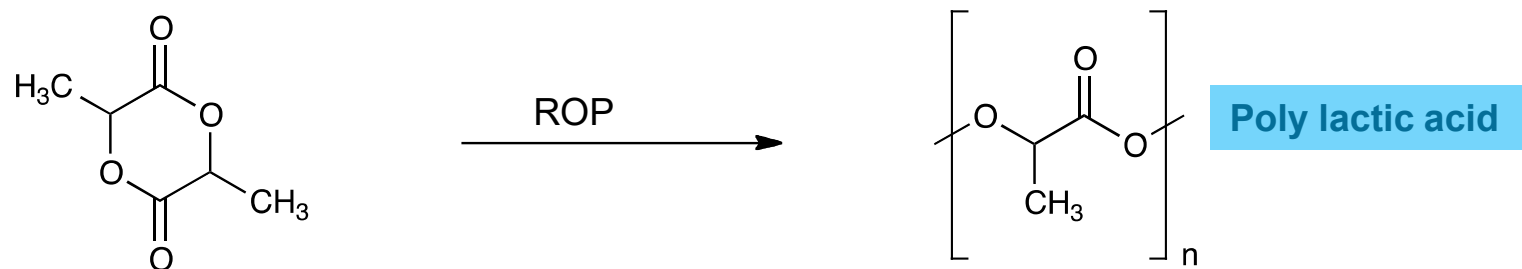
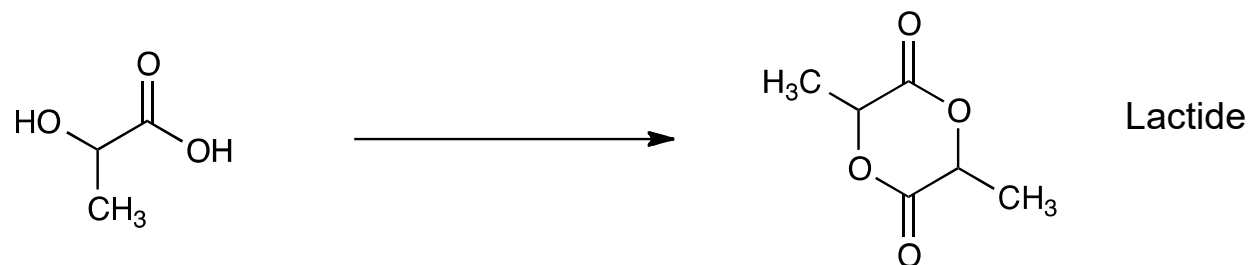
Renewable feedstock

- Starch (from corn, wheat, etc.) – is transformed to glucose
Fermentation to monomers (e.g. lactic acid) or PHA production
- Other feedstock for fermentative monomer production:
sucrose, glycerol, plant oils
cellulose raw material (C6 and C5 sugars)
- Direct access to monomers via extraction and processing of renewable feedstock (e.g. sebacic acid, biorefinery)

Polylactic acid (PLA)

- PLA is a melt processable thermoplastic polymer
- Completely based on renewable resources
- Manufacturing includes one fermentative step followed by several chemical transformations
- Source is corn starch, which is broken down to unrefined dextrose
- Sugar is then fermentative transformed to lactic acid

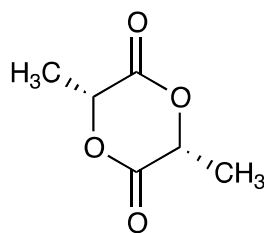
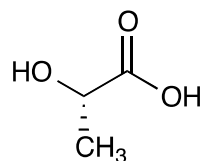
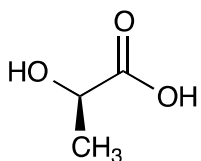
PLA: synthesis



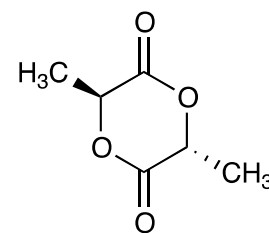
PLA: properties and application

- Due to its stereogenic center, lactic acid exists in 2 enantiomeric forms (D- and L) leading to 3 different lactide stereo isomers (D-, L-, meso-lactide)
- Depending on the relative amounts the crystallinity of the PLA is heavily influenced and by this the properties

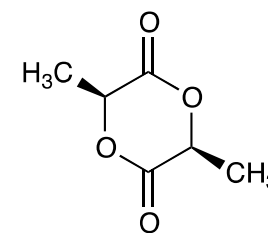
Lactic acid



Lactide



meso



PLA: properties and application

- PLA shows similar properties to Polystyrene
high stiffness and transparency makes it suitable for bottles, cups (cold drinks), stiff packages,
- Can be processed on standard equipment (IM, ISBM, BM, blown- and cast film, biaxially oriented films, paper coating but also durable applications)
- Limitation: low impact strength, poor barrier properties and low heat resistance

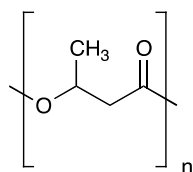
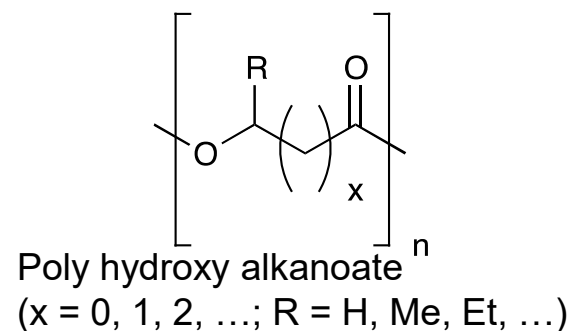
PLA: biodegradation

- Main degradation route for PLA is hydrolysis
- Takes place at high humidity and elevated temperatures
- Found in industrial composting not at home composting (temperatures too low)
- Fragments resulting from hydrolysis can be consumed by microorganisms

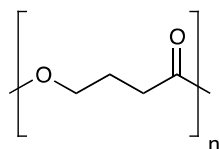


Poly hydroxy alkanoates (PHA)

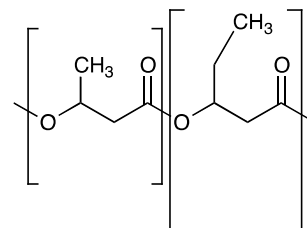
- Structurally related to PLA
- Polyester derived from hydroxyalkanoic acids which differ in chain length and in position of OH group



P(3-HB)
Poly(3-hydroxybutyrate)



P(4-HB)
Poly(4-hydroxybutyrate)



PHBV
Poly(hydroxybutyrate-co-valerate)

PHBH
Poly(hydroxybutyrate-co-hexanoate)

PHA: synthesis

- PHAs are derived from sugars, starch or fatty acids
- Hydrocarbons (CH_4) can be used for fermentative synthesis of PHB by methanotropic bacteria
- Polymer is made by microorganisms as carbon and energy storing substance
- Mass of the polymer in the microorganism can reach up to 90% of the dry weight of the cell mass
- Chemical structure depends on the bacterial species

PHA

Properties and applications

- PHAs due to their great structural variety have a broad range of possible applications

Biodegradation

- PHAs are accepted as food sources by many microbe species which break the polymer down by extracellular depolymerases

Producers

- Although PHB is known since 1926 commercial interest only began lately, only pilot plant quantities available

Bio degradable synthetic polyester

Aliphatic polyester

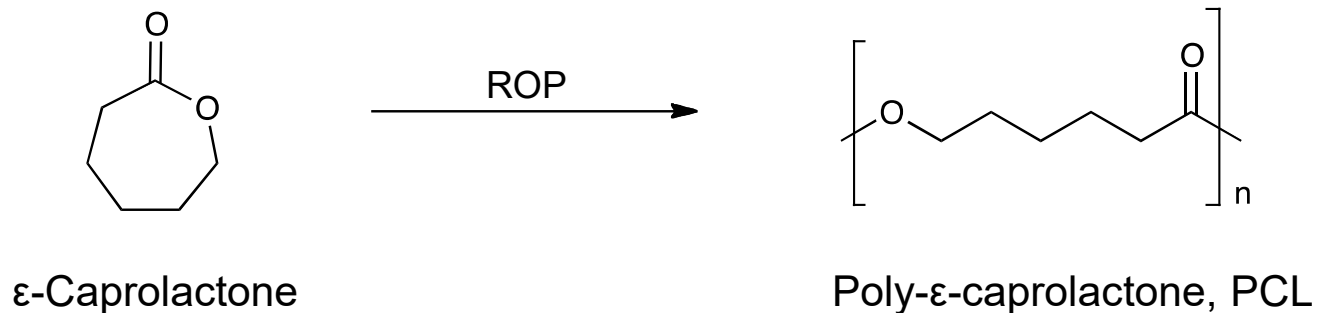
- Specially designed polymers keeping in mind biodegradability
- Fossil based monomeric building blocks
- Existing plants
- In addition good mechanical properties

Aliphatic/aromatic polyester

Bio degradable synthetic polyester

- There are 2 fundamentally different structures
- One is closely related to fermentative PHA - Lactones
- The other type is based on condensation reaction of aliphatic dicarboxylic acids and aliphatic glycols

Poly- ϵ -caprolactone (PCL): polyester produced in a ROP



Aliphatic Synthetic polyester

Condensates

- Condensation of aliphatic dicarboxylic acids
succinic acid, adipic acid, or combination of both
- And aliphatic glycols
ethylene glycol, 1,4-butane diol
- PES, PBS, PBSA

Properties and applications

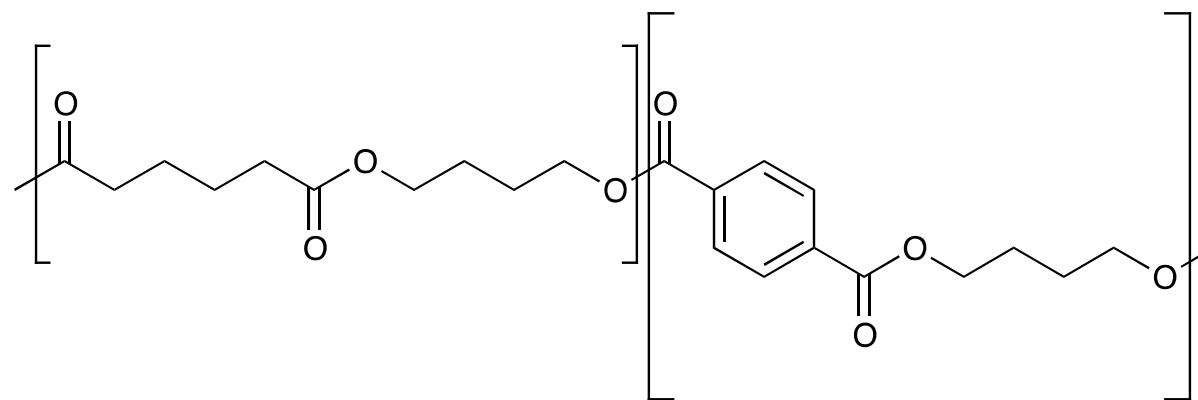
- Due to broad spectrum of monomers, properties can be adjusted
- The longer the dicarboxylic acid, the more flexible the material
- Drawback: prone to chemical hydrolysis (limitation in shelf life time)

Biodegradation

- In soil and compost degraded to CO₂ and water

Aliphatic aromatic polyester

- Good biodegradability of aliphatic polyester can be combined with the good mechanical strength of aromatic polyester
- Melt condensation of aliphatic (adipic) acids as well as aromatic dicarboxylic acids (terephthalic) with aliphatic glycols (1,4-butane diol)



Ecoflex, BASF

Aliphatic aromatic polyester

Properties and applications

- Strong and flexible material
- Mechanical properties comparable to PE
- Used as an “enabler” for starch, PLA, lignin or cellulose
- Biodegradable if aromatic content does not exceed a certain limit

Other biodegradable synthetic polymers

Polyethylene glycol (PEG)

- Water soluble polyether
- Mainly used in detergents and shampoos
- Aerobic and anaerobic biodegradable if molecular weight not too high

Polyvinyl alcohol (PVOH)

- Partly hydrolysis of polyvinyl acetate
- Soluble in water (dependent on degree of hydrolysis)
- Biodegradable in water and soils (flushable items)



Why is recycling of plastics so challenging?

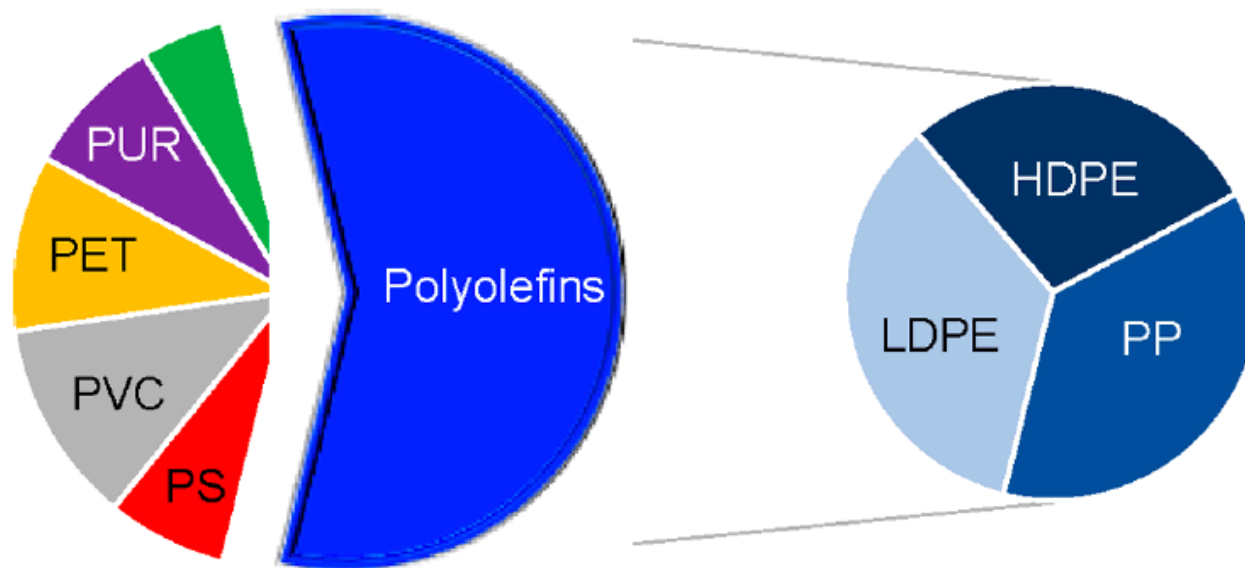
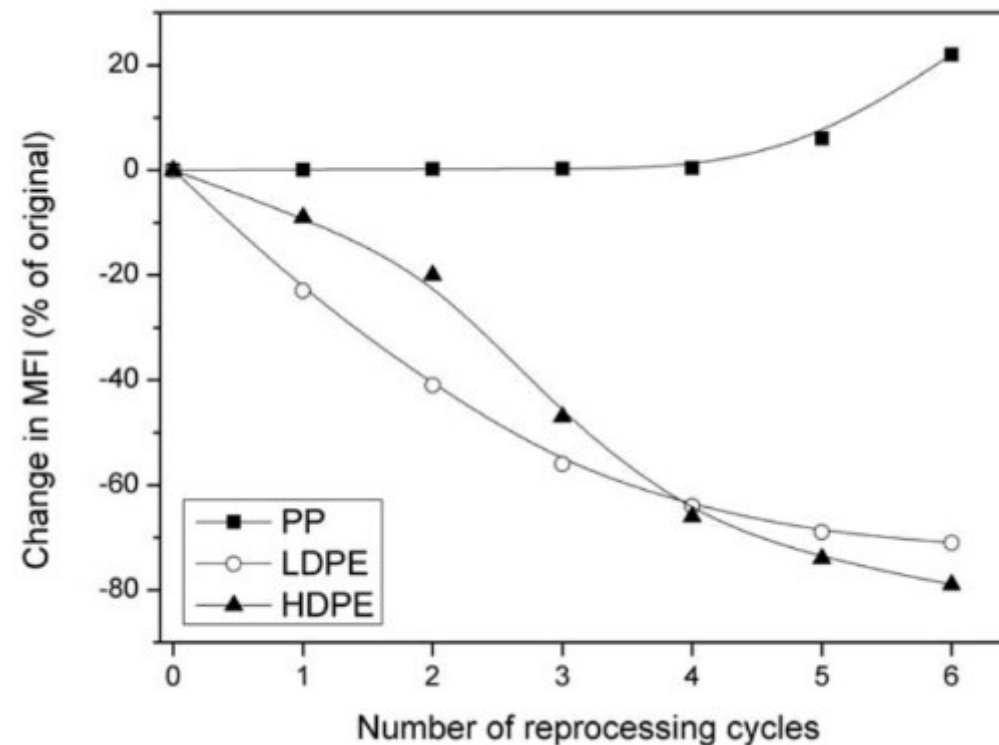
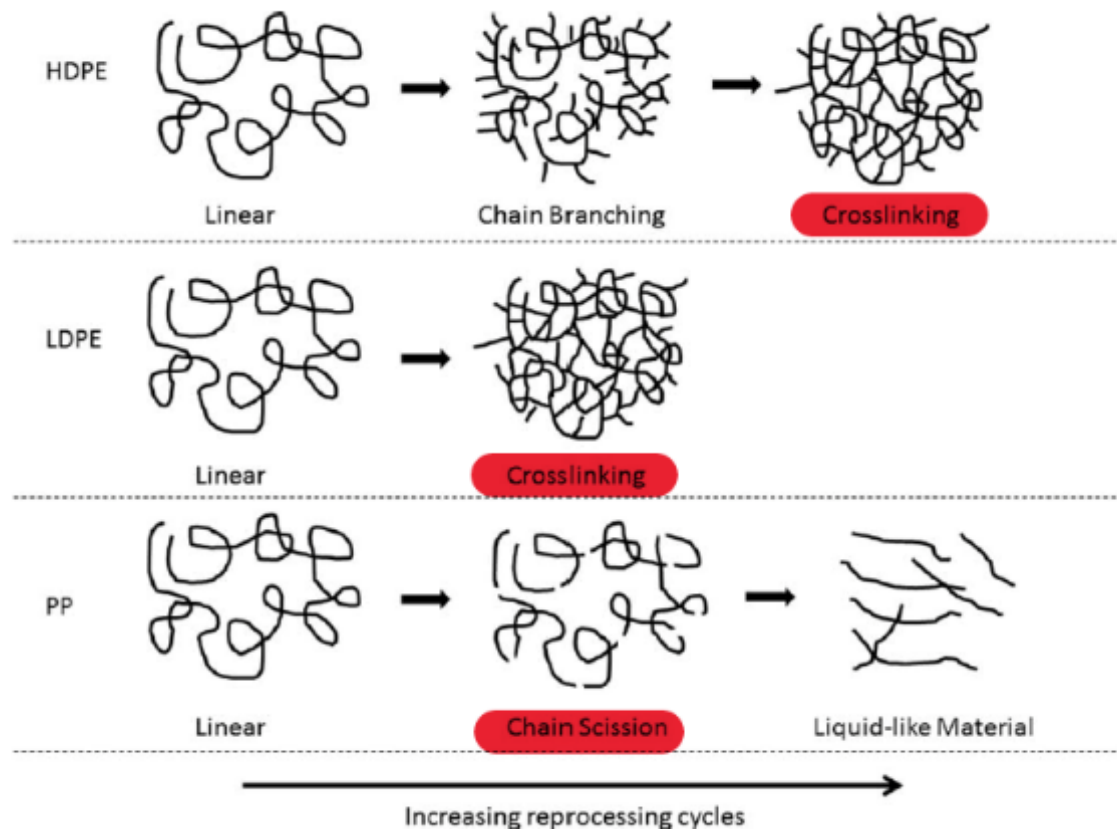


Figure 2. Composition of total polymer resin production from data for Europe, U.S., China, and India from 2002 to 2014 as tabulated in ref 4 (Table S2). More than 57% of the mass of plastic produced in this time period was polyolefins with the breakdown between LDPE (and LLDPE), HDPE and PP shown in the breakout. The green slice is all other polymers.

Changes in PO's due to mechanical reprocessing

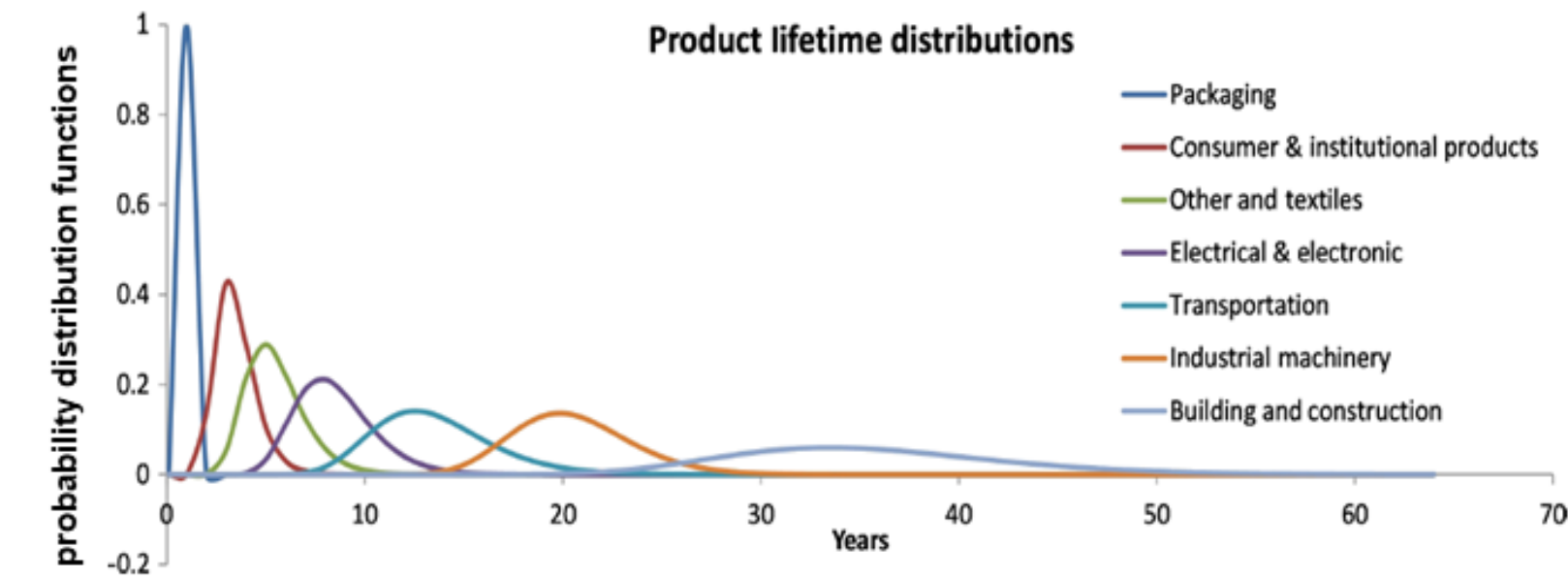
How long can we keep the material in the loop without compromising the quality of products?



S. Yin, R. Tuladhar, F. Shi, R.A. Shanks, M. Combe, T. Collister,
Poly. Eng. & Sci. 2015, 2907, DOI 10.1002/pen

Plastic product lifetime

- In Europe about 20 Mio t for packaging applications
- This is about 40 % of the polymer consumption
- Most of the polymer goes into short time use (less than 2 years)

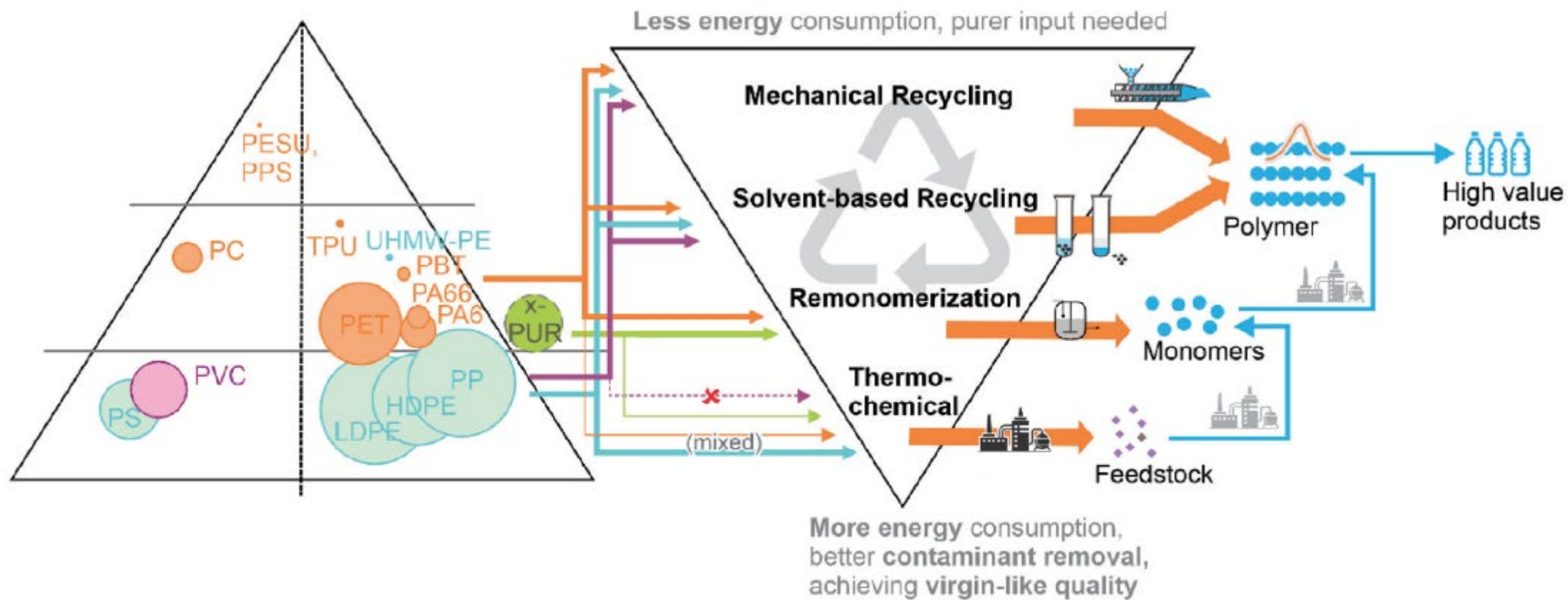


Source: Geyer, Jambeck, Law Sci. Adv. 2017; 3:e1700782

Recycling of plastics is challenging

- Although majority is based on polyolefins, complex situation
- Multilayer, multi-material structures
- Barrier layers, labels, glues, additives, fillers, pigments, modifiers
- Residues of use
- Ageing (degradation) products of all these materials
- Mixtures of plastics loose in properties
- Contaminations limit useability

Plastics recycling



H. Mangold, B. Vacano, Macromol. Chem. Phys. 2022, 2100488

Chemical Recycling

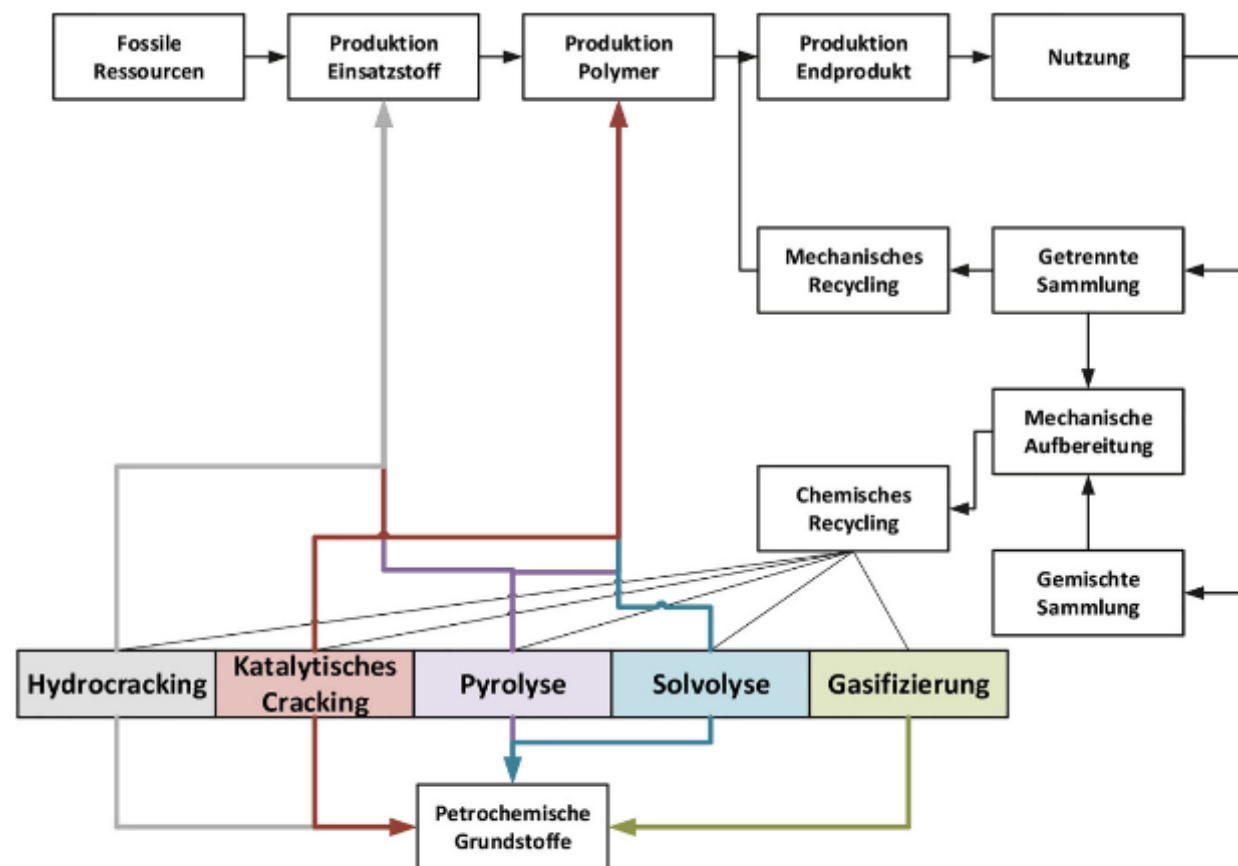


Abb. 3 Möglichkeiten des chemischen Recyclings im Lebenszyklus von Kunststoffprodukten

Quelle: Österr Wasser- und Abfallw 2020 · 72:47–60; <https://doi.org/10.1007/s00506-019-00628-w>

Chemical Recycling

- Polymers are broken down into basic components
- Available products
 - Monomers
 - Intermediates
 - Liquid product
 - Gases
- Advantages
 - Products streams can be purified → standard operations like distillation etc.
 - Afforded products can be reused to produce virgin materials
- Disadvantage
 - Costs
 - Availability of waste streams

Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

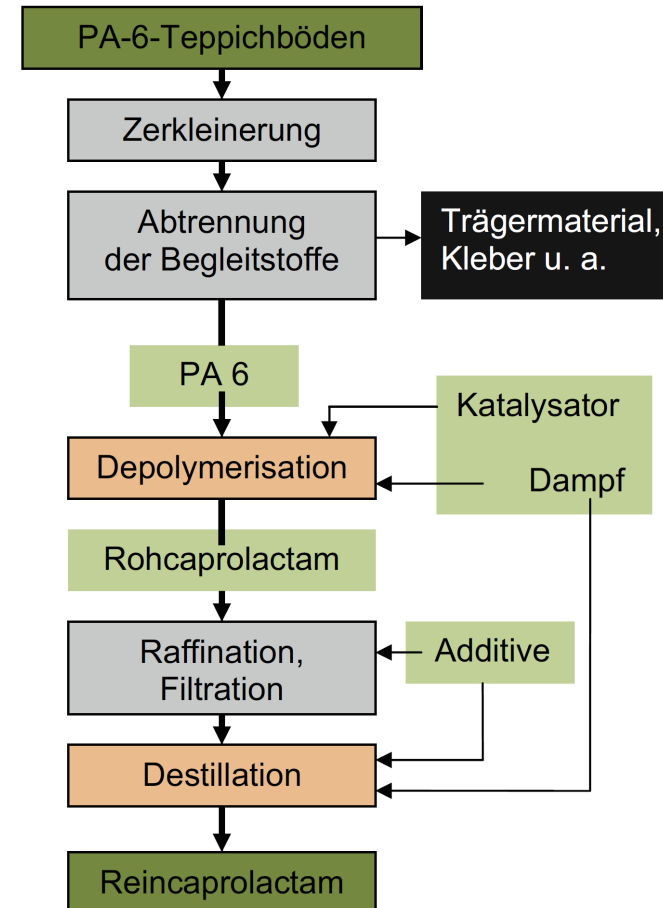
Processes used in chemical recycling

- Conversion into monomers
 - Hydrolysis
 - Solvolysis
 - Catalytic depolymerization
- Hydrogenation to liquid hydrocarbons
- Thermochemical conversion
 - Pyrolysis
 - Cracking
- Gasification to Syngas

Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Chemical Recycling of PA6

Hydrolysis of carpet waste to caprolactam



Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Chemical Recycling of PET

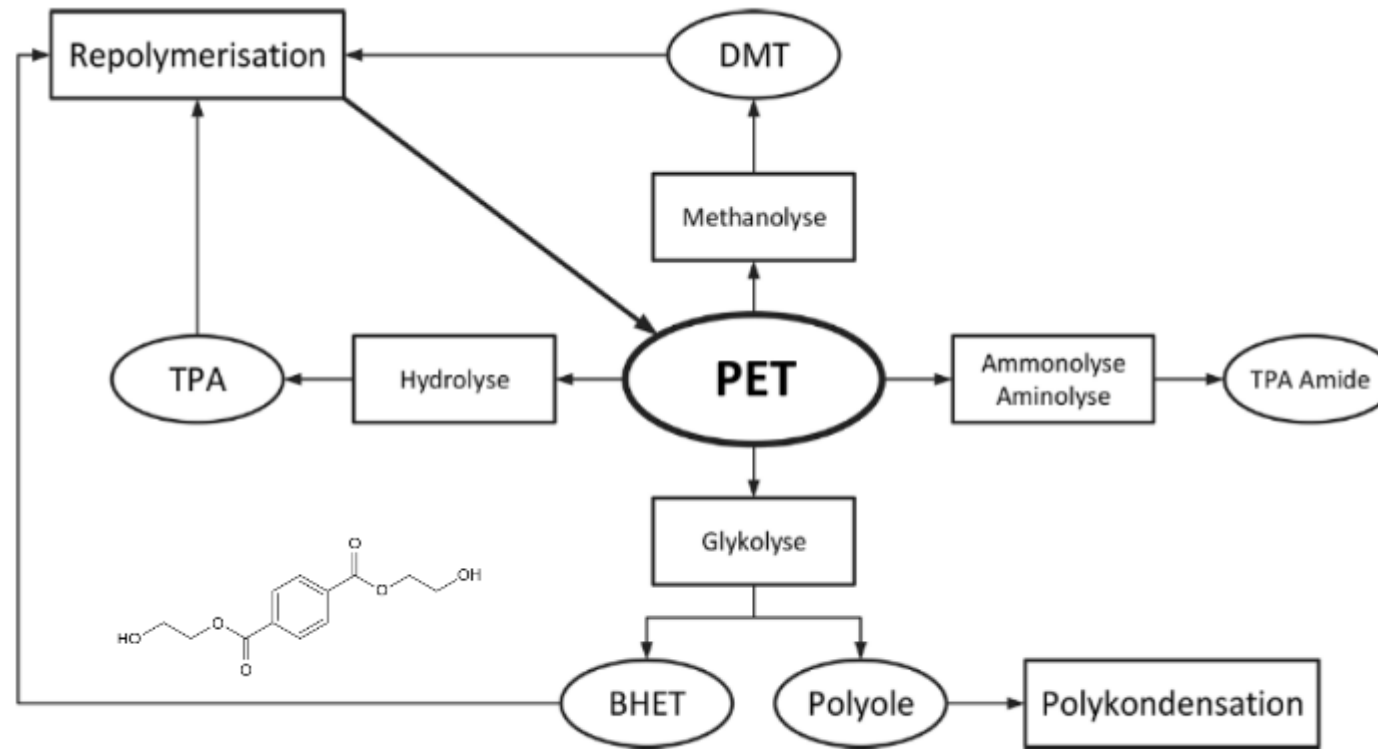
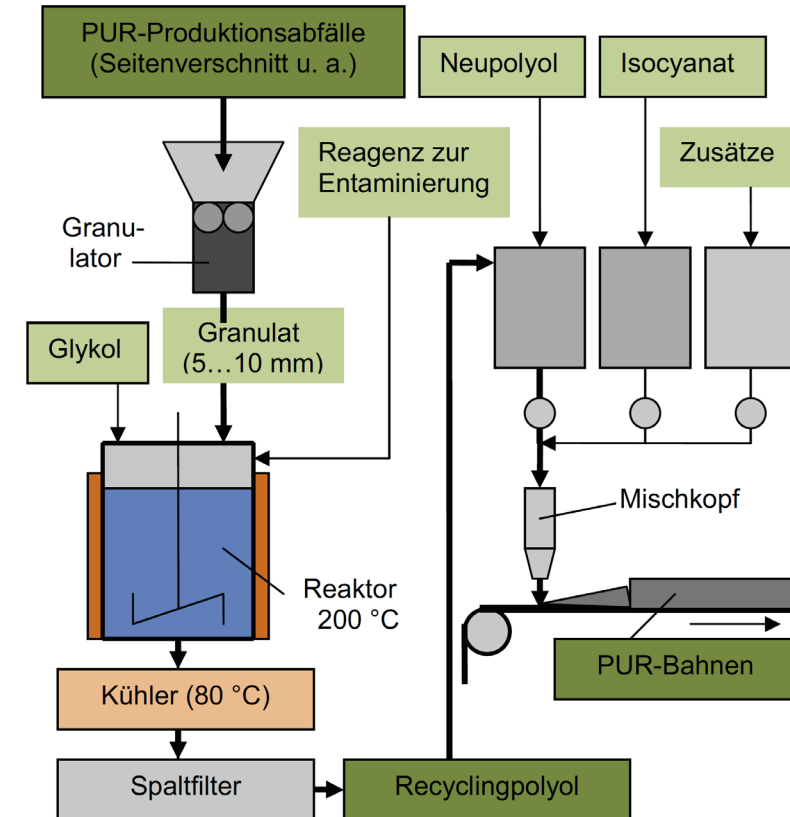
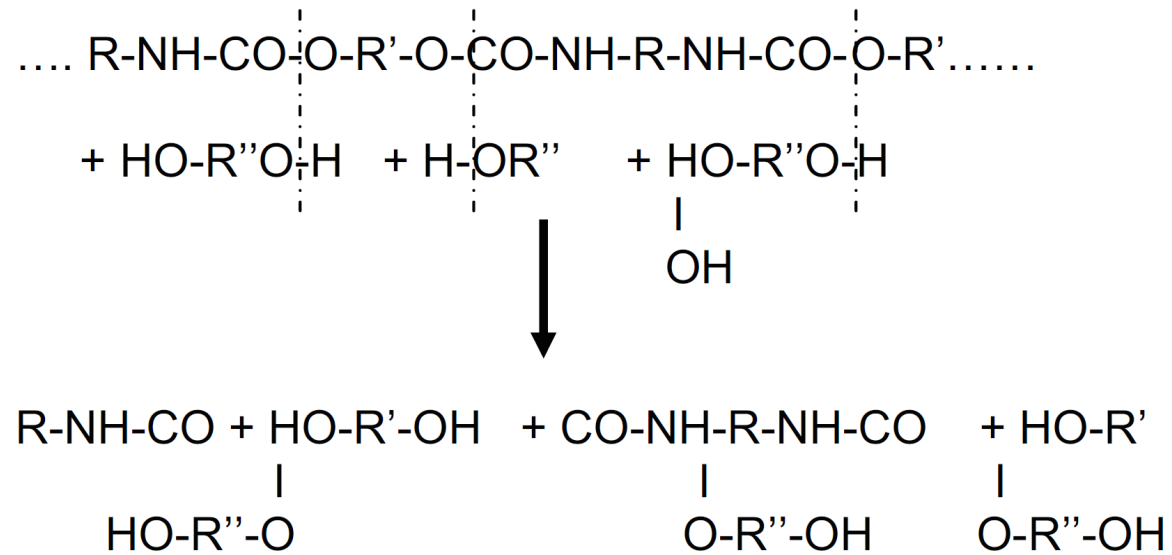


Abb. 4 Pfade der chemischen Depolymerisation von PET. (Nach Clarke et al. 1999)

Source: Österr Wasser- und Abfallw 2020 · 72:47–60; <https://doi.org/10.1007/s00506-019-00628-w>

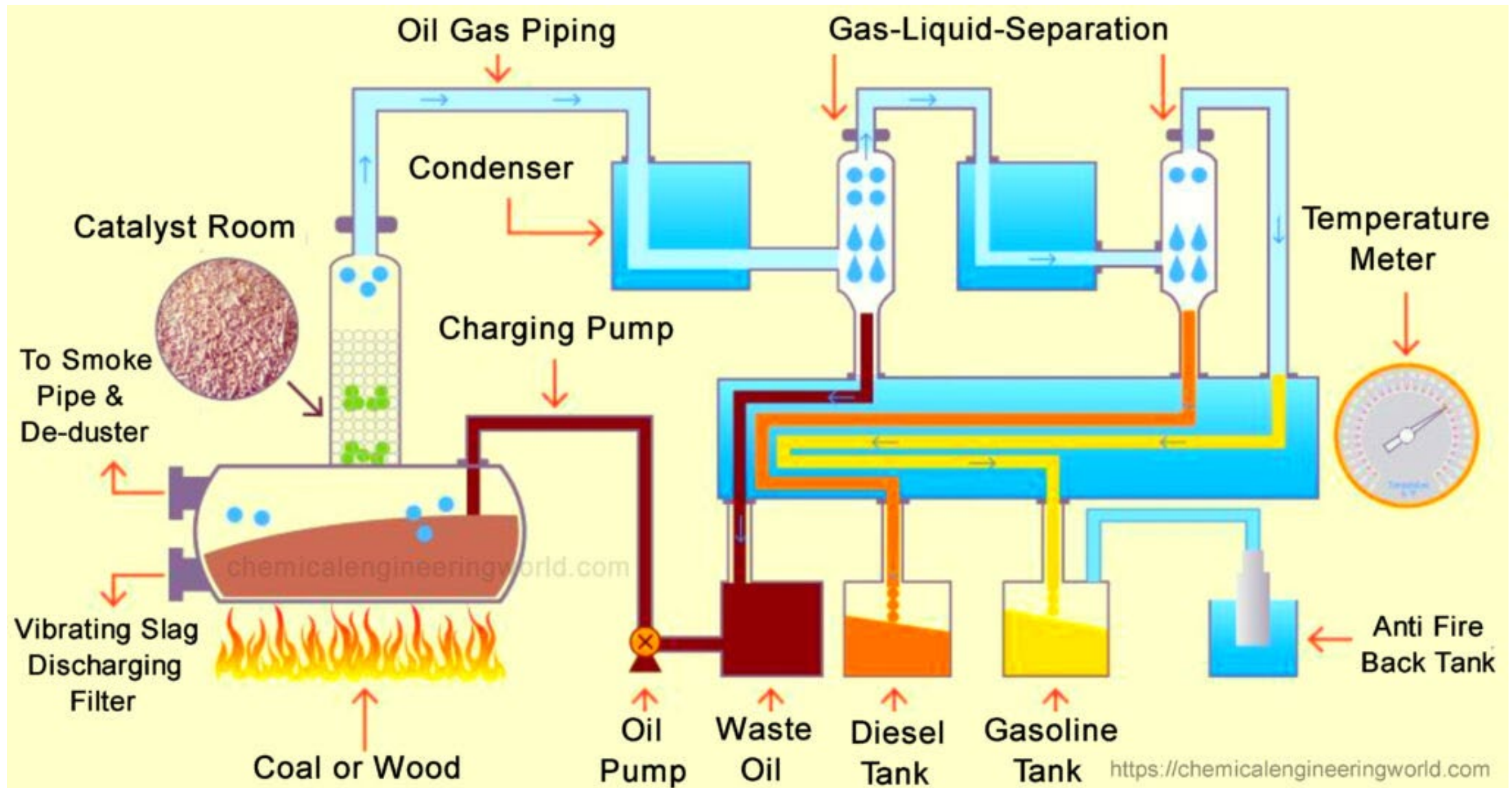
Chemical Recycling of PUR

Alcoholysis (glycolysis) of PUR production waste



Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Catalytic depolymerization



Hydrogen assisted depolymerization

Hydrogenation

- Step 1: Preheating of feed to about 400 °C under inert atmosphere
- Step 2: Hydrogenation
 - 200 - 250 bar
 - ca. 450 °C
 - H₂
- Scission of hydrocarbon chain leading to pure oil fractions - Syncrude
- PVC can be tolerated
- Economics difficult

Polyolefine: $-\text{CH}_2-\text{CH}_2-\text{CH}_2- \rightarrow -\text{CH}_3 + -\text{CH}_3-\text{CH}_2-$

Polyamid: $-\text{CH}_2-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\underset{\text{H}}{\underset{|}{\text{N}}}-\text{CH}- \rightarrow -\text{CH}_2-\text{CH}_3 + \text{H}_2\text{O} + \text{NH}_3 + \text{H}_3\text{C}-$

Polyester: $-\text{CH}_2-\text{C}-\text{O}-\text{CH}_2- \rightarrow -\text{CH}_3 + \text{CH}_4 + 2 \text{H}_2\text{O} + \text{H}_3\text{C}-$

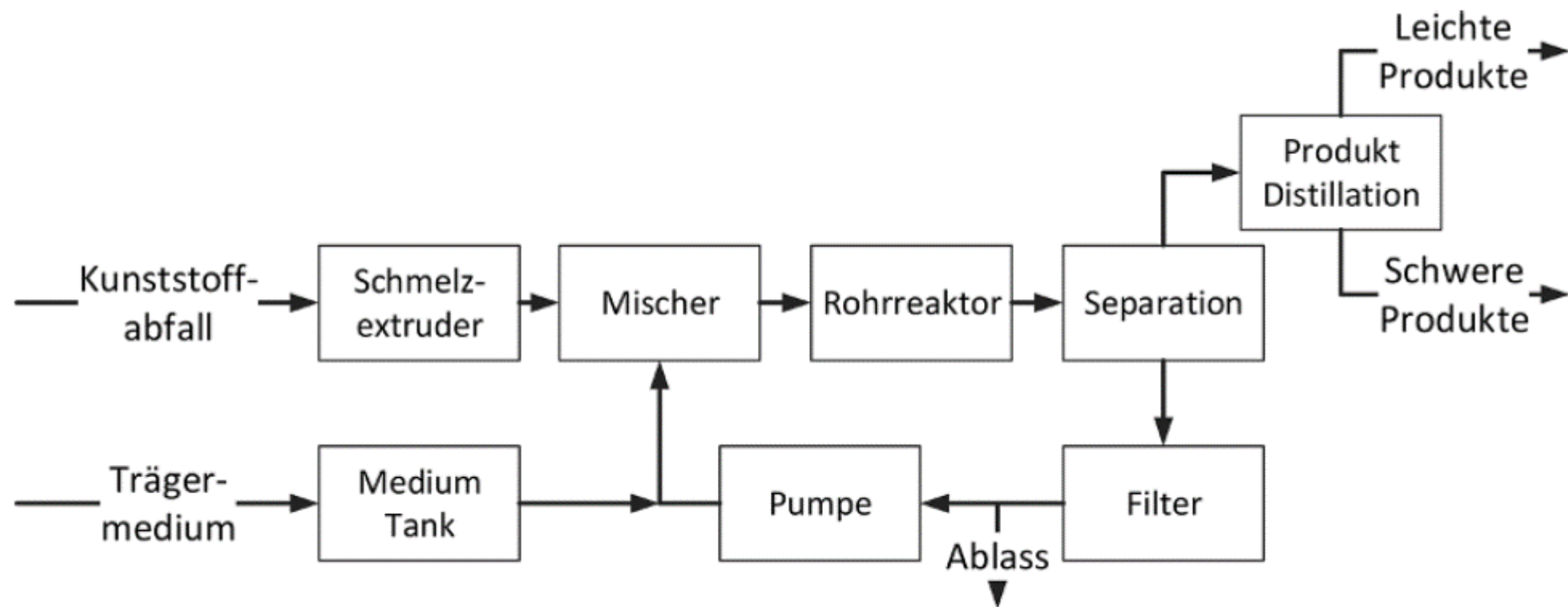
Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Pyrolysis of polymers

- Depolymerization of polymers at 500 - 800 °C under inert atmosphere
- Thermal decomposition of polymeric materials
- Products
 - Volatile products rich in H₂ (gas and oil)
 - Gas: 30 – 55 %
 - Oil: 40 – 50 %
 - Solid residues (coke), especially for rubber, PC and PF materials
- PVC is problematic

Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Pyrolysis of polymers: OMV ReOil process



Source: Österr Wasser- und Abfallw 2020 · 72:47–60; <https://doi.org/10.1007/s00506-019-00628-w>

Gasification of polymers

- Gasification of solid waste polymers
 - O_2
 - Water vapor
 - CO_2
 - H_2 (seldom)
- Product gas
 - CH_4
 - CO/H_2
- Gasification always comes along with incineration of polymer waste
- Gas products can be used to produce heat and/or electricity or use the Syngas for chemical processes (e.g. MeOH)

Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

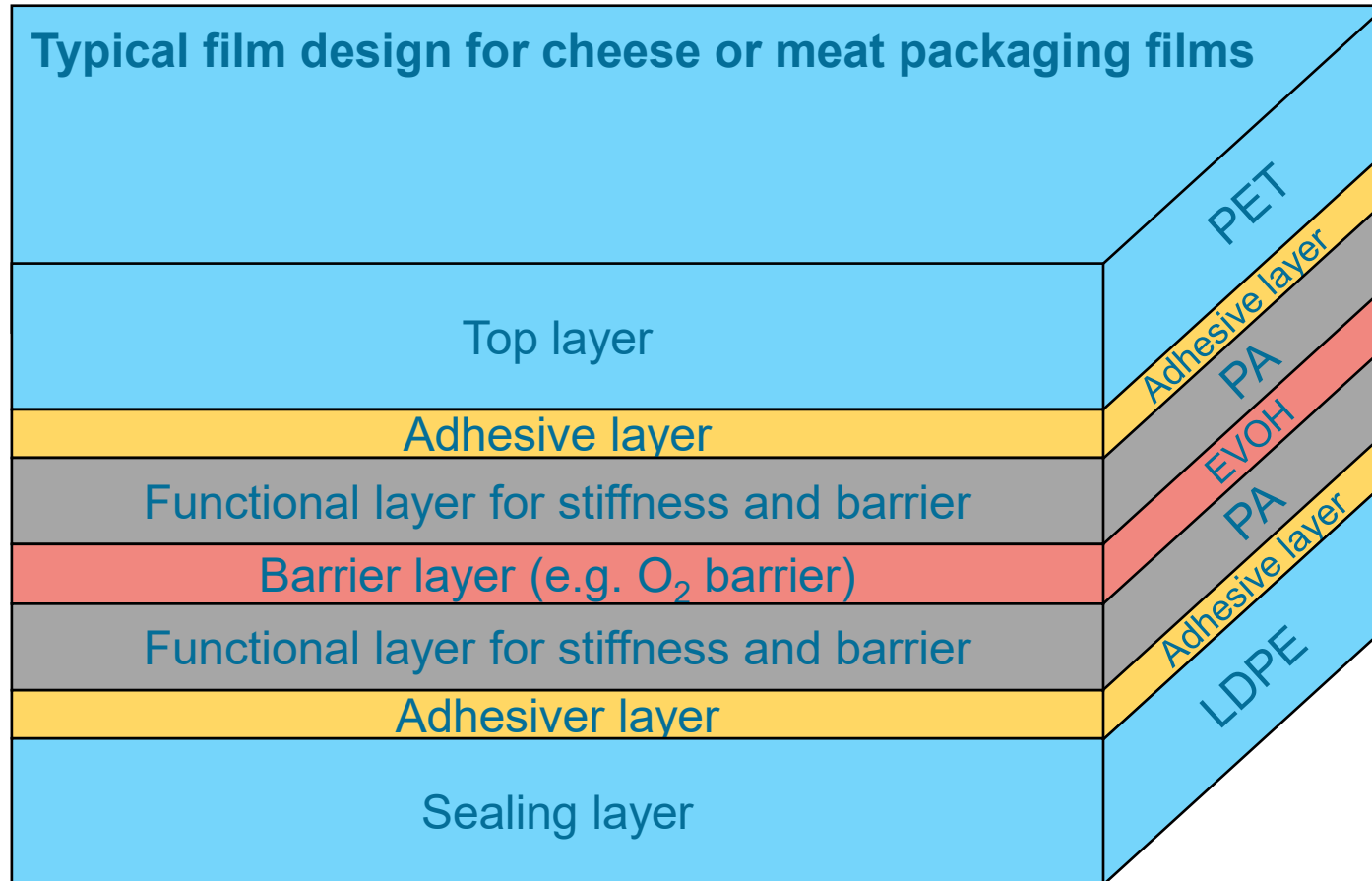
Selective dissolution of polymers

Products (waste streams) where a selective dissolution makes sense

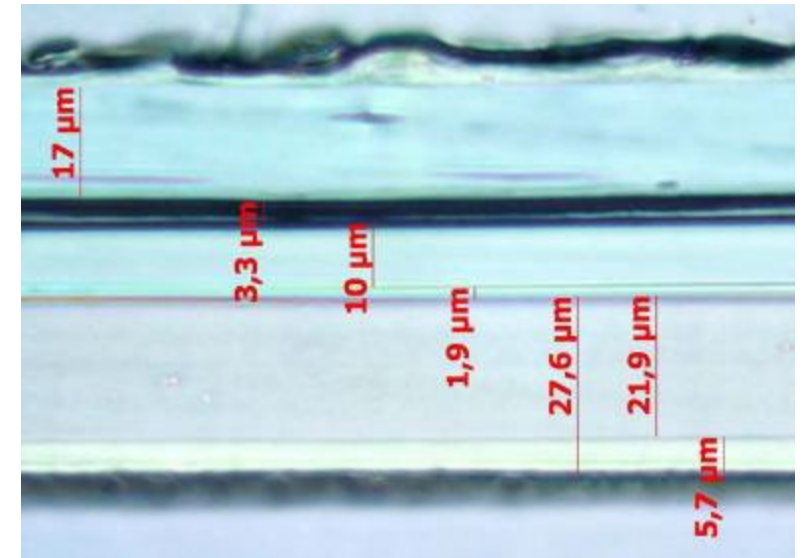
- Multilayer films
- PVC cable insulation (low voltage cables) or materials containing PVC
- ABS
- EPS
- Electronic waste – separation of halogen based flame retardants (PS, ABS)
 - Mobil phones: selective separation of PC and ABS
- Safety glass (Polyvinylbutyrate)
- Mixed PET materials

Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Multilayer film for food packaging



- Enabling functional properties like O₂ or H₂O barrier
- Prolonging shelf life time of food → reducing food waste
- Reducing packaging weight



Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Selective dissolution of polymers

Dissolution of plastics by means of organic solvents

- Typical solvents:
 - THF
 - Xylene
 - Methyl ethyl ketone (MEK)
 - Dichloromethane
 - Trichlorobenzene (TCB)
- Solubility depending on
 - Polymer
 - Temperature
- Selective dissolution by specific use of
 - Solvent
 - Temperature

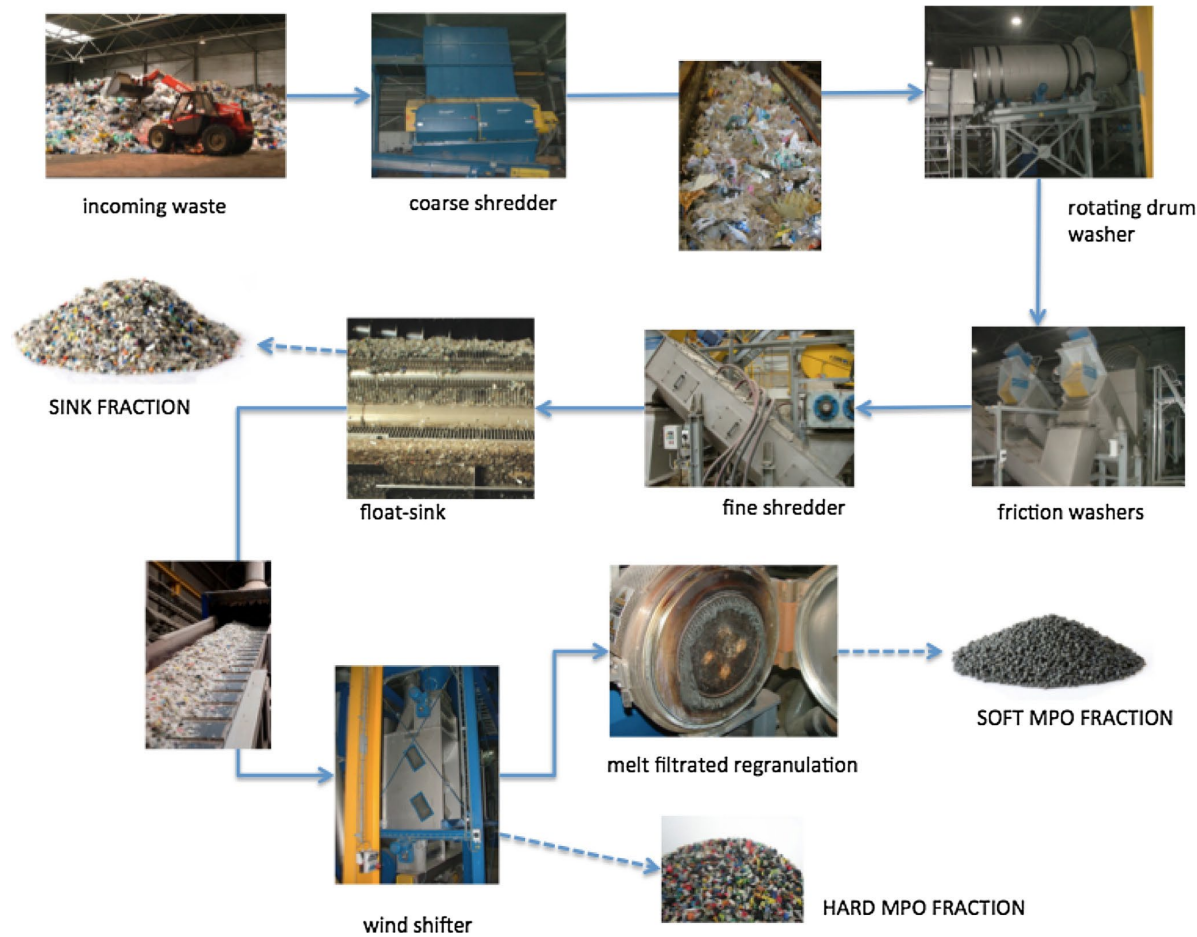
Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Selective dissolution of polymers

- PVC
 - THF and MEK at 25 °C
 - Xylene at 138 °C
- PS
 - Xylene at 25 °C
- PE-LD
 - Xylene at 75 °C
- PE-HD
 - Xylene at 105 °C
- PP
 - Xylene at 120 °C
- PET
 - Insoluble in THF and xylene

Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Mechanical recycling: process chain



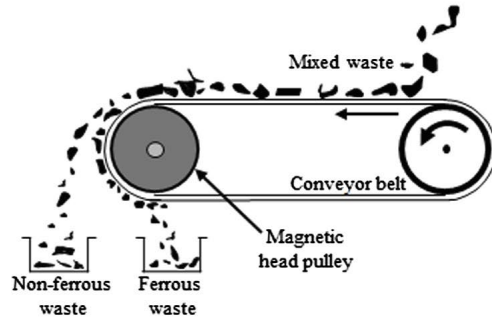
Possible equipment or process steps

- Size reduction (grinding)
- Sorting
 - Manual sorting, sieves, dry and wet density separation, metal separation, electrostatic separation, optical sorting, NIR sorting, Laser spectroscopic sorting
- Washing: Temp., medium (water, caustic, etc.), method (stirrer, friction, etc.)
- Drying (mechanical or thermal)
- Extrusion (degassing, filtration, granulation, etc.)
- Granulate post-treatment
 - Decontamination Aroma
 - Modification, etc.

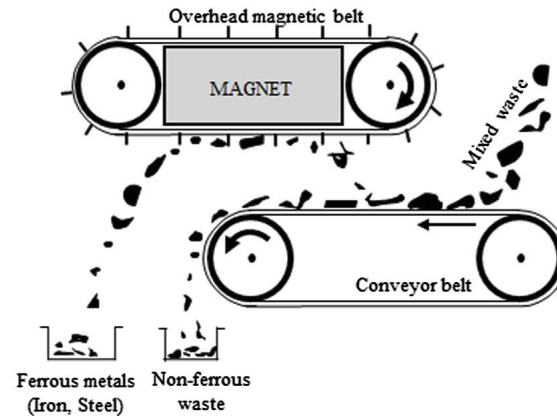
Source: K. Ragaert et al. / Waste Management 69 (2017) 24–58

Mechanical recycling: sorting

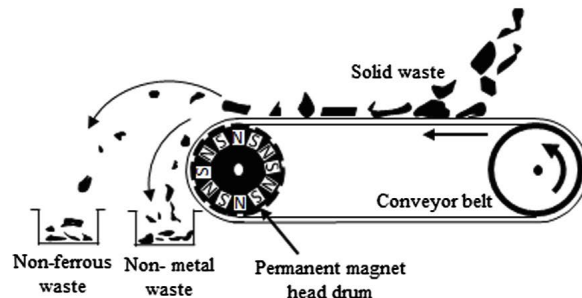
(e) Magnetic head pulley sort



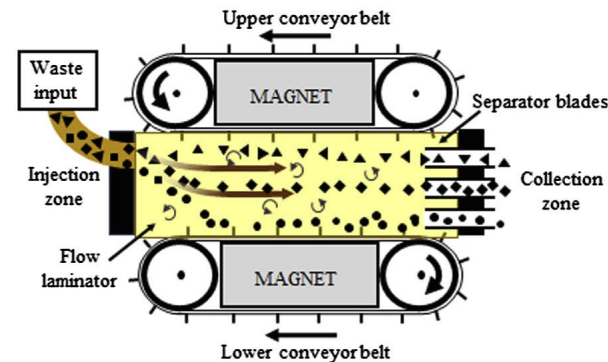
(f) Magnetic overhead belt sort



(g) Eddy current sort



(h) Magnetic density separation (MDS)

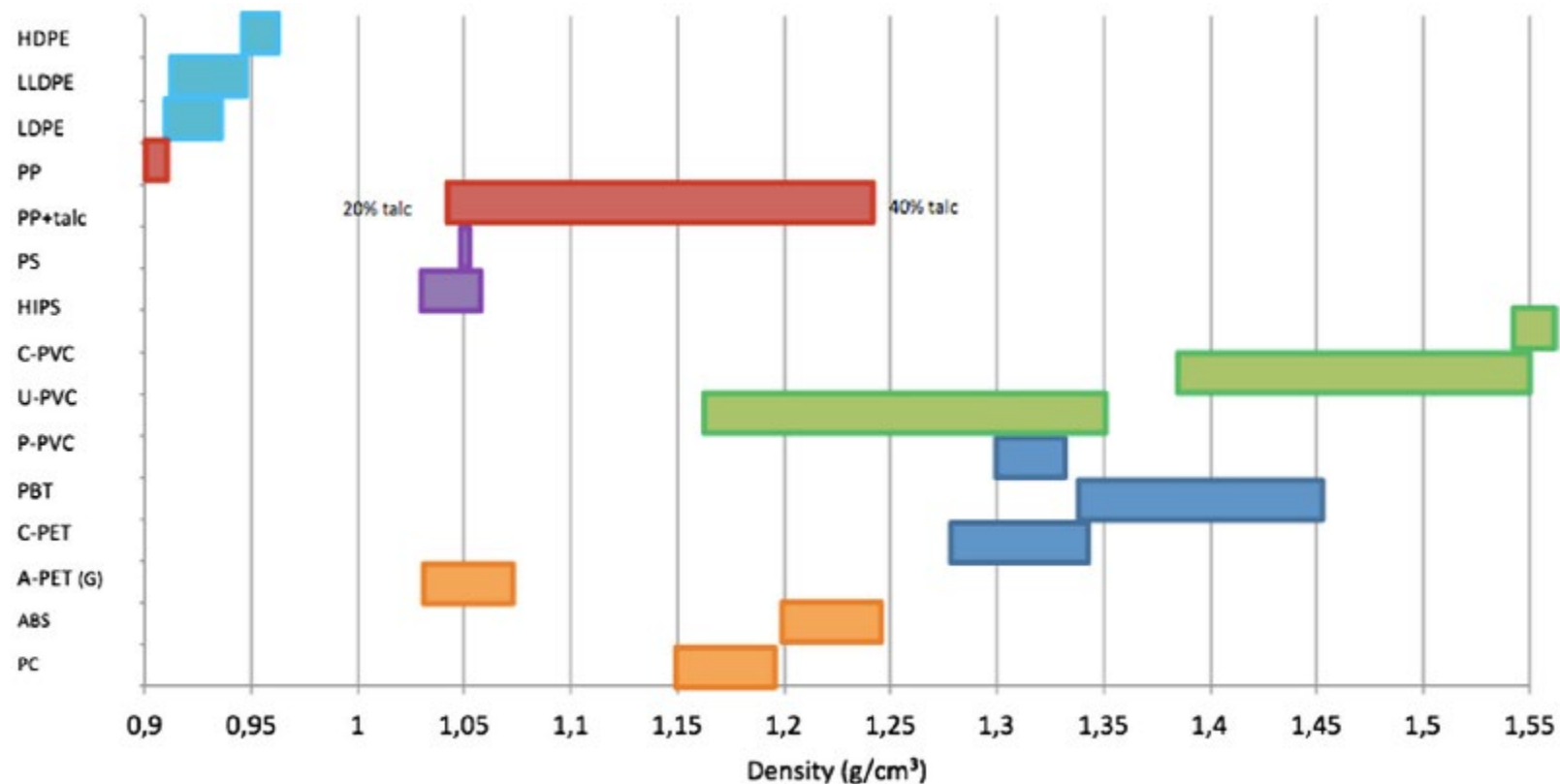


Sorting based on

- Magnetism (ferrous metals)
- Eddy current (non-ferrous metals)
- Density
- Magnetic density separation (MDS)

Source: S.P. Gundupalli et al. / Waste Management 60 (2017) 56–74

Mechanical recycling: sorting



Sorting based on density

- Swim sink sorting

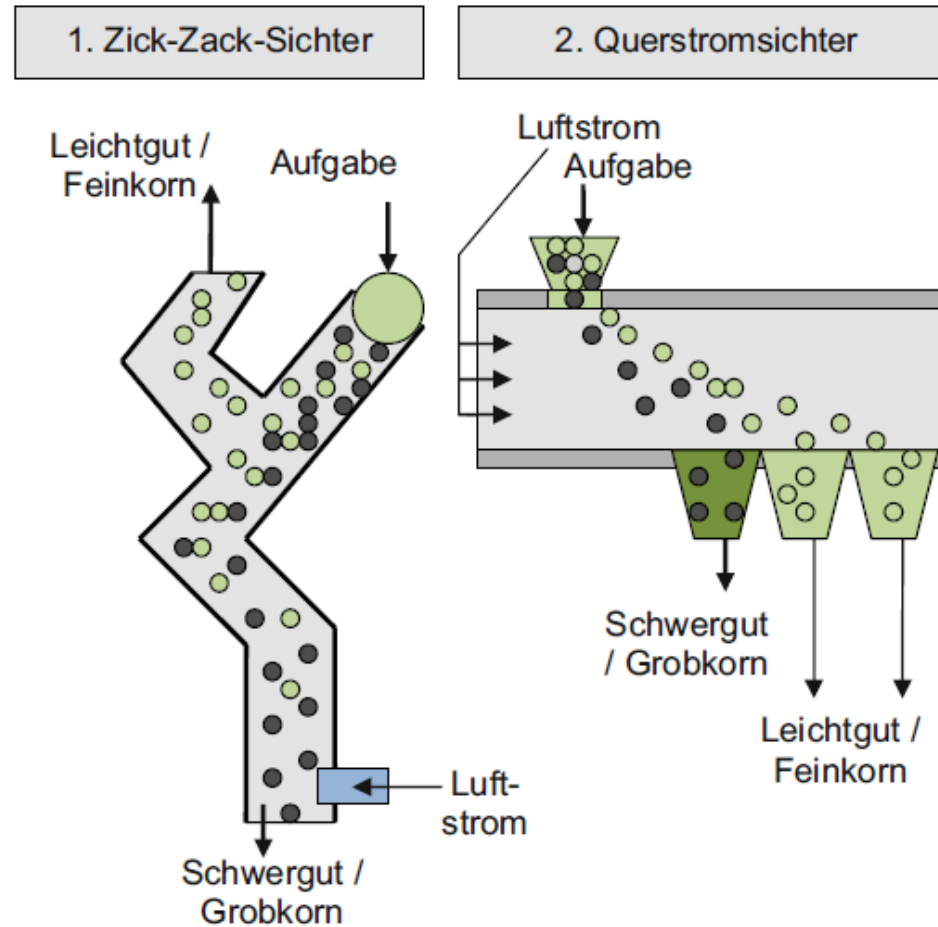
Problems

- Overlapping density ranges in plastics
- Addition of fillers & reinforcing materials complicates grading

Source: K. Ragaert et al. / Waste Management 69 (2017) 24–58

Mechanical recycling: sorting

Air classifier



Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

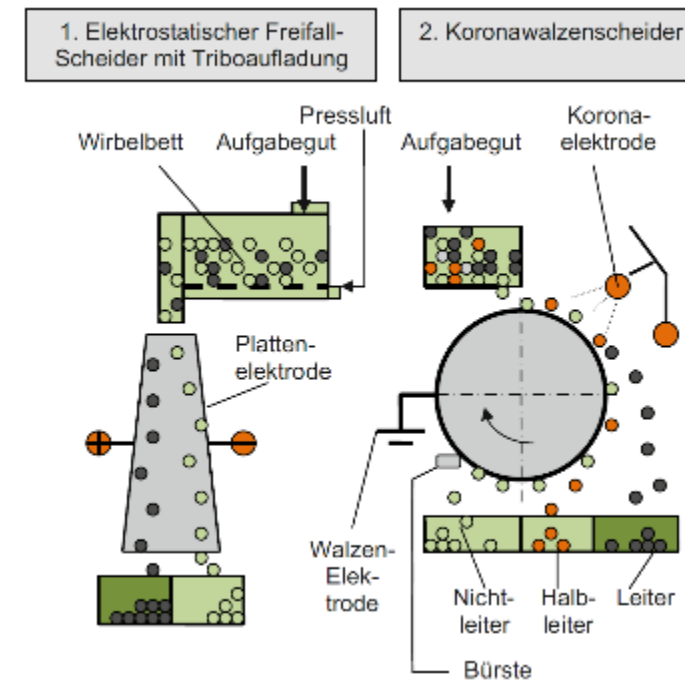
Electrostatic classification

Fields of application

- Plastic / plastic: separation of blends HDPE/PP, PET/PVC, PVC/rubber, PP/PS, ABS/PMMA, PVC/PE
- Metal / plastic: separation of metal particles in the range of a few hundred μm to less mm, complement to eddy current sorting in the fine particle range

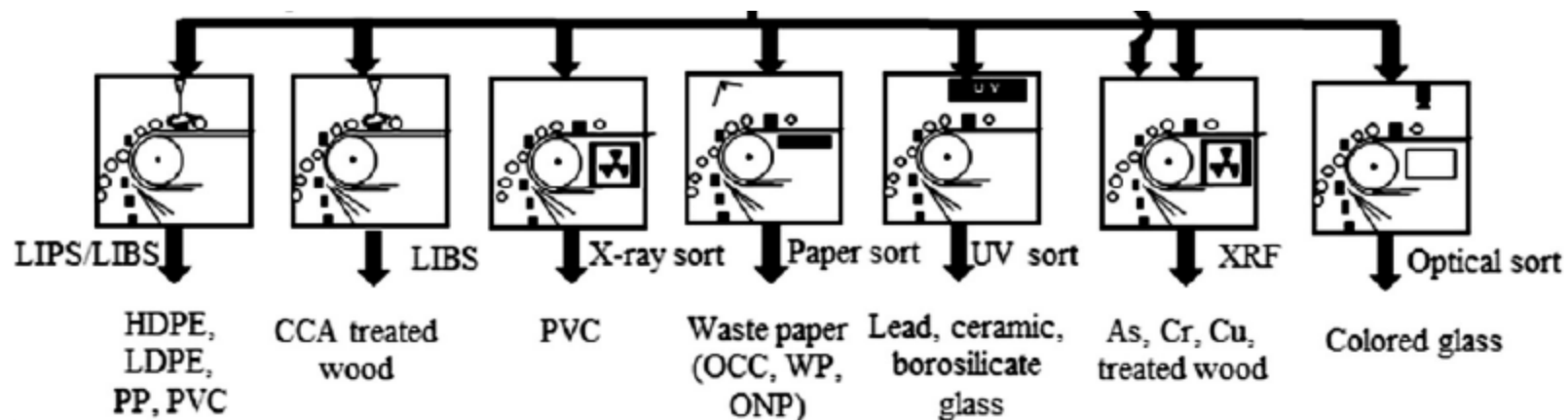
Tab. 3.8 Triboelektrische Aufladungsreihe von Kunststoffen nach Brück [3,21]

Ladungstendenz	Kunststoff	Kurzzeichen	polare Gruppe
+++	Polyethylenimin	PEI	-NH-
++	Polyethylenoxid	PEO	-O-
+	Polyurethan	PUR	N CO ₂
	Polymethylmethacrylat	PMMA	O CO
	Polycarbonat	PC	-O-CO ₂
	Celluloseacetat	CA	
	Polyamid	PA	-N-CO
	Polyacrylnitril	PAN	-CN
	Polystyrol	PS	
	Polyethylen	PE	
	Polypropylen	PP	
	Polyethylenterephthalat	PET	
	Chlorkautschuk	RUC	-Cl
-	Polyvinylidenchlorid	PVDC	Cl
--	Polyvinylchlorid	PVC	Cl
---	Polytetrafluorethylen	PTFE	-F



Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Sorting assisted by analytics

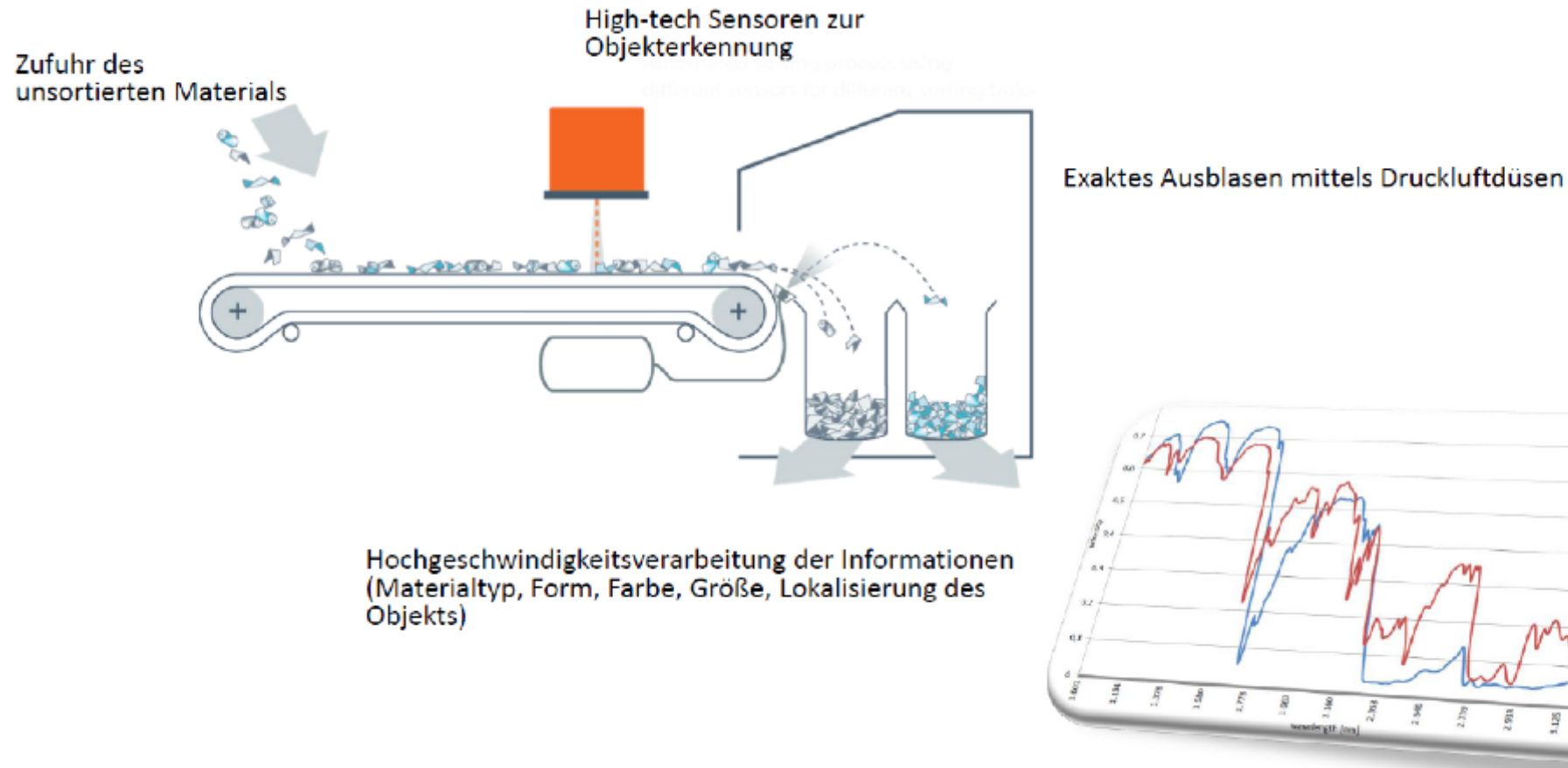


Sorting based on

- Radiation absorption or excitation
- Shape
- Size
- Color

Source: S.P. Gundupalli et al. / Waste Management 60 (2017) 56–74

Sorting assisted by NIR



Source: TOMRA, Vortrag Lisa Höflechner, Treffpunkt Kunststoffrecycling 2.0, 17.09.2020

Sorting assisted by analytics

- Near-infrared spectroscopy (NIR 800 - 2500 nm)
 - Determination of molecular composition by measuring the reflection radiation of excited molecules
 - Use for differentiation of plastic types (PET, PE, PP, PS, PVC, etc.)
- Visual spectrometry (VIS) incl. fluorescence measurement (UV-VIS)
 - Recognition of colors
- Color line scan camera (CCD camera)
 - Measurement of reflection and transmission
 - Evaluation according to colors (RGB), shape, size, surface structure
- Electromagnetic induction
 - Determination of conductivity of particles and thereby identification of metals
 - Metal detectors, electromagnetic sensors (EM), induction sensors (IND)
- X-ray transmission measurement (XRT) with spectral resolution (RSA)
 - Hard X-rays can be used to identify materials based on the atomic density of their elements
 - The use of two energy ranges allows the elimination of the influence of material thickness

Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Sorting assisted by analytics

- X-ray fluorescence spectrometry (XRF)
 - Measurement of an element specific fluorescence radiation (only for particles with centimeter dimensions).
 - Application for lead glass and glass ceramics
- Laser Induced Breakdown Spectroscopy (LIBS)
 - Point material evaporation and spectroscopy (elemental analysis)
 - Use for metals and metal alloys (differentiation of cast and wrought aluminum alloys, alloyed steels), black plastics, refractory materials
 - Only for materials that can be fully described by surface or near-surface characterizations
- Hyperspectral Imaging (HIS)
 - Divides spectrum into several hundred wavelength channels
 - VIS, NIR, MIR, Raman
 - Reflected, transmitted or emitted light is detected
 - Over entire conveyor bandwidth

Source: Martens, H. & Goldmann, D. (2016): Recyclingtechnik - Fachbuch für Lehre und Praxis, 2. Auflage, Springer Vieweg, DOI: 10.1007/978-3-658-02786-5

Mechanical recycling: washing



Washing lines

- For high quality plastic waste processing necessary

Variation possibilities

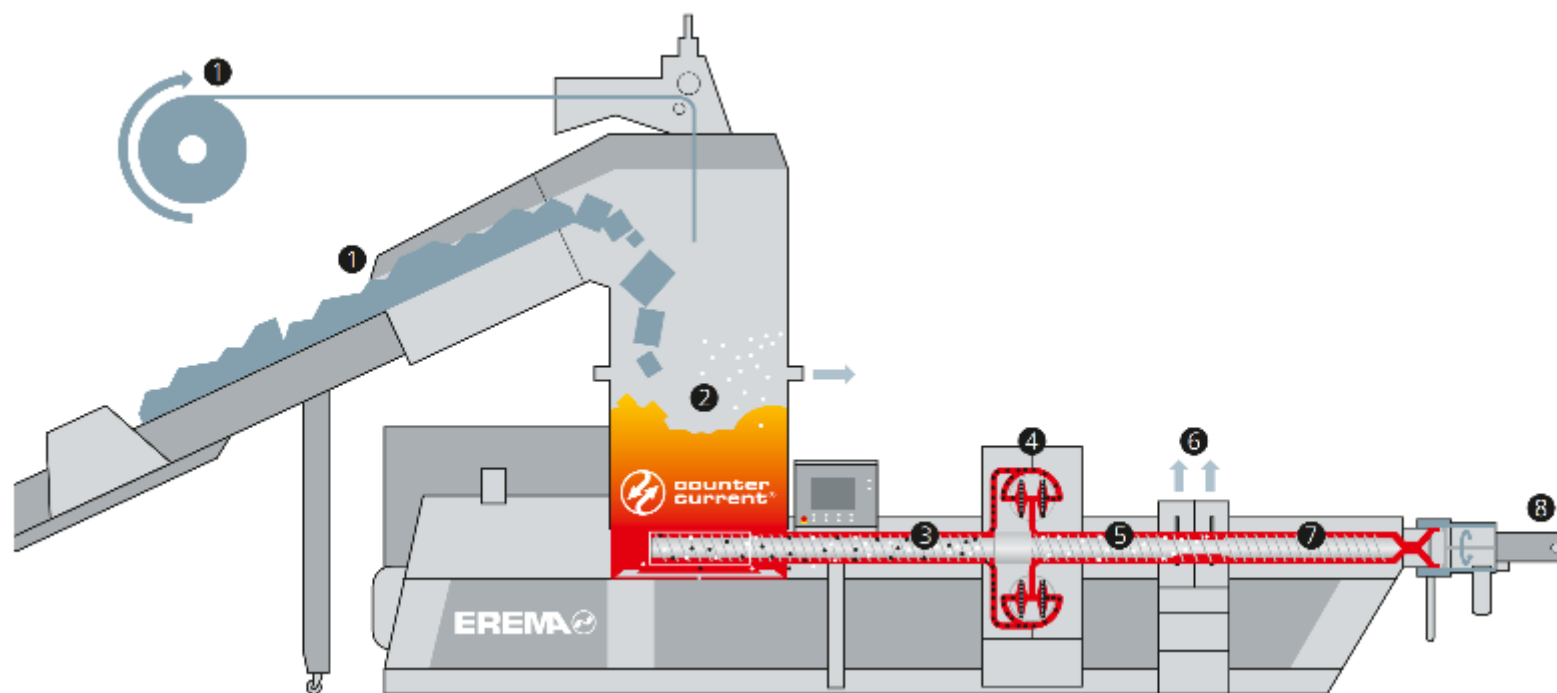
- Medium (e.g. water, detergent, solvent, etc.)
- Cold wash, hot wash or a mix
- Spray or rinse water
- Rotary drum washing
- Friction wash

Source: Lindner Washtech, Vortrag Marcel Willberg, Treffpunkt Kunststoffrecycling 2.0, 17.09.2020

Mechanical recycling: washing

Medium	Advantages	Disadvantages
Water	+ high cleaning capacity + simultaneous sorting possible + relatively cheap	- large quantities required - waste or waste water treatment - drying step required
Solvent (organic)	+ high cleaning capacity with regard to colors, apolar impurities, etc. + simultaneous sorting possible + selective solution (e.g. multilayer)	- high costs - environmental compatibility? - contamination by solvents
Dry cleaning	+ no waste or dirty water + no solvents + no drying	- limited cleaning efficiency

Mechanical recycling: material conversion



1. Feeding
2. Cutter compactor
3. Extruder screw
4. Fully automatic, self-cleaning filters
5. Final homogenization of the melt
6. Degassing zone
7. Discharge zone
8. Pelletizing zone

Source: Erema Group, www.erima.com, 2020; <https://www.youtube.com/watch?v=2mY1a9NrzVA&t=12s>

Mechanical recycling: post treatment

Decontamination

- Removal of highly volatile components
- Odor optimization

Condensation

- Adjustment and homogenization of molecular weight (viscosity number (IV))
- Working against material degradation
- Only possible with polycondensates

Material modification

- Stabilization
- Coloration
- Generation of blends with virgin materials
- etc.

Thanks for listening

